Baghdad Proposal

Urban railway and Iraqi manufacturing
Future national development

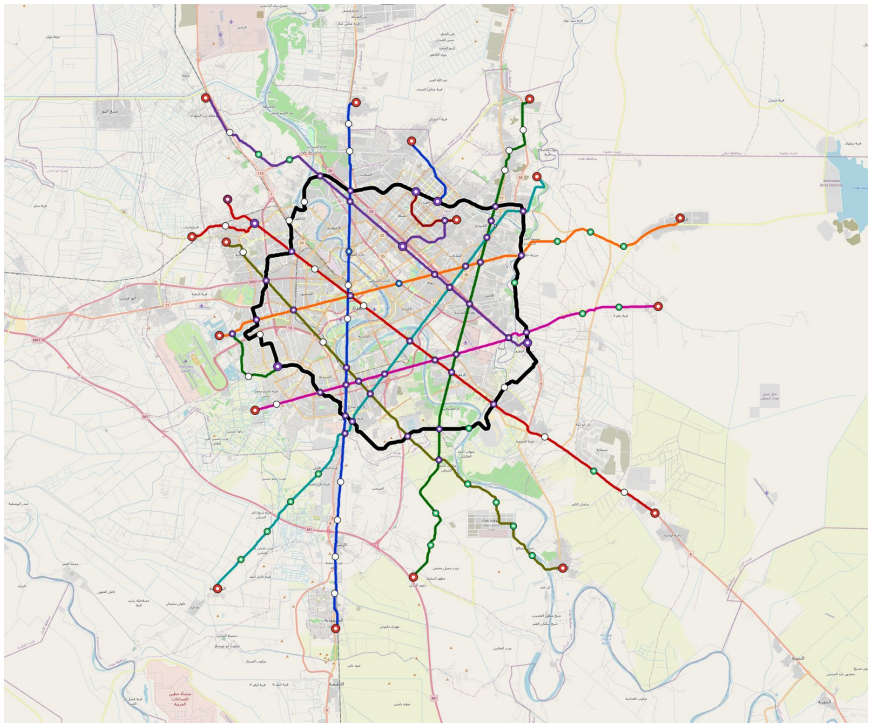
OpenSourceRail · Planning baseline 2026-10-09

Retained conventional wheel/rail comparator — six-car Baghdad planning

Configuration: retained Baghdad proposal. Financial scope: historical wheel/rail finance; not the current v0.5 maglev budget.

Prepared for the prospective Iraqi sponsor, Baghdad authorities, operating organisation and financing partners.

Baghdad finance only. National expansion is a separate future development programme. Construction and operating approval remain pending.



Baghdad planning network

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Baghdad Proposal — retained wheel/rail comparator

Source: cities/catalogue/west-asia/Iraq/Baghdad/BAGHDAD-PROPOSAL.md

Architecture: conventional wheel/rail. Configuration: retained six-car Baghdad proposal. Financial scope: historical wheel/rail programme; no acceptance or cost transfer to v0.5 ferrite maglev.

OpenSourceRail proposes an owner led feasibility and front end engineering programme for Baghdad, with Iraqi train manufacture and local infrastructure delivery. This proposal brings the Baghdad network, railway systems, operating organisation, delivery evidence and financing together, and sets out a separate path for future national development. It is addressed to the prospective Iraqi public sponsor, Baghdad authorities, operating organisation and financing partners; no appointment or financing commitment is asserted.

The latest programme recalculation sizes 3,523 permanent operating FTE and 14 line-local depots for all 809 six-car trainsets. With completed components bought, the base planning allowance is USD 8.795bn. The unquoted positive-margin local-production option gives a base planning allowance of USD 8.667bn, imported invoice exposure of USD 2.016bn and 23.27% USD capital intensity. It records IQD 0.000tn residual unsourced support after assumed facilities and retains IQD 0.000tn total debt at the horizon. The assumed facilities cover the modelled cash requirements and repay debt within this scenario; demand, funding and unpriced scope remain unaccepted. Financing alternatives remain sensitivities, with no lender commitment.

The earlier delivery-cost continuation remains a comparator: USD 8.135bn capital, before still-unpriced scope, and IQD 0.000tn unpaid gap debt. The original financial schedules below likewise remain controlled reference cases. Debt clearance in an older reference is not the current feasibility conclusion; higher fares and conditional additional funding are tested separately, with no adopted tariff or lender commitment.

The current planning network is 14 lines, 513.3 km of double track route, 159 stations and 809 six car trainsets. Regenerated original-reference accounting, including one final-assembly plant and its EPC, is USD 7.951 billion equivalent; it does not include the latest scope replacement. The direct government capital contribution remains 25% of each revised case. Imports are financed 50% government USD cash and 50% proposed Chinese USD credit; all remaining capital cash, bonds and bank debt are IQD.

The immediate decision proposed is to establish a sponsor, commission survey and demand work, develop the first operable line and plant packages, qualify suppliers and obtain executable financing terms. Construction and operating release require the recorded physical and approval gates. The current resource constrained plan reaches first line revenue in month 40 and full operation in month 81 after financial close. That long schedule is a material design and delivery problem to resolve; this proposal does not substitute a five year promise.

How to read the proposal

The current all-system design index lists actual geometry, component records, software allocations and open design gates. The city station federation places retained station geometry at the proposed city locations; elevations, orientations, interfaces and site acceptance remain provisional. These source-bound outputs replace stale generated views without turning planning evidence into manufacturing approval.

The first part states the integrated proposal and the decisions it needs. The next part prints the detailed network registers and six month financing schedules. The technical appendices reproduce every current Baghdad Markdown report, the national brief and selected shared standards. The supporting ZIP preserves the full controlled Baghdad files, every Baghdad financing spreadsheet, the operations payload, national city design inputs and the cited shared references. The source inventory and manifest identify exact file bytes.

Baghdad is the only financed city. National expansion is a future strategic option with its own budgets and approvals. The national totals and generic city pages retain original catalogue assumptions; the latest Baghdad study does not silently reprice or finance other cities. Procurement origin, loan currency and reporting currency are different measures. USD equivalents use the historical model anchor of IQD 1,300/USD; this is not a current execution quote. All prices, demand, debt terms and rights proceeds remain planning assumptions. Older generic foreign turnkey examples reproduced in source appendices are separate from the historical 148 km comparison and the scheduled Baghdad financing cases.

Latest scope, local production and matched funding cases

The 159 stations have two concurrent staff during two normal eight-hour shifts, giving 636 daily shift assignments. The retained 20.5-hour railway also funds 4.5 hours of late cover. Weekly/leave/training/sickness cover gives 1572 station FTE, included in the 3523 permanent railway FTE. Loaded annual payroll is IQD 83.415bn before subsequent OPEX inflation. The general pay floor is IQD 1,175,455/month: 1.5 times the 2021 employee median indexed to 2026 at an assumed 5% a year. Technical/supervisor/senior/director grades use 2.25/3/4/5 times the same proxy. This is not a newly observed national median and does not change the fare-income denominator.

Each line has one depot; all spare/reserve trains are included. Storage uses the actual 111 m train plus 10 m clearance; workshop bays follow bay-hour workload separately. The 14 sites total USD 289.645m and replace the old USD 8m allowance once. Itemised storage/workshop/access tracks, points, civil shells, process equipment, services, wash plants, wheel lathes, stores, rescue/isolation and retained energy stock are costed. Land/title, utilities, actual foundations and installed charging/grid upgrades remain open; launch/throat conflicts need an operating replay.

The order requires 9,708 bogies, 9,708 motor/inverter sets, 4,854 battery packs (1.092 GWh gross), 19,416 door cassettes and 29,124 window cassettes. Final assembly employs 1215 production and 183 support FTE for 59 scheduled paid months. Upstream facilities are sized to the train factory rate and price imported process machinery, local buildings/materials, residual imported inputs, qualification and their full paid establishment. Selected processes (bogie fabrication, motor assembly, battery-pack assembly, door manufacture) have positive whole-order margins under the unquoted assumptions. The remaining processes (window manufacture) stay bought in this case; their Baghdad-only whole-order margins are shown in the make/buy register. Imported cells/BMS, inverters, wheels/axles/bearings and other safety parts remain; no cell gigafactory or future national sales credit is assumed.

The main design uses the reworked central elevated routes and 56.73% elevated track across Baghdad. Further outer civil alternatives permit up to 65% elevated and screen 7.924 km of additional at-grade conversion around exceptional curves, reaching 58.27% elevated. These outer candidate intervals are not adopted or surveyed. Elevation alone removes no horizontal bend. Added viaduct carries its base allowance; search penalties are excluded from money and the earlier penalty-removal cases are retired. Special designs, actual foundations, land and consequential installed prices remain unresolved.

Matched case	CAPEX USD bn	USD intensity	Unsourced support IQD tn	Terminal all debt IQD tn	Defaulted junior vintages
additional_elevation_base_allowance	8.731	23.20%	0.000	0.000	0
construction_wage_content_stress	9.267	22.77%	0.000	0.000	0
integrated_fare_1_25_boardings	8.667	23.27%	0.000	0.000	0
integrated_fare_1_5_boardings	8.667	23.27%	0.406	10.554	0
integrated_fare_2_boardings	8.667	23.27%	12.070	13.000	0
local_all	8.677	23.12%	0.000	0.000	0
local_positive	8.667	23.27%	0.000	0.000	0
local_positive_commercial_gap	8.667	23.27%	0.000	0.000	0
local_positive_mezzanine	8.667	23.27%	0.000	0.000	81
local_positive_mezzanine_stress	8.667	23.27%	0.000	21.346	83
local_positive_raw_price_stress	8.770	23.74%	0.000	0.000	0
local_positive_supplier_delay	8.667	23.27%	0.000	0.000	0
revised_scope_buy	8.795	26.82%	0.000	0.000	0

All 13 cases have invoice-level capital registers, monthly native-currency cash/principal ledgers and six-month bond/loan placement schedules in the latest study and supporting archive. Government is 25% of total capital, with USD machinery/input downpayments on actual invoice dates and the balance allocated as local IQD appropriation. Only Chinese credit is USD debt; bonds, senior bank/gap debt and mezzanine are IQD. Six months of senior service are reserved from the first draw, with three months of OPEX and industrial working capital. This reserve policy differs from the older reference; matched cases are the valid comparison.

The capacity-led reference assumes one boarding per paid journey: a zero-transfer upper bound, not a surveyed demand forecast. The integrated-fare sensitivities use 1.25, 1.5 and 2 boardings per paid journey, charge each journey once, and retain service, energy, costs and non-fare receipts. At 1.5 boardings, the month-101 annualised operating balance before debt service is USD +140.284m. This annualises one nominal month and is not a calendar-year forecast. The demand handoff and same-alignment cost reconciliation retain the evidence needed to replace these assumptions. Population catchments alone create no fare uplift.

The following capital-only source table is for the latest positive-margin local-production senior case. Fees, interest, reserve funding and gap facilities are separate cashflows; this table reconciles to capital uses only. Ordinary and green bonds are separate placements within the same total funding requirement.

Latest capital source	Currency	Native amount	USD equivalent m
Government import cash	USD	1,008,216,375	1,008.216
Government local cash	IQD	1,505,948,192,086	1,158.422
Chinese capital credit	USD	1,008,216,375	1,008.216
Ordinary capital bonds	IQD	4,182,186,517,697	3,217.067
Green capital bonds	IQD	1,147,843,845,665	882.957
Senior bank capital credit	IQD	1,776,676,787,787	1,366.674
Conditional climate capital grant	IQD	32,500,000,000	25.000

The proposed IQD mezzanine replaces 10% of residual domestic capital borrowing, with 6% cash coupon, 4% PIK, 2% fee and a 15-year balloon. Cash coupons require senior DSCR of 1.20, funded reserves and genuine residual cash. Deferred coupons and PIK are debt. Unpaid balloons become overdue balances with separately disclosed simple-interest sensitivity, without automatic refinancing, conversion or interest-on-arrears. The case leaves IQD 0.000tn total debt and 81 defaulted draw vintages. It does not improve the unlevered company NPV of USD -3.246bn at the assumed discount rate. Grants, net development-rights receipts, green terms and concessional gap funding are uncommitted; an 8% gap-rate case exposes this dependency. Fares, fare/OPEX inflation, advertising, kiosk/rental receipts and inherited additional income are already included.

Baghdad network and population access

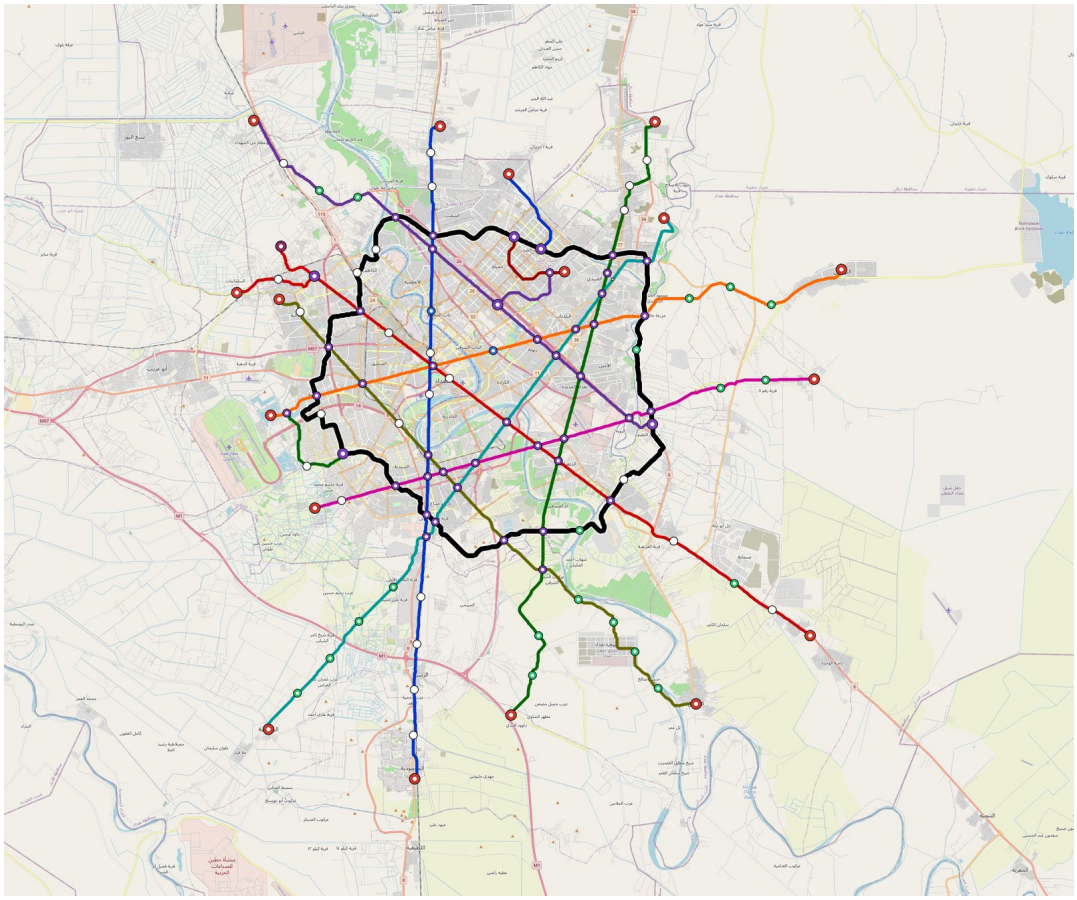
The design retains a planning population of 9,780,429. The former 40.5% routing-demand cell fraction is not population coverage and its resident multiplication is retired. Native WorldPop 2020 pixels give a bbox population of 5,985,974, with residents counted once in the union of station circles:

Radius (m)	Covered residents (2020)	Share of raster bbox population
500	472,960	7.9%
800	1,129,864	18.9%
1000	1,660,588	27.7%
1500	3,110,313	52.0%
2000	4,332,753	72.4%

These are potential radial catchments, not verified walksheds or unique passengers. The 1,500–2,000 m cases require actual walking/feeder provision. Population dates and boundaries differ from the catalogue; no rebasing or extra fare demand is assumed. River crossings, entrances, barriers and topography require validation. Reachable line pairs are 100.0% through intermediate lines; direct transfers cover 39.6%. Source counts, sensitivities and transfer paths.

The straight viaduct concept remains obstacle-unreleased. The beam/support/terrain register retains 879 nearby mapped footprints, including 110 with source height tags. It flags 60 provisional roof clashes and 635 unresolved-height checks. Low-roof overflight never clears the pier/foundation below; supports, tall-building clearance, terrain/grades, utilities, property/air rights and construction access require survey and redesign. Raising the deck needs grade-compliant approaches and revised capital; moving piers must use verified spans or an independently checked special crossing. Current financial figures include no unpriced adopted obstacle solution.

Line	Shape	Route km	Stations	Peak fleet	Total fleet	Opening month
line-1	radial	49.068	17	78	86	40
line-2	radial	52.851	14	86	95	50
line-3	radial	49.496	16	81	90	55
line-4	radial	41.932	12	69	76	59
line-5	radial	47.345	16	78	86	64
line-6	radial	52.503	16	86	95	69
line-7	radial	39.957	13	65	72	73
line-8	radial	48.137	15	80	89	78
line-9	ring	97.684	26	40	45	81
line-10	radial	6.804	3	13	15	41
line-11	radial	6.884	2	12	14	43
line-12	radial	6.586	3	13	15	42
line-13	radial	8.042	4	16	18	44
line-14	radial	6.047	2	11	13	44



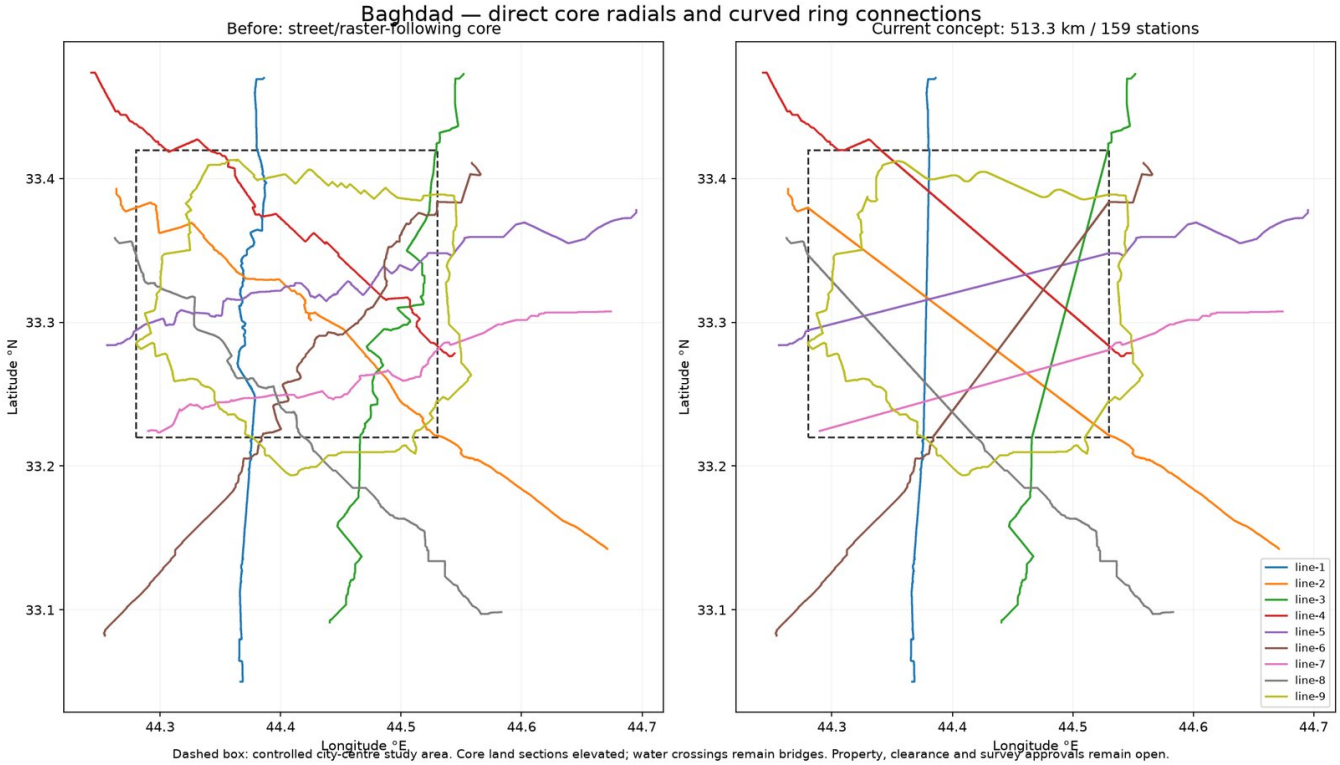
Baghdad network

Line names are controlled design identifiers. Public station names, route brands and final termini require owner approval. Chainages, coordinates, interchange platforms and fleet roles are printed in the network registers and regenerated from the adopted core corridor concept.

The service concept operates 05:30 to 02:00 with a three minute protected peak headway. The current scheduled journeys and fleet sizing are capacity led; accepted junction/authority capacity, ridership, a timetable, station crowding and degraded recovery need operating review. The 809 trainsets comprise peak, spare and cold reserve roles documented in the annex.

City-centre alignment rework

Within the controlled core rectangle (33.22–33.42°N, 44.28–44.53°E), radial corridors use straight tangents and ring connections use broad circular planning fillets where the water constraints permit. Detoured sections require a new curve and structural review. Core land sections are elevated; water crossings retain bridge classification. The final water-constrained core routes change from 301.670 to 286.225 km. This is the main design used by the updated station, fleet, civil, energy, depot, staffing, delivery and financing calculations. The earlier extra-viaduct cost-only case did not change geometry.



Before and after core routing

These corridors reserve no property or air rights. Obstacles, heritage/security constraints, surveyed clearances, pier access, utilities, transition curves, cant, vertical geometry and foundations require engineering and owner review. Outer approaches retain bends and exceptional elevated products. The coverage proxy must be measured before selecting the faster/straighter layout as an access solution; no old coverage score is carried forward.

Trains and imported component strategy

The Baghdad profile is metro 6car: 6 cars, 111 m body length, 720 passengers at nominal planning load including 120 seats, and 960 at short duration crush load. Each train has 1,350 kWh nameplate / 1,080 kWh usable LFP battery capacity, 12 traction controllers, 3,600 kW peak traction and a 50 C design ambient. These are reference profiles requiring supplier and physical qualification, not delivered fleet performance.

The original industrial scope is train assembly, body modules, fit out, wiring, coatings, inspection, testing and maintenance, with completed bogies, batteries, windows and doors bought. The latest positive-margin local-production option replaces selected completed imports with Chinese process machinery and residual inputs for Iraqi fabrication/assembly. Solar equipment, glazing and critical component inputs retain imports. Chinese supplier origin and export lender eligibility need evidence for every financed item; the entire imported basket is currently an unqualified scenario. Candidate CRRC equipment remains subject to competitive supplier selection, interface and safety qualification. There is no established CRRC partnership, quotation or endorsement.

Shared LM3 fabrication and first article documentation is reference process evidence for a three car platform. It does not qualify Baghdad's six car consist. The national programme should qualify the shared modules and then validate each consist and its interfaces, rather than treating a shared drawing as an accepted Baghdad train.

Civil infrastructure and stations

The civil screen contains 1,747 segments: 191.0 km at-grade, 291.2 km elevated, 31.1 km bridge. Route kilometres describe double track corridors; they are not track kilometres or a measurement of all sidings. Station products, 45 interchange complexes and 10 junction records are bound to the generated layout.

The proposed civil programme starts with survey control, utilities, property and access, geotechnical investigation, flood/drainage levels, alignment and station fit. Desktop soil inputs contain 7,287 sample locations and 65 missing profiles; they do not supply foundation bearing capacity, groundwater or deep stratigraphy. Structural calculations, spans, erection, movements and independent checking must follow route specific evidence. Generated alignment exports have unfitted curves, placeholder vertical profiles and undesigned cant.

Stations require accessible approaches, platforms, passenger information, fire and evacuation design, fare equipment, retail and advertising layouts, security, sanitation and maintenance access. Platform and station access standards must be

checked against the final six car envelope and passenger demand. Equipment and architecture references do not establish installed compliance.

The original USD 8m depot allowance is replaced by the regenerated full-fleet line depots in the latest recalculation above. Surveyed land, foundation/electrical interfaces, fire/security release and supplier quotations remain open. Workshop bays cannot be counted as overnight train parking. The earlier reference policy proposed two revenue trains at selected powered stations and line-local storage for remaining fleet; the latest study gives no capacity credit to station parking. Usable tracks, charging, protected morning release, evening repositioning and repeated-day replay still require acceptance. The depot and stabling appendices retain these failures explicitly.

Manufactured viaduct alternatives and installed cost

The manufactured-viaduct comparison covers Pi20 and Pi25 with two bearing/connection schemes, plus an OSR-US constrained-access option. Baghdad's infrastructure load seed now explicitly requires the complete 24-axle, 111 m six-car train, with supplier axle positions and loaded distribution still unresolved. A link slab retains independent girder-end bearings; shared bearings require a checked structural continuity connection and staged load path.

Every one of the 1127 elevated segments, including 49 exceptional segments, has a comparison record. The corrected elevated base allowance is USD 2.839bn. Search deterrents are excluded from monetary estimates. Unknown special/segmental increments remain outside the base allowance; removing the old segment-wide score establishes no realised saving. Realignment, station movements, ground/utility investigations and installed whole-life alternatives remain unaccepted.

Retaining simple-span bearings changes the existing periodic cost index from USD 9.748m/km to USD 10.198m/km. The USD 131.042m uniform whole-elevated-network difference is an unadopted rate illustration; Pi bearing quantities are not transferred to OSR-US/special designs. The lower existing rate remains conditional on an unaccepted structural scheme.

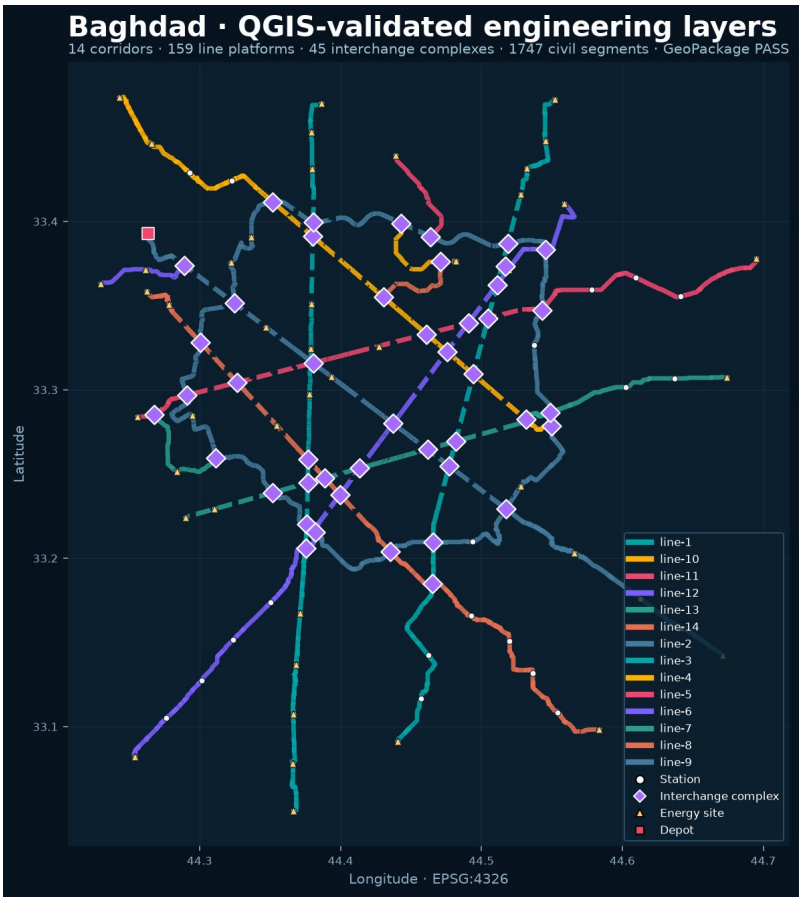
The separate financed sensitivity applies only to 269.129 km of standard Pi25, adding USD 121.108m direct and incremental EPC once. It gives USD 8.264bn capital and IQD 0.000tn terminal gap debt, with monthly/six-month financing recalculated under the same 25% government and USD/IQD rules. Existing civil invoice dates and origin proportions are inherited assumptions. Connection, finite end effects, foundations, actual import eligibility and other consequential costs remain unpriced; the original full-fleet case is preserved as a comparator.

Finite supports, unknown foundation lengths, complete member/hook mass gates, configured erection bids, per-item invoice currencies and eight production fronts are included in the supporting data. Installed-price totals remain unknown while scope is unpriced. Accepted complete double-track bays/week, first beam/pier/connection trials and independent design release must precede programme and budget selection. No literature savings or 30 m product is assumed.

Energy and desert operation

The current duty model schedules 6,278 one way journeys and about 215,989 train km per day, with 2,043.4 GWh annual traction demand. It assigns 46.4 MW station/depot PV, 316.0 MWh site storage, 278.0 MW connected charging and 1,019.7 MW dedicated solar. The modelled 0.0 GWh annual residual grid/PPA import is an energy accounting result; it does not establish uninterrupted operation or installed grid independence.

Storage endurance, adverse weather, PV land, heat/dust derating, losses, supplier fire separation, actual charging duty, protection and backup import need a time resolved operating appraisal. Snapshot solver passes and grid only diagnostics do not close these gates. Battery protection and cabin/battery thermal separation must be qualified at the declared ambient and duty. No claimed unlimited battery autonomy or accepted solar islanding is used to close financing.



Engineering map

Train control and operational safety

The current architecture reference is RFC 0033, TACS runtime and committed resource control. RFC 0032's separate authority/protection prototype is superseded. The design retains committed resource ownership in the interlocking, ordered consensus decisions, onboard movement authority, automatic train protection, station departure gates and final brake requests. OCC supplies service intentions and observations; ERP and supervision cannot bypass railway authority or clear protection latches.

The reference uses three static consensus voters and two logical protection channels. Hardware placement, independent power/network domains, sensing, brake outputs and common cause analysis need assessment. Two software channels do not establish independent safety hardware. Lost required permissions, stale/invalid evidence and missing physical proving retain restrictive protection. The runtime is a synthetic process reference and not a deployed Baghdad control system or safety certificate. Existing Baghdad timetable simulations do not constitute RFC 0033 physical acceptance.

Iraqi manufacture and local economic benefit

The Baghdad plant is sized from 4,854 vehicle modules. Its base allowance is USD 303.505m plus USD 21.245m EPC, counted once outside city CAPEX. The current plant construction assumption is 390 working days before train production. Factory siting, freight access, utilities, tooling, staff, production rate, quality capacity and six car qualification require their own approved business and delivery plan.

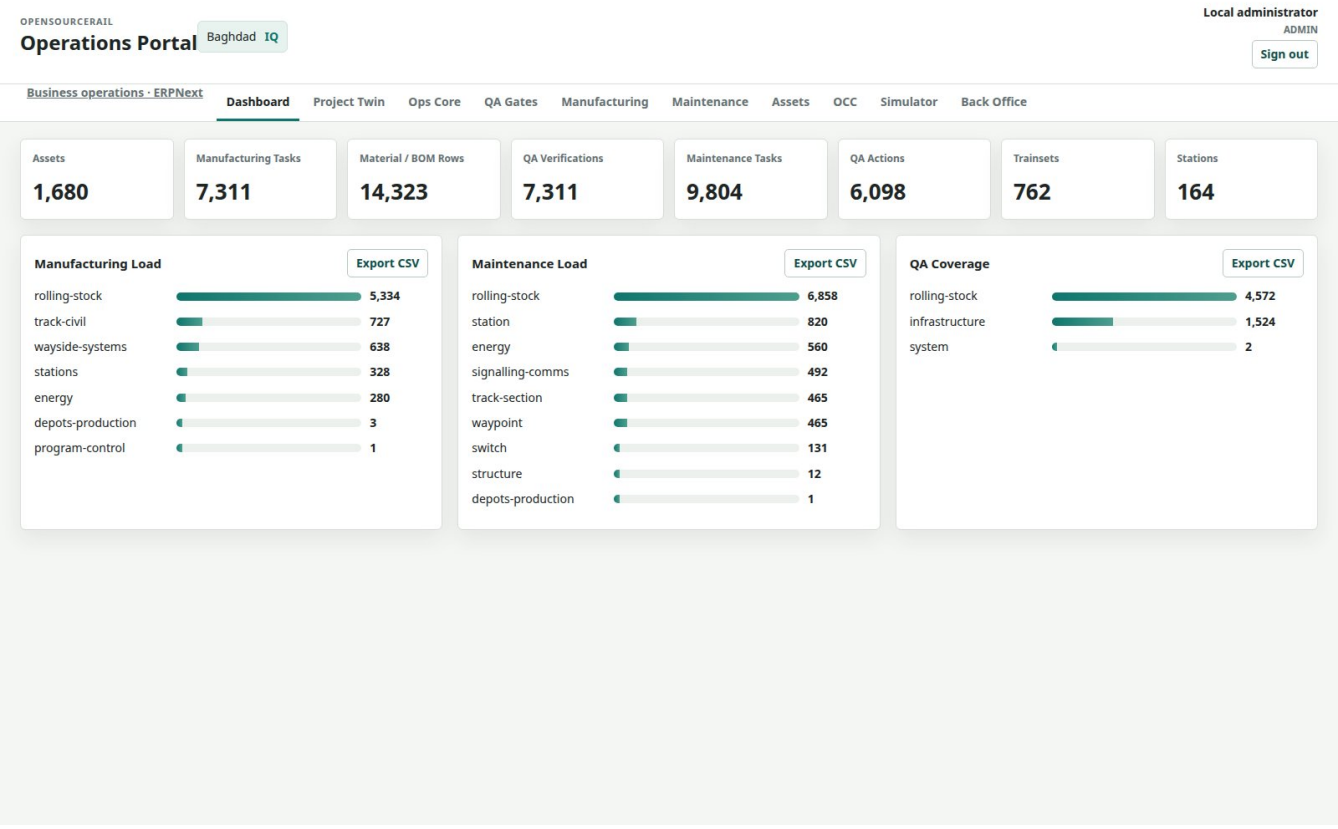
The original procurement allocation assigns USD 6.154bn equivalent locally. This is potential local expenditure, not payroll, GDP added or a guaranteed Iraqi content ratio. Local train and infrastructure work can retain skills, supplier income, repair capacity and spares knowledge. The latest process appraisal above includes whole-order labour, factory support/maintenance and make/buy margins; manufacturing sales to other customers still need a separate business case. No multiplier, tax recovery or future national plant profit is booked as project cash without evidence. Imported machinery and residual inputs retain foreign exchange and supply-chain exposure.

Operating organisation and digital management

The latest operating establishment is 3,523 FTE with IQD 83.415bn annual loaded payroll, replacing the original 3,422 FTE / IQD 50.286bn allowance. It includes OCC/remote assistance, fleet maintenance, infrastructure/energy, stations, passenger service, administration and training. It remains a planning workload/roster model; legal duties, appointments and competence require acceptance. Construction and factory populations are priced separately as explained above, with measured work hours, crew mixes and throughput still required.

Operating role	Required FTE	Wage grade
occ-lead	5	supervisor
dispatcher	70	technical
remote-assist	455	technical
station-lead	82	supervisor
platform-assistance	786	general
customer-service	786	general
station-cleaning	122	general
workshop-lead	42	supervisor
fleet-mechanical	262	technical
fleet-electrical	176	technical
fleet-finish-cleaning	396	general
infrastructure-lead	28	supervisor
civil-track	116	technical
solar-storage	62	technical
wayside-comms	49	technical
city-director	1	director
chief-engineer	1	senior
quality-safety	25	senior
training	26	senior
procurement-stores	20	technical
finance-people	13	technical

The Baghdad operating package contains 1,717 assets, 7,602 manufacturing/verification tasks, 14,463 material/procurement rows, 10,148 maintenance tasks and 6,362 QA actions. These are generated planning records. Actual purchase orders, execution, measurements and accountable release evidence remain distinct. The project twin, ERPNext/Frappe integration, supervision, QR identities, maintenance and advisory AI support business work; they do not issue movement or safety release authority.



Delivery and commissioning sequence

First obtain survey and demand inputs, freeze a viable first line and plant scope, and reconcile depot and energy duties. Qualify long lead components and the first six car train, then deliver infrastructure, energy, station systems and trained operating staff in accepted phases. Each line needs its own operating and safety acceptance before fare revenue is realised.

The city-sized plant becomes available after 18 months from NTP, followed by first-article qualification and finite six-car production cells. All 809 trains remain in scope. The calculated full fleet finishes alongside the overall infrastructure programme. Civil work and invoice milestones are rephased within the existing crew lanes and dependency graph. Line 1's infrastructure moves from day 194 to 672, reducing its fleet wait from 568 to 90 working days. Resources, durations and opening dates are preserved. Survey, land, utility, permit and contract approval remain necessary before deferring work. See the physical factory sizing, crews, test paths and capital reconciliation.

Current capital milestones span 82 months, based on 260 working days/year and 30 pre NTP working days. Conditional first/full network opening is month 40/81, including the separate three-month line commissioning allowance. Opening-weighted demand and the 25% fixed / 75% variable OPEX proxy require a surveyed phase-specific plan. Factory sizing and its explicit capital increase are included in the new monthly and six-month financing schedules; rates, physical qualification and delivery risk remain open.

The factory's final fleet margin is only 5 working days; test-path throughput margin is 2.71%. The frozen-resource delivery and financing study separates civil productivity from investment timing: a 1.0 cycle multiplier preserves the complete baseline schedule, faster cycles retain the rephased start floors, and earliest construction is a separate comparison. Removing spending delays at unchanged productivity must not be called a productivity financing penalty.

Recovery options price extra structural/electrical/composite or coordinated production shifts, hiring/training, supplier expediting and testing, with unchanged cell counts and indexed incremental payroll/nonlabour costs. Testing alone does not improve the 75% availability opening dates; upstream production still limits them. A separate temporary first-article facility sensitivity adds USD 37.450m capital plus support staffing and tests first opening in month 36; permanent acceptance paths, first-article qualification and full line fleets remain required. These options are unquoted deterministic comparisons, not adopted delivery commitments.

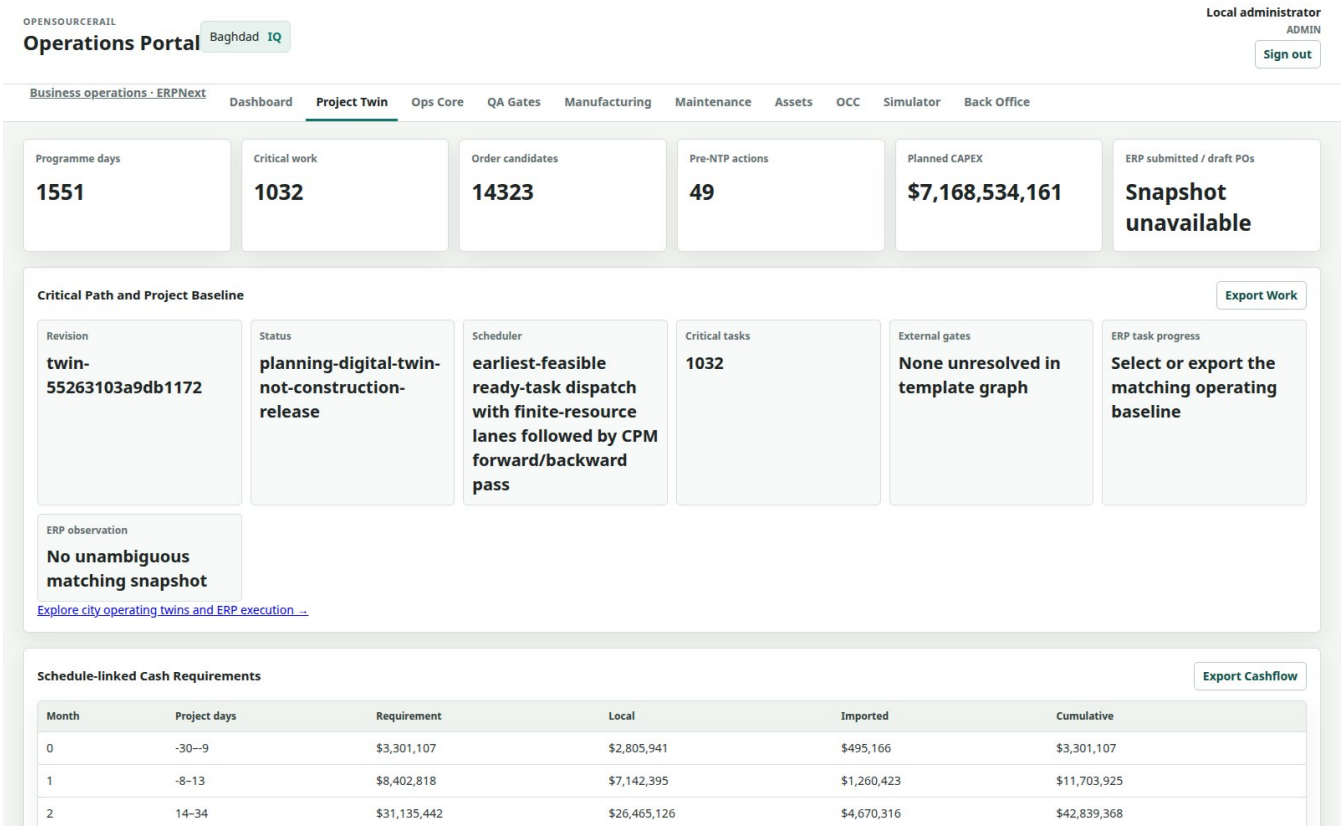
Shift compression now excludes fixed curing/bonding/inspection holds and the additional 60-day first-article qualification. The industrial qualification and funding-gate package adds a metric temporary-site layout, cell tooling/transfer interfaces, quantity-based RFQs reconciled inside the existing USD 35m direct allowance, and ten ERPNext evidence tasks with source-bound measurement templates and authenticated independent result verification. Named owners, measurements, quotations and signatures remain pending.

Funding interruptions halt procurement/construction/production and defer invoices, adding local remobilisation and carrying/prolongation costs. Recovered domestic placement and delayed export credit are conditional on re-placement; permanent refusal has no opening or debt-clearance date, and is never filled by an automatic government or gap-loan replacement. The separate first-section study uses 14.034 km and five actual line-1 stations, 16 already-planned six-car trains at six-minute headways, independent turnbacks/charging/maintenance and USD 29.96m extra capital. Conditional section service is month 36, with surveyed demand and physical acceptance still required; the full 809-train baseline remains unchanged. Earlier small-section fares alone do not establish better finance: the study includes advanced invoices, additional support costs, phase demand deducted from later full-line receipts, and native-currency reserves/debt.

Combined delay-cost cases add extended staffing, supervision, carrying, storage/insurance and construction prolongation allowances without buying baseline crew-months twice. The financial downside ladder tests 30% fewer paid trips, 25% lower retail/advertising receipts, 5% annual invoice escalation, 7% rail OPEX growth, no assumed green/grant/rights enhancements, core rates two percentage points higher and an 8% IQD gap facility limited to IQD 4tn. The joint case leaves IQD 12.690tn cumulative uncovered cash and IQD 0.000tn terminal gap debt. Uncovered cash is a missing funding requirement; it is not an additional government contribution or secured credit. Reported repayment in such a case is conditional on filling that gap. The physical qualification register remains entirely not demonstrated; no model run supplies measured production/civil evidence or lender commitments.

Scenario	First/full month	Peak IQD gap tn	Interest/fees USD eq bn	Debt cleared month
Baseline	40/81	2.158	3.504	203
Faster civil; retained starts	40/81	2.164	3.505	203
Earliest civil; original cycles	40/81	2.669	3.669	205
Earliest civil; faster cycles	40/80	3.057	3.798	207
75% availability, costed	42/89	2.510	3.589	208
75% availability + test shift	42/89	2.517	3.592	208
75% availability + all stages	40/81	2.226	3.532	204
Temporary first-article facility	36/81	1.951	3.495	201

Scenario	First/full month	Peak IQD gap tn	Interest/fees USD eq bn	Debt cleared month
Combined delays, costed	55/105	3.370	3.769	219
Joint downside	55/105	4.000	12.896	Unfunded



Baghdad project twin

Original reference capital and procurement origin

Capital scope	USD eq m	Imported USD m	Local USD eq m
civil	3,937.044	590.557	3,346.487
stations	977.000	195.400	781.600
depots	8.000	2.000	6.000
rolling stock	1,359.120	475.692	883.428
solar plant	815.740	367.083	448.657
signalling	25.667	12.833	12.833
charging microgrid	58.400	23.360	35.040
epc overhead	445.566	66.835	378.731
Baghdad plant	303.505	60.701	242.804
Plant EPC	21.245	3.187	18.059
Total Baghdad capital	7,951.287	1,797.648	6,153.639

The city and plant total includes the plant EPC once. Budgets are unquoted planning estimates; land, utilities, taxes/duties, escalation, contingency and accepted depot/site scope require closure. Loan principal repaid later is a financing cashflow, not additional construction CAPEX. Origin shares do not establish citizenship of vendors, employment or lender eligibility.

Original reference financing in USD and Iraqi dinars

Capital source	Currency	Native amount	USD equivalent m
bank credit	IQD	1,646,008,491,319	1,266.160
chinese export credit	USD	898,823,898	898.824
domestic bonds	IQD	4,938,025,473,956	3,798.481
government import cash	USD	898,823,898	898.824
government local cash	IQD	1,415,697,276,635	1,088.998

Government cash totals USD 1.988bn equivalent, exactly 25% of capital. Its USD 898.824m import cash is inside that limit. The same amount of Chinese USD debt covers the other half of imports. Remaining capital sources are IQD. Combined USD capital funding is USD 1.798bn (22.61%); IQD is 77.39%. Only Chinese credit is USD debt, while government also needs USD

cash for downpayments.

Facility	Currency	Rate assumption	Grace months per draw	Amortisation months	Arrangement fee
chinese export credit	USD	5%	48	180	1.0%
domestic bonds	IQD	8%	24	180	0.5%
bank credit	IQD	9%	0	84	1.0%

Bond and bank capital split the residual after government and Chinese credit 75:25. Interest, fees and reserve deposits are funded separately in the cash forecast. A direct 25% capital contribution does not cap sovereign bond liabilities, guarantees or all lifetime public risk. Ministry of Finance issuance powers, sponsor structure, investor mandates, IQD placement capacity, draw availability and creditor consent for city/plant pooling need confirmation. Chinese supplier and lender qualification remains pending. Terms are sensitivities, not loan offers or an approved appropriation.

The capital source table above is the reference allocation. The conditional blended/early repayment case below replaces part of those ordinary bonds with green bonds and adds a climate grant that reduces domestic borrowing. It is a separate capital mix with the same total uses, government cash and Chinese credit; the two tables must not be added together.

Conditional blended capital source	Currency	Native amount	USD equivalent m
government import cash	USD	898,823,898	898.824
government local cash	IQD	1,415,697,276,635	1,088.998
Chinese export credit	USD	898,823,898	898.824
Ordinary IQD bonds	IQD	3,727,194,709,487	2,867.073
Green IQD bonds	IQD	1,186,455,764,471	912.658
Bank capital credit	IQD	1,637,883,491,319	1,259.910
Conditional climate capital grant	IQD	32,500,000,000	25.000

The conditional capital grant replaces USD 25m equivalent of domestic borrowing. Additional development rights and new local operating receipts enter later project cash and are not counted as construction capital a second time. Supplemental gap draws pay financing/OPEX/reserve cash needs and are separate from both capital tables.

Delivery estimate, physical scope and operating establishment

The published original USD 7.951287bn equivalent remains a planning base with 12 explicit unpriced scope categories, not the amount proved sufficient to deliver service. The delivery reconciliation records quantity/rate/source/currency/price-date/inclusions/exclusions/estimator/uncertainty, depot/storage alternatives, six-car BOM/mass/axle/interfaces/labour/qualification, hourly energy, demand-led fleet requirements and workload/competence. Quotations, price dates and actual accountable appointments remain pending. Existing EPC, factory contingency, training/qualification and train QA/labour are not added twice. Uncalibrated correlated cost/delay quantiles remain separate sensitivities, not approved risk budgets.

The original USD 8m depot allowance does not fund the full current fleet, line-local storage and maintenance requirements. Quantity-based alternatives price storage tracks, points, drainage/access, workshop shells/equipment, line-local inspection/rescue/isolation/quarantine and full depot PV/storage. Gross reference budgets are USD 179.592m for workload bays and USD 182.583m retaining declared bays; land, foundations, utility diversion, installation and overlap with charging/EPC remain open. Their replacement illustrations remove the old allowance once, and neither is adopted into loans or original government funding. The existing failed morning-direction/stabling and conflict-aware access gates remain open.

Annual energy netting supplies no firm hourly charging guarantee. The synthetic hourly owned-solar reference requires 942.7 GWh purchases / USD 94.274m per year, before separately priced wheeling, balancing and connection. It also exposes 3.197 GWh unmet charging, so the assumed service is not fully delivered even with aggregate pooling. Weather, charger queues, feeder rights/outages, storage ageing and actual tariffs require measured per-site replay; these are sensitivity values, not a forecast or silently updated finance allowance. Owned, contracted and hybrid options use the same duty and no export income.

Reference workload cover totals 2,940 operating FTE / USD 39.353m loaded annual pay equivalent, against the original catalogue payroll allowance. Local pay, employer/rest terms and measured task hours remain unaccepted. Recruitment cohorts work backwards through joining, practical assessment/repeats and supervised authorisation; factory production payroll and temporary commissioning are separate. The real pilot roster has zero appointed workers and blocks every slot. Native HR/training/maintenance/manufacturing mappings and the read-only task eligibility preview preserve human work-release authority. Six additional 90-day evidence Tasks carry accountable functions, unquoted closure work budgets and independent exit criteria.

Opening fleet comparisons reduce service supply and require calibrated OD/access/fares and usable corridor/dependency evidence. They do not retain unchanged fares or announce an earlier opening from fewer trains alone. The continuation now reconciles these assumptions into explicit unquoted monthly cash sensitivities. Current review supersedes the historical 347-month figures; clean-checkout test bootstrap restores the exact archived operations input.

Latest per-site energy, funding and fare reconciliation

The new monthly and six-month funding ledgers replace allowances once and preserve the original financing reconstruction exactly. The full-fleet sensitivity has peak supplemental debt IQD 3.750tn, company cash NPV before finance USD -2.874bn, and no debt-clear month. Government is 25% of scenario capital; a separate case fixes its original absolute contribution. Imports retain 50% government USD / 50% Chinese USD loan, with bonds, bank/gap credit and receipts in IQD. Conditional grants, rights receipts, local income and cheap gap lending remain uncommitted; removing the additional income exposes unfunded cash.

Per-site reference energy requires 940.2 GWh imports, exposes 10.5 GWh unserved charging, and prices firm energy services/owned-plant maintenance at USD 140.310m/year. Physical grid and charger upgrades remain unpriced. Buying shortage energy in the financial sensitivity does not provide a physical connection or an accepted timetable. Depot layouts label individual tracks/slots; Iraqi yard-slab mould capacity meets cumulative installation dates at assumed cycles, while plant cost and actual curing/qualification remain open. The 23 child part RFQs sit inside the eight parent allocations; all 26 maintenance intervals, practical lesson cards and rest-limited anonymous slots retain their actual evidence gates.

Battery reserve contributions are already in rolling maintenance. Separate monthly restricted-cash ledgers add only inflation shortfalls at replacement, including in the reduced-fleet opening case, and prohibit spending reserve cash on early bond/loan repayment. Reduced fleet procurement also reduces service receipts; contracted solar moves plant capital to a provider whose resource costs remain visible.

Installed-site shortage diagnostics separate local PV from wheeled generation and trace every constrained hour. The installed energy/charger throughput case reduces fare and commercial receipts; it is an upper bound until actual charging events and timetable feasibility are accepted. Development and training mobilisation funds interim authorities from NTP, schedules external teachers, assessors and mentors before opening, and retains absent real appointments and equipment as explicit gates. The reduced-fleet factory replay removes deferred orders and rebuilds finite production queues, preserving expansion-capable plant/depot capital. Opening dates use the later of civil readiness and replayed fleet acceptance, with later dates propagated into all operating cashflows. No earlier receipts or cheaper factory are invented. All integrated sensitivities retain zero dividends; terminal project cash is not audited distributable profit.

Initial base fare IQD	Paid-demand multiplier	44-trip income share	Debt-clear month	Terminal gap IQD tn	Before-finance cash NPV USD bn
1500	0.962	13.4%	233	0.000	-2.403
1750	0.918	15.6%	207	0.000	-1.785
2000	0.882	17.8%	188	0.000	-1.192
2500	0.825	22.3%	164	0.000	-0.068
3000	0.781	26.7%	148	0.000	0.992

These fare sensitivities retain annual 5% fare/OPEX/income increases and variable pricing, with the existing uncalibrated price elasticity. Debt clearance is conditional on funding placement; a negative company cash NPV persists across these trials. The affordability share uses the historical income proxy, not disposable-income surveys. No higher fare, commercial first-corridor rank, operational release or accepted lease/loan is created by the calculation.

Early deficits and additional financing

The independent flat price reconstruction requires USD 3.194bn gross early additional cash and retains USD 7.867bn later. Their difference is USD -4673.113m net nominal deficit before pricing gap finance. These figures cannot be added to construction capital as though they were new infrastructure. Later revenue cannot fund an earlier payment without a priced and available facility.

The candidate replaces eligible ordinary capital bonds with IQD green debt at 4% plus arrangement and 0.5% annual guarantee charges; it also tests an uncommitted USD 25m equivalent climate grant, USD 300m equivalent net development rights and USD 25m equivalent annual additional net local receipts. A green label alone changes no coupon. Grants replace eligible domestic capital borrowing; guarantees enhance credit rather than provide cash. Supplemental IQD credit at 2% and a 0.5% draw fee is capped at IQD 13tn outstanding, not treated as proven market capacity. All these conditional sources require legal, donor, investor and valuation evidence.

Additional routes to qualify include climate/renewable energy grants or concessional finance, guaranteed IQD on lending, phased green bonds/sukuk, development rights and station land leases, telecom/fibre leases, naming rights, sponsorship, employer travel contracts and carefully priced concessions. Carbon receipts remain unbooked contingent upside. Upfront lease receipts cannot be added while the same future rents remain in revenue. Keep net proceeds after costs and any transferred liabilities; foreign development finance needs confirmed IQD on lending or a priced hedge to preserve the currency structure.

Financing and pricing sensitivity	Peak gap IQD tn	Uncovered IQD tn	Terminal gap IQD tn
blended candidate	3.384	0.000	0.000

Financing and pricing sensitivity	Peak gap IQD tn	Uncovered IQD tn	Terminal gap IQD tn
commercial gap credit	13.000	7.067	13.000
concessional gap credit	5.317	0.000	0.000
fare 5pct flat costs elastic	2.037	0.000	0.000
fare 5pct flat costs fixed demand	1.807	0.000	0.000
fare 5pct opex 5pct	2.102	0.000	0.000
fare 5pct opex 5pct income 2pct	2.302	0.000	0.000
fare 5pct opex 5pct rents indexed	2.063	0.000	0.000
fare 5pct opex 7pct	2.283	0.000	0.000
fixed fare 5pct opex	13.000	0.486	13.000
green concessional only	4.428	0.000	0.000
green label only	0.000	4.152	0.000
unfunded reference	0.000	4.152	0.000
variable fare 5pct opex 5pct	2.025	0.000	0.000
zero cost liquidity bound	4.152	0.000	0.000

Rail, property, energy and industrial financing redesign

The twelve-case financing redesign tests opening-linked principal, longer civil amortisation, a 15-year total insured green tenor and USD 1bn-equivalent total station rights replacing USD 300m, plus three retained viaduct-rental variants. Separate rail, solar, factory and developer monthly/native-currency accounts cancel PPA payments, train invoices, plant capacity fees and rights transfers on consolidation. The rental variants add a fifth borrower with zero extra operating-gap cap and no unapproved transfers to rail. Six-month placement envelopes and loan-vintage dates accompany every borrower. Fares/kiosks/advertising remain included; no future national order, surplus-power sale or unawarded climate grant services the integrated case.

The integrated sensitivity leaves peak aggregate IQD liquidity of IQD 2.311tn, cumulative missing funding IQD 0.000tn, and resource NPV USD -1.316bn, or USD -1.783bn after unverified public-land opportunity cost. Company cash balances and an assumed credit cap cannot establish bankability. Longer grace still pays interest, and changing finance terms leaves the core unlevered NPV unchanged.

Government remains 25% of the original rail/energy/factory capital. Import cash/Chinese credit still split 50:50 in USD; other credit, equity, fares, rights and payments are IQD. Private property construction is an additional explicitly priced scope, with its own IQD debt and equity. Optional indexed availability payments add public obligations outside the 25% capital limit; resource NPV cancels those government transfers. The combined downside retains delayed openings, weaker fare/nonfare demand, inflation and restricted financing, and adds weaker property sales/rights: IQD 30.574tn remains missing. Fifteen station candidates and six 90-day evidence workstreams have no invented title, valuation, investor term or named owner.

Iraqi mixed joint-stock holding and staged ordinary equity

The eighteen-case ordinary-equity study proposes a holding company with 100%-owned rail, energy, factory and station-development businesses, including retained viaduct premises. It replaces the prior subsidiary private equity rather than adding it twice. Government's original USD 1.970bn-equivalent cash remains 25% of original capital; converting it into shares creates zero new money. Same-price private subscriptions of USD 500m/1bn/2bn equivalent leave government ownership of 79.9%/66.5%/49.8%. IQD subscriptions, paid share registers, premium/dilution and fees are explicit. A secondary government share sale gives cash to the seller and zero to the company.

The $1bn primary sensitivity requests six founder calls and a conditional later issue at month 60, with 2% fees. Monthly paid state capital constrains actual settlement; listing at that date is unproven. Against a matched tax-stressed holding reference, peak IQD liquidity falls from 2.131tn to 1.539tn. Domestic principal waits for full-network opening. Subscription failure cannot silently expand agreed six-month capital-credit placements or draw operating rescue for construction: the next unfunded invoice is withheld and no opening is reported. Whole-programme subscription delay and joint downside have separate cash/debt ledgers. Government's USD cash and Chinese USD credit remain the import split; other money remains IQD.

The $1bn case's illustrative private equity return is 9.18% nominal IRR, compared with an assumed 15% hurdle, with first dividend month 191. Dividends require profit, cleared debt, intact reserves and no missing cash; shares have no guaranteed redemption. Tax, depreciation, property inventory, factory impairment, retained earnings and internal-charge eliminations reconcile in pro-forma accounts. The base tax sensitivity avoids assumed group loss relief; a separate aggregate-profit proxy is explicitly unqualified. Resource NPV after land remains USD -1.783bn; changing financing does not create additional resources or financial feasibility.

Current Iraqi corporate/admission rules require an accepted mandate, legal/capital approvals, eligible audited business/accounts and actual investors. The 2019 liabilities restriction also needs entity-level assessment: the consolidated screening proxy flags 33 months in the $1bn sensitivity. The structure therefore needs capitalisation/legal resolution before

execution. Incorporated status, listing approval, public-land equity valuation and independent audit are all absent. Six additional source-bound ERP evidence packages cover legal/tax, mandate, subscriptions, accounts, governance and admission. Open licences are retained; no exclusive design valuation, national train order or uncontracted export income is assumed.

Retained premises beneath suitable viaducts

The commercial-space register, open unit plan, pilot and rental model links 1,127 elevated segments / 291.2043 km to civil chainage, planning coordinates and existing track assets. A preliminary 25 m bay screen reserves approaches, support inspection zones and independent access, suggesting 907,380 m². Confirmed eligible area remains zero: surveyed height/footprint, ownership, street/utility access, fire/flood/impact protection and station-sale overlap are unaccepted. The 200,000 m² illustration has 0 m² unmapped and supplies no integrated lease cash. A 30-unit draft pilot and reusable independent 30 m² enclosure have no invented tenant, permit or live contract.

The medium 100,000 m² scenario adds USD 89.193m equivalent of indexed fit-out capital. Leasing follows civil/rail readiness, rent-free periods, occupancy ramp, tenant turnover and arrears. Occupied/vacant maintenance, insurance, 12-year refurbishment, depreciation and tax are visible. Security deposits are liabilities backed by restricted cash; they supply no revenue, capital or dividends. Existing kiosks are unchanged and station buildings assumed sold cannot also generate rent. Additional under-viaduct land/rights cost is unverified; no free land value is asserted.

Before-tax incremental rental resource NPV is USD 18.317m, with the lower-rent/prolonged-vacancy case USD -37.876m. In the wholly owned $1bn holding case, medium rents give private nominal IRR 9.28%, first dividend month 191 and peak IQD liquidity 1.576tn. Recurring rent adds value only after its resources/costs; it does not close the multibillion-dollar financial deficit.

The independent lender-policy comparison allows annual dividends with at least 1.30 historical/projected debt coverage, positive profit, completed capital, intact operating/tax/forward-service/renewal reserves and conservative leverage. These terms are unapproved and do not produce earlier distributions under the current tested assumptions: the non-rental first dividend is month 191. A separately labelled final-cash diagnostic leaves actual accounts unchanged: the original $1bn private IRR from ordinary dividends is 9.18%; the terminal-cash sensitivity reports 9.59%, with no guaranteed redemption, unvalued asset sale or quoted share-exit price. An undefined IRR establishes no positive investor return; missing funding prevents a completed-horizon cash exit.

Each pilot unit carries lease/meter/inspection/use fields, civil-parent links and five common physical hazard paths. Tenant fire, flooding, impact, blocked bearing/inspection access and utility faults can affect piers, spans and railway operation; controller redundancy provides no mitigation for these hazards. Six additional native ERP evidence drafts require actual site, civil/fire, market, quote, maintenance/lease and finance evidence before occupancy or construction release.

Passenger fares and other operating income

Base model average paid trip yield is IQD 1,317. At full steady operation, existing fares contribute IQD 869.296bn/year, shops/kiosks IQD 15.538bn and advertising IQD 23.180bn. These already reduce the funding gap. Rental occupancy is 88%, advertising 85%; prices, collection and customer demand are not verified Baghdad lease quotations. Commercial figures are gross receipts, with dedicated concession costs requiring appraisal.

Low capacity use assumes 1,807,920 paid trips/day, not unique travellers or a demand survey. The proposed fare design should evaluate concession funding, student and low income access, commuter caps, transfer integration, peak/off peak tiers and collection costs before adopting a tariff. Neither a fare increase nor future kiosk rent places the early financing by itself.

The base full network operating allowance totals USD 205.006m/year equivalent, before the separate inflation sensitivities. Existing maintenance includes a battery renewal reserve on a 12 year reference cycle and fixed asset renewal allowances; it is not an accepted lifecycle replacement plan. Do not add the same battery reserve as new CAPEX. Condition based renewal quantities, inflation, dedicated concession costs, tax and supplier maintenance terms need qualification.

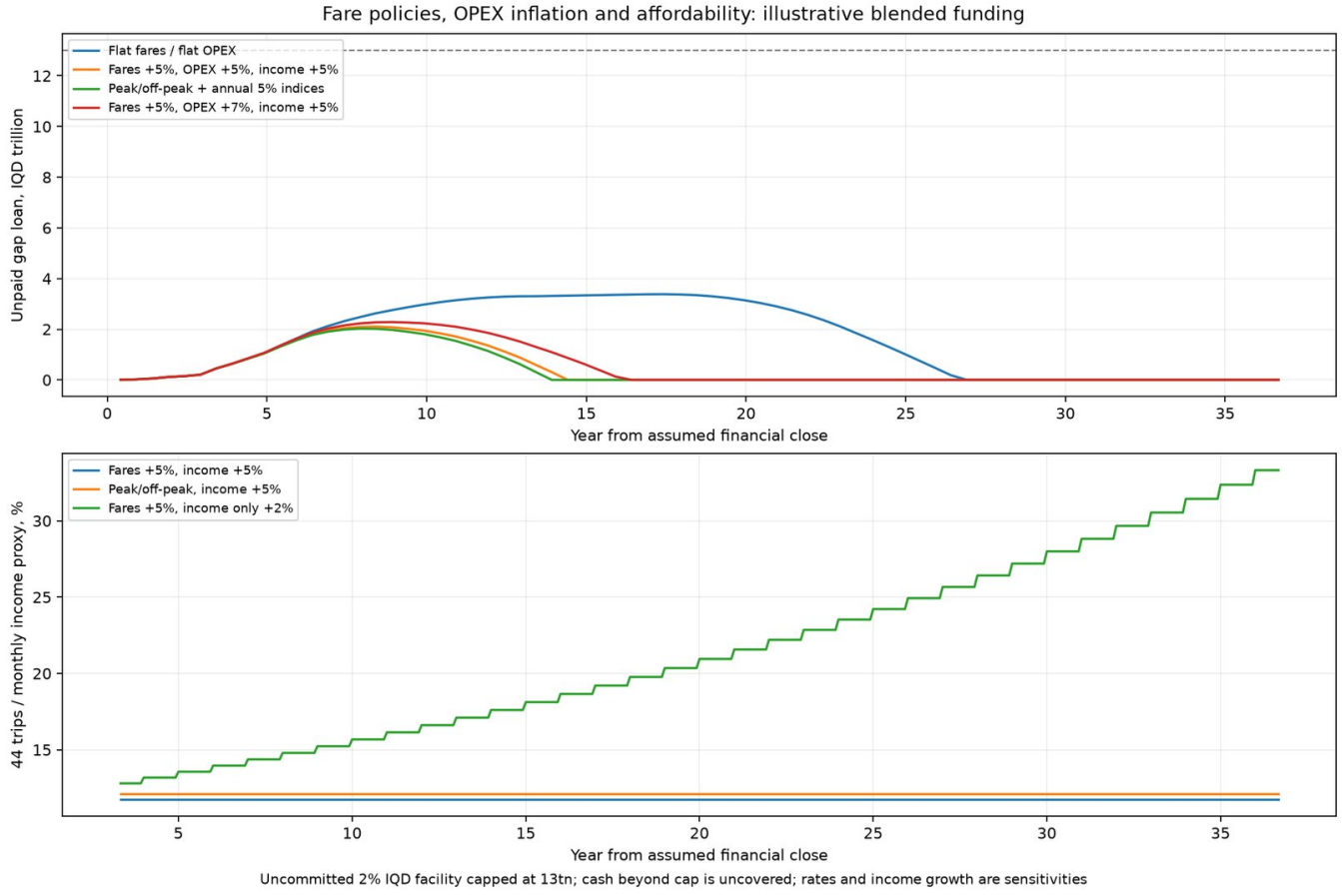
Base steady annual OPEX	USD equivalent m
civil station depot maintenance	98.441
labour	38.681
residual grid energy	0.000
rolling stock maintenance including battery renewal reserve	54.365
signalling maintenance	1.283
solar plant maintenance	12.236

Annual ticket increases and OPEX inflation

The requested paired sensitivity indexes fares, OPEX and income 5% annually from financial close. No tickets are sold before opening. Nominal average fares are IQD 1,525 at first opening and IQD 1,765 at full opening. With 5% income growth, 44 trips remain 11.7% of the income proxy. With only 2% income growth that burden reaches 13.96% at full opening and the assumed real price elasticity reduces paid trips.

Variable pricing tests 40% of baseline trips at 1.25 times the standard fare and 60% at 0.90 times it, with separate demand response and the same capacity limit. If OPEX grows 7% while fares/income grow 5%, the model leaves IQD 0.000tn uncovered cash and IQD 0.000tn terminal unpaid gap debt. This sensitivity still needs additional financing before later surpluses; repayment remains conditional on that funding being available. Existing rent is flat unless the rental indexation sensitivity is chosen; new net rights/receipt targets are held nominal.

The paired case's unlevered NPV is USD -1.670bn at 13.4% nominal discount, excluding new grant/rights/net income targets and with un-escalated capital. Paying debt under a nominal model is not evidence of positive discounted project value. CAPEX escalation, renewal inflation, future FX, floating rates and surveyed demand remain material appraisal work.



Fare and OPEX sensitivities

Surplus cash and early debt retirement

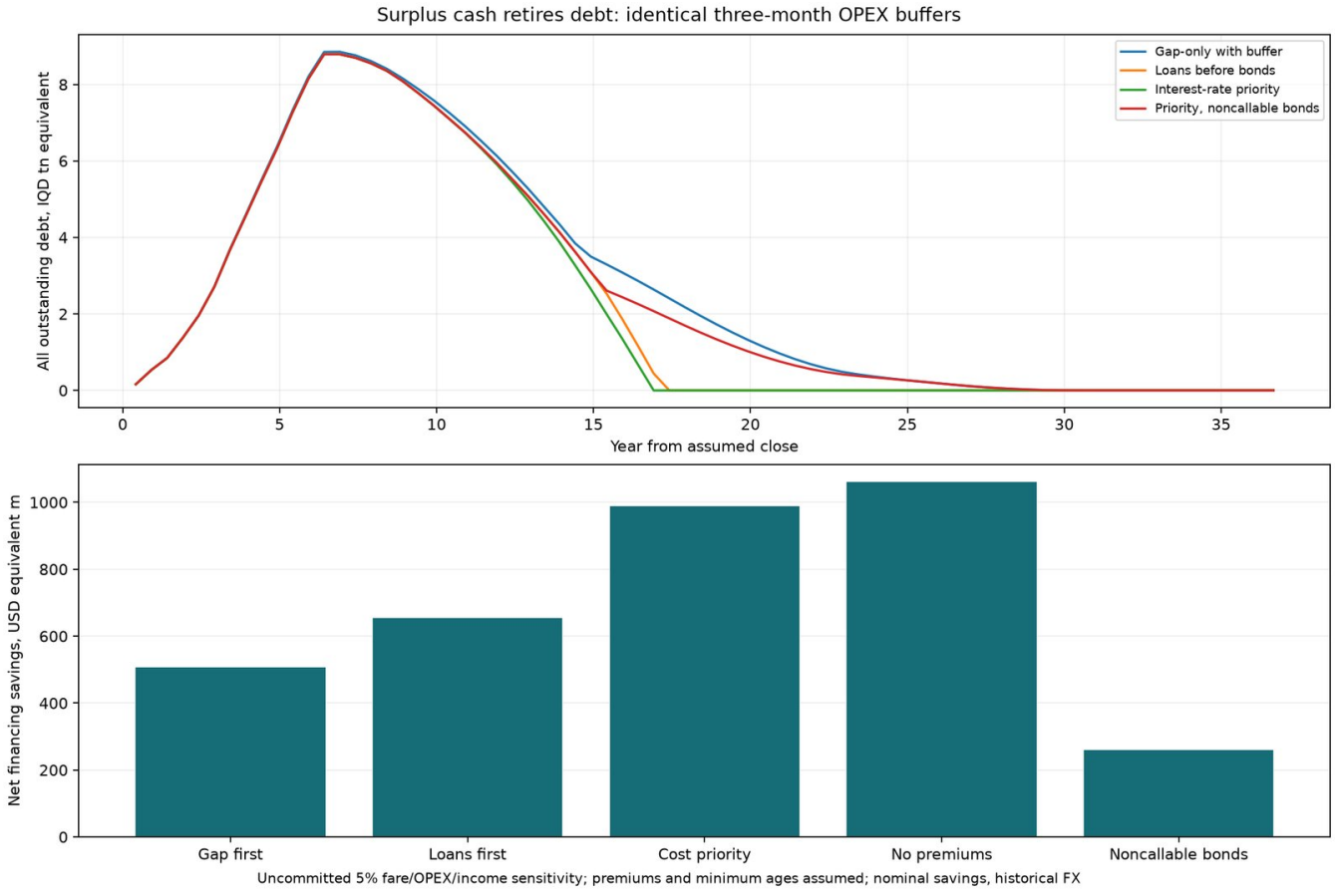
Surplus first pays OPEX, scheduled principal, interest and fees, then debt service reserves and three months of current OPEX. Voluntary payments cannot be funded by a new gap draw or uncovered cash in the same month. All comparisons below use the same buffers and 5% fare/OPEX/income inputs. Scheduled and voluntary principal are deducted once from native balances.

Repayment policy	All debt cleared month	Net saving vs buffered gap only USD eq m	Premium USD eq m
cost priority	203	988.754	56.843
cost priority noncallable bonds	369	260.705	12.395
cost priority zero premium	203	1,060.010	0.000
gap only buffered	369	0.000	0.000
gap then core	209	506.359	36.995
loans then bonds	207	653.848	42.809

Cost priority retires bank credit, ordinary bonds, Chinese credit, green bonds, then cheaper gap credit. Under assumed contractual rights it saves USD 988.754m equivalent after premiums and reports debt as cleared in month 203 versus cleared

in month 369 for buffered gap only. Loans first saves USD 653.848m. All retain IQD 0.386tn operating buffer at the horizon.

Premiums assume 1% bank/Chinese/green and 2% ordinary bonds, with minimum draw ages 6/12/24 months respectively. Eligible vintages are oldest first, retaining instalments and shortening maturity. Calls, notice, compensation, tax and market buyback prices need actual terms. The noncallable case makes no early bond payments; cost ordering is a heuristic rather than a globally best solution. Savings are nominal finance costs, not principal savings or present value wealth. Month numbers run from financial close, with no calendar commencement date assumed.

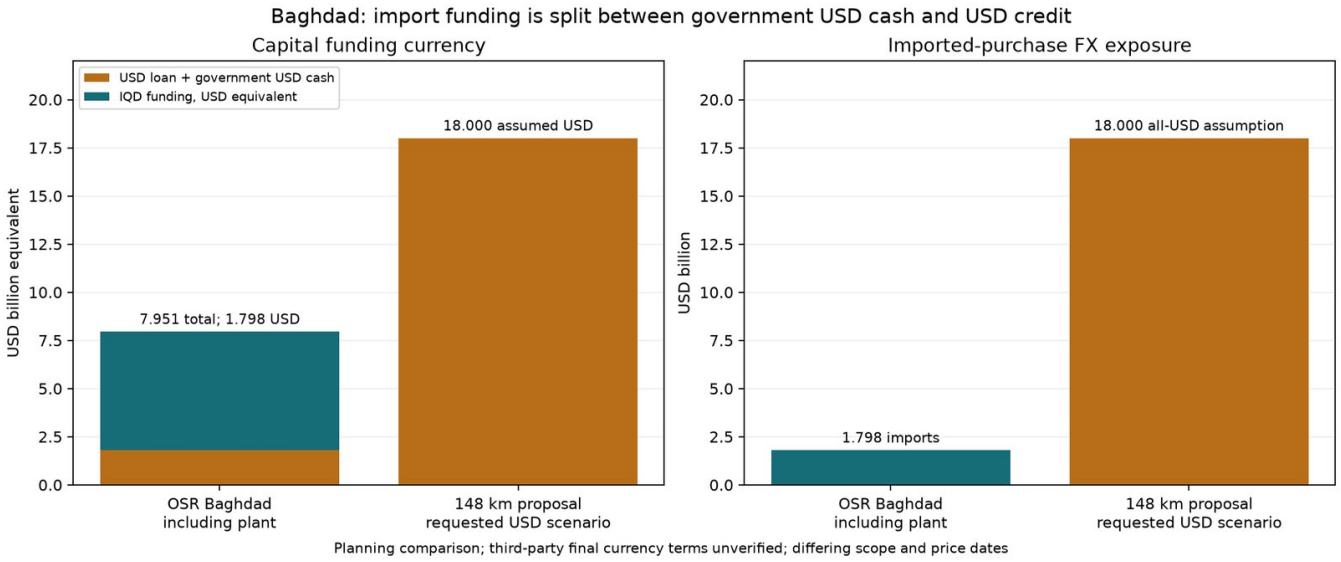


Early debt retirement

Historical Baghdad metro comparison

Measure	Latest positive-margin local-production Baghdad study	Historical proposal or requested comparator
Route km	513.3	148
Lines / stations	14 / 159	7 / 64
Capital USD equivalent bn	8.667	18.000 reported
Capital USD equivalent m / route km	16.88	121.62
USD capital funding bn	2.016	18.000 requested all USD scenario
Chinese USD debt bn	1.008	Final debt and government split unverified

The July 2024 reported estimate is a historical 148 km, USD 18bn scope. An entirely USD foreign loan/government cash basis is the requested comparator, not a verified financing contract. Under that assumption the latest study's USD capital requirement is 88.8% lower. Distinct scope, price date, tunnelling/structures, land, utilities, qualification and schedule prevent a like for like bid saving claim. Third party fares, actual financing and comparable population access are not established. Its Iraqi labour share cannot be assumed zero. The following retained figure illustrates the original reference allocation; the table above uses the revised study.



USD capital comparison

Future national development

Baghdad can establish manufacturing, maintenance, training, procurement and digital delivery capacity that later Iraqi cities can reuse. The existing catalogue contains 18 cities representing 29,491,199 planning residents, 4,851 trainsets and 20,161 vehicle modules across several standard families. These are proposed urban networks; national intercity rail and freight connections are a future feasibility topic, without routes, budgets, revenues or approvals in these totals.

Future catalogue city	Planning population	Fleet family	Route km	Trainsets	City CAPEX USD eq m
Baghdad	9,780,429	metro-6car	513.3	809	7,626.537
Basra	3,955,000	metro-6car	348.0	557	6,579.255
Sulaymaniyah	2,150,000	metro-4car	131.5	144	1,995.266
Erbil	1,952,000	metro-4car	152.3	261	2,351.004
Mosul	1,940,000	metro-4car	230.4	308	3,927.791
Kirkuk	1,780,000	metro-4car	174.0	235	2,810.111
Najaf	1,540,000	metro-4car	193.3	309	3,330.826
Karbala	1,390,000	metro-4car	184.4	253	2,836.168
Nasiriyah	705,000	light-metro-3car	57.6	184	1,091.657
Hillah	700,000	light-metro-3car	74.7	236	1,193.453
Amarah	660,000	light-metro-3car	55.3	177	1,129.751
Ramadi	525,000	light-metro-3car	54.9	181	978.492
Baqubah	470,000	light-metro-3car	76.7	243	1,324.146
Diwaniyah	440,000	light-metro-3car	62.3	198	1,066.909
Kut	410,000	light-metro-3car	53.1	166	1,052.838
Samawah	373,770	light-metro-3car	57.9	182	1,027.650
Duhok	360,000	light-metro-3car	68.4	213	1,084.332
Fallujah	360,000	light-metro-3car	60.2	195	1,058.900

The table retains generic catalogue appraisal scopes. Their arithmetic sums city capital to USD 42.465bn, and adds a shared factory at USD 939.223m plus EPC at USD 65.746m to reach USD 43.470bn equivalent. This is a retained comparator, not a current consolidated national budget. The historical Baghdad funding reference contains USD 303.505m of plant and USD 21.245m of EPC; its generic future-scope difference is USD 35.519bn, including USD 680.219m of additional shared-plant/EPC allowance. Generic imported/local procurement is USD 9.059bn / USD 34.411bn. These differences do not establish available finance or realised savings.

The current Iraq planning basis and national brief instead identify the retained USD 8.667bn Baghdad `local_positive` comparator, including its assembly/component facilities, and the separate unpriced accelerated scenario. Shared-factory reuse and upgrades require scope reconciliation before a national budget can be stated. Iraq already has many precast facilities and production expertise; beam-specific contracts and spare capacity remain unverified.

The additional national plant allowance is an unquoted planning envelope; the actual expansion, renewal and concurrent production programme remain unpriced. Intercity connections, research/training institutions, shared governance and capital acceleration also require separate budgets. Independent city-order factory plans do not prove simultaneous national throughput. The 18 city aggregate is not a five year delivery commitment. Future orders require a throughput/renewal study, scheduled allocation and separate appropriations; no national revenue or profit services Baghdad debt in this proposal.

National industrial and institutional programme

Create common interface standards and supplier quality processes; qualify body, interior and infrastructure suppliers in Iraq; develop local module assembly, battery service, tools, spares and repair capability. A shared apprenticeship and technician programme should cover fabrication, concrete/precast, rail installation, power electronics, charging, software, accessibility and accountable inspection. Set competency outcomes and measured hours before assigning job numbers or training budgets.

Establish a national asset and evidence framework with separate city ownership, budgets, installed configuration and operating approval. Shared procurement and maintenance standards may reduce duplicated development, but each city still needs its own survey, demand, fiscal powers, financial close and safety acceptance. National development should coordinate with existing railways, freight access and universities where feasible; institutional appointments and agreements remain proposed.

Conditional national sequence

Stage	Proposed work	Decision before investment
Baghdad foundation	Sponsor, surveyed first line, plant and six car qualification	Accepted scope, executable finance and physical evidence
Demonstration and repeatability	Prove operating and industrial outcomes; maintain Mosul and Samawah as planning examples	Evidence of safe service, actual cost, production and affordable demand
Next city feasibility	Evaluate regional demand, access, land, energy and local implementation capacity	City specific business case and legal/financial approval
Broader urban rollout	Allocate shared factory output by accepted opening dates; qualify family differences	Priced capacity expansion, renewal and approved city finance
Intercity and national services	Study railway interfaces, freight/logistics, training and shared standards	New scope, environmental approvals and independently funded appraisal

Mosul and Samawah are useful reference packages; their inclusion here is not a selected construction order. Basra, Erbil, Sulaymaniyah and the other cities require transparent prioritisation from actual need and readiness. Future national borrowing shares and USD/IQD exposure must be recalculated from each qualified import basket. Baghdad's 25% contribution and 50:50 import split are not silently imposed on future city budgets.

Ownership and decisions requested

The proposed public sponsor should establish the legal project vehicle and accountable budget authority. The city/operator should own demand, fares, service and operating competence. Manufacturing management should own supplier interfaces, tooling, work hours and first article evidence. Independent survey, civil, energy and safety assessors should control their acceptance evidence. Lenders and placement advisers should validate rights, currency, timing, fees and investor capacity. These are proposed functions, not appointments.

Commission a phased Baghdad feasibility package that closes the route, station, depot and energy scope, independently verifies demand and affordable fares, prices capital and lifecycle obligations, and obtains term sheets for the USD import and IQD domestic funding. Review the long production schedule and fund any acceleration explicitly. Present the accepted first phase to the sponsor and competent authorities before construction. Maintain the national chapter as a separately approved development framework.

Current Baghdad deployment gates

Gate	Status	Responsible function	Closure action
soil-desktop-inputs	closed	civil/geotechnical designer	Sample current station and civil geometry from the pinned soilDB source and retain means, uncertainty and nodata.
soil-coverage	open	geotechnical investigator	Fill mapped coverage gaps through local records and targeted sampling; keep estimated and measured records distinct.
control-processing	open	survey team	Receive survey control observations; process and check the project CRS, datum and residuals.
ground-model	open	survey team	Receive and inspect the terrain, point cloud and ground-model deliveries against independent control.
surveyed-alignment	open	alignment designer	Replace or confirm generated geometry with checked survey alignment and platform reconciliation.
route-station-fit	open	civil/station designer	Resolve utilities, land/access, flood levels, station access and construction staging against the current route.
drainage-ground	open	civil/geotechnical designer	Use the soil investigation plan to collect local geotechnics and groundwater, size foundations/treatment, and check drainage with local rainfall and surveyed levels.
structural-release	open	structural designer and checker	Complete route-specific support/span, foundation, movement and erection calculations and close checking comments.
model-timing-comparison	closed	simulation engineer	Reconcile current native reference journey times with SUMO using the same scenario dwells and route.
full-service-validation	closed	simulation engineer	Run current nominal and degraded full-service cases against the current scenario and simulator; retain failures and exact provenance.
operating-release	open	operator	Complete conflict-aware capacity and degraded-operation review and sign the bound operating evidence.
morning-fleet-allocation	open	service planner	Reconcile revenue fleet with required morning departures on each line; retain two station berths and same-line overflow storage.
continuous-stabling-replay	open	simulation engineer	Run the selected line-local station/depot candidate across consecutive evenings and synchronised morning starts.
stabling-physical-fit	open	track/station designer	Locate usable station and line-local depot tracks, shared charging, isolation, inspection access and protected morning release slots.
solar-storage-endurance	open	energy designer	Bind declared station/ROW/dedicated PV to site storage and actual charging duty, reconcile conversion losses and prove replenishment across adverse weather; specify residual backup duty explicitly.
depot-placement-and-budget	open	depot and cost designer	Place the declared PV/storage inventory within the controlled site layout and reconcile itemised installed costs and renewal scope with existing allowances.
workforce-duty-and-workload	open	city operating lead	Reconcile the funded role plan with named shift/leave cover, measured inspection and cleaning workloads, simultaneous service-depot arrivals and actual local employment inputs.

3 checks are closed and 14 deployment gates remain open. Manufacturing, system certification and donor/lender decisions have additional distinct gates. The Baghdad package is complete as a planning example and has operational release false. Synthetic simulation, generated registers and documentary completeness are not physical acceptance.

Detailed schedules and source appendices

The detailed engineering plan and component register expand Baghdad's six-car mechanical interfaces, civil works, Iraqi slab manufacture, missing parts, onboard power/wiring, software host allocation and ERP handover. The register carries 69 reference part rows and all 60 Rust software allocations, with unknown prices and supplier identities left open. It corrects the ST6 seat layout and electronics interface errors; it does not release shop drawings or add unpriced components to the accepted finance totals. Local ERP recovery and planning-contract checks are recorded separately from production readiness.

The following proposal annex prints every station, interchange, fleet role, energy site and the complete cost priority six month draw/repayment schedule. Civil segment chainages and junction details are in the attached registers and design. Full monthly

and alternative case ledgers remain in the supporting archive and repository. Technical annexes reproduce the city survey, ground, alignment, depot, stabling, delivery, deployment, finance and acceptance reports, followed by the current shared architecture and engineering references.

Source inventory and publication manifest identify exact inputs and outputs. The supporting data archive set begins with part 1; download every part listed in the publication manifest to obtain the controlled files and complete operations payload. Text evidence uses ZIP LZMA; extract with the repository's Python bootstrap or an archive reader supporting ZIP LZMA. The editable proposal, PDF and appendix source list can be regenerated with the repository's proposal builder.

External instrument and historical sources are retained from the financing baseline. Source retrieval attempts on 3 October 2026 for China Exim, the GCF Iraq page and the NIC notice returned a timeout or access denial; this proposal does not claim a new source verification or a new lending commitment. Detailed financing reports retain the original source URLs and their stated evidence limits.

Connected construction and battery scenario (2026-10-06)

The connected study reconciles actual island platforms, boarding faces, access equipment, running/station civil quantities, 18 launchers on two shifts, constrained supplier/logistics schedules, equipment reuse and sodium battery profiles. Its three chronological network energy cases show service shortfalls under the selected study inputs. Supplier contracts, structural/access releases and installed erection credits are absent; conditional dates and the unquoted $9m fleet allowance establish no accepted savings or opening. The financial figures above remain comparators pending scope-matched adoption.

Coordinated industrialisation and possible vendors

The industrialisation programme connects joint vehicle/support selection, staged bogie localisation, standard Pi20/Pi25 viaduct production, support releases, staffing and accepted-bay capacity. Its official-source vendor register provides possible component and equipment suppliers for configuration-specific comparison. Existing Iraqi precast facilities and expertise are a capability base requiring plant and process qualification. The delivery/service bridge retains the current financial comparator; missing quotations, commissioned vehicles, station/system handovers and paid-journey evidence prevent adoption of additional-elevation economics.

The coupled programme study uses completed train sections to reserve OD capacity, charges one fare across transfers and replaces electricity and distance-sensitive maintenance once in the retained financing engine. It reports conditional monthly receipts/debt and complete line-opening dependencies. Synthetic demand sensitivities, retained unquoted prices/opening months and missing first-article, survey and calibration evidence prevent financial or operating adoption. The existing financial comparators remain distinct.

Detailed civil construction methods (2026-10-08)

The civil works package and master plan use 25 m average spans for production/logistics sizing while retaining actual identified Pi20/Pi25 bays and unresolved closures. They connect workfront release, foundations, precast production, transport/storage, launcher erection/relocation, station structures/access, at-grade works, special bridges, utilities/drainage, depot/energy civil, inspection and reinstatement. Individual run/station registers and fourteen civil packages per line support finite-resource and complete-opening planning. Illustrative capacity and calendar lower bounds establish no contracted supply, accepted programme, installed saving or construction release.

Population-led route additions (2026-10-09)

The controlled planning inventory has 14 lines and 159 stations, including 5 additional residential lines. Actual emitted-station proximity reaches 27.7% of retained 2020 bbox residents, against 40.3% in the retained original inventory. Remaining gaps are explicit; this is not current census, surveyed walking coverage or observed fare demand. Fleet, depot, civil, energy, staffing and financing data is regenerated for the actual inventory; structural/site releases, suppliers and complete installed prices remain open. Route additions and actual station evaluation preserve the source-bound comparison.

Integrated junctions, residential access and line-based foundations (2026-10-08)

The coordinated planning package provides 14 line-specific assembly plans, 10,985 support foundation packets, 9,973 uniquely directed bay assemblies and separate station/bridge/depot/energy foundation scopes. It holds junction-affected member orders pending actual profiles, clearances and staged-load design. Bounded station complexes prevent transitive oversized transfer groups. The ring's spurious 40 m out-and-back path has been removed and dependent native evidence rerun.

Retained 2020 station-circle proximity is 27.7%; 40 infill candidates give a conditional 45.7% screen. About 2,173,888 currently unserved retained residents are outside the existing 1 km corridor-vertex screen. Source-linked branch/feeder investigation paths address priority gaps; they do not establish funded service, walksheds or extra fares. Companion planning data archive and offline support/junction viewer preserve the detailed new model separately from the original supporting archive. Ground design, supplier quotes and independent construction/operating approvals remain open.

Baghdad detailed schedules

Source: cities/catalogue/west-asia/Iraq/Baghdad/DETAILED-SCHEDULES.md

Station register

Coordinates and chainages are planning inputs. IDs are controlled; anchor names remain in the UTF-8 CSV rather than being adopted as public station names. Latitude/longitude are WGS84 degrees.

Station ID	Line	Chainage km	Latitude	Longitude	Platform m
line-1-2500-0775-s000000	line-1	0.000	33.049995	44.366651	121.0
line-1-2343-0771-s003515	line-1	3.515	33.078249	44.365791	121.0
line-1-2180-0774-s007014	line-1	7.014	33.107584	44.366436	121.0
line-1-2017-0784-s010529	line-1	10.529	33.136918	44.368585	121.0
line-1-1848-0797-s014034	line-1	14.034	33.167333	44.371379	121.0
line-1-1633-0817-s018544	line-1	18.544	33.206025	44.375677	121.0
line-1-1554-0819-s020393	line-1	20.393	33.220243	44.376106	121.0
line-1-1418-0822-s023138	line-1	23.138	33.244718	44.376751	121.0
line-1-1341-0823-s024686	line-1	24.686	33.258575	44.376966	121.0
line-1-1124-0828-s029068	line-1	29.068	33.297628	44.378040	121.0
line-1-0975-0831-s032084	line-1	32.084	33.324443	44.378685	121.0
line-1-0826-0834-s035089	line-1	35.089	33.351258	44.379330	121.0
line-1-0604-0839-s039571	line-1	39.571	33.391210	44.380404	121.0
line-1-0557-0840-s040519	line-1	40.519	33.399669	44.380619	121.0
line-1-0382-0837-s044110	line-1	44.110	33.431163	44.379975	121.0
line-1-0260-0833-s046583	line-1	46.583	33.453119	44.379115	121.0
line-1-0166-0866-s049067	line-1	49.068	33.470036	44.386206	121.0
line-10-0562-1131-s000000	line-10	0.000	33.398769	44.443154	121.0
line-10-0688-1260-s005702	line-10	5.702	33.376093	44.470875	121.0
line-10-0686-1312-s006804	line-10	6.804	33.376453	44.482050	121.0
line-11-0605-1228-s000000	line-11	0.000	33.391030	44.463998	121.0
line-11-0337-1112-s006884	line-11	6.884	33.439261	44.439071	121.0
line-12-0701-0415-s000000	line-12	0.000	33.373754	44.289289	121.0
line-12-0713-0287-s003217	line-12	3.217	33.371594	44.261782	121.0
line-12-0760-0138-s006586	line-12	6.586	33.363136	44.229763	121.0
line-13-1336-0518-s000000	line-13	0.000	33.259475	44.311423	121.0
line-13-1381-0389-s003238	line-13	3.239	33.251377	44.283702	121.0
line-13-1190-0317-s007937	line-13	7.937	33.285750	44.268229	121.0
line-13-1193-0313-s008042	line-13	8.042	33.285210	44.267370	121.0
line-14-0804-1072-s000000	line-14	0.000	33.355217	44.430475	121.0
line-14-0688-1260-s006047	line-14	6.047	33.376093	44.470875	121.0
line-2-1987-2192-s000000	line-2	0.000	33.142317	44.671157	121.0
line-2-1895-2056-s003494	line-2	3.494	33.158874	44.641931	121.0
line-2-1800-1920-s007001	line-2	7.001	33.175971	44.612706	117.0
line-2-1648-1703-s012612	line-2	12.612	33.203326	44.566074	121.0
line-2-1504-1477-s018846	line-2	18.846	33.229241	44.517507	121.0
line-2-1362-1289-s024085	line-2	24.085	33.254796	44.477107	121.0
line-2-1309-1219-s026100	line-2	26.100	33.264334	44.462064	121.0
line-2-1222-1104-s029320	line-2	29.320	33.279991	44.437351	121.0
line-2-1067-0900-s035129	line-2	35.129	33.307886	44.393513	121.0
line-2-1022-0840-s036713	line-2	36.713	33.315985	44.380619	121.0
line-2-0903-0683-s041144	line-2	41.144	33.337401	44.346881	121.0
line-2-0825-0580-s044026	line-2	44.026	33.351438	44.324747	121.0
line-2-0701-0415-s048669	line-2	48.669	33.373754	44.289289	121.0
line-2-0595-0296-s052851	line-2	52.851	33.392830	44.263716	121.0
line-3-2271-1120-s000000	line-3	0.000	33.091207	44.440790	121.0
line-3-2129-1197-s003513	line-3	3.513	33.116762	44.457337	117.0
line-3-1987-1220-s007007	line-3	7.007	33.142317	44.462279	117.0
line-3-1751-1233-s013011	line-3	13.011	33.184789	44.465073	121.0
line-3-1613-1236-s015796	line-3	15.796	33.209625	44.465718	121.0
line-3-1362-1289-s021278	line-3	21.278	33.254796	44.477107	121.0
line-3-1281-1310-s023072	line-3	23.072	33.269373	44.481620	121.0
line-3-1059-1369-s028013	line-3	28.012	33.309326	44.494299	121.0
line-3-0874-1418-s032165	line-3	32.165	33.342620	44.504828	121.0

Station ID	Line	Chainage km	Latitude	Longitude	Platform m
line-3-0764-1447-s034606	line-3	34.606	33.362416	44.511060	121.0
line-3-0691-1466-s036223	line-3	36.223	33.375553	44.515143	121.0
line-3-0628-1483-s037624	line-3	37.624	33.386891	44.518797	121.0
line-3-0465-1526-s041263	line-3	41.264	33.416226	44.528037	121.0
line-3-0378-1545-s043281	line-3	43.281	33.431883	44.532120	121.0
line-3-0288-1607-s046283	line-3	46.282	33.448080	44.545444	121.0
line-3-0151-1637-s049496	line-3	49.496	33.472735	44.551890	121.0
line-4-0146-0199-s000000	line-4	0.000	33.473635	44.242872	121.0
line-4-0298-0305-s004261	line-4	4.261	33.446280	44.265650	121.0
line-4-0395-0433-s007765	line-4	7.765	33.428823	44.293157	117.0
line-4-0420-0570-s011272	line-4	11.272	33.424324	44.322598	117.0
line-4-0491-0706-s015030	line-4	15.030	33.411547	44.351823	121.0
line-4-0604-0839-s018966	line-4	18.966	33.391210	44.380404	121.0
line-4-0804-1072-s025869	line-4	25.869	33.355217	44.430475	121.0
line-4-0927-1214-s030089	line-4	30.089	33.333081	44.460990	121.0
line-4-0985-1282-s032081	line-4	32.081	33.322643	44.475603	121.0
line-4-1059-1369-s034683	line-4	34.683	33.309326	44.494299	121.0
line-4-1209-1543-s039827	line-4	39.827	33.282331	44.531690	121.0
line-4-1230-1627-s041932	line-4	41.932	33.278552	44.549742	121.0
line-5-1199-0260-s000000	line-5	0.000	33.284131	44.255980	121.0
line-5-1193-0313-s001126	line-5	1.126	33.285210	44.267370	121.0
line-5-1190-0317-s001231	line-5	1.231	33.285750	44.268229	121.0
line-5-1129-0423-s003985	line-5	3.985	33.296728	44.291008	121.0
line-5-1086-0589-s007673	line-5	7.673	33.304467	44.326681	121.0
line-5-1022-0840-s013247	line-5	13.247	33.315985	44.380619	121.0
line-5-0967-1057-s018078	line-5	18.078	33.325883	44.427251	121.0
line-5-0927-1214-s021555	line-5	21.555	33.333081	44.460990	121.0
line-5-0891-1352-s024642	line-5	24.642	33.339560	44.490645	121.0
line-5-0874-1418-s026103	line-5	26.103	33.342620	44.504828	121.0
line-5-0853-1593-s029938	line-5	29.938	33.346399	44.542435	121.0
line-5-0843-1600-s030196	line-5	30.196	33.348199	44.543939	121.0
line-5-0781-1760-s034121	line-5	34.121	33.359356	44.578323	117.0
line-5-0740-1906-s037629	line-5	37.629	33.366735	44.609697	117.0
line-5-0803-2053-s041124	line-5	41.124	33.355397	44.641287	117.0
line-5-0677-2304-s047345	line-5	47.345	33.378073	44.695225	121.0
line-6-2322-0251-s000000	line-6	0.000	33.082029	44.254046	121.0
line-6-2194-0355-s003511	line-6	3.511	33.105064	44.276395	117.0
line-6-2069-0472-s007016	line-6	7.016	33.127560	44.301538	117.0
line-6-1936-0575-s010529	line-6	10.529	33.151496	44.323672	117.0
line-6-1813-0699-s014028	line-6	14.028	33.173631	44.350319	117.0
line-6-1633-0817-s018903	line-6	18.903	33.206025	44.375677	121.0
line-6-1580-0848-s020342	line-6	20.342	33.215563	44.382338	121.0
line-6-1457-0929-s023846	line-6	23.846	33.237699	44.399745	121.0
line-6-1370-0993-s026315	line-6	26.315	33.253356	44.413498	121.0
line-6-1222-1104-s030195	line-6	30.195	33.279991	44.437351	121.0
line-6-0985-1282-s036755	line-6	36.755	33.322643	44.475603	121.0
line-6-0891-1352-s039209	line-6	39.209	33.339560	44.490645	121.0
line-6-0764-1447-s042688	line-6	42.688	33.362416	44.511060	121.0
line-6-0713-1485-s044164	line-6	44.164	33.371594	44.519226	121.0
line-6-0647-1607-s047565	line-6	47.566	33.383472	44.545444	121.0
line-6-0495-1669-s052503	line-6	52.503	33.410827	44.558767	121.0
line-7-1069-2206-s000000	line-7	0.000	33.307526	44.674166	121.0
line-7-1073-2033-s003493	line-7	3.493	33.306806	44.636989	117.0
line-7-1102-1872-s007005	line-7	7.005	33.301587	44.602391	117.0
line-7-1185-1622-s012693	line-7	12.693	33.286650	44.548667	121.0
line-7-1209-1543-s014684	line-7	14.684	33.282331	44.531690	121.0
line-7-1281-1310-s019964	line-7	19.964	33.269373	44.481620	121.0
line-7-1307-1217-s022051	line-7	22.051	33.264694	44.461635	121.0
line-7-1370-0993-s027076	line-7	27.076	33.253356	44.413498	121.0
line-7-1402-0878-s029642	line-7	29.642	33.247597	44.388785	121.0
line-7-1418-0822-s030917	line-7	30.918	33.244718	44.376751	121.0
line-7-1451-0706-s033517	line-7	33.517	33.238779	44.351823	121.0
line-7-1504-0514-s037813	line-7	37.813	33.229241	44.310563	121.0

Station ID	Line	Chainage km	Latitude	Longitude	Platform m
line-7-1531-0418-s039957	line-7	39.957	33.224382	44.289934	121.0
line-8-2231-1783-s000000	line-8	0.000	33.098405	44.583265	121.0
line-8-2175-1647-s003504	line-8	3.504	33.108484	44.554039	117.0
line-8-2045-1565-s007016	line-8	7.016	33.131879	44.536418	117.0
line-8-1940-1488-s010527	line-8	10.527	33.150776	44.519871	117.0
line-8-1857-1362-s014032	line-8	14.032	33.165713	44.492794	117.0
line-8-1751-1233-s017689	line-8	17.689	33.184789	44.465073	121.0
line-8-1645-1094-s021347	line-8	21.347	33.203866	44.435203	121.0
line-8-1457-0929-s026861	line-8	26.861	33.237699	44.399745	121.0
line-8-1402-0878-s028547	line-8	28.547	33.247597	44.388785	121.0
line-8-1341-0823-s030364	line-8	30.364	33.258575	44.376966	121.0
line-8-1228-0719-s033825	line-8	33.825	33.278912	44.354617	121.0
line-8-1086-0589-s038140	line-8	38.140	33.304467	44.326681	121.0
line-8-0954-0467-s042166	line-8	42.166	33.328222	44.300463	121.0
line-8-0829-0364-s045862	line-8	45.862	33.350718	44.278329	121.0
line-8-0785-0290-s048137	line-8	48.137	33.358637	44.262427	121.0
line-9-0606-0636-s003011	line-9	3.011	33.390851	44.336781	121.0
line-9-0491-0706-s006266	line-9	6.266	33.411547	44.351823	121.0
line-9-0557-0840-s009730	line-9	9.730	33.399669	44.380619	121.0
line-9-0562-1131-s016676	line-9	16.676	33.398769	44.443154	121.0
line-9-0605-1228-s019019	line-9	19.019	33.391030	44.463998	121.0
line-9-0627-1485-s025111	line-9	25.111	33.387071	44.519226	121.0
line-9-0648-1607-s028125	line-9	28.125	33.383292	44.545444	121.0
line-9-0843-1600-s032244	line-9	32.244	33.348199	44.543939	121.0
line-9-0853-1593-s032502	line-9	32.502	33.346399	44.542435	121.0
line-9-0964-1569-s035131	line-9	35.131	33.326423	44.537278	117.0
line-9-1185-1622-s040057	line-9	40.057	33.286650	44.548667	121.0
line-9-1230-1627-s040998	line-9	40.998	33.278552	44.549742	121.0
line-9-1429-1525-s047139	line-9	47.139	33.242738	44.527822	121.0
line-9-1504-1477-s049136	line-9	49.136	33.229241	44.517507	121.0
line-9-1611-1366-s054147	line-9	54.147	33.209985	44.493654	117.0
line-9-1613-1236-s056763	line-9	56.763	33.209625	44.465718	121.0
line-9-1645-1094-s059868	line-9	59.868	33.203866	44.435203	121.0
line-9-1580-0848-s066616	line-9	66.616	33.215563	44.382338	121.0
line-9-1554-0819-s067411	line-9	67.411	33.220243	44.376106	121.0
line-9-1451-0706-s071327	line-9	71.327	33.238779	44.351823	121.0
line-9-1336-0518-s076879	line-9	76.879	33.259475	44.311423	121.0
line-9-1194-0440-s081155	line-9	81.154	33.285030	44.294661	121.0
line-9-1129-0423-s085079	line-9	85.079	33.296728	44.291008	121.0
line-9-0954-0467-s089572	line-9	89.572	33.328222	44.300463	121.0
line-9-0825-0580-s094205	line-9	94.205	33.351438	44.324747	121.0
line-9-0689-0569-s097310	line-9	97.310	33.375913	44.322383	121.0

Interchange register

Interchange	Lines	Latitude	Longitude
interchange-000	line-1, line-6	33.206025	44.375677
interchange-001	line-1, line-9	33.220243	44.376106
interchange-002	line-1, line-7	33.244718	44.376751
interchange-003	line-1, line-8	33.258575	44.376966
interchange-004	line-1, line-4	33.391210	44.380404
interchange-005	line-1, line-9	33.399669	44.380619
interchange-006	line-10, line-9	33.398769	44.443154
interchange-007	line-10, line-14	33.376093	44.470875
interchange-008	line-11, line-9	33.391030	44.463998
interchange-009	line-12, line-2	33.373754	44.289289
interchange-010	line-13, line-9	33.259475	44.311423
interchange-011	line-13, line-5	33.285480	44.267799
interchange-012	line-14, line-4	33.355217	44.430475
interchange-013	line-2, line-9	33.229241	44.517507
interchange-014	line-2, line-3	33.254796	44.477107
interchange-015	line-2, line-7	33.264514	44.461849
interchange-016	line-2, line-6	33.279991	44.437351
interchange-017	line-2, line-5	33.315985	44.380619
interchange-018	line-2, line-9	33.351438	44.324747
interchange-019	line-3, line-8	33.184789	44.465073
interchange-020	line-3, line-9	33.209625	44.465718
interchange-021	line-3, line-7	33.269373	44.481620
interchange-022	line-3, line-4	33.309326	44.494299
interchange-023	line-3, line-5	33.342620	44.504828
interchange-024	line-3, line-6	33.362416	44.511060
interchange-025	line-3, line-6	33.373574	44.517185
interchange-026	line-3, line-9	33.386981	44.519012
interchange-027	line-4, line-9	33.411547	44.351823
interchange-028	line-4, line-5	33.333081	44.460990
interchange-029	line-4, line-6	33.322643	44.475603
interchange-030	line-4, line-7	33.282331	44.531690
interchange-031	line-4, line-9	33.278552	44.549742
interchange-032	line-5, line-9	33.296728	44.291008
interchange-033	line-5, line-8	33.304467	44.326681
interchange-034	line-5, line-6	33.339560	44.490645
interchange-035	line-5, line-9	33.347299	44.543187
interchange-036	line-6, line-9	33.215563	44.382338
interchange-037	line-6, line-8	33.237699	44.399745
interchange-038	line-6, line-7	33.253356	44.413498
interchange-039	line-6, line-9	33.383382	44.545444
interchange-040	line-7, line-9	33.286650	44.548667
interchange-041	line-7, line-8	33.247597	44.388785
interchange-042	line-7, line-9	33.238779	44.351823
interchange-043	line-8, line-9	33.203866	44.435203
interchange-044	line-8, line-9	33.328222	44.300463

Fleet roles

Line	Peak	Rotation	Spare	Cold reserve	Total
line-1	78	0	7	1	86
line-2	86	0	8	1	95
line-3	81	0	8	1	90
line-4	69	0	6	1	76
line-5	78	0	7	1	86
line-6	86	0	8	1	95
line-7	65	0	6	1	72
line-8	80	0	8	1	89
line-9	40	0	4	1	45
line-10	13	0	1	1	15
line-11	12	0	1	1	14
line-12	13	0	1	1	15
line-13	16	0	1	1	18
line-14	11	0	1	1	13

Energy site register

Site station	Tier	PV kW	Storage kWh	Grid import kW	Charger kW
line-1-2500-0775-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-1-2343-0771-s003515	standard	300.0	2000.0	2000.0	2000.0
line-1-2180-0774-s007014	standard	300.0	2000.0	2000.0	2000.0
line-1-2017-0784-s010529	standard	300.0	2000.0	2000.0	2000.0
line-1-1848-0797-s014034	standard	300.0	2000.0	2000.0	2000.0
line-1-1633-0817-s018544	interchange	300.0	2000.0	2400.0	2000.0
line-1-1554-0819-s020393	interchange	300.0	2000.0	2400.0	2000.0
line-1-1418-0822-s023138	interchange	300.0	2000.0	2400.0	2000.0
line-1-1341-0823-s024686	interchange	300.0	2000.0	2400.0	2000.0
line-1-1124-0828-s029068	standard	300.0	2000.0	2000.0	2000.0
line-1-0975-0831-s032084	standard	300.0	2000.0	2000.0	2000.0
line-1-0826-0834-s035089	major	300.0	2000.0	2400.0	2000.0
line-1-0604-0839-s039571	interchange	300.0	2000.0	2400.0	2000.0
line-1-0557-0840-s040519	interchange	300.0	2000.0	2400.0	2000.0
line-1-0382-0837-s044110	standard	300.0	2000.0	2000.0	2000.0
line-1-0260-0833-s046583	standard	300.0	2000.0	2000.0	2000.0
line-1-0166-0866-s049067	terminal	300.0	2000.0	3200.0	2000.0
line-10-0562-1131-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-10-0688-1260-s005702	interchange	300.0	2000.0	2400.0	2000.0
line-10-0686-1312-s006804	terminal	300.0	2000.0	3200.0	2000.0
line-11-0605-1228-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-11-0337-1112-s006884	terminal	300.0	2000.0	3200.0	2000.0
line-12-0701-0415-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-12-0713-0287-s003217	standard	300.0	2000.0	2000.0	2000.0
line-12-0760-0138-s006586	terminal	300.0	2000.0	3200.0	2000.0
line-13-1336-0518-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-13-1381-0389-s003238	standard	300.0	2000.0	2000.0	2000.0
line-13-1190-0317-s007937	interchange	300.0	2000.0	2400.0	2000.0
line-13-1193-0313-s008042	terminal	300.0	2000.0	3200.0	2000.0
line-14-0804-1072-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-14-0688-1260-s006047	terminal	300.0	2000.0	3200.0	2000.0
line-2-1987-2192-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-2-1895-2056-s003494	standard	300.0	2000.0	2000.0	2000.0
line-2-1648-1703-s012612	standard	300.0	2000.0	2000.0	2000.0
line-2-1504-1477-s018846	interchange	300.0	2000.0	2400.0	2000.0
line-2-1362-1289-s024085	interchange	300.0	2000.0	2400.0	2000.0
line-2-1309-1219-s026100	interchange	300.0	2000.0	2400.0	2000.0
line-2-1222-1104-s029320	interchange	300.0	2000.0	2400.0	2000.0
line-2-1067-0900-s035129	standard	300.0	2000.0	2000.0	2000.0
line-2-1022-0840-s036713	interchange	300.0	2000.0	2400.0	2000.0
line-2-0903-0683-s041144	standard	300.0	2000.0	2000.0	2000.0
line-2-0825-0580-s044026	interchange	300.0	2000.0	2400.0	2000.0
line-2-0701-0415-s048669	interchange	300.0	2000.0	2400.0	2000.0
line-2-0595-0296-s052851	depot-main	5000.0	40000.0	12000.0	2000.0
line-3-2271-1120-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-3-1751-1233-s013011	interchange	300.0	2000.0	2400.0	2000.0
line-3-1613-1236-s015796	interchange	300.0	2000.0	2400.0	2000.0
line-3-1362-1289-s021278	interchange	300.0	2000.0	2400.0	2000.0
line-3-1281-1310-s023072	interchange	300.0	2000.0	2400.0	2000.0
line-3-1059-1369-s028013	interchange	300.0	2000.0	2400.0	2000.0
line-3-0874-1418-s032165	interchange	300.0	2000.0	2400.0	2000.0
line-3-0764-1447-s034606	interchange	300.0	2000.0	2400.0	2000.0
line-3-0691-1466-s036223	interchange	300.0	2000.0	2400.0	2000.0
line-3-0628-1483-s037624	interchange	300.0	2000.0	2400.0	2000.0
line-3-0465-1526-s041263	standard	300.0	2000.0	2000.0	2000.0
line-3-0378-1545-s043281	standard	300.0	2000.0	2000.0	2000.0
line-3-0288-1607-s046283	standard	300.0	2000.0	2000.0	2000.0
line-3-0151-1637-s049496	terminal	300.0	2000.0	3200.0	2000.0
line-4-0146-0199-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-4-0298-0305-s004261	standard	300.0	2000.0	2000.0	2000.0
line-4-0491-0706-s015030	interchange	300.0	2000.0	2400.0	2000.0

Site station	Tier	PV kW	Storage kWh	Grid import kW	Charger kW
line-4-0604-0839-s018966	interchange	300.0	2000.0	2400.0	2000.0
line-4-0804-1072-s025869	interchange	300.0	2000.0	2400.0	2000.0
line-4-0927-1214-s030089	interchange	300.0	2000.0	2400.0	2000.0
line-4-0985-1282-s032081	interchange	300.0	2000.0	2400.0	2000.0
line-4-1059-1369-s034683	interchange	300.0	2000.0	2400.0	2000.0
line-4-1209-1543-s039827	interchange	300.0	2000.0	2400.0	2000.0
line-4-1230-1627-s041932	terminal	300.0	2000.0	3200.0	2000.0
line-5-1199-0260-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-5-1193-0313-s001126	interchange	300.0	2000.0	2400.0	2000.0
line-5-1190-0317-s001231	interchange	300.0	2000.0	2400.0	2000.0
line-5-1129-0423-s003985	interchange	300.0	2000.0	2400.0	2000.0
line-5-1086-0589-s007673	interchange	300.0	2000.0	2400.0	2000.0
line-5-1022-0840-s013247	interchange	300.0	2000.0	2400.0	2000.0
line-5-0967-1057-s018078	major	300.0	2000.0	2400.0	2000.0
line-5-0927-1214-s021555	interchange	300.0	2000.0	2400.0	2000.0
line-5-0891-1352-s024642	interchange	300.0	2000.0	2400.0	2000.0
line-5-0874-1418-s026103	interchange	300.0	2000.0	2400.0	2000.0
line-5-0853-1593-s029938	interchange	300.0	2000.0	2400.0	2000.0
line-5-0843-1600-s030196	interchange	300.0	2000.0	2400.0	2000.0
line-5-0677-2304-s047345	terminal	300.0	2000.0	3200.0	2000.0
line-6-2322-0251-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-6-1633-0817-s018903	interchange	300.0	2000.0	2400.0	2000.0
line-6-1580-0848-s020342	interchange	300.0	2000.0	2400.0	2000.0
line-6-1457-0929-s023846	interchange	300.0	2000.0	2400.0	2000.0
line-6-1370-0993-s026315	interchange	300.0	2000.0	2400.0	2000.0
line-6-1222-1104-s030195	interchange	300.0	2000.0	2400.0	2000.0
line-6-0985-1282-s036755	interchange	300.0	2000.0	2400.0	2000.0
line-6-0891-1352-s039209	interchange	300.0	2000.0	2400.0	2000.0
line-6-0764-1447-s042688	interchange	300.0	2000.0	2400.0	2000.0
line-6-0713-1485-s044164	interchange	300.0	2000.0	2400.0	2000.0
line-6-0647-1607-s047565	interchange	300.0	2000.0	2400.0	2000.0
line-6-0495-1669-s052503	terminal	300.0	2000.0	3200.0	2000.0
line-7-1069-2206-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-7-1185-1622-s012693	interchange	300.0	2000.0	2400.0	2000.0
line-7-1209-1543-s014684	interchange	300.0	2000.0	2400.0	2000.0
line-7-1281-1310-s019964	interchange	300.0	2000.0	2400.0	2000.0
line-7-1307-1217-s022051	interchange	300.0	2000.0	2400.0	2000.0
line-7-1370-0993-s027076	interchange	300.0	2000.0	2400.0	2000.0
line-7-1402-0878-s029642	interchange	300.0	2000.0	2400.0	2000.0
line-7-1418-0822-s030917	interchange	300.0	2000.0	2400.0	2000.0
line-7-1451-0706-s033517	interchange	300.0	2000.0	2400.0	2000.0
line-7-1504-0514-s037813	standard	300.0	2000.0	2000.0	2000.0
line-7-1531-0418-s039957	terminal	300.0	2000.0	3200.0	2000.0
line-8-2231-1783-s000000	terminal	300.0	2000.0	3200.0	2000.0
line-8-1751-1233-s017689	interchange	300.0	2000.0	2400.0	2000.0
line-8-1645-1094-s021347	interchange	300.0	2000.0	2400.0	2000.0
line-8-1457-0929-s026861	interchange	300.0	2000.0	2400.0	2000.0
line-8-1402-0878-s028547	interchange	300.0	2000.0	2400.0	2000.0
line-8-1341-0823-s030364	interchange	300.0	2000.0	2400.0	2000.0
line-8-1228-0719-s033825	standard	300.0	2000.0	2000.0	2000.0
line-8-1086-0589-s038140	interchange	300.0	2000.0	2400.0	2000.0
line-8-0954-0467-s042166	interchange	300.0	2000.0	2400.0	2000.0
line-8-0829-0364-s045862	standard	300.0	2000.0	2000.0	2000.0
line-8-0785-0290-s048137	terminal	300.0	2000.0	3200.0	2000.0
line-9-0606-0636-s003011	standard	300.0	2000.0	2000.0	2000.0
line-9-0491-0706-s006266	interchange	300.0	2000.0	2400.0	2000.0
line-9-0557-0840-s009730	interchange	300.0	2000.0	2400.0	2000.0
line-9-0562-1131-s016676	interchange	300.0	2000.0	2400.0	2000.0
line-9-0605-1228-s019019	interchange	300.0	2000.0	2400.0	2000.0
line-9-0627-1485-s025111	interchange	300.0	2000.0	2400.0	2000.0
line-9-0648-1607-s028125	interchange	300.0	2000.0	2400.0	2000.0
line-9-0843-1600-s032244	interchange	300.0	2000.0	2400.0	2000.0
line-9-0853-1593-s032502	interchange	300.0	2000.0	2400.0	2000.0

Site station	Tier	PV kW	Storage kWh	Grid import kW	Charger kW
line-9-1185-1622-s040057	interchange	300.0	2000.0	2400.0	2000.0
line-9-1230-1627-s040998	interchange	300.0	2000.0	2400.0	2000.0
line-9-1429-1525-s047139	standard	300.0	2000.0	2000.0	2000.0
line-9-1504-1477-s049136	interchange	300.0	2000.0	2400.0	2000.0
line-9-1613-1236-s056763	interchange	300.0	2000.0	2400.0	2000.0
line-9-1645-1094-s059868	interchange	300.0	2000.0	2400.0	2000.0
line-9-1580-0848-s066616	interchange	300.0	2000.0	2400.0	2000.0
line-9-1554-0819-s067411	interchange	300.0	2000.0	2400.0	2000.0
line-9-1451-0706-s071327	interchange	300.0	2000.0	2400.0	2000.0
line-9-1336-0518-s076879	interchange	300.0	2000.0	2400.0	2000.0
line-9-1194-0440-s081155	standard	300.0	2000.0	2000.0	2000.0
line-9-1129-0423-s085079	interchange	300.0	2000.0	2400.0	2000.0
line-9-0954-0467-s089572	interchange	300.0	2000.0	2400.0	2000.0
line-9-0825-0580-s094205	interchange	300.0	2000.0	2400.0	2000.0
line-9-0689-0569-s097310	standard	300.0	2000.0	2000.0	2000.0

Six month capital and supplemental draw requirements

Cost priority with identical three month operating buffers. Settlement is monthly; six month rows are envelopes, not advance placements. Unrounded CSV values reconcile; displayed amounts are rounded. Tranches continue through the entire model horizon.

Months	Gov USD m	Gov IQD bn	China USD m	Ordinary bond IQD bn	Green IQD bn	Bank IQD bn	Gap IQD bn
0 to 5	12.71	35.27	12.71	104.13	0.00	34.71	4.98
6 to 11	32.04	79.08	32.04	240.40	0.00	80.13	17.16
12 to 17	22.28	66.12	22.28	192.21	0.00	64.07	34.98
18 to 23	41.20	107.37	41.20	320.86	1.05	107.30	58.05
24 to 29	61.05	108.22	61.05	302.83	58.53	120.45	68.59
30 to 35	91.69	125.17	91.69	317.07	140.48	152.52	93.89
36 to 41	93.19	130.29	93.19	324.43	147.41	157.28	294.25
42 to 47	94.06	133.32	94.06	332.58	147.75	160.11	195.59
48 to 53	95.55	132.44	95.55	330.07	151.13	160.40	215.35
54 to 59	92.92	131.79	92.92	329.49	145.24	158.24	215.54
60 to 65	98.05	135.98	98.05	340.93	153.07	164.67	259.24
66 to 71	93.72	127.41	93.72	317.20	149.16	155.45	235.99
72 to 77	68.21	100.06	68.21	266.26	90.05	118.77	209.27
78 to 83	2.14	3.20	2.14	8.73	2.60	3.78	133.96
84 to 89	0.00	0.00	0.00	0.00	0.00	0.00	74.65
90 to 95	0.00	0.00	0.00	0.00	0.00	0.00	38.61
96 to 101	0.00	0.00	0.00	0.00	0.00	0.00	8.37
102 to 107	0.00	0.00	0.00	0.00	0.00	0.00	0.00
108 to 113	0.00	0.00	0.00	0.00	0.00	0.00	0.00
114 to 119	0.00	0.00	0.00	0.00	0.00	0.00	0.00
120 to 125	0.00	0.00	0.00	0.00	0.00	0.00	0.00
126 to 131	0.00	0.00	0.00	0.00	0.00	0.00	0.00
132 to 137	0.00	0.00	0.00	0.00	0.00	0.00	0.00
138 to 143	0.00	0.00	0.00	0.00	0.00	0.00	0.00
144 to 149	0.00	0.00	0.00	0.00	0.00	0.00	0.00
150 to 155	0.00	0.00	0.00	0.00	0.00	0.00	0.00
156 to 161	0.00	0.00	0.00	0.00	0.00	0.00	0.00
162 to 167	0.00	0.00	0.00	0.00	0.00	0.00	0.00
168 to 173	0.00	0.00	0.00	0.00	0.00	0.00	0.00
174 to 179	0.00	0.00	0.00	0.00	0.00	0.00	0.00
180 to 185	0.00	0.00	0.00	0.00	0.00	0.00	0.00
186 to 191	0.00	0.00	0.00	0.00	0.00	0.00	0.00
192 to 197	0.00	0.00	0.00	0.00	0.00	0.00	0.00
198 to 203	0.00	0.00	0.00	0.00	0.00	0.00	0.00
204 to 209	0.00	0.00	0.00	0.00	0.00	0.00	0.00
210 to 215	0.00	0.00	0.00	0.00	0.00	0.00	0.00
216 to 221	0.00	0.00	0.00	0.00	0.00	0.00	0.00
222 to 227	0.00	0.00	0.00	0.00	0.00	0.00	0.00
228 to 233	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Months	Gov USD m	Gov IQD bn	China USD m	Ordinary bond IQD bn	Green IQD bn	Bank IQD bn	Gap IQD bn
234 to 239	0.00	0.00	0.00	0.00	0.00	0.00	0.00
240 to 245	0.00	0.00	0.00	0.00	0.00	0.00	0.00
246 to 251	0.00	0.00	0.00	0.00	0.00	0.00	0.00
252 to 257	0.00	0.00	0.00	0.00	0.00	0.00	0.00
258 to 263	0.00	0.00	0.00	0.00	0.00	0.00	0.00
264 to 269	0.00	0.00	0.00	0.00	0.00	0.00	0.00
270 to 275	0.00	0.00	0.00	0.00	0.00	0.00	0.00
276 to 281	0.00	0.00	0.00	0.00	0.00	0.00	0.00
282 to 287	0.00	0.00	0.00	0.00	0.00	0.00	0.00
288 to 293	0.00	0.00	0.00	0.00	0.00	0.00	0.00
294 to 299	0.00	0.00	0.00	0.00	0.00	0.00	0.00
300 to 305	0.00	0.00	0.00	0.00	0.00	0.00	0.00
306 to 311	0.00	0.00	0.00	0.00	0.00	0.00	0.00
312 to 317	0.00	0.00	0.00	0.00	0.00	0.00	0.00
318 to 323	0.00	0.00	0.00	0.00	0.00	0.00	0.00
324 to 329	0.00	0.00	0.00	0.00	0.00	0.00	0.00
330 to 335	0.00	0.00	0.00	0.00	0.00	0.00	0.00
336 to 341	0.00	0.00	0.00	0.00	0.00	0.00	0.00
342 to 347	0.00	0.00	0.00	0.00	0.00	0.00	0.00
348 to 353	0.00	0.00	0.00	0.00	0.00	0.00	0.00
354 to 359	0.00	0.00	0.00	0.00	0.00	0.00	0.00
360 to 365	0.00	0.00	0.00	0.00	0.00	0.00	0.00
366 to 371	0.00	0.00	0.00	0.00	0.00	0.00	0.00
372 to 377	0.00	0.00	0.00	0.00	0.00	0.00	0.00
378 to 383	0.00	0.00	0.00	0.00	0.00	0.00	0.00
384 to 389	0.00	0.00	0.00	0.00	0.00	0.00	0.00
390 to 395	0.00	0.00	0.00	0.00	0.00	0.00	0.00
396 to 401	0.00	0.00	0.00	0.00	0.00	0.00	0.00
402 to 407	0.00	0.00	0.00	0.00	0.00	0.00	0.00
408 to 413	0.00	0.00	0.00	0.00	0.00	0.00	0.00
414 to 419	0.00	0.00	0.00	0.00	0.00	0.00	0.00
420 to 425	0.00	0.00	0.00	0.00	0.00	0.00	0.00
426 to 431	0.00	0.00	0.00	0.00	0.00	0.00	0.00
432 to 437	0.00	0.00	0.00	0.00	0.00	0.00	0.00
438 to 440	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Six month early repayments and closing balances

Early principal is additional to the scheduled principal separately retained in the full ledger. Outstanding debt uses the final month of each period. USD equivalent aggregates convert Chinese native debt at the historical IQD 1,300 anchor. Premiums and operating buffers are distinct from principal.

Months	Core early principal USD eq m	Premium USD eq m	Gap principal IQD bn	All debt closing IQD tn eq	Cash closing IQD tn	OPEX buffer IQD bn
0 to 5	0.00	0.00	0.00	0.160	0.000	0.000
6 to 11	0.00	0.00	0.00	0.536	0.000	0.000
12 to 17	0.00	0.00	0.00	0.849	0.000	0.000
18 to 23	0.00	0.00	0.00	1.377	0.000	0.000
24 to 29	28.27	0.28	0.00	1.953	0.000	0.000
30 to 35	36.88	0.37	0.00	2.703	0.000	0.000
36 to 41	32.63	0.33	0.00	3.670	0.000	26.504
42 to 47	20.38	0.20	0.00	4.556	0.000	30.794
48 to 53	16.35	0.16	0.00	5.455	0.000	39.467
54 to 59	16.81	0.17	0.00	6.328	0.000	51.930
60 to 65	0.00	0.00	0.00	7.280	0.000	61.306
66 to 71	0.00	0.00	0.00	8.148	0.000	68.795
72 to 77	0.00	0.00	0.00	8.788	0.000	78.195
78 to 83	0.00	0.00	0.00	8.789	0.000	89.287
84 to 89	0.00	0.00	0.00	8.696	0.000	93.751
90 to 95	0.00	0.00	0.00	8.550	0.000	93.751
96 to 101	0.00	0.00	0.00	8.354	0.000	98.439
102 to 107	44.91	0.45	0.00	8.087	0.000	98.439
108 to 113	98.13	0.98	0.00	7.774	0.000	103.360

Months	Core early principal USD eq m	Premium USD eq m	Gap principal IQD bn	All debt closing IQD tn eq	Cash closing IQD tn	OPEX buffer IQD bn
114 to 119	127.21	1.27	0.00	7.444	0.000	103.360
120 to 125	158.49	2.49	0.00	7.091	0.000	108.528
126 to 131	170.03	3.40	0.00	6.729	0.000	108.528
132 to 137	204.44	4.09	0.00	6.330	0.000	113.955
138 to 143	230.01	4.60	0.00	5.908	0.000	113.955
144 to 149	266.41	5.33	0.00	5.448	0.000	119.653
150 to 155	296.91	5.94	0.00	4.962	0.000	119.653
156 to 161	340.20	6.80	0.00	4.433	0.000	125.635
162 to 167	375.78	7.37	0.00	3.875	0.000	125.635
168 to 173	430.12	4.30	0.00	3.268	0.000	131.917
174 to 179	458.60	4.59	0.00	2.648	0.000	131.917
180 to 185	371.86	3.72	162.06	1.996	0.000	138.513
186 to 191	0.00	0.00	644.90	1.352	0.000	138.513
192 to 197	0.00	0.00	675.86	0.676	0.000	145.439
198 to 203	0.00	0.00	675.68	0.000	0.014	145.439
204 to 209	0.00	0.00	0.00	0.000	0.733	152.710
210 to 215	0.00	0.00	0.00	0.000	1.459	152.710
216 to 221	0.00	0.00	0.00	0.000	2.213	160.346
222 to 227	0.00	0.00	0.00	0.000	2.974	160.346
228 to 233	0.00	0.00	0.00	0.000	3.763	168.363
234 to 239	0.00	0.00	0.00	0.000	4.560	168.363
240 to 245	0.00	0.00	0.00	0.000	5.387	176.781
246 to 251	0.00	0.00	0.00	0.000	6.222	176.781
252 to 257	0.00	0.00	0.00	0.000	7.089	185.621
258 to 263	0.00	0.00	0.00	0.000	7.964	185.621
264 to 269	0.00	0.00	0.00	0.000	8.872	194.902
270 to 275	0.00	0.00	0.00	0.000	9.789	194.902
276 to 281	0.00	0.00	0.00	0.000	10.741	204.647
282 to 287	0.00	0.00	0.00	0.000	11.702	204.647
288 to 293	0.00	0.00	0.00	0.000	12.700	214.879
294 to 299	0.00	0.00	0.00	0.000	13.707	214.879
300 to 305	0.00	0.00	0.00	0.000	14.753	225.623
306 to 311	0.00	0.00	0.00	0.000	15.809	225.623
312 to 317	0.00	0.00	0.00	0.000	16.905	236.904
318 to 323	0.00	0.00	0.00	0.000	18.012	236.904
324 to 329	0.00	0.00	0.00	0.000	19.161	248.749
330 to 335	0.00	0.00	0.00	0.000	20.322	248.749
336 to 341	0.00	0.00	0.00	0.000	21.527	261.187
342 to 347	0.00	0.00	0.00	0.000	22.744	261.187
348 to 353	0.00	0.00	0.00	0.000	24.007	274.246
354 to 359	0.00	0.00	0.00	0.000	25.283	274.246
360 to 365	0.00	0.00	0.00	0.000	26.608	287.958
366 to 371	0.00	0.00	0.00	0.000	27.946	287.958
372 to 377	0.00	0.00	0.00	0.000	29.335	302.356
378 to 383	0.00	0.00	0.00	0.000	30.738	302.356
384 to 389	0.00	0.00	0.00	0.000	32.195	317.474
390 to 395	0.00	0.00	0.00	0.000	33.667	317.474
396 to 401	0.00	0.00	0.00	0.000	35.194	333.348
402 to 407	0.00	0.00	0.00	0.000	36.738	333.348
408 to 413	0.00	0.00	0.00	0.000	38.340	350.015
414 to 419	0.00	0.00	0.00	0.000	39.959	350.015
420 to 425	0.00	0.00	0.00	0.000	41.640	367.516
426 to 431	0.00	0.00	0.00	0.000	43.338	367.516
432 to 437	0.00	0.00	0.00	0.000	45.101	385.892
438 to 440	0.00	0.00	0.00	0.000	45.991	385.892

Complete civil chainages · station names and attributes · junction records · complete repayment ledger

Baghdad detailed engineering and Iraqi manufacturing plan

Source: cities/catalogue/west-asia/Iraq/Baghdad/DETAILED-ENGINEERING.md

Baghdad uses a six-car, 111 m metro family. The next design stage retains ballastless slab track and expands the civil, mechanical, electronics and software packages into controlled interfaces, parts and manufacturing operations. The component register, parts CSV and source-bound JSON contain the calculated city quantities and open supplier positions. These are engineering references; production drawings, surveys, qualified suppliers and operational approvals remain required.

The current city allocation, six-car trainsets, cars, stations, energy sites, bogies and wheelsets are regenerated in the quantity register from the main design and scenario. Wheelsets are children of complete bogies, not extra purchases. Doors, cooling loops, fixtures, harnesses and equipment repeats are reference allocations that must pass the six-car family review. The detailed three-car LM3 manufacturing package is useful process evidence; it is not a released Baghdad production package and cannot be doubled to obtain qualified six-car interfaces.

Reference architecture and drawing package

The mechanical load path is wheelset to suspension/axlebox to bogie frame to secondary suspension/pivot to car underframe to coupler and body structure. Cladding attaches to the structural frame through controlled service rails and retention features; FRP panels do not replace the primary crash structure. Keep longitudinal crash loads, recovery loads, passenger fixture loads and equipment retention as separate calculation cases with a common drawing revision.

The electrical path is protected charging contact to traction battery/HV junction to inverters/motors and protected auxiliary DC/DC branches. Battery, inverter and motor cooling circuits require a coordinated heat balance. The nominal traction bus is 675 V and normal maximum 740 V; “800 V class” describes a component class, not the normal bus voltage. Coordination must include abnormal/transient limits, insulation monitoring, HV interlocks, precharge, discharge, isolation and emergency disconnect.

Controlled drawing	Required content	Reference data available	Release owner / remaining evidence
BG-M-001 Trainset arrangement	Six-car positions, couplers, gangways, swept and evacuation envelopes	111 m train envelope; six cars; five intercar interfaces; two ends	Rolling-stock authority: exact end/coupler lengths, curve sweep and tolerance stack
BG-M-002 Car structure	Underframe/side/roof members, joints, datums, lift/recovery points	18.5 m average module allocation; geometry is not a cut list	Structures: stress/fatigue/crash, weld map, material thicknesses, distortion and NDT
BG-M-003 Bogie interface	Pivots, spring/air attachments, wheel profile, brake/motor clearances	One powered and one trailer bogie per car reference	Supplier + dynamics authority: axle loads, suspension rates, hunting, curves and brake duty
BG-M-004 Battery and HV bay	Racks, crash retention, isolation, vent paths and removable pack access	225 kWh gross allocation per car; 1,350 gross / 1,080 usable per train	HV/battery authority: actual pack mass/dimensions, fault current, thermal propagation and retention
BG-M-005 Cooling and HVAC	Chiller/radiator/pump, hoses, expansion/bleed, ducts and condensate	Hot-desert 50 °C design ambient	Thermal authority: traction/aux duty, degraded operation, dust and heat rejection
BG-M-006 Door/glazing	Openings, leaves, locks, seals, glazing retention and maintenance access	Two double-leaf openings per side/car reference	Body + accessibility authority: family-specific layout, loading, escape and obstruction
BG-M-007 Harness and connector	Wire/terminal IDs, cavity views, cut lengths, routing, clamps and separation	HV/LV/data service zones; reference host counts	Electrical authority: current/voltage drop, insulation, EMC, connector and supplier pin freeze
BG-M-008 Interior/accessibility	Seats, rails, ramps/steps, lighting, fire barriers and service panels	720 nominal passengers, 120 seats, 960 crush per train	Interior authority: fixture loads, fire/smoke, accessibility and passenger evacuation
BG-C-001 ST6 and slipform section	Concrete geometry, cage, inserts, rail seats, drainage and transitions	Six-metre panel geometry and 20 seats	Track/civil authority: reinforcement, support stiffness, shrinkage, fasteners and actual geotechnical zones
BG-C-002 Viaduct and bridge interfaces	Pi20/Pi25, foundations, bearings, joints, CWR and walkway interfaces	Standard 20/25 m spans; bridge-specific exceptions	Structures: permanent/variable/thermal/seismic/braking, fatigue, scour and foundation release
BG-E-001 End electronics cabinet	Rails, protection, output chain, harness, thermal and enclosure	Corrected T-OBS capacity and exposed Pico bench pins	Electronics authority: actual vendor BOM, schematic, HAL, EMC and injected faults

Use train longitudinal datum X, track centre Y and surveyed rail datum Z for integration. Each supplier drawing must state its own mating datum and tolerance contribution. The track reference gauge is 1,435 mm between the running faces; rail-head width and inclination are needed to convert this into rail-centre/baseplate coordinates. Existing simplified CAD rail-line positions must not be used as drilling coordinates. A platform step/gap or bogie pivot offset copied from LM3 is an unresolved six-car interface until reviewed.

Mechanical manufacturing and missing kit content

Local work can cover steel structure fabrication, service rails, FRP moulding/trim, interiors, mounts, battery enclosures, harness manufacture, pipework and final integration. Initially import complete qualified bogies, motor/inverter packages, cells/qualified packs, safety glazing, doors and electronics where local process and product qualification is absent. A later local-content decision needs supplier drawings/licences, process capability, acceptance tests and economics; local fabrication is not evidence that a safety-critical design is accepted.

Work package	Make/buy and production route	Child content that must be explicit	Verification / hold point
Car frame	Iraqi cut/form/machine/fixture/weld	Plates/sections, brackets, joint preparation, isolation, drains and finish	Released datum/fixture, WPS, weld/NDT, dimensional and load inspection
FRP skin	Iraqi mould/layup/cure/trim/fit	Resin/fibre/core, inserts, retention clips, edge seals and fire-compatible finish	Laminate/fire/smoke, cure coupon, insert pullout and panel retention
Bogie	Complete supplier package first	Wheels/axles, bearings, suspension, dampers, brakes, grounding and guards	Family axle-load/dynamics/brake qualification and complete serial records
Battery/HV	Import qualified core; locally integrate after approval	Rack, fuses, contactors, precharge, disconnect, HVIL, BMS, sensors, vent/containment and service lift	Supplier duty, propagation, isolation, fault-current and retention tests
Thermal	Local formed tube/hose/brackets with sourced pumps/chillers	Fluid, filters, manifolds, isolation valves, drains, bleed/expansion and leak detection	Pressure/cleanliness, heat balance at hot ambient and failure response
Brake	Supplier-controlled complete design	Compressor, dryer, reservoir, valve blocks, parking brake, hoses and sensors	Pressure/leak, stop distance, adhesion/degraded modes and rescue
Doors/glass	Qualified sourced units with local frame/fitting	Lock/detection/obstruction, emergency release, drainage, seals, adhesive and retainers	Opening alignment, force/leak/retention and interlock to safety host
Wiring	Iraqi formboard/cut/crimp/test	Cable, contacts, mating halves, seals, glands, shielding, clamps and labels	Calibrated crimp/pull, cavity checks, continuity and insulation
Final vehicle	Local installation and configuration	Torque evidence, fluids, firmware, mass/CG, identification and handover records	Weighed family vehicle, static brake/door tests, route acceptance and traceability

The existing small-component standard, factory release, tooling, inspection and mass-closure modules remain useful. Their product IDs and drawing seeds must be cross-referenced to a Baghdad family-specific EBOM/MBOM before ERP Work Orders are released. No exact bolt torque, weld size, laminate layup or heat-exchanger area is invented here. Those values must close the applicable load, thermal or supplier calculation.

Slab track decision

Choose ballastless slab track as the reference permanent way. Use continuous slipforming for long, open, machine-accessible at-grade sections; use Iraqi-made OSR-ST6 single-track precast panels in constrained streets, utility/transition zones, short possessions, station/depot interfaces and replaceable sections. On standard elevated Pi20/Pi25 structures, use direct-fixation plinths above the structural flange with a thin alignment layer only where needed. Floating track is a location-specific vibration treatment, not a blanket addition.

Slab track is selected for the design package because it supports controlled alignment, modular installation and local concrete work. Lifecycle value must be tested against settlement risk, access, repairability, construction traffic, drainage and noise. It is not inherently cheaper on weak or uninvestigated ground. Keep ballasted/special transition arrangements available as engineered exceptions where life-cycle, geotechnical or depot requirements justify them.

ST6 single-track reference	Value / quantity
Dimensions	6,000 × 2,900 mm; 250 mm base; two 380 × 160 mm plinths
Double track	Two independently manufactured rows on 3,500 mm centres
Concrete, base and plinths	5.0796 m³ per panel; excludes service-trough envelope, bedding and steel
Bare concrete planning mass	12.699 t at 2,500 kg/m³; steel, inserts and lifting rig add weight
Reinforcement allowance	761.94 kg at 150 kg/m³; not a released bar schedule
Rail seats	20 per panel; 10 per rail, at X = 300 + 600 × i mm, i = 0…9
Repeated-module seat pitch	600 mm including the panel joint; maximum reference spacing remains 650 mm
Complete seat child allocation	One pad/baseplate kit, two clips, two lateral insulators, two anchor assemblies and one adjustment kit; supplier pattern to freeze
Bedding illustration	30 mm over 17.4 m² = 0.522 m³; actual support/grout design required
Lifting illustration	Four engineered anchors; supplier load sharing, strength, angle and rigging required

The previous CAD placed the first baseplate centre at the concrete edge, extending its 260 mm footprint beyond the panel. The corrected layout centres seats inside the module and keeps uniform pitch across repeated joints. Continuous-run planning quantities still use the existing 650 mm basis; precast method zones need their own 20-seat-per-panel schedule. Do not apply a single rounded per-km seat allowance as a fabrication order for every six-metre panel.

A complete track seat needs the matched rail, baseplate, resilient pad/layer, clips, insulators, anchors, shim/adjustment and tested fastening system. The legacy sleeper/clip CAD remains a ballasted alternative; it is not the default slab-track MBOM.

Source proprietary/safety-critical elastic parts and qualified anchors initially; manufacture concrete/cages, ducts/covers, moulds, jigs and appropriate metalwork locally. Local clip or pad manufacture requires licensing/material/process/fatigue and insulation qualification, not just matching the rendered shape.

Pandrol's published direct-fixation system provides an example of adjustable, resilient fastening for slab structures with system-specific performance and testing. This supports the architecture; it does not select Pandrol, approve an equivalent, transfer the vendor stiffness/adjustment values to OSR or establish Iraq pricing. Pandrol technical sheet.

Iraqi slab manufacturing line

Add a dedicated slab line alongside the shared Baghdad plant or a route-adjacent civil yard after a haulage/capacity comparison. Do not consume the same Pi-beam moulds, crane shifts or curing spaces twice in the production programme. A mould row must record mould ID, cycle, available shifts, crane allocation, rejected pieces and actual release strength. The previous 24-hour mould-cycle target remains a trial target; use the existing 48-hour planning basis until hot-weather trials substantiate an alternative.

1. Release panel/cage/insert drawings, supplier fastener pattern, concrete mix/exposure class and prototype inspection plan.
2. Qualify Iraqi cement, aggregate, water and admixtures for the site sulfate/chloride exposure. Check shrinkage, creep and curing rather than assuming a higher concrete grade solves durability.
3. Build surveyed reusable steel moulds, cage/weld jigs and insert/rail-seat gauges. Fit engineered lifting sockets, bonding/drain sleeves and unique panel identity.
4. Inspect cage cover, insert positions, cleanliness and mould dimensions before casting. Record supplier/batch and mixing conditions.
5. Place/vibrate the approved mix or use qualified self-compacting concrete. Protect from heat/wind, control moisture and record maturity against strength tests.
6. Release lifting only at the calculated strength. Survey rail-seat datums, flatness, geometry and cracks; record repair/rejection decisions against panel serials.
7. Transport with designed support points and route clearance; inspect after transport. Install on accepted formation/bed, grout under the released procedure and survey rail alignment.
8. Install the matched fastening system, complete CWR welding/stress management, verify gauge/cant/height and inspect bonding/insulation/drainage. Record the as-installed rail/fastener/panel configuration.
9. Perform proof loading/ride/geometry and monitored settlement trials before series deployment. Retain replaceability and maintenance-access demonstrations.

The line requires batching/aggregate handling, controlled curing, slab moulds, cage jigs, insert fixtures, maturity/strength testing, survey gear, lifting frames, certified rigging and loading/storage space. Iraqi labour can fabricate cages/moulds and operate casting, survey, installation and maintenance; specialist training, qualified lifting/welding and independent inspection need funded work packages. No construction-job count or local-content percentage is asserted without a priced resource and origin schedule.

A quantity example for one kilometre of double track wholly assigned to ST6 is 334 full panels = 2 × ceil(1,000/6), 6,680 seats and 1,696.5864 m³ of base/plinth concrete. These cover 1,002 m per track with end trimming/interface treatment unresolved. It is an example, not Baghdad's method allocation. If the surveyed constrained length is L on each of the two tracks, use 2 × ceil(L/6) and separately schedule short end pieces; do not infer L from all at-grade kilometres.

Civil interfaces that need completion

Interface	Required design information	Acceptance and cost implication
Formation/slab	Ground zones, groundwater, modulus, settlement, sulfate exposure, utility trenches	Settlement and transition design; preparation/ground improvement measured by zone
Drainage/flood	Catchments, design storm, outlet levels, maintenance access, debris and scour	Cable routes separated from wet channels; outfalls/attenuation/pumps counted where required
Viaduct/slab/CWR	Thermal range, neutral rail temperature, braking, deck movement and restraint	Rail–structure interaction analysis; no automatic expansion device at every deck gap
Bearings/continuity	Unit lengths, pier stiffness, seismic/thermal restraints, replacement/lift access	Supplier bearing envelope and certified reactions; quantity schedule by actual support
River/highway bridges	Survey/navigation, scour, traffic/diversion, foundations and special spans	Separate bridge design; standard Pi spans do not release major crossings
Elevated evacuation	Walkway/barrier loads, clear width, station access and rescue route	Outer 1 m cassette reference checked against obstruction and fire/evacuation rules
Station boarding	Six-car swept envelope, floor height, tolerance and accessibility	Surveyed step/gap and dwell/evacuation capacity; LM3 offsets cannot be copied blindly
Depot/stabling	Full-fleet circulation and storage, workshop equipment, wash/drainage/recovery	One depot per line is priced by fleet and workload in the current programme; workshop bays are not parking
Plant/logistics	Crane charts, rig loads, haul route, curing, power/water/waste and laydown	12.699 t concrete panel is not the certified shipping/lift weight; steel/rigging included

Keep the existing construction-method and foundation catalogs. Their “actual length and cost required” gates prevent generic deep-pile lengths from being adopted as surveyed Baghdad foundations. Release each geotechnical zone and map it to the alignment segments before recalculating quantities and installed costs.

Onboard electronics corrections

The revised T-OBS bench design and integration JSON close erroneous interface assumptions and add missing power, watchdog, relay-feedback, harness and external-controller blocks. They do not release a production board or pretend that a driver exists because its crate appears in the workspace.

Issue identified	Corrected design treatment
HDR-60-24 treated as 24 V to 5/12 V	It is AC/high-voltage DC input to 24 V output; use appropriately protected DC/DC branches onboard
USB isolator used as SPI / Pico device-to-device link	Use directionally correct SPI isolation with one controller and one peripheral
TPS3701 called dual heartbeat watchdog	Voltage detector only; separate channel heartbeat watchdogs and hardware inhibition
Pico GP30–39, ADC at GP14–17, native CAN/RMII	Exposed Pico pins and supported functions only; custom chip pinout separately frozen; external CAN/Ethernet controllers
ATECC608B driven through SPI	Selected breakout uses I2C; provisioning and exact variant confirmed
Digital ultrasonic echo sampled through ADS1115	Timed digital capture with voltage-level conversion; raw piezo requires an AFE/driver design
Radar said to ship M12 / USB-C	AWR1843BOOST EVM uses vendor J3 CAN, micro USB and 5 V barrel; add qualified adapter/harness
HAP claimed Gigabit/12 V PoE	Select HAP TX 100BASE-TX; vendor 9–18 V power separately; T1 interface differs
Ultrasonic face hidden behind radome	Separate acoustic apertures and tested radar/optical windows
Under-20-W host and 1-A input protection	71.5 W simultaneous output sizing envelope; 79.44 W at 90% efficiency; 99.31 W with 25% margin; 100 W reference capacity
Two relays asserted “always safe”	Require end-to-end polarity, feedback, welded-contact/driver faults and common-cause analysis

Capacity is not measured average power. Do not inflate train OPEX from a 100 W converter nameplate or retain a 40 W allowance as if it covers the updated candidate assembly. Production sensor and firmware selection must provide a measured duty cycle for the auxiliary energy/thermal model. Exact wire gauge, fuse, supply voltage range, bus topology, PCB layout, heat spreader and relay remain design-owned.

Primary specifications checked 3 October 2026: Mean Well, TI voltage detector, TI radar, Livox, Raspberry Pi carrier, Pico 2, Microchip secure element. Candidate lifecycle/price and rail suitability are not implied by a working datasheet link.

Software and hardware integration work

The component register verifies all 60 Rust workspace packages against `deployment/components.toml`. Existing control, diagnostics, AFC, CBM, station, supervision and identity software is retained. RFC0033 TACS runtime/resource control remains a process reference; it is not installed Baghdad protection equipment. A logical redundant process cannot substitute for independent power, I/O, sensing and qualified timing.

Work package	Required binding / acceptance
Device HAL	Board-revision ADC/timer/PIO, external CAN/Ethernet, GPIO defaults, watchdog and relay feedback; mock tests plus actual bench traces
Sensor ingress	Vendor framing, calibration, timestamps/sequence/CRC, data age, malformed frames, packet loss and common sensor faults

Work package	Required binding / acceptance
Protection output	Complete boot/reset/timeout/permission truth table and hardware watchdog inhibition; braking and traction supplier interfaces
Runtime deployment	Qualified host composition, task/WCET budgets, fault containment, resource authority, committed resource state and degraded recovery
Clock/network	Actual timestamp accuracy, redundant routes/switches, PTP loss, CAN termination, bandwidth and shared-path failure analysis
Target images	Reproducible signed build, configuration/board hash, driver versions, SBOM, secure boot/update/rollback and release evidence
Operations integration	Event provenance, CBM/Issue triage, maintenance states, trusted operator decisions and reporting; ERP/FUXA remain outside the safety authority
Ticketing/revenue	Actual validators/payment/acquirer contracts, offline/replay/refund behaviour, Arabic UX, tariff versions and daily settlement reconciliation
Cybersecurity/service	Key provisioning/revocation, account/role access, logs/storage limits, patch ownership and restore drills

The image cookbook explicitly records that production SD images are absent. Leave that fact visible. Add real driver/board work against each package and evidence gate; do not create empty drivers or announce “complete onboard firmware” from successful host-library tests. Hardware-in-loop must test a full protected output path using representative I/O and faults.

ERP recheck and detail handover

The city component compiler, city configuration validator and supervision asset validator pass for all 266 cities; supervision covers 249,114 equipment records. This proves reusable planning/asset contracts. Baghdad's enabled reusable profile currently contains the induction training programme; it does not automatically create production, purchasing, quality or maintenance documents.

The local ERP MariaDB service was repeatedly failing on 3 October 2026 because it could not create temporary InnoDB files under `/tmp`. The compose definition now gives the database a writable, bounded 512 MiB temporary filesystem with mode 1777. The database was recreated against its existing persistent volume; it became healthy and the ERP ping API returned HTTP 200. No city assets, Work Orders, stock movements or loan records were fabricated during this repair. Health does not replace backup/restore or load testing.

ERP handover	Existing capability	Inputs needed to activate the detailed package
Engineering parts / Item / BOM	Native Items, nested BOMs and version exposure/disposition	Exact family part/revision, UOM, make/buy, supplier alternatives, child content and approved production BOM
Production	Draft Work Orders and native execution/Job Cards	Approved BOM, city source/WIP/FG warehouses, workstations, routing, actual hours, dated capacity and operator permissions
Procurement / stock	Project material requests, purchasing, receipts, replenishment	Landed freight/duty/tax, lead time, MOQ, safety stock, serial/batch and warehouse availability; no invented stock
Quality	Draft native Quality Inspections and evidence references	Released characteristics, sampling and limits, gauges/calibration, inspector, receipt line and retained actual results
Maintenance / CBM	Asset Maintenance, Issue-driven repair preview and controlled draft repair	Commissioned serial asset, approved interval/team/date, spares and handback acceptance
Training	Reusable programme and native events	Competence criteria, assessor, actual attendance and role authorisation
Finance	Native currency/accounting and project budgets; proposal cashflow remains a separate planning model	Iraqi legal company, fiscal year, IQD chart/accounts, USD-import clearing, actual bank/loan/bond terms, treasury schedule and approvals
Lifecycle	Version/evidence/disposition and physical identity templates	Real HTTPS resolver, physical serial binding and verified evidence before printable deployment labels

Use the 69 engineering rows as reference requirements first. They are not automatically posted as live ERP Items or added to existing priced city materials; complete assemblies and their child content need a nested MBOM to avoid double counting. The generic ERP manufacturing/quality/maintenance contracts should remain disabled until actual approved inputs exist.

For Baghdad finance, keep the previously selected government share 25%, imported component cash/Chinese-credit split 50/50 and all other cash/debt/bonds in IQD. Confirm company currency IQD, actual FX conversion dates and USD exposures when live treasury accounting is configured. Imported procurement is not automatically all Chinese-origin credit-eligible; rails, fastening hardware, safety electronics and tooling need actual origin/loan-eligibility schedules. Foreign climate money is not IQD financing unless an on-lender or a priced hedge provides that currency structure.

Cost and release reconciliation

The parts register does not reprice individual civil or vehicle components. The separate city-sized factory calculation now replaces the four-slot production assumption with physical six-car cells, crews, tooling and test paths, with facility readiness at 18 months from NTP. Its larger physical plant allowance and EPC are included once in the Baghdad funding programme. It adds no second plant or future national-city debt. Existing civil and vehicle allowances already contain planning permanent-way and equipment scope; the component breakdowns clarify what a complete package must contain. Unknown detailed unit costs remain null/open, not zero or a claim that all omissions are covered. The depot, civil manufacturing line, full control-electronics integration, spares, qualification and measured auxiliary duty need RFQs and scope reconciliation before the current financing model becomes a tender-grade model.

For each priced assembly, mark child lines as included, separately payable or excluded; bind quantities, installed rates and origin to one revision. Compare the detail subtotal with its current allowance, update CAPEX/import split/EPC only for an accepted scope delta, then regenerate the Baghdad cashflow, tranche and debt sweep outputs together. A prototype/localisation plan must not be treated as an agreed Chinese loan, grant or bond subscription.

Release sequence: surveyed alignment/ground/utility zones to family and track interface freeze to priced complete BOM/MBOM to local manufacturing trials to hardware/software integration and fault tests to independent review to approved construction/production documents to field acceptance. This makes missing components actionable without marking them as manufactured or qualified.

Baghdad — Urban Rail Network

Source: cities/catalogue/west-asia/Iraq/Baghdad/README.md

> Primary v0.5 ferrite-maglev planning: Open the current city planning package. Formation, duty, fleet, charging, depot quantities and partial cost allowances are recomputed; physical acceptance remains open.
>
> The material below this notice is the retained v0.4 wheel/rail comparator. Its costs, fleet, geometry and acceptance do not transfer to maglev.

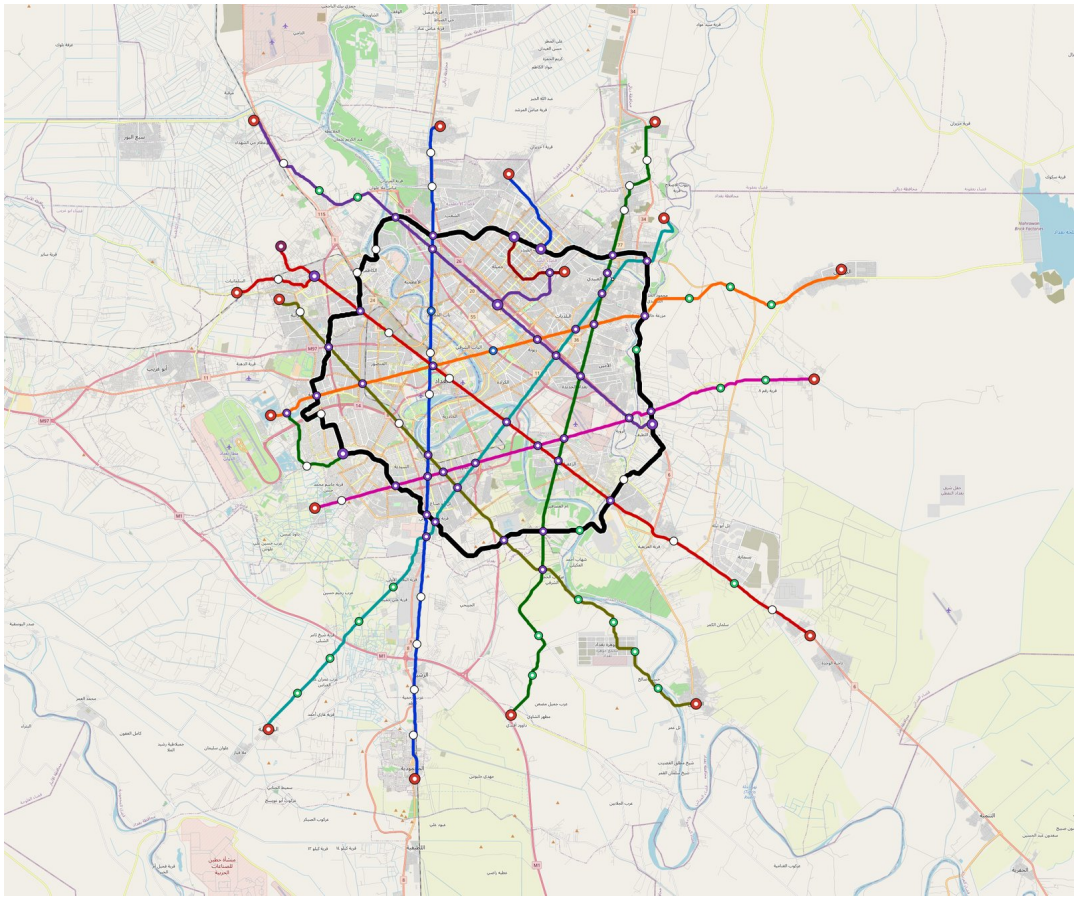
Current design coverage, geometry and open gates · Placed station IFC and map.

Country: IQ · Population: 9,780,429 · Current Iraq planning basis

Current planning basis: 2026-10-04 programme recalculation, `local_positive` conditional local-production case. The main route is the reworked city-centre elevated planning alignment; service remains a capacity-led assumption. Revised scope is unquoted and uncommitted; this is not a construction design or an operating release.

Connected construction and battery study, 2026-10-06 now reconciles island topology, 18 launchers/two shifts, supplier/logistics constraints, equipment cash and sodium alternatives. Its full-network energy duties report service shortfalls; supplier contracts, installed-rate credits and accessible-entrance coverage remain unqualified. The financial figures below are retained comparators and do not include an accepted accelerated-build saving. Base programme planning allowance is USD 8.667bn, including line-local depots, final assembly and selected upstream component plants. Assumed facilities cover the modelled cash deficits; their placement is uncommitted, and the case retains IQD 0.000tn terminal debt. Local special/segmental structures, installed grid/charging upgrades, actual foundations, land and utilities remain unpriced. Removing search penalties establishes no realised saving. Older catalogue financial passes establish arithmetic for their own assumptions, not viability of this revised scope.

Network



Baghdad rail network on OpenStreetMap

Lines / unique stations / interchanges: 14 / 159 / 45; Route length: 513.3 km double track; Direct transfers / reachable line pairs: 39.6% / 100.0%; Residents within 800 m radial station catchments: 1,129,864 (2020 raster; 18.9% of bbox).

Service span / peak headway: 05:30–02:00 / 3 min; Fleet: 809 × 6-car `metro-6car` trainsets (728 peak revenue); Peak network throughput: 403,200 passengers/hour; Practical service capacity: 3,615,840 passenger-trips/day.

Annual paid-trip planning range: 659.9–1055.8 M.

Lines

Line	Length	Stations	Trainsets	Termini
line-1	49.1 km	17	86	S Outer N Outer
line-2	52.9 km	14	95	SE Outer NW Mid
line-3	49.5 km	16	90	S Outer NE Outer
line-4	41.9 km	12	76	NW Outer E Mid
line-5	47.3 km	16	86	W Mid E Outer
line-6	52.5 km	16	95	SW Outer NE Mid
line-7	40.0 km	13	72	E Outer SW Mid
line-8	48.1 km	15	89	SE Outer NW Mid
line-9	97.7 km	26	45	NW Mid NW Mid
line-10	6.8 km	3	15	N Mid NE Mid
line-11	6.9 km	2	14	N Mid N Mid
line-12	6.6 km	3	15	NW Mid NW Outer
line-13	8.0 km	4	18	W Mid W Mid
line-14	6.0 km	2	13	N Inner NE Mid
Total	513.3 km	159 unique	809	

Population access uses retained native count pixels where available; radial catchments require pedestrian/feeder validation. The former demand-score resident proxy is retired. Access and transfers · Building, support and terrain clearance.

Energy

Scheduled service: 6,278 one-way journeys / 215,989 train-km/day; Annual traction demand: 2,043.4 GWh; Station/depot PV / storage: 46.4 MW / 316.0 MWh; Aggregate charging power: 278.0 MW.

Dedicated solar plant: 1,019.7 MW; Residual grid/PPA import: 0.0 GWh/yr; Worst powered-stop gap: line-6: 18.9 km / 305 kWh; Lowest traversal charging margin: line-11: 74 kWh. These energy quantities describe the current regenerated scenario. Zero annual residual grid import is an accounting balance, not accepted hourly autonomy. Depot charging/grid upgrades, duty and launch conflicts remain open.

Current capital and operating people

Revised capital scope	USD equivalent million
Civil and bearing allowance	4,058.152
Stations and core elevated access	1,269.500
Rolling stock	1,219.590
Final assembly and component plants	416.253
Line-local depots	289.645
Solar and charging	874.140
Signalling and programme overhead	539.271
Total programme	8,666.552

There are 14 depots, one per line, with 809 storage slots for all 111 m six-car trains plus clearance. Depot reference capital is USD 289.645m, replacing the old USD 8m once. Workshop bays are sized separately by workload. The current case gives no capacity credit to station stabling. Actual land, foundations, connected access, installed charging and morning launch acceptance remain open. Depot quantities · Items.

The operating establishment is 3,523 FTE, including 1,572 station-cover FTE. Two staff per station and two normal eight-hour shifts give 636 daily shift assignments; retained 20.5-hour service also funds late cover and weekly/leave/training/sickness relief. Loaded annual payroll is IQD 83.415bn, with subsequent OPEX inflation. General pay starts at IQD 1,175,455/month, 50% above the historical employee median indexed to 2026. Technical/supervisor/senior/director grades are separate; this is not a newly measured median. Roles and wages. Final assembly has 1215 production and 183 support FTE; selected upstream plants add 290 production and 45 support FTE. Their 59 paid production months are separate from permanent railway jobs. Construction crew screen is incomplete; contractor labour is already inside contract rates.

Iraqi manufacture and USD capital exposure

bogie: 9708 (Local process option); motor-inverter-set: 9708 (Local process option); battery-225kwh-pack: 4854 (Local process option); door-cassette: 19416 (Local process option); window-cassette: 29124 (Bought component).

Imported process machinery supports the selected Iraqi fabrication and assembly options shown above. Cells/BMS, wheels/axles/bearings, inverters and other critical inputs retain imports. Products with negative Baghdad-only whole-order margins remain bought in this case. Plant readiness/qualification within 18 months is assumed, not demonstrated. Make/buy appraisal.

Compared with the matched bought-component case, capital changes from USD 8.795bn to USD 8.667bn; imported invoice exposure changes from USD 2.358bn (26.82%) to USD 2.016bn (23.27%). The historical 148 km / USD 18bn proposal has a different scope and assumed all-USD funding; it is not a matched tender saving.

Current funding and cashflows

Government capital is exactly 25%. Imports use 50% government USD cash / 50% proposed Chinese USD credit. Only Chinese debt is USD; the rest of government cash, bonds, bank/gap credit and mezzanine is IQD. Government invoice downpayments are scheduled within the total 25% contribution. FX is the historical planning anchor of IQD 1,300/USD.

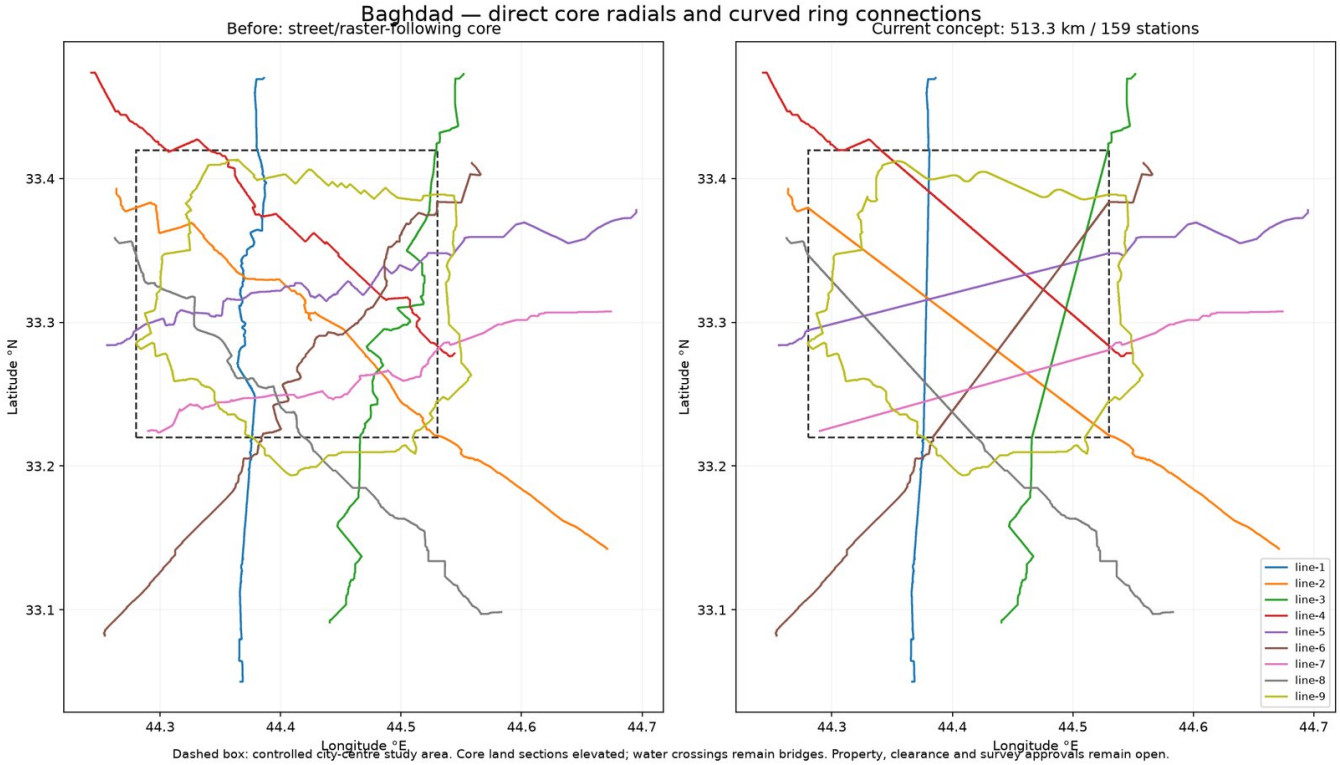
Capital-only source	Currency	Native amount
Government import cash	USD	1,008,216,375
Government local cash	IQD	1,505,948,192,086
Chinese capital credit	USD	1,008,216,375
Ordinary capital bonds	IQD	4,182,186,517,697
Green capital bonds	IQD	1,147,843,845,665
Senior bank capital credit	IQD	1,776,676,787,787
Conditional climate grant	IQD	32,500,000,000

Interest/fees, reserve cash, OPEX and gap facilities are additional cashflows, not capital added twice. Fares, kiosks, advertising, additional receipts and fare/OPEX indexation are included. Green/grant/rights terms and concessional gap credit remain uncommitted. Conditional first/full line revenue is month 40 / 81; physical and financing gates are open.

The tested IQD mezzanine leaves 81 defaulted draw vintages and increases terminal debt to IQD 0.000tn. It does not establish sustainable repayment. Monthly cashflow · Six-month bond/loan placements · All 13 cases · Cost and demand review.

City-centre elevated alignment

The main design uses straight core radial tangents and broad curved ring connections where the retained water evidence permits them, inside the explicit 33.22–33.42°N / 44.28–44.53°E study area. Shoreline detours retain their separate curve and structure review gates. Land sections there are elevated; water crossings remain bridges. The final water-constrained core routes change from 301.670 km to 286.225 km. The main network is 513.336 route km, with 56.73% elevated across the whole system. Maps, station placement, fleet, civil quantities, staff and finance use that reworked design. Analytical controls and limitations.



Earlier corridors and current central alignment

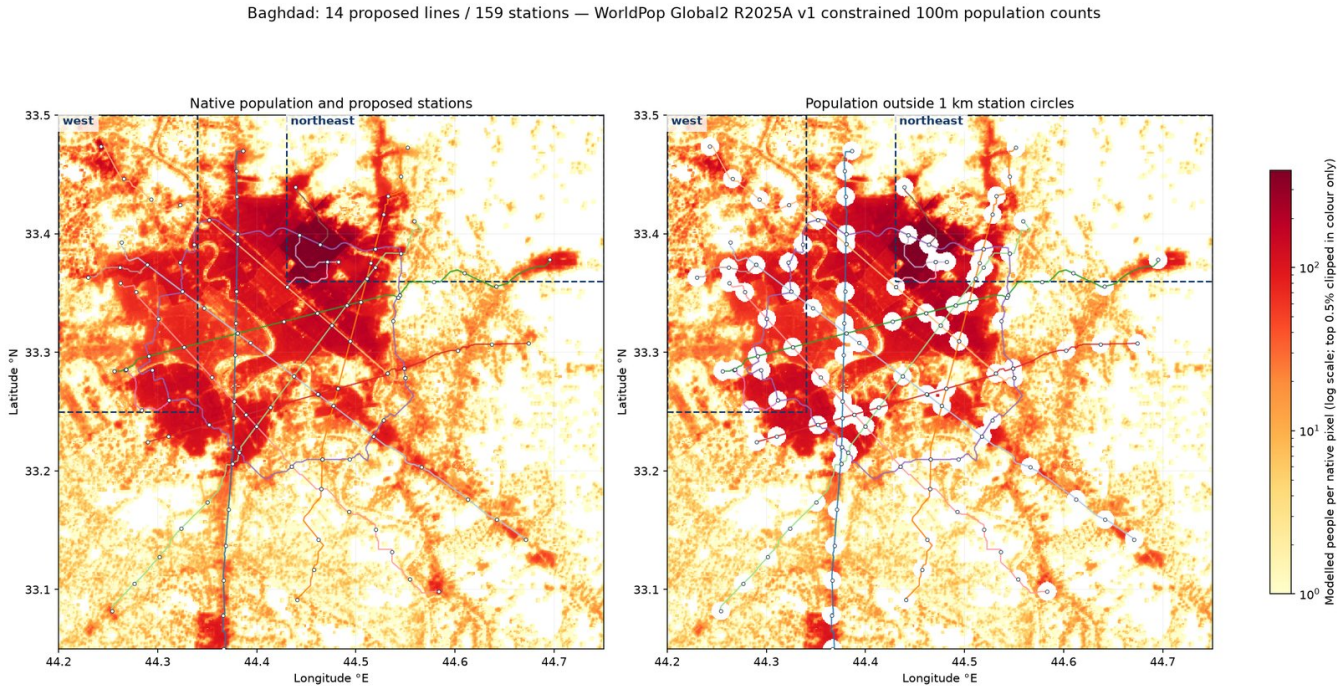
Property/air rights, obstacles, protected sites, surveyed heights, utilities, piers, foundations, transition curves and vertical alignment remain open. Existing outer approaches retain street/raster bends; exceptional geometry still requires realignment or special products. Additional outer grade-separation sensitivities add 7.924 km and retain current dates; they are not the adopted core geometry or an achieved routing-penalty saving.

Neighbourhood population and mapping sources

Native WorldPop 2026 counts guide the compact corridor additions, with separate GHSL 2025 and retained WorldPop 2020 comparisons. The 2026 epoch is an alpha modelled projection using older population inputs, not the 2024 census. Planning sectors are explicit rectangles, not administrative boundaries.

Sector	Modelled native population	Within 1 km of emitted stations	Outside station circles
northeast	2,197,150	38.4%	1,353,322
west	1,459,449	29.6%	1,027,430

Measured Baghdad traffic counts, speeds and OD demand remain unavailable. Pedestrian topology is screened on a dated OSM graph; station entrances, snap connections, sidewalks and accessible paths remain unverified. Station circles are potential radial access, not validated walking coverage or paid trips. Source comparisons, licences and neighbourhood gaps.



Population and underserved planning sectors

Evidence and regeneration

Complete proposal PDF · Editable proposal · Supporting data · Planning-only ERP drafts · Original catalogue finance · Original depot/stabling screen. Simulation, energy and asset evidence uses the current reworked geometry; the new staffing/depot/factory plans do not create accepted sites, appointed employees or operational assets.

Auto-planned by the OpenSourceRail design pipeline. Shared assumptions are in the deployment planning reference. Inputs: design, scenario, map, package evidence. City-local simulation, energy, GIS, operations and delivery retain their recorded planning/release gates. For presentation-only updates, run `.venv/bin/python tools/automation/publish-city-summary.py`; add `--check` to detect drift. The city regeneration pipeline uses this publisher. A stale scope study stops publication rather than silently restoring the original cost headline. Publication provenance is in publication-manifest.json.

Baghdad — population access and transfers

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/access/README.md

Controlled planning screen; population counts, passenger transfers and routing demand are separate measures.

Transfer measure	Value
Direct transfer line pairs	39.6%
Reachable line pairs, including intermediate lines	100.0%
Reachable with at most two transfers	87.9%
Connected line components	1
Maximum transfers within a connected component	4

A crossing creates no transfer unless passenger interchange membership is recorded. Physical access, waiting time, timetable and fares remain unverified.

Population access

Native 2020 bbox population: 5,985,974. Catalogue planning population: 9,780,429. Different dates and boundaries; no multiplication or rebasing of a demand score.

Radius	Residents in union of station circles (2020)	Share of raster bbox population
500 m	472,960	7.9%
800 m	1,129,864	18.9%
1000 m	1,660,588	27.7%
1500 m	3,110,313	52.0%
2000 m	4,332,753	72.4%

Excluded nodata pixels: 248,597; valid pixels: 107,803.

Retained source and attribution · Native count pixels.

Station circles use great-circle distances and count each native pixel once. They are potential radial access, not validated walking coverage: rivers, motorways, walls, hills and actual entrances can reduce access. Native pixel-centre assignment adds source-resolution uncertainty at catchment boundaries. The 1,500–2,000 m cases are extended access/feeder sensitivities; no feeder service or finance is assumed.

The legacy routing score (40.5%) measures high-demand cells close to tracks. It can be lower or higher than population access. Its former multiplication by city population is retired. Coverage alone never increases modeled paid trips or financial viability.

Machine-readable accounting and source hashes.

Baghdad Planning OSR-ALN Package

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/alignment/README.md

Deterministic alignment exports for every line in the current generated network.

File	Design line	Length	Stations
baghdad-line1.aln.toml	line-1	49,067.5 m	17
baghdad-line10.aln.toml	line-10	6,803.6 m	3
baghdad-line11.aln.toml	line-11	6,884.3 m	2
baghdad-line12.aln.toml	line-12	6,586.4 m	3
baghdad-line13.aln.toml	line-13	8,041.8 m	4
baghdad-line14.aln.toml	line-14	6,047.0 m	2
baghdad-line2.aln.toml	line-2	52,850.6 m	14
baghdad-line3.aln.toml	line-3	49,496.5 m	16
baghdad-line4.aln.toml	line-4	41,931.8 m	12
baghdad-line5.aln.toml	line-5	47,344.8 m	16
baghdad-line6.aln.toml	line-6	52,503.4 m	16
baghdad-line7.aln.toml	line-7	39,957.0 m	13
baghdad-line8.aln.toml	line-8	48,136.6 m	15
baghdad-line9.aln.toml	line-9	97,684.2 m	26

Status

These files are planning-only and not for construction. Horizontal control comes from the current WGS84 corridor and is projected to the local UTM zone. Circular curves and transitions are not fitted, the vertical profile is a zero-datum placeholder, and cant has not been designed. Survey, curve-fit, vertical-profile, cant, geotechnical, utility, property, drainage, and structural release gates therefore remain open.

Each OSR-ALN file records the SHA-256 hashes of `../../design.toml` and the city corridor GeoJSON used to generate it.

Baghdad detailed civil works plan

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/civil-works/README.md

Proposed planning methods; construction and complete opening remain unreleased.

Master plan connects detailed viaduct, station, other works, logistics, programme and inspection methods.

The 25 m average sizing basis reconciles to 9,973 identified catalogue bays, 404 unresolved specials/closures and 24.64 m actual running-span average. Orders retain span/track IDs and actual products.

Line	At grade km	Elevated km	Bridge km	Catalogue bays	Special spans	Stations
line-1	22.625	21.621	4.821	770	32	17
line-2	18.284	28.987	5.579	988	76	14
line-3	22.833	25.542	1.121	863	29	16
line-4	13.480	26.810	1.641	943	44	12
line-5	19.248	25.830	2.267	874	35	16
line-6	24.426	25.388	2.689	846	60	16
line-7	13.565	24.296	2.095	813	21	13
line-8	18.200	23.633	6.303	811	39	15
line-9	26.038	67.067	4.579	2280	61	26
line-10	0.000	6.804	0.000	238	2	3
line-11	2.828	4.057	0.000	153	1	2
line-12	5.207	1.380	0.000	46	1	3
line-13	4.299	3.743	0.000	123	2	4
line-14	0.000	6.047	0.000	225	1	2

Line quantities separate running structures from elevated station/transition scope. Run register retains the identified disconnected access/launcher boundary packages. Station packages retain individual layouts and equipment; civil segments retain every class/chainage.

Resources and logistics calculates required production positions, accepted/gross supply, concrete, cargo, complete transport cycles, vehicles, 10–15-bay buffers and six-day calendar lower bounds. These are illustrative requirements, not available supplier/route capacity or a completion forecast.

Package dependencies connects fourteen civil scopes per line to complete opening interfaces. Unprovided durations, installed prices and accepted dates stay unknown. Fleet, systems qualification, testing and authority approvals remain additional requirements.

Review register preserves open support/structure, capacity, buffer, station/special, stabling, quantity, calendar and cost findings. References are international method-development guidance; local adoption and actual site evidence are required.

Run `python tools/automation/civil-works-plan.py`; `--check` verifies deterministic bytes and source locks on a clean checkout. Manifest records content provenance independent of the containing commit.

Baghdad — viaduct obstacle clearance

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/clearance/README.md

Physical release blocked. This is a source-bound footprint and terrain screen of the existing planning route, not an obstacle-cleared alignment.

Mapped nearby footprints: 879; source heights: 110. Building source: retained; terrain: retained.

Screen	Status	Count
Beam/building	beam-roof-collision	60
Beam/building	conditional-roof-overflight	16
Beam/building	ground-collision	4
Beam/building	ground-lateral-clearance-conflict	2
Beam/building	height-unresolved	635
Beam/building	lateral-clearance-conflict	10
Beam/building	product-depth-unresolved	56
Reference support/foundation	foundation-footprint-conflict	550
Reference support/foundation	foundation-setback-conflict	99
Reference support/foundation	mapped-footprints-only-clear	14,223
Terrain	reference-gradient-exceeded	7,161
Terrain	reference-gradient-within-policy	6,369
Unresolved geometry/input	boundary-span-layout-unresolved	1,319
Unresolved geometry/input	special-product-depth-and-support-layout-unresolved	622

Counts are screening events per civil segment/reference support, not unique buildings, accepted piers or procurement quantities. Reference-gradient flags are DEM height differences between provisional supports; designed rail gradients are unavailable. Roof/canopy effects, raster quantisation, noise and real ground slope remain unseparated. These flags neither prove an unbuildable rail profile nor establish clearance.

Measures required for a buildable alignment

- Beam overflight: surveyed roof/plant heights plus ground datum, swept train/deck/walkway envelope, lateral clearance, fire/rescue access and property/air rights. A source height below a provisional soffit is conditional only; building floors are an uncertain height estimate.
- Tall obstacles: reroute, acquire/remove an approved obstacle, or raise the vertical alignment with gradient-compliant approaches and new pier/structure costs. Unknown heights never establish clearance.
- Supports: place the entire pier/pile-cap/foundation and construction-access footprint clear of buildings and services. A clear beam does not clear its supports. Move support lines along a tangent using verified Pi20/Pi25 spans, change foundations/cap arrangement, or design a independently checked special span; do not invent a 30 m catalogue beam.
- Terrain: survey a longitudinal/cross-section profile and test grades, vertical curves, terrain crests, flood levels and each erection stage. Straight beam soffits interpolate between supports; local ground peaks can still clash.
- Release: close every conflict and missing-height/mapping/terrain/utility finding against survey and independent engineering checks before accepting the corridor. Recalculate route, stations, supports, quantities, cost and service evidence for adopted changes.

The default screen uses a 9 m twin-track envelope plus 2 m lateral allowance, a 3 m foundation radius plus 2 m setback, 12 m reference rail height, 1.6 m conservative rail-to-soffit allowance, 2 m roof clearance and a 3.5% reference grade limit. The Pi structural depth is 1.155 m; the combined allowance leaves 0.445 m for an unverified track stack. Special products have unresolved depth and cannot pass a Pi roof-clearance check. Direct structural/foundation intersections are reported separately from clearance/setback conflicts. These are controlled screening assumptions requiring deployment-specific approval, not statutory clearances.

Summary and source hashes · Per-line beam, support and terrain events.

Baghdad connected construction and battery study

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/connected-build/README.md

Generated from RFC 0034; 2026-10-06. These are unquoted, conditional sensitivities, with no accepted supplier capacity or structural/access releases.

The package has 159 stations, 214 physical platforms and 354 boarding faces. Elevated islands use individually identified lifts, escalators, stairs and shafts; their flow/evacuation/one-lift-out approvals remain open. Station-specific passenger assignment accepts sourced OD legs, preserves paid journeys versus transfer boardings and links station accumulation to its declared service window. Empty evidence means unknown demand; 3,000 pax/h is a separately labelled stress case. Measured access throughput, assisted rescue and lift-out capacity remain open. Station quantities and access costs, track-spreading approaches and proposed entrances derive from the same layout. Coverage from accepted accessible entrances remains unknown; the existing centroid-based archive cannot close that assessment.

Running civil quantities and conditional deployment exclude station and spreading-approach zones. Identified planning spans and beam requirements now conserve interval lengths; non-catalogue closures are explicitly excluded from ordinary beam orders. Pier chainages are proposed and require survey/structural acceptance. The 28 workfronts share 18 configured launchers, with explicit predecessors for queued reuse. Each line has two candidate fronts where running work exists. Each disconnected section has a seven-day assumed relocation allowance; crossing feasibility remains unaccepted. A conditional case transfers available released radial launchers to additional ring fronts while retaining the same active machine/transport fleet. Added access and foundations are hypothetical, and transfer/mobilisation delays remain explicit. Two-shift 1.5, 1.8 and 2.0 bays/day sensitivities use the identified active launchers. Configured spare machines create no independent-front output; each case reports its actual fleet demand. Integer daily schedules are in `schedule-*.json.gz`. The explicit calendar gates working dates for casting, QA, transport, foundations and erection; night restrictions cap erection hours. Supplier-specific shifts, permits, relief and holidays remain unqualified. A six-day sensitivity is reported separately. The initial case completes its running bays in 1218 days under its explicit hypothetical supply/readiness inputs; special crossings, station structures, rolling stock and permits remain separate opening gates. The evidence-backed schedule reads the supplier and route registers and completes zero bays because accepted supplier capacity and released supports are absent.

Construction case	Running-span days	Launchers	Assumed transporters
initial-accelerated	1218	18	72
higher-productivity	1098	18	72
full-doubling	1022	18	72
initial-accelerated-reassigned	966	18	72
initial-six-day-calendar	1348	18	72

Supplier register records the sourcing basis that Iraq already has many precast facilities and production expertise. No named facility, available beam capacity or assumed contract is adopted. The logistics plan now feeds casting, acceptance holds, dispatch limits, gross vehicle/bridge loads, journeys, shared fleets and factory/front buffer limits into the same erection schedule. Rejected components require replacement; blocked storage pauses production. Each front reports days limited by casting, acceptance, transport, foundations or erection. The assumed pool is a resource requirement, not an identified supplier or confirmed spare capacity; real routes and contracts remain in separate registers.

Supplier quotation and scope requirements keep preliminary, firm and contracted prices separate from qualified factory output. Partial cashflow sensitivities link identified purchase payments to each conditional schedule at assumed 4/8/12% simple ACT/365 interest; complete CAPEX/finance and revenue remain unknown. The identified span register provides each catalogue beam requirement; no special span is ordered as an ordinary member.

Costs and cash sensitivities keep the $9m launcher purchase allowance and exclusions visible. Embedded erection in installed civil rates is unverified, so no net saving or complete financing total is claimed. Gross access equipment, maintenance, replacement, factory scope and night-shift sensitivities remain separate; rolling-stock/electronics factory allowances are retained. Payment dates are conditional cash planning, not purchase orders.

Energy comparisons cover three continuous service days with overnight state carry-over, and a study policy for grid replenishment of stationary storage within remaining site import capacity. They use explicit pack identities, mass-adjusted traction, shared chargers, setup time, temperature/SOC limits, seasonal solar/cleaning, hot auxiliaries, missed charges, outage/reserve duties and degradation. All three matched chemistry cases report minimum SOC and service shortfalls. The small onboard pack is a separate option. The full-network chronology uses declared stations, fleets, headways, dwell and site capacities with native section movement timing; it requires timetable qualification and makes no supplier chemistry claim. The fixed-assignment diagnostic retains its declared duties even when energy is missing. The separate coupled control study holds trains at their actual stations when reserve or discharge power is insufficient, continues charging and propagates their unavailability into later dispatches. It reports served/missed opportunities, distinct held journeys and hold minutes. Conflict authorities, berth/depot access, transient traction and field calibration remain unqualified.

The planning duty schedules 809 trainsets and 139 energy sites. There are 18 dispatches without a ready trainset. All matched cases below have service shortfalls under the selected hot-weather assumptions; none qualifies the declared service.

Case (lower-solar duty)	Minimum SOC	Unserved traction kWh	Distinct energy-affected journeys	Reserve-violation train-minutes
lfp-throughout	0.0%	676,286	8,883	2,005,400
sodium-stationary	0.0%	676,383	8,887	2,005,915
sodium-both	0.0%	4,024,245	17,031	3,057,440

ERP planning drafts identify reusable machines and distinct front/shift crews, with purchase, commissioning, inspection, maintenance, competence and transfer gates. Native Task/ToDo transition hooks require submitted allocation review, current controlled Employee/Asset evidence, shifts, leave, holidays, rest and exclusive allocations. Pack-specific commissioning bindings also feed the native onboard evaluator; study profiles cannot populate them. No live ERP allocation or battery commissioning is approved by this package. Cash purchase, project allocation and residual value are distinct.

Battery charge/thermal sensitivities sweep 180/360/540 kW and 25/45/50 °C on two-minute shared-site visits. These parameter probes expose the effect of charge maxima and inherited thermal/taper/efficiency assumptions; they are not supplier claims or full-network service acceptance.

Structural/handling stage screens report gross-section properties, selfweight, lift/storage/transport reactions, sling force and launcher-position envelopes. Prestress, reinforcement, release strengths, bearings, fatigue, stability and supplier charts remain unresolved; screens grant no release. Station area/capacity screens union shared-layout equipment footprints, report unoccupied envelopes and circulation lanes, and accept sourced criteria/lift rates. Evacuation, rescue, walking catchments and site approvals remain open.

Complete remaining-work register traces all 20 review areas, including rolling-stock/electronics release, redundancy, deployment, workforce, other cities and governance. Missing records and appointments stay open; input receipts cannot authenticate an authority or create acceptance.

Coordinated industrialisation links the vehicle/bogie interface freeze, staged local manufacture, ordinary/special viaduct packages, accepted production chain, urban access and delivered-service finance. HÜBNER/CRRC and alternative vendors remain proposed; exact products, rights, prices and approvals are open. The LM3 axle-load reference is a study comparator and does not replace this city's configured stock.

Manifest hashes every source and output and identifies the source revision and assumption register. Existing finance/proposal packages are retained comparators pending scope-matched adoption; this study is the current connected scenario, not an accepted replacement budget. Regenerate with `.venv/bin/python tools/automation/connected-build-study.py --city baghdad`; verify using `--check`.

Coordinated junction and residential interfaces bind crossings, shared corridors, bounded station complexes and expansion studies to this design. Per-line foundation and assembly plans identify every support, both beam parents, directed launcher sequence and junction-design hold. Proposed access/expansion does not adopt operating service or additional finance.

Baghdad cost correction and subsequent alignment reconciliation

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/cost-reconciliation/README.md

Base planning allowance; complete installed price and financing remain unresolved.

The original `d4cd3db2cd` monetary correction is verified against the retained same-alignment `a1a14f6d29` reference. The current design subsequently changed controlled fields lines, stations, fleets, depots. The latest values below therefore include separately recorded alignment effects; they are not all attributed to a same-geometry price correction. Route-search scores remain excluded from prices.

Measure	Reviewed baseline	Corrected base case
Programme allowance, USD bn	15.233	8.667
Unfunded support, IQD tn	27.330	0.000
Terminal debt, IQD tn	13.000	0.000
Operating FTE	3716	3523
Loaded annual payroll, IQD bn	86.303	83.415

The allowance correction is not a supplier saving, accepted alternative or complete delivery budget. Special/segmental increments, actual foundations, installed grid/charging upgrades, land, utilities and other closure items remain unpriced. Maintenance follows the corrected capital-percentage allowance; physical inspection, replacement and access workloads still require independent quantities and prices. The correction grants no lower staffing, traction duty or surveyed demand requirement.

The same-alignment corrected reference capital is USD 8.312371bn; the current alignment changes it by USD 354,180,934.18 in the unquoted model. The current route is 513.3355 km against 479.0124 km in that reference. The retained corrected snapshot preserves the original stage when an alignment revision exists; its bytes are also in the companion planning archive.

The original baseline bytes are retained in the source snapshot, so a fresh shallow checkout can reproduce this comparison offline. Month 101 comparisons in the source-bound reconciliation annualise one nominal model month, rather than predicting a calendar year. Paid journeys remain capacity-led; no population-to-fare uplift is adopted. The demand handoff, energy closure, clearance screen and current programme ledger retain their separate acceptance gates.

Coupled Baghdad programme study

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/coupled-programme/README.md

Unaccepted conditional study. Completed native-timed section events feed capacity-limited OD itineraries, one net reference fare per completed journey, grid purchases, distance-sensitive maintenance and the existing native-currency financing engine. Departures and whole train circuits are not paid passenger journeys. Short OD trips can complete on a train whose full duty is later held. Transfers consume seats on all legs and recognise one fare.

The controlled demand is a 25/50/75% capacity-reservation sensitivity at dispatch nodes; it is not surveyed OD demand. Ring destinations are halfway around the ring. The primary 50% case uses the same demand across all three unqualified energy profiles. Actual queues, refunds, income uptake, equipment calibration and representativeness over an operating year remain open. Money is reported to cents after full-precision calculation.

Summary links service, demand and financing. Three demand reports retain cohort outcomes and station boarding pulses. Three finance reports retain monthly receipts, OPEX, senior debt and supplemental support. The old ledger is reconstructed before any scope replacement. Electricity is replaced once; existing fixed solar/connection services and the battery reserve stay in scope. The residual rolling maintenance is treated as distance-sensitive solely as an uncalibrated sensitivity. Payroll and nonfare receipts remain explicit retained comparators. Additional vehicle, charger, station, special, installation and chemistry-specific renewal prices remain unknown. No revised national budget or financial baseline is adopted.

Complete line-opening milestones require running structures, stations, specials/transitions, track, energy, depots, fleet, testing and approvals. Missing dates and prices stay unknown; a running-span finish cannot become a complete railway opening. Known bottlenecks and cash subtotals are shown separately. A common financial-close anchor and actual handovers are missing, so the finance cases retain their older opening months as explicit conditional assumptions.

Workload and recruitment derives establishment from measured role-hours, concurrency and productive calendar hours, then works backwards through recruitment, training batches, supervised practice and assessment capacity. Empty measurement and readiness inputs do not become funded headcounts or appointments. Native Task guards check construction, manufacturing and maintenance authority, role quantities, live resource state, shifts, rest and future reservations. Maintenance on unavailable equipment requires current isolation evidence. Training attendance is not work authority.

Mass handoff keeps physical rows open. The reusable component model checks nonoverlapping included scope, uncertainty, locations, CG and individual axle reactions; no guessed CAD-envelope mass or selected vehicle startup is inserted. Supplier offers, first articles, surveyed yards, thermal calibration and accountable acceptance remain external evidence gaps.

Run `.venv/bin/python tools/automation/coupled-programme-study.py`; `--check` rejects drift. `--compare <previous summary.json>` writes a separate reviewable comparison of input locks and conclusion changes. Manifest binds source and output bytes; its revision is independent of its containing commit.

Baghdad organisation and design work

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery/README.md

> Original catalogue or earlier scope reference. The figures and policies below retain their original assumptions; they are not the latest Baghdad staffing, depot, procurement or funding basis. See the current Baghdad recalculation and current city summary. National/portfolio totals have not been repriced with that conditional Baghdad option.

3422 indicative operating FTE, including management; annual labour allowance $38,681,226. Group totals reconcile to the existing finance model. Individual role shares are editable planning allocations, not measured workload or an accepted duty roster.

GoA 4 control and remote assistance; local station batteries supply train top-ups. Two revenue trainsets per selected station with same-line depot overflow. Shared domestic design/software/factory capability serves city teams.

City operating organisation

Function	FTE	Reports to	Skills	Tasks
City accountable lead	1	Local owner board	Local operating governance; budget and emergency accountability	Own city delivery and operations; coordinate shared domestic capabilities
Engineering authority	5	City director	Systems integration; configuration and design checking	Own interfaces and technical concessions; commission competent discipline checks
Quality and safety assurance	9	Local owner board (independent escalation)	Independent inspection; hazard review; traceability	Audit evidence; raise stop-work/hold points independently of production
Training and competence assessors	9	City director	Practical assessment; instruction; competence records	Train local recruits and apprentices; witness competence before unsupervised work
Procurement and stores	10	City director	BOM traceability; repairable spares; supplier quality	Purchase controlled common parts; manage local repair loops and critical stock
Finance and people administration	9	City director	Payroll; cost control; roster and employment administration	Maintain local pay/leave inputs; reconcile labour budget and workload
Fleet workshop supervision	53	Chief engineer	Isolation authority; work planning; calibrated tooling	Allocate same-line workshop slots; control defect and release records
Mechanical technicians	184	Workshop lead	Wheels/brakes; joint inspection; controlled fastener assembly	Inspect bogies and brakes; replace modules; record torque/preload evidence
Electrical and battery technicians	158	Workshop lead	HV isolation; LFP diagnostics; controls and harness repair	Diagnose packs, HVAC and controllers; verify protective interlocks
Vehicle cleaning and finish repair	132	Workshop lead	GFRP/EPDM compatibility; coating preparation; controlled washing	Turnaround clean; inspect seals; repair finish damage under qualified process
Infrastructure work supervision	40	Chief engineer	Track access; lifting and electrical isolation coordination	Plan possessions; check civil and energy defect closure
Track and civil technicians	157	Infrastructure lead	Survey checks; track geometry; bearing/joint inspection	Inspect settlement, track, deck joints, drainage and walkways
Solar and station storage technicians	138	Infrastructure lead	PV DC safety; storage BMS; charger and protection testing	Maintain PV/storage/top-up equipment; measure soiling and energy balance
Wayside and communications technicians	59	Infrastructure lead	Fail-restrictive controls; networking; configuration control	Maintain point/crossing equipment where installed; test communications and recovery
Shift leadership and degraded-operation authority	79	City director	GoA 4 incident command; dispatch competence	Own service start; coordinate recovery and same-line depot releases
Line dispatch and control	211	OCC lead	Timetable supervision; protected movement authority	Monitor headways and station berths; authorise line-local recovery
Remote assistance	237	OCC lead	Vehicle fault diagnosis; passenger emergency response	Resolve remote-assist requests; coordinate on-site responders
Passenger service and fare support	201	City director	Accessible communication; fare-system support	Resolve passenger queries; verify fare equipment and complaints
Station cleaning and waste	164	Station lead	Product compatibility; wet-floor control; waste segregation	Clean measured public areas; record defects and wash-water disposal
Station team coordination	236	OCC lead	Accessible evacuation; platform protection	Allocate mobile teams; inspect opening and closing readiness
Platform and accessibility assistance	1330	Station lead	Accessibility assistance; crowd management; first aid	Assist boarding; manage obstructions and platform incidents

Recruit and apprentice locally; assess practical task competence, isolate/hold/release authority and refresher needs. Training duration follows the demonstrated skills gap, not a universal weeks-to-qualification claim.

Quality and safety staff can escalate holds directly to the local owner board. Functions can be combined where competence and cover permit. Management allocations are already inside the operating totals.

Roster closure: named duty and leave roster including service-depot turnaround crews; measured cleaning/inspection person-minutes and simultaneous arrivals; country/local pay and actual grade costs; local employment/rest requirements and night emergency cover.

Construction and domestic manufacturing work

Concurrent crews/workstations below come from the existing dated work orders, with inclusive finish days. Worker headcount awaits an explicit crew mix; named skills are not assumed to mean one person each. Factory capacity is shared across city orders. Peaks from different work centres need not occur together.

Work centre	Peak resources (unknown if unsized)	Peak task slots	First peak day	Resource-days	Skills/tasks
Baghdad six-car body installation	unknown	3	601	unknown	body-cell lead; composite fitter; quality technician / rs-25-clip-on-body
Baghdad six-car composite kits	unknown	10	597	unknown	composite cell lead; laminator; quality technician; trim technician / rs-23-moulded-body-modules
Baghdad six-car electrical integration	unknown	20	625	unknown	battery technician; electrical lead; quality technician; software technician / rs-30-traction-battery-control
Baghdad six-car fitout and static test	unknown	16	644	unknown	HVAC technician; door technician; fit-out lead; quality technician / rs-40-fitout-static-test
Baghdad six-car structural assembly	unknown	24	580	unknown	bogie technician; brake technician; fabrication lead; weld inspector / rs-20-carbody-bogie
Baghdad trainset acceptance bays	unknown	13	659	unknown	commissioning lead; quality manager; software technician; test operator / rs-50-dynamic-commissioning
Baghdad trainset kitting	unknown	4	556	unknown	production planner; quality technician; stores clerk / rs-10-material-kit
corridor civil crew	2	2	7	3214	civil supervisor; drainage technician; permit coordinator; plant operator; quality technician; safety officer; survey lead / trk-10-survey-utility, trk-20-subgrade-drainage
depot works	unknown	1	628	unknown	depot operations lead; depot supervisor; electrical supervisor; quality technician; systems technician; tooling technician / dep-10-building-plant, dep-20-charging-stabling
energy crew	unknown	2	529	unknown	PV technician; battery technician; commissioning lead; electrical supervisor; quality technician; software technician / en-10-energy-build, en-20-energy-commission
project controls	unknown	1	0	unknown	production planner; project controls lead; quality manager; safety assurance lead / sys-00-baseline-freeze
station civil crew	unknown	2	559	unknown	accessibility lead; quality technician; station civil supervisor; steelwork lead / st-10-civil-structure
station systems crew	unknown	2	609	unknown	electrical supervisor; fare-system technician; quality technician; systems technician / st-20-mep-passenger-systems

Work centre	Peak resources (unknown if unsized)	Peak task slots	First peak day	Resource-days	Skills/tasks
track crew	2	2	299	2629	quality technician; rail welder; survey technician; track supervisor / trk-30-trackform-rail
track/switch crew	unknown	2	240	unknown	quality technician; survey technician; switch technician; systems technician; track supervisor / sw-10-turnout-build, sw-20-install-proof
wayside bench	unknown	3	487	unknown	quality technician; software technician; systems technician / wpt-10-cabinet-bench
wayside systems crew	unknown	3	645	unknown	quality technician; radio technician; software technician; systems supervisor / wpt-20-install-commission

Civil and mechanical refinements

Climate: hot-desert; 28 °C reference, 50 °C upper ambient. Cold ambient: missing.

Free movement alphaLdeltaT. 8/10/12 microstrain/K are declared sensitivity cases, not measured concrete properties. Reference is the preset annual mean, not installation temperature; sunlit component temperatures and gradients require separate inputs.

Each elevated segment is separate; 25 m catalogue span, up to 4 spans/unit from the shared production/continuity templates, two tracks and two webs. Shared bearings require an unaccepted structural continuity connection; a simple-span link slab retains independent girder-end bearings. Rounded spans are a procurement screen, not surveyed support positions. Deck gaps count the existing catalogue unit interfaces; abutment details remain separate.

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-1 408.3–448.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 588.3–756.6 m	100.0	36	5	17.6 / 22.0 / 26.4
line-1 976.6–1036.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 1336.6–1376.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 1829.7–1849.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 1909.7–1929.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 1929.7–1978.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 2086.3–2146.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 2294.6–2334.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 2954.6–3014.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 3034.6–3114.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 3494.6–3562.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 3602.8–3631.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 3631.1–3651.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 3679.4–3739.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 3739.4–3779.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 3779.4–3799.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 4512.5–4552.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 4689.1–4729.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 5029.1–5069.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 5257.4–5317.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 6425.7–6505.7 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 7034.0–7094.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 7154.0–7222.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 7262.3–7322.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 7542.3–7602.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 7690.5–7790.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 8098.8–8138.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 8287.1–8307.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 8327.1–8347.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 8375.4–8395.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 8523.7–8543.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 8892.0–8912.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 8912.0–8952.0 m	50.0	12	1	8.8 / 11.0 / 13.2

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-1 8952.0–9028.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 9028.5–9048.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 9205.1–9253.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 9253.4–9281.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 9309.9–9358.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 9458.2–9526.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 9526.5–9554.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 9574.8–9634.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 9714.8–9774.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 9859.7–10007.9 m	100.0	32	4	17.6 / 22.0 / 26.4
line-1 10187.9–10236.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 10236.2–10256.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 10324.5–10344.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 10344.5–10384.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 10441.1–10461.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 10629.4–10709.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 10749.4–10877.6 m	100.0	32	4	17.6 / 22.0 / 26.4
line-1 11165.9–11225.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 11294.2–11314.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 11394.2–11414.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 11454.2–11682.5 m	100.0	52	7	17.6 / 22.0 / 26.4
line-1 11830.8–11870.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 12279.1–12299.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 14301.9–14321.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 15226.8–15295.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 15375.0–15475.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 15871.6–15891.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 16059.9–16079.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 16756.5–16804.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 16824.8–16844.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 17193.0–17213.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 17253.0–17273.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 17597.9–17626.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 17854.5–17874.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 17974.5–18034.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 18271.0–18299.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 18632.4–18660.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 18905.6–18933.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 19013.9–19033.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 19033.9–19062.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 19110.4–19138.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 19627.0–19667.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 19735.3–19755.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 19755.3–19795.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 19980.1–20000.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 20108.4–20128.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 20128.4–20176.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 20176.7–20196.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 20196.7–20225.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 20265.0–20293.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 20333.3–25134.7 m	100.0	968	144	17.6 / 22.0 / 26.4
line-1 25394.7–25554.7 m	100.0	36	5	17.6 / 22.0 / 26.4
line-1 25614.7–25674.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 25943.0–25963.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 26003.0–26083.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 26103.0–26123.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 26223.0–26503.0 m	100.0	60	9	17.6 / 22.0 / 26.4
line-1 26643.0–26831.3 m	100.0	40	6	17.6 / 22.0 / 26.4
line-1 26851.3–26871.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 26971.3–26991.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 27071.3–27191.3 m	100.0	28	3	17.6 / 22.0 / 26.4
line-1 27211.3–27271.3 m	75.0	16	2	13.2 / 16.5 / 19.8

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-1 27331.3–27391.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 27451.3–27671.3 m	100.0	48	6	17.6 / 22.0 / 26.4
line-1 28019.6–28619.6 m	100.0	120	18	17.6 / 22.0 / 26.4
line-1 28827.8–29047.8 m	100.0	48	6	17.6 / 22.0 / 26.4
line-1 29147.8–29207.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 29267.8–29287.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 29487.8–29607.8 m	100.0	28	3	17.6 / 22.0 / 26.4
line-1 29676.1–29736.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 29776.1–29796.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 29816.1–29936.1 m	100.0	28	3	17.6 / 22.0 / 26.4
line-1 29956.1–30036.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 30076.1–30136.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 30176.1–30216.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 30624.4–30784.4 m	100.0	36	5	17.6 / 22.0 / 26.4
line-1 30804.4–30824.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 30904.4–30944.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 30984.4–31024.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 31084.4–31264.4 m	100.0	40	6	17.6 / 22.0 / 26.4
line-1 31404.4–31584.4 m	100.0	40	6	17.6 / 22.0 / 26.4
line-1 31604.4–31684.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 31884.4–31924.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 32104.4–32184.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 32284.4–32344.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 32484.4–32552.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 32572.7–32612.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 32732.7–32792.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-1 33232.7–33252.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 33512.7–33761.0 m	100.0	52	7	17.6 / 22.0 / 26.4
line-1 33781.0–33921.0 m	100.0	32	4	17.6 / 22.0 / 26.4
line-1 33981.0–34081.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-1 34201.0–42755.534 m	100.0	1716	257	17.6 / 22.0 / 26.4
line-1 42755.534–42795.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 42948.7–42968.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 44130.1–44150.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-1 47299.8–47328.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-1 47376.4–47404.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 2228.8–2285.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 2446.8–2503.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 2608.2–2636.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 4334.7–4354.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 4572.7–4592.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 4649.3–4677.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 7743.1–7771.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 9424.2–9444.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 9549.1–9577.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 9673.9–9702.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 11717.8–11746.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 11919.3–11947.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 12035.8–12055.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 12478.7–12506.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 12788.4–12808.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 12904.9–12924.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 12953.2–12973.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 13106.4–13234.6 m	100.0	32	4	17.6 / 22.0 / 26.4
line-2 13234.6–13262.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 13339.5–13359.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 13359.5–13427.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 13447.8–13516.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 13556.1–13584.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 13604.3–13644.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 13644.3–13672.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 13672.6–13692.6 m	25.0	8	0	4.4 / 5.5 / 6.6

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-2 13692.6–13712.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 13740.9–13769.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 13994.1–14042.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 14388.6–14465.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 14485.2–14505.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 14593.5–14621.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 14886.6–14934.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 14954.9–15011.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 15172.9–15314.3 m	100.0	32	4	17.6 / 22.0 / 26.4
line-2 15515.7–15544.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 15620.6–15648.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 15773.7–15850.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 15918.6–15946.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 16051.7–16071.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 16120.0–16148.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 16188.3–16216.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 16341.4–16369.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 16699.4–16767.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 16824.2–16864.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 17225.7–17253.952 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 17253.952–19175.0 m	100.0	388	57	17.6 / 22.0 / 26.4
line-2 19251.6–27101.9 m	100.0	1576	236	17.6 / 22.0 / 26.4
line-2 27121.9–27141.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 27403.4–27423.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 27443.4–27519.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 27793.1–27841.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 27957.9–27986.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 28042.8–28119.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 28357.3–28433.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 28462.2–28538.7 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 28567.0–28643.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 28671.9–28691.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 28720.2–28901.6 m	100.0	40	6	17.6 / 22.0 / 26.4
line-2 28929.9–28949.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 28998.2–29086.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 29134.7–29183.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 29492.7–29512.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 29532.7–29552.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 29609.3–29697.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 29814.1–29842.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 29902.4–29930.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 29959.0–30075.6 m	100.0	28	3	17.6 / 22.0 / 26.4
line-2 30095.6–30115.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 30212.1–30232.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 30308.7–30328.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 30513.5–30541.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 30590.1–30675.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-2 30695.0–30751.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 31094.4–32700.9 m	100.0	328	48	17.6 / 22.0 / 26.4
line-2 33010.6–33215.5 m	100.0	48	6	17.6 / 22.0 / 26.4
line-2 33243.7–33263.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 33312.0–33360.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 33448.6–33633.4 m	100.0	40	6	17.6 / 22.0 / 26.4
line-2 33661.7–33778.3 m	100.0	28	3	17.6 / 22.0 / 26.4
line-2 33894.9–34079.7 m	100.0	40	6	17.6 / 22.0 / 26.4
line-2 34196.3–34381.1 m	100.0	40	6	17.6 / 22.0 / 26.4
line-2 34944.0–34964.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 34992.3–35108.8 m	100.0	28	3	17.6 / 22.0 / 26.4
line-2 35177.1–35282.0 m	100.0	28	3	17.6 / 22.0 / 26.4
line-2 35596.5–35624.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 35653.1–35701.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 35729.7–35834.5 m	100.0	28	3	17.6 / 22.0 / 26.4

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-2 35911.1–36177.4 m	100.0	56	8	17.6 / 22.0 / 26.4
line-2 36253.9–36282.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 36358.8–36387.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 36415.4–36435.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 36491.9–36645.1 m	100.0	36	5	17.6 / 22.0 / 26.4
line-2 36665.1–36829.9 m	100.0	36	5	17.6 / 22.0 / 26.4
line-2 36946.5–36994.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 37179.6–37529.3 m	100.0	72	10	17.6 / 22.0 / 26.4
line-2 37557.6–38160.5 m	100.0	128	18	17.6 / 22.0 / 26.4
line-2 38628.1–49816.374 m	100.0	2240	336	17.6 / 22.0 / 26.4
line-2 49816.374–49884.658 m	75.0	16	2	13.2 / 16.5 / 19.8
line-2 49884.658–49904.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 49904.7–49932.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 49972.9–50001.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 50274.4–50294.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-2 51489.8–51529.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-2 51669.8–51689.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 785.7–814.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 870.5–898.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 1060.2–1088.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 1116.8–1145.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 3436.5–3493.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 4211.0–4239.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 4584.2–4624.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 4624.2–4672.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 4740.7–4769.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 6777.0–6817.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 7995.2–8071.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-3 8100.0–8156.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 8689.2–8717.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 11571.9–11600.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 12084.4–12112.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 12714.2–12742.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 12990.7–13039.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 13559.0–13607.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 15115.6–15155.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 16663.9–16703.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 16972.1–18011.6 m	100.0	212	31	17.6 / 22.0 / 26.4
line-3 18216.4–36732.7 m	100.0	3708	555	17.6 / 22.0 / 26.4
line-3 36929.2–41733.2 m	100.0	968	144	17.6 / 22.0 / 26.4
line-3 42796.0–42816.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 42816.0–42836.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 43810.6–43838.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 43878.8–43907.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 43947.1–43967.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 44192.0–44220.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 44248.6–44268.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 44879.7–44899.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-3 46370.8–46410.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-3 46727.4–46787.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 48007.4–48067.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-3 48127.4–48187.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 0.0–80.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 140.0–160.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 240.0–268.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 268.3–296.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 421.4–469.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 546.3–622.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 642.8–731.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 1097.4–1117.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 1463.7–1492.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 2119.7–2167.9 m	50.0	12	1	8.8 / 11.0 / 13.2

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-4 2292.8–2369.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 2389.4–2437.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 2514.2–2542.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 2880.5–2908.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 3524.8–3544.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 3621.3–3641.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 4184.2–4212.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 6558.4–6586.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 7280.6–7308.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 7393.8–7413.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 7918.0–7946.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 8002.9–8031.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 8385.7–8414.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 8502.3–8522.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 8827.1–8855.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 9452.8–9532.834 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 9532.834–9972.834 m	100.0	92	13	17.6 / 22.0 / 26.4
line-4 9972.834–10041.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-4 10157.7–10186.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 10748.8–10797.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 10865.4–10885.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 11058.5–11098.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 11243.4–11271.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 11541.4–11686.2 m	100.0	32	4	17.6 / 22.0 / 26.4
line-4 12032.5–12060.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 12080.8–12100.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 12273.9–12302.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 13363.9–13383.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 13383.9–13412.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 13460.4–13508.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 13508.7–13528.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 13528.7–13625.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 13625.3–13645.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 13645.3–13778.4 m	100.0	32	4	17.6 / 22.0 / 26.4
line-4 14068.1–14144.7 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 14233.0–14261.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 14357.8–14406.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 14627.5–14732.4 m	100.0	28	3	17.6 / 22.0 / 26.4
line-4 14780.7–14829.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 14849.0–14933.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 15186.9–15215.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 15303.5–15388.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-4 15416.7–16640.3 m	100.0	248	36	17.6 / 22.0 / 26.4
line-4 16640.3–16660.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 16660.3–19284.2 m	100.0	528	78	17.6 / 22.0 / 26.4
line-4 19284.2–19304.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 19304.2–19996.8 m	100.0	140	21	17.6 / 22.0 / 26.4
line-4 19996.8–20016.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 20016.8–21868.2 m	100.0	376	56	17.6 / 22.0 / 26.4
line-4 21868.2–21888.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 21888.2–22629.0 m	100.0	152	22	17.6 / 22.0 / 26.4
line-4 22629.0–22649.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 22649.0–24512.1 m	100.0	376	56	17.6 / 22.0 / 26.4
line-4 24512.1–24532.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 24532.1–25212.9 m	100.0	140	21	17.6 / 22.0 / 26.4
line-4 25212.9–25232.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 25232.9–25985.5 m	100.0	156	23	17.6 / 22.0 / 26.4
line-4 25985.5–26005.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 26005.5–27084.3 m	100.0	220	33	17.6 / 22.0 / 26.4
line-4 27084.3–27104.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 27104.3–27788.6 m	100.0	140	21	17.6 / 22.0 / 26.4
line-4 27961.7–28231.4 m	100.0	56	8	17.6 / 22.0 / 26.4

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-4 28452.8–28557.7 m	100.0	28	3	17.6 / 22.0 / 26.4
line-4 28557.7–28577.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 28577.7–30440.8 m	100.0	376	56	17.6 / 22.0 / 26.4
line-4 30440.8–30460.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 30460.8–31201.6 m	100.0	152	22	17.6 / 22.0 / 26.4
line-4 31201.6–31221.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 31221.6–33073.0 m	100.0	376	56	17.6 / 22.0 / 26.4
line-4 33073.0–33093.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 33093.0–33785.5 m	100.0	140	21	17.6 / 22.0 / 26.4
line-4 33785.5–33805.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 33805.5–36429.4 m	100.0	528	78	17.6 / 22.0 / 26.4
line-4 36429.4–36449.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 36449.4–39444.5 m	100.0	600	90	17.6 / 22.0 / 26.4
line-4 39444.5–39464.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 39464.5–39665.9 m	100.0	48	6	17.6 / 22.0 / 26.4
line-4 39750.7–39779.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 40158.4–40186.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 40368.1–40416.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-4 40416.4–40436.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 40884.1–40904.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-4 41109.0–41149.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 0.0–20.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 48.3–68.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 68.3–156.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 656.6–684.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 684.9–744.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 744.9–773.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 773.1–878.0 m	100.0	28	3	17.6 / 22.0 / 26.4
line-5 878.0–986.3 m	100.0	28	3	17.6 / 22.0 / 26.4
line-5 986.3–1066.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 1066.3–1154.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 1154.6–1795.4 m	100.0	132	19	17.6 / 22.0 / 26.4
line-5 1795.4–1855.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 1855.4–1875.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 2479.7–2539.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 2539.7–2559.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 2857.6–12070.9 m	100.0	1848	276	17.6 / 22.0 / 26.4
line-5 12099.2–12737.2 m	100.0	132	19	17.6 / 22.0 / 26.4
line-5 12825.5–12962.1 m	100.0	32	4	17.6 / 22.0 / 26.4
line-5 12982.1–13022.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 13198.6–13395.2 m	100.0	40	6	17.6 / 22.0 / 26.4
line-5 13415.2–13483.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 13523.5–13631.8 m	100.0	28	3	17.6 / 22.0 / 26.4
line-5 13660.0–13836.6 m	100.0	40	6	17.6 / 22.0 / 26.4
line-5 13964.9–14013.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 14181.5–14269.7 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 14329.7–14378.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 14526.3–14642.9 m	100.0	28	3	17.6 / 22.0 / 26.4
line-5 14819.5–14859.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 14947.7–14976.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 15036.0–15064.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 15104.3–15192.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-5 15240.9–15389.2 m	100.0	32	4	17.6 / 22.0 / 26.4
line-5 15485.7–15545.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 15790.6–15838.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 16260.3–28663.5 m	100.0	2488	372	17.6 / 22.0 / 26.4
line-5 28663.5–28683.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 28683.5–28723.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 29003.5–29023.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 29323.5–29343.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 29608.4–29676.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 29676.7–29704.9 m	50.0	12	1	8.8 / 11.0 / 13.2

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-5 31470.9–31519.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 31815.7–31844.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 31844.0–31884.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 36294.8–36343.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 36511.4–36539.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 42526.1–42554.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 42610.9–42679.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-5 44267.1–44315.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 44665.1–44693.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 45003.1–45023.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 45268.0–45296.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-5 45296.2–45316.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 46222.5–46242.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-5 46550.8–46579.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 20.0–48.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 48.3–76.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 76.6–124.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 144.9–173.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 346.3–366.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 522.8–551.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 551.1–579.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 664.3–749.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-6 777.4–797.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 1552.8–1581.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 1657.6–1685.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 2611.0–2639.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 2982.2–3067.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-6 3067.0–3115.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 3596.1–3624.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 4088.7–4117.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 4278.4–4306.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 4363.2–4391.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 4807.5–4892.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-6 5251.8–5280.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 6127.1–6155.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 6692.8–6712.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 9111.3–9139.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 9394.2–9450.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 10924.9–10953.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 11405.8–11434.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 11462.3–11518.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 13745.1–13773.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 14395.6–14423.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 14811.6–14839.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 15114.4–15142.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 15465.6–15522.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 15550.4–15590.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 15860.1–15928.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 16499.5–16539.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 16759.5–16856.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-6 16912.7–16941.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 16989.2–17057.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 18353.5–18393.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 18393.5–18581.8 m	100.0	40	6	17.6 / 22.0 / 26.4
line-6 18581.8–18670.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-6 18670.1–19234.3 m	100.0	116	17	17.6 / 22.0 / 26.4
line-6 19234.3–19302.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 19302.6–19362.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 19362.6–19555.8 m	100.0	40	6	17.6 / 22.0 / 26.4
line-6 20028.9–20057.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 20490.3–20518.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 20980.0–21048.309 m	75.0	16	2	13.2 / 16.5 / 19.8

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-6 21048.309–27525.3 m	100.0	1300	195	17.6 / 22.0 / 26.4
line-6 27573.5–27630.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 27811.5–27839.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 28545.5–28573.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 28650.4–28726.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-6 28783.5–28936.6 m	100.0	36	5	17.6 / 22.0 / 26.4
line-6 28964.9–28993.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 29013.2–29041.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 29069.8–29098.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 29174.6–29202.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 29251.2–29279.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 29384.3–29517.5 m	100.0	32	4	17.6 / 22.0 / 26.4
line-6 29537.5–29670.6 m	100.0	32	4	17.6 / 22.0 / 26.4
line-6 29698.9–29803.7 m	100.0	28	3	17.6 / 22.0 / 26.4
line-6 30118.3–30146.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 30271.4–30328.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 30404.6–30509.4 m	100.0	28	3	17.6 / 22.0 / 26.4
line-6 30566.0–30824.0 m	100.0	56	8	17.6 / 22.0 / 26.4
line-6 30880.6–30900.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 30928.8–30957.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-6 30985.4–31300.0 m	100.0	68	9	17.6 / 22.0 / 26.4
line-6 31767.7–45184.7 m	100.0	2688	402	17.6 / 22.0 / 26.4
line-6 45212.9–46048.922 m	100.0	172	25	17.6 / 22.0 / 26.4
line-6 46048.922–46117.206 m	75.0	16	2	13.2 / 16.5 / 19.8
line-6 46117.206–46137.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 46277.2–46297.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 46757.2–46777.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-6 47585.5–47605.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 0.0–20.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 1036.6–1056.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 3493.1–3533.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 5141.4–5161.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 7085.1–7105.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 11133.9–11153.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 11173.9–11222.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 12516.1–12564.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 13002.4–13042.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 13255.5–13295.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 13432.1–13480.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 13577.0–13605.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 14026.7–14086.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 14086.7–14203.2 m	100.0	28	3	17.6 / 22.0 / 26.4
line-7 14203.2–14930.3 m	100.0	152	22	17.6 / 22.0 / 26.4
line-7 14930.3–14950.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 14950.3–14978.62 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 14978.62–14998.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 14998.6–15243.5 m	100.0	52	7	17.6 / 22.0 / 26.4
line-7 15331.8–22873.9 m	100.0	1512	226	17.6 / 22.0 / 26.4
line-7 23195.3–23323.6 m	100.0	32	4	17.6 / 22.0 / 26.4
line-7 23705.0–23725.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 23793.3–23841.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 23929.9–23949.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 24174.7–24194.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 24283.0–24451.3 m	100.0	36	5	17.6 / 22.0 / 26.4
line-7 24499.6–24567.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 24784.4–24832.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-7 24852.7–24872.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 25021.0–25089.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 25157.6–25225.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-7 25265.9–25285.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-7 25334.2–25422.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-7 25599.0–39908.8 m	100.0	2868	429	17.6 / 22.0 / 26.4

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-7 39908.8–39957.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 120.0–140.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 1178.0–1298.0 m	100.0	28	3	17.6 / 22.0 / 26.4
line-8 2573.4–2593.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 2689.9–2718.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 2718.2–2746.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 3225.9–3254.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 4354.5–4382.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 4499.3–4615.9 m	100.0	28	3	17.6 / 22.0 / 26.4
line-8 5122.2–5150.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 5869.3–5897.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 6159.0–6199.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 6259.0–6299.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 6359.0–6379.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 7443.8–7463.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 7503.8–7543.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 7723.8–7803.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-8 8223.8–8283.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 9500.4–9540.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 9838.4–9898.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 10991.5–11039.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 11684.1–11704.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 11704.1–11732.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 11760.6–11780.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 11905.5–11925.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 11953.8–12098.6 m	100.0	32	4	17.6 / 22.0 / 26.4
line-8 12146.9–12195.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 12561.5–12581.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 12629.7–12658.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 12706.3–12726.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 13016.0–13072.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 13112.6–13152.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 13520.9–13540.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 13765.7–13794.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 13794.0–13822.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 14012.0–14072.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 14140.3–14160.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 14248.6–14305.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 15144.0–15268.8 m	100.0	28	3	17.6 / 22.0 / 26.4
line-8 15288.8–15317.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 15337.1–15425.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-8 15425.4–15453.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 15453.7–15510.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 15623.4–15663.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 16102.8–16159.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 16405.6–16462.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 16575.3–16631.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 17669.3–17709.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 20555.5–20612.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 21633.7–21662.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 22429.1–22457.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 22565.7–22614.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 23741.1–23769.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 23874.2–30971.4 m	100.0	1420	213	17.6 / 22.0 / 26.4
line-8 31763.9–45297.509 m	100.0	2712	406	17.6 / 22.0 / 26.4
line-8 45297.509–45354.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 45438.9–45458.931 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 45458.931–45487.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 45563.8–45592.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 45861.8–45890.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 46074.9–46114.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 46191.5–46211.5 m	25.0	8	0	4.4 / 5.5 / 6.6

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-8 46239.8–46259.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-8 46288.0–46396.3 m	100.0	28	3	17.6 / 22.0 / 26.4
line-8 46436.3–46504.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 46581.2–46637.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 46657.8–46697.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 46854.3–46914.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-8 47002.6–47042.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 47279.2–47319.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-8 47991.7–48020.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 0.0–20.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 20.0–48.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 48.3–1200.8 m	100.0	236	35	17.6 / 22.0 / 26.4
line-9 1200.8–1220.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 1220.8–1349.1 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 1349.1–1369.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 1369.1–1409.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 1409.1–1437.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 1437.4–1514.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 1514.0–1554.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 1554.0–1832.0 m	100.0	60	9	17.6 / 22.0 / 26.4
line-9 1832.0–1892.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 1892.0–2112.0 m	100.0	48	6	17.6 / 22.0 / 26.4
line-9 2112.0–2228.5 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 2228.5–2256.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 2256.8–2341.7 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 2341.7–2398.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 2511.4–2591.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 2591.4–2731.4 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 2791.4–2811.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 2971.4–3031.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 3751.4–3771.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 3871.4–4071.4 m	100.0	40	6	17.6 / 22.0 / 26.4
line-9 4071.4–4099.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 4099.7–4372.8 m	100.0	56	8	17.6 / 22.0 / 26.4
line-9 4787.4–4815.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 4843.9–4985.3 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 4985.3–5053.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 5053.6–5161.9 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 5161.9–5326.8 m	100.0	36	5	17.6 / 22.0 / 26.4
line-9 5326.8–5403.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 5608.2–5696.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 5716.5–5736.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 5764.8–5913.0 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 6109.6–6137.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 6354.5–6382.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 6382.7–6402.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 6402.7–6442.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 6531.0–6819.3 m	100.0	60	9	17.6 / 22.0 / 26.4
line-9 6819.3–6847.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 6847.6–9778.4 m	100.0	592	88	17.6 / 22.0 / 26.4
line-9 9778.4–9798.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 9798.4–11954.4 m	100.0	436	65	17.6 / 22.0 / 26.4
line-9 11954.4–11982.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 11982.6–12448.9 m	100.0	96	14	17.6 / 22.0 / 26.4
line-9 12448.9–12468.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 12468.9–12818.6 m	100.0	72	10	17.6 / 22.0 / 26.4
line-9 12818.6–12838.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 12838.6–12886.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 12886.9–12906.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 12906.9–12955.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 12955.2–12995.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 12995.2–13135.2 m	100.0	32	4	17.6 / 22.0 / 26.4

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-9 13135.2–13203.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 13203.5–13368.3 m	100.0	36	5	17.6 / 22.0 / 26.4
line-9 13368.3–13388.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 13388.3–14362.3 m	100.0	196	29	17.6 / 22.0 / 26.4
line-9 14362.3–14382.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 14382.3–14887.1 m	100.0	108	15	17.6 / 22.0 / 26.4
line-9 14887.1–14915.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 14915.4–20452.7 m	100.0	1112	166	17.6 / 22.0 / 26.4
line-9 20452.7–20481.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 20481.0–21490.1 m	100.0	208	30	17.6 / 22.0 / 26.4
line-9 21490.1–21510.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 21510.1–22509.5 m	100.0	200	30	17.6 / 22.0 / 26.4
line-9 22774.4–23928.4 m	100.0	236	35	17.6 / 22.0 / 26.4
line-9 23928.4–23948.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 23948.4–24136.7 m	100.0	40	6	17.6 / 22.0 / 26.4
line-9 24136.7–24176.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 24176.7–26225.8 m	100.0	412	61	17.6 / 22.0 / 26.4
line-9 27270.6–27290.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 27538.9–27615.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 27615.5–27675.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 27675.5–27795.5 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 27795.5–27912.0 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 27912.0–27996.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 27996.9–28105.2 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 28105.2–28625.2 m	100.0	108	15	17.6 / 22.0 / 26.4
line-9 28773.5–28793.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 28813.5–28833.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 29213.5–29241.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 30046.6–30074.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 30074.9–30143.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 30239.7–30328.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 30617.7–30637.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 31342.6–31382.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 31795.7–31824.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 31944.0–32204.0 m	100.0	56	8	17.6 / 22.0 / 26.4
line-9 32204.0–32320.6 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 32320.6–33162.8 m	100.0	172	25	17.6 / 22.0 / 26.4
line-9 34371.9–34391.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 36052.8–36072.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 36474.2–36502.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 36679.0–36699.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 36699.0–36719.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 39548.4–39608.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 39608.4–40465.0 m	100.0	176	26	17.6 / 22.0 / 26.4
line-9 40465.0–40553.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 40553.3–40609.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 40609.8–40689.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 40689.8–41526.4 m	100.0	172	25	17.6 / 22.0 / 26.4
line-9 41606.4–41646.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 41956.1–41984.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 42871.5–42948.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 44242.9–44299.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 44327.7–44404.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 46386.2–46442.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 46744.2–46772.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 46772.5–46849.04 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 46849.04–47187.0 m	100.0	72	10	17.6 / 22.0 / 26.4
line-9 47207.0–47227.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 48019.6–48039.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 48059.6–48176.1 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 48196.1–50513.8 m	100.0	468	69	17.6 / 22.0 / 26.4
line-9 50513.8–50533.8 m	25.0	8	0	4.4 / 5.5 / 6.6

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-9 50773.8–50842.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 50898.6–50926.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 51336.6–51364.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 51364.9–51384.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 51564.9–51584.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 51584.9–51613.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 51613.2–51641.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 51689.8–51718.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 51786.3–51806.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 51806.3–51902.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 51902.9–51922.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 51991.2–52019.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 52019.5–52059.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 52059.5–52144.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 52172.6–52229.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 52229.2–52277.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 52866.6–52894.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 52991.4–53031.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 53998.5–54026.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 54106.8–54206.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 54246.8–54286.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 54466.8–54506.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 55963.4–55983.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 56263.4–57283.4 m	100.0	208	30	17.6 / 22.0 / 26.4
line-9 58802.8–58822.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 59202.8–59242.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 59242.8–59359.4 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 59359.4–60587.3 m	100.0	252	37	17.6 / 22.0 / 26.4
line-9 62031.6–62059.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 62671.0–62771.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 62811.0–62871.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 62979.3–63035.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 63749.8–63778.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 65822.0–65862.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 65890.3–65910.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 65950.3–66078.6 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 66078.6–67363.1 m	100.0	260	39	17.6 / 22.0 / 26.4
line-9 67363.1–67451.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 67451.4–67491.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 67491.4–67587.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 67587.9–67672.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 67672.8–67781.1 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 67781.1–67801.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 67801.1–67994.2 m	100.0	40	6	17.6 / 22.0 / 26.4
line-9 67994.2–68022.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 68022.5–68159.1 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 68159.1–68467.4 m	100.0	68	9	17.6 / 22.0 / 26.4
line-9 68467.4–68507.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 68507.4–68583.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 68583.9–68672.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 68672.2–68700.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 68700.5–68740.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 68740.5–68960.5 m	100.0	48	6	17.6 / 22.0 / 26.4
line-9 68960.5–69008.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 69008.8–69085.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 69085.3–69125.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 69125.3–69611.6 m	100.0	100	15	17.6 / 22.0 / 26.4
line-9 69611.6–69696.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 69696.5–69909.6 m	100.0	48	6	17.6 / 22.0 / 26.4
line-9 69909.6–69937.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 69937.9–70082.7 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 70082.7–70131.0 m	50.0	12	1	8.8 / 11.0 / 13.2

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-9 70131.0–70159.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 70159.3–70207.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 70207.6–70235.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 70235.9–70315.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 70315.9–70344.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 70344.2–70364.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 70364.2–70452.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 70452.5–70480.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 70480.7–70669.0 m	100.0	40	6	17.6 / 22.0 / 26.4
line-9 70669.0–70729.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 70729.0–70769.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 70769.0–70817.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 70817.3–71015.3 m	100.0	40	6	17.6 / 22.0 / 26.4
line-9 71015.3–71083.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 71083.6–71123.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 71123.6–71200.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 71200.2–71426.4 m	100.0	52	7	17.6 / 22.0 / 26.4
line-9 71426.4–71494.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 71494.7–71534.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 71534.7–71614.7 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 71614.7–71634.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 71634.7–71654.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 71654.7–71731.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 71731.3–71836.1 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 71836.1–71904.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 71904.4–71964.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 71964.4–72041.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 72041.0–72440.4 m	100.0	80	12	17.6 / 22.0 / 26.4
line-9 72440.4–72520.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 72520.4–72748.7 m	100.0	52	7	17.6 / 22.0 / 26.4
line-9 72748.7–72797.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 72797.0–73085.2 m	100.0	60	9	17.6 / 22.0 / 26.4
line-9 73085.2–73173.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 73173.5–73686.1 m	100.0	108	15	17.6 / 22.0 / 26.4
line-9 73686.1–73766.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 73766.1–73882.6 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 73882.6–73910.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 73910.9–74039.2 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 74039.2–74067.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 74067.5–74337.2 m	100.0	56	8	17.6 / 22.0 / 26.4
line-9 74337.2–74482.1 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 74482.1–74522.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 74522.1–74578.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 74578.6–74873.2 m	100.0	60	9	17.6 / 22.0 / 26.4
line-9 74873.2–74961.5 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 74961.5–75129.8 m	100.0	36	5	17.6 / 22.0 / 26.4
line-9 75129.8–75158.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 75158.0–75246.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 75246.3–75266.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 75266.3–75486.3 m	100.0	48	6	17.6 / 22.0 / 26.4
line-9 75486.3–75534.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 75534.6–75602.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 75602.9–75642.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 75642.9–75702.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 75702.9–75811.2 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 75811.2–75839.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 75839.5–75907.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 75907.7–75927.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 75927.7–75996.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 75996.0–76024.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 76024.3–76132.6 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 76132.6–76200.9 m	75.0	16	2	13.2 / 16.5 / 19.8

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-9 76200.9–76240.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 76240.9–76489.2 m	100.0	52	7	17.6 / 22.0 / 26.4
line-9 76489.2–76537.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 76537.5–76605.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 76605.7–76645.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 76645.7–76745.7 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 76745.7–76774.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 76774.0–76898.9 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 76898.9–76918.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 76918.9–77107.2 m	100.0	40	6	17.6 / 22.0 / 26.4
line-9 77107.2–77135.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 77135.4–77400.3 m	100.0	56	8	17.6 / 22.0 / 26.4
line-9 77400.3–77420.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 77420.3–77468.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 77468.6–77488.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 77488.6–77585.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 77585.1–77613.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 77613.4–77633.4 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 77633.4–77730.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 77730.0–77874.9 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 77874.9–77914.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 77914.9–77974.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 77974.9–78023.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 78023.1–78899.1 m	100.0	180	27	17.6 / 22.0 / 26.4
line-9 78899.1–78939.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 78939.1–78999.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 78999.1–79095.7 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 79095.7–79200.5 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 79200.5–79257.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 79257.1–79277.1 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 79277.1–79325.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 79325.4–79971.7 m	100.0	132	19	17.6 / 22.0 / 26.4
line-9 79971.7–79991.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 79991.7–80019.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 80019.9–80039.9 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 80039.9–80176.5 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 80176.5–80196.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 80196.5–80433.1 m	100.0	52	7	17.6 / 22.0 / 26.4
line-9 80433.1–80481.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 80481.4–80541.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 80541.4–80581.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 80581.4–80729.7 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 80729.7–80777.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 80777.9–80846.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 80846.2–80886.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 80886.2–81026.2 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 81026.2–81066.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 81066.2–81134.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 81134.5–81154.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 81154.5–81251.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 81251.1–81279.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 81279.4–81347.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 81347.6–81395.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 81395.9–82309.9 m	100.0	188	27	17.6 / 22.0 / 26.4
line-9 82309.9–82358.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 82358.2–82426.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 82426.5–82511.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 82511.3–82781.0 m	100.0	56	8	17.6 / 22.0 / 26.4
line-9 82781.0–82849.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 82849.3–82985.9 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 82985.9–83014.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 83014.2–83110.7 m	100.0	20	3	17.6 / 22.0 / 26.4

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-9 83110.7–83167.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 83167.3–83490.1 m	100.0	68	9	17.6 / 22.0 / 26.4
line-9 83490.1–83578.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 83578.4–83618.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 83618.4–83743.3 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 83743.3–83924.7 m	100.0	40	6	17.6 / 22.0 / 26.4
line-9 83924.7–83944.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 83944.7–84464.1 m	100.0	108	15	17.6 / 22.0 / 26.4
line-9 84464.1–84512.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 84512.4–84589.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 84589.0–84657.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 84657.2–84773.8 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 84773.8–84813.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 84813.8–84873.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 84873.8–84922.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 84922.1–85388.4 m	100.0	96	14	17.6 / 22.0 / 26.4
line-9 85388.4–85428.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 85428.4–85488.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 85488.4–85536.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 85536.7–86219.5 m	100.0	140	21	17.6 / 22.0 / 26.4
line-9 86219.5–86239.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 86239.5–86347.8 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 86347.8–86367.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 86367.8–86805.8 m	100.0	92	13	17.6 / 22.0 / 26.4
line-9 86805.8–86854.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 86854.1–86922.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 86922.3–86970.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 86970.6–87058.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 87058.9–87192.1 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 87192.1–87268.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 87268.6–87328.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 87328.6–87685.2 m	100.0	76	11	17.6 / 22.0 / 26.4
line-9 87685.2–87713.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 87713.5–87801.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 87801.8–87821.8 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 87821.8–87901.8 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 87901.8–87990.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 87990.0–88086.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 88086.6–88126.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 88126.6–88166.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 88166.6–88186.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 88186.6–88403.2 m	100.0	48	6	17.6 / 22.0 / 26.4
line-9 88403.2–88539.7 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 88539.7–88947.4 m	100.0	88	12	17.6 / 22.0 / 26.4
line-9 88947.4–89055.7 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 89055.7–89344.0 m	100.0	60	9	17.6 / 22.0 / 26.4
line-9 89344.0–89452.3 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 89452.3–89752.3 m	100.0	60	9	17.6 / 22.0 / 26.4
line-9 89752.3–89800.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 89800.6–89868.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 89868.9–89957.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 89957.1–89985.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 89985.4–90065.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 90065.4–90233.7 m	100.0	36	5	17.6 / 22.0 / 26.4
line-9 90233.7–90293.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 90293.7–90370.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 90370.3–90486.9 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 90486.9–90543.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 90543.4–90623.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 90623.4–90991.7 m	100.0	76	11	17.6 / 22.0 / 26.4
line-9 90991.7–91020.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 91020.0–91108.3 m	100.0	20	3	17.6 / 22.0 / 26.4

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-9 91108.3–91128.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 91128.3–91328.3 m	100.0	40	6	17.6 / 22.0 / 26.4
line-9 91328.3–91396.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 91396.6–91444.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 91444.8–91533.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 91533.1–91633.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 91633.1–91729.7 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 91729.7–91899.4 m	100.0	36	5	17.6 / 22.0 / 26.4
line-9 91899.4–91967.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 91967.7–92104.3 m	100.0	32	4	17.6 / 22.0 / 26.4
line-9 92104.3–92144.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 92144.3–93607.1 m	100.0	296	44	17.6 / 22.0 / 26.4
line-9 93607.1–93655.4 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 93655.4–93703.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 93703.7–93763.7 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 93763.7–93791.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 93791.9–93820.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 93820.2–93900.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 93900.2–93920.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 93920.2–93960.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 93960.2–93988.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 93988.5–94008.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 94008.5–94028.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 94028.5–94136.8 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 94136.8–94165.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 94165.1–95286.5 m	100.0	228	33	17.6 / 22.0 / 26.4
line-9 95286.5–95314.8 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 95314.8–95363.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 95363.1–95439.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 95439.6–95496.2 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 95496.2–95552.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 95552.8–95629.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 95629.4–95685.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 95685.9–95714.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 95714.2–95794.2 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 95794.2–96322.5 m	100.0	112	16	17.6 / 22.0 / 26.4
line-9 96322.5–96342.5 m	25.0	8	0	4.4 / 5.5 / 6.6
line-9 96342.5–96439.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 96439.1–96467.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 96467.3–96527.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-9 96527.3–96643.9 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 96643.9–96813.6 m	100.0	36	5	17.6 / 22.0 / 26.4
line-9 96813.6–96901.9 m	100.0	20	3	17.6 / 22.0 / 26.4
line-9 96901.9–96930.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 96930.2–97043.3 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 97043.3–97091.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 97091.6–97119.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 97119.9–97224.7 m	100.0	28	3	17.6 / 22.0 / 26.4
line-9 97224.7–97253.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 97253.0–97499.3 m	100.0	52	7	17.6 / 22.0 / 26.4
line-9 97499.3–97539.3 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 97539.3–97587.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 97587.6–97635.9 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 97635.9–97684.153 m	50.0	12	1	8.8 / 11.0 / 13.2
line-9 97684.153–97684.2 m	25.0	8	0	4.4 / 5.5 / 6.6
line-10 0.0–600.0 m	100.0	120	18	17.6 / 22.0 / 26.4
line-10 600.0–716.6 m	100.0	28	3	17.6 / 22.0 / 26.4
line-10 716.6–1140.8 m	100.0	88	12	17.6 / 22.0 / 26.4
line-10 1140.8–1249.1 m	100.0	28	3	17.6 / 22.0 / 26.4
line-10 1249.1–2029.1 m	100.0	160	24	17.6 / 22.0 / 26.4
line-10 2029.1–2145.7 m	100.0	28	3	17.6 / 22.0 / 26.4
line-10 2145.7–3506.8 m	100.0	276	41	17.6 / 22.0 / 26.4

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-10 3506.8–3631.6 m	100.0	28	3	17.6 / 22.0 / 26.4
line-10 3631.6–4271.6 m	100.0	132	19	17.6 / 22.0 / 26.4
line-10 4271.6–4388.2 m	100.0	28	3	17.6 / 22.0 / 26.4
line-10 4388.2–4925.6 m	100.0	112	16	17.6 / 22.0 / 26.4
line-10 4925.6–5033.9 m	100.0	28	3	17.6 / 22.0 / 26.4
line-10 5033.9–6482.2 m	100.0	292	43	17.6 / 22.0 / 26.4
line-10 6482.2–6570.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-10 6570.4–6655.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-10 6655.3–6755.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-10 6755.3–6803.574 m	50.0	12	1	8.8 / 11.0 / 13.2
line-10 6803.574–6803.6 m	25.0	8	0	4.4 / 5.5 / 6.6
line-11 0.0–953.4 m	100.0	196	29	17.6 / 22.0 / 26.4
line-11 953.4–1061.7 m	100.0	28	3	17.6 / 22.0 / 26.4
line-11 1061.7–1581.7 m	100.0	108	15	17.6 / 22.0 / 26.4
line-11 1581.7–1698.2 m	100.0	28	3	17.6 / 22.0 / 26.4
line-11 1698.2–2801.3 m	100.0	228	33	17.6 / 22.0 / 26.4
line-11 2801.3–2909.6 m	100.0	28	3	17.6 / 22.0 / 26.4
line-11 2909.6–3029.6 m	100.0	28	3	17.6 / 22.0 / 26.4
line-11 3029.6–3146.2 m	100.0	28	3	17.6 / 22.0 / 26.4
line-11 3146.2–3251.0 m	100.0	28	3	17.6 / 22.0 / 26.4
line-11 3251.0–3359.3 m	100.0	28	3	17.6 / 22.0 / 26.4
line-11 3359.3–3419.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-11 3419.3–3467.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-11 3467.6–3575.9 m	100.0	28	3	17.6 / 22.0 / 26.4
line-11 3575.9–3604.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-11 3604.2–4056.7 m	100.0	96	14	17.6 / 22.0 / 26.4
line-12 0.0–747.1 m	100.0	152	22	17.6 / 22.0 / 26.4
line-12 747.1–872.0 m	100.0	28	3	17.6 / 22.0 / 26.4
line-12 872.0–892.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-12 892.0–1093.4 m	100.0	48	6	17.6 / 22.0 / 26.4
line-12 1093.4–1153.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-12 1153.4–1209.9 m	75.0	16	2	13.2 / 16.5 / 19.8
line-12 1209.9–1351.371 m	100.0	32	4	17.6 / 22.0 / 26.4
line-12 1351.371–1379.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-13 0.0–20.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-13 20.0–40.0 m	25.0	8	0	4.4 / 5.5 / 6.6
line-13 40.0–96.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-13 96.6–209.7 m	100.0	28	3	17.6 / 22.0 / 26.4
line-13 209.7–258.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-13 258.0–318.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-13 318.0–431.1 m	100.0	28	3	17.6 / 22.0 / 26.4
line-13 431.1–705.7 m	100.0	56	8	17.6 / 22.0 / 26.4
line-13 705.7–754.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-13 754.0–814.0 m	75.0	16	2	13.2 / 16.5 / 19.8
line-13 814.0–870.5 m	75.0	16	2	13.2 / 16.5 / 19.8
line-13 870.5–1125.1 m	100.0	56	8	17.6 / 22.0 / 26.4
line-13 1125.1–1193.4 m	75.0	16	2	13.2 / 16.5 / 19.8
line-13 1193.4–1241.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-13 1241.7–1346.5 m	100.0	28	3	17.6 / 22.0 / 26.4
line-13 1346.5–1459.7 m	100.0	28	3	17.6 / 22.0 / 26.4
line-13 1459.7–1601.1 m	100.0	32	4	17.6 / 22.0 / 26.4
line-13 1601.1–1689.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-13 1689.4–1757.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-13 1757.6–1865.9 m	100.0	28	3	17.6 / 22.0 / 26.4
line-13 1865.9–1894.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-13 1894.2–1979.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-13 1979.1–2059.1 m	100.0	20	3	17.6 / 22.0 / 26.4
line-13 2059.1–2259.1 m	100.0	40	6	17.6 / 22.0 / 26.4
line-13 2259.1–2347.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-13 2347.4–2375.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-13 2375.6–2455.6 m	100.0	20	3	17.6 / 22.0 / 26.4
line-13 2455.6–2803.9 m	100.0	72	10	17.6 / 22.0 / 26.4

Elevated segment	Longest unit m	Bearings	Internal connections	Hot movement mm at 8 / 10 / 12 µstrain/K
line-13 2803.9–2920.5 m	100.0	28	3	17.6 / 22.0 / 26.4
line-13 2920.5–3005.3 m	100.0	20	3	17.6 / 22.0 / 26.4
line-13 3005.3–3053.6 m	50.0	12	1	8.8 / 11.0 / 13.2
line-13 3053.6–3170.2 m	100.0	28	3	17.6 / 22.0 / 26.4
line-13 3170.2–3198.5 m	50.0	12	1	8.8 / 11.0 / 13.2
line-13 3198.5–3266.8 m	75.0	16	2	13.2 / 16.5 / 19.8
line-13 3266.8–3323.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-13 3323.3–3496.5 m	100.0	36	5	17.6 / 22.0 / 26.4
line-13 3496.5–3573.0 m	100.0	20	3	17.6 / 22.0 / 26.4
line-13 3573.0–3742.7 m	100.0	36	5	17.6 / 22.0 / 26.4
line-14 0.0–1179.7 m	100.0	240	36	17.6 / 22.0 / 26.4
line-14 1179.7–1279.7 m	100.0	20	3	17.6 / 22.0 / 26.4
line-14 1279.7–1879.7 m	100.0	120	18	17.6 / 22.0 / 26.4
line-14 1879.7–1996.2 m	100.0	28	3	17.6 / 22.0 / 26.4
line-14 1996.2–2290.8 m	100.0	60	9	17.6 / 22.0 / 26.4
line-14 2290.8–2339.1 m	50.0	12	1	8.8 / 11.0 / 13.2
line-14 2339.1–2399.1 m	75.0	16	2	13.2 / 16.5 / 19.8
line-14 2399.1–2455.6 m	75.0	16	2	13.2 / 16.5 / 19.8
line-14 2455.6–2830.2 m	100.0	76	11	17.6 / 22.0 / 26.4
line-14 2830.2–2943.3 m	100.0	28	3	17.6 / 22.0 / 26.4
line-14 2943.3–3003.3 m	75.0	16	2	13.2 / 16.5 / 19.8
line-14 3003.3–3023.3 m	25.0	8	0	4.4 / 5.5 / 6.6
line-14 3023.3–3431.6 m	100.0	88	12	17.6 / 22.0 / 26.4
line-14 3431.6–3548.2 m	100.0	28	3	17.6 / 22.0 / 26.4
line-14 3548.2–3944.2 m	100.0	80	12	17.6 / 22.0 / 26.4
line-14 3944.2–4052.4 m	100.0	28	3	17.6 / 22.0 / 26.4
line-14 4052.4–4312.4 m	100.0	56	8	17.6 / 22.0 / 26.4
line-14 4312.4–4360.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-14 4360.7–4380.7 m	25.0	8	0	4.4 / 5.5 / 6.6
line-14 4380.7–4429.0 m	50.0	12	1	8.8 / 11.0 / 13.2
line-14 4429.0–4533.9 m	100.0	28	3	17.6 / 22.0 / 26.4
line-14 4533.9–4582.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-14 4582.2–4622.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-14 4622.2–4662.2 m	50.0	12	1	8.8 / 11.0 / 13.2
line-14 4662.2–4822.2 m	100.0	36	5	17.6 / 22.0 / 26.4
line-14 4822.2–4910.4 m	100.0	20	3	17.6 / 22.0 / 26.4
line-14 4910.4–4938.7 m	50.0	12	1	8.8 / 11.0 / 13.2
line-14 4938.7–5018.7 m	100.0	20	3	17.6 / 22.0 / 26.4
line-14 5018.7–6047.0 m	100.0	212	31	17.6 / 22.0 / 26.4

Mapped soil has 16 incomplete profiles. Prioritise investigation and settlement/drainage monitoring; mapped pH/texture do not determine bearing resistance, pile depth, sulfate/chloride exposure or a coating category.

Location-specific soil investigation plan. Flag counts: {"acidic-soil-durability-testing": 3, "coverage-gap": 16, "fine-soil-plasticity-and-shrink-swell-tests": 5377, "granular-density-and-groundwater-tests": 3198, "silt-moisture-frost-and-erosion-review": 3528}.

Joints, paints, finishes and cleaning

Steel, GFRP body and roof finish zones retain their shared qualified process. Protect electrical bond lands, seals, drains, PV glass and inspection areas. Radiative paint remains a trial option; no unverified cooling benefit enters the energy model.

Work package	Asset-level tasks generated
rs-finish-seal-joint	809
rs-controlled-wash	809
civil-joint-drain-finish	314
energy-soiling-cleaning	139

The operations bundle carries these inspection/cleaning triggers, owners and required work-order evidence. Cleaning intervals follow measured condition and access; material compatibility and wash-water handling must be qualified locally.

Release inputs still required:

- civil movement: complete deployment train axle positions and loaded distribution; component temperature extremes and erection temperature; measured concrete CTE/shrinkage/creep; selected connection load path, construction stages and bearing restraint; rail interaction, seismic and foundation movement; surveyed supports and abutment movement details.
- fasteners: joint loads and fatigue spectrum; fastener grade and proof load; actual finish/lubrication torque-tension trial; preload loss, slip/separation, bearing and thread checks.
- seals: supplier qualified compression range and ageing; actual thickness, land gap and tolerances; module thermal mismatch and vibration; water-ingress test after assembly and cleaning.
- finishes: local exposure and substrate preparation qualification; approved product system and batch TDS limits; DFT/adhesion/cure and fire/UV/wash qualification; masking, repair and inspection records.
- cleaning: measured washable area by material, not canopy area; trial person-minutes, water and safe access windows; detergent/nozzle/material compatibility; wash-water collection and local disposal route.

Structured report · Workforce spreadsheet · Deployment gates

Baghdad line-local depot and stabling quantity package

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-baseline/DEPOT-PACKAGE.md

> Original catalogue or earlier scope reference. The figures and policies below retain their original assumptions; they are not the latest Baghdad staffing, depot, procurement or funding basis. See the current Baghdad recalculation and current city summary. National/portfolio totals have not been repriced with that conditional Baghdad option.

The original allocation remains 270 station trains + 539 depot/storage trains = 809 and its morning-direction check remains failed. The 5 missing directional starts require an operating/stabling revision with conflict-aware evening returns, morning launches, recovery and degraded charging. This package does not turn storage capacity arithmetic into a physically accepted allocation.

Line	Station storage ID	Storage positions	Usable track m	Tracks / turnouts	Declared / workload bays
line-1	line-1-0166-0866-s049067	52	6,292	18 / 20	0 / 12
line-10	line-10-0686-1312-s006804	9	1,089	3 / 5	0 / 3
line-11	line-11-0337-1112-s006884	10	1,210	4 / 6	0 / 2
line-12	line-12-0760-0138-s006586	9	1,089	3 / 5	0 / 3
line-13	line-13-1193-0313-s008042	10	1,210	4 / 6	0 / 3
line-14	line-14-0688-1260-s006047	9	1,089	3 / 5	0 / 2
line-2	line-2-0595-0296-s052851	69	8,349	23 / 25	122 / 14
line-3	line-3-0151-1637-s049496	62	7,502	21 / 23	0 / 13
line-4	line-4-1230-1627-s041932	56	6,776	19 / 21	0 / 11
line-5	line-5-0677-2304-s047345	60	7,260	20 / 22	0 / 12
line-6	line-6-0495-1669-s052503	71	8,591	24 / 26	0 / 14
line-7	line-7-1531-0418-s039957	50	6,050	17 / 19	0 / 10
line-8	line-8-0785-0290-s048137	67	8,107	23 / 25	0 / 13
line-9	line-9-0689-0569-s097310	5	605	2 / 4	0 / 7

The planning arrangement uses at most three 121 m usable train slots per storage track, separate fan/throat turnouts and 5 m track spacing, with a 1.8 access/land multiplier. Track metres are single-track storage lengths, not route-km or workshop bays. Yard geometry is not located on a surveyed parcel. Storage charging shares existing powered sites; connecting tracks, peak chargers/feeders, protection, rescue access and quarantine need site engineering.

One alternative retains the declared 125 main-depot bays and zero workshop bays at eight line-local sites. Another derives line-local bays from an explicit 400 bay-hours/train/year sensitivity, 260 days/year, 16 hours/day and 70% bay availability. Neither is a released maintenance requirement. Bogie changes, heavy inspections, isolation, lifting, wash/cleaning, tools, stores and travel must establish actual bay occupation; person-hours are not bay-hours. The second alternative avoids assuming an unestablished interline move to the main workshop. Both price line-local inspection, rescue/isolation, quarantine, storage track/points, drainage/access, workshop shells/equipment and full depot PV/storage equipment quantities.

Detailed items show reference quantities and unquoted rates. The existing energy report's USD 5.93m increment above the catalogue equipment and 33,333 m² PV module requirement versus 4,000 m² canopy remain open. This package prices full 5 MW PV / 40 MWh equipment once; it does not also add the USD 5.93m delta. Equipment is not installed plant cost. PV placement, foundations/soil, battery fire spacing, DC protection, racking, utility connection and renewals are unpriced. Signed overlap decisions must reconcile the USD 8m depot and USD 67.6m charging allowances before adopting any replacement budget.

Baghdad delivery estimate and scope reconciliation

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-baseline/ESTIMATE.md

The published USD 7.951287bn equivalent is a base planning estimate, not a complete delivery budget. City subtotal is USD 7.626537bn and the one Baghdad factory including its EPC is USD 324.750m. Scope register retains quantity, implied rate, currency/source quality, inclusions, exclusions, estimator role and uncertainty. Every actual price date, quotation and named estimator remains unknown. Category implied rates reconcile the baseline; they are not surveyed rates.

Baseline WBS	Quantity	USD m equivalent	Evidence
civil	513.336	3,937.044	category-planning-estimate
stations	159.000	977.000	category-planning-estimate
depots	1.000	8.000	category-planning-estimate
rolling_stock	809.000	1,359.120	category-planning-estimate
solar_plant	1,019,675.493	815.740	category-planning-estimate
signalling	513.336	25.667	category-planning-estimate
charging_microgrid	1.000	58.400	category-planning-estimate
epc_overhead	1.000	445.566	category-planning-estimate
factory-direct	1.000	303.505	factory-sizing-allowance
factory-epc	1.000	21.245	factory-sizing-allowance

12 additional scope records retain null costs, not zero: land/title, utility diversions, duties/tax, owner costs, pre-opening people, six-car qualification, installed control/HIL, initial spares, working capital, civil/station investigations, task-derived maintenance and cyber deployment. Each requires a signed inclusion/quotation decision before financing is recalculated. Surveyed grade separation, foundations, standard spans, temporary works/erection, passenger access/evacuation, road interfaces and station functions must be investigated by corridor; civil and station standards alone do not resolve them.

EPC has identified delivery obligations, with individual priced contracts still null. Their sum cannot be asserted from the existing percentage. Factory contingency USD 50.584m, factory design/training/qualification USD 20.0m, routine train QA/labour/logistics and factory EPC are already embedded and must not be added twice. Initial stock, battery renewal cash dates, labour/maintenance contracts and working-capital reserves need account-level reconciliation.

Depot alternative	Gross priced reference USD m	Provisional replacement total USD bn
retained_declared_bays	182.583	8.125870
workload_bays	179.592	8.122879

Replacement illustrations subtract the old depot allowance once, then add the gross quantity-based reference. They remain incomplete: charging/PV/EPC overlaps and unpriced work are unresolved. No alternative is adopted into the published funding programme; government/import shares, loans and opening dates remain conditional on the original scope. Retained property additions remain their existing separate scenarios, not automatically added to this base.

The seeded correlated cost-and-delay sensitivity combines a shared risk quantile with idiosyncratic cost events and delay escalation. Illustrative known-scope P50 is USD 9.868bn and P80 USD 10.621bn. These are uncalibrated distributions, not an approved probabilistic budget. Unpriced scope is excluded rather than priced at zero; existing contingencies remain embedded; stress outputs are alternatives and cannot be added to the base. Calibrate from investigations, contract evidence, event-level dependencies and actual invoice dates. GAO estimating guidance supports technical baseline, WBS, data, alternatives, risk analysis and updates; it does not validate OSR's rates. Reference checked 4 October 2026.

Baghdad comparable opening fleet and corridor requirements

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-baseline/FIRST-PHASE.md

Demand is capacity-led in the existing model; population coverage is an access proxy. This line-specific supply comparison tests 6-minute opening peak service, 12-minute off-peak and 15-minute late service, 10% spares and retained cold reserves. Cycle time is inferred from the current peak fleet x minimum headway, not independently proved junction/turnback occupancy. Each line retains its published conditional full-line opening date.

Line	Opening month	Full fleet	Opening fleet	Deferred trains	Opening daily capacity
line-1	40	86	44	42	172,800
line-2	50	95	49	46	172,800
line-3	55	90	47	43	172,800
line-4	59	76	40	36	172,800
line-5	64	86	44	42	172,800
line-6	69	95	49	46	172,800
line-7	73	72	38	34	172,800
line-8	78	89	45	44	172,800
line-9	81	45	45	0	167,400
line-10	41	15	9	6	172,800
line-11	43	14	8	6	172,800
line-12	42	15	9	6	172,800
line-13	44	18	10	8	172,800
line-14	44	13	8	5	172,800

The reference opening fleet is 445 versus 809, deferring 364 trains / USD 611.520m gross equivalent, while requiring 0 extra train / USD 0.000m to meet the rounded spare policy on line 9. Net reduction is 364 trains at the current planning unit price. This is deferred procurement, not a validated saving or adopted service cut. Train-km and directional capacity fall with the new service; unchanged fares/revenue are never assumed. Expansion requires measured peak demand above 70% for 90 days, conflict-aware capacity, accepted spares and a newly financed delivery tranche.

Commercial corridor ranks, OD, access population and paid trips remain null. Compare actual origins/destinations/time, transfers vs paid boardings, accessibility/feeder catchments, fares, useful interchanges and standalone OPEX before choosing a first corridor. The existing 14.034 km outer-line demonstrator remains unproven commercially. Shared factory/civil/utility/control dependencies, accepted vehicles, staffing and operating release must accompany any opening. No earlier date or factory resize is asserted from vehicle subtraction alone. Existing acceleration/recovery cost-NPV trade-offs remain unchanged.

Baghdad chronological charging-energy and cost screen

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-baseline/HOURLY-ENERGY.md

The base traction intensity and timetable imply 2,043.4 GWh/year. The dedicated plant is 1,019.7 MW, plus existing on-site PV and 316 MWh storage. The base model's annual generation netting is not firm delivered charging. Ten percent of traction purchased at USD 0.10/kWh is USD 20.434m/year, an arithmetic sensitivity, not a forecast.

Hourly CSVs contain three 8,760-hour synthetic representative weather/age years for each of owned solar, contracted solar and additional hybrid storage. They use the same timetable-derived service duty, with overnight schedules split across midnight; charger loss, PV performance/wheeling loss, storage charge/discharge losses, SoC floor, age degradation, import limits and curtailment are explicit. A seven-day poor-production spell is included. Initial storage sits at its protected floor, supplying no free energy. End storage is recorded and not sold.

Synthetic case	Grid GWh	Unserved GWh	Annual energy USD m	Extra hybrid equipment USD m
synthetic_reference:owned_solar	942.7	3.197	137.978	0.0
synthetic_reference:contracted_solar	942.7	3.197	204.536	0.0
synthetic_reference:hybrid_storage	455.4	3.197	89.242	180.0
synthetic_poor_year:owned_solar	1071.2	3.433	146.444	0.0
synthetic_poor_year:contracted_solar	1071.2	3.433	193.304	0.0
synthetic_poor_year:hybrid_storage	807.4	3.399	120.070	180.0
synthetic_aged_year_10:owned_solar	960.4	3.197	139.747	0.0
synthetic_aged_year_10:contracted_solar	960.4	3.197	206.306	0.0
synthetic_aged_year_10:hybrid_storage	520.0	3.197	95.703	180.0

Electricity purchase, wheeling, balancing, capacity connection and owned-plant maintenance / contracted-solar payment are separate. PPA pays generated output even if curtailed; no surplus power sale is booked. The contracted case removes dedicated solar ownership capital and O&M, not existing station equipment. Hybrid storage adds its priced equipment once, with installed compound/protection, disposal and replacement costs still unquoted. Twenty-year indexed discounted cost is a partial comparison, not lifecycle bankability; no contract or tariffs are accepted. Unserved demand means the assumed service is unmet: no unchanged fares are credited to that service.

Aggregate connection and storage pooling are optimistic and unestablished. This screen does not supply site-specific wheeling rights or instantaneous charger arrival/queue proof. Measured multi-year weather/soiling, per-site feeder/charger duty, auxiliary loads, storage cycling/degradation, grid outages and binding backup contracts must be replayed before changing the funding programme's electricity allowance or selecting an ownership option.

Baghdad delivery reconciliation — 4 October 2026

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-baseline/README.md

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The original Baghdad estimate is USD 7.951287bn equivalent, with an incomplete delivery scope. This package turns the attached review into reproducible quantity/cost/workload comparisons and six budgeted evidence workstreams. It adds no assumed passengers, leases, supplier quotes, appointments or approval, and does not replace the published finance case before scope is accepted.

- Base estimate, exclusions, EPC and correlated cost/schedule sensitivity
- Line-local storage/depot quantities and alternative workshop scopes
- Six-car BOM, mass/axle ledger, interfaces, labour and qualification
- Chronological generation/charging/storage and ownership comparisons
- Opening fleet and first-corridor demand requirements
- Workload establishment, recruitment, curricula and eligibility pilot
- Baghdad-specific governance instance
- Six 90-day evidence workstreams and budgets

The governance instance is tailored to Baghdad with accountable functions and closure work, while 0/13 appointments, 0/11 systems and 0/8 gates remain ready/accepted. Names, appointment/competence evidence, legal entity and delegations remain pending. Role allocation is not a real appointment. Native ERP evidence Tasks remain open and do not approve budgets, manufacture or operation. Cybersecurity identity/secrets/updates/recovery/incident ownership and observation-only supervision boundaries retain their existing deployment gates.

Fresh-checkout validation: install the existing project test/render dependencies, then run `.venv/bin/python tools/automation/bootstrap_baghdad_tests.py` before the Baghdad tests. It selectively restores the exact ignored operations input from the tracked, hash-bound proposal archive, rejects drift/duplicate entries and preserves any differing workspace payload. It does not resynthesise the city or change controlled station IDs. Run the bootstrap with `--check` to verify only.

Regenerate `.venv/bin/python tools/automation/baghdad_delivery_baseline.py`, then the proposal. `--check` verifies all input/output hashes. None of the calculated alternatives has been accepted as an investment-ready budget.

- Rental fit-out allocation and inspection-interruption comparison

Rental delivery allocation sensitivities

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-baseline/RENTAL-DELIVERY.md

The existing medium portfolio retains identical physical fit-out and timing. These unquoted arrangements allocate costs and receipts; they create no extra funding, accepted tenant demand or macroeconomic saving.

Option	Landlord capital USD m	Partner capital USD m	Landlord before-tax NPV USD m	Partner value NPV USD m	Combined value NPV USD m
landlord_funded	89.193	0.000	17.295	0.000	17.295
tenant_funded_shell	53.516	35.677	19.000	-1.705	17.295
developer_concession	0.000	89.193	28.887	-11.592	17.295

Tenant-funded shells assume tenants pay 40% of fit-out/refurbishment for a 20% rent discount; their value column is avoided rent minus costs, not cash receipts or trading profit. The developer concession allocates 35% of collected rent and 20% of routine costs to the landlord; the developer funds fit-out/refurbishment. No concessions or subscriptions are signed.

All options lose two months of proportional rent every ten years after each cohort handover for structural inspection access. Routine costs continue. Maintenance work, compensation, tax, actual interruption dates and contract terms remain unpriced. Monthly ledgers and limitations keep each party and combined value separate. The existing property/finance cases remain unchanged. Actual market enquiries and a small accepted pilot must determine selection.

Baghdad controlled six-car planning product

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-baseline/SIX-CAR-BASELINE.md

Configuration M6-A-DRAFT, family metro-6car, covers 809 consists / 4854 cars at 111 m, 204 t controlled tare, 720 passengers / 960 crush, 1,350 kWh gross / 1,080 usable battery, 675 V nominal and 50 C HVAC ambient. Family BOM and interface/qualification register are separate from LM3's detailed 120-product manufacturing reference. Every quote, supplier part, mass evidence, production BOM and acceptance remains null/unaccepted. Common architectural ideas do not grant six-car applicability.

Six-car cost/mass category	Count / unit	Mass kg/consist	Allocated USD/consist
M6-carbody-structure	6 car	78,000	420,000
M6-bogie-running-gear	12 bogie	45,000	300,000
M6-traction-controls	12 controller	13,200	180,000
M6-battery-modules	6 225 kWh gross module	8,700	270,000
M6-interior-doors-glazing	6 car kit	42,000	240,000
M6-thermal-roof-services	6 car kit	12,000	90,000
M6-trainline-couplers	1 six-car system	1,500	70,000
M6-assembly-qa-logistics	1 consist	0	110,000

Cost allocations sum to the existing USD 1,680,000/train. These are a control allocation of the price ceiling, not supplier estimates. The category mass subtotal is 200,400 kg and unallocated controlled-tare reserve 3,600 kg; no weighed closure is claimed. There are 24 reference axles: average tare load 8.50 t and crush load 11.50 t using an explicit 75 kg/passenger assumption. Individual axle/bogie imbalance, payload distribution and structural limits require calculations and weighing. The system schedule has 24 door cassettes and 36 window cassettes. Five inter-car interfaces, trainline/brake propagation, accessible platform/door interfaces, redundant controls, HV/charging, thermal/fire/EMC/harness interfaces each carry an acceptance requirement.

Labour routes use the actual Baghdad factory stages, reference crew and cycle days. None is a measured six-car cycle. Rework, supplier queues and the current test-path throughput margin need trials; scaling an LM3 cycle is not evidence. Factory production payroll stays within train procurement, separate from railway operating payroll and factory capital. Work Order / Job Card references remain null until production BOMs, method revisions, calibrated tools, qualified operators and independent inspections exist.

The first-article qualification work breakdown has a USD 12.0m unquoted reference. Its incremental funding is null because overlap with the factory's USD 20.0m design/training/qualification allowance remains unresolved. No automatic extra qualification or train premium is added. Supplier quotes, a detailed child-part BOM, structural/braking/trainline/thermal testing, acceptance labour and series-release authority must replace the planning controls before manufacture.

Baghdad workload, recruitment and competence pilot

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-baseline/WORKFORCE-COMPETENCE.md

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The prior 3,422 FTE / USD 38.681m/year allowance is USD 941.97/person/month. The existing 21-role weighted split remains a budget allocation, not shift cover. This separate establishment derives posts/task hours, 1520 productive hours per FTE after leave, training, sickness and travel/handovers, integer cover and editable grade pay with employer/overtime allowances.

Function	Annual workload h	Concurrent posts	Reference FTE	Loaded annual IQD m
occ-lead	7,482	1	5	157.9
dispatcher	104,755	14	70	1,474.2
remote-assist	680,908	91	455	9,582.3
station-lead	119,720	16	82	2,590.4
platform-assistance	1,189,718	159	786	9,931.9
customer-service	299,300	40	203	2,565.1
station-cleaning	174,105	0	122	1,541.6
workshop-lead	58,240	14	42	1,326.8
fleet-mechanical	388,320	0	262	5,517.7
fleet-electrical	258,880	0	176	3,706.6
fleet-finish-cleaning	590,570	0	396	5,003.9
infrastructure-lead	29,120	14	28	884.5
civil-track	164,267	0	116	2,443.0
solar-storage	83,400	0	62	1,305.7
wayside-comms	63,600	0	49	1,031.9
city-director	1,520	1	1	31.6
chief-engineer	1,520	1	1	31.6
quality-safety	36,713	0	25	789.8
training	38,720	0	26	547.6
procurement-stores	29,040	0	20	421.2
finance-people	19,360	0	13	273.8

Reference total is 2,940 operating FTE, loaded payroll USD 39.353m/year equivalent. Quantities are explicit task-hour/post assumptions, not measured local work or accepted Iraqi pay/rest terms. Higher overtime, emergency cover, queues, simultaneous faults and actual maintenance could change them. The dedicated solar plant specialist workload is unpriced. Maintenance contracts and payroll must be reconciled to avoid adding contractor labour twice to existing envelopes. No revised OPEX is silently adopted.

Factory 1,044 direct staffed-shift positions are separate; their production payroll is already in train procurement. Construction crew counts remain unknown and separate, rather than scheduled task slots called people. Temporary commissioning has its own cash requirement.

Recruitment cohorts work backwards from each conditional line opening through requisition, selection, joining, training, first/repeated assessment and supervised experience. Offers allow 15% pre-join attrition and 80% first-pass / 75% repeat-pass assumptions. Hiring alone provides zero authorised productive capacity. Paid pre-opening training plus trainer/assessor reference cash is IQD 35.466bn; additional three-month commissioning payroll is IQD 1.327bn. These are distinct pre-opening uses, with overlap against EPC still unverified; rigs, agency fees and local terms remain unquoted. No trained/appointed person is invented.

Seven controlled curriculum drafts cover induction, OCC, stations/accessibility, fleet, civil/energy, factory and supervisors/assessors. Each lists prerequisites, method revision, equipment, practical exercises, critical assessment criteria, assessor requirements and retraining triggers. Arabic glossary is a draft requiring native technical review; full Arabic lessons are pending. Attendance, practical competence and scoped task authorisation are separate records.

The seven-day first-corridor OCC roster contains actual service-hour coverage requirements and zero named workers. Every slot is blocked pending appointments and task-specific authorisation. The executable eligibility evaluator checks worker record, assessed task/asset-family/location/method scope, validity, suspension, skill fade, availability, checked rest, access, permits, tools/materials, competent supervision and independent verification. Evaluate both at planning and task start; changed or expired evidence blocks eligibility. It creates no assignment or operational release.

Department owns outcomes/budget/risk; unit owns assets/backlog; crew owns executable access-window work; worker owns a scoped method/evidence; verifier owns independent acceptance/handback. Native ERP mappings retain Staffing

Plan/recruitment/Employee/onboarding/training/shift records, manufacturing Work Order/Job Card and maintenance Asset Maintenance/Asset Repair, with Project/Task for programme evidence. Dashboards track accepted output, blocks, competence expiry, shortages, overdue hazards and forecast cost; attendance or Task closure never counts as physical acceptance. Deploy permissions/recovery/rest rules against actual records before enabling assignment.

Frappe staffing documentation supports administrative records. ORR competence guidance is a useful reference for assessment/monitoring/reassessment, not Iraqi legal approval. References checked 4 October 2026.

Comparable standalone opening corridors

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-closure/CORRIDOR-COMPARISON.md

Each corridor pays for its own workload/shift cover including central leadership, existing maintenance/solar shares and per-site energy service assumptions. The price test uses the same current fare, 20% capacity-use sensitivity and existing station commercial revenue allocation. Debt, tax, land, parcel-specific works, interchange effects and grid upgrades remain outside this operating-only break-even. OD and access evidence remain null, so no commercial rank or first-corridor selection is fabricated.

Line	Fleet	FTE	OPEX USD m/year	Operating-neutral paid trips m/year	Neutral capacity use	Fare at 20% use IQD
line-1	44	250	22.943	19.498	30.9%	2036
line-2	49	238	24.551	21.640	34.3%	2260
line-3	47	245	23.427	20.161	32.0%	2105
line-4	40	202	19.949	17.468	27.7%	1824
line-5	44	237	22.393	19.141	30.3%	1999
line-6	49	249	24.791	21.507	34.1%	2246
line-7	38	210	19.141	16.486	26.1%	1722
line-8	45	231	22.764	19.691	31.2%	2056
line-9	45	333	40.140	34.805	57.0%	3752
line-10	9	69	3.990	3.383	5.4%	353
line-11	8	61	3.910	3.488	5.5%	364
line-12	9	69	3.909	3.303	5.2%	345
line-13	10	77	4.548	3.749	5.9%	391
line-14	8	61	3.612	3.195	5.1%	334

Detailed cases and investigation forms retain fares, affordability, OD/transfers, paid demand, pedestrian/feeder catchments, interchange and standalone debt-service fields. The affordability column prices 44 monthly trips against the historical model income proxy; it is not surveyed disposable income. Lower vehicle procurement reduces service/capacity; unchanged receipts and earlier openings are not assumed. Surveyed ground, utilities, foundations, drainage, erection/temporary works, evacuation/access and actual shared dependencies must accompany selection. A 10% civil opportunity is not claimed as a saving.

14 depot/storage design studies and Iraqi slab manufacture

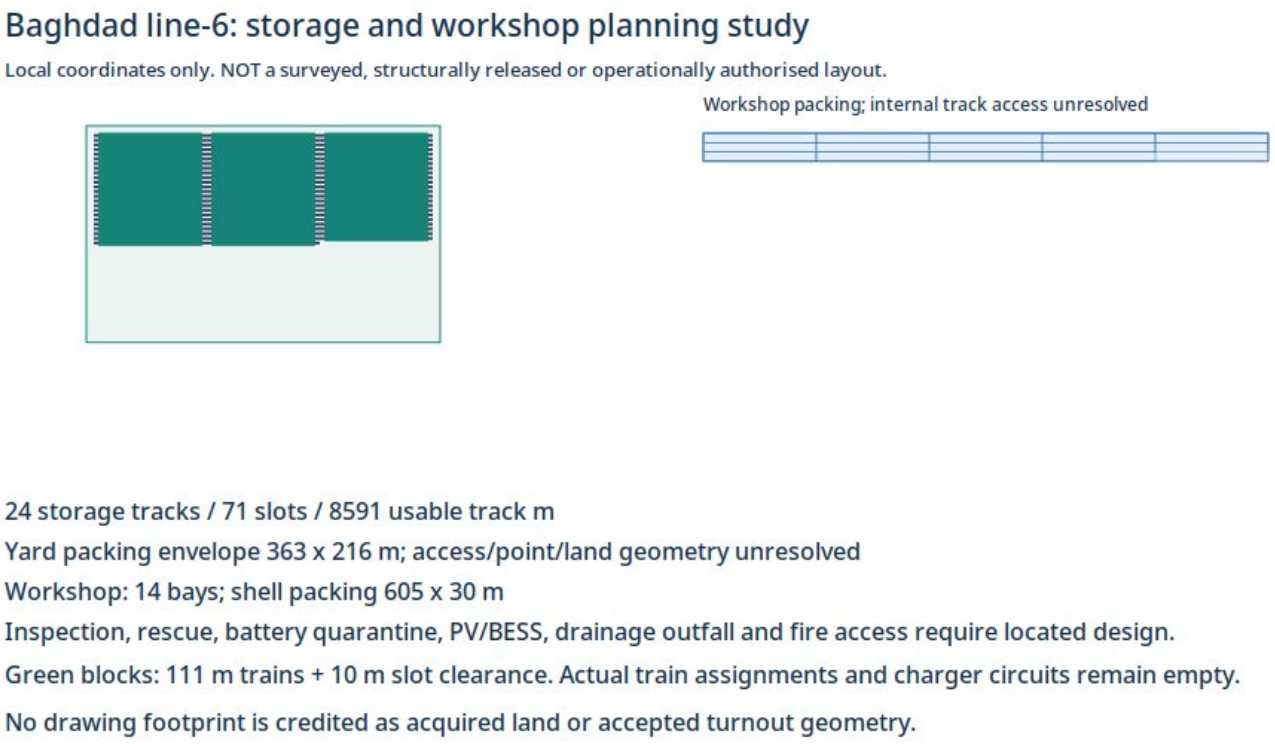
Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-closure/DEPOTS-SLAB-MANUFACTURE.md

Located track and slot register identifies 539 storage positions, 184 tracks, 65,219 usable track metres, 130,438 rail metres and 217,406 reference fixation-seat kits. 55,214.4 m³ concrete is a geometric reference for yard slabs, excluding foundations, reinforcement, grout/stray-current systems and point panels. Existing track/package costs are not supplemented by these quantities a second time.

Line	Slots	Storage tracks	Workshop bays	Layout
line-1	52	18	12	SVG
line-10	9	3	3	SVG
line-11	10	4	2	SVG
line-12	9	3	3	SVG
line-13	10	4	3	SVG
line-14	9	3	2	SVG
line-2	69	23	14	SVG
line-3	62	21	13	SVG
line-4	56	19	11	SVG
line-5	60	20	12	SVG
line-6	71	24	14	SVG
line-7	50	17	10	SVG
line-8	67	23	13	SVG
line-9	5	2	7	SVG

Drawings use local planning coordinates and distinguish storage slots from workshop bays. Rectangular packing can include unused end positions; it creates no acquired parcel or released alignment. Throat radii, bypass/escape tracks, drainage/outfalls, road/fire/evacuation access, utilities, lifting/inspection/rescue/quarantine areas and grid connection need actual surveys and supplier footprints. Line 6 requires about 33,333m² PV module area; no 4,000m² canopy is credited as adequate.

The yard-only slab production study requires 12 parallel mould cells at a one-working-day cycle/75% utilisation, ready month 18 and delivering two months before each opening. Its prefix-capacity programme includes cut/end panels and never casts beyond usable lengths. Moulds, curing/QA, batching, reinforcement, transport, lifting and installation require quantities/quotes and measured cycles. Factory CAPEX is null, not free. Main-line slab manufacture is a different existing production requirement; these cells cannot be represented as its total capacity. Iraqi casting/assembly remains a proposed procurement route, not a qualified plant.



Line 6 storage and workshop packing study

Startup alternatives compare the existing revenue fleet with the frozen original dispatch-point set, including 5 missing line-9 directions. Holding that set fixed requires 5 extra revenue trains and 1 extra spares: USD 10.08m vehicle reference, before consequential costs. Adding trains must not silently expand dispatch-point selection. Reserving fewer start points or buying trains requires a separate selected-start, cycle, turnback, depot/charging and funding replay; neither changes the baseline nor closes the existing failed gate.

Development leadership and training mobilisation

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-closure/DEVELOPMENT-TRAINING.md

Interim function	Start month	Operating handover month	Loaded IQD/month
city-director	0	40	2,632,500
chief-engineer	0	40	2,632,500
development-training-head	18	40	1,755,000

Interim development authorities begin at NTP. The training head mobilises at month 18; rigs must be ready by month 24, before operating cohorts join. Cohort lesson schedules and all operating roles distinguish teaching, initial/repeat assessment, supervised experience and handover. Named incumbents/successors, qualified assessors, equipment and signed handovers remain absent.

Monthly capacity and cash buys the existing contracted teaching/assessment allowance once. New interim leadership and external supervision add IQD 2.483bn before inflation, explicitly included in the integrated financing cases. Operating trainees supply zero qualified teaching/assessor/mentor capacity. These hour-derived minimum posts require actual competence, rig availability, observation ratios and daily/rest schedules; they do not establish an operating roster. The OCC pilot remains an anonymous rest-constrained example; all 21 roles have coverage and lesson requirements, with actual full rosters still unaccepted.

The native Employee attachment preview validates a controlled snapshot only. It returns `snapshot_eligible` for those assertions, keeps `eligible` and `live_start_check` false, and does not resolve current authoritative revocation. Real task start requires current permits, calibration, stock release, competence, availability and human work-release authority.

Field investigations, quotations and actual appointments

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-closure/FIELD-WORK-PACKETS.md

Scope inclusion forms, supplier requests, corridor investigations and six work packages turn unresolved items into defined data collection and independent exit criteria. No external contact has been made. Existing USD 360,000 estimator-work budget is an unquoted closure allowance with EPC overlap unknown, not committed funding or priced boreholes/equipment tests.

Repository artifacts are prepared; all six native evidence Tasks remain open, preserving actual owner/project/dates/status. Actual survey, qualified suppliers, OD/tenant enquiries, Iraqi appointments/approvals, signed finance and subscriptions remain pending. Honest nulls must be replaced by independently accepted evidence before those workstreams can close. GAO estimating guidance supports technical scope, WBS, data, alternatives/risk and actual-cost updates; it does not validate these rates or release the scheme.

Delivery-cost financial reconciliation

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-closure/FINANCE-RECONCILIATION.md

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These executable sensitivities preserve the original reference, then replace the original depot allowance once and replace labour/solar OPEX once. Incremental EPC is an explicit 7% allowance; qualification credit against the existing factory allowance is explicit. Paid training/assessment/supervised experience and additional commissioning specialists are separate pre-opening cash; no factory production wages are duplicated. Battery top-ups add only the shortfall above reserve contributions already in maintenance. Land, duties, utility diversions, spare inclusion, actual owner insurance and grid/charger upgrade capital remain unpriced.

Case	Capital USD bn	Government USD bn equivalent	Peak gap debt IQD tn	Unfunded support IQD tn	Debt clear without unfunded cash month	Company before-finance cash NPV USD bn
reference	7.951	1.988	2.158	0.000	203	-1.670
reconciled_full_fleet	8.135	2.034	3.750	0.000	259	-2.874
reconciled_fixed_original_government	8.135	1.988	3.813	0.000	260	-2.874
reconciled_without_uncommitted_income	8.135	2.034	5.091	0.000	278	-2.874
opening_fleet_supply_scaled	7.523	1.881	8.799	0.000	None	-4.051
contracted_solar	7.319	1.830	4.367	0.000	283	-2.943
installed_energy_supply_bound	8.135	2.034	6.329	0.000	336	-3.731
simple_span_bearing_index	8.264	2.066	3.894	0.000	262	-2.960

The fixed-government case retains the original absolute government contribution and reallocates its USD downpayment within that ceiling. Other cases use 25% of their own capital; increased appropriation is not committed. In every case imports are funded 50% government USD/50% proposed Chinese USD credit; ordinary/green bonds, bank and gap credit remain IQD. All debt/reserve/buffer/principal/cash residuals reconcile. Six-month bond units are placement requirements, not subscriptions. Climate/rights/additional local income remain uncommitted in conditional cases; the dedicated case removes all these targets and green pricing benefits.

Opening-fleet procurement follows the selected supply case and its extra spare. Factory cost/capacity is retained, and opening dates follow the later of civil readiness and replayed fleet acceptance; paid receipts scale with reduced timetable supply and ramps. Future expansion dates and funding remain null. Contracted solar removes city-owned plant CAPEX but keeps provider plant resources visible and charges generation; a private provider, rights, prices and firm deliverability are not established. Company cash NPVs are not like-for-like complete resource NPVs.

Summary and source-bound cases include full monthly native-currency ledgers, six-month bond/loan sale and repayment schedules, fees, grace, early principal, buffers and terminal debt/cash. Only genuine surplus after all obligations and buffers can fund contractual voluntary repayment; uncovered support cannot fund it. No new public subsidy or tariff adoption is claimed. Baseline fares and OPEX retain 5% growth and the existing elasticity/income assumptions.

Opening factory replay manufactures exactly 445 planned trains, including the extra line-9 spare, with rebuilt finite factory queues. Dropped orders are removed from manufacturing and invoice cash; original expansion-capable factory CAPEX is retained. Openings are the later of civil readiness and replayed fleet acceptance; later dates flow through receipts, payroll, reserves and the operating horizon. Depots retain full eventual-network capacity; staffing, maintenance, reserve and site energy follow the lower supply. All sensitivities retain zero modelled dividends; retained cash is not distributable profit without taxes, covenants and approvals.

The `simple_span_bearing_index` sensitivity removes the bearing reduction only from existing standard Pi25 length, then adds the declared incremental EPC once. It inherits original civil invoice dates and procurement-origin proportions as assumptions, keeps government at 25%, splits assumed imports 50% USD government cash / 50% Chinese USD credit, and retains all other funding in IQD. Monthly/six-month ledgers, interest, fees, reserves and early repayment are recalculated. This is an unquoted index counterfactual, not a bearing supplier offer; finite end effects, connection costs, actual import eligibility and consequential foundations remain unpriced. OSR-US and special segments do not inherit Pi25 bearing quantities.

Fare/demand/affordability sensitivities

Base fare IQD	Paid-demand multiplier	44-trip income share	Debt-clear month	Terminal gap IQD tn	Before-finance cash NPV USD bn
1500	0.962	13.4%	233	0.000	-2.403
1750	0.918	15.6%	207	0.000	-1.785
2000	0.882	17.8%	188	0.000	-1.192
2500	0.825	22.3%	164	0.000	-0.068
3000	0.781	26.7%	148	0.000	0.992

Full results change the base fare once, retain existing peak/off-peak pricing and 5% annual fare/OPEX/income growth, reduce paid trips with the existing uncalibrated elasticity, and scale existing station commercial receipts with demand. Debt clearance and cash NPV are different tests. Conditional green/grant/rights/local-income targets and cheap IQD gap credit remain uncommitted. An assumed higher fare is not a sustainable tariff until OD, distributional affordability and actual service/credit conditions are accepted. The 44-trip share uses the historical income proxy and base fare before variable-price weighting, not measured disposable income.

Task-derived maintenance and reserve cash

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-closure/MAINTENANCE-RENEWALS.md

Task workload covers daily/weekly/monthly inspections, every-roundtrip safety/diagnostic checks, wheel reprofile at 150,000km, bogie overhaul at 600,000km and body overhaul at ten years. Six-car quantities are 24 wheelsets and 12 bogies/train. Reference workload is 418,696 mechanical and 217,395 electrical person-hours/year, compared with existing workload allowances 388,320 / 258,880. Durations, defects, access queues, equipment, parts and non-fleet tasks remain unmeasured; lower reference hours do not authorise staff reductions.

Battery replacement stays on the existing 12-year cycle. Reference annual reserve is USD 18.203m within the existing USD 54.365m rolling maintenance envelope. Monthly reserve ledger separates indexed contributions, replacement withdrawals, required top-ups and closing restricted cash. At 5% inflation/zero reserve interest, reference top-ups total USD 327.476m nominal; terminal restricted cash is USD 670.268m. It cannot repay loans/bonds or become terminal sale income. The replacement bill is not added to annual reserve spending a second time.

Initial 2% battery-spare stock is USD 4.369m reference; existing allowance inclusion is unknown. Serial-specific service/mileage/cell-health dates must replace line-opening proxies. Native Asset Maintenance/Asset Repair, calibration/parts and independent handback records remain required. No actual asset, spare, Work Order or accepted Job Card is invented.

All 26 controlled interval records also retain station, civil/track/structures, energy, wayside, tooling, finish/joint/wash and soiling work. Condition-based tasks have no invented frequency or labour cost. Opening-fleet reserve cash uses 445 trainsets and adds its own indexed shortfalls to the reduced-supply financing case.

Baghdad remaining-work delivery packages

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-closure/README.md

> Original catalogue or earlier scope reference. The figures and policies below retain their original assumptions; they are not the latest Baghdad staffing, depot, procurement or funding basis. See the current Baghdad recalculation and current city summary. National/portfolio totals have not been repriced with that conditional Baghdad option.

This continuation completes repository calculations, concept layouts, RFQ/field forms, financial sensitivities and competence/recovery checks. It does not complete surveys, obtain quotes, appoint people, sign leases/loans or release a railway. The original financing and engineering examples remain their controlled baseline. All alternatives are unquoted/uncommitted and retain unknown costs.

- Depot track/slot layout and yard slab production
- Six-car child parts, interfaces and supplier request forms
- Per-site hourly energy and upgrade requirements
- Maintenance tasks, renewal dates and restricted reserve cash
- Comparable standalone corridor break-even and field investigations
- Reconciled monthly funding, six-month bonds/loans and repayment
- Practical training, department/unit allocation and rest-limited roster
- Remaining evidence and six native ERP work packages

Run `.venv/bin/python tools/automation/baghdad_delivery_closure.py`, then rebuild the proposal. `--check` rejects source/output drift. Existing `bootstrap_baghdad_tests.py` restores the exact test input. Native personal records and recovery receipts remain private in `var/erpnext/`.

Chronological site-limited energy and installed upgrades

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-closure/SITE-ENERGY.md

Each station retains its own storage capacity/power and grid limit. Timetable train-km allocate duty between lines; installed charger kW allocate duty within a line. This is a synthetic allocation, not measured charger events or a queue/feeder acceptance test. The utility plant is allocated by duty, on-site PV has no wheeling loss, and utility PV has the explicit loss/charge. Each node conserves hourly energy and preserves losses, curtailment, shortages and SoC.

Weather	Imports GWh	Unserved GWh	Firm energy services USD m/year	Required grid upgrade MW	Charger overload MW
reference	940.2	10.5	140.310	141.504	203.587
poor	1055.9	23.9	148.841	145.359	203.587
aged10	955.2	11.0	141.856	141.504	203.587

The firm-service cash sensitivity buys unmet energy and pays the increased connection charges. It requires physical connection and charger upgrades whose capital is unpriced. It does not claim a trip can be served by paying a bill. Site totals and reference line-hourly ledgers expose every node requirement. Whole-grid pooling is removed; within-line duty allocation, site transformer/feeder limits, rights/outages, charging times, installed storage ageing and weather still need measurements. Original annual-netting finance remains unchanged, with no export income or energy ownership proceeds invented.

All 28 aggregate shortage hours remain an explicit pooled comparator. All site-specific shortage hours, losslessly compressed expose grid/charger limits, opening/closing stored energy, local PV and remote generation before wheeling. Local PV never incurs remote wheeling losses/charges. The installed-throughput case reduces fare, station and additional commercial receipts; it is an energy upper bound, not a validated achieved timetable.

Six-car child parts and procurement/qualification work

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-closure/SIX-CAR-PROCUREMENT.md

Child parts expand the eight parent allocations without duplicating their USD 1,680,000/train budget. Body modules, 12 bogies, 24 wheelsets/48 wheels, battery interfaces, traction controllers, doors/windows, mounting rails, lighting and thermal/trainline interfaces derive from the actual six-car system counts. Harnesses, protection, braking, redundant control and installed sensing quantities remain null where the supplier/design schedule is absent. Price, mass, drawing and applicability are not inferred from the LM3 reference.

Supplier RFQ forms require drawing/revision, interface, price/currency/date, country/invoice origin, delivery, duties, warranty, payment terms, local labour and actual Chinese-credit eligibility. None has been sent or answered. Qualification work packages retain the existing USD 12m unquoted reference and unresolved overlap with the factory's USD 20m design/training allowance. Routine assembly/QA wages already in train procurement are not added to OPEX. Software/runtime/bench evidence remains observation/reference evidence until the target family and installed interfaces are qualified.

Workforce, practical training and a rest-limited pilot

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-closure/WORKFORCE-PILOT.md

The seven lesson cards provide practical scenarios, prerequisites/apparatus, critical observations, assessor authority, retraining and Arabic draft instructions. Attendance, practical assessment and task/asset/location authorisation stay separate. All actual candidate/assessor outcomes and native Arabic review remain null. The draft must not become an operational rulebook through generation.

Pilot schedule covers 143.5 OCC hours over seven days with 5 unnamed planning positions, maximum 8-hour shifts, minimum 11-hour rest and maximum 40 hours per rolling seven days. These are editable scheduling assumptions, not accepted Iraqi employment terms. No worker identity is invented; every slot remains ineligible until real Employee/assessment/authorisation/resources and a fresh start check exist. Leave/sickness/annual relief, supervised staffing, tools/permits and independent checks remain necessary.

Department/unit/role budgets, recruitment cohorts, task workload and native HR/Work Order/Job Card/Asset Maintenance/Asset Repair records remain linked to the existing workload package. Read-only administrative previews require native role/document permissions and a baseline-bound private employee packet; human work-release authority remains separate. Frappe staffing documentation describes the native administration; it does not appoint Baghdad staff.

Baghdad frozen-resource delivery and funding study

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/delivery-risk/README.md

These are deterministic disturbances, not P80/P90 dates or measured failure rates. All planned six-car trains must pass acceptance before their line opens. Selected stage cells, civil crew lanes and dispatch order remain fixed in every stress. No stress calls the capacity-sizing search.

The explicit segregated-path calendar books 16 exclusive running hours per train at the case availability, inside acceptance-bay occupation, with setup and clearance. This adds a conservative calendar audit to the published average-throughput check. Baseline lane order is preserved; a replanned optimal dispatch might recover some delay. Financial-close months include 30 pre-NTP working days and three commissioning months.

Case	First/full month	Peak IQD gap debt, tn	Interest/fees USD eq, bn	Debt cleared month
calendar_baseline	40/81	2.158	3.504	203
factory_delay_6months	46/86	2.504	3.572	208
supplier_shortage_6months	46/86	2.459	3.565	207
half_staff_late_12months	44/85	2.401	3.556	206
rework_10pct	41/84	2.314	3.544	205
one_test_path_outage_6months	40/82	2.280	3.532	205
civil_access_6months	42/87	2.109	3.485	204
availability_75pct	42/89	2.488	3.584	207
availability_65pct	44/98	2.858	3.665	212
combined	55/105	3.281	3.750	218
combined_second_test_shift	52/101	3.074	3.694	216
availability_75pct_second_test_shift	42/89	2.495	3.587	208
civil_cycles_20pct_faster	40/81	2.164	3.505	203
civil_earliest_unchanged_cycles	40/81	2.669	3.669	205
civil_earliest_20pct_faster	40/80	3.057	3.798	207
availability_75pct_costed	42/89	2.510	3.589	208
availability_75pct_test_shift_costed	42/89	2.517	3.592	208
hiring_ramp_costed	44/85	2.413	3.558	207
supplier_shortage_costed	46/86	2.473	3.568	207
availability_75pct_structural_shift	41/88	2.503	3.586	207
availability_75pct_electrical_shift	41/88	2.469	3.576	207
availability_75pct_composite_shift	42/87	2.448	3.576	207
availability_75pct_all_stage_shifts	40/81	2.226	3.532	204
recruitment_training_recovery	40/81	2.163	3.508	203
supplier_expedite_recovery	43/83	2.310	3.496	205
temporary_first_article	36/81	1.951	3.495	201
combined_delay_costs	55/105	3.370	3.769	219
combined_lower_demand	55/105	5.088	5.190	304
combined_escalation	55/105	4.491	5.123	246
combined_finance_downside	55/105	4.000	8.942	Unfunded
joint_downside	55/105	4.000	12.896	Unfunded
domestic_placement_interrupted_recovered	46/87	2.259	3.498	207
export_credit_delayed_recovered	46/87	2.110	3.443	207

Separate productivity from investment timing

The corrected civil_cycles_20pct_faster case retains every rephased start floor and changes only civil occupation durations. A 1.0 multiplier is tested as an exact schedule identity, including running-path reservations and line openings. Civil_earliest_unchanged_cycles removes the spending delays at original durations; civil_earliest_20pct_faster changes both. Their costs must not be attributed to productivity alone. Earlier completion may still advance completion/retention invoices even when mobilisation timing is retained. These are diagnostic cycle assumptions with no added crews or accepted acceleration price.

Recovery comparisons and priced assumptions

Factory cells and dispatch lane order remain fixed. Fixed curing, bonding, inspection and test holds are listed in baghdad-delivery-risk.toml as unqualified working-calendar equivalents. Shift compression applies only to the remaining staffed occupation; the additional 60-day first-article qualification is unchanged. At one shift the original schedule is preserved exactly. Cure elapsed hours and batch/test evidence must replace these assumed splits before adoption. The test-only option uses the same selected segregated paths with a second eight-hour test shift: 26 incremental staff and USD 0.75m direct lighting/training per selected path plus 7% EPC, and indexed payroll before and after fares. It does not repair upstream stage throughput. The single structural, electrical and composite options add four staffed hours/day to the named stage at unchanged bay count; the coordinated option applies this to all seven stages and funds the second test shift. Incremental production FTE is half each selected stage crew, paid at a 25% premium, with nonlabour shift costs equal to 25% of added payroll. Each stage adds a USD 1m installation/training allowance and USD 10,000 per added FTE plus EPC. These assumed shift efficiencies, relief and wage premiums need qualification.

Use availability_75pct_costed, hiring_ramp_costed and supplier_shortage_costed as matching delay-cost baselines for their respective recovery options. Recruitment recovery funds USD 10,000 per delayed half of the 1044 production positions plus EPC and tests a six-month rather than twelve-month staffing ramp. Supplier recovery charges an assumed IQD local logistics fee of 2% of city imported invoice value, indexed to payment, and tests a shortage cut from 130 to 65 working days; this is a causal scenario assumption, not a guaranteed delivery improvement. Foreign-currency freight reimbursement and invoice eligibility need quotations. Neither purchases another train nor credits an unspecified subsidy.

The temporary first-article option adds USD 35m direct facility/tooling and 7% EPC, ready at day 260 (12 planning months from NTP), plus 60 incremental support FTE until the permanent plant is ready. Only the already-planned prototype assembles there. Acceptance still waits for permanent bays and running paths at day 390, and series still waits for first-article qualification. All line fleets remain complete at opening. The temporary site and staff must have their own RFQs and acceptance; no shorter passenger section is folded into this option.

Delay costs and combined financial downside

Original delivery-only cases remain lower-bound comparisons. Combined_delay_costs additionally prices extended direct production staffing plus 100 support FTE at the existing Iraqi labour proxy; it compares staffed span lengths so a pure start delay does not buy the same crew-months twice. Plant storage/insurance/utilities carrying uses 1% of direct plant capital per extra/idle year excluding wages. Civil prolongation adds 150 supervision FTE and nonlabour site overhead at 0.5% of city capital per year beyond baseline civil completion. All these incremental operating costs are charged monthly at 5% annual indexing; baseline train labour/materials already inside procurement are not repeated. Rework prices 83 affected trains at USD 25,000 each. Monthly incremental-cost files reconcile each allowance to the case OPEX ledger. Contingency, tax and contractor claims remain unquoted; these are not a funded risk reserve.

The downside ladder isolates 30% fewer paid trips plus 25% lower existing retail/advertising receipts, 5% annual capital escalation applied to each invoice at actual payment month from financial close, and weaker financing. Weaker financing removes assumed green enhancement, climate grants, development rights and additional net income, increases core coupons two percentage points, and replaces the 2%/IQD 13tn gap sensitivity with 8% credit at a 1% draw fee and IQD 4tn maximum outstanding. Chinese credit and government import cash remain USD, all domestic credit/bonds/cash IQD; government remains 25% of escalated capital. Separately, domestic_placement_interrupted_recovered and export_credit_delayed_recovered stop procurement/construction/production for 130 working days and price local remobilisation plus prolongation; repayment is conditional on subsequently placing the refused debt. Permanent refusal has no opening or repayment date: see the funding gates and evidence execution package, including denied amounts, escrow requirements and six-month placement shortfalls. No gap facility is used to conceal a refused core source.

Lower demand with combined delays leaves IQD 0.000tn terminal gap debt. Joint_downside applies all of those assumptions together and 7% rail OPEX inflation: IQD 12.690tn cumulative uncovered cash and IQD 0.000tn terminal gap debt. Uncovered support is a balancing requirement, not an extra government appropriation, loan or cash source. Its presence blocks any unconditional repayment claim even if the simulated debt eventually amortizes. Interest totals in such cases also assume the missing cash is supplied; they do not establish an executable financed programme.

Base fare and OPEX sensitivities remain 5% from financial close, with existing kiosks/advertising, separate receipts and each line revenue ramp included. No tickets are sold before opening. Capital escalation is absent from reference cases and explicit in the named escalation cases. Green/grant/rights and all financing availability remain uncommitted. No future national city cashflow supports Baghdad debt.

Evidence needed before adopting a recovery plan

The qualification register assigns owners, required measurements/RFQs and acceptance rules for production, suppliers, recruitment, civil quantities, depot/site fit, demand, funding and operation. Every row remains not demonstrated. No measurements or quotations were obtained by running this model. The baseline fleet and test-path margins follow the current factory sizing; they require measured cycles and acceptance trials. Shorter passenger sections require a separate route, turnback, charging, depot, fleet duty and safety/operating acceptance study; the present comparisons retain full fleets and existing operating scope.

Regenerate after the city controls, factory plan and Baghdad funding programme are current with `.venv/bin/python tools/automation/baghdad_delivery_stress.py`, then regenerate the proposal. Verify source/output bindings with `--check`.

Scenario comparison · Machine-readable assumptions/results · Baseline half-year funding · Combined stress half-year funding · Joint downside half-year funding

Baghdad access, demand and paid journeys

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/demand-bridge/README.md

Status: survey handoff; no calibrated demand forecast or revenue uplift.

Native population pixels verified station entrances/walking times jobs/destinations and surveyed OD journeys route choice section/time train loads unique paid journeys collected fare receipts.

The retained access report counts radial unions in 2020 pixels. Rivers, motorways, walls, crossing availability and station vertical access need an entrance-based walking network. The retained OSM export contains selected arterial geometry and building/water polygons; it is not a complete pedestrian graph or entrance survey. Do not route through private premises, infer a crossing from coincident lines or count a 2 km feeder catchment as convenient walking coverage. Jobs, entrances and OD observations remain missing.

The current financial paid-trip assumption remains capacity-led. Shorter lines and different catchments do not automatically preserve or increase patronage. `baghdad_demand_bridge.py --od <survey.json>` validates explicit line paths against declared transfers and reports unique paid journeys, train boardings, line boardings and fare receipts separately. An integrated fare charges each journey once; only an explicitly declared boarding tariff charges its legs. The source and survey date are required. Aggregated annual line boardings are not peak section loads, unique people or a validated route-choice forecast.

Before adopting receipts, independently qualify the OD evidence and sampling/expansion weights, income/concession response, entrances and accessible walking paths, waiting/transfer time, route choice, peak section capacity, fare caps and collection losses. Re-size service/fleet/energy/OPEX together and regenerate the financial ledger. No survey, jobs, path, peak load or additional income is fabricated to close the gap.

Source-bound handoff and accounting.

Baghdad deployment gaps

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/deployment/README.md

3 closed checks; 14 open gates. Status follows the linked evidence; generating a receipt template does not count as receiving field data.

Soil inputs include a route/station investigation plan.

Gate	Status	Responsible function	Closure work
soil-desktop-inputs	closed	civil/geotechnical designer	Sample current station and civil geometry from the pinned soilDB source and retain means, uncertainty and nodata.
soil-coverage	open	geotechnical investigator	Fill mapped coverage gaps through local records and targeted sampling; keep estimated and measured records distinct.
control-processing	open	survey team	Receive survey control observations; process and check the project CRS, datum and residuals.
ground-model	open	survey team	Receive and inspect the terrain, point cloud and ground-model deliveries against independent control.
surveyed-alignment	open	alignment designer	Replace or confirm generated geometry with checked survey alignment and platform reconciliation.
route-station-fit	open	civil/station designer	Resolve utilities, land/access, flood levels, station access and construction staging against the current route.
drainage-ground	open	civil/geotechnical designer	Use the soil investigation plan to collect local geotechnics and groundwater, size foundations/treatment, and check drainage with local rainfall and surveyed levels.
structural-release	open	structural designer and checker	Complete route-specific support/span, foundation, movement and erection calculations and close checking comments.
model-timing-comparison	closed	simulation engineer	Reconcile current native reference journey times with SUMO using the same scenario dwells and route.
full-service-validation	closed	simulation engineer	Run current nominal and degraded full-service cases against the current scenario and simulator; retain failures and exact provenance.
operating-release	open	operator	Complete conflict-aware capacity and degraded-operation review and sign the bound operating evidence.
morning-fleet-allocation	open	service planner	Reconcile revenue fleet with required morning departures on each line; retain two station berths and same-line overflow storage.
continuous-stabling-replay (evidence missing)	open	simulation engineer	Run the selected line-local station/depot candidate across consecutive evenings and synchronised morning starts.
stabling-physical-fit	open	track/station designer	Locate usable station and line-local depot tracks, shared charging, isolation, inspection access and protected morning release slots.
solar-storage-endurance	open	energy designer	Bind declared station/ROW/dedicated PV to site storage and actual charging duty, reconcile conversion losses and prove replenishment across adverse weather; specify residual backup duty explicitly.
depot-placement-and-budget	open	depot and cost designer	Place the declared PV/storage inventory within the controlled site layout and reconcile itemised installed costs and renewal scope with existing allowances.
workforce-duty-and-workload	open	city operating lead	Reconcile the funded role plan with named shift/leave cover, measured inspection and cleaning workloads, simultaneous service-depot arrivals and actual local employment inputs.

city civil and operating deployment evidence; manufacturing and system certification remain in their own release registers.

baghdad depot scope reconciliation

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/depot-scope/README.md

> Original catalogue or earlier scope reference. The figures and policies below retain their original assumptions; they are not the latest Baghdad staffing, depot, procurement or funding basis. See the current Baghdad recalculation and current city summary. National/portfolio totals have not been repriced with that conditional Baghdad option.

Depot energy quantities reconciled: yes. Physical/cost/stabling closure: open.

The policy assigns two revenue trains per selected powered station for coordinated morning starts and the remaining fleet to storage on its own line. Depot storage tracks are sized separately from maintenance bays; see the station/depot allocation. The dispatch table below diagnoses the current simulator initialization; it is not a proposed overnight parking allocation or a requirement for more depots.

Depot station	PV kWp	Storage modules / kWh	Required / reference PV area m²	Additional equipment reference USD
line-2-0595-0296-s052851	5,000	80 / 40,000	33,333.3 / 4,000	5,930,000

The equipment reference is an unapproved sensitivity using existing repository rates. It is not added to CAPEX; allowance inclusion and installed scope remain unverified.

Initial dispatch station	Line	Trainsets	Train-body length m	Slot length with clearance m	Physical slots
line-1-2500-0775-s000000	line-1	43	4,773.0	5,203.0	unverified
line-1-0166-0866-s049067	line-1	43	4,773.0	5,203.0	unverified
line-2-1987-2192-s000000	line-2	48	5,328.0	5,808.0	unverified
line-2-0595-0296-s052851	line-2	47	5,217.0	5,687.0	unverified
line-3-2271-1120-s000000	line-3	45	4,995.0	5,445.0	unverified
line-3-0151-1637-s049496	line-3	45	4,995.0	5,445.0	unverified
line-4-0146-0199-s000000	line-4	38	4,218.0	4,598.0	unverified
line-4-1230-1627-s041932	line-4	38	4,218.0	4,598.0	unverified
line-5-1199-0260-s000000	line-5	43	4,773.0	5,203.0	unverified
line-5-0677-2304-s047345	line-5	43	4,773.0	5,203.0	unverified
line-6-2322-0251-s000000	line-6	48	5,328.0	5,808.0	unverified
line-6-0495-1669-s052503	line-6	47	5,217.0	5,687.0	unverified
line-7-1069-2206-s000000	line-7	36	3,996.0	4,356.0	unverified
line-7-1531-0418-s039957	line-7	36	3,996.0	4,356.0	unverified
line-8-2231-1783-s000000	line-8	45	4,995.0	5,445.0	unverified
line-8-0785-0290-s048137	line-8	44	4,884.0	5,324.0	unverified
line-9-0606-0636-s003011	line-9	23	2,553.0	2,783.0	unverified
line-9-0689-0569-s097310	line-9	22	2,442.0	2,662.0	unverified
line-10-0562-1131-s000000	line-10	8	888.0	968.0	unverified
line-10-0686-1312-s006804	line-10	7	777.0	847.0	unverified
line-11-0605-1228-s000000	line-11	7	777.0	847.0	unverified
line-11-0337-1112-s006884	line-11	7	777.0	847.0	unverified
line-12-0701-0415-s000000	line-12	8	888.0	968.0	unverified
line-12-0760-0138-s006586	line-12	7	777.0	847.0	unverified
line-13-1336-0518-s000000	line-13	9	999.0	1,089.0	unverified
line-13-1193-0313-s008042	line-13	9	999.0	1,089.0	unverified
line-14-0804-1072-s000000	line-14	7	777.0	847.0	unverified
line-14-0688-1260-s006047	line-14	6	666.0	726.0	unverified

- Operating PV/storage capacities are retained assumptions, not validated depot sizing. Reassess depot and station energy duties after distributed overnight placement and charging are represented.
- PV area is module area at the catalogue 0.15 kWp/m2, not gross land area; setbacks, access and packing require layout.
- Battery footprint and separation require supplier/fire-engineering data; no area or approval is invented.
- Dispatch queues are initial simulator placements, not an overnight operating plan. Slot lengths use one train plus 5 m at each end from RFC 0014.
- Workshop bays and passenger platforms are not credited as verified overnight stabling slots.

Required closure records: a supplier equipment/footprint schedule; located PV and battery layout; itemised budget with explicit baseline inclusions; a station-by-station healthy-fleet allocation with track IDs, usable lengths, assigned trainsets,

charger sharing, security/isolation and access; and coordinated morning departures with conflict-aware evening/morning repositioning evidence.

Baghdad detailed component register

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/detail/README.md

Generated from the current city design, six-car profile, engineering parts and software/ERP contracts. Engineering release remains false. Cost and finance rates remain unchanged; unknown unit costs are open rather than zero.

Reference allocation: 809 trainsets, 4,854 cars, 9,708 bogies, 159 stations, 139 energy sites and one shared plant.

See complete parts CSV, source-bound register, dimensioned ST6 reference and design and Iraqi manufacturing plan. The two-bogie-per-car allocation and repeated mechanical kit quantities are reference architecture, subject to Metro-6car family release. Nested wheelsets, clips and batteries are not extra charges on top of complete bought assemblies.

Track and parts quantities

The rail reference covers net mainline running length only. Actual procurement additionally needs alignment method zones, switches, stabling, depot track and cutting/weld/spares schedules. Slab panel totals remain unassigned until constrained zones are measured; rail-seat child quantities require a seat installation schedule.

ID	Part / assembly	Basis	Per basis	Network reference	Route
TR-01	60E1 reference running rail	route-km (m)	4000	2,053,342	import-or-qualified-local-source
TR-02	Resilient direct-fixation complete seat kit	rail-seat (each)	1	Open	import-first-local-assembly-later
TR-03	Elastic clips	rail-seat (each)	2	Open	import-qualified
TR-04	Rail pad / bonded resilient layer	rail-seat (each)	1	Open	import-qualified
TR-05	Lateral rail insulators	rail-seat (each)	2	Open	import-qualified
TR-06	Anchor assembly including insert, bolt and washer	rail-seat (set)	2	Open	local-inserts-after-qualification
TR-07	Vertical/lateral adjustment and levelling kit	rail-seat (set)	1	Open	local-machining-after-qualification
TR-08	OSR-ST6 precast slab panel	slab-panel (each)	1	Open	Iraqi-precast
TR-09	ST6 concrete, base and plinths only	slab-panel (m3)	5.0796	Open	Iraqi-batching
TR-10	ST6 reinforcement planning allowance	slab-panel (kg)	761.94	Open	Iraqi-cage-fabrication
TR-11	ST6 bedding / levelling grout	slab-panel (m3)	0.522	Open	local-qualified-material
TR-12	ST6 lifting sockets / engineered anchors	slab-panel (each)	4	Open	import-qualified-local-install
TR-13	Drainage channel and removable cover	route-km (m)	2000	1,026,671	Iraqi-precast
TR-14	Rail electrical continuity bond / monitored stray-current provision	project (set)	Open	Open	local-harness-import-terminals
TR-15	Turnout slab bearer, point drive, detection and locking kit	project (set)	Open	Open	mixed
TR-16	Rail/deck expansion interface and transition section	project (set)	Open	Open	mixed
TR-17	Utility ducts, draw pits, barriers and fencing	project (set)	Open	Open	Iraqi-manufacture
ME-01	Underframe and load-bearing car structure	car (each)	1	4,854	Iraqi-fabrication
ME-02	Modular FRP cladding, end frames and service rails	car (each)	1	4,854	Iraqi-composite-and-metalwork
ME-03	Powered bogie complete including frame, suspension and brakes	car (each)	1	4,854	import-first
ME-04	Trailer bogie complete including frame, suspension and brakes	car (each)	1	4,854	import-first
ME-05	Wheelset including wheels, axle, bearings and axleboxes	bogie (each)	2	19,416	included-in-bogie-import
ME-06	Motor and traction inverter/controller set	car (each)	2	9,708	import-qualified
ME-07	225 kWh nameplate traction battery allocation	car (each)	1	4,854	import-cells-local-pack-after-qualification
ME-08	Battery support, HV junction box, fuse, contactors and precharge	car (each)	1	4,854	mixed
ME-09	Battery/inverter liquid cooling loop	car (each)	1	4,854	local-assembly-import-pump-and-chiller

ID	Part / assembly	Basis	Per basis	Network reference	Route
ME-10	HV-to-LV DC/DC and battery-backed auxiliary distribution	car (each)	1	4,854	import-qualified-local-harness
ME-11	HVAC unit, ducts, filters, condensate and mounts	car (each)	1	4,854	mixed
ME-12	Passenger door package including leaves, drive, locks and detection	car (each)	4	19,416	import-qualified-local-fitting
ME-13	Side glazing, window frames, seals and adhesives	car (each)	Open	Open	import-glass-local-frame
ME-14	Intercar coupler, gangway, jumper and air/electrical interfaces	trainset (each)	5	4,045	mixed
ME-15	Train-end recovery coupler and sensor cowl	trainset (each)	2	1,618	mixed
ME-16	Brake supply/control and parking brake package	car (each)	1	4,854	mixed
ME-17	Roof PV, MPPT, mounts and protected DC wiring	car (each)	1	4,854	import-panels-local-mounts
ME-18	Interior, seats, grab rails, lighting and accessibility kit	car (each)	1	4,854	Iraqi-assembly
ME-19	Fire detection, alarm, segregation and suppression provisions	car (each)	1	4,854	mixed
ME-20	HV/LV/data harness, connector and clamp schedule	car (each)	1	4,854	Iraqi-harness-production
ME-21	Service consumables, fasteners, fluids and replacement spares	project (each)	Open	Open	mixed
EL-01	T-ECU/S end host assembly	trainset (each)	2	1,618	import-electronics-Iraqi-integration
EL-02	T-ECU/A application end host assembly	trainset (each)	2	1,618	import-electronics-Iraqi-integration
EL-03	T-OBS end host and external sensor assembly	trainset (each)	2	1,618	import-electronics-Iraqi-integration
EL-04	Safety evaluator MCU channel	t-obs (each)	2	3,236	import-electronics-Iraqi-integration
EL-05	Independent hardware heartbeat watchdog	t-obs (each)	2	3,236	import-electronics-Iraqi-integration
EL-06	Channel power conversion and voltage monitoring branch	t-obs (each)	2	3,236	import-electronics-Iraqi-integration
EL-07	Application compute module and carrier	t-obs (each)	1	1,618	import-electronics-Iraqi-integration
EL-08	Radar evaluation assembly and CAN interface adapter	t-obs (each)	1	1,618	import-electronics-Iraqi-integration
EL-09	Lidar HAP TX candidate and data/power harness	t-obs (each)	1	1,618	import-electronics-Iraqi-integration
EL-10	Stereo camera, carrier-compatible flex and rigid calibration bar	t-obs (each)	2	3,236	import-electronics-Iraqi-integration
EL-11	Ultrasonic sensing position	t-obs (each)	4	6,472	import-electronics-Iraqi-integration
EL-12	Isolated peer cross-check SPI link	t-obs (each)	1	1,618	import-electronics-Iraqi-integration
EL-13	Hardware series permission contacts and independent coil drivers	t-obs (each)	1	1,618	import-electronics-Iraqi-integration
EL-14	HV/LV input surge protection and branch fuse assemblies	t-obs (each)	1	1,618	import-electronics-Iraqi-integration
EL-15	Enclosure, heat spreader, glands, vents and grounding hardware	t-obs (each)	1	1,618	import-electronics-Iraqi-integration
EL-16	Train network switch, isolated CAN and external Ethernet controller set	trainset (each)	Open	Open	import-electronics-Iraqi-integration
EL-17	Train-to-ground radios, modem carrier, antennas and surge arresters	trainset (each)	Open	Open	import-electronics-Iraqi-integration
EL-18	W-SBC field cabinet and point/crossing I/O	project (each)	Open	Open	import-electronics-Iraqi-integration
EL-19	S-SBC station host and protected UPS/network enclosure	station (each)	1	159	import-electronics-Iraqi-integration
EL-20	Operator/server cluster, backup and disaster recovery	project (each)	Open	Open	import-electronics-Iraqi-integration

ID	Part / assembly	Basis	Per basis	Network reference	Route
CI-01	Pi20/Pi25 beam, pier cap, column and foundation package	project (each)	Open	Open	Iraqi-precast-and-site
CI-02	Walkway, barrier, cable tray and access cassette	project (each)	Open	Open	Iraqi-fabrication
CI-03	Station accessible platform, ramp/lift/stair and canopy package	station (each)	1	159	Iraqi-civil-and-metalwork
CI-04	Station fire, water, drainage, power and earthing services	station (each)	1	159	mixed
CI-05	Ticket validators, vending, gates and settlement terminals	station (each)	Open	Open	mixed
CI-06	Station PA/PIS, CCTV, help points and telecom cabinet	station (each)	1	159	mixed
CI-07	Charging contact, interlock, protection, DC/DC and site battery	site (each)	1	139	import-core-local-assembly
CI-08	Depot lifts, pits, wheel service, wash and recovery equipment	project (each)	Open	Open	mixed
CI-09	Precast plant moulds, cage jigs, batch/curing and survey tools	plant (each)	1	1	Iraqi-fabrication-import-special-tooling
CI-10	Rail weld/clip/anchor tools and track survey trolley	plant (each)	1	1	mixed
CI-11	Electrical bench, harness formboards and EMC/environmental test access	plant (each)	1	1	mixed

Corrected onboard power envelope

T-OBS simultaneous output allocation is 71.50 W; at 90% efficiency, input is 79.44 W. A 25% sizing margin requires 99.31 W, selecting a 100 W reference capacity. At the assumed 18 V minimum input, envelope current is 4.41 A, or 5.52 A including capacity margin. These are capacity assumptions, not energy-consumption measurements or qualified fuse/converter ratings.

Software host allocation

All 60 workspace packages have exactly one disposition record. This verifies inventory, not physical HAL, sensor driver, firmware timing or operational safety.

Component	Disposition	Hosts
osr-afc	composed-service	s-sbc
osr-afc-backoffice	deployable-endpoint	o-srv
osr-alignment	shared-library	design-tooling
osr-analytics	deployable-endpoint	o-srv
osr-ato	composed-service	t-ecu-a
osr-atp	composed-service	t-ecu-s
osr-aux-power	composed-service	t-ecu-a
osr-balise	composed-service	w-sbc
osr-bms	composed-service	t-ecu-s
osr-brake	composed-service	t-ecu-s
osr-cbm-backend	deployable-endpoint	o-srv
osr-cbm-onboard	composed-service	t-ecu-a
osr-city-studio	tooling	design-tooling
osr-consensus	composed-service	w-sbc, o-srv
osr-core	shared-library	all
osr-crypto	shared-library	all
osr-derailment	composed-service	t-ecu-s
osr-design	tooling	design-tooling
osr-door-control	composed-service	t-ecu-s
osr-energy-site	composed-service	w-sbc
osr-event-recorder	composed-service	t-ecu-a
osr-fire-safety	composed-service	t-ecu-s
osr-gui-shared	shared-library	o-srv, simulation
osr-historian	deployable-endpoint	o-srv
osr-hot-axle	composed-service	t-ecu-a

Component	Disposition	Hosts
osr-hot-axle-wayside	composed-service	w-sbc
osr-hvac	composed-service	t-ecu-a
osr-interlocking	composed-service	w-sbc
osr-intrusion-detect	composed-service	w-sbc
osr-level-crossing	composed-service	w-sbc
osr-lifecycle-identity	shared-library	o-srv, design-tooling
osr-lighting	composed-service	t-ecu-a
osr-obstacle-detect	composed-service	t-obs
osr-onboard-routing	composed-service	t-ecu-s
osr-passenger-assist	composed-service	t-ecu-a
osr-occ	deployable-endpoint	o-srv
osr-occ-gui	deployable-endpoint	o-srv
osr-odometry	composed-service	t-ecu-s
osr-pis-onboard	composed-service	t-ecu-a
osr-pis-station	composed-service	s-sbc
osr-proto	shared-library	all
osr-psd	composed-service	s-sbc
osr-ptp	shared-library	t-ecu-s, t-ecu-a, t-obs, w-sbc, s-sbc
osr-regen	composed-service	t-ecu-a
osr-routing	shared-library	o-srv, design-tooling
osr-safety-case	tooling	assurance-tooling
osr-secbus	shared-library	all
osr-selftest	deployable-endpoint	t-ecu-s, t-ecu-a, t-obs, w-sbc, s-sbc
osr-sim	simulation	simulation
osr-sim-gui	simulation	simulation
osr-station-scada	composed-service	s-sbc
osr-supervision-contract	shared-library	o-srv, simulation
osr-t2g	composed-service	t-ecu-a
osr-tcms	composed-service	t-ecu-a
osr-runtime	shared-library	design-tooling
osr-tcn	shared-library	t-ecu-s, t-ecu-a, t-obs
osr-traction	composed-service	t-ecu-s
osr-trainset-image	deployable-endpoint	t-ecu-s, t-ecu-a, t-obs
osr-tvm	composed-service	s-sbc
osr-wayside-points	composed-service	w-sbc

ERP coverage

Validated workflow contracts: assignment, budget, maintenance, manufacturing, quality, replenishment, service, training.

Enabled profile instances: training/city-induction. Other workflows exist in the app but require real deployment inputs before enablement. A valid reusable profile does not mean that production orders, maintenance assignments or capital budgets have been activated.

Profiles/templates only; legal company, physical bindings, approved schedules, suppliers, stock and production BOMs remain deployment-owned. Reference parts are not auto-created as live ERP Items.

Source hashes bind this report to its exact inputs. Rebuild with `.venv/bin/python engineering/baghdad_detail.py`; verify with `--check`.

Baghdad mixed joint-stock holding and ordinary-equity study

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/equity/README.md

> Original catalogue or earlier scope reference. The figures and policies below retain their original assumptions; they are not the latest Baghdad staffing, depot, procurement or funding basis. See the current Baghdad recalculation and current city summary. National/portfolio totals have not been repriced with that conditional Baghdad option.

As of 2026-10-04. This is a proposed Iraqi holding company with 100%-owned rail, station-development, energy and manufacturing subsidiaries, not an incorporated company, offering, approved listing or committed funding. Baghdad alone supplies the order book. Future national projects remain uncontracted and supply zero revenue, collateral and terminal sale proceeds.

Ownership, new money and currencies

Government still provides USD 1.987821803bn equivalent: 25% of the original USD 7.951287213bn rail/energy/factory capital, rather than 25% of shares or of the enlarged property programme. Reclassifying that existing contribution as ordinary equity generates zero additional cash. At the same subscription price, ownership is government cash divided by government plus private subscribed cash. Borrowing earns no ordinary shares.

Transaction	Gross primary USD m eq	Company net primary USD m eq	Final government shares	Final Iraqi ownership	Government secondary sale USD m eq
primary_500m	500.0	490.0	79.90%	93.97%	0.0
primary_1000m	1000.0	980.0	66.53%	89.96%	0.0
primary_2000m	2000.0	1960.0	49.85%	84.95%	0.0
primary_1000m_premium	1000.0	980.0	67.47%	90.24%	0.0
secondary_500m	0.0	0.0	74.85%	92.45%	500.0

The share unit is an illustrative IQD 1 par and IQD 1 subscription price, not an accepted company valuation or legal share denomination. The premium sensitivity prices the later 25% issue at IQD 1.2, recording the difference in share premium and reducing dilution. Private subscriptions are 70% Iraqi and 30% foreign as an explicit scenario. Every paid-capital month observes the model's 25% state floor and 51% Iraqi floor; actual eligibility and beneficial ownership require counsel. Pending private subscriptions wait for sufficient paid government shares instead of treating promised government capital as paid. No shares or cash are credited for undelivered subscriptions. Land/concession/open designs are not priced into paid-up equity.

The government_grant_reference and government_equity_reference have identical debt, cash and physical costs before distributions. Government grants are a separately identified capital reserve without shares; cash-equity contribution is paid-in ordinary equity. Guarantees and public-service appropriations are uncommitted and never booked as subscription cash. A secondary sale transfers existing government shares: the seller receives the price, the company receives zero, and borrowing requirements remain unchanged. The secondary case omits sale fees as an explicit sensitivity; no government sale is authorised.

Original-scope imported parts are USD 1.797647796bn: government USD cash USD 898.823898m and Chinese USD credit USD 898.823898m. Only Chinese credit is USD debt. Remaining government cash, share subscriptions, bonds, bank/gap credit, fares, property proceeds and dividends are IQD. USD figures for those items are reference conversions at 1,300 IQD/USD, without a dollar-indexed obligation or promised conversion. Original-scope USD capital intensity is 22.61%; local procurement prices still require actual imported-content and FX quotes. The additional USD 2.4bn-equivalent property construction is assumed IQD and makes physical group capital USD 10.351287213bn. It does not add government cash beyond the original 25% rule.

Conditional results and investor returns

Case	Actual gross primary USD m eq	Conditional first/full month	Peak IQD gap tn	Missing IQD cash tn	First dividend month	Private annual equity IRR
government_grant_reference	0.0	40/81	2.131	0.000	not reached in the modelled horizon	undefined
government_equity_reference	0.0	40/81	2.131	0.000	month 227	undefined
primary_500m	500.0	40/81	1.817	0.000	month 215	9.61%
primary_1000m	1000.0	40/81	1.539	0.000	month 191	9.18%
primary_2000m	2000.0	40/81	1.094	0.000	month 167	8.52%
primary_1000m_premium	1000.0	40/81	1.539	0.000	month 191	9.05%
secondary_500m	0.0	40/81	2.131	0.000	month 227	11.67%
undersubscribed_1000m	55.9	Blocked	0.000	0.000	not reached in the modelled horizon	undefined
failed_later_primary	750.0	Blocked	1.302	0.000	not reached in the modelled horizon	undefined
delayed_1000m	1000.0	46/87	1.583	0.000	month 191	9.42%
joint_downside_1000m	1000.0	55/105	4.000	21.943	not reached in the modelled horizon	undefined
joint_downside_undersubscribed	56.1	Blocked	0.000	0.000	not reached in the modelled horizon	undefined
aggregate_tax_proxy_1000m	1000.0	40/81	1.200	0.000	month 191	9.45%
coverage_dividends_1000m	1000.0	40/81	1.469	0.000	month 191	9.18%
rental_small_1000m	1000.0	40/81	1.553	0.000	month 191	9.19%
rental_medium_1000m	1000.0	40/81	1.576	0.000	month 191	9.28%
rental_medium_coverage_1000m	1000.0	40/81	1.482	0.000	month 191	9.27%
rental_medium_downside_1000m	1000.0	40/81	1.559	0.000	month 191	9.13%

The $1bn primary case places IQD 1.300tn gross and IQD 1.274tn net after 2% issue fees. Peak IQD liquidity falls from 2.131tn in the matched holding-company equity reference to 1.539tn. These references pool four wholly owned businesses and include a conservative tax stress. The rail-only planning baseline and independently financed SPV alternative retain their separate scope and cashflows. The original SPV private subscriptions of IQD 1.402tn are replaced, not added to holding subscriptions.

Private equity IRR for the $1bn case is 9.18% nominal, with a 15% hurdle NPV of USD -357.657m for the 70% Iraqi private cohort. Its first dividend is month 191; ordinary shares have no guaranteed redemption or principal payment. Nominal returns require assessment against general inflation and investment risk. An undefined IRR is not a zero return or evidence of an attractive investment. Actual ordinary dividends, retained profits, reserve balances, shareholder payments and ownership are exported monthly. IRR is undefined without positive shareholder receipts, or for ambiguous multi-sign cash flows; blocked-case NPV covers only funded investment to suspension and omits unresolved recovery/termination, so it is not a complete investor downside valuation.

Planned unlevered group resource NPV remains USD -1.316bn, or USD -1.783bn after the assumed unverified USD 600m land opportunity cost. It excludes debt/equity transfers, corporation tax and issuance fees and reconciles to the prior consolidated resource case. Debt replacement changes investor cash and finance costs, not this before-financing resource value. Blocked scenarios retain this planned-scope diagnostic while reporting only actual funded-prefix capital; no unbuilt assets or future earnings are presented as delivered. Financial resource NPV excludes wider social/economic benefits such as travel time and local employment, which need separate evidence.

Subscriptions, six-month placements and release gates

Founder/anchor subscriptions request 75% in six equal calls at months 0, 6, 12, 18, 24 and 30. The remaining 25% is a conditional primary issue at month 60, which may be private rather than listed; no admission is assumed. Actual settlements may extend beyond the requested dates to preserve the paid state floor. primary_1000m-six-months.csv reports actual subscriptions, net fees, restricted capital cash, government USD/IQD cash, native capital-credit draws, dividends and gap requirements every six months. Each monthly row reconciles physical invoices against capital sources; unspent subscription cash is restricted until construction invoices complete.

Net equity displaces the residual 25% bank/75% bond capital basket, bank first in each invoice. Domestic bank principal is amortised over 240 months and bond principal over 216 months, starting no earlier than full-network opening. This group-wide grace is more conservative than per-line cohorts and needs actual facility documents. Reference coupons stay 9%/8%; China retains 5%, 48-month draw grace and 180-month amortisation. Interest is paid throughout construction; arrangement fees,

undrawn Chinese commitment charges, prepayment lock-ins/premiums, six months of debt service reserve, three months of OPEX buffer and factory warranty cash are included. The separate IQD 2% gap sensitivity has a 0.5% draw fee and 13tn cap; downside uses 8%, 1% and 4tn. None is committed.

Undersubscription and failed later issues hold the fully subscribed case's six-month capital-debt envelopes fixed. If the missing shares require capital credit beyond that envelope, the next physical invoice is withheld and the case reports no opening. The operating gap facility cannot cover refused capital. Already funded costs and debt remain; there is no forgiveness or assumed rescue. Resumption requires newly executable finance, rephasing, priced hold/restart and physical acceptance; resolution costs are unmodelled. The six-month delay sensitivity instead moves financial-close-to-NTP work and subscriptions together, retains fixed nominal property sales and indexes operations from close; it is a conditional whole-programme deferral, not a priced mid-construction suspension. Joint downside uses the existing 55/106-month physical stress, fare/nonfare shortfall, higher rates/inflation and weaker property sales, without uncontracted extra income.

Retained viaduct rentals, dividend policies and terminal cash

The civil-linked rental portfolio adds distinct, retained under-viaduct enclosures to the existing station-area sales. No sold parcel also supplies rent; actual title/overlap, height, fire/access and local tenant demand remain unaccepted. The 200,000 m² illustration exceeds the reference geometry screen and is omitted from group funding. Smaller cases include their own IQD fit-out capital, collected rents, rent-free/occupancy ramps, arrears/write-offs, vacant costs, insurance, refurbishment and cohort depreciation. Existing kiosk income is unchanged. Cash corporate tax is calculated once inside the group model; the rental ledger's separate tax diagnostic is not added again. Tenant deposits remain matched restricted cash and liabilities, supplying zero capital or dividends. Government original cash and Chinese USD imports remain unchanged; fit-outs borrow the domestic residual and use the existing parent subscription, with no additional partner cash in this 100%-owned alternative. Rental cases extend the domestic principal gate to the latest fit-out handover if it follows full-network opening; the conditional railway opening dates remain unchanged.

The medium rental case gives peak IQD gap debt 1.576tn, first dividend month 191, private nominal IRR 9.28% and planned before-tax resource NPV after existing station land USD -1.765bn. Additional corridor rights/opportunity cost is unknown, so this is not a complete new-land appraisal. The lower-rent/prolonged-vacancy case retains its costs rather than assuming free premises. Annual rental resource gains and investor gains are separately quantified.

Default ordinary dividends still require full debt repayment. coverage-and-reserves is an unapproved lender-policy sensitivity: annual distributions require 12 actual and 12 projected months of coverage at least 1.30, completed capital, cumulative distributable-profit proxy, no missing cash/current gap draw, restricted capital exhausted, intact OPEX/warranty reserves, six months of forward scheduled debt service and 12 months of upcoming rental refurbishment. CFADS excludes property-sale receipts and adds back their transaction costs; debt coverage includes scheduled interest/principal/fees and gap interest/fees, excluding voluntary principal sweeps. Forecast debt uses actual remaining vintages without future prepayment; forecast tax conservatively assumes no depreciation/interest/loss relief. No upcoming construction draw is permitted by the completion gate. The revolving gap sensitivity has no stated contractual amortisation, so its interest enters coverage and any distribution still needs an actual maturity/covenant agreement.

Up to 50% of cash surplus is held for possible dividends before optional debt sweeps; the balance repays the most expensive eligible credit. Final distributions also preserve cash for accrued tax and a liabilities/book-equity ceiling of 3.0. These are conservative group proxies, not subsidiary upstream permission, a legal reserve calculation or signed lender consent. The default first dividend is month 191; the policy case first dividend is month 191, private IRR 9.18%. With medium rentals, its first dividend is month 191 and private IRR 9.27%. Monthly ledgers expose the actual coverage numerator/denominator, reserves, profit, dividends and debt balances. Earlier dividends reduce sweeps and can increase future finance costs; they are not new profit.

The original $1bn default case ends with USD 5.805bn-equivalent unrestricted cash. A separately labelled terminal_cash_sensitivity allocates available final cash after nonsecurity debt/tax liabilities, at current proportional shares. It changes no actual dividend, company cash or share count and assumes zero transaction cost as an explicit illustration. Private IRR including this unapproved final cash return is 9.59%, versus 9.18% from dividends alone. Restricted deposits, reserves, unresolved funding, unvalued property/rail/factory assets and quoted share-price appreciation are excluded. Failed/unfunded cases cannot present a completed-horizon cash exit. No redemption, liquidation authority or proceeds are guaranteed; counsel must assess capital reduction, tax, transaction and continuing-service obligations before any distribution of capital. Each case exports both ordinary returns and the independent cash-only diagnostic, avoiding double counting.

Pro-forma accounts and consolidation

The group eliminates PPA, internal train invoices/capacity fees and station-rights charges using the existing intercompany ledger. Factory input cash moves to group train assets at its actual purchase date; original manufacturing payroll/QA/overhead remains counted once. Subsidiary equity is replaced by parent subscriptions. Central treasury access, subsidiary distributions/guarantees and lender consent are assumptions requiring explicit agreements; 100% ownership alone does not permit unrestricted transfers. Business operating records remain separately attributable in group-resource-monthly.csv. These are cash-based pro-forma accounts, not audited statutory statements or proof of admission.

Rail, train, solar and factory assets depreciate over 30, 20, 25 and 20 years respectively, phased into operation; depreciation is a noncash charge. Factory remaining book assets are conservatively impaired at Baghdad backlog/warranty close; there are no later orders or salvage receipts. Property construction is inventory, released once to cost of sales as sold; sold space generates no extra rental revenue. Existing maintenance/battery renewals remain in cash OPEX, plus the earlier explicit solar/caretaker allowance; reserves are not spent twice. Land opportunity cost remains a separate in-kind appraisal exposure, not a cash injection, paid-up shares or free balance-sheet asset.

The 15% corporate rate is the Iraqi GCT headline non-oil rate, not a tax ruling. Base scenarios tax each positive annual business operating-profit proxy after assigned depreciation/COGS, retaining internal business charges solely for a conservative tax stress. They assume no parent-interest deductions, subsidiary loss netting, exemptions, refunds or carryforward relief. This yields IQD 7.347tn lifetime cash tax in the normal case. The separately labelled aggregate_tax_proxy_1000m sensitivity taxes positive annual consolidated profit including finance costs; group eligibility is unestablished. Neither is statutory taxable income. Customs, property-transfer taxes, transaction withholding, relief and deductible rates require a priced Iraqi tax opinion before these budgets are investment-ready. GCT corporate declaration guidance.

Annual tax is accrued at year end and paid next month, including final stub settlement. Net assets reconcile against nominal capital, share premium, issue-fee deductions, net retained profit and dividends; cash and book residuals must stay below USD 0.02/0.05. Missing cash is explicitly separate assumed unfunded support, never new government equity. Dividends require cumulative positive distributable-profit proxy, no missing funding, all debt repaid, no unused capital cash and intact OPEX/warranty/DSRA reserves. A further 10% positive-profit retention is contractual sensitivity, not the statutory reserve rule. Audited distributable profit, legal reserves, subsidiary upstreamability and board approval remain additional gates even when the model shows a possible dividend.

Iraqi legal, governance and admission gates

Companies Law Articles 7 and 7 bis support a mixed company with initial state capital at least 25%, holding subsidiaries and annual consolidated accounts. Mandate, concession, incorporation, capital approvals and enforceable public-service obligations still require the competent Iraqi bodies. Publish ordinary commercial contracts separately from affordability/service subsidies; optional future availability support requires an appropriation outside the 25% construction rule.

The official 2019 amendment, Articles 12 and 28 contains the relevant 51% Iraqi participation condition and a liabilities limit of 300% of capital plus other owners' rights, with adjustment powers. Application, current amendments, foreign-investor exceptions, entity-level calculation and permitted treasury structure require counsel. A consolidated screening proxy flags 33 months above that limit or with insufficient book equity in the $1bn scenario. It is not a legal compliance certificate; passing a share-floor check does not clear this debt constraint. The model is therefore not an executable capital structure until legal/capitalisation resolution is accepted.

Regular-market Regulation 6 normally requires two years/history and audits, at least 100 shareholders, equity at least paid-up capital, actual business and two years' distributable profit; a discretionary shorter history provision does not constitute admission. Secondary-market Regulation 15 provides a one-year activity/audit route and equity of at least 50% of paid-up capital, alongside approvals and disclosures. Apply the ISC September 2026 amendments and seek the regulator's current interpretation before fixing any offering date. Actual shareholder count, audits, eligible profits, applications and approvals are absent. All cases set company_incorporated and listing_approved false; month 60 is a funding assumption, not a listing prediction.

Use an independent board/audit committee, published related-party prices and conflicts policy, minority pre-emption/approval protections, consolidated and subsidiary disclosure, audited capital registers, procurement oversight and an explicit public-service contract. Government control and future retail ownership do not themselves demonstrate these controls. Open licences remain intact: investors own businesses, assets and enforceable contracts, without any valuation premium for exclusive ownership of openly licensed designs. New land/concession contributions require independent valuation, title and separate dilution treatment; none is included here.

ISC's 2025 market report records IQD 531.5485bn share turnover and IQD 23.673365tn market capitalisation. The matched holding-company reference's IQD 2.131tn peak gap is 4.01 times that annual turnover; turnover is not a ceiling on primary

subscriptions. The model therefore requires anchor placement evidence for each tranche rather than assumed retail absorption. The $1bn primary gross IQD 1.3tn is about 2.45 times that turnover, which is a scale comparison, not proof of capacity or incapacity. Sources checked 3 October 2026.

Evidence programme

Six new native ERP work packages require Iraqi corporate/tax counsel, public mandate and capital register, anchor/debt placement proof, audit/consolidation and dividend policy, minority/open-licence controls, and actual admission evidence. Owners, quotes, signatures and approvals remain pending. Due days are relative to an approved programme start. The ERPNext Task import contains draft descriptions and no invented live IDs; local import receipts remain private. Existing ten physical/funding qualification tasks and six financing-redesign tasks continue to apply. Model outputs do not accept any evidence or permit operation. No external investor solicitation, issuance or expenditure is authorised.

Regenerate: `.venv/bin/python tools/automation/baghdad_equity.py`; validate source/output hashes with `--check`. Commitments and accepted physical evidence are required before any construction/opening/dividend decision.

Baghdad factory sized to the city programme

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/factory/README.md

> Original catalogue or earlier scope reference. The figures and policies below retain their original assumptions; they are not the latest Baghdad staffing, depot, procurement or funding basis. See the current Baghdad recalculation and current city summary. National/portfolio totals have not been repriced with that conditional Baghdad option.

One factory is sized for 809 complete six-car trainsets / 4,854 cars, with physical cells, crews and test paths. Facility readiness is 18 months from notice to proceed (working day 390), replacing the old 24-month assumption. Financial close precedes NTP by 30 working days; these are conditional offsets, not dated construction commitments.

The unchanged infrastructure resource model finishes on working day 1640. The sized flow accepts the final train on day 1635, 5 working days earlier. Full-network commissioning therefore follows infrastructure completion, rather than waiting decades for four generic train-production slots. All 809 trains remain in the planned scope; there is no smaller opening-fleet substitution.

Flow and physical capacity

Cycles explicitly represent a whole six-car trainset, with structural, electrical and fit-out allowances doubled from the old generic cycle. One eight-hour shift and 85% productive availability are assumed. Occupation is ceil(productive cycle / availability); availability is not deducted again from that output. The first complete train belongs to the 809 and carries an additional 60-working-day qualification allowance. Every remaining material-kit task depends on that first-article acceptance; no series release is assumed before day 556.

Stage	Productive days	Occupied days	Cells/bays	Direct crew
Baghdad trainset kitting	3	4	4	12
Baghdad six-car structural assembly	24	29	24	432
Baghdad six-car composite kits	10	12	10	80
Baghdad six-car body installation	2	3	3	36
Baghdad six-car electrical integration	20	24	20	240
Baghdad six-car fitout and static test	16	19	16	192
Baghdad trainset acceptance bays	12	15	13	52

The limiting steady capacity is 215.2 trainsets / 1291 cars per working year. The resource scheduler simulates each train's linked stages, finite lanes and first-article gate. Cells are calculated from the city's fleet and the remaining infrastructure window, then increased to the first feasible balanced allocation; this is a reproducible capacity design, not a proof of minimum land or minimum cost.

Production bays use a 135 m by 6.5 m envelope for the 111 m consist, not a released building module. Composite and kitting cells have separate floor assumptions. Process and support space totals 107,582 m². The site screen is 36.9 hectares, including circulation/storage and 2 independently segregated 2.0 km test paths. These are site-reservation assumptions pending geometry, braking, fire, access, geotechnical and safe-operation design. Dynamic acceptance bays are not independent running tracks: the separate path calculation permits only 221.0 trainsets/year, based on 16 exclusive track-hours/train and 85% path availability. A shared route cannot be counted twice. The test-path margin above limiting production is only 2.71%; the final fleet has only 5 working days of schedule margin. See the frozen-resource disruption, costed second-shift recovery and civil acceleration study. Those stresses preserve the selected cells rather than resizing them to hide delay.

Direct cell staffing totals 1,044 positions per staffed shift before management, stores, maintenance, relief and shift coverage. It is a proposed resource requirement, not measured job creation. Manufacturing labour and materials are already in train procurement CAPEX; they are not added again to plant CAPEX or railway OPEX.

Line handover and the early-line constraint

Line	Full fleet	Infrastructure day	Fleet day	Opening month
line-1	86	672	762	40
line-10	15	690	780	41
line-12	15	708	798	42
line-11	14	725	815	43
line-13	18	747	837	44
line-14	13	762	852	44
line-2	95	877	967	50
line-3	90	986	1076	55
line-4	76	1078	1168	59
line-5	86	1182	1272	64
line-6	95	1296	1386	69
line-7	72	1383	1473	73
line-8	89	1491	1581	78
line-9	45	1640	1635	81

The factory completes alongside the overall city civil programme. Full-fleet openings are retained. Noncritical infrastructure now moves within its existing crew lanes and dependency graph towards fleet handover, with a 90-working-day target buffer; critical infrastructure retains its original completion dates. Line 1's infrastructure completion moves from day 194 to day 672, reducing the idle interval before its fleet from 568 to 90 working days. No rolling-stock date, task duration, resource count or opening date changes. This is a conditional investment-timing proposal: surveys, land, utilities, permits and contract dates require approval before deferring site work. Original and rephased line reconciliation preserves both sets of dates. Line priority follows the original infrastructure requirements, with actual asset identifiers preserved.

The opening calculation retains the separate three-month integrated commissioning allowance after line infrastructure, the full fleet and shared depot/control work. The 60-day first-article allowance and three-month line allowance cover different activities. Required approvals, surveys, physical tests and independent acceptance remain open; neither allowance constitutes accepted safety evidence or an approved construction calendar.

Eighteen-month facility construction sequence

Activity	Working days	Accountable role
Land/access, survey, design and approvals	60	Sponsor and owner engineer
Mobilisation, earthworks, foundations and underground services	70	Local civil contractor
Factory shell, lifting structure and utility backbone	70	Local structural and MEP contractors
Production cells, moulds, cranes and test-track construction	80	Production engineer and equipment suppliers
Equipment installation, calibration, staffing and training	70	Factory operator and quality manager
Factory process proving, safe-work and facility acceptance	40	Owner engineer and independent inspectors

These six sequential planning packages total 390 working days. Long-lead tooling, cranes, battery handling equipment and imported systems must be ordered early enough to install and prove within that window. Land/access and financing availability are required at NTP. A delay to those gates changes the programme; the schedule is not a contractor-backed promise.

Factory capital and cashflow

Physical budget allocation	USD million
buildings	75.307
serviced site	12.913
stage tooling	97.700
test tracks	12.000
shared utilities stores labs logistics	35.000
design training qualification	20.000
contingency	50.584

The physical envelope is USD 303.505 million direct, compared with the old module allowance of USD 291.240 million. The programme budgets the larger amount, USD 303.505 million plus USD 21.245 million EPC, counted once. The resulting capital increase is USD 13.123 million; added bays are not treated as free capacity. Rates and quantities are engineering allocations requiring Iraqi contractor/vendor quotations, not market-verified prices. Actual equipment origin must also reconcile to the programme's imported/local allowances.

Factory payments are re-timed to the 390-day facility programme and city train payments follow the new production tasks. Baghdad's monthly debt, interest, 5% fare/OPEX indexing, liquidity gaps, surplus repayments and six-month bond/loan requirements are recalculated from those dates. Government remains 25% of capital; import cash and Chinese credit retain the 50:50 USD split, with remaining funding in IQD. The existing aggregate plant import allowance remains 20% of direct plant capital, allocated within tooling/test equipment rather than buildings; supplier origin and the actual equipment mix require RFQs and eligibility checks. EPC retains its separate procurement-origin allowance. Future national cities do not enter this plant's production load or Baghdad cashflow.

Procurement and qualification requirements

Before release, obtain measured six-car labour routes and prototype cycle times; trainset-level supplier delivery schedules; released mould/fixture drawings and duplication plans; lifting/handling and HV/battery fire segregation; stores, quarantine and rework capacity; inspection/calibration capacity; crew recruitment and training; both independent test-path designs; utility availability; and a land/industrial permit package. Composite cells cannot be mistaken for the number of panel moulds: the final panel design and cure/release conditions determine mould duplication. The hypothetical whole-kit cycle must be validated against that tooling count. Existing civil precast/slab yards retain their own production resources and are not assigned to train bays.

The 85% factor is a deterministic allowance for non-productive time, not a quantified programme contingency. Supplier disruption, multi-month qualification, redesign and civil delays need risk scenarios before investment decisions. Expansion beyond Baghdad needs a new demand and capacity assessment; a national module budget is not a promise of parallel national production.

Cell and tooling register · 809 planned trainset acceptances · Line handover reconciliation · Machine-readable calculation · Factory capacity inputs

Regenerate after operations with `.venv/bin/python tools/automation/generate-factory-plan.py`; validate the same command with `--check`. Status: reference planning engineering, no construction or manufacturing release.

Baghdad — Iraq funding appraisal

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/finance/FUNDING-MODEL.md

> Original catalogue or earlier scope reference. The figures and policies below retain their original assumptions; they are not the latest Baghdad staffing, depot, procurement or funding basis. See the current Baghdad recalculation and current city summary. National/portfolio totals have not been repriced with that conditional Baghdad option.

Generated from current city CAPEX and procurement milestones. All facilities and appropriations remain uncommitted.

Government capital is 25% of total city CAPEX, before Chinese credit is deducted. The residual after that contribution and proposed Chinese proceeds is split 75% IQD bonds / 25% IQD bank credit. Imported purchases use 50% government USD cash and 50% proposed Chinese credit, assuming full-basket eligibility is later qualified; the remaining government contribution is IQD. Financing fees, construction interest, reserves and operating/debt support are additional funding requirements, not included in the 25% capital contribution. See the Baghdad-only programme for the uncovered cash requirements if no additional government cash is available.

Construction cash runs through month 82; full-network operations start in month 82 after retention. This conservative rollout follows the current resource-constrained CPM, not a five-year promise.

Capital sources and uses

USD is the comparison unit below. Chinese credit is USD debt; domestic bonds and bank credit are IQD debt. Baghdad imported purchases are funded 50% government USD cash and 50% proposed Chinese loan; the remaining government capital and local cash are IQD. The full Baghdad import basket is assumed eligible pending supplier-origin and lender qualification. Other city appraisals retain their own assumptions. Fares and local operating payments are budgeted in IQD.

Capital source	Contract / cash currency	Native amount at model FX	USD equivalent
chinese export credit	USD	866,880,047.75	866,880,047.75
domestic bonds	IQD	4,731,697,412,249.30	3,639,767,240.19
bank credit	IQD	1,577,232,470,749.77	1,213,255,746.73
government import cash	USD	866,880,047.75	866,880,047.75
government local cash	IQD	1,351,680,586,280.91	1,039,754,297.14
Total city capital uses	Mixed	—	7,626,537,379.57

Chinese buyer credit is proposed for eligible Chinese component invoices only. Government contributes its configured capital share, including the eligible-invoice downpayment; IQD bonds and bank credit finance the residual. The conditional ledger also calculates cash needed for fees, construction interest, reserves and operating/debt shortfalls. That additional support is uncommitted and is an unfunded requirement if Baghdad public cash is capped at its 25% capital contribution.

Proposed instruments

Instrument	Currency	Rate assumed	Grace from draw	Repayment
chinese export credit	USD	5.0%	48 months	180 months
domestic bonds	IQD	8.0%	24 months	180 months
bank credit	IQD	9.0%	0 months	84 months

Chinese component allocation

These allocations divide existing imported budgets; they are not additional costs, supplier quotes or confirmed origin.

Bucket	Component	Assumed eligible invoice USD
civil	other_imports_pending_lender_and_origin_qualification	590,556,608.25
stations	other_imports_pending_lender_and_origin_qualification	195,400,000.00
depots	other_imports_pending_lender_and_origin_qualification	2,000,000.00
rolling_stock	bogies	190,276,800.00
rolling_stock	traction_batteries	142,707,600.00
rolling_stock	windows	23,784,600.00
rolling_stock	doors	47,569,200.00
rolling_stock	other_imports_pending_lender_and_origin_qualification	71,353,800.00
solar_plant	solar_panels_and_inverters	367,083,177.55
signalling	other_imports_pending_lender_and_origin_qualification	12,833,386.00
charging_microgrid	solar_panels_and_inverters	8,176,000.00
charging_microgrid	station_batteries	9,344,000.00

Bucket	Component	Assumed eligible invoice USD
charging_microgrid	other_imports_pending_lender_and_origin_qualification	5,840,000.00
epc_overhead	other_imports_pending_lender_and_origin_qualification	66,834,923.70

Annual cash requirements

Year 1 begins at assumed financial close; NTP follows 30 working days later. Amounts below are USD millions. DSCR excludes subsidy; public cash includes capital, fees, interest, reserves and support.

Year	CAPEX	Revenue	OPEX	Debt service	Public cash	DSCR before support
1	319.74	0.00	0.00	10.17	91.75	0.00
2	674.06	0.00	0.00	47.54	219.55	0.00
3	1329.10	0.00	0.00	122.77	462.24	0.00
4	1560.09	0.00	0.00	242.13	640.60	0.00
5	1566.91	0.00	0.00	375.12	775.33	0.00
6	1577.50	0.00	0.00	518.61	921.53	0.00
7	599.13	58.21	34.17	642.28	1104.51	0.04
8	0.00	378.34	205.01	681.51	520.88	0.25
9	0.00	552.96	205.01	695.76	347.23	0.50
10	0.00	698.47	205.01	676.85	168.32	0.73
11	0.00	698.47	205.01	638.55	123.21	0.77
12	0.00	698.47	205.01	591.60	74.58	0.83
13	0.00	698.47	205.01	543.50	25.75	0.91
14	0.00	698.47	205.01	504.07	0.32	0.98
15	0.00	698.47	205.01	499.66	5.61	0.99
16	0.00	698.47	205.01	499.66	6.20	0.99
17	0.00	698.47	205.01	499.66	6.20	0.99
18	0.00	698.47	205.01	492.58	1.16	1.00
19	0.00	698.47	205.01	465.02	0.00	1.06
20	0.00	698.47	205.01	410.94	0.00	1.20
21	0.00	698.47	205.01	326.82	0.00	1.51
22	0.00	698.47	205.01	233.15	0.00	2.12
23	0.00	698.47	205.01	130.53	0.00	3.78
24	0.00	698.47	205.01	42.37	0.00	11.65
25	0.00	698.47	205.01	16.40	0.00	30.09
26	0.00	698.47	205.01	1.62	0.00	304.26
27	0.00	698.47	205.01	0.00	0.00	—
28	0.00	698.47	205.01	0.00	0.00	—
29	0.00	698.47	205.01	0.00	0.00	—
30	0.00	698.47	205.01	0.00	0.00	—
31	0.00	698.47	205.01	0.00	0.00	—
32	0.00	698.47	205.01	0.00	0.00	—
33	0.00	698.47	205.01	0.00	0.00	—
34	0.00	698.47	205.01	0.00	0.00	—
35	0.00	698.47	205.01	0.00	0.00	—
36	0.00	698.47	205.01	0.00	0.00	—
37	0.00	582.06	170.84	0.00	0.00	—

Sensitivities

Scenario	Peak annual public cash USD m	Minimum operating DSCR	Peak uncovered monthly capital USD m
base low demand	1104.51	0.04	0.00
high capacity use	1071.07	0.09	0.00
capex plus 25 percent	1386.64	0.03	0.00
demand minus 40 percent	1127.79	0.00	0.00
iqd depreciation 35 percent	932.77	0.03	0.00
commissioning delay two years	1017.20	0.03	0.00
china credit unavailable	1028.52	0.04	18.14
four year bullet bonds	1862.45	0.02	0.00
government payment delay six months	1291.63	0.04	15.65
interest plus three points	1307.66	0.03	0.00
combined downside	1276.42	0.00	0.00

Assumptions and evidence

- Working-day milestones use 260 working days/year and 30 working days before NTP; holidays and an approved local calendar are pending.
- No signed Chinese loan, supplier-origin qualification, guarantee, bond mandate, bank facility, appropriation or tax/duty assessment.
- Chinese origin and lender eligibility remain unqualified proxies. Default advance is 85%; Baghdad assumes all imports can qualify for 50% USD loan / 50% government USD cash. Additional imported categories require lender and supplier approval; neither assumption is a verified lender rule.
- Domestic bonds assume a proposed MoF sovereign IQD programme; municipal borrowing powers and market demand are not assumed.
- Long amortising domestic bonds are an appraisal target; the four-year bullet sensitivity exposes redemption without automatic refinancing.
- Construction cohorts need separately approved facilities. A long rollout does not imply a lender offers decades of draw availability.
- Interest is paid on drawn native balances; mid-month draw convention, monthly repayment, fees paid by government; no interest capitalisation.
- All cash is nominal; no inflation/escalation in base. IQD revenue and OPEX are translated at scenario FX; imported energy/parts OPEX needs a separate currency survey.
- Unquoted CAPEX budgets remain nominal USD planning values. IQD tranches and appropriations convert at draw-date FX; no automatic local supplier price benefit from depreciation is assumed.
- Income used for labour and fares is the retained country-finance planning proxy, not a verified current Iraqi household median or an agreed wage/fare contract.
- Government payment delays and unavailable China credit produce explicit capital cash gaps; no committed bridge credit is assumed.
- Restricted DSRA targets six times current monthly service and uses available project cash before requesting support. Support cash is a required contribution, not a commitment; repayments and reserve balances are conditional on it. Baghdad programme caps government capital at 25% and separately discloses the additional unfunded requirement. Future-service covenant testing is pending.
- Battery renewal reserve remains inside existing rolling-stock maintenance OPEX; no second battery CAPEX is added.
- Demand is capacity-led, not a surveyed forecast. Passing reconciliation does not demonstrate affordability or bankability.

Long construction schedules can leave much of the debt already repaid by government before full-network opening. An operating DSCR above one therefore does not establish self-financing or remove the construction-period public cash burden. Uncovered capital gaps are conditional funding requirements, not actual cash receipts.

See editable assumptions, monthly cashflow, annual cashflow, and machine-readable model.

The IMF Article IV provides the historical FX anchor. CBI describes its role as fiscal agent for MoF bonds. China Exim describes export buyer credit; numeric project terms remain assumptions.

Conditional phased-opening sensitivity

The base above withholds all fares until every capital milestone. This separate sensitivity opens completed lines after the editable commissioning lag and gates every line on shared system/depot work. The city-sized Baghdad production plant is unavailable for its first 390 working days (18 months from NTP), followed by the scheduled first-article qualification gate. Opening also requires actual civil, train, depot, power and safety acceptance, which is not established by this calculation.

Revenue and variable OPEX use each line's share of controlled trainsets. Each line has its own 50% / 75% / 100% revenue ramp. Fixed OPEX is 25% of the full-network annual budget from first opening; remaining OPEX scales with opened fleet. These are explicit uncalibrated sensitivity assumptions, not a ridership survey or verified staffing plan.

First / last planned opening: month 40 / 81 from financial close (month zero). City additional funding requirement: USD 2,780.13 million. The plant remains separate. Future surplus is retained; gross additional support is not net lifetime loss or discounted cost.

Line	Opening month	Revenue / variable OPEX share
line-1	40	10.63%

Line	Opening month	Revenue / variable OPEX share
line-10	41	1.85%
line-11	43	1.73%
line-12	42	1.85%
line-13	44	2.22%
line-14	44	1.61%
line-2	50	11.74%
line-3	55	11.12%
line-4	59	9.39%
line-5	64	10.63%
line-6	69	11.74%
line-7	73	8.90%
line-8	78	11.00%
line-9	81	5.56%

Phased monthly cashflow · phased annual cashflow

Baghdad financing reconciliation and six-month placement programme

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/finance/FUNDING-RECONCILIATION.md

> Original catalogue or earlier scope reference. The figures and policies below retain their original assumptions; they are not the latest Baghdad staffing, depot, procurement or funding basis. See the current Baghdad recalculation and current city summary. National/portfolio totals have not been repriced with that conditional Baghdad option.

Scope: Baghdad city and one Baghdad manufacturing plant. Samawah, Mosul and every other city are excluded. Month zero is an assumed financial close, not an approved date. All financing and additional revenues remain uncommitted.

The numbers that must add up

The original capital sources and uses reconcile. The confusion was combining upfront capital, repayment of that same capital debt, gross liquidity injections and later retained cash. The revised schedule also spreads city EPC overhead across direct works instead of paying the entire programme allowance against the initial five-day baseline-freeze task. Capital cost is unchanged. An independent implementation rebuilds every debt vintage, interest charge, principal payment and financing fee from native draws; it does not copy the reported support total.

Capital source	Native currency	Native amount, billion	USD equivalent, million
chinese export credit	USD	0.898824	898.824
government import cash	USD	0.898824	898.824
government local cash	IQD	1,415.697277	1,088.998
domestic bonds	IQD	4,938.025474	3,798.481
bank credit	IQD	1,646.008491	1,266.160
Capital uses / sources	Mixed	—	7,951.287

Government cash remains 25% of capital, with government USD import cash inside that contribution. Imports are financed 50% government USD cash / 50% Chinese USD loan. Every other loan and bond is IQD. An international green lender must provide confirmed IQD on-lending or a priced currency hedge to preserve that strategy. A green label does not supply additional money.

Lifetime operating / finance reconciliation	USD equivalent, million
Fare and existing nonfare receipts	21,638.485
OPEX	6,591.251
Repayment of originally drawn capital principal	5,963.465
Interest on original capital financing	4,367.991
Original financing fees	42.664
Net nominal lifetime liquidity deficit	-4,673.113
Gross extra cash required as bills fall due	3,193.940
Later unrestricted cash retained if those requirements are met	7,867.053

USD 3,193.940m gross injections minus USD 7,867.053m retained cash equals USD -4,673.113m net deficit. Capital purchases are already financed by their capital sources; principal repayments belong to the lifetime cash ledger and must not be added to the construction budget again. Net reserve deposits are zero at the end. This is a nominal identity, not discounted viability, and assumes city/plant cash can legally be pooled

A zero-interest, zero-fee revolving bridge is only a mathematical lower bound. Real gap funding must charge interest and fees on the borrowing needed to carry early deficits, repay itself from later surplus and show any terminal debt. Later surplus cannot be counted both as retained cash and as loan repayment.

Tickets, kiosk rents and advertising already included

Full-network steady annual receipts	IQD billion	USD equivalent, million
Passenger tickets	869.296	668.689
Station shops and kiosk leases	15.538	11.952
Advertising space	23.180	17.831
Total existing operating receipts	908.015	698.473

Tickets use 659.89m annual paid trips at IQD 1,317 average yield. This is assumed capacity use, not calibrated demand. Shops/kiosks use 37,232 m² at IQD 39,520/m²/month and 88% occupancy; advertising uses 6,572 boards at IQD 345,800/month and 85% occupancy. Rates derive from the retained income proxy, not Baghdad lease quotes. These are gross

rental/advertising receipts; dedicated concession costs, tenant demand and collection losses require appraisal.

All three receipts enter the existing operating cashflows and therefore already reduce the financing gap. The phased sensitivity scales the combined receipts by opened-fleet share and the separate 50% / 75% / 100% revenue ramp for each line. Retail and advertising demand may follow different ramps in practice. Proposed naming rights, telecom leases, parking, development proceeds or new commercial receipts must be genuinely additional and net of costs; existing rents cannot be booked again or sold upfront while also retained in future revenue.

What gap finance actually does

The illustrative maximum outstanding facility is IQD 13.000 trillion (USD 10.000bn equivalent). This is a size assumption, not proof that the Iraqi banking market can place it. All available surplus after core obligations and reserves is swept to its repayment. No refinancing or debt forgiveness is assumed at the final maturity.

Scenario	Gross new liquidity draws, IQD tn	Peak facility balance, IQD tn	Uncovered cash beyond facility, IQD tn	Terminal unpaid facility, IQD tn
unfunded reference	0.000	0.000	4.152	0.000
green label only	0.000	0.000	4.152	0.000
zero cost liquidity bound	4.152	4.152	0.000	0.000
commercial gap credit	13.000	13.000	7.067	13.000
concessional gap credit	5.318	5.317	0.000	0.000
blended candidate	3.574	3.384	0.000	0.000
green concessional only	4.433	4.428	0.000	0.000

Reference and label-only cases have no gap facility: their uncovered column is the required external cash contribution, not an approved government top-up. With a facility, uncovered cash excludes the separate terminal loan balance; both need a solution. Gross draws can exceed the maximum outstanding commitment when a revolving facility is repaid and redrawn. Conditional ledgers do not establish that an unfunded payment occurred.

Commercial gap credit assumes 8% interest and 1.0% draw fees. Concessional gap credit assumes 2% interest and 0.5% draw fees. Neither is a term sheet or an available GCF line. Funding old debt-service deficits is not automatically eligible green use of proceeds.

Explicit blended sensitivity

Candidate climate expenditure is USD 2,233.26m of electric rolling stock, solar and charging provision, including USD 1,367.13m assigned locally. Climate benefit, supplier qualification and green eligibility require independent review. Candidate green bonds replace only domestic bonds attributable to those expenditures; civil works and unrelated overhead are not automatically relabelled.

The blended sensitivity replaces that eligible bond portion with proposed IQD green bonds/sukuk at 4%, 48 months' grace and 240 months' amortisation, with 0.75% arrangement fees and 0.5%/year enhancement fees on outstanding principal. The lower rate depends on a concessional or guarantee structure; a green label alone retains the original 8% terms and changes no financial outcome.

Additional targets are a USD 25m climate capital grant translated into IQD receipts, replacing domestic borrowing against eligible local capital after month 24; USD 300m net development-rights proceeds in 6 six-month instalments from month 24; and USD 25m/year genuinely new net local receipts at full opening, scaled with the opened fleet. These are editable targets, not appraisals, GCF approvals, developer contracts or statutory levies. Existing rents, advertising and fares cannot be counted again; transaction/development costs and displaced future rents must be deducted. Public land contribution also needs economic valuation.

Solving the model rather than declaring the targets sufficient requires approximately USD 0.00m/year additional NET receipts (IQD 0.00bn/year at full opening) to eliminate both uncovered cash and the terminal facility under the other blended assumptions. This is a break-even funding requirement, not a forecast or a recommended passenger fare increase. Any covenant, inflation, FX or demand change can raise it.

Variable tickets, annual price changes and OPEX inflation

The following cases retain the same conditional blended capital/grant/rights/2% IQD facility assumptions. They change fares, OPEX and income separately. A 5% tariff increase with flat costs is an optimistic sensitivity; the paired 5% OPEX case is the requested inflation comparison. None is an adopted fare, a lender commitment or an Iraqi inflation forecast.

Indices compound in annual steps from financial close, including the years before first opening. No tickets are sold before service starts; the nominal launch price reflects elapsed inflation. Existing retail/advertising are held flat unless the case explicitly indexes those rents. New net income and rights targets remain fixed nominal amounts. CAPEX escalation and future

FX changes still require a separate full nominal appraisal.

Demand responds to fare relative to the indexed income proxy with an illustrative elasticity of -0.30, except the explicitly fixed-demand cases. Income growth is a separate assumption, not evidence that wages will keep pace. Discount-led demand growth is capped at 2.0 times the baseline, within the original practical capacity. There is no calibrated Baghdad peak-switching or household demand model.

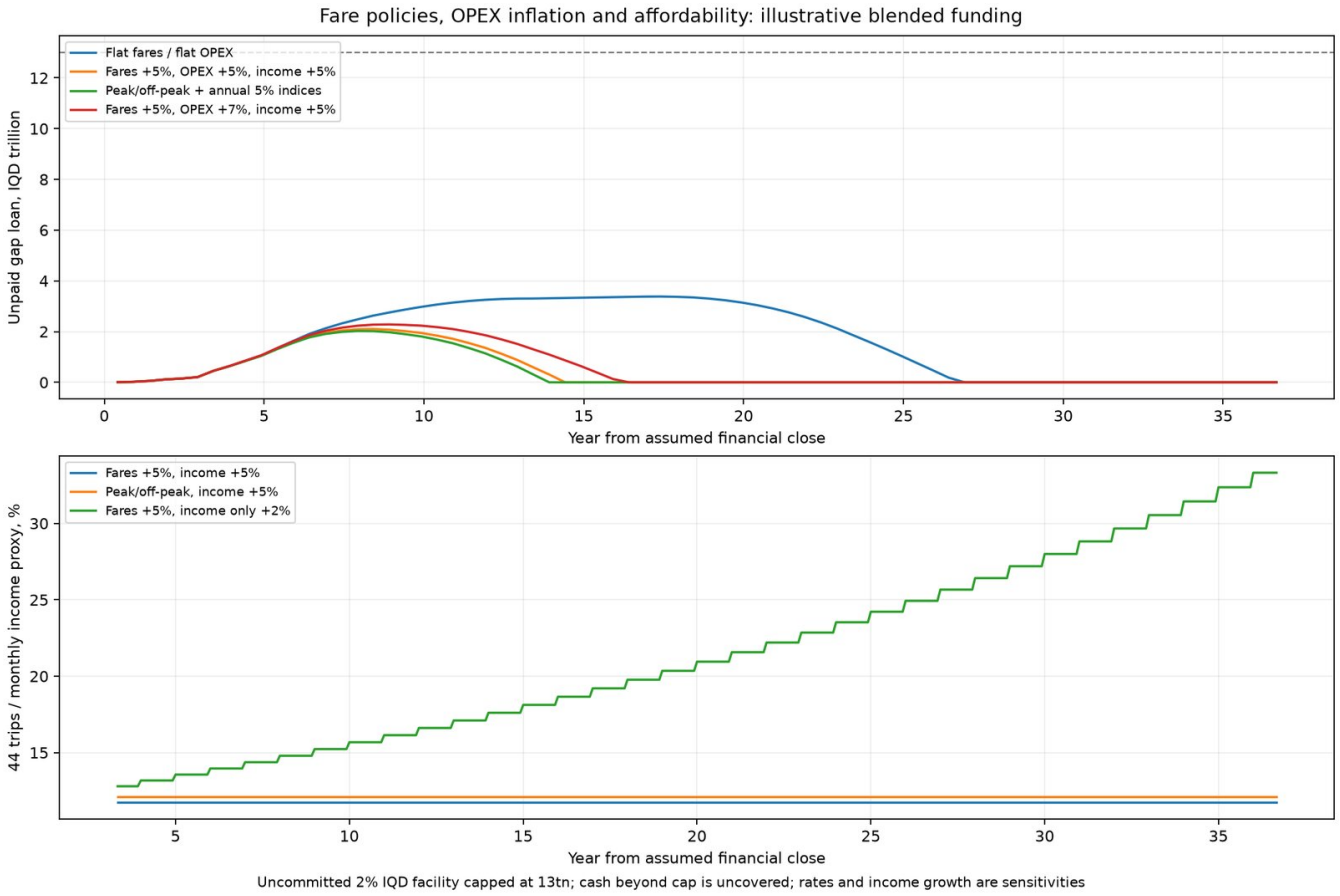
Variable tickets assume 40% of baseline trips in a peak tier at 1.25 times the standard fare, and remaining trips at 0.90 times it. Each tier gets its own demand response before revenues are combined. Concessions, commuter caps, integrated transfers and off-peak eligibility require an actual tariff design and funding contract; their costs are not silently waived.

Pricing sensitivity	Fare / OPEX / income growth	Full-opening average fare, IQD	44 trips / income at full opening	Peak facility, IQD tn	Uncovered cash, IQD tn	Terminal debt, IQD tn
fare 5pct flat costs fixed demand	5% / 0% / 0%	1,765	15.7%	1.807	0.000	0.000
fare 5pct flat costs elastic	5% / 0% / 0%	1,765	15.7%	2.037	0.000	0.000
fixed fare 5pct opex	0% / 5% / 5%	1,317	8.8%	13.000	0.486	13.000
fare 5pct opex 5pct	5% / 5% / 5%	1,765	11.7%	2.102	0.000	0.000
fare 5pct opex 7pct	5% / 7% / 5%	1,765	11.7%	2.283	0.000	0.000
fare 5pct opex 5pct income 2pct	5% / 5% / 2%	1,765	14.0%	2.302	0.000	0.000
variable fare 5pct opex 5pct	5% / 5% / 5%	1,822	12.1%	2.025	0.000	0.000
fare 5pct opex 5pct rents indexed	5% / 5% / 5%	1,765	11.7%	2.063	0.000	0.000

In the paired 5% case, the first line's average nominal fare is IQD 1,525 in month 40; at full opening in month 81 it is IQD 1,765. Its maximum supplemental facility is IQD 2.102tn, with IQD 0.000tn uncovered cash and IQD 0.000tn unpaid at the end. The case repays the facility within the model horizon under these conditional assumptions. Early borrowing still needs placed facilities.

At an 5% general-price assumption, the 8% real discount assumption becomes 13.4% nominal. The paired-case unlevered NPV is USD -1.670bn equivalent before grants/new rights/net-income targets and with capital still un-escalated. Large distant nominal balances are not present-value wealth or proof of project viability.

If income rises with fares, the modelled commuting share stays broadly constant. With income rising only 2%, the same ticket policy becomes progressively less affordable and reduces paid trips. If OPEX rises 7% against 5% fares, the tested financing again leaves uncovered requirements and terminal debt. Maintaining an affordable real tariff, collecting revenue and placing the required early IQD facility matter as much as the nominal annual increase.



Fare and OPEX inflation sensitivities

Paired 5% six-month funding requirements · paired monthly prices and affordability · variable-ticket funding requirements · variable monthly tickets

The initial real-fare multiplier required to remove both terminal debt and uncovered cash under paired 5% indices and the other blended assumptions is approximately 1.000 times the existing baseline. This is a conditional break-even sensitivity, not a tariff recommendation; it does not guarantee the facility can be placed or that households' incomes will grow.

Surplus cash and early repayment

These cases use identical paired 5% fare/OPEX/income assumptions and the same conditional blended capital sources. Each pays OPEX, scheduled principal, interest and fees, then funds the debt-service reserve and a three-month current-OPEX buffer before voluntary repayment. Borrowed gap proceeds and uncovered external cash cannot fund early payments. The extra buffer remains cash held at the terminal horizon and earns no interest.

Compare strategies against buffered gap-only, rather than attributing the buffer change to repayment savings. Loans-first pays bank credit, Chinese credit and gap credit before bonds. Cost priority pays bank (9%), ordinary bonds (8%), Chinese credit (5%), green bonds (4% plus 0.5% annual guarantee), then gap credit (2%). It is an interest-rate heuristic, not a proof of the globally best strategy.

Surplus strategy	All debt cleared, month from close	Finance cost saving vs buffered gap-only, USD equivalent m	Early-payment premiums, USD equivalent m	Peak gap debt, IQD tn	Terminal unrestricted cash, IQD tn
gap only buffered	369	0.000	0.000	2.208	44.706
gap then core	209	506.359	36.995	2.208	45.364
loans then bonds	207	653.848	42.809	2.158	45.556
cost priority	203	988.754	56.843	2.158	45.991
cost priority zero premium	203	1,060.010	0.000	2.156	46.084
cost priority noncallable bonds	369	260.705	12.395	2.158	45.045

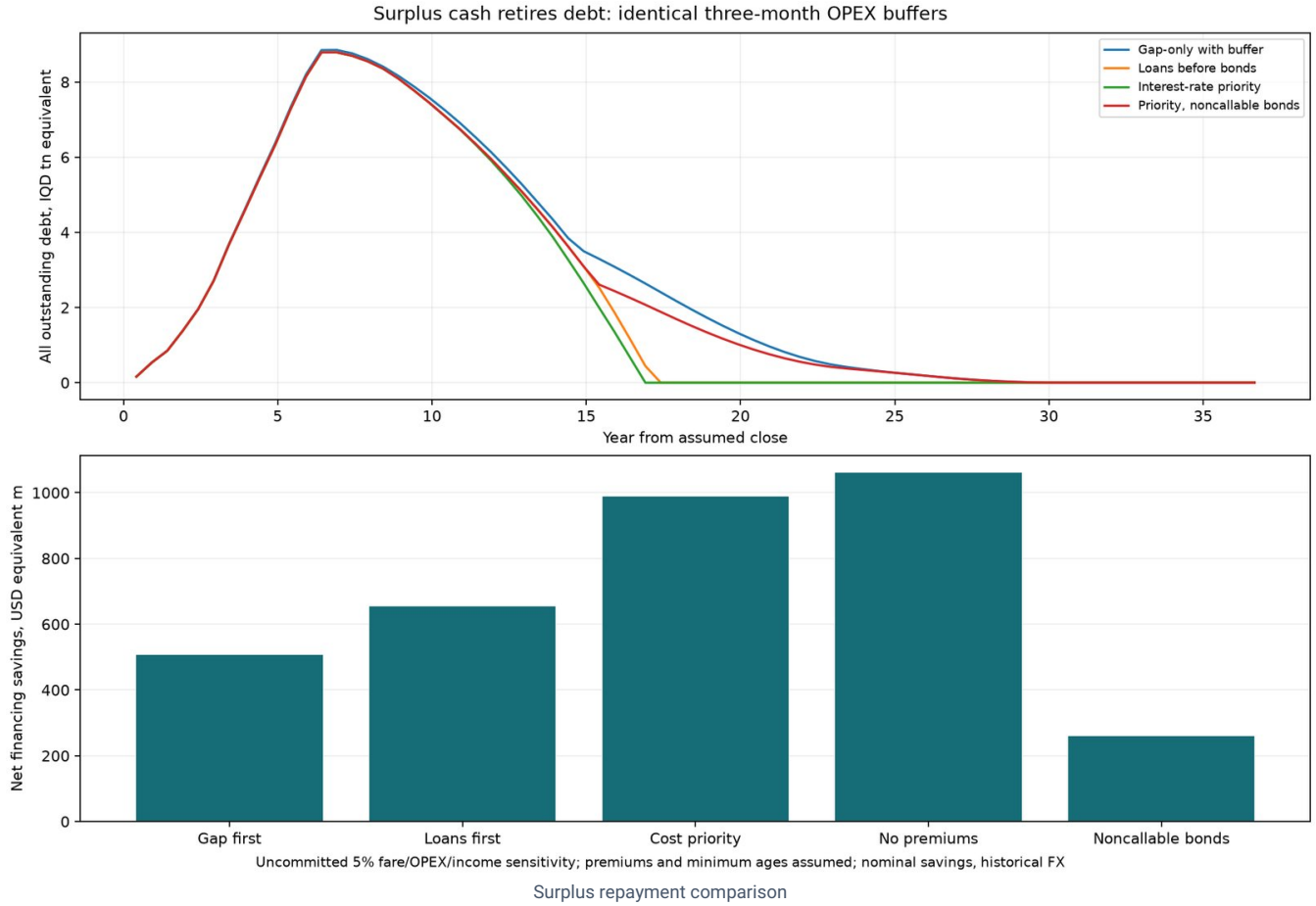
Cost-priority debt is cleared in month 203; buffered gap-only debt is cleared in month 369. Net nominal financing savings are USD 988.754m equivalent, after USD 56.843m assumed early-payment premiums. Savings include core interest, annual green guarantee charges and supplemental interest/draw fees; they exclude principal, which is returned once. All cases retain IQD 0.386tn operating buffer separately from unrestricted cash. There is no additional government contribution above 25% of CAPEX in these cases.

Assumed premiums are 1% of bank/Chinese/green principal and 2% of ordinary bond principal. A minimum draw age of 6 months (bank), 12 (Chinese) and 24 (both bonds) prevents immediate issue-and-redemption. Eligible vintages are repaid oldest first; contractual instalments are kept and maturity shortens. Notice, issuer call rights, investor consent, buyback price, remaining-maturity compensation, tax and FX require actual agreements. Noncallable-bond sensitivity makes no voluntary bond payments; zero-premium sensitivity removes only the assumed premium, retaining minimum ages.

Facility	Buffered gap-only final principal payment month	Cost-priority final principal payment month	Early principal, native currency	Premium, native currency
bank credit	165	122	IQD 636.771bn	IQD 6.368bn
domestic bonds	285	167	IQD 2,547.814bn	IQD 50.956bn
chinese export credit	309	175	USD 592.389m	USD 5.924m
green bonds	369	184	IQD 887.075bn	IQD 8.871bn

The gap facility's final principal payment is in month 203. Later surplus remains unrestricted cash after debt retirement. Each month's native debt balance equals prior balance plus draws minus scheduled and early principal. Each six-month closing balance is its final month's balance; payments, premiums, interest and reserve movements are period sums. Contractual repayment windows in tranche files describe the original draw terms; the actual final-payment months above incorporate early repayment.

Monthly cost-priority repayments · six-month repayment tranches · loans-first tranches · all six complete calculations

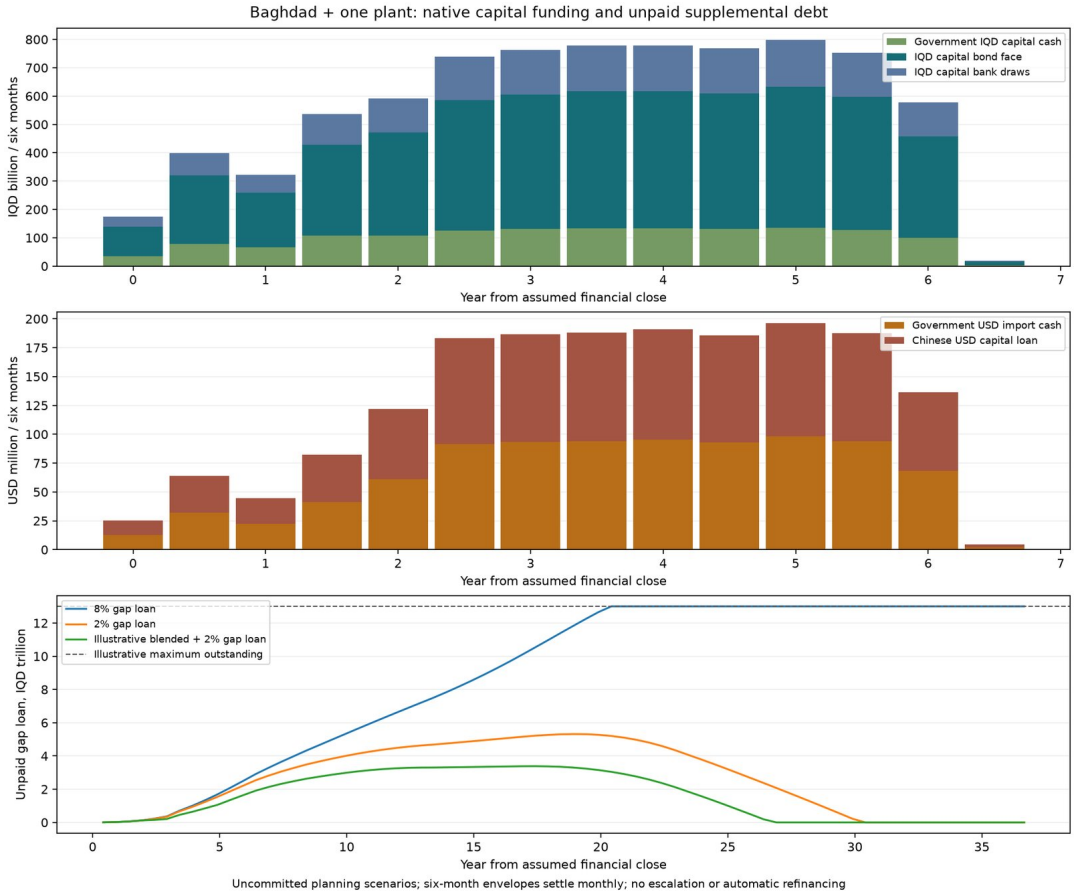


Surplus repayment comparison

Loan prepayment can carry redeployment/unwind charges: the World Bank Treasury FAQ illustrates the concept, without establishing Chinese or Iraqi loan terms. The US Treasury buyback FAQ and published purchase results illustrate issuer authority and priced buybacks rather than automatic par redemption; they do not supply an Iraqi legal mandate.

This repayment allocation does not change the unlevered project NPV or validate demand. Nominal savings over several decades are not present-value gains. CAPEX escalation, FX, lifecycle replacement, floating-rate risk and the uncommitted early funding still require appraisal.

Six-month bond sales and capital loan requirements



Native six-month capital funding and supplemental loan balances

Each half-year is a placement/draw envelope, with monthly settlement against capital milestones. Bonds assume par proceeds and an illustrative IQD 1,000,000 denomination. Face-value requirements below use unrounded values; unit counts in CSV round upwards and separately disclose the resulting indicative face amount. Sale discounts, issuance fees and investor capacity need actual bookbuilding. Selling every half-year amount upfront changes interest/carry and grace clocks and is not simulated here.

The government USD cash and Chinese USD loan are separate; government local cash, bonds, bank draws and uncovered liquidity are native IQD. Chinese facility commitment fees retain the existing annual procurement-cohort assumption; six-month envelopes do not assert a new signing schedule. Grace starts on each actual monthly settlement. First/last repayments for a half-year series therefore differ by draw date.

Tranche	Months from close	Government USD m	Government IQD bn	Chinese loan USD m	Bond face IQD bn	Bank loan IQD bn	Additional cash need IQD bn
BGD-H001	0–5	12.71	35.27	12.71	104.13	34.71	4.94
BGD-H002	6–11	32.04	79.08	32.04	240.40	80.13	16.96
BGD-H003	12–17	22.28	66.12	22.28	192.21	64.07	34.43
BGD-H004	18–23	41.20	107.37	41.20	321.91	107.30	56.92
BGD-H005	24–29	61.05	108.22	61.05	362.54	120.85	98.95
BGD-H006	30–35	91.69	125.17	91.69	460.44	153.48	121.11
BGD-H007	36–41	93.19	130.29	93.19	474.88	158.29	320.33
BGD-H008	42–47	94.06	133.32	94.06	483.38	161.13	256.29
BGD-H009	48–53	95.55	132.44	95.55	484.31	161.44	283.82
BGD-H010	54–59	92.92	131.79	92.92	477.73	159.24	296.42
BGD-H011	60–65	98.05	135.98	98.05	497.15	165.72	313.81
BGD-H012	66–71	93.72	127.41	93.72	469.42	156.47	310.71
BGD-H013	72–77	68.21	100.06	68.21	358.14	119.38	306.16
BGD-H014	78–83	2.14	3.20	2.14	11.38	3.79	244.86
Total capital envelopes	—	898.82	1415.70	898.82	4938.03	1646.01	—

Displayed two-decimal values can differ from full-precision totals by rounding. The six-month CSVs continue through the entire operating/repayment horizon, including periods with no new capital but interest, principal, liquidity draws, reserve movements and cash sweeps. Every tranche independently reconciles its capital sources and uses.

Reference six-month requirements · blended six-month requirements · commercial gap-credit six-month requirements · independent machine-readable calculation

Financing routes to investigate in Iraq

Route	What it can finance	Cashflow treatment and next requirement
IQD green bonds / green sukuk	Qualified electric transport, solar and charging capital	Replace conventional debt. Define eligible use, segregate proceeds, obtain independent review and report allocation/impact. Sukuk requires an approved asset/usufruct, legal and Sharia structure; the model does not assume a qualified sukuk issue.
GCF / GEF preparation and climate capital support	Climate studies, proven incremental mitigation/adaptation expenditure and eligible capital	Grants reduce debt only when approved; concessional debt still repays. Seek an accredited entity and Iraq NDA support with an emissions/resilience baseline. No grant is booked in the reference.
CBI renewable-energy initiative	Qualifying solar installations under the latest participating-bank rules	Obtain bank confirmation of borrower/project eligibility, current caps and terms. This existing Iraqi route is not assumed available for the whole metro or booked as a new capital source.
MDB / IFC credit enhancement and IQD syndication	Mobilising Iraqi banks and institutional bond investors	Guarantees improve placement or terms, supply no standalone cash, cost fees and may create sovereign contingent liabilities. Confirm IQD availability, hedge pricing and lender limits.
Station-area joint development and development rights	A contracted contribution towards stations/access or unrestricted net proceeds	Independently value parcels and rights, confirm title and revenue powers, tender transparently, deduct costs and future rent foregone. Early rights proceeds can reduce peak liquidity more than late rents.
Parking/congestion proceeds, employer transport subscriptions and business improvement charges	Incremental local recurring cash after authorised net collection costs	Separate each revenue stream from existing fare/nonfare receipts; validate legal powers, demand, affordability, exemptions and enforcement. No unverified tax power is assumed.
Solar/BESS project company, energy-service or availability contracts	Private financing of an accepted energy package	Remove transferred CAPEX and related city debt once; add the PPA/lease/service payments and guarantees. Compare their NPV with existing electricity/OPEX to avoid a free-CAPEX saving. No assumed saving is booked here.
Manufacturer / tooling leases and supplier finance	Qualified equipment and assembly capacity	Deferred invoices or leases are debt-like obligations with priced future payments. Do not duplicate Chinese invoice credit. Factory customer receipts and actual margins need a consolidated business case.
Diaspora and institutional IQD issues	Local capital subscription within the bond programme	These are purchaser channels for existing bond face, not another capital source. Disclose investor FX risk, maturity and redemption; no automatic rollover or compulsory pension purchase.
Carbon / results-based climate receipts	Verified additional emission reductions after delivery	Treat as contingent upside until methodology, monitoring, ownership and an offtake are contracted; no carbon revenue in the reference or blended case.

Priority: first align civil spending and production with accepted openings, then seek eligible cheaper/longer capital financing and contractual early rights proceeds. Obtain a funded recurring-revenue plan before borrowing a multi-trillion-dinar gap facility. Green finance and creative structures do not cure an unfunded timing mismatch or unsupported demand on their own.

Primary evidence checked 3 October 2026

The GCF Iraq country profile lists two projects with USD 49.5m total GCF finance and USD 5.2m approved readiness support when checked; these are existing country programmes, not Baghdad rail allocations. The Iraq country programme provides national priorities. An additional USD 25m application target is therefore a sensitivity requiring a new proposal, not money already available.

GCF investment framework and financial instruments support the grant/concessional/guarantee concepts and require viable revenue-generating activities for loans. ICMA Green Bond Principles require a transparent use-of-proceeds process, allocation controls and impact reporting; they do not promise a reduced coupon.

The CBI renewable-energy amendment of April 2025 establishes a local financing route to investigate for eligible energy packages. Its limits and bank eligibility cannot be extrapolated into a multi-trillion-dinar metro facility. IsDB green/sustainability sukuk guidance supplies a structuring reference, not a Baghdad issuance mandate.

World Bank guarantee programme supports foreign/local-currency obligations subject to project and sovereign approvals. IFC local-currency syndications explains domestic parallel loans and hedge structures; it does not establish an IQD loan offer for OSR. IFC Iraq investment activity establishes an existing country presence, not project eligibility or earmarked money.

World Bank land-value-capture guidance and innovative revenues provide structuring examples, not Iraq borrowing/tax authority or a Baghdad parcel valuation.

World Bank integrated-fare guidance supports examining peak/off-peak, transfer and integrated ticket structures. Its Bogotá affordability study illustrates the need for household-specific demand evidence; neither calibrates the -0.30 sensitivity for Baghdad or endorses a 5% annual increase.

Remaining modelling and term-sheet limits

- Six-month bond placement envelopes settle monthly at par; they are not advance-funded bond sales. Advance issuance changes carry, reserve and interest costs and needs a separate cash forecast.
- Full unrounded placement values reconcile; illustrative IQD 1 million unit counts round upwards and are not executable orders. No auction price, investor demand or mandate is established.
- A green label alone uses unchanged domestic pricing. The lower coupon/longer tenor and 0.5% annual guarantee fee are explicit sensitivities, not donor or lender terms.
- Climate capital grants replace domestic capital borrowing only, against candidate local expenditure. They do not pay old debt interest, imported invoices already funded 50:50, or increase government cash above 25%.
- Incremental land/rights/local receipts are zero in the reference and must be genuinely additional to existing fare, retail, rent and advertising forecasts. Net proceeds after costs, title, statutory powers and independent valuation require qualification.
- The supplemental IQD facility pays its own interest and origination fees. Surplus is swept to repayment; terminal balances are exposed rather than assumed refinanced or forgiven.
- The illustrative IQD facility has a 13 trillion maximum outstanding commitment; residual cash requirements beyond that cap remain explicitly uncovered. No committed lender or future market capacity is assumed.
- Supplemental facilities have no extra DSRA or undrawn commitment fee and no fixed amortisation before the terminal maturity; term sheets can materially increase costs. Long draw availability and the terminal balloon are uncommitted assumptions.
- City/plant cash is pooled subject to legally permitted transfers and creditor consent. Factory income/OPEX remain outside the capital-only plant appraisal; no free factory profit is credited.
- All non-Chinese debt is modelled in IQD. Foreign-funded concessional programmes need confirmed IQD on-lending or a priced hedge; an unhedged USD development loan would change the currency strategy.
- The reference has constant nominal fares, OPEX and FX. Dedicated pricing cases explicitly index fares, OPEX and income independently from financial close. Capital escalation, FX inflation pass-through and separate lifecycle replacement inflation remain outside this sensitivity; full nominal appraisal must include them.
- Pricing cases retain fixed nominal core and gap debt terms; supplier terms, floating rates and imported OPEX exposure need qualification. New net revenue targets are held nominal rather than assumed inflation-protected.
- Price elasticity is an uncalibrated response to fare relative to indexed income; no transfer, peak switching or low-income household demand survey is modelled. Off-peak trip increases are bounded by the same practical capacity, not new physical capacity.
- Early repayment is a contractual sensitivity, not an exercised call right. Bank/Chinese loan eligibility, bond call or voluntary buyback rights, notice, price and premiums require actual terms. The noncallable case forbids all early bond principal. Illustrative minimum draw ages are 6 months for bank credit, 12 for Chinese credit and 24 for bonds; these are not actual covenants.
- Cost priority is a coupon/guarantee-fee heuristic, not a globally optimal treasury strategy. Premiums, tax, remaining maturities, future draws and FX can change the best order; no refinancing or capital-issue cancellation is modelled.
- A three-month OPEX buffer is funded before discretionary repayment in every early-policy comparison. It is retained at the terminal horizon and earns no income. Original baseline cases have no such additional buffer; savings are compared with a buffered gap-only case as well as the original.

Editable options · existing capital assumptions · original city appraisal · Baghdad-only programme

Baghdad financing redesign — 2026-10-04

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/financing-redesign/README.md

> Original catalogue or earlier scope reference. The figures and policies below retain their original assumptions; they are not the latest Baghdad staffing, depot, procurement or funding basis. See the current Baghdad recalculation and current city summary. National/portfolio totals have not been repriced with that conditional Baghdad option.

Twelve executable alternatives use 14 lines, 513.3355 km, 159 stations and 809 six-car trains. Financial close, title, investor equity, insurance eligibility and physical acceptance remain unestablished. The existing baseline remains the planning reference. Conditional full-line openings are 40, 41, 43, 42, 44, 44, 50, 55, 59, 64, 69, 73, 78, 81 months; the independently operable first section is a separately costed sensitivity, not silently added to these full-network cases.

Results and what they mean

Case	Peak aggregate IQD liquidity tn	Cumulative missing IQD funding tn	Terminal IQD gap debt tn	Lifetime interest/fees USD bn eq	Consolidated resource NPV USD bn eq
reference	2.158	0.000	0.000	3.504	n/a
opening_linked_debt	1.462	0.000	0.000	4.035	n/a
development_rights_1bn	1.100	0.000	0.000	3.019	n/a
insured_15year_debt	1.712	0.000	0.000	4.519	n/a
integrated	2.311	0.000	0.001	4.428	-1.316
integrated_low_ppa	2.078	0.000	0.001	3.906	-1.316
integrated_high_ppa	2.608	0.000	0.001	5.154	-1.316
integrated_public_availability	1.947	0.000	0.001	3.588	-1.316
integrated_joint_downside	3.400	30.574	3.400	18.540	-5.348
integrated_rental_small	2.311	0.050	0.001	4.447	-1.321
integrated_rental_medium	2.311	0.040	0.001	4.454	-1.298
integrated_rental_medium_downside	2.311	0.340	0.001	4.493	-1.354

Next financing priority: obtain opening-linked IQD terms and independently valued station proceeds, then compare quoted PPA prices against continued ownership. The integrated high-PPA structure shifts substantial cash to the energy company while rail remains short; private equity needs an acceptable return as well as debt repayment. The tested property margin improves resource NPV before land cost, but the assumed in-kind land cost more than offsets that improvement. None of these scenarios establishes financial feasibility.

Summary with exact figures and appraisal definitions; every case has a complete JSON model, monthly native-currency CSVs, six-month placements and contractual loan-vintage dates. Missing funding must be filled before any reported conditional repayment date is credible. Independently funded entities cannot use another company's cash without a separately approved agreement. Liquidity caps are allocated once: 65% rail, 10% energy, 5% factory and 20% development. The sum is IQD 13.0tn reference; the downside divides the existing IQD 4tn cap in the same proportions.

Integrated borrower	Capital uses USD m eq	Private equity IQD bn	Peak gap IQD tn	Missing funding IQD tn	Terminal gap IQD tn
rail	6810.797	0.000	1.968	0.000	0.000
energy	815.740	212.093	0.016	0.000	0.000
factory	324.750	84.435	0.290	0.000	0.001
development	3400.000	1105.000	0.368	0.000	0.000

These are pre-tax sensitivities. Corporate tax, duties, VAT, sponsor equity hurdle rates, dividend restrictions, lender DSCR and termination liabilities need quoted/legal terms. Surplus is swept after monthly OPEX, scheduled debt service, reserve funding and the existing lock-ins/premiums. A positive cash balance or an annuity model does not establish bankability. The factory has no assumed national orders or residual asset sale.

Opening-linked credit and insured green instruments

Bank amortisation becomes 240 months and ordinary bonds 216 months at the unchanged 9%/8% reference rates. Monthly draw cohorts use the latest opening of the assets included in that cohort; shared contracts use full opening. This conservative aggregation needs actual line-linked facility documents. Interest is paid throughout grace, funded by the separate IQD liquidity account if cash is insufficient; no interest is silently forgiven or capitalised. Arrangement/commitment charges, buffer and surplus rights remain priced. Chinese USD loan terms remain unchanged.

Loan-vintage and six-month CSVs show actual draw, first principal and final contractual principal dates. Cheaper/longer debt changes liquidity and finance cost, while core unlevered NPV stays USD -1.670bn. Core excludes grant, development rights

and new net receipt targets.

ICIEC's official product describes sovereign/sub-sovereign sukuk with coverage up to 15 years. The tested insured instrument has grace plus amortisation capped at 180 months from each draw, replacing the previous 48+240-month assumption in that sensitivity. A 6% coupon, 0.75% annual insurance and 1% annual IQD conversion/on-lending cost are assumptions, not insurer pricing or proof that an IQD issue is eligible. No climate grant is booked in insured/integrated cases. Obtain written eligibility, binding terms and a priced local-currency route before using an institution's name in funding commitments.

Station-area partnership and public land

15 station candidates are selected only by network archetypes, with actual station IDs/coordinates. Parcel title, legal ownership, survey area, encumbrances, independent values, displacement and developer interest are null/unverified. No selection is a claimed commercial valuation. First investigate 10–20 sites through independent market/title/safeguards work and competitive long leases or voluntary pooling.

USD 1bn total net development-rights replaces USD 300m; the increase is USD 700m. Six equal tranches begin month 24, every six months. Receipts are IQD using the model's 1,300 IQD/USD reference; they are not USD debt or dollar-indexed obligations. In the integrated sensitivity developers finance both rights and buildings. The hypothesis is 1.2m m² land, FAR 2.5, 3m m² buildings at USD 800/m², USD 4.2bn sales in months 60–96 and 5% sales costs. There is no rent revenue additionally booked against sold floor space. Public land's assumed USD 600m opportunity cost is an in-kind exposure, never spendable cash. All prices are unverified and require an Iraqi valuation/tax review. The downside halves rights, reduces sales 30%, delays both 24 months and escalates construction.

Rights payments are a developer use and a rail receipt. They cancel in consolidated appraisal; they cannot also be counted as programme profit. Developer equity is 25% of its rights/build uses, and its borrowing is IQD bank credit. Property debt/transaction costs and missing funding are included in its own ledger. World Bank land-value guidance and MTR's model provide precedents, not Baghdad valuations.

Energy and manufacturing contracts

Dedicated solar capital is transferred once, without inventing an additional EPC removal. Charging/microgrid equipment and its existing EPC remain rail assets. The railway pays a PPA of IQD 104/kWh, indexed 5% annually; the annual planning purchase is 1.942bn kWh, USD 155.345m equivalent at reference prices. Dispatch/load follows open-line weights, not fare demand. Energy receives exactly what rail pays. Existing solar maintenance transfers once; a further 0.5% of solar capital/year is an explicit renewal-spending sensitivity. Timetable-derived annual generation is not 8760-hour dispatch proof: grid backup, curtailment, storage adequacy and metered purchases require qualification. Export sales are zero, with no signed buyer. Low/high PPA cases redistribute company cash and leave the consolidated resource NPV unchanged. World Bank PPA guidance explains the contractual financing route.

Energy and factory each test 20% private IQD equity. Government remains 25% of original physical rail/energy/factory capital. The factory receives Baghdad's existing train invoices and pays the same resource budgets one month earlier, exposing working capital. Original train prices already include manufacturing labour, QA, ordinary overhead and warranty; 1,044 production staff wages are not added again. An additional USD 250,000-equivalent capacity fee per train, annually indexed and paid in IQD, is a separately visible investor-return sensitivity. It cancels on consolidation but increases rail's operating obligations. Warranty cash is locked at 5% of cumulative train payments until 12 months after the final invoice, then released; it is cash restriction, not a duplicate warranty expense. Fifteen caretaker staff and 0.25% plant carrying cost are added during that tail. Plant costs, working-capital funding, interest and outstanding debt at backlog close remain visible in integrated.json. New orders and salvage are zero until contractual.

At the illustrative capacity fee, factory cash remaining at contract/warranty close is USD 0.000m against USD 64.950m invested equity. The 15% equity-hurdle NPV is USD -56.391m. Debt repayment therefore does not establish an investable factory partnership. Developer equity-hurdle NPV is USD -194.940m. These close-date distribution diagnostics deduct any remaining debt and disclose missing funding; no dividend or sponsor return is assumed to be committed. Competitive pricing, equity terms and a bankable capacity contract require further work.

Appraisal and public exposure

Three retained under-viaduct rental variants add a fifth independently financed entity to the integrated case: small, medium and lower-rent/prolonged-vacancy medium. The civil-linked register, pilot unit, costs, leases and hazards have no accepted site/demand evidence. External rent adds revenue; actual collection follows fit-outs, physical line availability, initial rent-free periods, occupancy ramp, arrears/recovery, tenant turnover and lease-end refunds. Fit-out capital, occupied/vacant maintenance/insurance, 12-year refurbishment and a standalone 15% cash-tax proxy are included. No station sale parcel also earns rent; no existing kiosk income is duplicated. Tenant deposits are restricted cash and matching liabilities, never construction finance or revenue.

The rental partner supplies 25% fit-out capital as private IQD equity and the residual is IQD bank capital, with opening-linked principal and 240-month amortisation. It has zero new operating-gap capacity: missing construction interest, fees or reserves remains an explicit funding requirement. The original four borrowers keep their existing allocated caps; combined cap remains 13tn. Rental transfers to rail are zero until an approved agreement exists, so rail cannot silently use the new company's cash. Each borrower exports monthly native ledgers, six-month placements and loan vintages. These ring-fenced structures are alternative to the wholly owned holding rental cases, whose subscriptions replace rather than add partner equity.

Consolidated resource NPV adds collected rent less its OPEX and physical fit-out capital, before financing/tax, to the original resource case. A separately labelled after-rental-tax diagnostic deducts rental tax once; the original four businesses retain their existing before-tax basis. Additional rental-site rights/land opportunity cost is unknown, so the existing station-land subtraction is not a complete new-site appraisal. The 200,000 m² area illustration is blocked as unmapped and supplies no integrated cash.

Core rail/energy/factory unlevered NPV: USD -1.670bn. Rail entity NPV including its PPA, capacity fees and internal rights: USD -1.508bn. Consolidated resource NPV including property sales/building costs and additional renewals/caretaking: USD -1.316bn; including land opportunity cost: USD -1.783bn. Debt/equity/grants are financing transfers and excluded from resource NPV; intercompany transfers cancel to the cent. Nominal discount is 13.4%, combining 8% real and 5% general inflation.

Government original-capital cash is USD 1.988bn equivalent (25%). Its USD import cash is USD 898.824m and Chinese USD credit is USD 898.824m, splitting imports 50:50. All other debt, private equity, property payments, fares and subsidy payments are IQD. Original programme USD capital intensity therefore remains 22.61%; SPV transfers do not remove those USD import requirements. Property construction is assumed contracted in IQD; actual imported content/FX exposure needs quotes.

Public cash capital PV is USD 1.273bn; in-kind land PV USD 0.467bn. The optional availability case pays USD 100m equivalent/year at full opening, indexed 5% and phased by available lines; its PV is USD 0.794bn. It sits outside the 25% capital limit and needs appropriation, service performance terms and affordability review. Recipient NPV adds public cash explicitly; the public-plus-private resource appraisal cancels that transfer. Contingent bond/guarantee/termination exposures remain unquantified rather than assumed zero. World Bank public-support guidance describes these obligations.

Existing fares and kiosk/advertising receipts are retained with their original demand/price/OPEX indices. Integrated cases omit the old USD 25m grant and USD 25m/year new-net-income targets. Fibre, naming rights, parking and employer contracts require incremental net quotes; prepaid fares change timing of existing receipts and do not create revenue twice. At 13.4%, USD 50m/year for 20 end-of-year payments has PV USD 342.962m at the beginning of the payment period, lower when discounted back to close.

Iraqi financing routes and 90-day evidence programme

The World Bank's USD 930m national railway project is a separate Umm Qasr–Mosul programme through Baghdad, not metro money. IFC's September 2025 Iraqi announcement combines investments and engagements across industrial/property/financial partnerships, not an available USD 1bn fund for this proposal. ICIEC's 2025 impact report records reinsurance for Iraqi substations; it does not commit rail credit. Sources checked 3 October 2026. GCF's published Iraq country programme and transport programme establish potential application routes, not an award. Climate funding requires emissions baselines, eligibility, accredited-entity engagement and actual terms; no grant is presumed.

Six workstreams cover lender terms, land/valuation, PPA, standalone factory, existing independently operable phase, and independent model/public-affordability review. Due days are relative to an actual approved start. Named people, legal opinions, quotes and accepted evidence remain pending. ERPNext Task JSON is a draft native-import package with stable evidence subjects, not live IDs or accepted evidence. Local deployments record actual IDs privately and preserve assignments/status when refreshing descriptions. No external lender/developer contact, issuance or expenditure is authorised by these model files.

Later construction commitments require executable native-currency funding, separately accepted physical evidence and a conservative demand case. Integrated joint downside retains the existing 55/106-month physical stress and adds weaker property conditions: missing funding is IQD 30.574tn, terminal gap debt IQD 3.400tn. The proposed structure is a market-test programme, not a claim that the financing gap is solved.

Regenerate with `.venv/bin/python tools/automation/baghdad_financing_redesign.py`; validate exact inputs/outputs with `--check`. No new financial commitments or physical release are implied.

Baghdad mapping inputs and neighbourhood access

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/mapping/README.md

Source-selected planning evidence. Population counts remain native; no averaging across datasets or scaling to the catalogue total.

Population source	Sector	Native population	Within 1 km of a station	Outside station circles
worldpop-2026	city	8,597,980	30.0%	6,014,797
worldpop-2026	northeast	2,197,150	38.4%	1,353,322
worldpop-2026	west	1,459,449	29.6%	1,027,430
ghsl-2025	city	7,347,247	30.7%	5,090,413
ghsl-2025	northeast	1,159,005	35.6%	745,888
ghsl-2025	west	1,908,545	33.2%	1,274,202
worldpop-2020	city	5,985,974	27.7%	4,325,387
worldpop-2020	northeast	665,396	35.3%	430,612
worldpop-2020	west	1,673,741	25.1%	1,252,936

Sector rectangles are explicit planning areas, not official neighbourhood boundaries. Station circles are radial screens, not surveyed walksheds.

The selected WorldPop 2026 epoch is an alpha modelled projection, not a current census. GHSL 2025 is a separate model comparison. Potential graph access within 1 km: 11.5%, with 105/116 unique stations snapped within 150 m. Snaps, entrances, foot permissions, sidewalks and accessible paths remain unverified. Roads connect only through shared OSM node IDs; paths outside the retained bbox are unavailable.

Dated street extract: 2026-10-09; 308,905 pedestrian graph nodes, 362,348 edges; 53,174 mapped building ways. Incomplete building mapping never means an empty neighbourhood.

Measured traffic counts, speeds and origin–destination data are unavailable. Iraq is absent from current TomTom classic and Orbis Traffic API coverage tables. No measured traffic benefit or passenger demand is inferred.

Selected sources, licences and sector policy · Machine-readable report and hashes · Street geometry · Research and integration policy.

Baghdad programme recalculation

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/programme-recalculation/README.md

Controlled planning basis, 2026-10-04; unquoted and uncommitted. This is the latest staffing/depot/local-production review. Existing cost and financing cases remain historical comparators; no survey, manufacturing qualification or lender commitment is created.

Operating people and wages

The 159 stations require two concurrent staff during two normal eight-hour shifts: 636 daily shift assignments, not a headcount of people working every day. Retaining 20.5 service hours also requires 4.5 hours of separately funded late cover per station. Weekly cover, leave, training, sickness and handovers give 1572 station-cover FTE, within 3523 permanent operating FTE. Full-operation loaded payroll is IQD 83.415bn/year (USD 64.165m equivalent), indexed thereafter with OPEX.

The employee median is IQD 614,000 from the 2021 Labour Force Survey, cited in the IMF 2023 report. Applying an assumed 5% annual index to 2026 gives IQD 783,637; the general-worker floor is IQD 1,175,455/month, 50% higher. Technical, supervisor, senior and director roles use 2.25/3/4/5 times that indexed proxy. This is a historical observation plus an editable index, not a measured current national median. Employer costs and overtime are priced separately. Roles and recruitment/cohorts retain appointment and competence gates.

ERP planning drafts map the revised Staffing Plan, depot Project/Asset budgets and factory Workstations to native business records; they create no Employee, appointment, transaction or approval.

Construction workers, factory production/support workers and permanent operators are separate populations. Construction applies explicit trade-crew allowances to actual work-package resource lanes/dates, counting overlapping use of the same lane once; its peak is 429 FTE. It is not a measured complete contractor crew schedule; revised depot and upstream factory construction crews still require their own work-package schedules. Construction labour is already inside contract rates; its unknown payroll-content overlap remains unpriced, preventing a duplicate charge. A separate wage-content stress increases assumed labour content (20% of affected contracts) by 50%, with incremental EPC once; it is an overlap sensitivity, not proof of contractor wage contents. Existing assembly wages are replaced using an explicitly assumed embedded payroll credit; component process labour is inside component unit prices. Paid production capacity above those productive hours is added separately, never charged twice.

The final-assembly establishment is 1215 production FTE plus 183 support FTE. The scheduled order requires 59 paid months from month 18 through month 76; payroll follows this calendar, rather than dividing the order by a theoretical steady-state rate and losing ramp-up/idle months. Each upstream plant also funds its whole production establishment for that period, with the productive labour already in unit prices deducted once. These jobs are additional to permanent railway operations; no national order or permanent post-order subsidy is assumed.

One depot per line, full-size trains and fleet

There are 14 line-local depots, providing 809 storage slots for all six-car revenue/spare/reserve trains. Slots use the actual 111 m train plus 10 m clearance. Workshop bays follow annual bay-hour demand and usable bay capacity; parking slots are not maintenance bays. Station storage is an option without any capacity credit in this case. Full overnight-to-morning launch and throat conflicts remain unvalidated.

Line	Trainsets/storage slots	Tracks	Workshop bays	Land envelope ha	Reference USD m
line-1	86	29	12	11.50	27.045
line-2	95	32	14	12.72	36.828
line-3	90	30	13	11.94	28.589
line-4	76	26	11	10.40	24.991
line-5	86	29	12	11.50	27.045
line-6	95	32	14	12.72	30.328
line-7	72	24	10	9.62	23.282
line-8	89	30	13	11.94	28.559
line-9	45	15	7	6.32	17.210
line-10	15	5	3	2.57	9.624
line-11	14	5	2	2.45	8.335
line-12	15	5	3	2.57	9.624
line-13	18	6	3	2.89	9.879
line-14	13	5	2	2.45	8.305

Total depot reference: USD 289.645m, replacing the original USD 8m exactly once. Storage, workshop and access tracks, points, drainage/access, workshop shell/equipment/services, yard lighting/fire services, wash plant, wheel lathe,

offices/stores and rescue/quarantine are visible. Access-track length is an editable allowance, not a connected site alignment. Existing depot PV/BESS is retained once. Land/title, utility relocation, actual soil/foundations, installed charger/grid upgrades and tax/duties remain unpriced. Item register.

Core elevated stations

The reworked core has 84 station platforms with raised-structure/access scope. The register replaces their catalogue allowance with a separately stated elevated reference, adding USD 292.500m direct, with incremental EPC and operating maintenance once. Elevated interchange allowances are already present where catalogued; they receive no duplicate uplift. Other core stations use a 30% structure allowance plus USD 1m for vertical access. These are unquoted allowances, not released multi-level junction or station designs. Core station register.

Iraqi component manufacture

Product	Network quantity	Required units/year	Cells	Production/support FTE	Factory capital USD m	Unit make/buy USD	Whole-order margin USD m
bogie	9708	2582	30	140/21	42.568	15156/25000	41.595
motor-inverter-set	9708	2582	20	70/11	29.253	11078/15000	2.088
battery-225kwh-pack	4854	1291	8	38/6	25.577	29828/45000	42.977
door-cassette	19416	5164	12	42/7	15.351	2023/3333	6.269
window-cassette	29124	7746	9	32/5	22.392	687/1111	-14.256

Make/buy inputs price imported process machinery, Iraqi buildings/site work, qualification, residual imported inputs, local materials, graded labour, process overhead, fixed support and plant maintenance. Cells are sized to the existing train factory's required production rate, not a small demonstration line. The order is 1.092 GWh of gross onboard packs. Battery pack assembly is evaluated; imported cells/BMS remain. Bogie wheels/axles/bearings, motor inverters/magnets, door safety electronics and glazing feedstock also retain imports. Local manufacture does not mean zero USD input.

The buy case itemises these five imported completed products; the old blanket vehicle import percentage is retained only for other parent scope. This prevents replacing already-local scope with a second import credit. The all-product and positive-margin selections are separate unquoted cases. Whole-order margins are before finance/tax/risk; vendor prices, license, QA, process yield, supply commitments and first articles must validate them. No future national order pays Baghdad debt or makes an uneconomic line look profitable. Serial component qualification is assumed within the 18-month readiness target, not proven. A six-month supplier delay shifts rolling-stock invoices and opening/service cash, extends the full operating horizon and prices idle production payroll. A 25% raw-input price stress preserves the same selected facilities. Rejected/reworked product still needs a measured production replay. Existing final assembly tooling remains priced; upstream machinery is added without an unproven overlap credit.

Paid journeys and transfers

Revenue remains a capacity-led sensitivity rather than a surveyed OD forecast. The reference uses one boarding per paid journey, a zero-transfer upper-bound assumption. Integrated-fare sensitivities use 1.25, 1.5 and 2 boardings per journey: only fare receipts are divided; kiosk/rental/advertising receipts, service, staffing and energy are retained. These are uncalibrated factors, not estimates of Baghdad travel. Physical access and surveyed OD/section loads must qualify any adopted demand forecast. Demand handoff.

More elevation and fewer bends

The current main design adopts the straight central elevated alignment. The screening policy allows up to 65% elevated and at least 25% at grade, subject to site/design acceptance. Baghdad currently has 56.73% elevated. Investigation windows around all 49 exceptional segments add 7.924 km of candidate at-grade conversion, reaching 58.27%; overlapping intervals are merged and existing viaduct/bridge lengths excluded. Approach length is at least 300.0 m from assumed height/gradient.

Elevation alone removes no horizontal bend. Wider-radius geometry, station moves, ROW, vertical alignment, ramps, egress, ground/utility evidence, crossings and whole-life costs must be designed together. The additional-elevation case includes only its base allowance; routing scores are excluded from money and the old 25%/50% penalty-removal cases are retired. Special designs and consequential installed costs remain unknown. The added standard civil allowance uses the conservative simple-span bearing index. Candidate intervals.

Funding and mezzanine comparison

Government capital remains exactly 25% of the scenario total. Imported invoices receive 50% government USD cash / 50% Chinese USD credit. Remaining government cash, bonds, senior bank/gap credit and mezzanine are IQD. Government USD payment dates follow machinery/input invoices; the remaining appropriation is allocated proportionally to local invoices, so a machinery-heavy month need not be falsely limited to 25% government cash. Chinese supplier origin and export-credit eligibility are assumptions requiring vendor/lender confirmation. China Exim describes buyer credit for Chinese products, technologies and services; machinery eligibility is not a loan commitment (official product description).

The positive-margin local-production senior case's capital-only sources are shown below. They sum to its capital uses; gap credit, interest/fees, reserve funding and operating receipts are additional cashflows in its monthly ledger and six-month placement schedule. Ordinary and green bonds are separate placements, never added again to a combined bond figure. Bond face units are IQD 1m; rounded placement envelopes are not additional cash raised.

Capital source	Currency	Native amount	USD equivalent m
Government import cash	USD	1,008,216,375	1008.216
Government local cash	IQD	1,505,948,192,086	1158.422
Chinese capital credit	USD	1,008,216,375	1008.216
Ordinary capital bonds	IQD	4,182,186,517,697	3217.067
Green capital bonds	IQD	1,147,843,845,665	882.957
Senior bank capital credit	IQD	1,776,676,787,787	1366.674
Conditional climate capital grant	IQD	32,500,000,000	25.000

Mezzanine replaces 10% of domestic residual capital borrowing; it is not extra capital on top of the uses. The IQD case has 6% cash coupon, 4% PIK, 2% arrangement fee and a 15-year balloon from each draw. Deferred cash coupon and PIK increase outstanding debt until maturity. Thereafter the overdue principal is retained with separately disclosed simple 10% contractual-interest sensitivity; arrears are not compounded and the balloon is not silently extended. Cash junior payments require senior DSCR of 1.20, funded reserves and actual residual cash; new gap draws/unfunded support cannot pay junior debt. No conversion, fresh equity or automatic refinancing is invented. Unpaid maturity balances/defaults remain visible. Mezzanine is structurally junior to senior debt and senior to equity, as described by UNCITRAL; the rates here are project sensitivities.

Matched case / six-month placement schedule	CAPEX USD bn	USD capital intensity	Peak IQD gap tn	Terminal all debt IQD tn	Unfunded IQD tn	Junior defaulted vintages
revised_scope_buy	8.795	26.82%	5.109	0.000	0.000	0
local_all	8.677	23.12%	5.059	0.000	0.000	0
local_positive	8.667	23.27%	5.033	0.000	0.000	0
local_positive_mezzanine	8.667	23.27%	4.267	0.000	0.000	81
local_positive_commercial_gap	8.667	23.27%	12.041	0.000	0.000	0
local_positive_mezzanine_stress	8.667	23.27%	9.117	21.346	0.000	83
additional_elevation_base_allowance	8.731	23.20%	5.114	0.000	0.000	0
local_positive_raw_price_stress	8.770	23.74%	5.128	0.000	0.000	0
local_positive_supplier_delay	8.667	23.27%	5.366	0.000	0.000	0
construction_wage_content_stress	9.267	22.77%	5.927	0.000	0.000	0
integrated_fare_1_25_boardings	8.667	23.27%	9.220	0.000	0.000	0
integrated_fare_1_5_boardings	8.667	23.27%	13.000	10.554	0.406	0
integrated_fare_2_boardings	8.667	23.27%	13.000	13.000	12.070	0

The 4 positive-margin process options reduce capital from USD 8.795bn to USD 8.667bn, and imported invoice exposure from USD 2.358bn to USD 2.016bn. Half of that exposure is government USD cash and half Chinese USD credit; all other capital funding is IQD. The historical 148 km / USD 18bn third-party benchmark has a different scope and assumed full foreign-currency financing; this is a planning comparison, not a like-for-like tender saving. The reworked 513.3 km network's 40.5% legacy routing-demand cell fraction is not population access. Its former resident multiplication is retired; see the native population and transfer audit.

Current financial conclusion: the positive-margin senior case records IQD 0.000tn of residual unsourced support after assumed facilities and retains IQD 0.000tn of debt at the horizon. Its company NPV before finance is USD -3.246bn at the assumed nominal discount rate. Adding mezzanine does not change the underlying operating return: it leaves IQD 0.000tn of

total debt and 81 unpaid junior vintages. It is not recommended as a cure for the funding deficit. Fare and OPEX indexation, kiosks/advertising and inherited additional receipts are already included; new verified capital, affordable revenue or accepted scope savings are still needed.

All cases use the same revised staff, depot and indexed fare/OPEX assumptions. Six months of scheduled senior service are reserved from the first draw; three months of OPEX plus explicit industrial working capital are restricted. This revised reserve policy differs from older reference cases, so compare matched cases in this table when judging mezzanine. Concessional 2% gap funding, grants/rights/additional income and enhanced green coupons are uncommitted; the 8% gap and high mezzanine-rate case expose that dependence. There are zero dividends. Cash/principal/PIK identities, maturity risk, monthly draws and six-month bond denominations are retained for every case.

Regenerate with `.venv/bin/python tools/automation/baghdad_programme_recalculation.py`; verify with `--check`. Complete installed scope, construction wage-content adjustment, local-pay survey, validated demand/timetable and supplier/lender agreements remain open.

Baghdad qualification, funding gates and first operating section

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/qualification/README.md

This package turns the review findings into executable draft work. It supplies no physical acceptance, supplier quotation, named owner assignment or financing commitment. The full 809-train Baghdad baseline uses the current generated route and opening schedule.

Temporary first-article industrial package

Site block layout, quantities and reconciliation, and RFQ schedule specify six prototype cells, 4,960.0 m² process floor, 6,696.0 m² total covered floor and 16,740.0 m² fenced-site allowance. The metric 180 x 93 m envelope is a reference arrangement, not a selected or surveyed site. Access/fire widths, hazardous materials, drainage, foundations, heavy-haul route, crane coverage and utilities require detailed vendor/civil drawings. No production acceptance paths are counted at this temporary site.

The quantity-based direct subtotal is USD 19.873m; contingency USD 3.975m; allocated total USD 23.848m. USD 11.152m remains unallocated within the existing USD 35m allowance. EPC remains USD 2.45m. Nothing in this package is added a second time to the USD 37.45m recovery case. Rates are editable assumptions, not received quotations or realised savings.

Vendor interfaces: bogie/body datum and lifting reactions; composite mould temperature/cure traceability and ventilation; door/window fit and tolerance stack; battery transport SOC/fire segregation and HV isolation; electrical test connectors and calibrated instruments; ERP lot/serial traveller and revision interfaces. RFQ returns must identify imported tooling versus Iraqi buildings/labour, actual origin, tax, delivery slots and financed invoice eligibility; the recovery model's 20% tooling/import fraction remains unqualified until these are reconciled.

Transfer sequence: permanently identify TFA-C01 through C06 tooling in the asset register; release the prototype traveller and first-article configuration; move the complete six-car train on certified frames/haulage with clearance and escort approval to permanent acceptance at day 390; retain all temporary tools until prototype acceptance and production continuity are secured; lock out, pack, transfer, reinstall and recalibrate each cell into its named permanent work centre; validate first-off dimensions/HV tests before release. Permanent plant tooling is already fully budgeted: temporary equipment is incremental and any reuse credit needs an approved asset/cost change, never an automatic reduction.

Readiness gates: survey/land and supplier interfaces before procurement; civil/fire/utility/lifting acceptance and calibrated equipment before day-260 assembly; permanent segregated test paths and bays before day-390 dynamic tests; the additional 60 working-day first-article qualification remains fixed; no series release before signed acceptance. Day 260/390 and month-36 first service are conditional dates, with independent ERP evidence tasks below.

Executable evidence work and ERPNext

Ten source-bound work packages map every delivery qualification row to an accountable role, measurement fields, units, numeric criteria, source content revision, ERPNext Task subject and result template. A named accountable person is deliberately pending rather than fabricated. Task import JSON and CSV are drafts; they are not live ERP records or accepted evidence. Reconcile them with the ERP app's `qualification.import_tasks` method; preview is the default. Assign actual users/owners through ERP and preserve assignment when refreshing descriptions.

Populate each BAG-EVID result template using real measurements and raw evidence file checksums. Use `.venv/bin/python tools/automation/baghdad_qualification.py --verify-result RESULT --envelope ENVELOPE --signature SIGNATURE --policy TRUSTED_POLICY`. The verifier rejects null/wrong-unit/out-of-threshold measurements, missing raw files, source changes, absent or unauthorized review and self-review. It reuses the subsystem qualification's externally controlled Ed25519 reviewer policy and envelope (`osr-subsystem-review/1`), with evidence fingerprint = SHA256 of the exact result bytes, configuration fingerprint = package source revision, required role = assessor, acceptance reference and expiry. Trusted reviewer identities/keys must be supplied independently; the generator provides no signing key and no acceptance. An accepted evidence package still does not authorize public operation.

Coverage ratios require item/lot/phase-level underlying files and independent scope review, not an unsupported declaration of 1. Production records must separate elapsed cure clocks and inspection holds from staffed hours. Baseline throughput and the shifted recoveries remain unmeasured. Funding evidence needs the appropriation, actual native-currency term sheets, investor/underwriter placement mandates, collateral/guarantees, maturity/fees, and every six-month sources-and-uses gate, including reserves and interest. Demand work must include a surveyed OD matrix, affordable fare/elasticity tests and net kiosk/advertising concessions after vacancy, collection and management costs.

Denied and interrupted funding

Funding gate cases and each placement-monthly/six-month CSV distinguish requested source face, available fraction, refused amount, invoices held and undrawn capital capacity reserved for deferred invoices. Recovered domestic placement assumes half the ordinary IQD bond placement fails for months 29–34; all unfinished procurement, civil work, production and running tests pause at working day 600 for 130 working days. Recovered export credit pauses the programme at NTP for 130 working days and requires the refused initial USD credit to be placed before imported procurement restarts. The costed schedules and financing are in the delivery-risk cases `domestic_placement_interrupted_recovered` and `export_credit_delayed_recovered`. A USD 10m equivalent IQD local remobilisation assumption plus 7% EPC is added once, with staff/site carrying costs charged in the six suspension months for the already-mobilised domestic case. An export refusal at proposed NTP shifts the entire unstarted programme without buying another construction/production staffed span; only the quoted-later remobilisation/reservation allowance is added. Cure/inspection restart requirements could cost more and need vendor evidence. Longer delay, storage damage, cancellation penalties, FX changes and lender acceleration/default are outside these recovery prices.

The placement files diagnose the original invoice request. Recovered cashflow files retime actual invoice milestones using the held schedule; they must not be added together. The available-capital column is an undrawn capacity requirement, not received or free operating cash. Recovered cases retain uncommitted liquidity for interest and hold costs; it never replaces the refused core capital. Investor commitment/carry fees on undrawn domestic placement are unquoted and excluded. Partial grants/green placements already in the baseline remain assumed; refusing ordinary bonds does not prove green placements will succeed. Bank refusal can be run with the same gate mechanism. Recovered dates and repayment require renewed placement and restart appropriations, which remain uncommitted. Permanent domestic/export refusal has no operating opening or debt-clearance date. Existing debt is not forgiven; cancellation, default/workout, liquidation proceeds and prior-creditor claims require a separate resolution model and are not represented by zero balances. Government is not automatically increased beyond 25%; the original import split stays 50% government USD / 50% Chinese USD loan; only actual import invoices can qualify. Domestic cash, bonds, bank/green/gap credit remain IQD, while foreign donor receipts would have separately identified FX conversion requirements.

Gate	Currency	Refused amount	Recovery assumed
domestic_half_placement_recovered	IQD	160,988,490,050.52	True
domestic_half_placement_permanent	IQD	1,311,563,838,269.65	False
export_unavailable_recovered	USD	12,710,257.32	True
export_denied_permanent	USD	898,823,898.22	False

Separate independently operable first-section option

First-section duty, assets and finance takes the actual first five line-1 stations, from the Mahmudiya terminal to line-1-1848-0797-s014034, spanning 14.034 km. This outer corridor is a demonstrator candidate, not a proven best catchment. It needs its own OD/accessibility study and comparison against better central corridors. Existing station/track/wayside tasks alone are pulled forward within the frozen lane/predecessor graph; their already-budgeted capital is retimed, not purchased again. Other route quantities and the 809-train city fleet remain unchanged.

At the reference six-minute headway and 40 km/h average running speed, the round trip is 74.10 minutes including intermediate dwells and ten-minute charging/turnaround at each end. It requires 13 peak + 2 spare + 1 cold reserve = 16 existing six-car trains. Allocated train IDs are in the data file and must all pass acceptance before section operation. A six-minute section service is an explicit early-phase option rather than the ultimate line's three-minute design demand.

Each end needs 2 independent 121 m clear berths for the 111 m train, No.9 crossover/points, a safeguarded 185 m terminal arrangement, 2 x 2,000 kW chargers and 4,000 kW feeder capacity. Dwell plus clearance occupies each berth for 11 minutes against a 12-minute two-berth arrival cycle. These are geometry/utility requirements pending surveyed access, swept path, braking, interlocking and electrical studies; a generic standard station cannot silently serve as the new terminal.

Round-trip traction plus the conservative auxiliary proxy is 587.9 kWh, compared with 600.0 kWh delivered terminal charge at 90% efficiency. Battery definitions are 1,350 kWh nameplate / 1,080 kWh usable from the controlled six-car profile. The conservative section-specific 40% floor is applied to usable energy, leaving 648 kWh; one-leg duty fits that window. Normal charging margin is only 2.05%. A 10% reduction in delivered charging gives -8.15% balance and is not qualified. Roof PV and intermediate charging are excluded. The battery acceptance power, HVAC electrical demand, gradients, ambient heat, degraded charging and rescue case require measured qualification; the small energy margin is not operational robustness.

The section RFQs provide four incremental packages excluding already-budgeted route, station and fleet equipment. Reuse one original charger per end and add two berth chargers in total, upgrading each end to 4 MW. Independent maintenance/stabling, battery quarantine, lifting, rescue and road access avoid reliance on the distant line-6 depot. The provisional 3.5 ha maintenance site has eight stabling roads of 250 m usable length (two 111 m trains per road plus 28 m clearance) and two 135 x 6.5 m maintenance bays; site rights, turnout ladders, fire/access geometry and measured depot duty

remain open. Turnback/platform adaptation USD 5m, charging/grid USD 8m, maintenance/access USD 12m and controls/acceptance USD 3m give USD 29.96m including EPC, assuming 20% imported direct scope pending RFQs. The section has its own reference 200 operating FTE and USD 10.445m/year OPEX before indexation, replacing the whole-network fixed OPEX before month 41. It covers 16 hours/day over 330 days/year, USD 0.10/kWh grid electricity with no PV generation credit, Iraqi wage proxies, proportional existing civil/solar/signalling maintenance and train maintenance including the existing battery reserve, plus 2% maintenance on additional support assets. These cost/demand/utility assumptions need quotations and surveys. Separately, sixty temporary construction/qualification support FTE are indexed and charged from support readiness until the full line opens, then integrated into the baseline allowance; they are not counted as section operating staff. Civil/support acceptance at day 560 is an additional unqualified support-package assumption, not an existing accepted site.

The first 16 accepted trains finish at day 677; selected civil at day 652; independent support at day 560. The resulting conditional section opening is month 36, including the same three-month commissioning allowance; complete line opening stays month 40. No passengers or fares occur before that opening. Section paid demand is only half the smaller of fleet-capacity and route-length allocation proxies, weight 0.009889, deducted from later full-line demand to avoid double counting. This is a scenario, not population coverage evidence. Opening acceptance must cover section-specific turnbacks/chargers/depot, full allocated section fleet, isolation from works, evacuation/rescue, software configuration, hazards and independently signed operator/authority release; the full-line BAG-EVID-009 remains separate.

Monthly finance, six-month tranches and incremental cost ledger include early baseline invoice timing, new support capital, indexed support payroll, fares/nonfare ramp, fixed/variable OPEX, reserves, native debt and prepayments. Peak gap debt is IQD 2.222tn; clearance month 204. Its unlevered NPV is USD -1.715bn. Compare against the current baseline IQD 2.158tn peak / month 203 clearance / USD -1.670bn NPV. At the conservative demand proxy, mature section receipts are only USD 6.907m/year versus its USD 10.445m OPEX before indexing/ramp: it is not independently self-financing. Earlier service requires an explicit operating bridge; the extra facilities and advanced invoices must be justified by measured access benefits, demand or lower costs.

Financial feasibility and funding evidence

The baseline remains conditionally financeable under uncommitted cheap IQD gap credit and green/rights/receipt assumptions. Its unlevered financial NPV is USD -1.670bn at 13.4% nominal (8% real plus 5% inflation), before additional grant/rights/receipt targets. Clearing debt through later nominal fare growth is not the same as positive discounted investment value. Feasibility data supplies an illustrative annual external-benefit threshold of USD 210.3m in reference prices, phased with baseline openings and indexed 5%, whose discounted value would offset that financial deficit. This is not a completed economic appraisal: economic CBA must remove fares/taxes/financing transfers, use shadow resource costs, count time/safety/emissions benefits without double counting, and assess displacement/opportunity cost of labour rather than call every wage a net benefit.

Additional funding routes to investigate are competitively leased station land/development rights, employer travel contracts, advertising/kiosk concessions, utility/telecom corridor leases, climate grant/guarantee applications, and Iraqi institutional or retail IQD placements. Their net contract proceeds, authority, collection costs, timing and senior-creditor covenants are required before cash enters the ledger; a loan/guarantee is financing, not new operating revenue. The existing USD 300m net rights and USD 25m/year additional net receipts are targets, not a valuation or signed concession, and must not be added again as these same mechanisms. Replicable land value capture depends on land rights, planning powers and market demand (World Bank financing guidance). Iraq's GCF country programme establishes a project-development framework, not a Baghdad rail award or a 2% IQD facility. No fund application, lender term sheet or placement was obtained in this work.

Compare liquidity, repayment and discounted value separately

Financial comparison data uses each case's actual capital and indexed operating ledger, excluding debt and new grant/rights/receipt targets from unlevered NPV. All dates require physical acceptance and financing availability.

Case	First/full month	Capital USD bn	NPV USD bn	Peak IQD gap tn	Debt cleared month
calendar_baseline	40/81	7.951	-1.670	2.158	203
temporary_first_article	36/81	7.989	-1.637	1.951	201
availability_75pct_costed	42/89	7.951	-1.750	2.510	208
availability_75pct_all_stage_shifts	40/81	7.966	-1.706	2.226	204
domestic_placement_interrupted_recovered	46/87	7.962	-1.611	2.259	207
export_credit_delayed_recovered	46/87	7.962	-1.501	2.110	207
independent_first_section	36/81	7.981	-1.715	2.222	204

Temporary first-article production opens the complete first line earlier and lowers peak gap/repayment time, yet worsens discounted reference investment value after its additional capital/early operating costs. The independent shorter section also

adds capital and runs a conservative standalone operating deficit. Neither is adopted merely because fares arrive sooner. Coordinated shifts should be compared with the matching 75% availability case, not assumed to save money against the stronger baseline. Funding-delay NPV can look less negative because nominal un-escalated purchases are deferred; this discounting effect is not a benefit of denied credit. Real delay appraisal needs contract escalation, FX, cancellation/default costs and lost passenger benefits. Existing escalation/downside scenarios demonstrate their materiality.

Rebuild after delivery stresses with `.venv/bin/python tools/automation/baghdad_qualification.py`; validate with `--check`. Physical evidence and finance gates remain open.

Baghdad operations cross-check

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/simulation/operations-crosscheck.md

- Status: running-time-screen-passed-awaiting-junction-evidence
- Automatic running-time cross-check: passed
- Retained full-service replay matches current inputs/tools: yes
- Service execution basis: exact-local-executable
- Junction occupancy evidence: pending
- Authority accepted: no

Line	OSR reference	SUMO mean	Difference	Tolerance	Result
line-1	4350.0 s	4386.0 s	36.0 s	652.5 s	pass
line-2	4701.9 s	4769.0 s	67.1 s	705.3 s	pass
line-3	4469.5 s	4521.0 s	51.5 s	670.4 s	pass
line-4	3685.7 s	3752.5 s	66.8 s	552.9 s	pass
line-5	4275.4 s	4347.5 s	72.1 s	641.3 s	pass
line-6	4726.2 s	4790.5 s	64.3 s	708.9 s	pass
line-7	3526.0 s	3565.0 s	39.0 s	528.9 s	pass
line-8	4327.7 s	4384.0 s	56.3 s	649.2 s	pass
line-9	9081.6 s	9324.0 s	242.4 s	1362.2 s	pass
line-10	469.2 s	461.0 s	-8.2 s	120.0 s	pass
line-11	332.9 s	322.0 s	-10.9 s	120.0 s	pass
line-12	459.4 s	451.0 s	-8.4 s	120.0 s	pass
line-13	660.2 s	657.0 s	-3.2 s	120.0 s	pass
line-14	294.9 s	284.0 s	-10.9 s	120.0 s	pass

> The automatic result is a deterministic planning-model timing comparison, not proof of safe headways, signalling performance or junction capacity.

> Junction occupancy must be checked in an independently reviewed conflict-capable model and the operator or authority must sign the bound evidence before operational release.

Remaining junction gate

- independently reviewed junction-occupancy evidence not received

Remaining acceptance gate

- signed operational acceptance record not received

Baghdad civil soil screening

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/soil/README.md

5,669 route/station sample locations; 5,653 complete profiles; 16 profiles with missing data.

Published soilDB 2020–2022 predictions, 120 m pixels, 0–30 / 30–60 / 60–100 cm depths. Mean and p16–p84 are retained for clay, sand, silt, bulk density, organic carbon and pH.

The samples, source receipt, map and civil investigation plan are reproducible planning inputs. Each station and civil segment has an investigation scope. These records do not close the surveyed ground-model or foundation-design gates.

Investigation trigger	Sample locations
acidic-soil-durability-testing	3
coverage-gap	16
fine-soil-plasticity-and-shrink-swell-tests	5377
granular-density-and-groundwater-tests	3198
silt-moisture-frost-and-erosion-review	3528

- No bearing capacity, CBR, friction angle, cohesion, groundwater, contamination, sulfate/chloride or deep stratigraphy is inferred from these maps.
- Desert, water, urban fill and other nodata remain unknown; no nearest-pixel or climate-based substitution.
- Texture fractions are independent predictions; they are not renormalised or converted to a geotechnical soil class.
- Prioritisation thresholds are editable OSR screening rules, not statutory limits or evidence of a hazard.
- No excavation depths, foundation dimensions, treatment quantities or civil costs are changed from pedological predictions.

Source: OpenLandMap soilDB, Hengl et al., DOI 10.5194/essd-2025-336, CC BY 4.0. Exact source revision, URLs and raster metadata are retained in the receipt.

Sampled property ranges

Ranges below span the sampled locations; they are not a city-wide characteristic soil value. The uncertainty envelope spans the lowest p16 to highest p84.

Depth	Property	Unit	Mean range	Uncertainty envelope	Available locations
0..30cm	clay	%	16–32	2–42	5653
0..30cm	sand	%	24–58	5–84	5653
0..30cm	silt	%	26–45	8–59	5653
0..30cm	bd.core	kg/m3	1410–1510	1210–1730	5653
0..30cm	soc	g/kg	2.3–6.4	0.5–14.8	5653
0..30cm	ph.h2o	pH	7.5–8.4	6.6–9.2	5653
30..60cm	clay	%	17–33	1–43	5653
30..60cm	sand	%	26–57	5–92	5653
30..60cm	silt	%	24–42	5–60	5653
30..60cm	bd.core	kg/m3	1410–1580	1170–1810	5653
30..60cm	soc	g/kg	1.6–3.9	0–13.1	5653
30..60cm	ph.h2o	pH	7.5–8.7	6.2–10.2	5653
60..100cm	clay	%	17–34	2–44	5653
60..100cm	sand	%	27–58	7–92	5653
60..100cm	silt	%	22–41	2–58	5653
60..100cm	bd.core	kg/m3	1400–1630	1200–1940	5653
60..100cm	soc	g/kg	1.5–3.2	0–8.1	5653
60..100cm	ph.h2o	pH	7.5–8.8	5.1–10.3	5653

Station and depot overnight allocation

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/stabling/README.md

> Original catalogue or earlier scope reference. The figures and policies below retain their original assumptions; they are not the latest Baghdad staffing, depot, procurement or funding basis. See the current Baghdad recalculation and current city summary. National/portfolio totals have not been repriced with that conditional Baghdad option.

The hybrid allocation below remains a planning candidate; physical release is open.

Plan: 270 trainsets at stations + 539 at depots = 809 total. Two revenue trainsets per selected station support coordinated morning starts; the remaining revenue trains and reserves stay at storage on their own line.

Allocation check: FAIL. Depot stabling positions are planning requirements, separate from workshop bays. Physical release remains open.

Depot/storage station	Line	Site basis	Stabling positions required	Usable slot length m	Workshop bays
line-1-0166-0866-s049067	line-1	storage-at-existing-powered-service-point	52	6,292.0	0
line-10-0686-1312-s006804	line-10	storage-at-existing-powered-service-point	9	1,089.0	0
line-11-0337-1112-s006884	line-11	storage-at-existing-powered-service-point	10	1,210.0	0
line-12-0760-0138-s006586	line-12	storage-at-existing-powered-service-point	9	1,089.0	0
line-13-1193-0313-s008042	line-13	storage-at-existing-powered-service-point	10	1,210.0	0
line-14-0688-1260-s006047	line-14	storage-at-existing-powered-service-point	9	1,089.0	0
line-2-0595-0296-s052851	line-2	declared-depot	69	8,349.0	122
line-3-0151-1637-s049496	line-3	storage-at-existing-powered-service-point	62	7,502.0	0
line-4-1230-1627-s041932	line-4	storage-at-existing-powered-service-point	56	6,776.0	0
line-5-0677-2304-s047345	line-5	storage-at-existing-powered-service-point	60	7,260.0	0
line-6-0495-1669-s052503	line-6	storage-at-existing-powered-service-point	71	8,591.0	0
line-7-1531-0418-s039957	line-7	storage-at-existing-powered-service-point	50	6,050.0	0
line-8-0785-0290-s048137	line-8	storage-at-existing-powered-service-point	67	8,107.0	0
line-9-0689-0569-s097310	line-9	storage-at-existing-powered-service-point	5	605.0	0

Line	Location	Type	Direction	Role	Trainsets
line-1	line-1-0166-0866-s049067	station	reverse	revenue	2
line-1	line-1-0260-0833-s046583	station	forward	revenue	1
line-1	line-1-0260-0833-s046583	station	reverse	revenue	1
line-1	line-1-0382-0837-s044110	station	forward	revenue	1
line-1	line-1-0382-0837-s044110	station	reverse	revenue	1
line-1	line-1-0557-0840-s040519	station	forward	revenue	1
line-1	line-1-0557-0840-s040519	station	reverse	revenue	1
line-1	line-1-0604-0839-s039571	station	forward	revenue	1
line-1	line-1-0604-0839-s039571	station	reverse	revenue	1
line-1	line-1-0826-0834-s035089	station	forward	revenue	1
line-1	line-1-0826-0834-s035089	station	reverse	revenue	1
line-1	line-1-0975-0831-s032084	station	forward	revenue	1
line-1	line-1-0975-0831-s032084	station	reverse	revenue	1
line-1	line-1-1124-0828-s029068	station	forward	revenue	1
line-1	line-1-1124-0828-s029068	station	reverse	revenue	1
line-1	line-1-1341-0823-s024686	station	forward	revenue	1
line-1	line-1-1341-0823-s024686	station	reverse	revenue	1
line-1	line-1-1418-0822-s023138	station	forward	revenue	1
line-1	line-1-1418-0822-s023138	station	reverse	revenue	1
line-1	line-1-1554-0819-s020393	station	forward	revenue	1
line-1	line-1-1554-0819-s020393	station	reverse	revenue	1
line-1	line-1-1633-0817-s018544	station	forward	revenue	1
line-1	line-1-1633-0817-s018544	station	reverse	revenue	1
line-1	line-1-1848-0797-s014034	station	forward	revenue	1

Line	Location	Type	Direction	Role	Trainsets
line-1	line-1-1848-0797-s014034	station	reverse	revenue	1
line-1	line-1-2017-0784-s010529	station	forward	revenue	1
line-1	line-1-2017-0784-s010529	station	reverse	revenue	1
line-1	line-1-2180-0774-s007014	station	forward	revenue	1
line-1	line-1-2180-0774-s007014	station	reverse	revenue	1
line-1	line-1-2343-0771-s003515	station	forward	revenue	1
line-1	line-1-2343-0771-s003515	station	reverse	revenue	1
line-1	line-1-2500-0775-s000000	station	forward	revenue	2
line-10	line-10-0562-1131-s000000	station	forward	revenue	2
line-10	line-10-0686-1312-s006804	station	reverse	revenue	2
line-10	line-10-0688-1260-s005702	station	forward	revenue	1
line-10	line-10-0688-1260-s005702	station	reverse	revenue	1
line-11	line-11-0337-1112-s006884	station	reverse	revenue	2
line-11	line-11-0605-1228-s000000	station	forward	revenue	2
line-12	line-12-0701-0415-s000000	station	forward	revenue	2
line-12	line-12-0713-0287-s003217	station	forward	revenue	1
line-12	line-12-0713-0287-s003217	station	reverse	revenue	1
line-12	line-12-0760-0138-s006586	station	reverse	revenue	2
line-13	line-13-1190-0317-s007937	station	forward	revenue	1
line-13	line-13-1190-0317-s007937	station	reverse	revenue	1
line-13	line-13-1193-0313-s008042	station	reverse	revenue	2
line-13	line-13-1336-0518-s000000	station	forward	revenue	2
line-13	line-13-1381-0389-s003238	station	forward	revenue	1
line-13	line-13-1381-0389-s003238	station	reverse	revenue	1
line-14	line-14-0688-1260-s006047	station	reverse	revenue	2
line-14	line-14-0804-1072-s000000	station	forward	revenue	2
line-2	line-2-0595-0296-s052851	station	reverse	revenue	2
line-2	line-2-0701-0415-s048669	station	forward	revenue	1
line-2	line-2-0701-0415-s048669	station	reverse	revenue	1
line-2	line-2-0825-0580-s044026	station	forward	revenue	1
line-2	line-2-0825-0580-s044026	station	reverse	revenue	1
line-2	line-2-0903-0683-s041144	station	forward	revenue	1
line-2	line-2-0903-0683-s041144	station	reverse	revenue	1
line-2	line-2-1022-0840-s036713	station	forward	revenue	1
line-2	line-2-1022-0840-s036713	station	reverse	revenue	1
line-2	line-2-1067-0900-s035129	station	forward	revenue	1
line-2	line-2-1067-0900-s035129	station	reverse	revenue	1
line-2	line-2-1222-1104-s029320	station	forward	revenue	1
line-2	line-2-1222-1104-s029320	station	reverse	revenue	1
line-2	line-2-1309-1219-s026100	station	forward	revenue	1
line-2	line-2-1309-1219-s026100	station	reverse	revenue	1
line-2	line-2-1362-1289-s024085	station	forward	revenue	1
line-2	line-2-1362-1289-s024085	station	reverse	revenue	1
line-2	line-2-1504-1477-s018846	station	forward	revenue	1
line-2	line-2-1504-1477-s018846	station	reverse	revenue	1
line-2	line-2-1648-1703-s012612	station	forward	revenue	1
line-2	line-2-1648-1703-s012612	station	reverse	revenue	1
line-2	line-2-1895-2056-s003494	station	forward	revenue	1
line-2	line-2-1895-2056-s003494	station	reverse	revenue	1
line-2	line-2-1987-2192-s000000	station	forward	revenue	2
line-3	line-3-0151-1637-s049496	station	reverse	revenue	2
line-3	line-3-0288-1607-s046283	station	forward	revenue	1
line-3	line-3-0288-1607-s046283	station	reverse	revenue	1
line-3	line-3-0378-1545-s043281	station	forward	revenue	1
line-3	line-3-0378-1545-s043281	station	reverse	revenue	1
line-3	line-3-0465-1526-s041263	station	forward	revenue	1
line-3	line-3-0465-1526-s041263	station	reverse	revenue	1
line-3	line-3-0628-1483-s037624	station	forward	revenue	1
line-3	line-3-0628-1483-s037624	station	reverse	revenue	1
line-3	line-3-0691-1466-s036223	station	forward	revenue	1
line-3	line-3-0691-1466-s036223	station	reverse	revenue	1
line-3	line-3-0764-1447-s034606	station	forward	revenue	1
line-3	line-3-0764-1447-s034606	station	reverse	revenue	1

Line	Location	Type	Direction	Role	Trainsets
line-3	line-3-0874-1418-s032165	station	forward	revenue	1
line-3	line-3-0874-1418-s032165	station	reverse	revenue	1
line-3	line-3-1059-1369-s028013	station	forward	revenue	1
line-3	line-3-1059-1369-s028013	station	reverse	revenue	1
line-3	line-3-1281-1310-s023072	station	forward	revenue	1
line-3	line-3-1281-1310-s023072	station	reverse	revenue	1
line-3	line-3-1362-1289-s021278	station	forward	revenue	1
line-3	line-3-1362-1289-s021278	station	reverse	revenue	1
line-3	line-3-1613-1236-s015796	station	forward	revenue	1
line-3	line-3-1613-1236-s015796	station	reverse	revenue	1
line-3	line-3-1751-1233-s013011	station	forward	revenue	1
line-3	line-3-1751-1233-s013011	station	reverse	revenue	1
line-3	line-3-2271-1120-s000000	station	forward	revenue	2
line-4	line-4-0146-0199-s000000	station	forward	revenue	2
line-4	line-4-0298-0305-s004261	station	forward	revenue	1
line-4	line-4-0298-0305-s004261	station	reverse	revenue	1
line-4	line-4-0491-0706-s015030	station	forward	revenue	1
line-4	line-4-0491-0706-s015030	station	reverse	revenue	1
line-4	line-4-0604-0839-s018966	station	forward	revenue	1
line-4	line-4-0604-0839-s018966	station	reverse	revenue	1
line-4	line-4-0804-1072-s025869	station	forward	revenue	1
line-4	line-4-0804-1072-s025869	station	reverse	revenue	1
line-4	line-4-0927-1214-s030089	station	forward	revenue	1
line-4	line-4-0927-1214-s030089	station	reverse	revenue	1
line-4	line-4-0985-1282-s032081	station	forward	revenue	1
line-4	line-4-0985-1282-s032081	station	reverse	revenue	1
line-4	line-4-1059-1369-s034683	station	forward	revenue	1
line-4	line-4-1059-1369-s034683	station	reverse	revenue	1
line-4	line-4-1209-1543-s039827	station	forward	revenue	1
line-4	line-4-1209-1543-s039827	station	reverse	revenue	1
line-4	line-4-1230-1627-s041932	station	reverse	revenue	2
line-5	line-5-0677-2304-s047345	station	reverse	revenue	2
line-5	line-5-0843-1600-s030196	station	forward	revenue	1
line-5	line-5-0843-1600-s030196	station	reverse	revenue	1
line-5	line-5-0853-1593-s029938	station	forward	revenue	1
line-5	line-5-0853-1593-s029938	station	reverse	revenue	1
line-5	line-5-0874-1418-s026103	station	forward	revenue	1
line-5	line-5-0874-1418-s026103	station	reverse	revenue	1
line-5	line-5-0891-1352-s024642	station	forward	revenue	1
line-5	line-5-0891-1352-s024642	station	reverse	revenue	1
line-5	line-5-0927-1214-s021555	station	forward	revenue	1
line-5	line-5-0927-1214-s021555	station	reverse	revenue	1
line-5	line-5-0967-1057-s018078	station	forward	revenue	1
line-5	line-5-0967-1057-s018078	station	reverse	revenue	1
line-5	line-5-1022-0840-s013247	station	forward	revenue	1
line-5	line-5-1022-0840-s013247	station	reverse	revenue	1
line-5	line-5-1086-0589-s007673	station	forward	revenue	1
line-5	line-5-1086-0589-s007673	station	reverse	revenue	1
line-5	line-5-1129-0423-s003985	station	forward	revenue	1
line-5	line-5-1129-0423-s003985	station	reverse	revenue	1
line-5	line-5-1190-0317-s001231	station	forward	revenue	1
line-5	line-5-1190-0317-s001231	station	reverse	revenue	1
line-5	line-5-1193-0313-s001126	station	forward	revenue	1
line-5	line-5-1193-0313-s001126	station	reverse	revenue	1
line-5	line-5-1199-0260-s000000	station	forward	revenue	2
line-6	line-6-0495-1669-s052503	station	reverse	revenue	2
line-6	line-6-0647-1607-s047565	station	forward	revenue	1
line-6	line-6-0647-1607-s047565	station	reverse	revenue	1
line-6	line-6-0713-1485-s044164	station	forward	revenue	1
line-6	line-6-0713-1485-s044164	station	reverse	revenue	1
line-6	line-6-0764-1447-s042688	station	forward	revenue	1
line-6	line-6-0764-1447-s042688	station	reverse	revenue	1
line-6	line-6-0891-1352-s039209	station	forward	revenue	1

Line	Location	Type	Direction	Role	Trainsets
line-6	line-6-0891-1352-s039209	station	reverse	revenue	1
line-6	line-6-0985-1282-s036755	station	forward	revenue	1
line-6	line-6-0985-1282-s036755	station	reverse	revenue	1
line-6	line-6-1222-1104-s030195	station	forward	revenue	1
line-6	line-6-1222-1104-s030195	station	reverse	revenue	1
line-6	line-6-1370-0993-s026315	station	forward	revenue	1
line-6	line-6-1370-0993-s026315	station	reverse	revenue	1
line-6	line-6-1457-0929-s023846	station	forward	revenue	1
line-6	line-6-1457-0929-s023846	station	reverse	revenue	1
line-6	line-6-1580-0848-s020342	station	forward	revenue	1
line-6	line-6-1580-0848-s020342	station	reverse	revenue	1
line-6	line-6-1633-0817-s018903	station	forward	revenue	1
line-6	line-6-1633-0817-s018903	station	reverse	revenue	1
line-6	line-6-2322-0251-s000000	station	forward	revenue	2
line-7	line-7-1069-2206-s000000	station	forward	revenue	2
line-7	line-7-1185-1622-s012693	station	forward	revenue	1
line-7	line-7-1185-1622-s012693	station	reverse	revenue	1
line-7	line-7-1209-1543-s014684	station	forward	revenue	1
line-7	line-7-1209-1543-s014684	station	reverse	revenue	1
line-7	line-7-1281-1310-s019964	station	forward	revenue	1
line-7	line-7-1281-1310-s019964	station	reverse	revenue	1
line-7	line-7-1307-1217-s022051	station	forward	revenue	1
line-7	line-7-1307-1217-s022051	station	reverse	revenue	1
line-7	line-7-1370-0993-s027076	station	forward	revenue	1
line-7	line-7-1370-0993-s027076	station	reverse	revenue	1
line-7	line-7-1402-0878-s029642	station	forward	revenue	1
line-7	line-7-1402-0878-s029642	station	reverse	revenue	1
line-7	line-7-1418-0822-s030917	station	forward	revenue	1
line-7	line-7-1418-0822-s030917	station	reverse	revenue	1
line-7	line-7-1451-0706-s033517	station	forward	revenue	1
line-7	line-7-1451-0706-s033517	station	reverse	revenue	1
line-7	line-7-1504-0514-s037813	station	forward	revenue	1
line-7	line-7-1504-0514-s037813	station	reverse	revenue	1
line-7	line-7-1531-0418-s039957	station	reverse	revenue	2
line-8	line-8-0785-0290-s048137	station	reverse	revenue	2
line-8	line-8-0829-0364-s045862	station	forward	revenue	1
line-8	line-8-0829-0364-s045862	station	reverse	revenue	1
line-8	line-8-0954-0467-s042166	station	forward	revenue	1
line-8	line-8-0954-0467-s042166	station	reverse	revenue	1
line-8	line-8-1086-0589-s038140	station	forward	revenue	1
line-8	line-8-1086-0589-s038140	station	reverse	revenue	1
line-8	line-8-1228-0719-s033825	station	forward	revenue	1
line-8	line-8-1228-0719-s033825	station	reverse	revenue	1
line-8	line-8-1341-0823-s030364	station	forward	revenue	1
line-8	line-8-1341-0823-s030364	station	reverse	revenue	1
line-8	line-8-1402-0878-s028547	station	forward	revenue	1
line-8	line-8-1402-0878-s028547	station	reverse	revenue	1
line-8	line-8-1457-0929-s026861	station	forward	revenue	1
line-8	line-8-1457-0929-s026861	station	reverse	revenue	1
line-8	line-8-1645-1094-s021347	station	forward	revenue	1
line-8	line-8-1645-1094-s021347	station	reverse	revenue	1
line-8	line-8-1751-1233-s017689	station	forward	revenue	1
line-8	line-8-1751-1233-s017689	station	reverse	revenue	1
line-8	line-8-2231-1783-s000000	station	forward	revenue	2
line-9	line-9-0491-0706-s006266	station	forward	revenue	1
line-9	line-9-0491-0706-s006266	station	reverse	revenue	1
line-9	line-9-0557-0840-s009730	station	forward	revenue	1
line-9	line-9-0557-0840-s009730	station	reverse	revenue	1
line-9	line-9-0562-1131-s016676	station	forward	revenue	1
line-9	line-9-0562-1131-s016676	station	reverse	revenue	1
line-9	line-9-0605-1228-s019019	station	forward	revenue	1
line-9	line-9-0605-1228-s019019	station	reverse	revenue	1
line-9	line-9-0606-0636-s003011	station	forward	revenue	1

Line	Location	Type	Direction	Role	Trainsets
line-9	line-9-0606-0636-s003011	station	reverse	revenue	1
line-9	line-9-0627-1485-s025111	station	forward	revenue	1
line-9	line-9-0627-1485-s025111	station	reverse	revenue	1
line-9	line-9-0648-1607-s028125	station	forward	revenue	1
line-9	line-9-0648-1607-s028125	station	reverse	revenue	1
line-9	line-9-0843-1600-s032244	station	forward	revenue	1
line-9	line-9-0853-1593-s032502	station	forward	revenue	1
line-9	line-9-0853-1593-s032502	station	reverse	revenue	1
line-9	line-9-1129-0423-s085079	station	forward	revenue	1
line-9	line-9-1129-0423-s085079	station	reverse	revenue	1
line-9	line-9-1185-1622-s040057	station	forward	revenue	1
line-9	line-9-1185-1622-s040057	station	reverse	revenue	1
line-9	line-9-1194-0440-s081155	station	forward	revenue	1
line-9	line-9-1194-0440-s081155	station	reverse	revenue	1
line-9	line-9-1230-1627-s040998	station	forward	revenue	1
line-9	line-9-1230-1627-s040998	station	reverse	revenue	1
line-9	line-9-1336-0518-s076879	station	forward	revenue	1
line-9	line-9-1336-0518-s076879	station	reverse	revenue	1
line-9	line-9-1429-1525-s047139	station	forward	revenue	1
line-9	line-9-1429-1525-s047139	station	reverse	revenue	1
line-9	line-9-1451-0706-s071327	station	forward	revenue	1
line-9	line-9-1451-0706-s071327	station	reverse	revenue	1
line-9	line-9-1504-1477-s049136	station	forward	revenue	1
line-9	line-9-1504-1477-s049136	station	reverse	revenue	1
line-9	line-9-1554-0819-s067411	station	forward	revenue	1
line-9	line-9-1554-0819-s067411	station	reverse	revenue	1
line-9	line-9-1580-0848-s066616	station	forward	revenue	1
line-9	line-9-1613-1236-s056763	station	forward	revenue	1
line-9	line-9-1613-1236-s056763	station	reverse	revenue	1
line-9	line-9-1645-1094-s059868	station	forward	revenue	1
line-9	line-9-1645-1094-s059868	station	reverse	revenue	1
line-1	line-1-0166-0866-s049067	depot	—	revenue	44
line-1	line-1-0166-0866-s049067	depot	—	spare	7
line-1	line-1-0166-0866-s049067	depot	—	cold_reserve	1
line-2	line-2-0595-0296-s052851	depot	—	revenue	60
line-2	line-2-0595-0296-s052851	depot	—	spare	8
line-2	line-2-0595-0296-s052851	depot	—	cold_reserve	1
line-3	line-3-0151-1637-s049496	depot	—	revenue	53
line-3	line-3-0151-1637-s049496	depot	—	spare	8
line-3	line-3-0151-1637-s049496	depot	—	cold_reserve	1
line-4	line-4-1230-1627-s041932	depot	—	revenue	49
line-4	line-4-1230-1627-s041932	depot	—	spare	6
line-4	line-4-1230-1627-s041932	depot	—	cold_reserve	1
line-5	line-5-0677-2304-s047345	depot	—	revenue	52
line-5	line-5-0677-2304-s047345	depot	—	spare	7
line-5	line-5-0677-2304-s047345	depot	—	cold_reserve	1
line-6	line-6-0495-1669-s052503	depot	—	revenue	62
line-6	line-6-0495-1669-s052503	depot	—	spare	8
line-6	line-6-0495-1669-s052503	depot	—	cold_reserve	1
line-7	line-7-1531-0418-s039957	depot	—	revenue	43
line-7	line-7-1531-0418-s039957	depot	—	spare	6
line-7	line-7-1531-0418-s039957	depot	—	cold_reserve	1
line-8	line-8-0785-0290-s048137	depot	—	revenue	58
line-8	line-8-0785-0290-s048137	depot	—	spare	8
line-8	line-8-0785-0290-s048137	depot	—	cold_reserve	1
line-9	line-9-0689-0569-s097310	depot	—	spare	4
line-9	line-9-0689-0569-s097310	depot	—	cold_reserve	1
line-10	line-10-0686-1312-s006804	depot	—	revenue	7
line-10	line-10-0686-1312-s006804	depot	—	spare	1
line-10	line-10-0686-1312-s006804	depot	—	cold_reserve	1
line-11	line-11-0337-1112-s006884	depot	—	revenue	8
line-11	line-11-0337-1112-s006884	depot	—	spare	1
line-11	line-11-0337-1112-s006884	depot	—	cold_reserve	1

Line	Location	Type	Direction	Role	Trainsets
line-12	line-12-0760-0138-s006586	depot	—	revenue	7
line-12	line-12-0760-0138-s006586	depot	—	spare	1
line-12	line-12-0760-0138-s006586	depot	—	cold_reserve	1
line-13	line-13-1193-0313-s008042	depot	—	revenue	8
line-13	line-13-1193-0313-s008042	depot	—	spare	1
line-13	line-13-1193-0313-s008042	depot	—	cold_reserve	1
line-14	line-14-0688-1260-s006047	depot	—	revenue	7
line-14	line-14-0688-1260-s006047	depot	—	spare	1
line-14	line-14-0688-1260-s006047	depot	—	cold_reserve	1

Native hybrid candidate unavailable: morning station allocation is incomplete.

Station-only native benchmark

The runnable scenario below tests station holding and restart behaviour. It does not yet execute the station/depot allocation above or depot yard movements. Its station overflow is a diagnostic result, not the overnight design allocation.

Operating allocation: 809 trainsets at 139 stations; largest initial station queue 10. Physical release: open.

This candidate preserves all non-fleet scenario inputs and the existing fleet counts/service windows. It enables station holding and 150 kW top-up to 95% SoC, subject to shared site limits. Existing canonical simulation evidence still describes the retained endpoint-dispatch scenario.

Fleet roles: 728 revenue, 67 spare, 14 cold reserve. Reserves are held out of routine dispatch.

Two-train station-capacity check: FAIL. Selected stations provide 278 positions; 531 fleet positions exceed station-only provision. The initial allocation exceeds the limit at 114 stations. Four-berth reference platforms do not override the two-train provision.

Line	Station	Direction	Role	Initial trainsets	Verified track slots
line-1	line-1-2500-0775-s000000	forward	revenue	3	pending
line-1	line-1-2343-0771-s003515	forward	revenue	3	pending
line-1	line-1-2343-0771-s003515	reverse	revenue	3	pending
line-1	line-1-2180-0774-s007014	forward	revenue	3	pending
line-1	line-1-2180-0774-s007014	reverse	revenue	3	pending
line-1	line-1-2017-0784-s010529	forward	revenue	3	pending
line-1	line-1-2017-0784-s010529	reverse	revenue	3	pending
line-1	line-1-1848-0797-s014034	forward	revenue	3	pending
line-1	line-1-1848-0797-s014034	reverse	revenue	3	pending
line-1	line-1-1633-0817-s018544	forward	revenue	3	pending
line-1	line-1-1633-0817-s018544	reverse	revenue	3	pending
line-1	line-1-1554-0819-s020393	forward	revenue	3	pending
line-1	line-1-1554-0819-s020393	reverse	revenue	3	pending
line-1	line-1-1418-0822-s023138	forward	revenue	3	pending
line-1	line-1-1418-0822-s023138	reverse	revenue	2	pending
line-1	line-1-1341-0823-s024686	forward	revenue	2	pending
line-1	line-1-1341-0823-s024686	reverse	revenue	2	pending
line-1	line-1-1124-0828-s029068	forward	revenue	2	pending
line-1	line-1-1124-0828-s029068	reverse	revenue	2	pending
line-1	line-1-0975-0831-s032084	forward	revenue	2	pending
line-1	line-1-0975-0831-s032084	reverse	revenue	2	pending
line-1	line-1-0826-0834-s035089	forward	revenue	2	pending
line-1	line-1-0826-0834-s035089	reverse	revenue	2	pending
line-1	line-1-0604-0839-s039571	forward	revenue	2	pending
line-1	line-1-0604-0839-s039571	reverse	revenue	2	pending
line-1	line-1-0557-0840-s040519	forward	revenue	2	pending
line-1	line-1-0557-0840-s040519	reverse	revenue	2	pending
line-1	line-1-0382-0837-s044110	forward	revenue	2	pending
line-1	line-1-0382-0837-s044110	reverse	revenue	2	pending
line-1	line-1-0260-0833-s046583	forward	revenue	2	pending
line-1	line-1-0260-0833-s046583	reverse	revenue	2	pending
line-1	line-1-0166-0866-s049067	reverse	revenue	2	pending
line-1	line-1-1418-0822-s023138	reverse	spare	1	pending
line-1	line-1-1341-0823-s024686	forward	spare	1	pending
line-1	line-1-1341-0823-s024686	reverse	spare	1	pending

Line	Station	Direction	Role	Initial trainsets	Verified track slots
line-1	line-1-1124-0828-s029068	forward	spare	1	pending
line-1	line-1-1124-0828-s029068	reverse	spare	1	pending
line-1	line-1-0975-0831-s032084	forward	spare	1	pending
line-1	line-1-0975-0831-s032084	reverse	spare	1	pending
line-1	line-1-0826-0834-s035089	forward	cold_reserve	1	pending
line-2	line-2-1987-2192-s000000	forward	revenue	4	pending
line-2	line-2-1895-2056-s003494	forward	revenue	4	pending
line-2	line-2-1895-2056-s003494	reverse	revenue	4	pending
line-2	line-2-1648-1703-s012612	forward	revenue	4	pending
line-2	line-2-1648-1703-s012612	reverse	revenue	4	pending
line-2	line-2-1504-1477-s018846	forward	revenue	4	pending
line-2	line-2-1504-1477-s018846	reverse	revenue	4	pending
line-2	line-2-1362-1289-s024085	forward	revenue	4	pending
line-2	line-2-1362-1289-s024085	reverse	revenue	4	pending
line-2	line-2-1309-1219-s026100	forward	revenue	4	pending
line-2	line-2-1309-1219-s026100	reverse	revenue	4	pending
line-2	line-2-1222-1104-s029320	forward	revenue	4	pending
line-2	line-2-1222-1104-s029320	reverse	revenue	4	pending
line-2	line-2-1067-0900-s035129	forward	revenue	4	pending
line-2	line-2-1067-0900-s035129	reverse	revenue	3	pending
line-2	line-2-1022-0840-s036713	forward	revenue	3	pending
line-2	line-2-1022-0840-s036713	reverse	revenue	3	pending
line-2	line-2-0903-0683-s041144	forward	revenue	3	pending
line-2	line-2-0903-0683-s041144	reverse	revenue	3	pending
line-2	line-2-0825-0580-s044026	forward	revenue	3	pending
line-2	line-2-0825-0580-s044026	reverse	revenue	3	pending
line-2	line-2-0701-0415-s048669	forward	revenue	3	pending
line-2	line-2-0701-0415-s048669	reverse	revenue	3	pending
line-2	line-2-0595-0296-s052851	reverse	revenue	3	pending
line-2	line-2-1067-0900-s035129	reverse	spare	1	pending
line-2	line-2-1022-0840-s036713	forward	spare	1	pending
line-2	line-2-1022-0840-s036713	reverse	spare	1	pending
line-2	line-2-0903-0683-s041144	forward	spare	1	pending
line-2	line-2-0903-0683-s041144	reverse	spare	1	pending
line-2	line-2-0825-0580-s044026	forward	spare	1	pending
line-2	line-2-0825-0580-s044026	reverse	spare	1	pending
line-2	line-2-0701-0415-s048669	forward	spare	1	pending
line-2	line-2-0701-0415-s048669	reverse	cold_reserve	1	pending
line-3	line-3-2271-1120-s000000	forward	revenue	4	pending
line-3	line-3-1751-1233-s013011	forward	revenue	4	pending
line-3	line-3-1751-1233-s013011	reverse	revenue	4	pending
line-3	line-3-1613-1236-s015796	forward	revenue	3	pending
line-3	line-3-1613-1236-s015796	reverse	revenue	3	pending
line-3	line-3-1362-1289-s021278	forward	revenue	3	pending
line-3	line-3-1362-1289-s021278	reverse	revenue	3	pending
line-3	line-3-1281-1310-s023072	forward	revenue	3	pending
line-3	line-3-1281-1310-s023072	reverse	revenue	3	pending
line-3	line-3-1059-1369-s028013	forward	revenue	3	pending
line-3	line-3-1059-1369-s028013	reverse	revenue	3	pending
line-3	line-3-0874-1418-s032165	forward	revenue	3	pending
line-3	line-3-0874-1418-s032165	reverse	revenue	3	pending
line-3	line-3-0764-1447-s034606	forward	revenue	3	pending
line-3	line-3-0764-1447-s034606	reverse	revenue	3	pending
line-3	line-3-0691-1466-s036223	forward	revenue	3	pending
line-3	line-3-0691-1466-s036223	reverse	revenue	3	pending
line-3	line-3-0628-1483-s037624	forward	revenue	3	pending
line-3	line-3-0628-1483-s037624	reverse	revenue	3	pending
line-3	line-3-0465-1526-s041263	forward	revenue	3	pending
line-3	line-3-0465-1526-s041263	reverse	revenue	3	pending
line-3	line-3-0378-1545-s043281	forward	revenue	3	pending
line-3	line-3-0378-1545-s043281	reverse	revenue	3	pending
line-3	line-3-0288-1607-s046283	forward	revenue	3	pending
line-3	line-3-0288-1607-s046283	reverse	revenue	3	pending

Line	Station	Direction	Role	Initial trainsets	Verified track slots
line-3	line-3-0151-1637-s049496	reverse	revenue	3	pending
line-3	line-3-1613-1236-s015796	forward	spare	1	pending
line-3	line-3-1613-1236-s015796	reverse	spare	1	pending
line-3	line-3-1362-1289-s021278	forward	spare	1	pending
line-3	line-3-1362-1289-s021278	reverse	spare	1	pending
line-3	line-3-1281-1310-s023072	forward	spare	1	pending
line-3	line-3-1281-1310-s023072	reverse	spare	1	pending
line-3	line-3-1059-1369-s028013	forward	spare	1	pending
line-3	line-3-1059-1369-s028013	reverse	spare	1	pending
line-3	line-3-0874-1418-s032165	forward	cold_reserve	1	pending
line-4	line-4-0146-0199-s000000	forward	revenue	4	pending
line-4	line-4-0298-0305-s004261	forward	revenue	4	pending
line-4	line-4-0298-0305-s004261	reverse	revenue	4	pending
line-4	line-4-0491-0706-s015030	forward	revenue	4	pending
line-4	line-4-0491-0706-s015030	reverse	revenue	4	pending
line-4	line-4-0604-0839-s018966	forward	revenue	4	pending
line-4	line-4-0604-0839-s018966	reverse	revenue	4	pending
line-4	line-4-0804-1072-s025869	forward	revenue	4	pending
line-4	line-4-0804-1072-s025869	reverse	revenue	4	pending
line-4	line-4-0927-1214-s030089	forward	revenue	4	pending
line-4	line-4-0927-1214-s030089	reverse	revenue	4	pending
line-4	line-4-0985-1282-s032081	forward	revenue	4	pending
line-4	line-4-0985-1282-s032081	reverse	revenue	4	pending
line-4	line-4-1059-1369-s034683	forward	revenue	4	pending
line-4	line-4-1059-1369-s034683	reverse	revenue	4	pending
line-4	line-4-1209-1543-s039827	forward	revenue	3	pending
line-4	line-4-1209-1543-s039827	reverse	revenue	3	pending
line-4	line-4-1230-1627-s041932	reverse	revenue	3	pending
line-4	line-4-1209-1543-s039827	forward	spare	1	pending
line-4	line-4-1209-1543-s039827	reverse	spare	1	pending
line-4	line-4-1230-1627-s041932	reverse	spare	1	pending
line-4	line-4-0146-0199-s000000	forward	spare	1	pending
line-4	line-4-0298-0305-s004261	forward	spare	1	pending
line-4	line-4-0298-0305-s004261	reverse	spare	1	pending
line-4	line-4-0491-0706-s015030	forward	cold_reserve	1	pending
line-5	line-5-1199-0260-s000000	forward	revenue	4	pending
line-5	line-5-1193-0313-s001126	forward	revenue	4	pending
line-5	line-5-1193-0313-s001126	reverse	revenue	4	pending
line-5	line-5-1190-0317-s001231	forward	revenue	4	pending
line-5	line-5-1190-0317-s001231	reverse	revenue	4	pending
line-5	line-5-1129-0423-s003985	forward	revenue	4	pending
line-5	line-5-1129-0423-s003985	reverse	revenue	3	pending
line-5	line-5-1086-0589-s007673	forward	revenue	3	pending
line-5	line-5-1086-0589-s007673	reverse	revenue	3	pending
line-5	line-5-1022-0840-s013247	forward	revenue	3	pending
line-5	line-5-1022-0840-s013247	reverse	revenue	3	pending
line-5	line-5-0967-1057-s018078	forward	revenue	3	pending
line-5	line-5-0967-1057-s018078	reverse	revenue	3	pending
line-5	line-5-0927-1214-s021555	forward	revenue	3	pending
line-5	line-5-0927-1214-s021555	reverse	revenue	3	pending
line-5	line-5-0891-1352-s024642	forward	revenue	3	pending
line-5	line-5-0891-1352-s024642	reverse	revenue	3	pending
line-5	line-5-0874-1418-s026103	forward	revenue	3	pending
line-5	line-5-0874-1418-s026103	reverse	revenue	3	pending
line-5	line-5-0853-1593-s029938	forward	revenue	3	pending
line-5	line-5-0853-1593-s029938	reverse	revenue	3	pending
line-5	line-5-0843-1600-s030196	forward	revenue	3	pending
line-5	line-5-0843-1600-s030196	reverse	revenue	3	pending
line-5	line-5-0677-2304-s047345	reverse	revenue	3	pending
line-5	line-5-1129-0423-s003985	reverse	spare	1	pending
line-5	line-5-1086-0589-s007673	forward	spare	1	pending
line-5	line-5-1086-0589-s007673	reverse	spare	1	pending
line-5	line-5-1022-0840-s013247	forward	spare	1	pending

Line	Station	Direction	Role	Initial trainsets	Verified track slots
line-5	line-5-1022-0840-s013247	reverse	spare	1	pending
line-5	line-5-0967-1057-s018078	forward	spare	1	pending
line-5	line-5-0967-1057-s018078	reverse	spare	1	pending
line-5	line-5-0927-1214-s021555	forward	cold_reserve	1	pending
line-6	line-6-2322-0251-s000000	forward	revenue	4	pending
line-6	line-6-1633-0817-s018903	forward	revenue	4	pending
line-6	line-6-1633-0817-s018903	reverse	revenue	4	pending
line-6	line-6-1580-0848-s020342	forward	revenue	4	pending
line-6	line-6-1580-0848-s020342	reverse	revenue	4	pending
line-6	line-6-1457-0929-s023846	forward	revenue	4	pending
line-6	line-6-1457-0929-s023846	reverse	revenue	4	pending
line-6	line-6-1370-0993-s026315	forward	revenue	4	pending
line-6	line-6-1370-0993-s026315	reverse	revenue	4	pending
line-6	line-6-1222-1104-s030195	forward	revenue	4	pending
line-6	line-6-1222-1104-s030195	reverse	revenue	4	pending
line-6	line-6-0985-1282-s036755	forward	revenue	4	pending
line-6	line-6-0985-1282-s036755	reverse	revenue	4	pending
line-6	line-6-0891-1352-s039209	forward	revenue	4	pending
line-6	line-6-0891-1352-s039209	reverse	revenue	4	pending
line-6	line-6-0764-1447-s042688	forward	revenue	4	pending
line-6	line-6-0764-1447-s042688	reverse	revenue	4	pending
line-6	line-6-0713-1485-s044164	forward	revenue	4	pending
line-6	line-6-0713-1485-s044164	reverse	revenue	4	pending
line-6	line-6-0647-1607-s047565	forward	revenue	4	pending
line-6	line-6-0647-1607-s047565	reverse	revenue	3	pending
line-6	line-6-0495-1669-s052503	reverse	revenue	3	pending
line-6	line-6-0647-1607-s047565	reverse	spare	1	pending
line-6	line-6-0495-1669-s052503	reverse	spare	1	pending
line-6	line-6-2322-0251-s000000	forward	spare	1	pending
line-6	line-6-1633-0817-s018903	forward	spare	1	pending
line-6	line-6-1633-0817-s018903	reverse	spare	1	pending
line-6	line-6-1580-0848-s020342	forward	spare	1	pending
line-6	line-6-1580-0848-s020342	reverse	spare	1	pending
line-6	line-6-1457-0929-s023846	forward	spare	1	pending
line-6	line-6-1457-0929-s023846	reverse	cold_reserve	1	pending
line-7	line-7-1069-2206-s000000	forward	revenue	4	pending
line-7	line-7-1185-1622-s012693	forward	revenue	4	pending
line-7	line-7-1185-1622-s012693	reverse	revenue	4	pending
line-7	line-7-1209-1543-s014684	forward	revenue	4	pending
line-7	line-7-1209-1543-s014684	reverse	revenue	4	pending
line-7	line-7-1281-1310-s019964	forward	revenue	3	pending
line-7	line-7-1281-1310-s019964	reverse	revenue	3	pending
line-7	line-7-1307-1217-s022051	forward	revenue	3	pending
line-7	line-7-1307-1217-s022051	reverse	revenue	3	pending
line-7	line-7-1370-0993-s027076	forward	revenue	3	pending
line-7	line-7-1370-0993-s027076	reverse	revenue	3	pending
line-7	line-7-1402-0878-s029642	forward	revenue	3	pending
line-7	line-7-1402-0878-s029642	reverse	revenue	3	pending
line-7	line-7-1418-0822-s030917	forward	revenue	3	pending
line-7	line-7-1418-0822-s030917	reverse	revenue	3	pending
line-7	line-7-1451-0706-s033517	forward	revenue	3	pending
line-7	line-7-1451-0706-s033517	reverse	revenue	3	pending
line-7	line-7-1504-0514-s037813	forward	revenue	3	pending
line-7	line-7-1504-0514-s037813	reverse	revenue	3	pending
line-7	line-7-1531-0418-s039957	reverse	revenue	3	pending
line-7	line-7-1281-1310-s019964	forward	spare	1	pending
line-7	line-7-1281-1310-s019964	reverse	spare	1	pending
line-7	line-7-1307-1217-s022051	forward	spare	1	pending
line-7	line-7-1307-1217-s022051	reverse	spare	1	pending
line-7	line-7-1370-0993-s027076	forward	spare	1	pending
line-7	line-7-1370-0993-s027076	reverse	spare	1	pending
line-7	line-7-1402-0878-s029642	forward	cold_reserve	1	pending
line-8	line-8-2231-1783-s000000	forward	revenue	4	pending

Line	Station	Direction	Role	Initial trainsets	Verified track slots
line-8	line-8-1751-1233-s017689	forward	revenue	4	pending
line-8	line-8-1751-1233-s017689	reverse	revenue	4	pending
line-8	line-8-1645-1094-s021347	forward	revenue	4	pending
line-8	line-8-1645-1094-s021347	reverse	revenue	4	pending
line-8	line-8-1457-0929-s026861	forward	revenue	4	pending
line-8	line-8-1457-0929-s026861	reverse	revenue	4	pending
line-8	line-8-1402-0878-s028547	forward	revenue	4	pending
line-8	line-8-1402-0878-s028547	reverse	revenue	4	pending
line-8	line-8-1341-0823-s030364	forward	revenue	4	pending
line-8	line-8-1341-0823-s030364	reverse	revenue	4	pending
line-8	line-8-1228-0719-s033825	forward	revenue	4	pending
line-8	line-8-1228-0719-s033825	reverse	revenue	4	pending
line-8	line-8-1086-0589-s038140	forward	revenue	4	pending
line-8	line-8-1086-0589-s038140	reverse	revenue	4	pending
line-8	line-8-0954-0467-s042166	forward	revenue	4	pending
line-8	line-8-0954-0467-s042166	reverse	revenue	4	pending
line-8	line-8-0829-0364-s045862	forward	revenue	4	pending
line-8	line-8-0829-0364-s045862	reverse	revenue	4	pending
line-8	line-8-0785-0290-s048137	reverse	revenue	4	pending
line-8	line-8-2231-1783-s000000	forward	spare	1	pending
line-8	line-8-1751-1233-s017689	forward	spare	1	pending
line-8	line-8-1751-1233-s017689	reverse	spare	1	pending
line-8	line-8-1645-1094-s021347	forward	spare	1	pending
line-8	line-8-1645-1094-s021347	reverse	spare	1	pending
line-8	line-8-1457-0929-s026861	forward	spare	1	pending
line-8	line-8-1457-0929-s026861	reverse	spare	1	pending
line-8	line-8-1402-0878-s028547	forward	spare	1	pending
line-8	line-8-1402-0878-s028547	reverse	cold_reserve	1	pending
line-9	line-9-0606-0636-s003011	forward	revenue	1	pending
line-9	line-9-0606-0636-s003011	reverse	revenue	1	pending
line-9	line-9-0491-0706-s006266	forward	revenue	1	pending
line-9	line-9-0491-0706-s006266	reverse	revenue	1	pending
line-9	line-9-0557-0840-s009730	forward	revenue	1	pending
line-9	line-9-0557-0840-s009730	reverse	revenue	1	pending
line-9	line-9-0562-1131-s016676	forward	revenue	1	pending
line-9	line-9-0562-1131-s016676	reverse	revenue	1	pending
line-9	line-9-0605-1228-s019019	forward	revenue	1	pending
line-9	line-9-0605-1228-s019019	reverse	revenue	1	pending
line-9	line-9-0627-1485-s025111	forward	revenue	1	pending
line-9	line-9-0627-1485-s025111	reverse	revenue	1	pending
line-9	line-9-0648-1607-s028125	forward	revenue	1	pending
line-9	line-9-0648-1607-s028125	reverse	revenue	1	pending
line-9	line-9-0843-1600-s032244	forward	revenue	1	pending
line-9	line-9-0853-1593-s032502	forward	revenue	1	pending
line-9	line-9-0853-1593-s032502	reverse	revenue	1	pending
line-9	line-9-1185-1622-s040057	forward	revenue	1	pending
line-9	line-9-1185-1622-s040057	reverse	revenue	1	pending
line-9	line-9-1230-1627-s040998	forward	revenue	1	pending
line-9	line-9-1230-1627-s040998	reverse	revenue	1	pending
line-9	line-9-1429-1525-s047139	forward	revenue	1	pending
line-9	line-9-1429-1525-s047139	reverse	revenue	1	pending
line-9	line-9-1504-1477-s049136	forward	revenue	1	pending
line-9	line-9-1504-1477-s049136	reverse	revenue	1	pending
line-9	line-9-1613-1236-s056763	forward	revenue	1	pending
line-9	line-9-1613-1236-s056763	reverse	revenue	1	pending
line-9	line-9-1645-1094-s059868	forward	revenue	1	pending
line-9	line-9-1645-1094-s059868	reverse	revenue	1	pending
line-9	line-9-1580-0848-s066616	forward	revenue	1	pending
line-9	line-9-1554-0819-s067411	forward	revenue	1	pending
line-9	line-9-1554-0819-s067411	reverse	revenue	1	pending
line-9	line-9-1451-0706-s071327	forward	revenue	1	pending
line-9	line-9-1451-0706-s071327	reverse	revenue	1	pending
line-9	line-9-1336-0518-s076879	forward	revenue	1	pending

Line	Station	Direction	Role	Initial trainsets	Verified track slots
line-9	line-9-1336-0518-s076879	reverse	revenue	1	pending
line-9	line-9-1194-0440-s081155	forward	revenue	1	pending
line-9	line-9-1194-0440-s081155	reverse	revenue	1	pending
line-9	line-9-1129-0423-s085079	forward	revenue	1	pending
line-9	line-9-1129-0423-s085079	reverse	revenue	1	pending
line-9	line-9-0954-0467-s089572	forward	spare	1	pending
line-9	line-9-0954-0467-s089572	reverse	spare	1	pending
line-9	line-9-0825-0580-s094205	forward	spare	1	pending
line-9	line-9-0825-0580-s094205	reverse	spare	1	pending
line-9	line-9-0689-0569-s097310	forward	cold_reserve	1	pending
line-10	line-10-0562-1131-s000000	forward	revenue	4	pending
line-10	line-10-0688-1260-s005702	forward	revenue	3	pending
line-10	line-10-0688-1260-s005702	reverse	revenue	3	pending
line-10	line-10-0686-1312-s006804	reverse	revenue	3	pending
line-10	line-10-0688-1260-s005702	forward	spare	1	pending
line-10	line-10-0688-1260-s005702	reverse	cold_reserve	1	pending
line-11	line-11-0605-1228-s000000	forward	revenue	6	pending
line-11	line-11-0337-1112-s006884	reverse	revenue	6	pending
line-11	line-11-0605-1228-s000000	forward	spare	1	pending
line-11	line-11-0337-1112-s006884	reverse	cold_reserve	1	pending
line-12	line-12-0701-0415-s000000	forward	revenue	4	pending
line-12	line-12-0713-0287-s003217	forward	revenue	3	pending
line-12	line-12-0713-0287-s003217	reverse	revenue	3	pending
line-12	line-12-0760-0138-s006586	reverse	revenue	3	pending
line-12	line-12-0713-0287-s003217	forward	spare	1	pending
line-12	line-12-0713-0287-s003217	reverse	cold_reserve	1	pending
line-13	line-13-1336-0518-s000000	forward	revenue	3	pending
line-13	line-13-1381-0389-s003238	forward	revenue	3	pending
line-13	line-13-1381-0389-s003238	reverse	revenue	3	pending
line-13	line-13-1190-0317-s007937	forward	revenue	3	pending
line-13	line-13-1190-0317-s007937	reverse	revenue	2	pending
line-13	line-13-1193-0313-s008042	reverse	revenue	2	pending
line-13	line-13-1190-0317-s007937	reverse	spare	1	pending
line-13	line-13-1193-0313-s008042	reverse	cold_reserve	1	pending
line-14	line-14-0804-1072-s000000	forward	revenue	6	pending
line-14	line-14-0688-1260-s006047	reverse	revenue	5	pending
line-14	line-14-0688-1260-s006047	reverse	spare	1	pending
line-14	line-14-0804-1072-s000000	forward	cold_reserve	1	pending

Reference platform capacity comparison

402 trainsets exceed the reference platform envelope, requiring 48,642.0 m of additional usable slots under this initial allocation. This does not establish the location or need for new infrastructure; existing sidings require evidence before crediting them.

Station	Trainsets	Reference platform berths	Beyond envelope	Additional usable slot length m
line-1-0166-0866-s049067	2	2	0	0.0
line-1-0260-0833-s046583	4	2	2	242.0
line-1-0382-0837-s044110	4	2	2	242.0
line-1-0557-0840-s040519	4	4	0	0.0
line-1-0604-0839-s039571	4	4	0	0.0
line-1-0826-0834-s035089	5	2	3	363.0
line-1-0975-0831-s032084	6	2	4	484.0
line-1-1124-0828-s029068	6	2	4	484.0
line-1-1341-0823-s024686	6	4	2	242.0
line-1-1418-0822-s023138	6	4	2	242.0
line-1-1554-0819-s020393	6	4	2	242.0
line-1-1633-0817-s018544	6	4	2	242.0
line-1-1848-0797-s014034	6	2	4	484.0
line-1-2017-0784-s010529	6	2	4	484.0
line-1-2180-0774-s007014	6	2	4	484.0
line-1-2343-0771-s003515	6	2	4	484.0
line-1-2500-0775-s000000	3	2	1	121.0

Station	Trainsets	Reference platform berths	Beyond envelope	Additional usable slot length m
line-10-0562-1131-s000000	4	2	2	242.0
line-10-0686-1312-s006804	3	2	1	121.0
line-10-0688-1260-s005702	8	4	4	484.0
line-11-0337-1112-s006884	7	2	5	605.0
line-11-0605-1228-s000000	7	2	5	605.0
line-12-0701-0415-s000000	4	2	2	242.0
line-12-0713-0287-s003217	8	2	6	726.0
line-12-0760-0138-s006586	3	2	1	121.0
line-13-1190-0317-s007937	6	4	2	242.0
line-13-1193-0313-s008042	3	2	1	121.0
line-13-1336-0518-s000000	3	2	1	121.0
line-13-1381-0389-s003238	6	2	4	484.0
line-14-0688-1260-s006047	6	2	4	484.0
line-14-0804-1072-s000000	7	2	5	605.0
line-2-0595-0296-s052851	3	2	1	121.0
line-2-0701-0415-s048669	8	4	4	484.0
line-2-0825-0580-s044026	8	4	4	484.0
line-2-0903-0683-s041144	8	2	6	726.0
line-2-1022-0840-s036713	8	4	4	484.0
line-2-1067-0900-s035129	8	2	6	726.0
line-2-1222-1104-s029320	8	4	4	484.0
line-2-1309-1219-s026100	8	4	4	484.0
line-2-1362-1289-s024085	8	4	4	484.0
line-2-1504-1477-s018846	8	4	4	484.0
line-2-1648-1703-s012612	8	2	6	726.0
line-2-1895-2056-s003494	8	2	6	726.0
line-2-1987-2192-s000000	4	2	2	242.0
line-3-0151-1637-s049496	3	2	1	121.0
line-3-0288-1607-s046283	6	2	4	484.0
line-3-0378-1545-s043281	6	2	4	484.0
line-3-0465-1526-s041263	6	2	4	484.0
line-3-0628-1483-s037624	6	4	2	242.0
line-3-0691-1466-s036223	6	4	2	242.0
line-3-0764-1447-s034606	6	4	2	242.0
line-3-0874-1418-s032165	7	4	3	363.0
line-3-1059-1369-s028013	8	4	4	484.0
line-3-1281-1310-s023072	8	4	4	484.0
line-3-1362-1289-s021278	8	4	4	484.0
line-3-1613-1236-s015796	8	4	4	484.0
line-3-1751-1233-s013011	8	4	4	484.0
line-3-2271-1120-s000000	4	2	2	242.0
line-4-0146-0199-s000000	5	2	3	363.0
line-4-0298-0305-s004261	10	2	8	968.0
line-4-0491-0706-s015030	9	4	5	605.0
line-4-0604-0839-s018966	8	4	4	484.0
line-4-0804-1072-s025869	8	4	4	484.0
line-4-0927-1214-s030089	8	4	4	484.0
line-4-0985-1282-s032081	8	4	4	484.0
line-4-1059-1369-s034683	8	4	4	484.0
line-4-1209-1543-s039827	8	4	4	484.0
line-4-1230-1627-s041932	4	2	2	242.0
line-5-0677-2304-s047345	3	2	1	121.0
line-5-0843-1600-s030196	6	4	2	242.0
line-5-0853-1593-s029938	6	4	2	242.0
line-5-0874-1418-s026103	6	4	2	242.0
line-5-0891-1352-s024642	6	4	2	242.0
line-5-0927-1214-s021555	7	4	3	363.0
line-5-0967-1057-s018078	8	2	6	726.0
line-5-1022-0840-s013247	8	4	4	484.0
line-5-1086-0589-s007673	8	4	4	484.0
line-5-1129-0423-s003985	8	4	4	484.0
line-5-1190-0317-s001231	8	4	4	484.0
line-5-1193-0313-s001126	8	4	4	484.0

Station	Trainsets	Reference platform berths	Beyond envelope	Additional usable slot length m
line-5-1199-0260-s000000	4	2	2	242.0
line-6-0495-1669-s052503	4	2	2	242.0
line-6-0647-1607-s047565	8	4	4	484.0
line-6-0713-1485-s044164	8	4	4	484.0
line-6-0764-1447-s042688	8	4	4	484.0
line-6-0891-1352-s039209	8	4	4	484.0
line-6-0985-1282-s036755	8	4	4	484.0
line-6-1222-1104-s030195	8	4	4	484.0
line-6-1370-0993-s026315	8	4	4	484.0
line-6-1457-0929-s023846	10	4	6	726.0
line-6-1580-0848-s020342	10	4	6	726.0
line-6-1633-0817-s018903	10	4	6	726.0
line-6-2322-0251-s000000	5	2	3	363.0
line-7-1069-2206-s000000	4	2	2	242.0
line-7-1185-1622-s012693	8	4	4	484.0
line-7-1209-1543-s014684	8	4	4	484.0
line-7-1281-1310-s019964	8	4	4	484.0
line-7-1307-1217-s022051	8	4	4	484.0
line-7-1370-0993-s027076	8	4	4	484.0
line-7-1402-0878-s029642	7	4	3	363.0
line-7-1418-0822-s030917	6	4	2	242.0
line-7-1451-0706-s033517	6	4	2	242.0
line-7-1504-0514-s037813	6	2	4	484.0
line-7-1531-0418-s039957	3	2	1	121.0
line-8-0785-0290-s048137	4	2	2	242.0
line-8-0829-0364-s045862	8	2	6	726.0
line-8-0954-0467-s042166	8	4	4	484.0
line-8-1086-0589-s038140	8	4	4	484.0
line-8-1228-0719-s033825	8	2	6	726.0
line-8-1341-0823-s030364	8	4	4	484.0
line-8-1402-0878-s028547	10	4	6	726.0
line-8-1457-0929-s026861	10	4	6	726.0
line-8-1645-1094-s021347	10	4	6	726.0
line-8-1751-1233-s017689	10	4	6	726.0
line-8-2231-1783-s000000	5	2	3	363.0
line-9-0491-0706-s006266	2	4	0	0.0
line-9-0557-0840-s009730	2	4	0	0.0
line-9-0562-1131-s016676	2	4	0	0.0
line-9-0605-1228-s019019	2	4	0	0.0
line-9-0606-0636-s003011	2	2	0	0.0
line-9-0627-1485-s025111	2	4	0	0.0
line-9-0648-1607-s028125	2	4	0	0.0
line-9-0689-0569-s097310	1	2	0	0.0
line-9-0825-0580-s094205	2	4	0	0.0
line-9-0843-1600-s032244	1	4	0	0.0
line-9-0853-1593-s032502	2	4	0	0.0
line-9-0954-0467-s089572	2	4	0	0.0
line-9-1129-0423-s085079	2	4	0	0.0
line-9-1185-1622-s040057	2	4	0	0.0
line-9-1194-0440-s081155	2	2	0	0.0
line-9-1230-1627-s040998	2	4	0	0.0
line-9-1336-0518-s076879	2	4	0	0.0
line-9-1429-1525-s047139	2	2	0	0.0
line-9-1451-0706-s071327	2	4	0	0.0
line-9-1504-1477-s049136	2	4	0	0.0
line-9-1554-0819-s067411	2	4	0	0.0
line-9-1580-0848-s066616	1	4	0	0.0
line-9-1613-1236-s056763	2	4	0	0.0
line-9-1645-1094-s059868	2	4	0	0.0

- Initial queues are balanced across powered station/direction choices; this is not a surveyed parking layout.

- Declared spares and cold reserves remain parked and charge; automatic substitution, defect routing and maintenance release are not modelled.
- Returning trains hold at selected powered stations after service ends; no train is teleported back to its initial placement.
- Reference platform berths are an optimistic length/count comparison, not verified parking capacity; dedicated sidings and workshop bays receive no automatic credit.
- Station berths, crossovers and shared junction conflicts are outside the simplified interstation movement-authority graph.
- Existing depot/station energy quantities and service schedules are preserved, not accepted as correctly sized.

Regenerate this report and its local runnable scenario with:

```
.venv/bin/python tools/automation/generate-stabling-plan.py --design 'cities/catalogue/west-asia/Iraq/Baghdad/design.
toml'
```

The scenario is generated under `build/engineering/stabling/`; its hash is recorded in `summary.json`. Source-linked operating replay evidence, where available, is in `operating-screen.json` beside this report.

Baghdad survey-control processing

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/survey/control-processing-readiness.md

- Status: awaiting-field-data
- RTKLIB runtime: not probed
- Processing completed: no
- Technical screen passed: no
- Survey authority accepted: no

> The automated result is a reproducibility and solution-quality screen. It is not a network adjustment, survey certification or permission to use the control for design or construction.

> Only the appointed deployment survey authority can accept the control network after redundant occupations, independent checks, network adjustment, datum/geoid review and monument documentation.

Current gates

- Missing processing roles: rover_observation, base_observation, navigation, rtklib_configuration
- Duplicate processing roles: none
- Processing-configuration findings: 0
- Missing authority-review roles: rover_observation, base_observation, navigation, rtklib_configuration, field_log, monument_schedule, network_adjustment_report, authority_acceptance_record

Controlled-storage workflow

RINEX observations and navigation data remain in access-controlled project storage. Git records only the receipt manifest, file hashes, requirements and non-sensitive processing/acceptance summaries.

1. Put received files in access-controlled project storage.
2. Add one `SUR-CTRL` manifest row per file, including `file_role`, metadata and SHA-256.
3. Freeze the required RTKLIB settings, then run this processor; the four processing roles must occur exactly once.
4. Review RTKLIB outputs, complete independent checks and network adjustment, then add the controlled acceptance record.
5. Re-run; only the appointed survey authority's explicit record can close the authority gate.

Baghdad drainage and ground-design gate

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/survey/drainage-ground-readiness.md

- Status: awaiting-drainage-ground-evidence
- Lines/stations: 14 / 159
- SWMM/ground technical screen passed: no
- OpenGeoSys required by received decision: no
- Authority accepted: no

> The automated gate reruns the received SWMM model, checks continuity and hydraulic performance for normal/blockage/backwater scenarios, reconciles exact reviewed assets/supports, and checks numerical site comparisons for catalogue foundation/ground-treatment selections. It does not validate rainfall, flood levels, soil parameters, hydraulic boundaries, structural capacity, settlement predictions or construction suitability.

> The drainage and geotechnical engineers of record, asset owner, relevant authority and information manager must accept the project inputs, calculations, selected variants, residual risks and immutable evidence record.

Current gates

- Missing technical roles: ground_model_readiness, route_station_fit_readiness, accepted_hydrology_basis, swmm_model, swmm_processing_report, geotechnical_ground_model, foundation_ground_schedule, ground_design_verification_report, groundwater_coupling_decision, civil_asset_register
- Duplicate roles: none
- Authority findings:
  - drainage/ground acceptance record not received

Desktop soil inputs and route/station investigation priorities are available. These do not replace the received geotechnical ground model.

Controlled workflow

Keep borehole logs, laboratory results, sensitive asset data, native calculations and large model results in controlled project storage. Git carries requirements, receipts, hashes, reproducible exchange inputs and non-sensitive summaries.

1. Accept the ground model, route fit, hydrology basis and geotechnical model.
2. Run the project SWMM model; retain its input, report, source hashes and continuity results.
3. Size a checked catalogue foundation or ground-treatment system for every line and station scope.
4. Record whether groundwater/coupled analysis is warranted; require OpenGeoSys evidence only when the reviewed triggers say yes.
5. Close residual risks and obtain the signed drainage/geotechnical acceptance record.

Baghdad field-evidence brief

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/survey/field-evidence-brief.md

Deterministic pre-mobilisation requirements generated from the shared field-evidence template. It issues the information request; it does not claim that field data or approvals exist.

- Brief status: brief-issued-awaiting-signatures
- Mobilisation authorized: no
- Field evidence accepted: no
- Horizontal CRS: EPSG:32638 candidate UTM zone 38N; survey authority to confirm or replace
- Vertical datum: `authority-to-confirm-before-mobilisation`
- Canonical requirements: `lib/templates/field-evidence.toml`

> The appointed survey authority, architect/engineer of record, operator and information manager approve this brief, responsibility matrix, CRS, vertical datum, accuracy schedule, access plan and data rights before field capture.

Required packages

ID	Dataset	Owner role	Delivery
SUR-CTRL	Primary and secondary survey control	appointed survey authority	RINEX plus field logs, adjustment report, coordinate schedule and signed control report
SUR-TOPO	Corridor and site topographic survey	survey delivery lead	GeoPackage 3D points/lines/polygons plus LandXML terrain and accepted PDF plans
SUR-UTIL	Above- and below-ground utilities	utility survey lead	GeoPackage 3D utility features, investigation records, photographs and confidence/quality attributes
SUR-LAND	Property, right-of-way and access constraints	client land and legal lead	GeoPackage boundaries/rights plus controlled title, consent and access register
SUR-FLOOD	Hydrology, flood and drainage evidence	drainage and flood lead	GeoPackage water/flood/outfall layers plus rainfall, level, model and authority records
GEO-GI	Ground investigation and laboratory testing	geotechnical engineer of record	AGS or equivalent structured logs, coordinates, samples/tests, factual report and interpretive design report
OPS-WORKSHOP	Existing workshop and depot capability audit	operator engineering lead	asset/capability register, measured layouts, utilities, photographs and gap assessment
OPS-FLEET	Existing fleet and operating interface audit	operator rolling-stock lead	fleet register, measured interface sheets, duty/condition records and migration constraints
SUR-IMAGE	Georeferenced photography and photogrammetry	reality-capture lead	source-image manifest and hashes, camera logs, GCP/check points, orthophoto/DSM and processing report
SUR-SCAN	Point-cloud and dimensional interface capture	reality-capture lead	E57/LAS/LAZ plus station setup, targets/control, registration and CloudCompare QA report
INF-GOV	Information ownership, security and acceptance	client information manager	responsibility matrix, naming/revision rules, common-data-environment plan, licence/consent register and acceptance record

Scope and acceptance

SUR-CTRL — Primary and secondary survey control

Stable primary/secondary monuments covering every corridor, station, depot, bridge, energy site and construction compound.

Provisional accuracy/quality requirement: Primary control <=10 mm horizontal and <=15 mm vertical expanded uncertainty; interface control <=5 mm relative to accepted project control. Survey authority must approve or replace these project targets.

Acceptance evidence:

- approved projected CRS, epoch, units, geoid model and vertical datum
- instrument/antenna serials and calibration status
- redundant occupations and independent checks

- least-squares residuals, covariance/uncertainty and rejected-observation record
- signed monument descriptions and protection/inspection plan

SUR-TOPO — Corridor and site topographic survey

Full corridor, station, depot, access, watercourse, road, structure, compound and lifting-route coverage with breaklines and feature coding.

Provisional accuracy/quality requirement: Hard-surface/detail points <=25 mm horizontal and <=20 mm vertical; general ground <=50 mm. Survey authority must approve class-by-feature targets and check density.

Acceptance evidence:

- feature code and layer dictionary
- check-point residual report
- coverage/void map and inaccessible-area register
- capture date/epoch and change notes
- signed completeness review against route limits

SUR-UTIL — Above- and below-ground utilities

Electricity, telecoms, water, sewer, drainage, fuel, irrigation and unknown services within the works influence zone.

Provisional accuracy/quality requirement: Every feature carries horizontal/vertical uncertainty and detection/verification method; trial holes or records alone must not be silently treated as surveyed truth.

Acceptance evidence:

- records search and authority contacts
- detection method, equipment and calibration
- surface-feature correlation
- verification exposure/trial-hole schedule where design risk warrants
- unresolved and abandoned-service issue register

SUR-LAND — Property, right-of-way and access constraints

Ownership, occupation, easements, road reserve, temporary access, compounds, lifting routes and acquisition constraints.

Provisional accuracy/quality requirement: Boundary geometry remains indicative until reconciled to authoritative records and verified by the competent land authority.

Acceptance evidence:

- authoritative source and retrieval date
- boundary/status confidence
- owner/occupier and consent state
- temporary/permanent requirement
- dispute and encroachment register

SUR-FLOOD — Hydrology, flood and drainage evidence

Catchments, channels, historical levels, culverts, outfalls, tailwater, groundwater indicators and exceedance routes affecting the works.

Provisional accuracy/quality requirement: Levels use the accepted vertical datum; rainfall return periods and climate allowances require local authority confirmation.

Acceptance evidence:

- source dates and return periods
- surveyed channel/outfall/culvert levels
- historical event and community evidence

- blockage/tailwater assumptions
- authority-approved design storm and discharge constraints

GEO-GI — Ground investigation and laboratory testing

Boreholes, CPTu, test pits, groundwater monitoring and laboratory programme zoned for at-grade, elevated, station, depot and energy works.

Provisional accuracy/quality requirement: Investigation spacing/depth and tests are risk-based and approved by the geotechnical engineer after desk study and walkover.

Acceptance evidence:

- desk study, walkover and conceptual ground model
- location/elevation tied to accepted control
- factual logs and chain of custody
- strength, settlement, corrosivity and groundwater parameters with uncertainty
- signed geotechnical design recommendations and residual risks

OPS-WORKSHOP — Existing workshop and depot capability audit

Buildings, pits, cranes, lifting, machine tools, power, compressed air, fire systems, stores, access and workforce capability.

Provisional accuracy/quality requirement: Critical lifting, clearance and interface dimensions are measured to the tolerance required by the selected equipment supplier.

Acceptance evidence:

- asset identity, condition and certification
- measured capacity/clearance and utility availability
- repair/reuse/replace decision
- production and maintenance throughput
- operator-approved upgrade work packages

OPS-FLEET — Existing fleet and operating interface audit

Existing vehicles, gauges, couplers, wheel/rail, depot interfaces, rescue, operating rules and reusable components relevant to the pilot.

Provisional accuracy/quality requirement: Safety and physical interfaces use independently checked measurements and controlled vehicle/configuration identity.

Acceptance evidence:

- vehicle/configuration and maintenance history
- measured gauge/coupler/wheel/depot interfaces
- operating duty and failure records
- rescue/recovery constraints
- signed compatibility and migration decisions

SUR-IMAGE — Georeferenced photography and photogrammetry

Continuous corridor and detailed station/depot/structure/utility-context coverage, subject to permissions and privacy controls.

Provisional accuracy/quality requirement: Ground sampling distance and independent check-point limits are agreed by use case; neither EXIF position nor self-reported processing precision is acceptance evidence.

Acceptance evidence:

- capture plan, permissions and privacy controls
- camera/lens identity and calibration

- GCP versus independent check-point separation
- processing software/version/settings
- check-point residuals, voids, artefacts and obscuration map

SUR-SCAN — Point-cloud and dimensional interface capture

Complex structures, station/depot interfaces, constrained utilities and clearance-critical areas selected by the design team.

Provisional accuracy/quality requirement: Registration RMS is reported but not used alone; independent control/check residuals and local interface comparisons govern acceptance.

Acceptance evidence:

- scanner identity/calibration and setup log
- target/control schedule
- registration method and residuals
- independent check points and cloud-to-cloud sections
- density, range, noise, void and movement/change register

INF-GOV — Information ownership, security and acceptance

All raw and derived field information, personal/sensitive content, third-party records and authoritative project derivatives.

Provisional accuracy/quality requirement: Not applicable; this package governs provenance, access, revision, retention and authorized use.

Acceptance evidence:

- owner, producer, checker and accepting authority per dataset
- licence, reuse, publication and personal-data constraints
- file naming, metadata, revision and immutable hash rules
- backup, retention and access-control plan
- signed receipt/rejection/reissue workflow

Data handling and handoff

Keep raw survey, imagery, scan, utility and personal data in access-controlled project storage; Git contains requirements, manifests, hashes, accepted summaries and non-sensitive derivatives only.

Complete `survey-input-manifest.csv` for every delivery. Each accepted row must name the producer/checker, CRS and vertical datum, capture date, controlled project-storage path, SHA-256 digest and acceptance state. RTKLIB processing configurations/status/residual outputs, QGIS/CloudCompare/ODM settings and signed review records accompany their respective source data.

Baghdad surveyed-ground-model gate

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/survey/ground-model-readiness.md

- Status: awaiting-ground-model-data
- Technical inspection completed: no
- Technical screen passed: no
- Survey authority accepted: no

> Automated inspection checks file integrity, GeoPackage structure, independent-check residuals and processing-report completeness. It does not prove feature completeness, datum correctness or fitness for design.

> Only the appointed survey authority and information manager can accept the master ground model after control, coverage, void, epoch, uncertainty, CRS and vertical-datum review.

Current gates

- Missing technical roles: control_acceptance_report, topographic_features, terrain_dtm, checkpoint_residuals, odm_processing_report, orthophoto, registered_point_cloud, cloudcompare_qa_report, void_register, master_ground_model
- Duplicate roles: none
- Missing authority-review roles: control_acceptance_report, topographic_features, terrain_dtm, checkpoint_residuals, odm_processing_report, orthophoto, registered_point_cloud, cloudcompare_qa_report, void_register, master_ground_model, ground_model_acceptance_record
- Authority-record findings:
  - ground-model acceptance record not received

Controlled workflow

Keep source imagery, raw scans and intermediate dense clouds in access-controlled project storage. Git records requirements, file hashes, non-sensitive QA summaries and accepted derivatives only.

1. Complete accepted survey control before registering terrain or clouds.
2. Process imagery in OpenDroneMap and registration/comparison in CloudCompare; preserve settings, source/output hashes and QA reports.
3. Build the federated master GeoPackage in the approved horizontal CRS and vertical datum.
4. Add every derivative to the shared receipt with its immutable hash and independent checker.
5. Run inspection, resolve residual/coverage findings, and obtain the controlled authority acceptance record.

Baghdad route and station fit gate

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/survey/route-station-fit-readiness.md

- Status: awaiting-route-fit-evidence
- Lines: 14
- Stations: 159
- Technical screen passed: no
- Coordinated authority acceptance: no

> The automated gate proves that every current line and station is covered by hash-locked, internally consistent route-fit evidence and that no unresolved critical/high issue is hidden. It does not determine land rights, utility position, flood risk, accessibility, traffic safety, lifting feasibility or possession sufficiency.

> Only the relevant utility, land, drainage, highway, operations and design authorities can accept their evidence, followed by the engineer of record, operator and information manager accepting the coordinated route-fit record.

Current gates

- Missing technical roles: ground_model_readiness, surveyed_alignment_readiness, utility_coordination, land_access, flood_drainage, station_access, intercity_yard_integration, road_traffic_management, construction_logistics, possession_staging, route_fit_verification_report, route_fit_issue_register
- Duplicate roles: none
- Authority findings:
  - route-fit acceptance record not received

Controlled workflow

Store titles, personal data, utility-sensitive records, permits and native studies in the controlled project environment. Git carries requirements, empty receipts, immutable hashes and non-sensitive acceptance summaries.

1. Accept survey control, the ground model and every surveyed alignment.
2. Register checked utility, land, flood, station-access, yard/intercity, road, logistics and possession evidence.
3. Reconcile exactly one resolved result for every current line and station, with source hashes.
4. Close or formally transfer all high/critical issues in the controlled issue register.
5. Obtain the coordinated authority acceptance record against the immutable evidence hashes.

Baghdad structural-release gate

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/survey/structural-release-readiness.md

- Status: awaiting-structural-evidence
- Lines: 14
- Technical screen passed: no
- Authority accepted: no

> The automated gate checks actual solver output files and numerical limits, reviewed asset/support coverage, load-case/result reconciliation and independent-check completeness. It does not validate modelling assumptions, codes, loads, mesh quality, reinforcement, prestress, soil springs, fatigue detail, seismic response or structural safety.

> Only the structural and geotechnical engineers of record, independent checker, asset owner and approving authority can release the project structures in a signed controlled record.

Current gates

- Missing technical roles: drainage_ground_readiness, structural_design_basis, structural_asset_schedule, load_case_register, opensees_model, opensees_report, calculix_input, calculix_report, structural_verification_report, independent_check_record, civil_asset_register
- Duplicate roles: none
- Authority findings:
  - structural acceptance record not received

Controlled workflow

1. Accept drainage/ground design and freeze the project structural basis and load combinations.
2. Schedule every span, pier, abutment, foundation and special structure against line chainage.
3. Preserve OpenSees global/seismic/soil-spring and CalculiX component input/output hashes and convergence evidence.
4. Reconcile foundation, wind, seismic, fatigue and bearing/movement results per scheduled asset.
5. Close independent-check comments and obtain the signed structural release.

Baghdad surveyed-alignment gate

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/survey/surveyed-alignment-readiness.md

- Status: awaiting-surveyed-alignments
- Lines expected: line-1, line-2, line-3, line-4, line-5, line-6, line-7, line-8, line-9, line-10, line-11, line-12, line-13, line-14
- Technical screen passed: no
- Authority accepted: no

> The automated gate proves receipt integrity, OSR-ALN hard-gate compliance, explicit horizontal/vertical/cant content, line and station reconciliation, deterministic LandXML re-import, and recorded interface tolerances. It does not establish that the survey, fit, clearances, possessions or construction design are correct.

> Only the appointed alignment designer, survey authority, track engineer and information manager can accept an alignment for design use in a controlled signed record.

Current gates

- Missing technical roles: *:ground_model_readiness, *:interface_verification_report, line-1:surveyed_osr_aln, line-1:landxml_export, line-1:landxml_roundtrip_report, line-2:surveyed_osr_aln, line-2:landxml_export, line-2:landxml_roundtrip_report, line-3:surveyed_osr_aln, line-3:landxml_export, line-3:landxml_roundtrip_report, line-4:surveyed_osr_aln, line-4:landxml_export, line-4:landxml_roundtrip_report, line-5:surveyed_osr_aln, line-5:landxml_export, line-5:landxml_roundtrip_report, line-6:surveyed_osr_aln, line-6:landxml_export, line-6:landxml_roundtrip_report, line-7:surveyed_osr_aln, line-7:landxml_export, line-7:landxml_roundtrip_report, line-8:surveyed_osr_aln, line-8:landxml_export, line-8:landxml_roundtrip_report, line-9:surveyed_osr_aln, line-9:landxml_export, line-9:landxml_roundtrip_report, line-10:surveyed_osr_aln, line-10:landxml_export, line-10:landxml_roundtrip_report, line-11:surveyed_osr_aln, line-11:landxml_export, line-11:landxml_roundtrip_report, line-12:surveyed_osr_aln, line-12:landxml_export, line-12:landxml_roundtrip_report, line-13:surveyed_osr_aln, line-13:landxml_export, line-13:landxml_roundtrip_report, line-14:surveyed_osr_aln, line-14:landxml_export, line-14:landxml_roundtrip_report
- Duplicate roles: none
- Authority findings:
  - alignment acceptance record not received

Controlled workflow

Keep signed source surveys, native authoring files and large point clouds in controlled project storage. Git carries the empty receipt, requirements, reproducible checks and non-sensitive accepted exchange derivatives.

1. Accept survey control and the surveyed ground model.
2. Fit each line in the confirmed CRS/datum; explicitly issue horizontal, vertical and cant schedules.
3. Export one OSR-ALN and LandXML file per line and record the deterministic re-import hash and comparison tolerances.
4. Reconcile every platform and record yard, turnout and clearance dispositions per line.
5. Run this inspection and obtain the controlled multi-discipline acceptance record.

Baghdad manufactured-viaduct comparison

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/viaduct-comparison/README.md

Status: controlled planning package; no construction design, quotations or selected corridor.

Compare Pi20, Pi25 and constrained-access OSR-US for the same complete double-track scope. Supplier bids must include fabrication, transport, erection, temporary works, foundations, traffic management, connections, track/egress, checking and maintenance. No 30 m product is introduced.

The investigation candidate is line-3 chainage 18216.4–36732.7 m (18516.3 m), the longest current Pi25 run. It is not a surveyed or selected alignment. The separate 1,000 m comparison below includes both end supports; support ground zones and deep-element lengths are unknown.

Package	Bays	Supports	Bearings	Beam/segment lifts	Bare member kg	Installed USD/double-track m
Pi20-simple-span-link-slab	50	51	400	100	59950.0	Unpriced
Pi20-structural-continuity	50	51	252	100	59950.0	Unpriced
Pi25-simple-span-link-slab	40	41	320	80	74937.5	Unpriced
Pi25-structural-continuity	40	41	200	80	74937.5	Unpriced
US25-segmental	40	41	None	800	None	Unpriced

Pi25 has only 62.5 kg bare margin beneath 75 t; Pi20 has 50 kg beneath its 60 t product target. Complete member and hook masses remain unknown. Segmental concrete and support/bearing geometry require their own supplier design; Pi quantities are not transferred. The current OSR-US coordination envelope is 5.12 m wide, exceeding the primary 3 m shipping target. Shorter/lighter segments do not solve a width restriction without redesign or a reviewed oversize route.

Connections and quantities

A link slab can remove an expansion joint while simply supported girders retain their bearings. The shared-bearing scheme needs a structural connection and checked staged load path. A finite 40-span Pi25 corridor has 320 simple-span bearings versus 200 in the structural alternative. The periodic planning index also uses 320 versus 200, but its ten expansion gaps differ from the nine gaps between finite units. Both schemes need CWR/thermal/braking/seismic/foundation interaction, fatigue, replacement access and connection prices.

Under the existing index only, restoring simple-span bearings changes USD 9.748m/km to USD 10.198m/km: USD 131.042m as a uniform-rate illustration over all current elevation. This is an unadopted counterfactual before connection/end-effect/EPC costs, not a priced scope or a transfer of Pi bearings to segmental/special designs.

The financed bearing sensitivity applies only to 269.129 km of existing standard Pi25: USD 121.108m direct plus the existing incremental EPC rate once. Its capital is USD 8.264bn and terminal gap debt IQD 0.000tn. Monthly/six-month cash, debt/reserves/early repayment are recalculated under unchanged government and currency rules. Original civil invoice timing and origin shares are inherited assumptions, not supplier evidence; all consequential costs remain open.

Alignment and budget boundary

The register covers all 1127 elevated segments, including 49 individual special reviews. Its standard-rate base allowance is USD 2.839bn. Search deterrents are excluded from monetary allowances. This base allowance is not a complete installed price or an achieved saving; special/segmental increments remain unknown. Each special segment has wider-curve, station-move, right-of-way, segmental, land/utility, traffic and whole-life comparison fields. No alternative is accepted and original capital/debt figures are preserved.

The individual special reviews are ranked by local search-deterrent weighted metres to direct investigation effort. This score is not money or an achieved saving. Radius comes from the local controlled geometry screen; it is not a surveyed radius. The Pi20 chord screen shows where shortening alone still fails the existing catalogue allowance. Nearby station IDs support bounded station/right-of-way studies.

Special priority	Line/chainage m	Length m	Search-deterrent weighted m	Inferred radius m	Pi20 chord screen
1	line-9 1220.8–1349.1	128.3	3452.3	51.0	False
2	line-9 80039.9–80176.5	136.6	1284.2	79.6	False
3	line-9 94028.5–94136.8	108.3	1184.6	78.9	False
4	line-9 86239.5–86347.8	108.3	1178.0	78.9	False
5	line-9 76774.0–76898.9	124.9	1042.5	91.9	False

Special priority	Line/chainage m	Length m	Search-deterrent weighted m	Inferred radius m	Pi20 chord screen
6	line-9 71534.7–71614.7	80.0	985.3	70.7	False
7	line-1 3679.4–3739.4	60.0	840.7	70.7	False
8	line-9 69008.8–69085.3	76.5	817.4	79.6	False
9	line-9 84512.4–84589.0	76.6	809.7	85.4	False
10	line-10 6655.3–6755.3	100.0	797.6	85.0	False

A further 20% reduction in beam concrete changes the existing index by only USD 0.450m/km before any offsetting prestress, reinforcement or fabrication changes. No literature percentage is applied again.

Production, procurement and first article

comparison.json contains per-support foundation records, five complete-package quantity/rate forms, equipment-release fields, all eight production fronts and first-article hold points. installed-cost-rfq.csv retains invoice currency, origin, quote date and exclusions. Unknown costs remain null, not zero; partial totals cannot produce an installed price. IQD quotes stay IQD, with historical FX used only for reporting. Loan eligibility requires actual invoices and lender terms.

Measure casting, strength release, column/cap connections, transport, erection, closure curing, survey/tolerance correction, track/egress and acceptance in crew-hours per complete double-track bay. The rate is the slowest front, not the beam lift count. Parallel moulds and fronts, shifts, closures, 10–15-bay foundation/accepted-stock buffers and deck-supply loads require a resource-loaded programme and trials. The existing 48-hour casting planning cycle remains until the 24-hour target is physically qualified.

Issue competing ground-crane/portal and overhead-launcher bids for the same route. Include configured charts, pads/ground improvement, mobilisation, closures, launch/deck reactions and recovery. Use conventional prestressed concrete first; compare UHPC connections and special steel/composite crossings where measured whole-life resources justify them. Carbon percentages are not cost savings.

Build and measure the first beam and representative pier/cap/connection only after design release. Record accepted complete bays/week, labour, rejected/reworked pieces and installed cost. Supplier bids, boreholes, utilities, demand counts, actual production trials, signed checks and funding remain external work.

Review scope and evidence

The older review of `135e249e` predates the integrated USD 8.135bn full-fleet sensitivity and its IQD 0.000tn terminal gap debt. Depot/workforce/site-energy reconciliation, supplier RFQ forms, opening-fleet replay, rental alternatives and clean-checkout bootstrap already exist in delivery closure. This package closes repository comparison gaps; it does not replace supplier offers, measured demand or a complete delivery budget.

- FHWA field-cast UHPC connections: Link slabs can join simple-span decks while independent bearings remain; highway precedent only (checked 2026-10-04).
- FHWA prefabricated bridge selection framework: Compare access, supplier capability, connections and the complete installation system (checked 2026-10-04).
- Chennai Metro construction methods: Precast substructure and advance investigation are precedents, not Iraqi production rates or quotations (checked 2026-10-04).

Baghdad retained under-viaduct rental portfolio — 2026-10-04

Source: cities/catalogue/west-asia/Iraq/Baghdad/engineering/viaduct-rentals/README.md

> Original catalogue or earlier scope reference. The figures and policies below retain their original assumptions; they are not the latest Baghdad staffing, depot, procurement or funding basis. See the current Baghdad recalculation and current city summary. National/portfolio totals have not been repriced with that conditional Baghdad option.

Rental income is a conditional addition to the existing station-development sale case, with different assets and external tenants. No site is commercially accepted, surveyed, leased or funded. Network Rail's arch portfolio provides a precedent for railway SME premises. Places for London's guidance emphasises access, services, insulation, fit-out responsibilities and permitted uses. These UK precedents do not establish Iraqi demand, prices or building permissions. Sources checked 4 October 2026.

Geometry, eligibility and pilot

The controlled design has 1,127 elevated civil segments totalling 291.2043 km, with median length 48 m. Each register entry carries its civil index, line/chainage, planning coordinates, existing parent track assets and nearest station. 25 segments lack a materialised parent track link and need asset-register reconciliation; no ID is invented. Coordinates interpolate the corridor and are not surveyed footprints. Clear height, frontage, ownership, title, utilities, permitted use and street/rail access remain null or unaccepted.

The reference screen excludes segments shorter than 75 m, reserves 10 m at each approach and counts complete 25 m planning bays. Each bay tests three independent 30 m² internal units, with 2 m protected support-end inspection zones and 1.2 m independent access. It suggests 907,380 m², before unknown road crossings, utility space, pier positions, access, ownership, flooding, station-sale overlap and fire exclusions. Confirmed eligible area is zero. Elevation alone never grants eligibility. Pier height is not usable clear height beneath a cap/deck.

The 200,000 m² illustration exceeds this screen by 0 m² and is unmapped-area-blocked, with no lease receipts in executable integrated cases. It requires a different surveyed footprint/reference layout; there is no automatic widening of a viaduct. Small/medium models allocate assumed area against identified civil segments and phase construction by their infrastructure dates. These are conditional surveys-to-test, not valuations. Thirty draft pilot units use one long candidate selected by station-chainage proximity and bay count, without a claim of commercial desirability. Market-test 20–50 units through comparable IQD rents, tenant interest, permissions and costed fit-outs before expansion. No external contact has been made.

Open reference unit and costs

Unit plan gives a 6 × 5 m internal module in a 6.4 × 5.4 m independent shell, requiring 3.2 m measured clear height. Parts and allowances total USD 21,000 equivalent per 30 m² unit: USD 18,000 direct, USD 2,000 design/shared services and USD 1,000 contingency, contracted in IQD. These are planning allowances, not quotes. Walls/roof, thermal/acoustic insulation, drainage, slab, independent foundations, doors/windows, separate metered/isolation services, fire/egress equipment, vehicle protection and removable inspection panels are included. Actual soil, materials, fire protection and service capacity may change cost materially. No structure is dimensioned from rental value alone; compare net value against costs before modifying supports or spans. Open licences remain intact.

The shell is independent of railway piers/caps/deck: no drilling, welding or load attachment. Preserve bearing/jacking, drainage and electrical access, surveyed support clearances and maintenance easements. Independent utility meters are paid directly by tenants in the model; no utility reimbursement is booked as extra profit. Retail, services and studios are candidate uses only; workshops, cooking, hot works, fuel/solvent storage and battery activities require specific assessments. There is no generic permission for a use because it fits the floor area.

Illustrations versus priced cash-flow scenarios

Area illustration	Lettable m²	Monthly rent USD eq/m²	Occupancy	Annual landlord NOI USD m eq
small	40,000	10	70%	2.52
medium	100,000	15	80%	10.80
large	200,000	20	85%	30.60

These illustrations deduct 25% of occupied receipts and exclude capital, financing and tax. For the medium illustration, 30 annual receipts, first at month 48, rising 5% each year, discounted at 13.4%, have PV USD 79.405m. Explicit payment dates matter: this is receipts-only PV, not project NPV or the phased portfolio forecast.

Executable rental scenario	Status	Fit-out capital USD m eq	First conditional cash month	Resource NPV before tax USD m eq	Standalone after-tax NPV USD m eq
small	conditional-unverified-retained-portfolio	36.988	43	-4.907	-5.964
medium	conditional-unverified-retained-portfolio	89.193	43	18.317	11.312
large	conditional-unverified-retained-portfolio	181.023	43	96.389	74.259
medium_downside	conditional-unverified-retained-portfolio	89.193	55	-37.876	-38.358

Monthly ledgers phase fit-outs after the corresponding infrastructure-completion month and delay tenant handover until railway opening or fit-out completion, whichever is later. Six-month enclosures, three initial rent-free months and an 18-month occupancy ramp precede steady receipts. The last fractional module is constructed in full; spare floor area receives no rent. Rent and OPEX escalate 5% from financial close, and unquoted capital escalates 5% to each invoice. The downside uses rent USD 9/m², 55% target occupancy, 12 extra months of vacancy, a 36-month ramp, 8% arrears and 7% OPEX inflation.

Market vacancy is separate from modeled tenant turnover (10%/year, three-month re-letting vacancy and one-month free concession). Three-percent arrears have 50% recovery after six months; unrecovered debt is written off, and final unresolved receivables have zero recoverable value. Revenue is cash collected in these pro-forma sensitivity accounts; statutory accrual/tax treatment needs advice. Occupied maintenance/insurance uses 25% of reference occupied face rent with its own OPEX index, including during rent-free concessions, plus USD 10/m²/year for vacant-unit maintenance/insurance. Costs persist through concessions or defaults. Major refurbishment is additional cash expense at 20% of reference fit-out every 12 years, with OPEX inflation; it is not also capitalised. Existing station kiosk receipts are unchanged.

Security deposits of two months' indexed face rent are assumed topped up/refunded with occupancy. They are separately restricted cash and tenant liabilities, with zero revenue, zero construction funding and no shareholder distribution. Ending leases refund all deposits. No upfront lease premium or sold rental parcel is booked. Parcel overlap with the existing 15 station sale candidates is unknown and must be accepted as disjoint before either case can become executable. Current site-right cash is explicitly assumed zero, with additional land opportunity cost unknown: full site economics cannot be concluded from the reported before-land NPV. No USD credit or additional government cash is assumed for fit-outs.

Assets depreciate over 20 years from each cohort handover. Standalone tax applies the illustrative 15% positive annual cash-profit-after-depreciation proxy, with no loss relief or interest deductions, paid the next month; it is not an Iraqi tax ruling. The holding model instead includes the same portfolio inside its separate-business tax stress, so standalone tax is not deducted twice. The project horizon caps late cohorts' rental years and leaves no unsupported terminal sale/rent beyond the railway forecast. These cases improve only to the extent that collected rent exceeds all additional resources and costs.

Financing, dividends and terminal cash

Financing redesign adds a separately financed retained-property entity with 25% private partner fit-out equity and IQD construction bank debt. Its operating-gap capacity is zero; missing carry/fees/reserves are exposed rather than adding an uncommitted facility to the existing 13tn cap. Internal distributions are zero pending lender consent. External tenant cash adds group revenue; future property-to-rail transfers would cancel exactly. The holding-equity study uses the same resources with a 100%-owned business, replacing partner equity with the parent subscription case and borrowing the residual in IQD. These are alternative ownership/funding structures, not cumulative sources.

The holding study separately compares all-debt-repaid dividends with an unapproved coverage/reserve/profit-based policy, and reports a diagnostic of proportional final unrestricted cash net of liabilities. Cash at the horizon is not a guaranteed redemption, quoted share price or assumed sale of railway/property assets. Do not confuse that diagnostic with actual company distributions or count cash twice. Company accounts remain separately reconciled.

Digital twin, common hazards and evidence

Every draft pilot unit links its civil segment/parent track, proposed enclosure design, unknown lease and tenant IDs, meters, isolation, inspection rights, permitted use and hazard records. The native ERP mapping uses existing Asset/Contract/Customer/Sales Invoice/Payment Entry/Asset Maintenance/Issue types when actual records and permissions exist; null IDs are not live leases, meters or safety records. No fake tenant, permit, contract or asset is created from an unverified footprint.

Tenant fire, flooding/drainage, delivery impact, blocked inspection access and utility/attachment faults propagate from enclosure to pier/foundation, bearings and shared spans and can affect railway operations. Redundant onboard controllers provide no mitigation for these common physical hazards. Proposed inspections include pre-lease engineering acceptance, monthly access/drain/isolation checks, post-incident checks, annual structure-linked inspection and controlled jacking/maintenance closures; durations and legal/fire requirements need actual assessment. Native maintenance/Issue records must trace the landlord obligation to the railway asset and escalation owner. Occupancy and any railway restriction

require the authorised engineer's decision, not automatic evidence acceptance by this model.

Six 90-day evidence packages require surveyed eligibility and rights/no-overlap, independent civil/fire/utility access acceptance, market-tested pilot leases, supplier fit-out/refurbishment quotes, native lease/meter/maintenance controls, and independent cash/tax/funding/distribution review. Named people, surveys, quotes and leases remain pending. Native ERP Task imports are open drafts; no evidence or operating release is granted.

Regenerate `.venv/bin/python tools/automation/baghdad_viaduct_rentals.py`, then financing redesign, equity and proposal; validate source/output hashes with `--check`.

Baghdad Acceptance And Accreditation Evidence Basis

Source: cities/catalogue/west-asia/Iraq/Baghdad/operations/acceptance-evidence-report.md

This report is the generated evidence basis for acceptance, trial-running readiness, and accreditation review. It is not itself an approval or certificate; it is the traceability index that shows which assets, manufacturing packages, BOM/material refs, QA hold points, evidence records, release authorities, and predecessor controls must be closed.

Summary

Item	Count / Status
Assets in register	1,717
Manufacturing schedule rows	7,602
Manufacturing material/BOM rows	14,463
Manufacturing QA verification rows	7,602
Construction QA action rows	6,362
Maintenance handover schedule rows	10,148
Manufacturing rows with material refs	7,602 / 7,602
Manufacturing rows with verification refs	7,602 / 7,602
Manufacturing rows linked to QA actions	7,602 / 7,602
Unresolved external predecessors	0

Material / BOM Basis

Source	Rows
`project_kit`	14,463

Rolling-stock rows link to the generated rolling-stock BOM and COTS fit-out BOM. Infrastructure rows use controlled `project_kit:*` refs until detailed civil/station/energy BOMs are added.

QA Gate Coverage

QA gate	Verification rows
`qa-00-design-freeze`	1
`qa-10-carbody-structure`	2,427
`qa-11-bogie-wheelset`	809
`qa-12-traction-brake-battery`	809
`qa-13-passenger-systems`	809
`qa-15-first-article-trainset`	809
`qa-20-survey-geotech`	145
`qa-21-earthworks-drainage`	145
`qa-22-trackform-rail`	441
`qa-24-stations-depots-plant`	320
`qa-25-power-energy`	279
`qa-26-wayside-comms-safety`	608

Acceptance Control Logic

- Every manufacturing package has a controlled material/BOM row set.
- Every manufacturing package has a QA verification row.
- Every verification row links to a generated QA action by `qa_uid`.
- Resolved predecessor ids are generated for schedule blocking.
- The portal blocks successor manufacturing work until predecessor work orders are closed with pass evidence.
- The portal blocks manufacturing closeout until the selected work order has pass evidence.
- Failed evidence creates a defect/NCR and puts the work order on hold.
- SQLite-backed Ops Core stores work orders, inspections, defects/NCR, and audit records for handover.

Remaining External Release Gates

This generated evidence basis organizes acceptance work, but it does not close release gates that require independent review, field data, hardware bring-up, supplier freeze, first-article tests, or authority acceptance. The open gates remain:

Gate family	Still required before accreditation / revenue release
Independent safety assessment and residual risk	Named ISA/assessor review, action log, residual-risk acceptance, and deployment-specific ALARP/tolerability decision.
Formal safety integration	Consensus refinement proof from TLA+ to implementation behavior, plus signed safety-log integration tests for forged, replayed, stale, and unknown-issuer entries.
Pilot hardware evidence	Exact COTS BOM freeze, wiring/harness maps, enclosure/mounting, power/thermal margins, SD image checksums, self-test logs, bench/safety evidence, and commissioning records for T-ECU/S, T-ECU/A, T-OBS, W-SBC, and S-SBC.
Rolling-stock production release	Supplier envelope freeze, EN structural/FEA reports, weld/WPS/NDT packages, manufacturing drawings, flat patterns/NC output, harness routing, weight/balance, first-car build hold point, and first-article inspection.
Field validation	Obstacle/intrusion sensor calibration and representative hot-weather, dust/soiling, night, and rain datasets with false-positive/false-negative analysis.
Charging and site energy	Site-specific solar yield, grid interconnect, charger thermal study, protection settings, utility approval, and train charging interface tests.
Operations validation	Operator workshops, translated/deployment rulebook where needed, competence records, emergency exercises, maintenance access trials, and trial-running records.

Those items are tracked in the certification release gap register, control-electronics release checklist, and rolling-stock v2 release checklist. The acceptance matrix should be treated as the evidence index that collects those closures, not as the closure itself.

Review Artifacts

- Tracked compact asset register: `baghdad-assets.csv`
- Tracked operations manifest: `baghdad-operations-manifest.json`

The following high-volume files are regenerated in the local city package and intentionally excluded from Git. Their names are recorded for handover without presenting unavailable GitHub links:

- `baghdad-operations.json.gz`
- `baghdad-acceptance-evidence-matrix.csv`
- `baghdad-manufacturing-schedule.csv`
- `baghdad-manufacturing-materials.csv`
- `baghdad-manufacturing-verification.csv`
- `baghdad-qa-register.csv`
- `baghdad-maintenance-schedule.csv`

Accreditation Use

The evidence matrix can be filtered by asset, package, QA gate, release authority, or material source. During acceptance, each row should be matched to completed Ops Core evidence, defects/NCR status, and the release authority signoff before the related asset is accepted into trial running or passenger operation.

Iraq funding programme and Baghdad planning status

Source: cities/catalogue/west-asia/Iraq/IRAQ-FUNDING-PROGRAMME.md

Current Iraq planning basis — 2026-10-06

Current country basis · Baghdad summary · Connected design, production and energy study.

The retained `local_positive` Baghdad financial comparator is USD 8.667bn, with IQD 0.000tn unsourced support and IQD 0.000tn terminal debt. All debt clears at model month 295, conditional on the recorded funding assumptions. Its prices and facilities are unquoted and uncommitted. It does not price the complete accelerated construction and battery scenario.

The accelerated scenario's complete CAPEX, financing, net island savings and accepted opening dates remain unknown. Its USD 9m / 18-launcher allowance covers purchase cash only; freight, assembly, commissioning, transporters, lifting frames, temporary supports, spares, crews and relocations require separate scope-matched prices. The earlier delivered-and-commissioned interpretation is not used. No launcher saving is added to an installed civil rate before removing a verified matching embedded allowance.

Iraq has existing precast facilities and expertise; qualify contracted civil supply and price tooling/upgrades. Baghdad trainset assembly and component-process allowances remain in the financial comparator. Additional national plant capacity and transfer costs require separate scope reconciliation.

The coordinated industrialisation programme joins vehicle interfaces, staged bogie localisation and the standard viaduct system. Its possible vendor register records official-source component and equipment leads for comparison. Iraq already has many precast facilities and relevant expertise; plant capability, compatible tooling and available production slots require qualification. No listed vendor is a selected supplier or a manufacturing-rights agreement.

The coupled service–OD–cost–finance study now derives capacity-limited passenger itineraries from completed train sections and carries the same grid purchases and train distance into conditional monthly finance. Its demand and retained prices/opening months are explicit sensitivities. Complete line openings additionally require stations, specials, track, energy, depots, fleet, testing and approvals; missing dates or prices remain unknown. No coupled case is an accepted forecast or an adopted national budget.

No current consolidated national budget or funded rollout is established. Other Iraqi city catalogue estimates and the earlier generic foreign-turnkey calculation remain separately identified comparators. Their differences are scenario arithmetic, not realised capital or interest savings. Baghdad's later scope cannot be mixed into an older national aggregate and presented as one complete budget.

Construction sensitivities

Case	Days for modelled running bays	Scope
full-doubling	1022	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates
higher-productivity	1098	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates
initial-accelerated	1218	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates
initial-accelerated-reassigned	966	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates
initial-six-day-calendar	1348	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates

These are conditional resource schedules. Station structures, special/closure spans, approved pier locations, delivery access and release evidence remain opening gates. Empty supplier/contract registers indicate missing qualifying evidence; they do not imply that Iraqi industry lacks capability.

Chronological energy study

Case, lower-solar duty	Minimum SOC	Unserved traction kWh
lfp-throughout	0.0%	676,286
sodium-both	0.0%	4,024,245
sodium-stationary	0.0%	676,383

The fixed-assignment diagnostic above uses both ring directions, declared dispatch stations and native rest-to-rest section timing. The current screen spans 4,440 continuous model minutes (3.08 days), including overnight state carry-over and a

study strategy permitting grid replenishment within remaining site import capacity. Charge-rate, thermal and life inputs are unqualified profiles, so these results do not establish a chemistry advantage or suitability.

The separate coupled energy/service controller keeps reserve-limited trains at their actual stations, continues charging, and reflects their availability in subsequent dispatch opportunities.

Controlled case	Completed journeys	Missed dispatch opportunities	Distinct energy-held journeys
lfp-throughout	17,491	909	706
sodium-both	8,536	9,605	2,761
sodium-stationary	17,492	908	719

Geometry/operating-distance findings remain recorded. Conflict-capable timetable, berth/depot access, transient traction and operational multi-day acceptance remain open. Current station passenger demand is unknown where no surveyed station/OD assignment has been entered; the common 3,000 pax/h value is retained only as a separate stress case.

Historical funding calculation — retained comparator

The calculation below retains the earlier USD 7.581bn scope and its own assumptions. Its debt-clearance months and financial differences do not describe the current `local_positive` comparator or the unpriced accelerated scenario.

Scope: Baghdad and one manufacturing plant sized for Baghdad. Samawah, Mosul and every other city are excluded. All funding is proposed and uncommitted.

Independent reconciliation, priced gap finance and six-month tranches

Capital sources still total USD 7,951.287m. The independent pooled-cash reconstruction requires USD 3,193.940m of gross extra cash and retains USD 7,867.053m later, leaving a USD -4,673.113m nominal net deficit before pricing additional gap finance. These three amounts answer different questions. The original capital principal is repaid in the lifetime cash ledger, not added to CAPEX a second time. City EPC is now spread over direct works; the initial baseline-freeze task no longer receives the full programme overhead allowance.

Detailed arithmetic and Iraqi financing routes · six-month reference bond/loan requirements · six-month blended candidate · independent calculation.

Green debt replaces qualified conventional borrowing; grants replace domestic capital debt; guarantees supply credit enhancement rather than cash. The new sensitivity charges supplemental IQD funding for interest and fees, sweeps later surplus to repayment, and exposes cash beyond its illustrative IQD 13tn cap and any unpaid terminal loan. The earlier gross cash-support figures below assume external contributions; they are not a priced bridge-loan requirement.

Priced gap-finance sensitivity	Gross gap draws, IQD tn	Uncovered cash, IQD tn	Unpaid terminal loan, IQD tn
unfunded reference	0.000	4.152	0.000
green label only	0.000	4.152	0.000
zero cost liquidity bound	4.152	0.000	0.000
commercial gap credit	13.000	7.067	13.000
concessional gap credit	5.318	0.000	0.000
blended candidate	3.574	0.000	0.000
green concessional only	4.433	0.000	0.000
fare 5pct flat costs fixed demand	2.003	0.000	0.000
fare 5pct flat costs elastic	2.232	0.000	0.000
fixed fare 5pct opex	13.185	0.486	13.000
fare 5pct opex 5pct	2.293	0.000	0.000
fare 5pct opex 7pct	2.471	0.000	0.000
fare 5pct opex 5pct income 2pct	2.491	0.000	0.000
variable fare 5pct opex 5pct	2.216	0.000	0.000
fare 5pct opex 5pct rents indexed	2.254	0.000	0.000

The candidate mix is uncommitted: eligible IQD green bonds at an assumed 4% plus enhancement fees; USD 25m equivalent climate capital grant; USD 300m equivalent net development-rights proceeds; and USD 25m equivalent annual new net local receipts at full opening. The grant, valuation, legal powers and IQD concessional facility need evidence. Existing rents/fare receipts cannot be counted again. With constant nominal fares and OPEX, a terminal unpaid loan means that sensitivity has not achieved self-financing.

Additional pricing and OPEX inflation sensitivities

The requested paired sensitivity increases fares and OPEX 5% annually from financial close, while testing income growth separately. At 5% income growth and assumed -0.30 real-price elasticity, the illustrative blended case peaks at IQD 2.102tn supplemental debt and ends with IQD 0.000tn unpaid. The case repays the facility within the model horizon under these conditional assumptions. Placed early financing, candidate grant/rights receipts, fixed nominal debt terms and income growth remain assumptions. It is not a committed funding outcome.

Average nominal tickets move from IQD 1,525 at first opening to IQD 1,765 at full opening; 44 trips remain 11.7% of the indexed income proxy when incomes grow 5%. Separate cases test 2% income growth, 7% OPEX inflation, peak/off-peak tiers, fixed demand, and rental indexation. Capital escalation and future FX changes remain outside these sensitivities.

Paired 5% six-month financing · monthly tickets and affordability · variable-ticket six-month financing. The detailed report contains the NPV, all assumptions and downside cases.

Surplus cash and early repayment

These cases use identical paired 5% fare/OPEX/income assumptions and the same conditional blended capital sources. Each pays OPEX, scheduled principal, interest and fees, then funds the debt-service reserve and a three-month current-OPEX buffer before voluntary repayment. Borrowed gap proceeds and uncovered external cash cannot fund early payments. The extra buffer remains cash held at the terminal horizon and earns no interest.

Compare strategies against buffered gap-only, rather than attributing the buffer change to repayment savings. Loans-first pays bank credit, Chinese credit and gap credit before bonds. Cost priority pays bank (9%), ordinary bonds (8%), Chinese credit (5%), green bonds (4% plus 0.5% annual guarantee), then gap credit (2%). It is an interest-rate heuristic, not a proof of the globally best strategy.

Surplus strategy	All debt cleared, month from close	Finance cost saving vs buffered gap-only, USD equivalent m	Early-payment premiums, USD equivalent m	Peak gap debt, IQD tn	Terminal unrestricted cash, IQD tn
gap only buffered	369	0.000	0.000	2.208	44.706
gap then core	209	506.359	36.995	2.208	45.364
loans then bonds	207	653.848	42.809	2.158	45.556
cost priority	203	988.754	56.843	2.158	45.991
cost priority zero premium	203	1,060.010	0.000	2.156	46.084
cost priority noncallable bonds	369	260.705	12.395	2.158	45.045

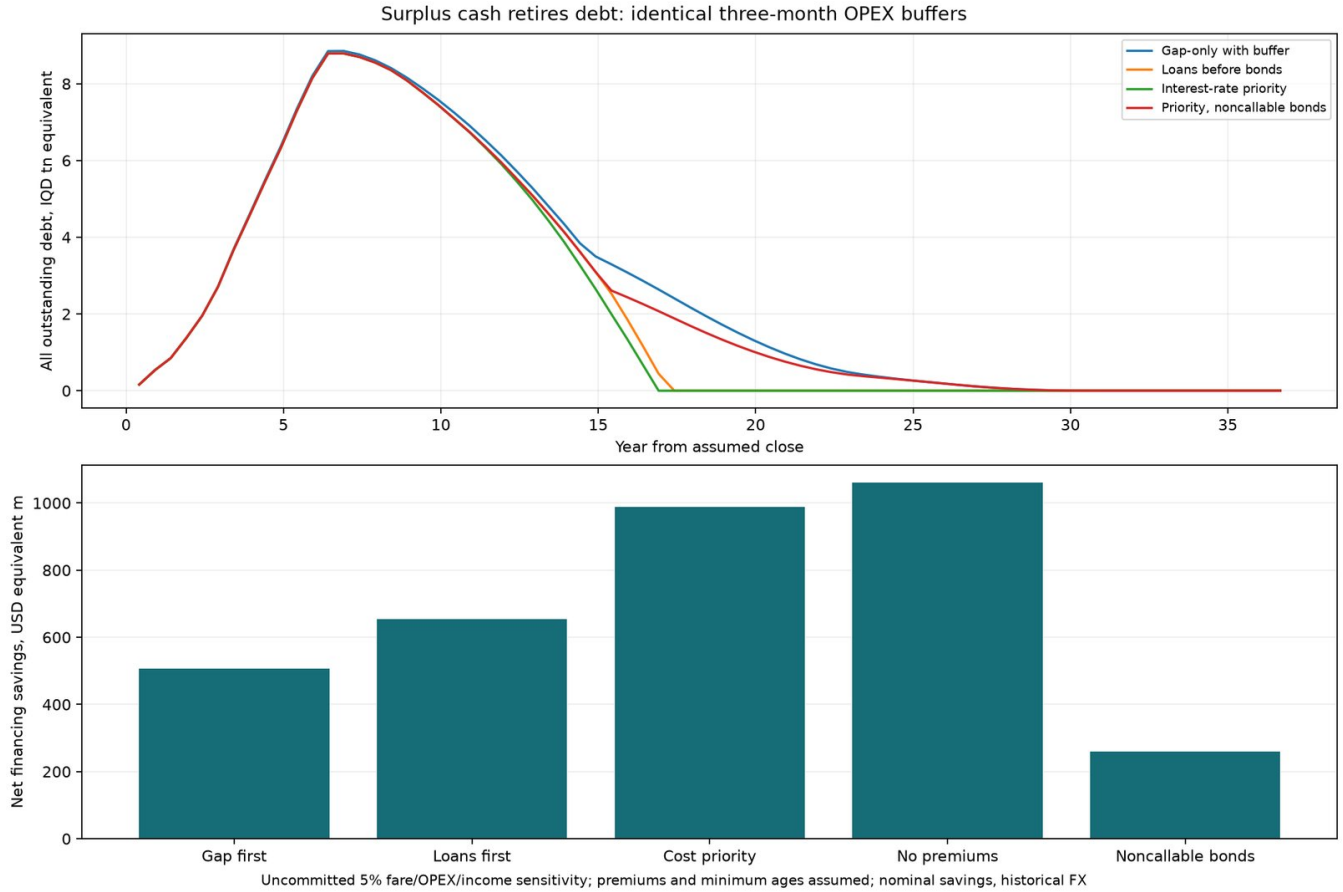
Cost-priority debt is cleared in month 203; buffered gap-only debt is cleared in month 369. Net nominal financing savings are USD 988.754m equivalent, after USD 56.843m assumed early-payment premiums. Savings include core interest, annual green guarantee charges and supplemental interest/draw fees; they exclude principal, which is returned once. All cases retain IQD 0.386tn operating buffer separately from unrestricted cash. There is no additional government contribution above 25% of CAPEX in these cases.

Assumed premiums are 1% of bank/Chinese/green principal and 2% of ordinary bond principal. A minimum draw age of 6 months (bank), 12 (Chinese) and 24 (both bonds) prevents immediate issue-and-redemption. Eligible vintages are repaid oldest first; contractual instalments are kept and maturity shortens. Notice, issuer call rights, investor consent, buyback price, remaining-maturity compensation, tax and FX require actual agreements. Noncallable-bond sensitivity makes no voluntary bond payments; zero-premium sensitivity removes only the assumed premium, retaining minimum ages.

Facility	Buffered gap-only final principal payment month	Cost-priority final principal payment month	Early principal, native currency	Premium, native currency
bank credit	165	122	IQD 636.771bn	IQD 6.368bn
domestic bonds	285	167	IQD 2,547.814bn	IQD 50.956bn
chinese export credit	309	175	USD 592.389m	USD 5.924m
green bonds	369	184	IQD 887.075bn	IQD 8.871bn

The gap facility's final principal payment is in month 203. Later surplus remains unrestricted cash after debt retirement. Each month's native debt balance equals prior balance plus draws minus scheduled and early principal. Each six-month closing balance is its final month's balance; payments, premiums, interest and reserve movements are period sums. Contractual repayment windows in tranche files describe the original draw terms; the actual final-payment months above incorporate early repayment.

Monthly cost-priority repayments · six-month repayment tranches · loans-first tranches · all six complete calculations



Surplus repayment comparison

Loan prepayment can carry redeployment/unwind charges: the World Bank Treasury FAQ illustrates the concept, without establishing Chinese or Iraqi loan terms. The US Treasury buyback FAQ and published purchase results illustrate issuer authority and priced buybacks rather than automatic par redemption; they do not supply an Iraqi legal mandate.

This repayment allocation does not change the unlevered project NPV or validate demand. Nominal savings over several decades are not present-value gains. CAPEX escalation, FX, lifecycle replacement, floating-rate risk and the uncommitted early funding still require appraisal.

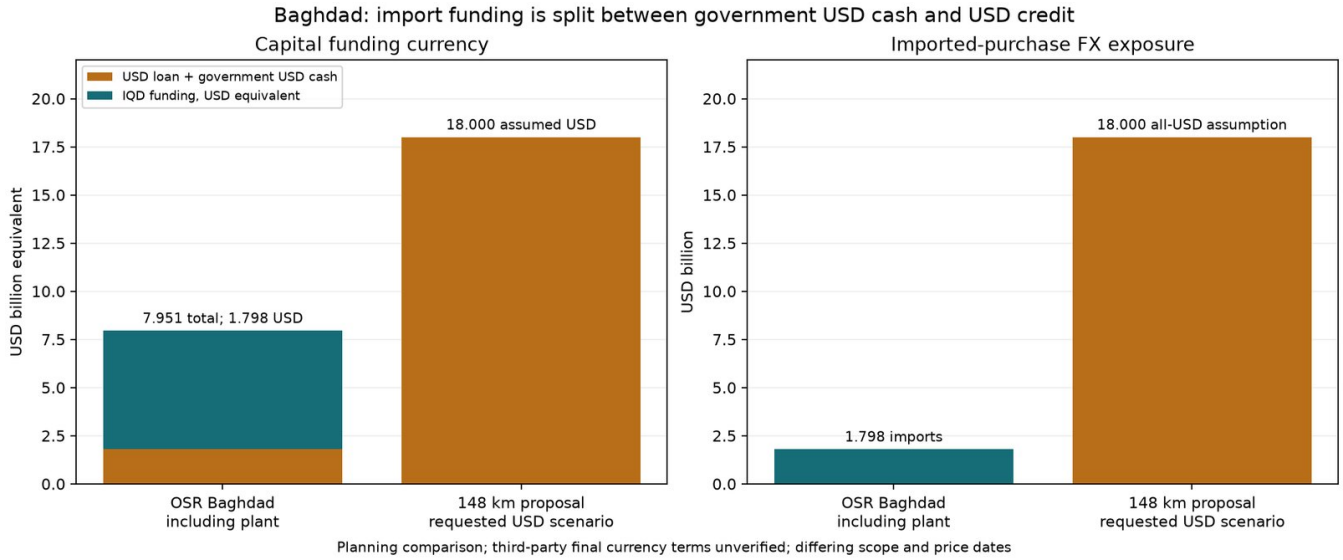
Consolidated sources and uses

Government contributes 25% of total capital uses, including the plant and EPC. Imported purchases are split 50% government USD cash and 50% proposed Chinese USD credit, assuming the full imported basket can qualify. That USD government cash is inside the 25% total contribution; the rest of the government contribution is IQD. The remaining balance after those two sources is split 75% domestic IQD bonds and 25% IQD bank term credit.

Proposed capital source	Currency	Native amount, billions	USD equivalent, million	Share of total capital
chinese export credit	USD	0.899	898.82	11.30%
government import cash	USD	0.899	898.82	11.30%
government local cash	IQD	1,415.697	1,089.00	13.70%
domestic bonds	IQD	4,938.025	3,798.48	47.77%
bank credit	IQD	1,646.008	1,266.16	15.92%
Total capital uses	Mixed	—	7,951.29	100%

Baghdad city CAPEX is USD 7,626.54 million. The plant is counted once at USD 303.50 million plus USD 21.25 million EPC. Its sizing basis is 4,854 vehicle/car modules from Baghdad's controlled fleet and car count. Imported tooling receives proposed Chinese credit within that plant budget. The physical cell/floor/tooling envelope replaces the smaller module allowance where necessary; both remain unquoted engineering assumptions.

Currency and USD capital intensity



Funding currency and imported-purchase FX comparison

Only Chinese credit is USD-denominated debt: USD 898.82 million. Government additionally provides USD 898.82 million cash for imports. Combined USD capital funding is 22.61% of uses; the remaining 77.39% is IQD government cash, IQD bonds and IQD bank credit. Revenue and local OPEX are also budgeted in IQD. At the historical planning conversion of 1,300 IQD/USD, the local government cash allowance is IQD 1.416 trillion, alongside its separate USD import cash. Together they equal 25% of total capital. USD columns are comparison equivalents, not a requirement to borrow or appropriate those domestic amounts in dollars.

Funding currency and procurement currency differ. Estimated imported purchases total USD 1,797.65 million (22.61% of capital), including plant imports. The government supplies half directly in USD cash; proposed Chinese credit supplies half as USD debt. The full import pool, including categories beyond the initially selected solar, bogies, batteries, windows, doors and tooling, is assumed lender-eligible pending origin qualification. If that expanded basket cannot qualify, its loan funding is uncovered; it is not automatically replaced by IQD bank credit or another grant. IQD borrowing reduces the revenue/debt currency mismatch; it does not remove Chinese debt-service FX risk, imported maintenance costs, domestic interest, inflation or placement constraints.

Eligibility sensitivity: the original named-component categories cover USD 849.64 million of invoices. At a 50% advance, they alone support USD 424.82 million of proposed credit. If the additional imported categories cannot qualify, USD 474.00 million of planned capital loan proceeds remains uncovered, in addition to the base support requirement below. Even the original categories remain unqualified; this screen is not a lender approval. No extra government contribution is assumed to replace that missing loan.

What the 25% government limit leaves unfunded

The 25% capital contribution is USD 1,987.82 million. If that is also the limit on all public cash, the model leaves USD 4,013.14 million of additional funding requirements across the full construction, operating and debt horizon. These are interest, fees, reserve and operating/debt cash needs after modelled revenue; they are not additional approved government contributions.

Peak annual capital contribution is USD 394.38 million. Peak annual additional funding requirement is USD 986.21 million. If all additional support were provided publicly, conditional lifetime public cash would be USD 6,000.96 million, with a combined annual peak of USD 1,135.99 million. This exceeds the requested contribution and is shown only to expose the funding gap.

Across the 36.8-year nominal model horizon, additional requirements divide into USD 1,888.68 million before Baghdad's full-network opening, USD 1,699.66 million during Baghdad operations/debt tail, and USD 424.80 million for plant capital financing. Principal repayment before opening is part of the pre-opening requirement. This base withholds all fares until full-network completion; the separate phased sensitivity below tests earlier revenue. Gross additional support is a liquidity requirement over time, not net lifetime loss: later retained cash cannot repay earlier obligations without an approved bridge.

The programme CSVs distinguish the capped capital contribution, additional funding requirement and conditional payment requirements. Underlying city and factory ledgers calculate the support needed to pay scheduled obligations; their positive reserve balances and completed repayments are conditional on that support being raised. They are not a cash-solvent forecast under the public cash cap. No bridge, equity investor, rollover or extra appropriation is invented to close the gap.

Cash and contractual structure

1. An Iraqi public sponsor seeks MoF authority for proposed sovereign IQD bonds through the established MoF/CBI issuance route, subject to legal review and placement. Municipal borrowing powers are not assumed. The 25% limit is the direct capital contribution, not a limit on sovereign bond liabilities, guarantees or lifetime public exposure.
2. Chinese export buyer credit is allocated to qualified Chinese invoices for solar equipment, bogies, batteries, windows, doors and Baghdad plant tooling. The requested 50% loan advance is matched by 50% government USD cash. All other imported categories also require Chinese supplier-origin and lender qualification before the full-basket financing assumption is valid. Noneligible imports still need foreign currency. Eligibility and origin require supplier evidence.
3. Government's 25% capital contribution covers the planned eligible downpayment; bonds and bank credit cover the remaining capital balance. Fees, construction interest and reserve funding are separate cash requirements, not silently capitalised or counted as capital-source proceeds.
4. Each draw starts its own grace and repayment clock. Planned receipts match procurement milestones; no actual payment or lender commitment is asserted. Chinese credit uses USD, bonds and bank credit IQD. Terms remain unconfirmed appraisal assumptions: 5%/48-month grace/180-month repayment, 8%/24/180, and 9%/0/84 respectively.
5. Revenue pays OPEX first and debt service next; subsidy is excluded from DSCR. Additional support shown by the city and plant models is an unfunded requirement under the cap. The factory model covers capital financing only; manufacturing income and factory OPEX need a separate business case.
6. Declined Chinese finance, delayed contributions, FX changes and bullet redemption are separately tested in the Baghdad appraisal. None creates an automatic facility or refinancing source.

Rollout and financial close

Baghdad's current capital milestones span 82 calendar months. Full-network fare revenue begins after all budget milestones. Lowering the capital grant increases borrowing and cash-service needs; it does not shorten this schedule. The city-sized Baghdad plant becomes available after 390 working days (18 months from NTP), followed by the separately scheduled first-article series-release gate. The separate phased sensitivity uses this integrated planning schedule; actual plant and line acceptance and a commissioned operating baseline remain pending.

Required before financial close: approved sponsor and borrowing powers; appropriation limited to the agreed government contribution; a funded solution for the additional cash gap; supplier quotations and origin evidence; signed term sheets, guarantees and insurance; IQD placement/redemption plan; lender draw windows and FX access; surveyed demand/fare policy; tax, duty, land and utility pricing; accepted resource calendar and reserve covenants.

Model and evidence

Programme monthly cashflow · annual cashflow · machine-readable programme · Baghdad city appraisal · editable assumptions. Other Iraqi city appraisals remain standalone examples outside this programme.

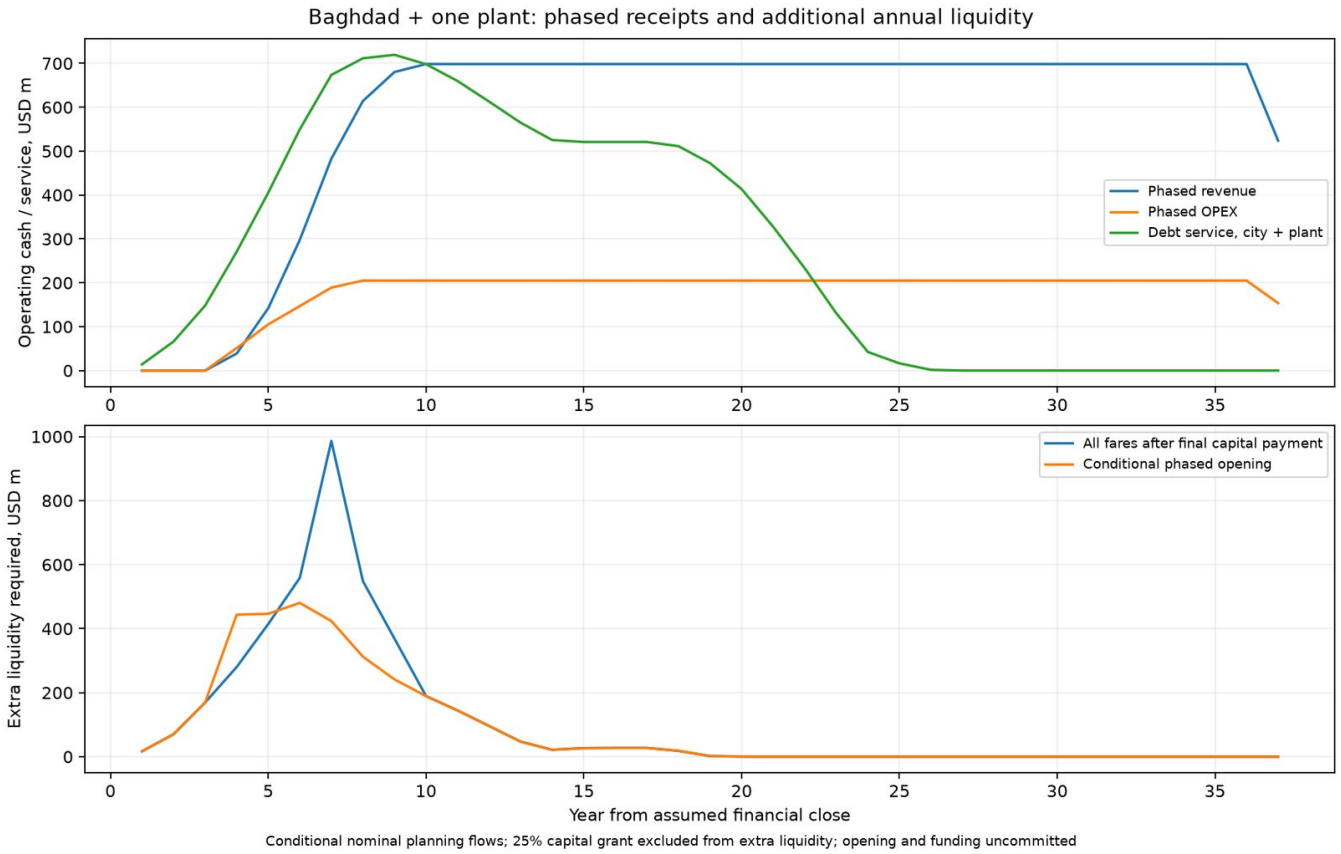
IMF 2025 Article IV supplies the historical 2024 FX anchor, not a current dealing rate. CBI fiscal-agent description and Injaz issue example support sovereign-bond context. China Exim export buyer credit supports the instrument concept; numeric terms and eligibility are unconfirmed assumptions.

Recalculation: conditional phased opening

The earlier USD 9.890 billion result combined whole-fleet stage batching, no fares before the final capital payment and reserve deposits always funded with additional public cash. Ready-task dispatch now pipelines work within the same resource limits, gates train production on the new plant, and funds reserves from available project cash before seeking support. The full-network-only result above remains a conservative comparator.

With line openings tied to completed line and shared/depot work plus a 3-month commissioning allowance, first revenue begins in month 40, and all nine lines operate from month 81 after financial close. These dates are conditional planning milestones, not authorisation to run trains.

Each line's controlled trainset share allocates revenue and variable OPEX; its own 50% / 75% / 100% ramp applies. From first opening, 25% of full-network OPEX is fixed and 75% scales with opened fleet. This proxy can misstate early demand, central staffing and transfer benefits; a surveyed phase-specific operating plan is required. The same nominal fares, capital total, debt terms and 25% grant apply in every case.



Phased operating cash and additional liquidity

Opening / demand scenario	Additional funding beyond 25% capital, USD bn	Conditional lifetime public cash, USD bn
All fares after final capital payment	4.013	6.001
Phased: commissioning delay two years	4.192	6.180
Phased: demand minus 40 percent	7.076	9.063
Phased: low demand	3.205	5.193

Line	Planned opening month	Fleet / variable-cost share
line-1	40	10.63%
line-10	41	1.85%
line-11	43	1.73%
line-12	42	1.85%
line-13	44	2.22%
line-14	44	1.61%
line-2	50	11.74%
line-3	55	11.12%
line-4	59	9.39%
line-5	64	10.63%
line-6	69	11.74%
line-7	73	8.90%
line-8	78	11.00%
line-9	81	5.56%

All these are gross nominal liquidity contributions, conditional on funding. Later operating surplus is retained and is not netted against earlier required injections. The phased city ledger ends with USD 7.878 billion of unrestricted cash; no distribution or return to the sponsor is assumed. The plant ledger has no manufacturing income/OPEX: it remains a capital-financing allowance. A 25% direct grant also does not cap sovereign IQD bond liabilities or guarantees.

Phased programme monthly cashflow · phased programme annual cashflow

Comparison with the historical 148 km third-party proposal

The July 2024 report, citing the prime minister's adviser, gives USD 18 billion, 148 km, seven lines and 64 stations. The NIC procurement notice describes DBOMFT and requires bidders to submit a funding plan. Neither establishes a final all-USD financing contract. The 100% USD foreign-loan/government-cash comparator below is the user's requested scenario, not a verified split of debt and grants. The older USD estimate has not been restated to a common price date.

Measure	Current OpenSourceRail Baghdad plan	Historical proposal / requested USD scenario
Route km (not track km)	513.3	148
Lines / stations	14 / 159	7 / 64
Capital uses, USD equivalent bn	7.951, including one plant	18.000, reported estimate
Capital intensity, USD equivalent m / route km	15.49	121.62
USD-denominated capital funding, USD bn	1.798, half loan / half government cash	18.000, assumed USD cash + loans
IQD-denominated capital funding, IQD tn	8.000	Zero in requested scenario
Imported-purchase FX exposure, USD bn	1.798	18.000 only under all-USD purchase assumption
USD purchase intensity, USD m / route km	3.50	121.62 under that assumption
Contracted fares / debt terms	Planning fare and uncommitted facilities	Not established by these sources

The OSR plan has 3.47 times the route length and 2.48 times the stations. Its planning capital estimate is 55.8% below USD 18 billion. Under the requested all-USD funding scenario, USD-denominated capital funding is 90.0% lower; under an all-USD purchase scenario, imported-purchase exposure is 90.0% lower. These measure different exposures and are not interchangeable debt-service savings.

The planning comparison is not a like-for-like qualified bid: route geometry, tunnelling/structures, land, utilities, taxes/duties, contingency, escalation, supplier qualification and acceptance maturity differ or remain unresolved. The older report's four-year completion expectation also differs substantially from OSR's current 82-month capital schedule. A lower capital estimate does not establish an earlier or more sustainable delivered service.

Fares, population access and financial sustainability

The modelled average paid-trip fare is IQD 1,317 (USD 1.01 comparison equivalent), denominated in IQD rather than automatically indexed to the USD loan. It derives from the retained USD 380/month income proxy, equivalent to IQD 494,000/month at the planning anchor. Thirty paid trips consume 8.0% of that proxy income; 44 commuter trips cost IQD 57,963, or 11.7%. This is an average-trip yield assumption, not an adopted tariff or unlimited monthly pass; concessions, transfers and family affordability need an explicit tariff and household survey.

Low/high capacity-use cases assume 1,807,920 / 2,892,672 paid trips/day, not unique people or surveyed demand. Low-case steady annual revenue is IQD 908.0 billion; OPEX is IQD 266.5 billion. At that trip volume, the OPEX-only neutral fare is IQD 345, with nonfare receipts held constant. It excludes capital, debt and reserve funding.

In the first complete operating year, the 50% revenue ramp produces 329.9 million paid trips. Holding those trips and nonfare receipts fixed, covering OPEX, scheduled debt service and fees requires approximately IQD 3,417 per paid trip, excluding reserve deposits and factory debt. This is a cash threshold, not a recommended tariff; higher fares can reduce demand. Minimum annual operating DSCR before support is 0.04. Higher utilisation improves later cashflow but cannot fund the long pre-opening debt-service period. Neither this fare nor IQD denomination closes the additional funding gap shown above.

The design retains a planning population of 9,780,429. The legacy 40.5% routing-demand cell fraction is not a population percentage; its former multiplication into a resident estimate is retired. See the native population-count and multi-hop transfer audit for radius sensitivities, date, bbox denominator and access limitations. The older report's 80% city-coverage ambition uses no published comparable denominator or access method; it cannot establish that either plan reaches more residents. The 159 station platforms and 14 routes offer a network to test, not proof of superior walking coverage.

Direct Iraqi labour and industrial benefits

Approximately USD 6.154 billion equivalent (77.4%) of programme procurement is assigned to local suppliers under the editable origin assumptions. This is potential local expenditure, not wages, GDP added or guaranteed Iraqi content. City civil works alone allocate USD 3.346 billion locally; rolling stock allocates USD 883.43 million locally, with plant local expenditure separately counted.

Local train assembly and fabrication create work in body modules, fit-out, wiring, coatings, systems integration, inspection and maintenance while bogies, batteries, windows and doors remain imported inputs requiring supplier and process qualification. Infrastructure work supports Iraqi concrete/precast production, civil erection, stations, utilities, solar installation and supervision. Tooling, training, process qualification and supplier access can leave reusable industrial capacity, shorter repair chains and retained skills after construction. Local employment and supplier income circulate in IQD and can generate Iraqi tax receipts; no multiplier or tax recovery is booked without evidence.

The operating allowance supports 3,422 indicative FTE with annual labour cost of IQD 50.29 billion. These are operating positions, not construction or manufacturing job counts. Construction employment requires validated work hours, productivity, wage rates, shift cover and local-content contracts; no fabricated job total is assigned. The third-party proposal may also use Iraqi civil labour, so its local share cannot be assumed zero.

A sustainable appraisal must demonstrate phased service before full-network completion, realistic paid demand and affordable tariffs, funded debt/interest/reserves within the public limit, placed IQD facilities, supplier and labour qualification,

lifecycle replacement funding and an accepted environmental/physical design. Current software and ledger checks establish planning consistency, not those outcomes.

Iraq National OpenSourceRail Strategy

Source: cities/catalogue/west-asia/Iraq/NATIONAL-BRIEF.md

Current Iraq planning basis — 2026-10-06

Current country basis · Baghdad summary · Connected design, production and energy study.

The retained `local_positive` Baghdad financial comparator is USD 8.667bn, with IQD 0.000tn unsourced support and IQD 0.000tn terminal debt. All debt clears at model month 295, conditional on the recorded funding assumptions. Its prices and facilities are unquoted and uncommitted. It does not price the complete accelerated construction and battery scenario.

The accelerated scenario's complete CAPEX, financing, net island savings and accepted opening dates remain unknown. Its USD 9m / 18-launcher allowance covers purchase cash only; freight, assembly, commissioning, transporters, lifting frames, temporary supports, spares, crews and relocations require separate scope-matched prices. The earlier delivered-and-commissioned interpretation is not used. No launcher saving is added to an installed civil rate before removing a verified matching embedded allowance.

Iraq has existing precast facilities and expertise; qualify contracted civil supply and price tooling/upgrades. Baghdad trainset assembly and component-process allowances remain in the financial comparator. Additional national plant capacity and transfer costs require separate scope reconciliation.

The coordinated industrialisation programme joins vehicle interfaces, staged bogie localisation and the standard viaduct system. Its possible vendor register records official-source component and equipment leads for comparison. Iraq already has many precast facilities and relevant expertise; plant capability, compatible tooling and available production slots require qualification. No listed vendor is a selected supplier or a manufacturing-rights agreement.

The coupled service–OD–cost–finance study now derives capacity-limited passenger itineraries from completed train sections and carries the same grid purchases and train distance into conditional monthly finance. Its demand and retained prices/opening months are explicit sensitivities. Complete line openings additionally require stations, specials, track, energy, depots, fleet, testing and approvals; missing dates or prices remain unknown. No coupled case is an accepted forecast or an adopted national budget.

No current consolidated national budget or funded rollout is established. Other Iraqi city catalogue estimates and the earlier generic foreign-turnkey calculation remain separately identified comparators. Their differences are scenario arithmetic, not realised capital or interest savings. Baghdad's later scope cannot be mixed into an older national aggregate and presented as one complete budget.

Construction sensitivities

Case	Days for modelled running bays	Scope
full-doubling	1022	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates
higher-productivity	1098	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates
initial-accelerated	1218	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates
initial-accelerated-reassigned	966	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates
initial-six-day-calendar	1348	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates

These are conditional resource schedules. Station structures, special/closure spans, approved pier locations, delivery access and release evidence remain opening gates. Empty supplier/contract registers indicate missing qualifying evidence; they do not imply that Iraqi industry lacks capability.

Chronological energy study

Case, lower-solar duty	Minimum SOC	Unserved traction kWh
lfp-throughout	0.0%	676,286
sodium-both	0.0%	4,024,245
sodium-stationary	0.0%	676,383

The fixed-assignment diagnostic above uses both ring directions, declared dispatch stations and native rest-to-rest section timing. The current screen spans 4,440 continuous model minutes (3.08 days), including overnight state carry-over and a

study strategy permitting grid replenishment within remaining site import capacity. Charge-rate, thermal and life inputs are unqualified profiles, so these results do not establish a chemistry advantage or suitability.

The separate coupled energy/service controller keeps reserve-limited trains at their actual stations, continues charging, and reflects their availability in subsequent dispatch opportunities.

Controlled case	Completed journeys	Missed dispatch opportunities	Distinct energy-held journeys
lfp-throughout	17,491	909	706
sodium-both	8,536	9,605	2,761
sodium-stationary	17,492	908	719

Geometry/operating-distance findings remain recorded. Conflict-capable timetable, berth/depot access, transient traction and operational multi-day acceptance remain open. Current station passenger demand is unknown where no surveyed station/OD assignment has been entered; the common 3,000 pax/h value is retained only as a separate stress case.

City estimates and their scope

| City | Population | Fleet | City-case capital | Assumed imported purchases | Basis | | --- | ---: | ---: | ---: | ---: | --- | | Baghdad | 9,780,429 | 809 | $8.67 B | $2.02 B | Retained local_positive comparator; accelerated scope unpriced | 14 | 5 | 40.3% | 27.7% | | Basra | 3,955,000 | 557 | $6.58 B | $1.40 B | Catalogue base estimate; scope and field releases open | 11 | 4 | 42.9% | 38.3% | | Sulaymaniyah | 2,150,000 | 144 | $2.00 B | $398.7 M | Catalogue base estimate; scope and field releases open | 6 | 2 | 35.5% | 31.1% | | Erbil | 1,952,000 | 261 | $2.35 B | $489.5 M | Catalogue base estimate; scope and field releases open | 8 | 3 | 31.8% | 29.8% | | Mosul | 1,940,000 | 308 | $3.93 B | $763.6 M | Catalogue base estimate; scope and field releases open | 9 | 3 | 40.6% | 41.0% | | Kirkuk | 1,780,000 | 235 | $2.81 B | $556.2 M | Catalogue base estimate; scope and field releases open | 8 | 3 | 41.2% | 41.1% | | Najaf | 1,540,000 | 309 | $3.33 B | $674.7 M | Catalogue base estimate; scope and field releases open | 9 | 3 | 33.3% | 46.2% | | Karbala | 1,390,000 | 253 | $2.84 B | $567.5 M | Catalogue base estimate; scope and field releases open | 9 | 3 | 46.1% | 42.0% | | Nasiriyah | 705,000 | 184 | $1.09 B | $221.0 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 41.8% | 34.8% | | Hillah | 700,000 | 236 | $1.19 B | $250.8 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 29.0% | 37.7% | | Amarah | 660,000 | 177 | $1.13 B | $224.6 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 37.4% | 35.1% | | Ramadi | 525,000 | 181 | $978.5 M | $204.3 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 30.0% | 36.9% | | Baqubah | 470,000 | 243 | $1.32 B | $275.3 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 37.2% | 33.8% | | Diwaniyah | 440,000 | 198 | $1.07 B | $222.2 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 30.6% | 49.1% | | Kut | 410,000 | 166 | $1.05 B | $209.6 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 40.8% | 42.7% | | Samawah | 373,770 | 182 | $1.03 B | $211.9 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 40.9% | 47.8% | | Duhok | 360,000 | 213 | $1.08 B | $235.9 M | Catalogue base estimate; scope and field releases open | 4 | 1 | 46.8% | 53.3% | | Fallujah | 360,000 | 195 | $1.06 B | $218.6 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 35.4% | 30.3% |

These case envelopes have different appraisal scopes. They are not summed into a current national project budget. Baghdad's included assembly/component facilities must be reconciled with any national shared-factory allowance before aggregation.

Shared assumptions and interpretation are in the deployment planning reference.

Regeneration

Run `python3 tools/automation/generate-national-briefs.py --country IQ` after regenerating the Baghdad study, then publish the current city/country documents and proposal archive.

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Population-led additional lines and country regeneration

Source: docs/civil/population-led-network-regeneration.md

The controlled route inventory now includes additional lines selected from residential gaps, rather than leaving all additions in a separate sensitivity map. Each added route connects to a retained corridor at an actual common grid cell. Near-ring diversions preserve selected residential sites and avoid new self-intersections, protected cells, unknown water and unapproved long crossings. Native crossing, platform, terminal and bounded-interchange checks still apply.

The shared policy uses retained native 2020 population counts, a working 80% station-circle target and a 1 km radius. Native station placement follows the local demand bands: 1.6 km in the core, 3 km in the urban area and 7 km on suburban/outer approaches. Sampled planning stops do not override these bands. Ordinary neighbours within an interchange's local spacing envelope are removed after the actual platforms and operational endpoints are established, provided removal preserves the outer station-gap limit. Distinct physical crossings remain explicit site-review exceptions.

New corridors must be at least 4 km long. Additions are limited to half the retained trunk count (rounded up), at most eight lines, and 25% of the retained route length. At least 45% of each added corridor's sampled routing cells must be 1.2 km away from the committed network, allowing shared junction approaches while rejecting parallel feeders with duplicate catchments. Population gains are recomputed after every addition. Retained DEM slope penalizes hills and excludes missing terrain from new routing; core straightening cannot introduce a steeper terrain conflict or cross protected/unknown cells. Water crossing bounds, native geometry and connectivity checks still apply. These are ground-terrain planning screens, not surveyed track grades. Country and city finance inputs remain local; Baghdad retains its selected-population and sector policy.

The network map shows one main junction symbol on an actual platform for each bounded interchange complex, with smaller dots at other platform locations. Connecting strokes identify complex membership schematically; they do not depict surveyed walking paths or rail switches. Platforms and their line chainages remain separate engineering records.

After final junction and bank consolidation, a separate spacing pass restores only intervals above the outer 7 km limit, selecting dry cells on the existing route. A hard final check includes the closing interval of rings. Local stops and mandatory crossing platforms keep their positions; unresolved excess gaps stop generation instead of being hidden in a map.

Closed out-and-back excursions on retained radial routes are removed before stations and junctions are generated. This preserves the endpoints and uses existing route edges, avoiding additional shortcuts through terrain or water. The alignment report records each removed retrace; final route quantities, stations, coverage and costs follow the emitted inventory.

Controlled revision and actual coverage

Each city retains `engineering/alignment/network-expansion-baseline.json.gz`, preserving the original design bytes and their hash. Its `residential-line-expansion.json` identifies added line IDs, priority cells, actual corridor connections, route quantities, controlled original-line junction changes and unresolved site/geometry reviews. Mandatory native geometry checks can reject a candidate; a failed diversion never creates a fictional transfer.

Selection credits retained baseline stops and explicit residential priority sites. Ordinary sampled stations are not credited as delivered coverage because native spacing and junction consolidation can remove them. The separate `residential-expansion-evaluation.json` audits the actual emitted station coordinates after generation, using the union of native population-count pixels. Remaining gaps and a missed target are reported. A target is never declared reached from a renderer, a routing-demand score or a sum of overlapping district circles.

Where population evidence is missing or contradicts the controlled source/jurisdiction finding, population-led additions remain unavailable. Those cities still receive the current route, junction, civil, fleet, depot and economic logic. No residents or coverage fraction are invented. Existing unsuitable geometry remains an explicit structural/profile/access review where a controlled replacement is infeasible.

Complete regeneration

1. Rework controlled core geometry, connect reviewed junctions and add population-led routes; retain immutable baselines and constraints.
2. Emit actual native stations, bounded complexes, civil classifications, fleet/depot quantities, operating overrides and scenarios. New native spacing owns placement; the older infill overlay becomes a comparator and records unresolved gap reviews.
3. Refresh actual local civil costing before freezing the design and scenario inputs. Execute the full nominal day and degraded native software cases, retaining their actual source-bound outputs and receipts.

4 Regenerate SUMO/GIS/energy, desktop ground and site-evidence packages, timing references, operations, finance, depots, stabling, factory, delivery and deployment. Reconcile operations and finance in dependency order; a funding source mismatch stops publication.

5 Audit emitted-station population and connectivity, geography and clearance. Update city packages, country briefs, portfolio, construction/assembly, readiness, catalogue audits and publications from the final inventory.

6 Verify source/output hashes, complete archives, links, file limits, repository health and generated drift before merging.

Current-source generation is the default for every city. Routine refreshes preserve existing controlled routes and station identities; new cities use their own current planner and local evidence. Use `tools/automation/regenerate-all.sh --resynthesise-design --prepare-only --jobs N` when deliberately replacing existing layouts. The batch receipt explicitly says engineering refresh is required; it does not mark a planning package complete. `--current-design-logic` remains accepted for existing callers. The complete run reruns engineering and native validation, or verifies native CI artifacts from the exact frozen source revision, and writes a source-bound `engineering/design-system/` report with unresolved design and acceptance gates. A current report does not establish manufacturing completion.

Finite construction plant

Two candidate erection fronts per line do not create two additional purchased machines. The connected construction study assigns fronts to a finite launcher pool, with the preceding assignment as a required dependency. Support-release teams also follow finite pool queues and working calendars. Optional ring transfers use only the final donor assignment; a machine cannot be taken from a queued line.

Per-line foundation, bay, beam and launcher records retain actual identities and directed installation order. Soil investigation, foundation selection, loading combinations, vertical profiles, access/property rights, temporary works, suppliers, prices and independent construction/operating acceptance remain required. Station-circle proximity is not current census, surveyed walking access, observed ridership or funded passenger service.

Line-based foundation and viaduct assembly planning

Source: docs/civil/line-based-foundation-and-assembly-planning.md

The integrated Baghdad package is the line-specific implementation of the civil master plan. Its construction plans for every current line, support register, bay register, directed launcher fronts, junction packages and stage library share actual line/span/component identities. The offline viewer lets reviewers select a line, zoom to individual supports and inspect the coordinate, adjacent spans, desktop investigation reference and unresolved design values.

One physical support, one foundation packet

Every unique running support has one `FND:<support-id>` record, a proposed route-interpolated position, line chainage and the adjoining bay IDs. Adjacent bays reference that packet; they do not create two footings at the same support. Geographic positions remain planning datums requiring survey control, utilities and property/access release. Coincident positions on different lines enter a common junction/site review; a shared foundation is never assumed merely from XY coincidence.

Station/concourse/shaft, approach, water bridge, depot/pit and energy-site foundations are separate design scopes in the other-structure register. Their counts and geometry remain unknown where no structural/site design exists. Reconcile station/running support ownership before adding quantities; do not price both as independent foundations for the same load path.

Ground investigation and design brief

The support packet links its nearest same-line desktop-soil sample by chainage, including the offset and investigation flags. Those mapped topsoil properties do not establish deep stratigraphy, bearing capacity, groundwater, settlement or pile length. Use them to organise field investigation, not to select a footing.

Design task	Required support/zone evidence
Survey and existing assets	Surveyed support/track coordinates and elevations, utilities/trial holes, property, adjacent buildings and traffic/access
Ground and groundwater	Borehole/CPT/test-pit programme, verified stratigraphy, groundwater and laboratory classification/strength/compressibility/durability; zone limits and uncertainty
Load transfer	Selected beam/continuity scheme, columns/caps/bearings, full train and lateral/dynamic actions, launcher/delivery/asymmetric stages, wind/seismic/collision/flood/scour
Foundation options	Shallow footing where verified competent ground permits; bored shaft for suitable constrained/vibration conditions; driven bent where installation is acceptable; pile group for selected lateral/scour/ground cases
Geometry and verification	Actual element count/size/depth, socket/cap/column details, settlement/rotation limits, installation and proof/integrity/load testing, independent checking
Construction interface	Working platform/excavation/dewatering/slurry/spoil method, reinforcement/connection, concrete/curing, tolerance/inspection and support-ready release

The register reports an unqualified simple-span beam-deadload screen from complete-member study masses: two single-track beams per bay, half of each at each end. It excludes cap/column, train, continuity redistribution, construction combinations and lateral/environmental actions. It is not an adopted foundation axial load or allowable bearing pressure; complete design axial/lateral/moment values remain unknown. Special-member mass also remains unknown.

Prepare a foundation design and installation method for each verified ground/load zone. Trial and test representative foundations before repetition, with a reviewed extension to unlike zones. Select concrete, reinforcement, corrosion/durability and test criteria through the project design basis. Actual lengths, drilling/driving logs, concrete volumes, specimens/maturity, integrity/load tests, dimensions and NCRs become installed records. No universal six-metre pile or generic foundation-concrete allowance is used.

Directed workfront plan

Two candidate launcher fronts per line are inherited from the running-bay sensitivity, with finite launcher-pool predecessors for queued reuse. Positive fronts install toward increasing chainage; negative fronts begin at the high-chainage end and install backward. The existing scheduler interprets direction internally; the new exports make the execution and dispatch order explicit. All identified catalogue spans are assigned to one front and one sequence position. Special spans retain a separately engineered package and create no ordinary Pi25 order.

For each front, release actual access/assembly space, working platforms, configured machinery, delivery route, competent rested crew and the first consecutive ready supports. Maintain 10–15 accepted bays of support and beam readiness only where continuous run and storage capacity permit. Factory slots, inspection, accepted stock, trailers, receiving plant and installation appointments follow the directed SKU/component order. Shared equipment/crews cannot be reserved to two fronts simultaneously.

Disconnected intervals, stations and specials require a checked passage or dismantle/transport/reassembly/recommission method. The predecessor is the previous executed bay, not simply the preceding ascending row in a database. Foundations used by adjoining bays share the same stage packet. Released radial launchers can transfer to additional ring fronts only after access, foundations, supply, crews and recommissioning are evaluated together.

Assembly-stage dependencies

Stage	Entry and exit
F0–F3 foundation	Survey/utility/ground/load design trial and temporary-platform release actual selected installation accepted tests/dimensions/NCR closure
P0–B0 support	Column/cap and connection/temporary restraint surveyed bearing seats and actual next-stage loading release
M0–M3 manufacturing	Frozen shop/handling configuration mould/cage/prestress/cast/curing separate strength gates and inspection accepted serialized member
H0–H1 delivery	Actual vehicle/route/load restraint and appointment receiving/damage survey, designed storage and reservation
E0 preparation	Both support B0 packets, both beam H1 records, predecessor/relocation and relevant junction J2 release
E1 first beam	Configured controlled lift, seating, balance and required lateral/torsional restraint
E2 paired bay	Second controlled lift, securing, surveyed geometry and bearing contact
E3 connections	Selected connection/joint/strength, bearing movement and NCR acceptance; no shared bearing reduction without structural continuity
E4 advance	Accepted current-stage path, checked advance/delivery loads, next supports, weather and protected access
E5 following trades	Walkway/barrier, waterproofing/drainage, track interface, permitted loads/access and signed handover

The stage library stores these reusable chains; the bay register binds them to actual component and parent IDs. Junction-affected member orders remain on design hold until level/clearance/support/stage releases close. Installation is never inferred from an accepted factory beam or a completed documentary task.

Inspection and complete opening

Apply the site ITP and handover plan. Record source configuration, personnel/equipment authority, measured results, test criteria and signatures before closing each hold. After transport damage, abnormal loading, changed temporary works, utilities or adverse weather, inspect and re-release affected stages.

Complete line opening requires stations, specials/transitions, track, energy, depot/stabling, commissioned fleet, integrated testing and approvals in addition to running-bay completion. The plan does not adopt an opening date, field foundation quantity, installed quote or paid-passenger uplift from illustrative capacity or coverage.

Network junction and residential integration

Source: docs/civil/network-junction-and-residential-integration.md

The integrated network model coordinates line crossings, shared corridors, station complexes and underserved residential areas with line-specific foundation/assembly packages. A visually overlapping route is not automatically a rail junction, walkable transfer or completed civil connection.

Crossing and overlap packages

Situation	Required planning and design treatment
Two independent rail lines cross	Grade-separated structure with actual rail levels, gradients, clearances, support/cap and construction load paths; a passenger interchange is a separate access decision
Routes occupy a common XY corridor	Compare a coordinated four-track footprint with laterally separate guides and shared station/access structure, or reviewed separate vertical levels; do not duplicate coincident decks or footings
A line revisits the same location	Preserve distinct chainage legs, then choose a real grade-separated passage or reroute; do not collapse both visits into one node or imply a switch
A short A–B–A ring spur	Remove the spurious one-cell reversal without introducing a new land/water shortcut; regenerate route chainages, stations, civil quantities, scenario and dependent evidence
An apparent near-miss interchange	Preserve platform positions; identify the actual concourse/street/vertical connection, barriers, rights, walking distance and timing before acceptance

The Baghdad ring repair removes its 20 m outward-and-return spur, shortening the planning route by 40 m. It does not erase a district-serving branch or insert an unreviewed chord. The source repair leaves longer excursions and supplied protected cells for explicit review.

Every junction design package identifies participating lines and chainage legs, affected spans and support packets, design alternatives and the J0–J3 arrangement/profile/site/stage/handover sequence. Upper/lower rail levels, vertical clearance and graded approaches stay unselected until a real profile and independent structural review exist. Ordinary member orders that touch the interface are held until J2; current drawings and existing route-km budgets cannot establish an accepted junction cost or final geometry.

Bounded station complexes

Grouping uses the maximum pairwise platform separation, with a 600 m planning screen. A chain of nearby stations does not make its distant ends one complex, and an additional same-line amalgamation cannot silently enlarge the diameter. Original oversized groups remain in the review register with the bounded replacement concepts and their actual platform IDs. No platform is shifted to an averaged centre point.

The diameter is a geometry screen, not a walking path. Rivers, roads, walls, level changes, stairs/lifts and entry permissions need actual accessible routes and transfer times. A close pair can still fail. Concourse/platform capacity, outage operation, assisted rescue, fire, PSD/train interfaces and structural loading use the installed arrangement. The proposed grouping does not alter the native timetable or invent a physical track switch.

Population-led coverage

Use retained native 2020 WorldPop count pixels and count each person once in a union of distance circles. Keep the population year, bbox and nodata explicit; never multiply a routing-demand score by a current city-population headline. The coverage model separates existing station proximity, corridor proximity, infill sensitivity and branch/feeder investigation.

Infill candidates sit on current lines and are ranked by incremental previously unserved retained population, subject to minimum same-line spacing and ring wraparound. Candidate stations require water/footprint, property, access, platform/approach, timetable/fleet/energy and complete installed-cost checks. A population-rich point is not a released station site.

Large residential gaps outside current corridors cannot be solved by infill alone. The priority areas and expansion corridors identify source-linked district/nearby place names, existing station connection, route/access gaps, and an arterial investigation path. Compare an engineered rail branch/extension with a frequent feeder and accessible transfer. Road vertices define an undirected preliminary graph; traffic direction, stopping sites, grade connections, bridge limits and permissions are unverified. Do not issue driver navigation or a rail alignment from this screen.

The combined 500 m feeder-stop and 1 km rail-station circle cases are conditional sensitivities, not measured walking coverage or funded service. Shared candidate road segments are counted once in the corridor-edge register. Priority-area circles can overlap and are never summed as unique residents. Actual travel-time/OD, affordability, service/headway, capacity, operating cost and finance remain required before selecting and adopting expansion.

ITDP People Near Transit provides a population/proximity assessment reference; ITDP physical integration addresses interchange/access development. WorldPop's model-method description explains the constrained/unconstrained dataset assumptions. These references do not turn retained circles or road lines into accepted Baghdad walksheds.

Catalogue review and adoption

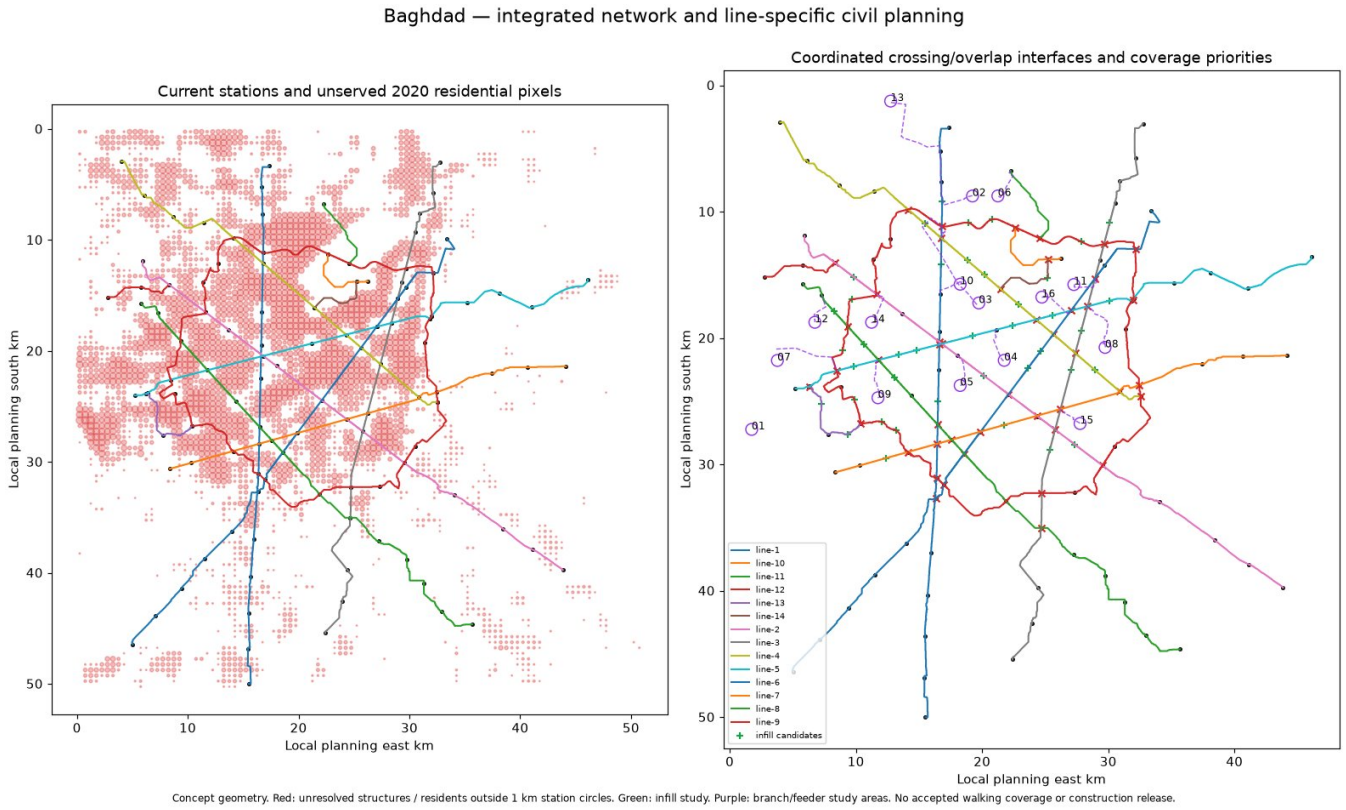
The 266-city audit reports declared transfer components, self-intersecting lines, cross-line overlap pairs, oversized interchange groups and retained 1 km population fractions. Missing geometry/population stays unavailable. Audit findings require local geometry, site and service investigation; they are not automatically deployed changes to every city.

Use one controlled adoption revision for selected corridors, junctions and stations. Regenerate design/scenario, civil and mechanical quantities, foundation/assembly, geography/clearance, energy/service, finance, staffing, deployment and publication in dependency order. Execute actual native nominal/degraded cases for changed inputs, preserve source-bound receipts, and reopen affected independent/site gates. The original supporting archive and new companion planning archive keep the earlier comparator and new model accessible without exceeding the repository's per-file limit.

Baghdad integrated network, foundations and assembly plan

Source: engineering/network-planning/baghdad/README.md

Coordinated design-development concept; field construction and operating adoption remain open.



Network interfaces and residential gaps

Interactive line / support / junction viewer lets reviewers zoom, select a line and inspect individual support packets and interface records offline.

One source-bound asset graph connects each line, actual span, shared support, beam, launcher front, junction interface, station complex and residential intervention. The current native timetable/finance remains a comparator; this package does not claim new operating service or accepted structural profiles.

The plan assigns 10,985 support-specific foundation packets, 9,973 catalogue bay assemblies, 404 specials and 28 workfronts sharing 18 configured launchers. 10 fronts have preceding asset assignments; a front record creates no additional machine. Support packets have coordinates, chainage, adjacent spans, desktop-soil investigation references and required loads/tests. Pile type/count/depth, groundwater and bearing capacity remain unselected until field design.

Line	Foundation packets	Catalogue bay assemblies	Special packages	Candidate infill
line-1	918	770	32	3
line-2	1165	988	76	3
line-3	932	863	29	4
line-4	1040	943	44	5
line-5	952	874	35	7
line-6	974	846	60	2
line-7	867	813	21	2
line-8	915	811	39	3
line-9	2423	2280	61	8
line-10	242	238	2	0
line-11	155	153	1	0
line-12	48	46	1	0
line-13	127	123	2	2
line-14	227	225	1	1

Junction and overlap coordination

46 crossing/shared-corridor/self-crossing interfaces need explicit structural treatment. 0 original interchange groups exceed the 600 m full-diameter screen. The new bounded grouping prevents transitive long-distance complexes and preserves platform coordinates. It does not create a track switch, measured walking path or accepted vertical profile.

Integrated geometry and complex register · GIS junction/residential layer · Foundation coordinate conflicts.

Residential coverage and expansion

Retained 2020 population within 1 km station circles is 27.7%. 40 population-ranked infill candidates give a conditional radial sensitivity of 45.7%. 2,173,888 currently unserved retained residents are also over 1 km from sampled existing corridors; infill cannot close those gaps. 16 named/located priority areas therefore require branch/extension or funded feeder/access studies. Area circles can overlap and are never summed as unique residents.

These are native population-count and distance screens, not current census, surveyed walksheds, ridership or funded service. Candidate sites need water/footprint/property/access, station structure, timetable, fleet, energy and full installed cost review.

Expansion corridor plans provide 15 source-bound routes along the retained arterial graph, tied to existing station connections, with access gaps and proposed stopping sites. Compare rail branches with frequent feeders; traffic direction, bridge grades/loads, railway curvature and walk routes are unverified. The combined infill/feeder 500 m stop-circle sensitivity is 51.4%; the rail 1 km sensitivity is 58.8%. Shared corridor edges count overlapping candidate road segments once. No corridor is adopted or funded.

Foundation and assembly execution

Foundation register has one row per unique proposed support. Span assembly register has identified component pairs and both foundation parents, actual line/run, unique front assignment, sequence and predecessor. Launcher fronts preserve disconnected runs and relocation gates. Stage library defines foundation/column/cap/bearing, factory/delivery, first/second beam securing, advance and handover dependencies. Specials and junctions use their own design/erection chains.

Decode compressed CSV using `python -c "import gzip; print(gzip.open('foundation-register.csv.gz','rt').read())"`. Coordinates are interpolated from the retained planning route; desktop soil does not select a foundation, and no pile depth or allowable bearing value is invented.

Complete per-line summary · Wider catalogue audit · Catalogue findings and provenance.

Companion data archive contains this planning package and new controlled inputs. The original Baghdad proposal archive retains the baseline city/engineering input evidence. Use both for the full planning review; the companion does not imply physical release.

Run `python tools/automation/integrated-network-plan.py`; `--check` validates source/output hashes. `--retain-context` creates the immutable retained OSM road/name input from the cached extraction. Native route and financial assumptions change only through a subsequent fully regenerated, validated adoption revision.

line-1 — foundation and assembly construction plan

Source: engineering/network-planning/baghdad/line-1-construction-plan.md

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 49.068 km. 918 unique proposed support packets; 770 identified catalogue bay assemblies and 32 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-1-front-1	launcher-01	initial mobilisation	1	408.3	385
line-1-front-2	launcher-02	initial mobilisation	-1	44150.1	385

Foundation workface plan

Filter the foundation register by `line-1`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-1-front-1	1	line-1-run-0001-span-0001	OSR-Pi20	FND:line-1-support-000000408300	FND:line-1-support-000000428300	mobilisation
line-1-front-1	2	line-1-run-0001-span-0002	OSR-Pi20	FND:line-1-support-000000428300	FND:line-1-support-000000448300	line-1-run-0001-span-0001
line-1-front-1	3	line-1-run-0002-span-0001	OSR-Pi25	FND:line-1-support-000000588300	FND:line-1-support-000000613300	line-1-run-0001-span-0002
line-1-front-1	4	line-1-run-0002-span-0002	OSR-Pi25	FND:line-1-support-000000613300	FND:line-1-support-000000638300	line-1-run-0002-span-0001
line-1-front-1	5	line-1-run-0002-span-0003	OSR-Pi25	FND:line-1-support-000000638300	FND:line-1-support-000000663300	line-1-run-0002-span-0002
line-1-front-1	6	line-1-run-0002-span-0004	OSR-Pi25	FND:line-1-support-000000663300	FND:line-1-support-000000688300	line-1-run-0002-span-0003
line-1-front-2	1	line-1-run-0114-span-0001	OSR-Pi20	FND:line-1-support-000044130100	FND:line-1-support-000044150100	mobilisation
line-1-front-2	2	line-1-run-0113-span-0001	OSR-Pi20	FND:line-1-support-000042948700	FND:line-1-support-000042968700	line-1-run-0114-span-0001
line-1-front-2	3	line-1-run-0112-span-0082	OSR-Pi20	FND:line-1-support-000042754500	FND:line-1-support-000042774500	line-1-run-0113-span-0001
line-1-front-2	4	line-1-run-0112-span-0081	OSR-Pi25	FND:line-1-support-000042729500	FND:line-1-support-000042754500	line-1-run-0112-span-0082
line-1-front-2	5	line-1-run-0112-span-0080	OSR-Pi25	FND:line-1-support-000042704500	FND:line-1-support-000042729500	line-1-run-0112-span-0081
line-1-front-2	6	line-1-run-0112-span-0079	OSR-Pi25	FND:line-1-support-000042679500	FND:line-1-support-000042704500	line-1-run-0112-span-0080

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-4761735b935e, crossing-596aa3d726d4, crossing-9090e87568d6, crossing-92b255c18da1, crossing-d02bcca1ecd2, crossing-e1b0b7941672, crossing-efa108fc0f05, shared-corridor-ea625bfc9e95.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-1-037500, infill-line-1-042500, infill-line-1-026500.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-2 — foundation and assembly construction plan

Source: engineering/network-planning/baghdad/line-2-construction-plan.md

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 52.851 km. 1,165 unique proposed support packets; 988 identified catalogue bay assemblies and 76 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-2-front-1	launcher-03	initial mobilisation	1	2228.8	494
line-2-front-2	launcher-04	initial mobilisation	-1	51689.8	494

Foundation workface plan

Filter the foundation register by `line-2`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-2-front-1	1	line-2-run-0001-span-0001	OSR-Pi25	FND:line-2-support-000002228800	FND:line-2-support-000002253800	mobilisation
line-2-front-1	2	line-2-run-0002-span-0001	OSR-Pi25	FND:line-2-support-000002446800	FND:line-2-support-000002471800	line-2-run-0001-span-0001
line-2-front-1	3	line-2-run-0004-span-0001	OSR-Pi20	FND:line-2-support-000004334700	FND:line-2-support-000004354700	line-2-run-0002-span-0001
line-2-front-1	4	line-2-run-0005-span-0001	OSR-Pi20	FND:line-2-support-000004572700	FND:line-2-support-000004592700	line-2-run-0004-span-0001
line-2-front-1	5	line-2-run-0008-span-0001	OSR-Pi20	FND:line-2-support-000009424200	FND:line-2-support-000009444200	line-2-run-0005-span-0001
line-2-front-1	6	line-2-run-0013-span-0001	OSR-Pi20	FND:line-2-support-000012035800	FND:line-2-support-000012055800	line-2-run-0008-span-0001
line-2-front-2	1	line-2-run-0101-span-0001	OSR-Pi20	FND:line-2-support-000051669800	FND:line-2-support-000051689800	mobilisation
line-2-front-2	2	line-2-run-0100-span-0002	OSR-Pi20	FND:line-2-support-000051509800	FND:line-2-support-000051529800	line-2-run-0101-span-0001
line-2-front-2	3	line-2-run-0100-span-0001	OSR-Pi20	FND:line-2-support-000051489800	FND:line-2-support-000051509800	line-2-run-0100-span-0002
line-2-front-2	4	line-2-run-0099-span-0001	OSR-Pi20	FND:line-2-support-000050274400	FND:line-2-support-000050294400	line-2-run-0100-span-0001
line-2-front-2	5	line-2-run-0097-span-0042	OSR-Pi20	FND:line-2-support-000049889800	FND:line-2-support-000049909800	line-2-run-0099-span-0001
line-2-front-2	6	line-2-run-0097-span-0041	OSR-Pi20	FND:line-2-support-000049869800	FND:line-2-support-000049889800	line-2-run-0097-span-0042

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-1c3badfbb529, crossing-5564106417eb, crossing-6644faf065db, crossing-778f06209d30, crossing-868cd72fc737, crossing-9eea764fde61, crossing-c9cacdc51c5b, shared-corridor-ea625bfc9e95.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-2-022000, infill-line-2-032000, infill-line-2-046500.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-3 — foundation and assembly construction plan

Source: engineering/network-planning/baghdad/line-3-construction-plan.md

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 49.496 km. 932 unique proposed support packets; 863 identified catalogue bay assemblies and 29 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-3-front-1	launcher-05	initial mobilisation	1	3436.5	431
line-3-front-2	launcher-06	initial mobilisation	-1	48187.4	432

Foundation workface plan

Filter the foundation register by `line-3`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-3-front-1	1	line-3-run-0005-span-0001	OSR-Pi25	FND:line-3-support-000003436500	FND:line-3-support-000003461500	mobilisation
line-3-front-1	2	line-3-run-0007-span-0001	OSR-Pi25	FND:line-3-support-000004584200	FND:line-3-support-000004609200	line-3-run-0005-span-0001
line-3-front-1	3	line-3-run-0007-span-0002	OSR-Pi20	FND:line-3-support-000004609200	FND:line-3-support-000004629200	line-3-run-0007-span-0001
line-3-front-1	4	line-3-run-0007-span-0003	OSR-Pi20	FND:line-3-support-000004629200	FND:line-3-support-000004649200	line-3-run-0007-span-0002
line-3-front-1	5	line-3-run-0009-span-0001	OSR-Pi20	FND:line-3-support-000006777000	FND:line-3-support-000006797000	line-3-run-0007-span-0003
line-3-front-1	6	line-3-run-0009-span-0002	OSR-Pi20	FND:line-3-support-000006797000	FND:line-3-support-000006817000	line-3-run-0009-span-0001
line-3-front-2	1	line-3-run-0040-span-0003	OSR-Pi20	FND:line-3-support-000048167400	FND:line-3-support-000048187400	mobilisation
line-3-front-2	2	line-3-run-0040-span-0002	OSR-Pi20	FND:line-3-support-000048147400	FND:line-3-support-000048167400	line-3-run-0040-span-0003
line-3-front-2	3	line-3-run-0040-span-0001	OSR-Pi20	FND:line-3-support-000048127400	FND:line-3-support-000048147400	line-3-run-0040-span-0002
line-3-front-2	4	line-3-run-0039-span-0003	OSR-Pi20	FND:line-3-support-000048047400	FND:line-3-support-000048067400	line-3-run-0040-span-0001
line-3-front-2	5	line-3-run-0039-span-0002	OSR-Pi20	FND:line-3-support-000048027400	FND:line-3-support-000048047400	line-3-run-0039-span-0003
line-3-front-2	6	line-3-run-0039-span-0001	OSR-Pi20	FND:line-3-support-000048007400	FND:line-3-support-000048027400	line-3-run-0039-span-0002

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-0fac546553f2, crossing-1fc40bbf9e02, crossing-42edbb5cfef1, crossing-658b4261f614, crossing-66c0e3e30d82, crossing-778f06209d30, crossing-86eb625bd399, crossing-bde9f0d16859.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-3-030000, infill-line-3-039500, infill-line-3-026500, infill-line-3-019500.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-4 — foundation and assembly construction plan

Source: engineering/network-planning/baghdad/line-4-construction-plan.md

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 41.932 km. 1,040 unique proposed support packets; 943 identified catalogue bay assemblies and 44 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-4-front-1	launcher-07	initial mobilisation	1	240.0	471
line-4-front-2	launcher-08	initial mobilisation	-1	41149.0	472

Foundation workface plan

Filter the foundation register by `line-4`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-4-front-1	1	line-4-run-0001-span-0001	OSR-Pi25	FND:line-4-support-000000240000	FND:line-4-support-000000265000	mobilisation
line-4-front-1	2	line-4-run-0002-span-0001	OSR-Pi25	FND:line-4-support-000000421400	FND:line-4-support-000000446400	line-4-run-0001-span-0001
line-4-front-1	3	line-4-run-0003-span-0001	OSR-Pi25	FND:line-4-support-000000546300	FND:line-4-support-000000571300	line-4-run-0002-span-0001
line-4-front-1	4	line-4-run-0003-span-0002	OSR-Pi25	FND:line-4-support-000000571300	FND:line-4-support-000000596300	line-4-run-0003-span-0001
line-4-front-1	5	line-4-run-0004-span-0001	OSR-Pi25	FND:line-4-support-000000642800	FND:line-4-support-000000667800	line-4-run-0003-span-0002
line-4-front-1	6	line-4-run-0004-span-0002	OSR-Pi20	FND:line-4-support-000000667800	FND:line-4-support-000000687800	line-4-run-0004-span-0001
line-4-front-2	1	line-4-run-0053-span-0002	OSR-Pi20	FND:line-4-support-000041129000	FND:line-4-support-000041149000	mobilisation
line-4-front-2	2	line-4-run-0053-span-0001	OSR-Pi20	FND:line-4-support-000041109000	FND:line-4-support-000041129000	line-4-run-0053-span-0002
line-4-front-2	3	line-4-run-0052-span-0001	OSR-Pi20	FND:line-4-support-000040884100	FND:line-4-support-000040904100	line-4-run-0053-span-0001
line-4-front-2	4	line-4-run-0051-span-0002	OSR-Pi20	FND:line-4-support-000040393100	FND:line-4-support-000040413100	line-4-run-0052-span-0001
line-4-front-2	5	line-4-run-0051-span-0001	OSR-Pi25	FND:line-4-support-000040368100	FND:line-4-support-000040393100	line-4-run-0051-span-0002
line-4-front-2	6	line-4-run-0049-span-0188	OSR-Pi25	FND:line-4-support-000039568400	FND:line-4-support-000039593400	line-4-run-0051-span-0001

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-27b707b08a14, crossing-42edbb5cfef1, crossing-4761735b935e, crossing-69a6a5b5c4a4, crossing-6f9463d5811f, crossing-954a77cefa6e, crossing-a79096d48aef, crossing-dff9d62fd115.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-4-028000, infill-line-4-022000, infill-line-4-037000, infill-line-4-024000, infill-line-4-017000.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-5 — foundation and assembly construction plan

Source: engineering/network-planning/baghdad/line-5-construction-plan.md

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 47.345 km. 952 unique proposed support packets; 874 identified catalogue bay assemblies and 35 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-5-front-1	launcher-09	initial mobilisation	1	656.6	437
line-5-front-2	launcher-10	initial mobilisation	-1	46242.5	437

Foundation workface plan

Filter the foundation register by `line-5`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-5-front-1	1	line-5-run-0001-span-0001	OSR-Pi25	FND:line-5-support-000000656600	FND:line-5-support-000000681600	mobilisation
line-5-front-1	2	line-5-run-0001-span-0002	OSR-Pi25	FND:line-5-support-000000681600	FND:line-5-support-000000706600	line-5-run-0001-span-0001
line-5-front-1	3	line-5-run-0001-span-0003	OSR-Pi25	FND:line-5-support-000000706600	FND:line-5-support-000000731600	line-5-run-0001-span-0002
line-5-front-1	4	line-5-run-0001-span-0004	OSR-Pi25	FND:line-5-support-000000731600	FND:line-5-support-000000756600	line-5-run-0001-span-0003
line-5-front-1	5	line-5-run-0001-span-0005	OSR-Pi25	FND:line-5-support-000000756600	FND:line-5-support-000000781600	line-5-run-0001-span-0004
line-5-front-1	6	line-5-run-0001-span-0006	OSR-Pi25	FND:line-5-support-000000781600	FND:line-5-support-000000806600	line-5-run-0001-span-0005
line-5-front-2	1	line-5-run-0042-span-0001	OSR-Pi20	FND:line-5-support-000046222500	FND:line-5-support-000046242500	mobilisation
line-5-front-2	2	line-5-run-0041-span-0001	OSR-Pi25	FND:line-5-support-000045268000	FND:line-5-support-000045293000	line-5-run-0042-span-0001
line-5-front-2	3	line-5-run-0040-span-0001	OSR-Pi20	FND:line-5-support-000045003100	FND:line-5-support-000045023100	line-5-run-0041-span-0001
line-5-front-2	4	line-5-run-0038-span-0001	OSR-Pi25	FND:line-5-support-000044267100	FND:line-5-support-000044292100	line-5-run-0040-span-0001
line-5-front-2	5	line-5-run-0037-span-0002	OSR-Pi20	FND:line-5-support-000042635900	FND:line-5-support-000042655900	line-5-run-0038-span-0001
line-5-front-2	6	line-5-run-0037-span-0001	OSR-Pi25	FND:line-5-support-000042610900	FND:line-5-support-000042635900	line-5-run-0037-span-0002

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-1c303c088c01, crossing-4e4707f7a13b, crossing-596aa3d726d4, crossing-66c0e3e30d82, crossing-6f9463d5811f, crossing-868cd72fc737, crossing-ec8cd65ba79b, shared-corridor-5c1c3ab579e9, shared-corridor-96ce2230245a.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-5-028000, infill-line-5-011000, infill-line-5-019500, infill-line-5-016000, infill-line-5-005500, infill-line-5-009500, infill-line-5-023000.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-6 — foundation and assembly construction plan

Source: engineering/network-planning/baghdad/line-6-construction-plan.md

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 52.503 km. 974 unique proposed support packets; 846 identified catalogue bay assemblies and 60 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-6-front-1	launcher-11	initial mobilisation	1	20.0	423
line-6-front-2	launcher-12	initial mobilisation	-1	46777.2	423

Foundation workface plan

Filter the foundation register by `line-6`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-6-front-1	1	line-6-run-0001-span-0001	OSR-Pi20	FND:line-6-support-000000020000	FND:line-6-support-000000040000	mobilisation
line-6-front-1	2	line-6-run-0001-span-0002	OSR-Pi20	FND:line-6-support-000000040000	FND:line-6-support-000000060000	line-6-run-0001-span-0001
line-6-front-1	3	line-6-run-0001-span-0003	OSR-Pi20	FND:line-6-support-000000060000	FND:line-6-support-000000080000	line-6-run-0001-span-0002
line-6-front-1	4	line-6-run-0001-span-0004	OSR-Pi20	FND:line-6-support-000000080000	FND:line-6-support-000000100000	line-6-run-0001-span-0003
line-6-front-1	5	line-6-run-0003-span-0001	OSR-Pi20	FND:line-6-support-000000346300	FND:line-6-support-000000366300	line-6-run-0001-span-0004
line-6-front-1	6	line-6-run-0004-span-0001	OSR-Pi25	FND:line-6-support-000000522800	FND:line-6-support-000000547800	line-6-run-0003-span-0001
line-6-front-2	1	line-6-run-0068-span-0001	OSR-Pi20	FND:line-6-support-000046757200	FND:line-6-support-000046777200	mobilisation
line-6-front-2	2	line-6-run-0067-span-0001	OSR-Pi20	FND:line-6-support-000046277200	FND:line-6-support-000046297200	line-6-run-0068-span-0001
line-6-front-2	3	line-6-run-0066-span-0036	OSR-Pi25	FND:line-6-support-000046087900	FND:line-6-support-000046112900	line-6-run-0067-span-0001
line-6-front-2	4	line-6-run-0066-span-0035	OSR-Pi25	FND:line-6-support-000046062900	FND:line-6-support-000046087900	line-6-run-0066-span-0036
line-6-front-2	5	line-6-run-0066-span-0034	OSR-Pi25	FND:line-6-support-000046037900	FND:line-6-support-000046062900	line-6-run-0066-span-0035
line-6-front-2	6	line-6-run-0066-span-0033	OSR-Pi25	FND:line-6-support-000046012900	FND:line-6-support-000046037900	line-6-run-0066-span-0034

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-1c303c088c01, crossing-27b707b08a14, crossing-526dbc821706, crossing-658b4261f614, crossing-7bba02156870, crossing-8431d2567c58, crossing-9090e87568d6, crossing-9eea764fde61, crossing-c15126eb0d40.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-6-033000, infill-line-6-035000.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-7 — foundation and assembly construction plan

Source: engineering/network-planning/baghdad/line-7-construction-plan.md

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 39.957 km. 867 unique proposed support packets; 813 identified catalogue bay assemblies and 21 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-7-front-1	launcher-13	initial mobilisation	1	1036.6	406
line-7-front-2	launcher-14	initial mobilisation	-1	39723.9	407

Foundation workface plan

Filter the foundation register by `line-7`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-7-front-1	1	line-7-run-0001-span-0001	OSR-Pi20	FND:line-7-support-000001036600	FND:line-7-support-000001056600	mobilisation
line-7-front-1	2	line-7-run-0002-span-0001	OSR-Pi20	FND:line-7-support-000005141400	FND:line-7-support-000005161400	line-7-run-0001-span-0001
line-7-front-1	3	line-7-run-0003-span-0001	OSR-Pi20	FND:line-7-support-000007085100	FND:line-7-support-000007105100	line-7-run-0002-span-0001
line-7-front-1	4	line-7-run-0004-span-0001	OSR-Pi20	FND:line-7-support-000011133900	FND:line-7-support-000011153900	line-7-run-0003-span-0001
line-7-front-1	5	line-7-run-0005-span-0001	OSR-Pi25	FND:line-7-support-000011173900	FND:line-7-support-000011198900	line-7-run-0004-span-0001
line-7-front-1	6	line-7-run-0006-span-0001	OSR-Pi20	FND:line-7-support-000013002400	FND:line-7-support-000013022400	line-7-run-0005-span-0001
line-7-front-2	1	line-7-run-0033-span-0068	OSR-Pi25	FND:line-7-support-000039698900	FND:line-7-support-000039723900	mobilisation
line-7-front-2	2	line-7-run-0033-span-0067	OSR-Pi25	FND:line-7-support-000039673900	FND:line-7-support-000039698900	line-7-run-0033-span-0068
line-7-front-2	3	line-7-run-0033-span-0066	OSR-Pi25	FND:line-7-support-000039648900	FND:line-7-support-000039673900	line-7-run-0033-span-0067
line-7-front-2	4	line-7-run-0033-span-0065	OSR-Pi25	FND:line-7-support-000039623900	FND:line-7-support-000039648900	line-7-run-0033-span-0066
line-7-front-2	5	line-7-run-0033-span-0064	OSR-Pi25	FND:line-7-support-000039598900	FND:line-7-support-000039623900	line-7-run-0033-span-0065
line-7-front-2	6	line-7-run-0033-span-0063	OSR-Pi25	FND:line-7-support-000039573900	FND:line-7-support-000039598900	line-7-run-0033-span-0064

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-66746bfdc84f, crossing-7bba02156870, crossing-80b4593a8386, crossing-bde9f0d16859, crossing-c7d529a495dc, crossing-c9cacdc51c5b, crossing-dff9d62fd115, crossing-e1b0b7941672.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-7-035500, infill-line-7-025000.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-8 — foundation and assembly construction plan

Source: engineering/network-planning/baghdad/line-8-construction-plan.md

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 48.137 km. 915 unique proposed support packets; 811 identified catalogue bay assemblies and 39 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-8-front-1	launcher-15	initial mobilisation	1	120.0	405
line-8-front-2	launcher-16	initial mobilisation	-1	47319.2	406

Foundation workface plan

Filter the foundation register by `line-8`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-8-front-1	1	line-8-run-0001-span-0001	OSR-Pi20	FND:line-8-support-000000120000	FND:line-8-support-000000140000	mobilisation
line-8-front-1	2	line-8-run-0002-span-0001	OSR-Pi25	FND:line-8-support-000001178000	FND:line-8-support-000001203000	line-8-run-0001-span-0001
line-8-front-1	3	line-8-run-0002-span-0002	OSR-Pi25	FND:line-8-support-000001203000	FND:line-8-support-000001228000	line-8-run-0002-span-0001
line-8-front-1	4	line-8-run-0002-span-0003	OSR-Pi25	FND:line-8-support-000001228000	FND:line-8-support-000001253000	line-8-run-0002-span-0002
line-8-front-1	5	line-8-run-0002-span-0004	OSR-Pi25	FND:line-8-support-000001253000	FND:line-8-support-000001278000	line-8-run-0002-span-0003
line-8-front-1	6	line-8-run-0002-span-0005	OSR-Pi20	FND:line-8-support-000001278000	FND:line-8-support-000001298000	line-8-run-0002-span-0004
line-8-front-2	1	line-8-run-0064-span-0002	OSR-Pi20	FND:line-8-support-000047299200	FND:line-8-support-000047319200	mobilisation
line-8-front-2	2	line-8-run-0064-span-0001	OSR-Pi20	FND:line-8-support-000047279200	FND:line-8-support-000047299200	line-8-run-0064-span-0002
line-8-front-2	3	line-8-run-0063-span-0002	OSR-Pi20	FND:line-8-support-000047022600	FND:line-8-support-000047042600	line-8-run-0064-span-0001
line-8-front-2	4	line-8-run-0063-span-0001	OSR-Pi20	FND:line-8-support-000047002600	FND:line-8-support-000047022600	line-8-run-0063-span-0002
line-8-front-2	5	line-8-run-0062-span-0003	OSR-Pi20	FND:line-8-support-000046894300	FND:line-8-support-000046914300	line-8-run-0063-span-0001
line-8-front-2	6	line-8-run-0062-span-0002	OSR-Pi20	FND:line-8-support-000046874300	FND:line-8-support-000046894300	line-8-run-0062-span-0003

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-0fac546553f2, crossing-3ba74f14bd23, crossing-66746bfdc84f, crossing-8431d2567c58, crossing-ec8cd65ba79b, crossing-efa108fc0f05, shared-corridor-481ca8ac8acb.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-8-040000, infill-line-8-036000, infill-line-8-044000.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-9 — foundation and assembly construction plan

Source: engineering/network-planning/baghdad/line-9-construction-plan.md

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 97.684 km. 2,423 unique proposed support packets; 2,280 identified catalogue bay assemblies and 61 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-9-front-1	launcher-17	initial mobilisation	1	0.0	1140
line-9-front-2	launcher-18	initial mobilisation	-1	97660.1	1140

Foundation workface plan

Filter the foundation register by `line-9`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-9-front-1	1	line-9-run-0001-span-0001	OSR-Pi25	FND:line-9-support-000000000000	FND:line-9-support-000000025000	mobilisation
line-9-front-1	2	line-9-run-0001-span-0002	OSR-Pi25	FND:line-9-support-000000025000	FND:line-9-support-000000050000	line-9-run-0001-span-0001
line-9-front-1	3	line-9-run-0001-span-0003	OSR-Pi25	FND:line-9-support-000000050000	FND:line-9-support-000000075000	line-9-run-0001-span-0002
line-9-front-1	4	line-9-run-0001-span-0004	OSR-Pi25	FND:line-9-support-000000075000	FND:line-9-support-000000100000	line-9-run-0001-span-0003
line-9-front-1	5	line-9-run-0001-span-0005	OSR-Pi25	FND:line-9-support-000000100000	FND:line-9-support-000000125000	line-9-run-0001-span-0004
line-9-front-1	6	line-9-run-0001-span-0006	OSR-Pi25	FND:line-9-support-000000125000	FND:line-9-support-000000150000	line-9-run-0001-span-0005
line-9-front-2	1	line-9-run-0083-span-0006	OSR-Pi20	FND:line-9-support-000097640100	FND:line-9-support-000097660100	mobilisation
line-9-front-2	2	line-9-run-0083-span-0005	OSR-Pi20	FND:line-9-support-000097620100	FND:line-9-support-000097640100	line-9-run-0083-span-0006
line-9-front-2	3	line-9-run-0083-span-0004	OSR-Pi25	FND:line-9-support-000097595100	FND:line-9-support-000097620100	line-9-run-0083-span-0005
line-9-front-2	4	line-9-run-0083-span-0003	OSR-Pi25	FND:line-9-support-000097570100	FND:line-9-support-000097595100	line-9-run-0083-span-0004
line-9-front-2	5	line-9-run-0083-span-0002	OSR-Pi25	FND:line-9-support-000097545100	FND:line-9-support-000097570100	line-9-run-0083-span-0003
line-9-front-2	6	line-9-run-0083-span-0001	OSR-Pi25	FND:line-9-support-000097520100	FND:line-9-support-000097545100	line-9-run-0083-span-0002

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-1c3badfbb529, crossing-1fc40bbf9e02, crossing-3ba74f14bd23, crossing-4e4707f7a13b, crossing-526dbc821706, crossing-532e0b6bb63c, crossing-6644faf065db, crossing-80b4593a8386, crossing-86eb625bd399, crossing-92b255c18da1, crossing-954a77cefa6e, crossing-a79096d48aef, crossing-c15126eb0d40, crossing-c4ea0b93414e, crossing-c6c859f1b2b3, crossing-c7d529a495dc, crossing-d02bcca1ecd2, shared-corridor-481ca8ac8acb, shared-corridor-5c1c3ab579e9.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-9-014500, infill-line-9-012000, infill-line-9-087500, infill-line-9-079000, infill-line-9-073500, infill-line-9-092000, infill-line-9-023000, infill-line-9-075000.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-10 — foundation and assembly construction plan

Source: engineering/network-planning/baghdad/line-10-construction-plan.md

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 6.804 km. 242 unique proposed support packets; 238 identified catalogue bay assemblies and 2 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-10-front-1	launcher-01	line-1-front-1	1	210.5	119
line-10-front-2	launcher-02	line-1-front-2	-1	6572.7	119

Foundation workface plan

Filter the foundation register by `line-10`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-10-front-1	1	line-10-run-0001-span-0001	OSR-Pi25	FND:line-10-support-00000210500	FND:line-10-support-00000235500	mobilisation
line-10-front-1	2	line-10-run-0001-span-0002	OSR-Pi25	FND:line-10-support-00000235500	FND:line-10-support-00000260500	line-10-run-0001-span-0001
line-10-front-1	3	line-10-run-0001-span-0003	OSR-Pi25	FND:line-10-support-00000260500	FND:line-10-support-00000285500	line-10-run-0001-span-0002
line-10-front-1	4	line-10-run-0001-span-0004	OSR-Pi25	FND:line-10-support-00000285500	FND:line-10-support-00000310500	line-10-run-0001-span-0003
line-10-front-1	5	line-10-run-0001-span-0005	OSR-Pi25	FND:line-10-support-00000310500	FND:line-10-support-00000335500	line-10-run-0001-span-0004
line-10-front-1	6	line-10-run-0001-span-0006	OSR-Pi25	FND:line-10-support-00000335500	FND:line-10-support-00000360500	line-10-run-0001-span-0005
line-10-front-2	1	line-10-run-0002-span-0027	OSR-Pi20	FND:line-10-support-00006552700	FND:line-10-support-00006572700	mobilisation
line-10-front-2	2	line-10-run-0002-span-0026	OSR-Pi20	FND:line-10-support-00006532700	FND:line-10-support-00006552700	line-10-run-0002-span-0027
line-10-front-2	3	line-10-run-0002-span-0025	OSR-Pi20	FND:line-10-support-00006512700	FND:line-10-support-00006532700	line-10-run-0002-span-0026
line-10-front-2	4	line-10-run-0002-span-0024	OSR-Pi25	FND:line-10-support-00006487700	FND:line-10-support-00006512700	line-10-run-0002-span-0025
line-10-front-2	5	line-10-run-0002-span-0023	OSR-Pi25	FND:line-10-support-00006462700	FND:line-10-support-00006487700	line-10-run-0002-span-0024
line-10-front-2	6	line-10-run-0002-span-0022	OSR-Pi25	FND:line-10-support-00006437700	FND:line-10-support-00006462700	line-10-run-0002-span-0023

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-057c84aa0956, crossing-c6c859f1b2b3.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: no qualifying candidate under this screen.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput.
Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-11 — foundation and assembly construction plan

Source: engineering/network-planning/baghdad/line-11-construction-plan.md

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 6.884 km. 155 unique proposed support packets; 153 identified catalogue bay assemblies and 1 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-11-front-1	launcher-03	line-2-front-1	1	210.5	76
line-11-front-2	launcher-04	line-2-front-2	-1	4035.5	77

Foundation workface plan

Filter the foundation register by `line-11`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-11-front-1	1	line-11-run-0001-span-0001	OSR-Pi25	FND:line-11-support-00 0000210500	FND:line-11-support-00 0000235500	mobilisation
line-11-front-1	2	line-11-run-0001-span-0002	OSR-Pi25	FND:line-11-support-00 0000235500	FND:line-11-support-00 0000260500	line-11-run-0001-span-0001
line-11-front-1	3	line-11-run-0001-span-0003	OSR-Pi25	FND:line-11-support-00 0000260500	FND:line-11-support-00 0000285500	line-11-run-0001-span-0002
line-11-front-1	4	line-11-run-0001-span-0004	OSR-Pi25	FND:line-11-support-00 0000285500	FND:line-11-support-00 0000310500	line-11-run-0001-span-0003
line-11-front-1	5	line-11-run-0001-span-0005	OSR-Pi25	FND:line-11-support-00 0000310500	FND:line-11-support-00 0000335500	line-11-run-0001-span-0004
line-11-front-1	6	line-11-run-0001-span-0006	OSR-Pi25	FND:line-11-support-00 0000335500	FND:line-11-support-00 0000360500	line-11-run-0001-span-0005
line-11-front-2	1	line-11-run-0001-span-0153	OSR-Pi25	FND:line-11-support-00 0004010500	FND:line-11-support-00 0004035500	mobilisation
line-11-front-2	2	line-11-run-0001-span-0152	OSR-Pi25	FND:line-11-support-00 0003985500	FND:line-11-support-00 0004010500	line-11-run-0001-span-0153
line-11-front-2	3	line-11-run-0001-span-0151	OSR-Pi25	FND:line-11-support-00 0003960500	FND:line-11-support-00 0003985500	line-11-run-0001-span-0152
line-11-front-2	4	line-11-run-0001-span-0150	OSR-Pi25	FND:line-11-support-00 0003935500	FND:line-11-support-00 0003960500	line-11-run-0001-span-0151
line-11-front-2	5	line-11-run-0001-span-0149	OSR-Pi25	FND:line-11-support-00 0003910500	FND:line-11-support-00 0003935500	line-11-run-0001-span-0150
line-11-front-2	6	line-11-run-0001-span-0148	OSR-Pi25	FND:line-11-support-00 0003885500	FND:line-11-support-00 0003910500	line-11-run-0001-span-0149

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-c4ea0b93414e.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: no qualifying candidate under this screen.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput.
Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-12 — foundation and assembly construction plan

Source: engineering/network-planning/baghdad/line-12-construction-plan.md

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 6.586 km. 48 unique proposed support packets; 46 identified catalogue bay assemblies and 1 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-12-front-1	launcher-05	line-3-front-1	1	210.5	23
line-12-front-2	launcher-06	line-3-front-2	-1	1355.5	23

Foundation workface plan

Filter the foundation register by `line-12`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-12-front-1	1	line-12-run-0001-span-0001	OSR-Pi25	FND:line-12-support-00000210500	FND:line-12-support-00000235500	mobilisation
line-12-front-1	2	line-12-run-0001-span-0002	OSR-Pi25	FND:line-12-support-00000235500	FND:line-12-support-00000260500	line-12-run-0001-span-0001
line-12-front-1	3	line-12-run-0001-span-0003	OSR-Pi25	FND:line-12-support-00000260500	FND:line-12-support-00000285500	line-12-run-0001-span-0002
line-12-front-1	4	line-12-run-0001-span-0004	OSR-Pi25	FND:line-12-support-00000285500	FND:line-12-support-00000310500	line-12-run-0001-span-0003
line-12-front-1	5	line-12-run-0001-span-0005	OSR-Pi25	FND:line-12-support-00000310500	FND:line-12-support-00000335500	line-12-run-0001-span-0004
line-12-front-1	6	line-12-run-0001-span-0006	OSR-Pi25	FND:line-12-support-00000335500	FND:line-12-support-00000360500	line-12-run-0001-span-0005
line-12-front-2	1	line-12-run-0001-span-0046	OSR-Pi20	FND:line-12-support-00001335500	FND:line-12-support-00001355500	mobilisation
line-12-front-2	2	line-12-run-0001-span-0045	OSR-Pi25	FND:line-12-support-00001310500	FND:line-12-support-00001335500	line-12-run-0001-span-0046
line-12-front-2	3	line-12-run-0001-span-0044	OSR-Pi25	FND:line-12-support-00001285500	FND:line-12-support-00001310500	line-12-run-0001-span-0045
line-12-front-2	4	line-12-run-0001-span-0043	OSR-Pi25	FND:line-12-support-00001260500	FND:line-12-support-00001285500	line-12-run-0001-span-0044
line-12-front-2	5	line-12-run-0001-span-0042	OSR-Pi25	FND:line-12-support-00001235500	FND:line-12-support-00001260500	line-12-run-0001-span-0043
line-12-front-2	6	line-12-run-0001-span-0041	OSR-Pi25	FND:line-12-support-00001210500	FND:line-12-support-00001235500	line-12-run-0001-span-0042

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-5564106417eb.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: no qualifying candidate under this screen.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput.
Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-13 — foundation and assembly construction plan

Source: engineering/network-planning/baghdad/line-13-construction-plan.md

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 8.042 km. 127 unique proposed support packets; 123 identified catalogue bay assemblies and 2 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-13-front-1	launcher-07	line-4-front-1	1	210.5	61
line-13-front-2	launcher-08	line-4-front-2	-1	3719.0	62

Foundation workface plan

Filter the foundation register by `line-13`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-13-front-1	1	line-13-run-0001-span-0001	OSR-Pi25	FND:line-13-support-00000210500	FND:line-13-support-00000235500	mobilisation
line-13-front-1	2	line-13-run-0001-span-0002	OSR-Pi25	FND:line-13-support-00000235500	FND:line-13-support-00000260500	line-13-run-0001-span-0001
line-13-front-1	3	line-13-run-0001-span-0003	OSR-Pi25	FND:line-13-support-00000260500	FND:line-13-support-00000285500	line-13-run-0001-span-0002
line-13-front-1	4	line-13-run-0001-span-0004	OSR-Pi25	FND:line-13-support-00000285500	FND:line-13-support-00000310500	line-13-run-0001-span-0003
line-13-front-1	5	line-13-run-0001-span-0005	OSR-Pi25	FND:line-13-support-00000310500	FND:line-13-support-00000335500	line-13-run-0001-span-0004
line-13-front-1	6	line-13-run-0001-span-0006	OSR-Pi25	FND:line-13-support-00000335500	FND:line-13-support-00000360500	line-13-run-0001-span-0005
line-13-front-2	1	line-13-run-0002-span-0011	OSR-Pi20	FND:line-13-support-00003699000	FND:line-13-support-00003719000	mobilisation
line-13-front-2	2	line-13-run-0002-span-0010	OSR-Pi25	FND:line-13-support-00003674000	FND:line-13-support-00003699000	line-13-run-0002-span-0011
line-13-front-2	3	line-13-run-0002-span-0009	OSR-Pi25	FND:line-13-support-00003649000	FND:line-13-support-00003674000	line-13-run-0002-span-0010
line-13-front-2	4	line-13-run-0002-span-0008	OSR-Pi25	FND:line-13-support-00003624000	FND:line-13-support-00003649000	line-13-run-0002-span-0009
line-13-front-2	5	line-13-run-0002-span-0007	OSR-Pi25	FND:line-13-support-00003599000	FND:line-13-support-00003624000	line-13-run-0002-span-0008
line-13-front-2	6	line-13-run-0002-span-0006	OSR-Pi25	FND:line-13-support-00003574000	FND:line-13-support-00003599000	line-13-run-0002-span-0007

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-532e0b6bb63c, shared-corridor-96ce2230245a.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-13-006000, infill-line-13-001500.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput.
Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-14 — foundation and assembly construction plan

Source: engineering/network-planning/baghdad/line-14-construction-plan.md

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 6.047 km. 227 unique proposed support packets; 225 identified catalogue bay assemblies and 1 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-14-front-1	launcher-09	line-5-front-1	1	210.5	112
line-14-front-2	launcher-10	line-5-front-2	-1	5815.5	113

Foundation workface plan

Filter the foundation register by `line-14`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-14-front-1	1	line-14-run-0001-span-0001	OSR-Pi25	FND:line-14-support-00000210500	FND:line-14-support-00000235500	mobilisation
line-14-front-1	2	line-14-run-0001-span-0002	OSR-Pi25	FND:line-14-support-00000235500	FND:line-14-support-00000260500	line-14-run-0001-span-0001
line-14-front-1	3	line-14-run-0001-span-0003	OSR-Pi25	FND:line-14-support-00000260500	FND:line-14-support-00000285500	line-14-run-0001-span-0002
line-14-front-1	4	line-14-run-0001-span-0004	OSR-Pi25	FND:line-14-support-00000285500	FND:line-14-support-00000310500	line-14-run-0001-span-0003
line-14-front-1	5	line-14-run-0001-span-0005	OSR-Pi25	FND:line-14-support-00000310500	FND:line-14-support-00000335500	line-14-run-0001-span-0004
line-14-front-1	6	line-14-run-0001-span-0006	OSR-Pi25	FND:line-14-support-00000335500	FND:line-14-support-00000360500	line-14-run-0001-span-0005
line-14-front-2	1	line-14-run-0001-span-0225	OSR-Pi20	FND:line-14-support-00005795500	FND:line-14-support-00005815500	mobilisation
line-14-front-2	2	line-14-run-0001-span-0224	OSR-Pi20	FND:line-14-support-00005775500	FND:line-14-support-00005795500	line-14-run-0001-span-0225
line-14-front-2	3	line-14-run-0001-span-0223	OSR-Pi20	FND:line-14-support-00005755500	FND:line-14-support-00005775500	line-14-run-0001-span-0224
line-14-front-2	4	line-14-run-0001-span-0222	OSR-Pi20	FND:line-14-support-00005735500	FND:line-14-support-00005755500	line-14-run-0001-span-0223
line-14-front-2	5	line-14-run-0001-span-0221	OSR-Pi25	FND:line-14-support-00005710500	FND:line-14-support-00005735500	line-14-run-0001-span-0222
line-14-front-2	6	line-14-run-0001-span-0220	OSR-Pi25	FND:line-14-support-00005685500	FND:line-14-support-00005710500	line-14-run-0001-span-0221

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-057c84aa0956, crossing-69a6a5b5c4a4.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-14-004500.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput.
Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

Coordinated industrialisation programme

Source: engineering/industrialisation/README.md

Status: proposed programme; products, rights, commercial terms and physical acceptance remain open. Vehicle platform, bogie localisation and the viaduct system share one configuration and evidence chain. HÜBNER and CRRC are proposed discussion partners; no exact product or manufacturing-transfer agreement is selected. Controlled programme.

Vehicle configuration and civil demand

Vehicle-to-civil trace preserves the current three-car, six-bogie/twelve-axle reference. Its 78,750 kg tare is an unclosed planning control. Equal car loading, midpoint CG and equal bogie axle sharing are explicitly assumed. Bogie locations follow the existing CAD rule. The shared-bogie alternative has an explicit four-support graph with unknown redesign masses. Buying an articulation creates no bogie-count, mass, energy or cost reduction. Exact joint/bogie selection, low-floor clearances, yaw/pitch/roll, stiffness/damping, fatigue/crash and maintenance/recovery require a joint interface freeze.

The same axle pattern feeds sampled static Pi20/Pi25 live-load demand. Dynamic augmentation, permanent capacity, prestress, bearings and launcher loads need independent design. Identified Baghdad spans remain the quantity authority. No vehicle/civil loading baseline is overwritten.

Bogie localisation and first articles

First article repeat assembly with qualified bought-in subcomponents released frame cutting/forming/welding/machining/NDT/coating further component localisation where economics justify qualification. Initially buy precision articulation assemblies and make approved surrounding adapters locally. Assess existing Iraqi fabrication and machining processes. Prioritise fixtures, metrology, weld quality, inspection and test equipment. Transfer rights, fees, tooling, qualification, imported components, labour, rejects, warranty and repeat prices have separate scopes.

Civil catalogue and urban delivery

Two single-track full-span beams share a pier: 25 m primary, 20 m secondary, and separately engineered exceptional crossings. Standardise column/cap connections, bearing/jacking/lifting seats, direct-fixation datums, removable walkway/service cassettes and island-station transitions. Ground zones, depth, capacity, utility clearances and working platforms remain site-specific.

Support packets and urban controls keep foundation mobilisation and launcher support acceptance distinct. Compare precast columns/caps with reusable-form in-situ columns/precast caps by measured accepted supports/week and complete installed cost. Require inspectable connections, temporary stability, grout and verified release strength. Maintain shop/pedestrian/emergency access, appointment-based deliveries/spoil, booked lifting exclusions and prompt reinstatement. Lane-hours, access interruptions and occupation days are unioned by resource and bound to accepted completed bays; absent observations stay unknown.

Production chain and 18-launcher scenario

Production balance treats one accepted double-track bay as the planning unit. Eighteen active independent fronts at 1/1.5/2 bays per working day give 18/27/36 bays, 36/54/72 beams and 450/675/900 m under the illustrative all-25 m assumption. Actual 20/25 m mixes and special spans retain identified quantities. At 54 beams/day, 48-hour occupation needs 108 casting positions before availability/reject allowances; a long-line bed can hold several positions. A five-working-day buffer needs 270 beams. None is a production commitment.

Released foundations accepted piers accepted serialized beams transport erection completed-bay acceptance limit output together. Casting alone cannot release a bay. Mobilise only against a sufficiently long released run, qualified supply, access, authorised crew and compatible inspected machinery. Coordinate existing Iraqi precast facilities and expertise; allocate qualified products/slots and add only demonstrated necessary tooling/capacity.

Work, training, programme gates and finance

Linked work packages and ERP handoffs connect the existing supplier, civil and execution structures. The first 30 and 30–90 day windows are proposed planning targets, followed by accepted pilot trials, controlled ramp-up and up-to-18-machine deployment. Responsible roles are named; appointments, funded recruitment and dated rosters remain open. Training attendance is separate from assessed equipment-specific work authority. Future reviewed reservations must undergo the same live machinery/worker exclusivity checks.

Installed costs and delivered-service finance separates priced scopes and missing amounts. The Baghdad delivery/finance bridge joins each line's conditional running-span date, retained finance opening month/fleet deadline, controlled required fleet and energy/service diagnostics. A common financial-close anchor, supplier vehicle deliveries and accepted station/special/system handovers are still missing. Dispatch completion does not automatically scale paid journeys. Running-bay completion is not railway revenue opening. Unique paid journeys/collection and operating acceptance are required before revised receipts can be considered. Transfers do not multiply paid journeys. Further elevation comparisons expose marginal civil reference allowances for explicit scope sensitivities, with complete installed costs, disruption values, financing and net benefit unknown. No additional alignment, beam order, price or accelerated saving is adopted.

Possible component and equipment vendors

Official manufacturer product and reference pages; narrow published scope only. Historic references are not current quotations. No supplier was contacted. Related group brands are not independent offers. Product-family listings establish neither OSR fit nor regional availability, rights, prices, capacity or acceptance. Gangways are not automatically structural joints; segmental gantries are not whole-beam launchers. Machine-readable candidates · RFQ comparison CSV.

Possible vendor	Scope	Official product/reference source
HÜBNER	structural-articulation	FXpro, FXcore and vehicle-specific adapters
HEMSCHEIDT Engineering	upper-articulation	HEUAS upper articulation systems
CRRC Zhuzhou	bogie-transfer	Vehicle and localisation engineering discussion
Siemens Mobility	complete-bogie	MoComp bogies and rail components
Alstom	complete-bogie	Flexx Urban / Flexx bogie portfolio
Lucchini RS	wheelset	Rail wheelsets, wheels and axles
GHH-BONATRANS	wheelset	Complete wheelsets and wheel/axle components
SKF	bearing-axlebox	Axlebox bearings, bearing units and monitoring
Continental	suspension	Rail secondary air springs
Knorr-Bremse	braking	Metro bogie brakes, controls and air supply
Wabtec / Faiveley	braking	Transit bogie brakes
ABB	traction-power	AMXM rail traction motors
Ingeteam	traction-power	INGETRAC traction converters
Voith	gearbox-coupling	Rail gear units and drive couplings
Dellner	gangway-coupler	Gangways, automatic and semi-permanent couplers
Forsee Power	traction-battery	PULSE PLUS RAIL high-power LTO battery
Schunk Transit Systems	charging-contacts	Rail contact/grounding and charging technology
Merak	hvac	Rail heating ventilation and air conditioning
Liebherr	hvac	Passenger and cab rail HVAC
IFE	doors	Rail entrance systems
duagon	control-electronics	Embedded computing, gateways and safe control systems
Kontron	control-electronics	Railway embedded computing / KBox R-101
HARTING	connectors-harness	Rail power and data connectors and cables
Moxa	networking	ToughNet TN-4500A rail Ethernet switches
DEAL	whole-beam-erection	Launching girders for full-span erection
DLT Engineering	whole-beam-erection	Bespoke full-span erection gantry engineering
Cimolai Technology	special-segmental-erection	Segment launching gantry
Freyssinet	prestressing-bearings	Prestressing, bearing and connection systems
mageba	bearings-joints	RESTON-SPHERICAL structural bearings
MAURER	bearings-joints	Structural bearings and expansion joints
Elematic	precast-tooling	Frame line for precast columns and beams
BAUER Maschinen	foundation-equipment	BG drilling-rig family

Regeneration

Run `.venv/bin/python tools/automation/industrialisation-study.py`; add `--check` for source/output drift. Manifest records exact inputs and output bytes. Missing supplier or engineering evidence never becomes acceptance.

RFC 0033 — TACS runtime and committed resource control

Source: docs/rfcs/0033-tacs-runtime-and-resource-control.md

Status: proposed development target; software evidence generated/unreviewed. Supersedes: RFC 0032's separate authority/protection implementation. Deployment: synthetic reference only; sectional-authority pilot retained.

Responsibility and interface map

Function	Decision owner	Physical or deployment boundary
Shared railway identities/configuration	`osr-core::deployment`	Exact model-byte hash; frozen startup parameters; designed revision distinct from installed and approved records
Authoritative resource lifecycle	`osr-interlocking::resources`, `resource_log`, `DerivedState`	Only verified committed `ResourceControl` entries mutate ownership
Ordered authoritative decisions	Existing `osr-consensus::step`	Three static voters; independent power/network placement must be designed and assessed
Physical proving	Identified point/station I/O hosts	Asset-scoped occupancy, integrity, points, restrictions and qualified no-reentry proof; synthetic ports in the reference
Onboard movement authority	Existing `compute_self_ma_from_state`	Validated committed view, exact frozen model, bounded full train footprint and evidence age
Speed and emergency protection	Existing `osr-atp::atp_evaluate`	Grade-adjusted illustrative braking profile; measured braking/adhesion/latency remains open
Mission and station requests	Existing `osr-ato::ato_evaluate`, `station_step`	Doors, charger isolation, connector clearance and energy independently gate departure
Final brake request	Existing `osr-brake::brake_evaluate`	Application request to separately qualified safety-output channel; software host is not qualified output hardware
Authentication and message contracts	`osr-proto`, `osr-crypto`, `osr-secbus`, consensus proposal verifier	Registered issuer, asset permissions, configuration/session/sequence/time checks before state effects
Process execution and persistence	Thin `osr-runtime` hosts	Framed local ports, event scheduling, disk writes before outbound effects, conservative cold start
OCC	Service intentions and observations	No authority, ownership release or latch-clear bypass
ERPNext/FUXA	Existing observation/maintenance integrations	Observation only; no new railway command interface

There is one MA implementation and one ATP implementation, executed in each logical protection channel. Resource contracts from the earlier prototype move into existing crates; `osr-tacs` and its parallel MA/protection algorithms are retired. `osr-runtime` contains orchestration and adapters. Route selection remains intent, per `osr-onboard-routing`; agreement cannot supersede missing permission or physical proving. Logical peer observations may traverse common infrastructure; no direct radio or `osr-t2t` crate is added.

Configuration and communications

`railway-model.json` schema `osr-tacs-railway/2` supplies resource IDs, existing section IDs, conflicts, extents, asset identities, train/session/formation parameters, station locations, static voters and role-scoped public keys. `deployment.json` is an exact-byte-hash-bound generated startup export. `FrozenRailway` rejects unsupported versions, invalid topology/identities, changed startup parameters and operational activation. This source is a designed synthetic configuration; `installed_state=not-installed` and `approved_operating_profile=prototype-unapproved` are separate explicit facts. Changes require a new bundle and restart, never live digital-twin mutation.

The reference transport uses signed `osr-runtime/1` packets, postcard encoding, existing signed consensus proposals and explicit internal interlocking entries. Legacy protobuf/postcard `osr-proto::Entry` remains a separate versioned format; it must not be deserialised directly into internal entries. Float-based legacy position inputs require checked unit conversion at an explicitly selected adapter. The runtime path publishes integer measured speed and bounded, non-zero uncertainty. Unsupported/malformed packets fail closed.

The envelope binds configuration hash, registered session, monotonic sequence, original observation time and message kind. Limits: 512 KiB application payload, 768 KiB signed payload, 4 MiB framed command, 16 publications per batch, 12,000 log entries and a 256-message reference queue. One-second input freshness and 100 ms maximum configured clock error are prototype assumptions; PTP and hardware timing must establish their operational applicability. Future-dated packets are rejected. Send/receive counters persist across restart. Asset permissions are separate from signature validity. All load-bearing events, including positions, use `Safety`. The legacy simulator consensus publisher also classifies occupancy-changing position reports as `Safety` and retains its last committed view when quorum confirmation is unavailable. No advisory route/position can unlock track.

The published deterministic reference keys are test fixtures. Operational provisioning, protected key storage, rotation, revocation distribution, offline replacement and recommissioning require a controlled maintenance pack. Changing keys or sessions changes the configuration hash; interoperability stops until the complete bundle is reconciled. Revoked participants

are rejected. Crash-fault Raft is assumed: a signed member's committed-prefix assertion is not a Byzantine quorum certificate. Authenticated erroneous sensors and common software, configuration, power and clock causes remain FMEA concerns.

Ownership and evidence freshness

Unknown qualified clear available reserve/prove/lock occupy explicit withdrawal qualified no-reentry/clearance available. Identified occupied startup admission can bind the proved train/session when no conflicting owner exists; it cannot transfer another owner's lock. Leases time out to `ReleasePending`, retaining their owner. Restart increments the committed controller epoch, retains owners as unknown and quarantines empty clearance for the maximum prior permission plus clock error. Identified retained owners can be reconciled without transferring protection. Blocked possessions cannot be cleared by these recovery APIs.

An empty detector does not prove cancellation safe. Qualified release requires bounded full rear clearance with integrity, or committed withdrawal and a proved stop/protected approach that prevents old-permission re-entry. These are physical interface assumptions to validate. Platform resources remain owned at arrival. Depot/work resource is explicitly protected; rescue, coupled formations, turnbacks, arbitrary topology and large fleets remain separate packages.

MA expiry is the minimum of the new MA window and the original position, local proof and held permission deadlines. Recalculation cannot refresh evidence. Every traversed section must have the correct active owner/configuration/epoch, current point proof and integrity. Local full-footprint uncertainty checking adds a restrictive gate before ATP. Recovery requires stopped, valid local facts and a privileged local procedure, never a dashboard acknowledgement. Civil conditions enter through asset-scoped signed restrictions/closures; urgent protection must have an engineered path independent of alarm viewing. Removal requires qualified inspection/handback, not an ERP status change.

Persistence and process reference

`osr-consensus::disk::DiskJournal` has one OS-locked writer. Frames carry node, configuration, term, vote, complete log deltas, commit index, application state and replay fences, with length and CRC bounds. Every host persists and fsyncs file and directory before exposing messages or actuator requests. Torn/corrupt journals inhibit startup; repair never silently discards an accepted lock. Storage compaction atomically replaces a full-log checkpoint after fsync. Logical Raft snapshotting/log truncation is not implemented. Capacity exhaustion inhibits further publication; offline servicing requires retained-protection and configuration reconciliation. A power interruption during rename/fsync requires storage/hardware qualification; SIGKILL tests are software evidence.

The default Raft batch remains one entry. The reference host uses bounded contiguous batches of twelve; Rust handles each entry under the same checked prefix semantics. This is outside the one-entry formal refinement and needs independent refinement review. Quorum confirmation records fresh per-peer responses in the current term, including empty-log heartbeats; stale historical replication indices cannot refresh a quorum. Durable reboot restores no leader role, quorum confirmation or volatile acknowledgements.

Run `python3 tools/automation/tacs_reference.py --output /tmp/reference.json`. The coordinator spawns four `osr-train-agent` processes (A/B for each train) and six `osr-wayside-agent` processes: three voters, point I/O and west/east station I/O. Charger requests and local proving pass through the station process ports; their asset-scoped interlocks can inhibit power and cannot grant movement. The depot/work resource is committed blocked and remains blocked across expiry and controller restart. Two further `osr-safety-output` processes each host the paired `osr-brake::dual::DualGuard` comparator. Both channels must agree to permit; either emergency, missing/stale/replayed request, disagreement or unhealthy feedback latches an emergency output with zero torque. The reference feeds at 50 ms, before the 100 ms output deadline. A late feed cannot replace an overdue request to hide a missed interval; only stopped, privileged, fresh recovery can clear the latch. The deterministic twin explicitly selects `--clock virtual`. The default output host now has an independent monotonic-clock sampling thread, bounded input parsing and timestamped output snapshots. Idle/partial input and blocked consumption cannot suspend that sampler. Source issue timestamps and process-clock epochs remain distinct from receipt time. Native process and expiry tests are software evidence; independent physical timers, scheduling bounds, source authentication and de-energised hardware outputs remain unqualified. See the runtime contract. Processes exchange signatures through asynchronous bounded queues; publication does not wait for commitment. Simulation truth enters only own-train sensors and local infrastructure proving ports, never the MA state. Real journals and process kill/restart are used. The same frozen bundle feeds every process. The physics fixture checks its supported extents, conflict identities, formation/braking and berth/energy parameters before startup; unsupported bundles require a new calibrated fixture. `osr-sim`'s legacy synchronous cluster publisher remains isolated from this reference; it is not claimed as the deployed asynchronous path. Full integration of its scenarios/physics, shared clocks, radio/T2G/TCN failover and direct-peer performance is future work.

Station sequence is stop/secure exchange/charge isolate physical departure proof ready. The reference measures synthetic energy and closed-loop speed; per-car BMS/thermal/door/PSD/traction integrations remain pending HIL. Failed

charging can hold service without erasing track ownership. Emergency departure cannot bypass a connected charger or unlocked door. Missing application output must trip a separately qualified deadline/output channel; the isolated reference output ports model deadlines and latches but do not qualify that hardware function.

Default redundant protection and platform review follow-up

The executable policy sets two logical channels, two-out-of-two permission and either-channel trip as the development default for train protection, point proving/locking, crossings, doors/platforms, charging isolation and obstacle protection. Loss of either channel stops the affected function. Automatic single-channel operation or takeover is excluded. Availability after a channel failure requires a separately assessed architecture. Three Raft voters replicate authoritative state; they are not protection outputs. The conservative deployable pilot profile and its release gates remain in force.

Each train A/B host uses a separate OS process, journal and restored committed view. The binary requires `--safety-channel A|B`; a persisted journal cannot change channel on restart. A publishes the train's network proposals; B evaluates the same committed domain inputs without publishing duplicate proposals. Loss of A does not authorise B takeover. Pair comparison uses exact synthetic cycle sequence/time, brake and torque agreement. It has no end-to-end source-issued output identity yet. Shared software, sensor fixture, host, test keys, coordinator clock and comparator are explicit common causes. Physical power/output separation, channel-local clocks and independent feedback must be engineered and qualified. Only train protection's logical pair is executed; the policy explicitly keeps all five other local pairs open. Two processes are not installed board quantities.

The 2 October 2026 review of `17b31f055c9e16b5439c18b4a4d83f004890bcad` identified a stale FMEA applicability check and a concealed deadline gap. This change rejects stale failure subjects as well as stale configuration baselines, retains the late-feed regression and adds missing-A, missing-B, disagreement, replay and feedback-failure process cases. Each new case must first permit motion, actually inject its fault, trip within one 50 ms reference step, stop with retained protection and remain latched. These are virtual-time software criteria, not measured hardware response times. The existing TLC abstraction covers resource ownership only; it does not prove the new paired output function.

Next work closes named integration gaps in this order. Owners and reviewers are roles to appoint; acceptance is still open.

Work	Owner / independent reviewer	Required exit evidence
Autonomous watchdog and output paths	Embedded/electronics / safety-I/O reviewer	Coordinator/application/comparator freeze and kill, delayed scheduling, clock rollback/skew, power loss and stuck feedback; measured traction inhibition and effective braking without a new application command
Sensors and emergency union	Train integration / protection reviewer	Typed adapters to existing localisation, fire, derailment, obstacle and BMS/thermal evaluators; wheel slip/bias and simultaneous resource faults independently reach the final output; shared bad sensors do not establish independence
Repeated station missions	ATO/station integration / operations reviewer	Explicit mission/stop identity and arrival/departure transitions; at least three stops, short charge, delayed doors, turnback/resume, per-car charging and no moving door release
Remaining local protection pairs	Wayside/station / safety-I/O reviewer	Point/crossing proving, door/PSD, charger isolation and obstacle channels; independent release and inspected handback
Calibrated plant and city conversion	Simulation/configuration / braking and site reviewers	Sealed city revision converted to supported resources and shared `osr-sim` plant with mass/load, adhesion, gradients, build-up/delay, thermal/energy limits; full LM3 stopping envelope under uncertainty sweeps
Duty, recovery and storage	Simulation/runtime / operations and storage reviewers	Representative city's station capacity, overnight storage, morning dispatch, charging outage/recovery; separate rescue/tow/split/reverse/handover missions and sustained duty without capacity exhaustion or unbounded restart; safe snapshots/replay fences retain locks
Formal and scenario evidence	Assurance / independent formal reviewer	Property-to-code/test mapping, overlapping multi-resource routes, sequences/epochs/crash points, counterexample replay and specific hazard activation/deadline/excursion/queue/CPU/storage/recovery metrics
Integration and lifecycle quantities	Platform/hardware / configuration and cost reviewers	Observation-only A/B health/discrepancy/feedback/age/degraded/revision data through supervision gateway to FUXA/ERP; no acknowledgement resets; actual serial/power/firmware/calibration/replacement records and matched-service redundancy/spares/training/assessment costs
Physical release	Hardware/operations / independent assessor	HIL, measured braking/integrity/radio, power interruption, closed-track recovery and maintenance drills before profile approval

`station_step(Ready, fresh berthed/secured arrival)` currently remains Ready; the host targets the final stop. Nominal arrival therefore does not demonstrate another passenger exchange or a multi-station mission. The bounded two-train fixture retains geometry checks; expanding it cannot substitute for calibrated city conversion. Civil closures must bind precise operational resources and inspected handback separately from ERP repair completion. No duplicated ERP/GUI or

comfort software is implied by protection redundancy. Commodity hosts, the existing small Rust evaluators, reusable local controllers and open replacement/ calibration procedures remain the implementation approach. Hardware quantities and hot-standby/two-MCU mappings must be reconciled at the hardware freeze.

Evidence and staged delivery

The shared connected-engineering compiler carries part subassembly controller function train/infrastructure hazard control evidence operating response. Source changes stale execution hashes and propagate through configuration-bound claims and blocked release decisions. TLC checks a bounded resource abstraction with qualified physical proving assumptions; it is neither a Rust refinement proof nor an unbounded liveness result. G0–G4 decisions remain blocked pending review and measurements.

Owners/reviewers below are accountable roles to appoint, not fabricated named approvals. No package is accepted merely because a generated checkbox is true.

Package	Technical owner	Independent reviewer	Exit evidence / dependency
1 Architecture	Control architect	Independent railway safety engineer	Reviewed responsibility map; current baseline
2 Shared configuration	Configuration/lifecycle engineer	Independent configuration auditor	Identity/schema/mixed-version tests; 1
3 Resource/durability	Interlocking/consensus engineer	Independent formal/storage reviewer	Lifecycle, crash and restart evidence; 2
4 Authority/protection	ATP/localisation engineer	Independent braking engineer	Freshness/uncertainty/integrity properties; 2–3
5 Communications	Communications/security engineer	Independent security/timing reviewer	Replay/loss/restart/vendor-neutral transport tests; 2
6 Runtime	Embedded runtime engineer	Independent safety-I/O reviewer	Multi-process reference and physical power-loss tests; 3–5
7 Station	Station/energy integration engineer	Independent doors/charging reviewer	Complete station-cycle and per-car tests; 4–6
8 Twin/campaign	Simulation engineer	Independent test engineer	Reproducible raw results; starts after 2
9 FMEA/GSN	Assurance engineer	Independent hazard reviewer	Connected change impact and applicability; continuous from 1
10 HIL/hardware	Electronics integration engineer	Independent hardware assessor	Measured output/timing/fault records; 6–9
11 Closed track	Test/operations engineer	Independent field safety reviewer	Braking, integrity, recovery drills; 10
12 Profile release	Railway release authority	Independent assessor	Named gates, rollback and approved installed bundle; 9–11

Hardware preparation proceeds with software and assurance using existing `t-ecu-s`, `t-ecu-a`, `t-obs`, `w-sbc`, `s-sbc` families. Each pack needs exact module/SKU/substitutes, wiring/connectors/enclosure, power/thermal/environmental budgets, fail-safe outputs/watchdogs, calibration, controlled firmware, fixtures, bench records, and complete-unit replacement/recommissioning instructions. Existing COTS and custom-board release tracks both remain valid; safety-channel selection follows `control-electronics/safety-controller-selection.md`. This change supplies no invented SKUs, installed serials or physical test records.

Cost comparison uses equivalent service and availability, separately counting train controllers, static voters, non-voting sites, radios/antennas/power, retained detection, installation, spares/calibration/maintenance, integration, assessment, recovery/disruption and local/import content. No equipment removal or city-model savings is authorised without replacement function and evidence. The reference's ten processes are software roles, not a bill of materials. Cost quantities/prices remain deployment-specific inputs; an unknown input must keep the comparison incomplete. Track duplicate safety implementations, configuration variants, core dependencies and physical interfaces alongside money.

RFC 0034 — Connected station, production and battery model

Source: docs/rfcs/0034-connected-station-production-and-battery-model.md

Status: proposed planning design; 2026-10-06. The draft mentioned against `5f5d9fd` was not present in this checkout; this RFC reconciles the requested position with the current source rather than assuming that an unpublished text is authoritative. No vendor quotation, structural release, railway approval, construction contract or funding commitment is created.

The shared model joins station topology, civil quantities, supplier and logistics contracts, shift scheduling, scope reconciliation, battery profiles and construction fleet/crew contracts. Baghdad is a generated application of these shared contracts, not a separate set of design rules.

Related decisions: RFC 0010 stations, 0011 civil, 0021 battery traction, 0024 thermal, 0026 charging, 0028 construction QA, 0029 maintenance, 0030 manufacturing, and 0002 energy sizing.

Shared decisions

Subject	Proposed position	Release boundary
Ordinary elevated station	One central island, two boarding faces and two tracks	Site exceptions require a recorded reason; accessibility, fire, passenger-flow and structure qualification
Running spans	Existing single-track OSR-Pi20/Pi25 family	Prestressing and all permanent/construction stages require structural release
Erection	Whole-beam launcher on constrained continuous fronts; mobile cranes where access is advantageous	Separate equipment, lifting and closures approval
Segmental	Separate geometry/transport option	Match-casting, temporary suspension, joints, stressing and tendon protection; separate gantry and productivity
Baghdad	18 launchers, two separate crews/shifts per day	User-selected sensitivity; independent access and ready supports required
Precast supply	Existing qualified Iraqi factories plus necessary tooling/bed upgrades	Current contracted capacity, delivery and acceptance evidence; candidate capability grants no capacity
Batteries	LFP reference plus separate stationary/onboard sodium-ion study profiles	Cell/pack, electrical, thermal, life, software calibration and charging qualification
Equipment reuse	Individually identified transferable construction fleet	Compatibility, availability, shipping and recommissioning checks in each city

Every new input carries units, source, date, confidence and qualification. User-selected scenarios, calculated quantities, documented historical capability and supplier quotations are different evidence kinds. Stable input-content revisions and file hashes identify generated packages, including modified working-tree inputs, without depending on their containing commit. Unknown installed prices, reaction designs or spare capacities remain unknown; zero is reserved for a measured/declared absence.

Station and civil contract

Elevation is independent of archetype. `major` is a physical island, not two side platforms. Each platform, face, track direction, boarding datum and door/PSD side comes from one layout. Stacked interchanges have individual platform levels and explicit concourse transfer connections. CAD, IFC, assembly datums and the EBOM consume this contract. A physical platform count never substitutes for boarding faces or train berths.

The initial elevated island study has two lifts, two escalators, two lift shafts and two separated stair routes. It is not an approved universal minimum. Equipment has stable identities, served levels, envelopes, entrances and concourse connections. The layout rejects overlapping equipment and insufficient clear widths. Bidirectional flow, one-lift-unavailable operation, independent evacuation, assisted rescue, road barriers and additional street equipment remain separate assessments. Additional equipment/entrances can be supplied by site configuration.

Access CAPEX uses installed item quantities and associated works; energy, annual maintenance and replacement have separate outputs. Net island savings deduct wider deck, changed supports and spreading approaches. Unknown structural costs prevent a net saving claim.

Station decks, widened pier caps and foundations never inherit running-viaduct width. Approach offsets use a quintic transition and a curvature screen, followed by alignment/cant/gradient qualification. Construction requires wider launcher support and clearance checks, delivery access, released supports and an erection sequence. Changed entrances have geographic coordinates and trigger a coverage review; existing centroid coverage cannot be certified for newly proposed entrances.

Manufacturing, suppliers and erection

Pi-beam specifications record concrete, steel, prestressing and inserts; manufactured study mass and centre of gravity; lifting/storage/transport positions; strength gates; bearing/restraint interfaces; tolerances; and acceptance records. The 60/75-tonne figures are bare sections. Study suspended load includes local diaphragms, reinforcement, prestressing, embedded items and lifting gear. Equipment approval fails until the arrangement and strength/tolerance records are released. Launcher support reactions, asymmetric placement torsion and delivery-vehicle deck loads have separate construction-stage screening outputs.

The capacity register records products, prestressing capability, beds, moulds, handling dimensions/mass, demonstrated cycles, total versus contracted output, commitments, storage/dispatch, prices, terms, qualification and upgrades. Iraq already has many precast facilities and production expertise; this is the user-provided sourcing basis. No named facility or unverified spare capacity is adopted. Each factory must enter the shared register with product-specific evidence and contracted capacity before it can supply the programme. Slabs, walkways, ducts and suitable pier parts can be allocated separately; aggregate allocations cannot exceed a product's contracted capacity.

The shared chronological supply chain now feeds contracted completions, acceptance holds, manufacturing rejects, dispatch and transport directly into erection. Factory buffers count both awaiting-acceptance and accepted stock. Full buffers pause casting without banking unused whole production capacity; rejected or damaged beams create replacement demand while installed quantities remain fixed. Shared transport fleets cannot be multiplied across routes. In-transit loads reserve front storage and receive credit only after arrival; an available qualified alternative route can serve a blocked primary route.

Delivery routes constrain journey/window/loading/unloading times, trailers, payload/bridge limits, long-load/turning access, accepted/rejected components, order and buffer space, plus alternative routes/suppliers. Unknown/rejected route approval supplies zero output. Front start/end chainage, direction, launcher, delivery access, support readiness, interruptions, start/finish/relocation dates and work intervals are explicit. “Two per line” initializes the scenario; assignments may change with lengths/readiness. Shared obstructed routes and sequential erection paths cannot receive simultaneous output credit.

The simulator counts integer beams and complete paired-track bays. Daily production is productive shift time divided by a complete bay cycle, with handovers, maintenance, weather, windows, ramp-up, station interruptions and relocations. Extra launchers on one sequential front are spares, not extra independent paths. Casting, acceptance, transport and released supports cap progress. Each front reports the model days limited by each resource, and the package separately records production, acceptance, shipment, arrival and rejection totals. The conditional pool describes a capacity requirement and requires explicit study opt-in; it is not inserted into the real supplier register. The shared `precast-logistics.json` register contains actual route evidence when supplied. Fractional 1.5/1.8/2.0 bays/launcher/day are averages, implying 54/64.8/72 beams/day across 18 machines. Two shifts do not automatically double output. A conditional forecast assumes the necessary supply; the evidence-backed forecast remains unreleased when no suppliers/contracts/supports are qualified.

Costs, equipment and people

Make/buy separates delivered purchases, project tooling, upgrades/beds, storage/dispatch and new factories. Supplier amortisation and project tooling cannot both be charged. Rolling-stock assembly and electronics factory allowances remain until their scope is reconciled; replacing civil supply is not permission to delete them.

The civil calibration is edited and the cost file regenerated. Supplied products, erection equipment, transport, temporary works, labour/supervision, night shifts, inspection, maintenance/relocation, traffic management and supports have explicit study allocations. Station exceptions and special structures stay separate. The $9m central purchase allowance for 18 whole-beam launchers remains unquoted, with transporter, lifting gear, freight/duties, assembly, commissioning, supports, spares, crews, relocation and station-modification exclusions. It is not a segmental-gantry quote. The installed civil rate's embedded erection allowance remains unverified: purchase expenditure creates no additional claimed saving until an explicit scope-matched removal is justified.

ERP drafts identify each launcher, transporter and lifting frame as an Asset, with purchase/commissioning records, operating configurations, inspection/maintenance, spares, competence, location, availability, allocation and dismantle/ship/recommission tasks. Cash purchase, project cost allocation and residual value stay separate; initial Baghdad cash is not eliminated by reuse. Transfers cannot overlap allocations and require subsequent-city compatibility.

Department unit crew qualified worker task is a validated chain. Each front and shift requires supervision/lifting, operators, riggers/signalling, survey/QA, mechanical/electrical maintenance, delivery coordination and factory acceptance. Workers cannot fill both shifts concurrently. Handovers, relief/leave coverage, competence expiry, supervised commissioning and equipment-specific assessment are explicit; unfilled drafts do not grant permission to work.

Battery and chronological energy contract

Profiles specify supplier/cell identity and evidence revision, gross/usable kWh, complete mass/volume, electrical configuration/range, temperature/SOC/age-dependent charge and discharge limits, efficiency/cooling, life, installed/replacement price and qualification. Catalogue sodium values are illustrative pack sensitivities, not asserted vendor performance. Vehicle tare changes by the difference from the included LFP pack; that mass propagates to traction energy and costs. Cases hold usable capacity and service constant: LFP throughout, sodium stationary/LFP onboard, sodium both. The small onboard pack remains a separate duty-cycle option.

The network duty uses declared fleets, dispatch termini, headways, station distances, dwell and installed site capacities, with an explicit average-speed timing assumption and stock-readiness checks. Minute chronology allocates concurrent arrivals across shared cabinet power, subtracts setup time, applies charge taper/temperature/SoH, accounts for regen acceptance, seasonal solar/cleaning, hot-weather auxiliaries, missed visits, outages, reserves, degradation and replacement. Report minimum SOC, unserved duties and missed charges, not annual totals alone. The study needs a calibrated timetable and full seasonal chronological validation before service acceptance.

`osr-bms` requires an explicit commissioned profile with matching pack, chemistry, evidence revision, qualification and valid calibration. A chemistry-string change cannot select or derive sodium protection from LFP defaults. Existing trip, latch, cooldown, off-gas/fire isolation and derating behaviour are preserved and shared by the validated evaluator. Planning profiles are not BMS commissioning profiles.

Regeneration and acceptance

The outstanding review is traced in `remaining-work.json`, covering all 20 software, engineering, commercial, workforce, deployment and governance areas. Source-hashed records may be entered through `connected-delivery-evidence.json`; a receipt is not authority authentication or acceptance. No supplier, appointment or qualification is invented. The user-provided Iraqi precast capability basis remains qualitative, without selecting a named facility.

`beam-stage-assurance.json` adds gross-section properties, uniform-selfweight moments, lift/store/transport support reactions, sling forces and launcher-position envelopes. Geometry comes from the canonical Pi family. These are statics screens with explicit mass/support/dynamic assumptions. Prestress, reinforcement, connections, material/release strength, fatigue/vibration, anchors, bearings, stability and manufacturer load charts remain required for an independently checked design.

`station-capacity.json` calculates the union of shared-layout equipment envelopes on each platform and its remaining area and contiguous lane. Lift-out capacity and circulation/density comparisons use source-entered criteria; an empty register yields unknown throughput. Multilevel demand allocation, street barriers, surveyed entrance walking routes, evacuation/rescue and station/launcher structures require site assessment.

`procurement-evidence.json` separates dated preliminary, firm and contracted quotations from factory capacity and purchase commitment. Its catalogue beam requirements reference identified span components; special spans remain outside ordinary orders. Tooling already amortised in a unit price cannot be added as project tooling. The USD 9m allowance remains purchase cash only. `cashflow-sensitivities.json` links those identified payments to each conditional schedule at assumed 4/8/12% simple ACT/365 interest with full borrowing of the identified cash. Unknown costs, scope-removal credit, complete financing and accepted revenue remain null, so these sensitivities cannot be published as complete capital or net savings.

The generic front splitter can reassign the available radial donors in another city without requiring six donors or fabricating 18 active fronts. Configured spare launchers add no independent output. Other-city runs remain independent, unqualified planning scenarios; transfer from Baghdad is not authorised before its critical path and compatibility/recommissioning are accepted.

ERP execution records now bind lifecycle release to the actual project/front. Required transfers need transfer, compatibility and recommissioning evidence. Actual work start/close timestamps are controlled; active work cannot be reopened to hide an allocation, and cancellation requires reviewed handback. Completed/cancelled work history continues to participate in worker-rest checks. These business controls grant no railway engineering acceptance.

The identified span register partitions running intervals at millimetre precision into proposed 20 m/25 m catalogue spans and explicitly unresolved special spans. Ordinary beam requirements have stable component and span IDs; special spans receive no ordinary-beam order or invented design. Proposed supports require survey and structural acceptance. Disconnected sections incur a stated relocation allowance instead of receiving continuous launcher progress. Fixed allocation and a conditional six-machine transfer to additional ring fronts use the same 18 launchers, contracted-throughput requirement and 72-transporter fleet. Added ring access and support readiness are assumptions, not feasibility evidence.

Ring stock serves both headings at the declared dispatch stations, including controlled wrap distances in reverse. The energy study continues through three service days without resetting train or storage state. A declared study strategy permits grid replenishment within each site's remaining import capacity. Native section timing comes from `osr-sim`'s scenario loader and rest-to-rest movement model. The fixed-assignment diagnostic and coupled dispatch controller are separate: the controller keeps an energy-held train at its actual station, continues charging and removes it from subsequent dispatch availability until recovery. It reports distinct held journeys, hold minutes and missed opportunities separately. Minute discretisation, berth/depot access, transient traction, conflict authorities and field calibration remain open; no service or chemistry acceptance follows from these screens.

Explicit working calendars now gate erection, casting, acceptance, deliveries and foundation release on closed days. Holidays, dated maintenance, available windows and permission for the erection night shift are configurable. A six-day sensitivity retains the same fleet and installed quantities. Daily supplier-specific shifts, permitted night work and named relief rosters need actual evidence; no statutory holiday list or permission is invented.

Native simulator commissioning uses `[[battery_commissioning]]` records, each naming train, profile ID, pack identity, chemistry, evidence revision, nominal pack voltage, model capacity and a complete `CommissioningProfile`. Every train must be covered if any binding is supplied; duplicate packs, mismatched chemistry/capacity and unqualified profiles fail. The onboard evaluator uses the selected immutable calibration, cell count and voltage and exposes identity in its report. Legacy absent bindings remain labelled by zero commissioned trains and are synthetic planning shadows. Catalogue chemistry profiles contain no accepted commissioning records.

ERP native Task/ToDo transitions now require a submitted `OSR Construction Release`, reviewed by a user with HR and project authority roles. Evidence is private, task-bound and hashed. Controlled Employee authorisations and Asset lifecycle records require appropriate HR/project roles to change and are re-read under resource locks. Work checks active company/department identity, unrevoked competence, asset configuration/commissioning, inspection/maintenance coverage, unresolved repairs, native shifts, approved leave and holiday lists, reviewed night permission, relief/handover, overlap and rest across completed shifts. Completing work additionally requires a reviewed completion record. Revoked reviews and changed attachments cannot grant work. Installation and local tests establish code paths, not a deployed workforce, actual competence or physical railway release.

Station-specific assignment accepts sourced OD cohorts with declared station legs, headings, transfers and a peak window. It counts each paid journey once, additional transfer boardings separately, section loads and station waiting accumulation against that window's declared service. Empty inputs produce unknown demand. The common 3,000 pax/h input is a separately labelled stress case; measured lift-out throughput, circulation, rescue and entrance-based catchments remain evidence gates.

The Iraq national brief and funding document use one current country basis. The USD 8.312bn `local_positive` comparator is distinct from the unpriced accelerated case and the historical USD 7.581bn funding calculation. No debt-clearance month, national budget or realised saving is claimed for the current case. Existing Iraqi precast facilities and expertise are acknowledged without selecting a named facility or assuming spare contracted capacity.

Run `.venv/bin/python tools/automation/connected-build-study.py --city baghdad --refresh-cad` and `--check`. Regenerate station kits, native FreeCAD station libraries and IFCs with their established generators, and generate `civil-cost-model.toml` from calibration. The connected package contains quantities, fronts, resource and procurement/payment sensitivities, conditional opening dates, battery/operating/replacement costs, financing gaps, proposed entrance geography, ERP asset/crew drafts and a hashed source/output manifest. Prior adopted cost/finance/coverage packages remain identified comparators while unresolved supplier, structural and access inputs prevent promotion to an accepted budget or programme.

Regression gates: one island/two faces; access equipment matches geometry; shifts change resources not physical quantities; no erection beyond accepted/delivered beams or supports; contracted supplier limits; mutually exclusive access and asset transfers; battery mass/energy/cost propagation; and source/assumption provenance. The implementation sequence is station/civil topology, suppliers/logistics/shifts, costs/reuse, batteries and city regeneration.

Distributed Onboard Control Profile

Source: docs/certification/distributed-onboard-control-profile.md

Status: development target; not the deployable pilot baseline.

This profile reduces dependence on centralised wayside computation and permits continued route selection during a communications outage. It does not remove the safety functions commonly implemented by interlocking, ATS or ATP. Those functions may move between onboard and distributed equipment only after a hazard analysis, independent assessment and deployment approval demonstrate equivalent or better safety.

The deployable reference remains the conservative sectional-authority profile in pilot-signalling-profile.md.

Executable train-centred milestone

RFC 0033 and the TACS assurance package use the existing committed interlocking MA and ATP/ATO/brake components through thin, authenticated process hosts. The reference includes A/B agents for each of two trains, three static voters, point I/O and two station/charging interfaces. The development default requires both protection channels to permit and either to trip; loss stops the affected function. Train logical pairs are executed; point/crossing/door/charger/obstacle pairs and physical independence remain open. Raft replication cannot substitute for safety outputs. Generated software evidence remains unreviewed. HIL, measured braking/integrity, field recovery, independent assessment and a named approved installed bundle are required to change the deployable profile. The conservative pilot remains the default.

Three independent route inputs

The onboard selector compares:

1. a signed, versioned plan stored onboard;
2. route and position inferred from onboard sensors against immutable topology;
3. an authenticated network command from operations control.

An input is eligible only while trusted, fresh and internally valid. Exactly matching route, next-section and plan-epoch values from two eligible sources are required. If all three disagree, only one source is eligible, or the two available sources disagree, the result is `Hold`. A network outage can produce `OnboardAutonomous` only when the onboard plan and sensor-derived input agree.

Route selection is not permission to move. Every accepted selection carries an explicit `movement_authority_required` condition and is passed through the independent authority, localisation, speed-envelope and emergency-brake path.

Functions that must remain

Any implementation must retain, even if equipment placement changes:

- movement-authority generation and expiry;
- train separation and opposing-move exclusion;
- route locking and detected/locked point position;
- occupancy or safety-integrity localisation;
- overspeed supervision and emergency braking;
- intrusion/obstacle gates required by the operating envelope;
- degraded-mode supervision, event recording and accountable override.

“Eliminating interlocking/ATS/ATP” is therefore interpreted as reducing duplicated central equipment, not deleting their safety outcomes. A model, sensor fusion algorithm or radio link cannot independently claim equivalence.

Narrow single-track operation

Short single-track sections are permitted only on exclusive guideway and with passing loops at passenger stops or protected portals. Entry requires locked route exclusivity, detected points, clear occupancy, a valid authority and a proved timetable/degraded-mode strategy. Loss of required state prevents entry; it does not rely on two trains negotiating informally over the radio.

Recovery sites and shunt robots

The reference planner reserves a recovery siding at or immediately beyond every third passenger station. Each site has a protected mainline interface, independent stop detection and remote isolation. Its small shunt robot is remote-controlled, has local emergency stop and fail-held braking, and cannot autonomously enter the main line. Radio loss inhibits traction. Mainline entry, towing and return to service remain controlled recovery operations.

Evidence gates

Before this profile can replace any pilot equipment, the project must freeze the safety allocation and demonstrate source independence/common-cause analysis, target-hardware timing, localisation accuracy, radio-loss behaviour, point and occupancy interfaces, cybersecurity, HIL fault injection, shadow running, representative field trials, recovery drills and independent safety assessment. The deterministic report records design coherence; it does not close these physical or regulatory gates.

Rolling-stock top-down / bottom-up design system

Source: docs/rolling-stock/design-system.md

This is the design authority layer above the parametric CAD. It gives the project a systemic way to move from whole-train intent down to bought-in parts, fabricated parts, subassemblies, final assemblies, and then back up through scoring and evidence.

The rule of thumb is simple:

1. Start top-down with requirements: platform fit, axle load, range, traction margin, thermal margin, cost, mass, manufacturability, and safety evidence.
2. Decompose into the owned design hierarchy: external components, fabricated parts, subassemblies, final assemblies.
3. Iterate candidate definitions automatically inside an explicit design space.
4. Promote only feasible candidates into FreeCAD/FEM regeneration.
5. Freeze only when the requirements scorecard, supplier evidence, CAD, FEM, BOM, and manufacturing plan agree.

Hierarchy

Layer	Meaning	Examples	Source of truth
Top-level requirements	What the train must achieve	platform margin, range, axle load, cost, mass, traction/HVAC margin	`osr_mech.design_definition.REQUIREMENTS`
External components	Bought-in or BID/SOURCE modules	traction motors, HVAC, batteries, doors, glazing, sensors, couplers	`ExternalComponent` records plus supplier evidence
Fabricated parts	Parts OSR owns and regional workshops make	shell, bogie frame, motor cradle, cowl rings, articulation adapters	`FabricatedPart` records plus CAD templates
Subassemblies	Integration units that can be inspected	car module, powered bogie, trailer bogie, cowl, articulation, roof services	`Subassembly` records plus FreeCAD review groups
Final assemblies	Released trainset configurations	promoted light-metro 3-car trainset	`FinalAssembly` records plus generated CAD/FEM/BOM

Automated iteration

The first implementation is deterministic and dependency-free:

```
tools/automation/design-iterate.sh
```

It exhausts the declared discrete design space for the default `light-metro-3car` family, evaluates every candidate, and writes:

- `design/component-catalogue/catalog/design-system/design-iteration.json`
- `design/component-catalogue/catalog/design-system/design-iteration-summary.md`

To turn the winning candidate into a buildable product tree and review the current CAD/BOM baseline against it:

```
tools/automation/buildable-trainset.sh
```

That writes the buildable manifest and buildability review under `design/component-catalogue/catalog/buildable-trainset/`.

To run a bounded exploratory pass:

```
tools/automation/design-iterate.sh --family metro-4car --max-iterations 250
```

The current scorecard is intentionally planning-grade. It is good enough to rank candidate architecture choices before spending FreeCAD/FEM time, but it is not a certification result. As the project matures, replace or augment the simple metrics with:

- FreeCAD-derived mass properties and package clearances;
- CalculiX margins from `tools/automation/freecad-generate.sh --fem`;
- supplier quotations and evidence status;
- lifecycle energy and maintenance models;
- risk penalties from the hazard log and compliance matrix.

Promotion loop

Use the generated best candidate as a design review input, then run:

```
tools/automation/freecad-generate.sh --models --assemblies --fem
tools/automation/freecad-generate.sh --screenshots --station-scenes
```

If CAD/FEM/supplier evidence rejects the candidate, update the component catalogue or constraints in `osr_mech.design_definition`, rerun `tools/automation/design-iterate.sh`, and promote the next best feasible candidate. That is the closed loop: requirements drive definitions, definitions drive CAD/FEM, evidence updates definitions.

RFC 0011 — Civil Infrastructure Design Standard

Source: docs/rfcs/0011-civil-infrastructure-design-standard.md

Status: Draft — planning only, no structural drawings ship with this RFC Date: 2026-04-22 Updated: 2026-08-17 — OSR Rapid Viaduct Kit correction Depends on: RFC 0008 Rolling-Stock Reference Design, RFC 0009 Track Design Standard, RFC 0010 Station Design Standard

1. Summary

OpenSourceRail commits to two civil classes only — at-grade and elevated — with bridges for water crossings as a narrow third class. No tunnels. Every corridor the auto-gen pipeline produces, and every hand-authored corridor the operator commits, will sit on one of:

Class	Share (target)	Description
at-grade	≥ 70 %	Track sits on prepared subgrade at natural ground level (possibly with minor cut/fill ≤ 3 m). The cheapest and fastest class by 1–2 orders of magnitude.
elevated	≤ 25 %	Track sits on a viaduct structure 5–12 m above ground (precast OSR-Pi20/Pi25 decked beams, standardised span).
bridge	≤ 5 %	Over a water crossing or deep gap. Separately engineered special span with site-specific substructure.

The fourth class the first-pass auto-gen inference used — tunnel (cut-and-cover and bored) — is removed from the catalogue. Where an alignment previously would have gone underground, it now goes elevated over the same footprint, or deviates around the constrained zone on at-grade. Operators who deliberately want a tunnel are forking the project; OpenSourceRail upstream does not carry tunnel designs, tunnel cost models, or tunnel-specific signalling / ventilation / evacuation engineering.

2. Non-goals

- Not a tunnel standard. By construction. The project's cost model, the simulator's energy budget, the sensor-tuning for GNSS + balise, and the operations rulebook (RFC 0013) are all calibrated against above-ground operation. Going underground reopens every one of those calibrations.
- Not a geotechnical rulebook. Soil survey, pile type, foundation depth, seismic detailing — owned by the deploying operator's civil team. This RFC fixes the functional envelope (span range, clearance, load class) not the method.
- Not a bridge design manual. Spans above 30 m, water crossings, turnouts, and exceptional crossings use a separately engineered segmental, I-girder, or steel-composite product. They are not a stretched version of the standard full-span trough.
- Not a land-acquisition policy. How a deployment acquires right-of-way (ROW) — expropriation, purchase, existing-rail corridor reuse, new-build — is municipal work.
- Not a standards body. We reference EN 1990 (Eurocode basis), EN 1991-2 (traffic loads on bridges / viaducts), EN 1998 (seismic), UIC 777-2 (infrastructure gauge), local building codes where deployed. We do not publish new ones.

3. Why no tunnels

A planning-grade cost comparison for a 1 km segment in typical dense-urban conditions across the target regions:

Class	OSR target / retained benchmark (USD/km)	Build time	Post-build maintenance
At-grade dedicated ROW	2.584 M / 3.0 M	6–12 months	Low — ballastless slab/embedded trackform
Elevated viaduct	9.748 M / 12.0 M	12–18 months	Medium — bearing + expansion-joint inspection
Bridge over water	18.0 M / 18.0 M	18–24 months	Medium — same as viaduct + scour inspection
Cut-and-cover tunnel	60–120 M conventional	30–48 months	High — ventilation + pumping + egress drills
Bored tunnel	90–200 M conventional	48–72 months	High — same as above, plus tunnel-boring-machine commissioning

Numbers are planning-grade, calibrated against generated deployment instances including Samawah context (RFC 0003 §2.3).

A tunnel segment is 10–40× the CAPEX and 4–8× the build-time of an equivalent at-grade segment. It also brings a permanent operational overhead: mechanical ventilation, flood pumping, fire suppression, emergency evacuation plans, and radio-coverage retrofit. Every one of those is an additional subsystem the deploying operator must staff and maintain.

The mission is "urban rail affordable enough that a developing nation can finance it domestically." A corridor with any tunnelling in it is outside that envelope. The design decision is: if the alignment can't go at-grade or elevated, the alignment should deviate around the constraint. Even at a 20 % longer corridor, the detour is cheaper than the tunnel it avoids.

4. At-grade — the 70 % case

4.1 Envelope

- Subgrade: compacted fill or in-situ soil graded to +/-25 mm of design. Capping layer of 150 mm crushed stone, then ballastless reinforced slab / embedded direct-fixation plinths for the running rails. The default urban trackform avoids ballast migration, tamping cycles, and loose aggregate at crossings.
- Earthworks: cut / fill up to 3 m depth only. Beyond 3 m the segment is either:
  - Re-classified to elevated (bridge over the gap / viaduct over the hill), or
  - Deviated around the constraint.
- Drainage: transverse slope >= 2 % either side; longitudinal channels every 50 m; culverts at every flow-path crossing. Drainage is not optional: a flooded slab undermines fastener integrity, corrodes embedded hardware, and damages the subgrade.
- Fencing: dedicated ROW is fenced the full length on both sides. No uncontrolled pedestrian crossings. Level crossings only where a city street intersects and an underpass / overpass is prohibitively expensive. See RFC 0012 for the level-crossing equipment standard.
- ROW width: minimum 10.5 m for double-track at-grade (two tracks × 3.5 m wide plus 1 m each side clearance plus fencing allowance); 7 m for single-track.

4.1.1 Narrow old-town single-track option

Some historic cores and market streets cannot fit a 10.5 m dedicated double-track reservation without demolition, but also do not justify viaduct intrusion. OpenSourceRail therefore permits a constrained single-track at-grade segment as an at-grade variant, not a fourth civil class.

Rules:

- Use only for short constrained sections where double track would require demolition or permanent removal of the street's essential pedestrian/service function.
- Fit the running way inside the 7 m single-track envelope, with protected pedestrian edges and no mixed running with road traffic.
- Provide double-track passing loops only at stations, portals, or immediately outside the constrained street.
- Interlocking treats the whole single-track section as one opposing-move exclusion zone unless a deployment-specific signalling design divides it into shorter protected blocks.
- Timetables must prove the desired headway through the single-track section before the option is accepted. At 3-minute peak headway, a single-track section longer than roughly 800-1,000 m will usually force either lower frequency, directional tidal operation, or a bypass.
- Emergency walkways, evacuation gates, drainage, charging isolation, and intrusion detection remain mandatory; the narrow option saves width, not safety systems.

The auto-generation pipeline should prefer this option before elevated only when a tagged old-town / constrained-street corridor is short, station-loop spacing is feasible, and the resulting timetable still meets the line's published service class. Otherwise it should stay double-track or go elevated/deviate.

4.2 When at-grade is inappropriate

Even with the default bias to at-grade, certain conditions push toward elevated:

- The alignment crosses a river wider than 30 m bridge.
- The alignment crosses a highway or mainline rail at grade that cannot be broken by a level crossing (too many crossings per hour would make the crossing barrier close more than open) elevated.

- The alignment runs through a dense built-up core where ROW acquisition would require demolition of more than 3 buildings per 100 m of corridor elevated or deviate.
- The alignment runs along a street where at-grade light-rail running would conflict with road vehicles at every junction elevated or route to a parallel street.

The auto-gen pipeline's civil inference (§6) now maps these conditions to `Elevated` where it previously would have emitted `BoredTunnel`.

5. Elevated — the 25 % case

5.1 Envelope

- Structure: paired single-track prestressed decked pi-beams. OSR-Pi25 is the primary product and OSR-Pi20 is the closure/utility- avoidance product. Each beam is 2.9 m wide with stems beneath the rail lines; independent outer walkway/barrier cassettes produce an 8.5–9.0 m controlled twin-track envelope. The catalogue contains no 30 m Pi beam.
- Piers: single-column reinforced concrete, 1.5 m × 2.0 m rectangular, heights 5–12 m, with a 6.5–7.5 m hollow/precast-shell cap. Foundation type comes from the soil/access catalogue and actual deep-element lengths.
- Deck clearance: 5.0 m vertical under the girder soffit over any road; 5.5 m over a truck route; 7.5 m over a rail corridor (UIC 777-2 minimum).
- Pier spacing: 25 m nominal, using 20 m closure bays where constrained. Spans above 30 m use OSR-SP or another separately engineered structure.
- Bearings: scheduled fixed/guided/free bearings. A semi-continuous internal pier has four bearings in one longitudinal line; an expansion-unit boundary has eight in two independent lines; an end support has four. Permanent checked jacking interfaces permit replacement.
- Movement: four normal spans form a short semi-continuous unit using a released reinforced diaphragm or link slab. Expansion gaps occur every 80–125 m, never as one kilometre-long continuous structure. The project issues a bearing/movement schedule and continuous-welded-rail interaction, braking, temperature, seismic and foundation-flexibility analyses.
- Parapets: 1.4 m solid concrete panel, continuous, with a handrail on the inspection walkway.
- Walkway: separate outer inspection/escape cassette, 1.0 m clear, escape ramps every 250 m down to at-grade.
- Acoustic screens: fit only where the project noise model demonstrates a requirement; use replaceable secondary panels rather than widening the primary member by default.
- PV integration: concentrate PV at stations, depots and canopies. Normal viaduct spans do not carry a default PV attachment.

5.2 Rationale for one reference girder

- One long-line mould family across the whole project. The precast yard casts one Pi-beam section with movable Pi20/Pi25 stop ends; every normal span of an OpenSourceRail deployment uses that shape. This is the biggest single CAPEX / schedule lever in elevated construction and matches the "one spares pool, one CAD reuse" principle from RFC 0008 / 0009 / 0010.
- Double-track substructure, repeated single-track beams. One shared hollow/precast-shell cap carries two identical Pi-beams at 3.5 m centres. A single-track initial line still builds the shared substructure and may defer the second girder until capacity demands it.
- Short semi-continuity, not kilometre-long continuity. Transport and erect ordinary individual beams, then connect four spans with a released link slab or diaphragm. This removes most deck gaps and one bearing line at ordinary internal supports while retaining manageable thermal units.

6. Bridge — the 5 % case

- Reference class: separately engineered OSR-SP segmental, I-girder, or steel-composite span. The standard full-span U catalogue ends at 30 m.
- Pier: when a pier has to sit in the water, foundation is a reinforced-concrete caisson or a pile group with a protection cutwater. Scour assessment per EN 1991-2 §13 with a 100-year design flood.

- Navigation clearance: per the navigable waterway's published class. The operator commissions the bridge design and the standard OSR-U catalogue ends at the approach where the site demands a non-catalogue structure.
- Thermal: project-specific bearing, joint, and rail/bridge interaction design for the complete temperature, creep, shrinkage, and seismic range.

7. Stations-as-structures

Station archetypes from RFC 0010 map to civil class as follows:

Archetype	Typical civil class	Notes
`halt`	at-grade	No canopy PV. Single side platform on direct-ground footing.
`standard`	at-grade	Solar canopy + side platforms.
`major`	at-grade	Island platform, larger canopy.
`interchange`	elevated + at-grade	One line is on the viaduct, the other at-grade. The two platforms stack vertically at the junction. Never two elevated lines stacked — the second level's wind and seismic loads rule that out of the upstream catalogue.
`terminal`	at-grade	End-of-line; turnback tracks sit at grade.
`depot-terminal`	at-grade	Yard footprint is large; elevating it is prohibitively expensive.

Underground stations are not in the catalogue. The previous draft of RFC 0010 used the word "stacked" to describe `interchange` layouts; this RFC clarifies that stacked means vertical separation between at-grade and elevated levels at the junction, not between two underground levels.

8. Auto-gen pipeline — the civil inference, revised

The civil-class inference in `crates/osr-routing/src/civil.rs` currently classifies a cell based on the underlying cost surface. Under the no-tunnel invariant:

- Cells with cost ≤ 40 at-grade (unchanged).
- Cells with cost 40..100 at-grade (was already; unchanged).
- Cells with cost 100..400 (water) bridge (unchanged).
- Cells with cost ≥ 400 (building / dense built-up) elevated (was `BoredTunnel` now elevated).

This aligns with the rule: "if the corridor can't be at-grade, it goes above, not below."

Two downstream consequences, surfaced in `design-quality.yaml`:

1. Elevated fraction as a soft gate. If the share of elevated structure for a line exceeds 30 %, the quality file raises a warning. Designs with > 50 % elevated either need deviation or are inherently expensive and should prompt review.
2. No tunnel class. Quality output reports only at-grade, elevated, and bridge fractions.

9. Cost-anchor lookup

Total CAPEX drives the national-treasury conversation. The auto-gen pipeline emits a full `[costs]` block per city in `design.toml`, broken down by bucket below.

The source currency is USD because the marketplace and country-finance templates quote in USD. Generated `*_eur` fields are converted reporting views at 0.92 USD->EUR. Non-civil units and retained civil benchmarks live in `capex-costs.toml`. The active civil rates are generated from `civil-cost-calibration.toml` and CAD quantities into `civil-cost-model.toml`; this RFC keeps the narrative summary.

9.1 Civil works (USD/km × civil mix)

Class	Active USD planning target	Retained benchmark
at-grade	2 584 000 / route-km	3 000 000 / route-km
elevated	9 748 000 / route-km	12 000 000 / route-km
bridge	18 000 000 / route-km	18 000 000 / route-km
elevated-interchange premium	4 500 000 / site	—

9.2 Stations (RFC 0010 archetype catalogue)

Archetype	USD planning-grade
`halt`	600 000
`standard`	2 500 000
`major`	4 500 000
`terminal`	4 500 000
`depot-terminal`	5 000 000
`interchange`	8 000 000
`interchange-elevated`	12 000 000

The elevated-junction premium in §9.1 covers the viaduct + upper deck at interchanges; the table above covers the at-grade lower platform, ground-level pedestrian access, and protected crossing equipment where needed. Overbridges, lifts, and stairs are not the standard station assumption; they are carried only by elevated/stacked sites or local road-barrier overrides.

9.3 Depots (RFC 0014 archetype catalogue)

Archetype	USD planning-grade
`main-heavy`	8 000 000
`secondary-medium`	3 000 000
`layup-minimal`	400 000

These rates assume RFC 0014 distributed overnight stabling: healthy trainsets may sleep at powered passenger stations and layups, so the main-heavy no longer carries parking capacity for every train every night and `layup-minimal` is a powered-station stabling/isolation kit rather than a small standalone yard.

9.4 Rolling stock (RFC 0008 local-owner production × fleet count)

Rolling-stock city CAPEX uses a local-owner trainset-family planning unit, not an inflated per-car multiplier and not the raw marketplace BOM floor. The unit includes direct material, local assembly/labour, nominal per-train QA/acceptance evidence, and modest local handover logistics. Fixtures/tooling live in the railway production plant line; warranty, spares, and routine commissioning support are OPEX.

Family	USD / trainset local-owner unit
`urban-shuttle-1car`	280 000
`tram-2car`	560 000
`light-metro-3car`	900 000
`metro-4car`	1 120 000
`metro-6car`	1 680 000

Fleet count = peak-revenue + spare + cold-reserve per RFC 0014 §4.

9.5 Railway production plant

Each country carries one lean shared railway production-plant setup allowance for all city fleets it will build or assemble locally. Cities do not duplicate this asset in city CAPEX. The national factory is sized to the largest single-city fleet programme, costed per supported vehicle/car module, reused through a phased rollout, and kept separate from rolling-stock procurement.

Item	Planning-grade basis
Base local plant setup	60 000 USD / vehicle-car module
High sensitivity check	120 000 USD / vehicle-car module

The allowance covers reusable 1 m fiberglass panel moulds, clip/drill gauges, basic steel fixtures, plant services, commissioning bay setup, material handling, and production-readiness work. It excludes a full-length body mould and production adhesive cure hall. Six parallel two-person crews can install and release a three-car exterior body in one eight-hour shift after painted-frame dimensional release; doors, glazing, equipment, bogies, commissioning, homologation, and first-article testing remain separate work. Distributed overnight stabling reduces the number of depot-centred commissioning and parking bays needed in the production/depot complex. A three-module increment of supported factory capacity therefore carries 180 000 USD of base plant allowance, while the high sensitivity is 360 000 USD.

9.6 Systems

Item	Planning-grade basis
Residual train-control wayside (RFC 0015 GoA 4)	50 000 USD / route-km
Station/depot charging microgrid interfaces	120 000–1 000 000 USD / stop, by station archetype

RFC 0015 mandates battery-electric operation with station charging: there is no overhead catenary, third rail, feeder substation, or route traction-power budget. The energy infrastructure bucket covers the conductive chargers, local switchgear, inverter interface, and station/depot microgrid tie-in. PV/storage quantities are sized in the energy-site catalogue rather than as a continuous per-km rail-power system.

9.7 EPC overhead

Integration + project management is 7 % of the subtotal across §9.1 – §9.6.

Rates are deployment-localised via the `country-costs.toml` template (labour, materials, finance cost adjustments). The base rates above are what the upstream catalogue quotes; actual numbers in any given deployment can be ± 40 %.

`total_usd` is the procurement-basis headline per city. `total_eur` is also emitted for compatibility with earlier generated schemas.

10. Constructability constraints

- Concrete supply. Precast concrete yards exist in every target region; the Pi section needs a supplier-designed railway prototype and a reusable long-line mould fabricated from established formwork systems (PERI, Meva, Doka). Operators without a nearby precast yard bootstrap one — a 5 km² yard with 20 formworks produces enough girders for 50 km of viaduct in 18 months.
- Steel reinforcement. All target regions have domestic rebar mills in the 12–40 mm range we use.
- Post-tensioning tendons. Commodity; local supply exists in every target region.
- Lifting and transport. Pi25 is 2.9 m wide and targets 65–75 t. The project must release the route, transporter, lifting points, portal/launcher or actual crane load chart, synchronized lift controls, ground pressure, wind limits and contingency landing method.
- Launching. Where full-span access is constrained, OSR-US uses match-cast 2.5–3.0 m segments, post-tensioning, and an overhead launcher with specialist geometry, epoxy-joint, tendon-grouting, and staged-load QA.

No restricted-export equipment. No bespoke machinery. The construction envelope matches what a mid-size regional civil contractor can mobilise.

11. Pitfalls and decisions

- No tunnels is the non-negotiable decision. An operator who believes their corridor absolutely requires a 3 km tunnel under a city centre has exactly two options: (a) deviate the alignment; (b) fork the project. Upstream stays on the at-grade / elevated / bridge envelope.
- Choose the civil mix from the actual corridor. Use at grade where land, access and geometry permit. Controlled central viaduct concepts trade higher civil costs for more direct routes and fewer street conflicts. Report the resulting quantities, property needs and whole-scope costs; the earlier 70% at-grade aspiration is retained as a comparison assumption.
- Use the controlled viaduct families. OSR-Pi20/Pi25 decked beams are the primary products; OSR-US match-cast segments and OSR-SP special crossings address project constraints. Product geometry, complete lift masses, load combinations and independent checking remain release gates.
- Special crossings are separate products. OSR-Pi25 is not stretched to 40 m. OSR hands off to OSR-SP or a local engineering tender at the approach.
- Noise mitigation is structural, not operational. Acoustic screens on viaducts > operational speed restrictions. A 2 m concrete panel costs almost nothing and avoids a permanent speed-limit loss. Operators sometimes skip this; upstream builds it in.
- Seismic detailing is per deployment. The reference viaduct envelope handles PGA ≤ 0.3 g; above that, the deployment's structural team ups the rebar + bearing spec per EN 1998. The shape stays the same.

12. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	`osr-routing::civil` updated to drop tunnel classes (in this RFC's commit)	v0
v2	Emit `[costs]` block (at_grade / elevated / bridge / total_eur) per city in the auto-gen output, using the §9 rate table (done 2026-04-22)	v1
v2.1	Extend `[costs]` to the full CAPEX stack — stations (RFC 0010 archetypes), depots (RFC 0014 archetypes), rolling stock (RFC 0008 families), GoA 4 onboard autonomy/residual wayside train control, station/depot charging microgrids, and EPC overhead. USD is the procurement basis and EUR values are explicit converted views.	v2
v2.2	Move the source cost basis to USD direct-procurement planning rates, add `*_usd` fields, keep `*_eur` mirrors, and drop EPC overhead to 7 % (done 2026-06-04).	v2.1
v2.3	Add the railway production-plant setup bucket, now 60 000 USD per vehicle/car module with 120 000 USD retained as a high sensitivity check after 1 m clip-on fiberglass body and distributed-stabling scope reductions (updated 2026-08-15).	v2.2
v2.4	Replace duplicated city production plants with one shared national factory sized to the largest city fleet programme; city CAPEX is zero for this bucket and national briefs carry the asset once. Add imported-value and local-capital financing boundaries.	v2.3
v3	Parametric OSR-Pi20/Pi25 decked beams, transportable OSR-ST6 panels, 3.5 m elevated track centres, 6.5–7.5 m hollow caps, soil/access foundation catalogue, geometry-derived quantities and resource-driven production equations are controlled under `design/component-catalogue/src/osr_mech/civil/`. Legacy U-troughs remain special products; OSR-US and OSR-SP cover exceptional geometry. The controlled design-basis, load, egress, movement, erection, first-article, foundation, production and quantity packages live under `docs/civil/`.	v0
v4	Surveyed civil design package for the first pilot line, generated from the current city model	v3, RFC 0003
v5	First-article viaduct erected for a deployment instance	v3

13. Relationship to existing work

- `lib/templates/structures.toml` — the Lego-block schema. v1 of this RFC removes the `cut-and-cover-tunnel` and `bored-tunnel` entries.
- `crates/osr-routing/src/civil.rs` — the inference function. v1 of this RFC collapses the top threshold to `Elevated` (was `BoredTunnel`) and removes the `CutAndCoverTunnel` and `BoredTunnel` variants from `CivilClass`.
- `crates/osr-design/src/emit.rs` — the quality YAML emitter. v1 stops counting tunnel length (always zero now) and adds the elevated-fraction soft gate.
- RFC 0009 §5 and RFC 0010 §13 reference tunnel alignment / stacked-underground interchange respectively — both amended in this commit to match the no-tunnel invariant.
- `lib/templates/platform-doors.toml` references "tunnel stations"; that line is updated to describe full-height PSDs as an optional upgrade for climate-sealed stations regardless of civil class.

14. Open questions

1. Grade-separated level crossings — a road underpass below an at-grade rail corridor counts as a structure; does it get its own civil class? v1 folds it into the road authority's scope, not rail's.
2. Cut-and-fill extent limits. 3 m is arbitrary. Real deployments in hilly terrain may need 5 m cuts. Revisit with a hilly-pilot case study.
3. Acoustic screen effectiveness in hot climates. Concrete screens expand/contract; joints need seasonal inspection. How does that feed the CBM schedule? Candidate `osr-cbm-*` extension.
4. Wooden / bamboo viaduct for ultra-low-cost deployments? Real regional-rail projects in parts of South and Southeast Asia have used timber structures historically. Out of scope upstream; a local fork can layer that on.
5. Green-roof viaducts — planting integrated into the deck for heat-island mitigation. Candidate v4 addendum.

15. Done criteria

- [x] Two classes only — at-grade + elevated — with bridges narrow (§1)
- [x] No-tunnel rationale explicit with cost comparison (§3)
- [x] At-grade envelope specified (§4)
- [x] Elevated envelope specified, one reference girder (§5)
- [x] Bridge class as a viaduct variant (§6)
- [x] Stations-as-structures mapping (§7)
- [x] Civil inference updated (§8)
- [x] Cost anchors named (§9)
- [x] Constructability constraints stated (§10)
- [x] Pitfalls + alternatives (§11)
- [x] Rollout ordered (§12)
- [x] Relationship to existing code + templates (§13)

The next session picks up at v2 — cost estimates in auto-gen output (a short follow-on to the v1 tunnel-removal already landed in this commit).

RFC 0014 — Depot Design Standard

Source: docs/rfcs/0014-depot-design-standard.md

Status: Draft — planning only, no site drawings ship with this RFC Date: 2026-04-22 Depends on: RFC 0008 Rolling-Stock Reference Design, RFC 0010 Station Design Standard, RFC 0011 Civil Infrastructure Design Standard, RFC 0013 Operations Rulebook

1. Summary

The current city catalogue adopts one full-fleet planning depot per line, sized to that line's complete stock and actual train length. Storage tracks and workload-sized maintenance bays are separate quantities. Depot PV/storage is priced once; property, access alignment, yard topology, installed quotations and morning launch capacity remain releases. The controlled generic quantities are in `city-operating-scope.toml` and each city's `engineering/line-depots/summary.json`. Baghdad retains its dedicated revised scope and financing ledger.

The distributed-stabling pattern below is retained as the earlier reference and simulation comparison. Its passenger-platform parking assumptions do not reduce the current full-fleet depot capital requirement.

1.1 Retained distributed-stabling reference

OpenSourceRail commits to three depot archetypes covering the full range of deployment sizes. Every deployment has exactly one main depot plus optional secondaries and layups. The main depot co-locates with a `depot-terminal` station (RFC 0010 §1).

The default operating pattern is distributed overnight stabling: clean, telemetry-healthy trainsets may finish service at powered passenger stations or layups near their first morning trip instead of all running empty to a terminal depot. This reduces depot stall requirements, avoids wasteful dead mileage, and lets first departures begin across the line at the same time. The main-heavy remains the maintenance authority. Defective trains are held and recovered to a service point on their own line; maintenance planning must not assume an interline rail connection. Two revenue trains per selected station form the morning launch stock; remaining revenue trains and reserves stable at a powered service point on their own line. Storage tracks are sized separately from workshop bays, using reusable depot/layup blocks.

Archetype	Catalogue max stalls	Heavy maintenance	Overhaul throughput	Notes
`main-heavy`	20	Yes — overhaul bay, wheelset lathe	≤ 40 trainsets	Larger workshops require a controlled site-specific extension
`secondary-medium`	12	Light only — brake disc, door actuator, HVAC modules	—	Second depot on a long line
`layup-minimal`	6	None — overnight stabling only	—	Remote-terminal layup

The catalogue max is the reference layout envelope. The per-deployment workshop requirement comes from §4 (`max(4, ceil(fleet × 0.15))`). It is not an overnight parking allowance. Requirements beyond that envelope need an explicit site design; a template ceiling must not silently truncate them.

Plus one optional type for edge cases:

Archetype	Purpose
`training-wing`	Co-located operations-training facility — OCC dispatcher-console simulators, maintenance-technician workshop mock-ups, and recovery-mode-crew briefing rooms. OSR is GoA 4 driverless (RFC 0015), so there are no revenue-service drivers to train — the wing exists for dispatchers, maintainers, station staff, and the recovery-mode crew who operate the manual-fallback joystick (RFC 0015 §8.2). Opt-in on `main-heavy`; not a standalone depot.

All depots are at-grade. Per RFC 0011 §1 depots are never elevated — the footprint is too large to economically put on viaduct. A deployment that can't site its depot at-grade has an alignment problem, not a depot problem.

2. Non-goals

- Not a workshop management system. How the depot schedules its work — CMMS, parts management, labour planning — is owned by the operator and overlays with the `osr-cbm-backend` output.
- Not an off-line overhaul facility. Major overhauls (wheel reprofiling at 150 000 km, bogie overhaul at 600 000 km, body overhaul at 10 years) happen in the `main-heavy` depot's overhaul bay. Deployments without a main-heavy either contract overhaul out or depot-trip their fleet to a sister network.
- Not a yard-design manual. Track yard geometry within the depot boundary uses RFC 0009's `standard-urban` or (for large shunting yards) a depot-specific geometry variant. Detailed yard layout per facility.

- Not an HR / workforce document. Shift patterns and headcount from the ops rulebook (RFC 0013) and the workforce template.
- Not a brownfield-rehabilitation scope. Where a deployment inherits an existing rail workshop (Samawah, Khartoum-Atbara, Karachi Cantt, Maputo Machava, etc.), RFC 0027 governs the conversion of the existing site into an OSR depot. This RFC's archetypes are the greenfield envelope; the brownfield path swaps greenfield depot CAPEX (use the current cost template and reconcile energy inclusions) for workshop rehabilitation (~$0.5–2 M building + $1–3 M new OSR-specific tooling) — an order-of-magnitude saving that the per-deployment plan should commit to whenever assets are real.

3. Why three archetypes

Depots are where the simplicity bet pays off most. The three- archetype set covers:

- `main-heavy` — the one non-negotiable facility per deployment, where heavy maintenance happens.
- `secondary-medium` — for deployments whose geography makes defect and light-maintenance access to the main depot impractical.
- `layup-minimal` — storage at a line's powered service location for stock beyond the two-train station provision; no separate heavy workshop or interline connection is implied.

The catalogue provides reference envelopes. A workshop exceeding the reference bay count needs a controlled extension at the deployment main depot, including access, equipment and staffing capacity.

4. Fleet-sizing formula

Historical distributed-stabling comparison. The workshop and overnight allocation formulas below belong to the earlier reference. Current full-fleet line depots use the workload, storage and price controls linked in §1; the current Samawah quantities are generated in RFC 0003.

Workshop bays are sized for concurrent maintenance and inspection. Two revenue sets stable at each selected powered passenger station where fleet permits; remaining revenue trains and reserves use separate depot stabling tracks:

```
main_depot_workshop_bays = max(4, ceil(total_fleet × 0.15))
secondary_depot_bays = site-specific exception
service_rotation = 0  # depot service is prohibited in peak windows
total_fleet = peak_revenue_trainsets + service_rotation + spare + cold_reserve
station_overnight_positions = 2 × unique_selected_station_count
depot_overnight_trainsets = total_fleet - station_overnight_trainsets
```

The two-train station provision is an overnight allocation constraint. Allocate remaining revenue trains and all reserves to same-line storage, preserving total fleet inventory. Depot storage positions and usable track lengths are sized separately from workshop bays. The earlier Samawah comparison allocated 40 trains to stations and 68 to storage on their own lines: 37 on Line 1, 16 on Line 2 and 15 on Line 3. These comprise 57 revenue, eight spare and three cold reserve trains. The storage requirement totals 4,046 m of usable slots at 59.5 m per train, separately from the existing 17 main-depot workshop bays. Lines 2 and 3 reuse their powered service terminals. Local storage layouts need detailed design; interline depot connections are not part of the operating requirement.

- `peak_revenue_trainsets` = what the schedule requires at peak. Sized from the physical round-trip cycle vs. the peak headway:

  ```
  traversal_energy_kwh = line_length_km × cars × 2.4 × 1.25 × 1.25
  charging_dwell_s = ceil_30s(3600 × traversal_energy_kwh / sum(charger_kw))
  charging_dwell_s = clamp(charging_dwell_s, 120, 600)
  added_dwell_min  = charging_stops × max(charging_dwell_s / 60 - 1, 0)
  stop_count       = station_count - 2  # radial; station_count for a ring
  one_way_min      = (line_length_km / cruise_speed_kmh) × 60
                     + stop_count × stop_penalty_min
  radial_cycle_min = 2 × one_way_min + 2 × 3 + 2 × added_dwell_min
  ring_cycle_min   = one_way_min + added_dwell_min
  peak_revenue_trainsets = max(2, ceil(cycle_min × recovery_factor / headway_min))
  ```

  Current native emitter parameters:

Family	Cruise km/h	Stop penalty min	Recovery factor	500 kW cabinets per powered station
tram-2car	48	1.45	1.10	1
light-metro-3car	62	1.35	1.62	1
metro-4car	72	1.30	1.30	3
metro-6car	72	1.30	1.45	4

  Radials use a three-minute peak headway and a three-minute turnback at each end; rings use a six-minute peak headway. The stop penalty includes the one-minute reference dwell, and only charging dwell above that reference is added. The energy balance uses a 25% climate uplift and a separate 25% charging margin. The larger recovery factors include the emitter's retained onboard-energy resilience allowance; they are planning assumptions requiring city-specific service and energy evidence, not a uniform 10% timetable margin.

These parameters replace the former written 35 km/h proxy and uniform 1.10 recovery factor. Existing city fleet inventories remain declared inputs; reconcile regeneration changes with timetable, energy, cost and physical stabling evidence before adopting a different count. This formula does not enforce two trainsets per station. Station capacity must be checked separately.

- `spare` = max(1, floor(peak_revenue_trainsets / 10)), for planned maintenance rotation.
- `cold_reserve` = 1 per line, for unplanned incidents per RFC 0013 §7.
- `service_rotation` = zero dedicated sets. Depot service is prohibited in the 3-minute peak windows and uses surplus peak-fleet capacity when the timetable relaxes to 6 or 12 minutes.

The 12-minute service happens at one designated service depot per line only in lower-frequency windows. During 07:00–09:00 and 15:00–17:00, trains make the normal quick terminal turnback, can use the platform connector, and may deplete the battery further while the 20% section-entry reserve remains mandatory. Interior cleaning, exterior/running-gear walk-around, door, coupler and emergency-equipment release checks, fault-log download, and a 150 kW low-C recharge run concurrently. A red defect removes the set from rotation and uses a planned spare; a clear inspection releases it straight back to revenue service. The opposite terminal may still top up or stable a train, but does not repeat the full inspection on the same round trip.

The 15% workshop concurrency allowance covers overhaul, defect repair, incoming inspection, and planned maintenance. Growth beyond the 20-bay catalogue envelope requires a controlled workshop extension; it does not restore fleet-wide depot parking.

Distributed overnight stabling does not increase the service-rotation fleet. End-of-service dispatch assigns healthy trainsets to powered stations with CCTV monitoring, remote isolation, and at least low-C charging access capped by installed equipment. The 150 kW rate is a top-up target, not a minimum station rating. Morning dispatch releases those sets from their stabled stations so outer stops do not wait for trains to run out from the terminal depot. The main-heavy stall formula remains sized for maintenance, inspection, overhaul, spares, and growth rather than for parking every healthy set every night.

Historical Samawah comparison (September 2026, 3-min peak headway):

- The earlier figures are retained in the historical Samawah reference. The current city summary adopts the full-fleet line-depot requirement.
- The September 2026 design has fleets of 53, 28 and 27 trainsets: 108 total. The depot reconciliation derives initial placements and slot-length requirements from the stored scenario. Physical overnight allocation remains unverified.
- `design.toml` emits the depot archetype, terminal assignment, and `fleet_stalls` as workshop/inspection bays from the same run. Older 1.25× fleet-parking examples and hand-calculated 6/4/8 trainset examples were removed from the docs and must not be copied into current deployment documents.

The emitter (RFC 0014 v2) computes this from the design.toml's `fleet_sizing` block.

5. `main-heavy` envelope

5.1 Functions

- Workshop holding for inspection, scheduled work and defective trainsets (bay count per §4); overnight fleet allocation requires a separate plan.
- Daily inspection of every incoming trainset.
- Weekly inspection (7-day service).
- 30-day inspection (A-class service per EN 50126).
- Light repair (door actuator, HVAC module, brake-disc replacement, battery module swap).
- Heavy overhaul: bogie swap, wheel reprofiling on depot lathe, body-work repaint.
- Wash: automated two-pass wash track.
- Battery-module inspection, coolant service, and sealed-module exchange.

5.2 Physical layout

Area	Target footprint (per consist bay)
Stabling track	1× consist length + 5 m clearance at each end
Daily-inspection bay	Roofed, pit-equipped, overhead power, compressed air, data sync drop (TCN-E monitoring)
Weekly bay	Pit + roof walkway for upper-body access
Overhaul bay	Pit + overhead crane (40 t) + wheel-lathe trench
Wash track	Two-pass automatic with water recycling
PV canopy	4 000 m² reference allowance; full operating PV inventory needs additional located area (§5.3)
Classroom + office	≥ 200 m² for operators + trainers
Stores	≥ 600 m² with rack storage for spare modules

Total `main-heavy` footprint: ~5 000 m² gross for a 10-stall facility, ~8 000 m² for a 20-stall. These historical workshop envelopes do not include the complete operating PV/storage inventory and cannot be used as total deployment land requirements.

5.3 Energy

- `depots.toml` selects the canonical `energy-sites.toml` tier. `main-heavy` uses `depot-main`: 5,000 kWp PV and 40,000 kWh storage, comprising 80 × 500 kWh stationary modules. Scenario, BOM and layout requirements resolve the same inventory; explicit site overrides must remain traceable. These are retained operating assumptions, not validated depot sizing. Reassess energy duties with distributed overnight station charging before freezing procurement or depot land requirements.
- The old 4,000 m² / 600 kWp / 2,000 kWh scope is a reference allowance. At the catalogue density of 0.15 kWp/m², 5,000 kWp requires 33,333 m² of modules, before access, setbacks and packing. Additional land/canopy and the battery compound are not yet located by the reference CAD.
- Annual PV generation must come from the city energy study with declared weather and loss assumptions. A nameplate rating alone is not annual yield.
- Stationary storage feeds the RFC 0021 DC charging system. Supplier dimensions, conversion/protection equipment, fire separation and installed cost require explicit reconciliation with the fixed depot allowance.
- Grid import/export must satisfy the declared connection limits and utility study. Per RFC 0002, every depot is one physical `osr-energy-site`, including when co-located with a passenger station.

5.4 Signalling

- A single W-SBC plus an interlocking zone boundary at the depot entry.
- Inside the depot, speed limit 15 km/h, manual operation under mode M3 (RFC 0013 §5).
- Shunting signals at each stall entry — mechanical dwarf signals driven by `osr-wayside-points` extensions.
- No ATP envelope inside the depot; `osr-atp` disables below 15 km/h in depot-tagged geofence, per a safety-bounded configuration. The operational safety envelope inside the depot is line-of-sight + hand-signal coordination.

5.5 Access

- Personnel access: turnstile-gated with ID-card entry, separate from the fleet access.
- Fleet access: one track connection to the line, TPS-style merge per RFC 0012 §4.3 (trains go in — it's a trailing move).
- Road access: for parts delivery, emergency services, depot staff parking. Minimum one 4 m-wide road connection.
- Rail access for deliveries of new trainsets is via the same line connection (new rolling stock rolls in on its own wheels).

6. `secondary-medium` envelope

- Functions: overnight stabling for ≤ 16 trainsets, daily inspection, weekly inspection, light module swap.
- No overhaul bay. No wheel lathe. No bogie-lift. Anything heavier is trips back to the main-heavy.
- Footprint: ~2 500 m² for 8 stalls.
- Reference PV canopy: 1,500 m² / 225 kWp, with 500 kWh baseline storage.

- Operating `depot-secondary` tier: 1,500 kWp PV and 5,000 kWh storage (10 × 500 kWh modules); additional placement and installed cost remain open.
- Signalling: same as main-heavy but smaller zone.
- Classroom: none. Reuse the nearest `major` station's community room if needed.

7. `layup-minimal` envelope

- Functions: overnight stabling, low-C top-up, cleaning, and a ground-level safety-release walk-around. Trainsets drive in at end of service, can pass through the same 12-minute turnaround service if the site is designated for its line, sit overnight, and drive out at morning service start. There is no pit inspection, repair work, or heavy maintenance.
- Footprint: ~1 000 m² for 4 stalls.
- PV canopy: 600 m² 90 kWp.
- Operating `depot-layup` tier: one 150 kWh stationary package for opportunity charging during dwell; the main-heavy is primary.
- Train top-up: 150 kW low-C per occupied stabling road, used while a set is held between duties and overnight; passenger-platform fast charging remains a separate terminal function.
- Signalling: single W-SBC at the layup entry; rest is manual line-of-sight.
- No personnel on-site overnight; the OCC monitors via CCTV.

8. Integration with the auto-gen pipeline

Today's emitter picks one `depot-terminal` archetype (`crates/osr-design/src/emit.rs`, `compute_archetypes`, priority rule). That choice is where the main-heavy sits. v2 of the emitter extends this:

1. Main depot: co-located with the longest radial line's far terminal. The station archetype is `depot-terminal`; the depot archetype is `main-heavy`.
2. Powered station stabling: every other terminal and selected intermediate powered stations receive healthy sets overnight; they do not emit depots.
3. Secondary/layup exception: emit only after a site-specific security, utility, access, or defect-recovery study shows station stabling is not viable.

Every depot writes a record into `design.toml`:

```
[[depots]]
station          = "line-a-east"
archetype        = "main-heavy"
fleet_stalls     = 4  # illustrative workshop bays; calculate per §4
energy_site_tier = "depot-main"  # optional: selected by the archetype
```

The scenario generator resolves depot energy tiers into `[[sites]]` and operating service settings into the simulator input. The simulator (`osr-sim`) models service countdowns and energy flows. A future physical distributed-stabling model must satisfy these requirements:

- Charges each trainset at depot pad power while held and overnight.
- Treat powered passenger stations and layups as valid overnight stabling points when the distributed-stabling policy is enabled in `lib/templates/depots.toml`.
- Holds a returning train for the configured clean/inspect/recharge slot at one designated depot per line and emits service-start/service-complete evidence.
- Preserves the zero dedicated service-rotation policy and distinguishes planned spares and cold reserve from revenue fleet requirements.
- Feeds the depot battery bank into the network-wide energy balance.

8.1 Executable station-stabling candidate

The simulator supports `station_stabling = true` in scenario `[[fleets]]`, with selected powered-station `dispatch_points`. Stations may use solar/storage without grid import. Waiting trains use up to 150 kW, capped by installed station and shared cabinet/source limits, for low-C top-up to 95% SoC; after closing, returning trains hold at selected stations and resume under schedule, energy and movement-authority gates. This does not require returning healthy trains to the main depot. Before departure, a distributed train must retain the 20% reserve over the route to the next selected charging station, including unpowered stops and terminal reversals. A failed pad or unavailable energy source excludes a destination. A grid outage alone does not disable a battery-backed charger. The gate checks current source availability; it does not reserve future energy against other arrivals. Actual charging debits shared storage, and each next departure rechecks the onboard reserve. Continuous replay verifies the duty. Optional `spare_count` and `cold_reserve_count` are included in each fleet's total inventory and default to zero. Candidates use the city's declared counts; these trains remain parked and can charge without routine dispatch.

The distributed-stabling workflow generates separate candidates for all catalogue cities and records the Samawah overnight comparison, checking each planned line/station/direction at opening. Plans compare allocated train lengths with the reference platform envelope. The planning/replay acceptance gate also limits overnight allocation to two trainsets per selected station, including reserves. Shared station IDs count once; a four-berth reference platform does not override this provision. The simulator exposes over-capacity queues for diagnosis, and the combined replay fails even when holding, charging and morning departures otherwise pass. Verified physical slots, security, inspection release, reserve activation, defective-train routing, full-day energy duty and deployment acceptance remain open.

9. Pitfalls and decisions

- At-grade only, per RFC 0011. Depots are large; elevating the footprint is prohibitively expensive. The auto-gen emitter rejects depot siting on an elevated section.
- One reference `main-heavy` everywhere. Operators with unique geography (coastal, mountainside) will push for bespoke depots. The upstream catalogue stays at one shape; the deployment's civil team handles the site-specific adaptation.
- Wheel lathe at the main-heavy only. Decentralising lathes is tempting (saves depot-trip time) but the operator depth for wheel-reprofiling is thin in target regions; centralising at one site lets that thin team do all the work.
- No pit at `layup-minimal`. A pit adds cost, safety maintenance, and lighting — just for overnight parking. Skip.
- Depot simulation in `osr-sim`. The sim models depot top-up while held plus an explicit service countdown at the designated depot. Cleaning, release inspection, diagnostics, and recharge are concurrent; deeper maintenance remains a separate availability/CMMS model.
- Distributed stabling is not distributed maintenance. Overnight passenger-station stabling is allowed only for healthy sets with remote isolation and CCTV. Any trainset needing inspection beyond the daily release walk-around, fault rectification, wheel work, or battery-module exchange goes to the main-heavy.

10. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	Shared energy tier, BOM quantities and layout requirements reconciled (2026-09-09); full equipment placement and installed cost remain open	v0
v2	Retained Rust seed and distributed-stabling studies; catalogue post-processing now prices one full-fleet depot per line	v0, RFC 0010 v2
v3	`osr-sim` turnaround-service state (clean / inspect / diagnostics / recharge), service countdown and event evidence (done 2026-08-12)	v2
v4	Accepted site/access/topology plans for all current Samawah line depots, with separately verified launch and maintenance capacity	Current full-fleet scope, v3
v5	Reference depot CAD under CERN-OHL-S v2	v4

11. Relationship to existing work

- `lib/templates/depots.toml` — schema this RFC ratifies.
- `crates/osr-design/src/emit.rs` — auto-gen emitter. `depot-terminal` detection today; the `[[depots]]` block emission is v2.
- `crates/osr-energy-site` — every depot is one `osr-energy-site` instance; existing integration holds.

- `crates/osr-sim` — modest extension for v3; charging semantics already work.
- RFC 0010 `depot-terminal` archetype pairs 1:1 with this RFC's `main-heavy` or a mix of main-heavy + layup.

12. Open questions

1. Co-located vs detached training wing. Some operators prefer the training simulator in a quiet city building (closer to drivers' homes). The catalogue default is co- located with the main depot; per-deployment override.
2. Decentralised battery swapping. A future opt-in where trainsets exchange their battery pack at depot instead of recharging — halves the charging-pad occupancy. Out of scope upstream until battery-swap mechanical designs mature.
3. Night-shift head count. Ops rulebook §4.5 (RFC 0013) fixes the shift structure; what's the minimum overnight staff headcount at a `layup-minimal`? Currently zero (CCTV-monitored); confirm when the first `layup-minimal` deploys.
4. Depot expansion path. A `secondary-medium` that outgrows its 8 stalls — does it upgrade to `main-heavy` in place, or is a second-depot-nearby preferred? Likely the latter given construction-on-live-facility complexity.

13. Done criteria

- [x] Three archetypes committed (§1)
- [x] Fleet-sizing formula (§4)
- [x] Main-heavy envelope (§5) + secondary-medium (§6) + layup-minimal (§7)
- [x] Auto-gen pipeline integration (§8)
- [x] Pitfalls (§9)
- [x] Rollout ordered (§10)
- [x] Relationship to existing software + templates (§11)

Next: locate the full PV/storage inventory, reconcile installed cost, and provide track-by-track overnight allocation and conflict-aware run-in/run-out evidence. The quantity contract alone does not satisfy those release gates.

RFC 0013 — Operations Rulebook

Source: docs/rfcs/0013-operations-rulebook.md

Status: Draft — planning only Date: 2026-04-22 Depends on: RFC 0008 Rolling-Stock Reference Design, RFC 0009 Track Design Standard, RFC 0010 Station Design Standard, RFC 0011 Civil Infrastructure Design Standard, RFC 0012 Switches & Crossings

1. Summary

Every OpenSourceRail deployment inherits one baseline operations rulebook — the standing rules for drivers, dispatchers, station staff, and maintenance workers. The rulebook is deliberately short (target ≤ 60 pages in the rendered PDF) and shared across deployments. Operators adapt it for national regulatory overlays but do not re-invent it.

This RFC fixes the rulebook's table of contents and the governing rule-granularity principle. The rule text itself lives in markdown under `docs/operations/` and is owned by a standing committee once the project has one; until then, the project maintainer ratifies additions.

2. Non-goals

- Not a national regulatory document. Every country that adopts OSR has a national rail safety authority (NRSA or equivalent) that must accept the rulebook. This RFC produces a baseline that a deployment's local legal counsel adapts.
- Not a driver training manual. Training content comes from the rulebook, but the pedagogy (lesson plans, sim scenarios, exam questions) is a per-deployment deliverable handled by workforce materials (`lib/templates/workforce.toml`).
- Not a safety-case evidence file. The rulebook is an operational artefact; the safety case (`docs/safety-case/`) references rules as supporting evidence, not the other way around.
- Not an incident-response playbook. Playbooks are scenario- specific and owned per-deployment (fires, passenger illness, derailment, signal failure); this RFC specifies the categories each playbook must exist for, not the specific response.

3. The simplicity principle

Every rule in the rulebook must pass this test:

> A new driver who has never seen this system before can read the whole rulebook in one morning and have an accurate enough mental model to operate safely under supervision by the end of the day.

This means:

- Every rule is one sentence, one decision. No compound rules.
- Every rule has exactly one author (the project maintainer until the standing committee exists).
- Every rule is numbered; operators cite rules by number in incident reports.
- Every rule has a why: a short justification paragraph. No rule is an unexplained ritual.

The target is a ≤ 60-page PDF. Rail industry rulebooks that hit 200 + pages are artefacts of 150 years of accretion — we're not accruing that.

4. Table of contents — one section per role

The rulebook is organised by role, not by scenario. A driver reads one section; a dispatcher reads another; the sections overlap only where the roles interact.

4.1 Dispatcher rulebook (target ≤ 15 pages)

Section	Scope
S1	Shift start — OCC login, fleet inventory check, schedule ready
S2	Routine dispatching — route grants, timetable adherence, headway maintenance
S3	Incident handling — single-train fault, section unavailable, mass-casualty
S4	Weather and environmental — dust storms, heatwave, flood warning
S5	Maintenance interlock — work-on-track grants, blocking a section from revenue
S6	Shift end — handover, outstanding-event log, system state summary
S7	Intrusion response (RFC 0016) — track-patrol dispatch, return-to-service gating, sustained-Unknown escalation

4.2 Station staff rulebook (target ≤ 10 pages)

Section	Scope
T1	Station opening — power up, PIS check, fare-gate test
T2	Passenger boarding — platform supervision, PSD status, wheelchair assist
T3	Passenger incidents — medical, fare dispute, unattended item
T4	Fare enforcement — inspector on-board, non-payment, appeal
T5	Station closure — evacuation, power down

4.3 Maintenance worker rulebook (target ≤ 10 pages)

Section	Scope
M1	Depot safety — hazard awareness, PPE, lone-worker protocol
M2	Work-on-track authorisation — formal blocks, protection signals, access to active ROW
M3	Routine track inspection — walking the line, visual checks, GPM/RECD recording-car cadence (per RFC 0009 §9)
M4	Switch maintenance — blade lubrication, detection-sensor cleaning, 30-day check
M5	Fleet maintenance at depot — inspection intervals (daily / 7-day / 30-day / overhaul)
M6	Incident site access — derailment response, fire site, crane operations
M7	Wayside sensor maintenance (RFC 0016) — 30-day walkthrough, post-weather inspection, sensor replacement + calibration

4.4 Control-centre rulebook (target ≤ 5 pages)

Section	Scope
C1	OCC watch roles — dispatcher, supervisor, engineer-on-call
C2	Communications — radio discipline, PA authority, external-party notifications
C3	Shift handover — written log, outstanding items, emergency-on-call list

Total target: 40 pages across four sections.

5. Degraded-mode operations — the important part

Modern CBTC metros routinely run in "normal mode" (full ATO under a fresh MA) and have short, mission-critical manual-restoration procedures when the signalling degrades. OpenSourceRail keeps exactly three degraded modes:

Mode	Trigger	Operator action
M1 — Automated restricted MA	ATO is healthy and ATP can enforce a shortened MA	OCC authorises reduced-speed automatic movement; ATP remains fully active and speed is capped at 80 % of normal.
M2 — Controlled stop	Consensus cannot issue a trustworthy MA	Every affected train remains stopped while OCC establishes a work block and restores the signalling quorum. No written order may bypass ATP.
M3 — Protected depot shunt	Movement is required inside an isolated depot worksite	An authenticated local maintenance pendant requests automatic movement at ≤ 15 km/h under ATP, direct CCTV observation, and a locked work zone.

No intermediate modes. No "sort-of-ATP" states. Either the safety envelope holds (modes M1 / M3) or the train remains stopped (mode M2). This is a forcing decision: ambiguous degraded modes are the dominant cause of metro-world incidents.

6. Weather-dependent rules

Target regions routinely experience weather conditions that challenge legacy rolling stock:

- Dust storm (haboob): visibility < 100 m automatic speed cap at 50 % (`osr-ato` enforces). Dispatcher issues a network-wide speed restriction at the OCC. Passenger announcements via `osr-pis-onboard`. See also `lib/templates/climate-adapters.toml`.
- Heatwave (> 45 °C sustained): rail-expansion risk triggers a pre-emptive 15 % speed cap on `mainline-mixed` and `standard-metro` presets with continuous-welded-rail. Switch detection sensors recalibrate at dawn.
- Flooding: any rainfall > 50 mm / h triggers immediate inspection of at-grade segments; the dispatcher can preventively block sections via `osr-interlocking`'s maintenance-override. Elevated segments per RFC 0011 §5 are unaffected by surface flooding — deliberate design bet.
- Lightning: nearest-strike distance < 2 km Cat.22 5G backhaul and LoRa backhaul are both watched for outage; if both fail for > 10 s the operator notifies the OCC and cooperates with `osr-atp`'s default fail-restrictive MA.

7. Incident categorisation

Every incident fits one of six categories. This drives the notification chain and the regulatory report:

Category	Example	Dispatcher action
I1 — fleet self-recovery	Door interlock fault on a single train, resolved at next terminal	Log in event recorder; notify depot; no regulatory report.
I2 — service disruption	Single-train immobile; next train coupled to rescue it	Passenger PA, timetable recovery plan; 24h incident report.
I3 — short safety event	Emergency brake triggered by spurious trip; no injury	Log, investigate root cause, 72h report.
I4 — passenger injury	Slip, fall, medical emergency on board	Medical call, depot handover; formal incident report to NRSA within 7 days.
I5 — collision / derailment	Train-to-train, train-to-obstacle, train off-rail	Full network shutdown of affected line; NRSA notification within 2 hours; incident-site preservation.
I6 — mass-casualty / fire	Onboard fire not extinguished, large-scale passenger injury	Immediate evacuation protocol; emergency services; NRSA + national media escalation.

The categories map directly to `osr-event-recorder` tags, so the post-incident analysis starts from the same data the categories describe.

8. Training and recurrent qualification

- Initial driver qualification: 4 weeks classroom + 2 weeks simulator + 4 weeks supervised revenue service. Final qualification exam in written + simulator + on-road segments.
- Recurrent: 2 days refresher every 12 months + 1 simulator session per quarter.
- Degraded-mode drills: every driver runs modes M1 / M2 / M3 in the simulator every quarter. The `osr-sim` sim is the primary training platform.
- Dispatcher qualification: 8 weeks OCC-side classroom + simulator + supervised desk time.
- Maintenance: role-specific certification per EN 50126 / per national NRSA rules.

Training content is a deployment deliverable; this RFC fixes the minimum hours + structure.

9. Auto-gen implications

The rulebook does not directly feed the corridor-generation pipeline — but it does feed two outputs:

1. Depot capacity: `osr-sim`'s fleet-sizing needs to include enough spare trains for the rulebook's degraded-mode operations. A deployment with 10 trainsets on rostered revenue needs ≥ 1 spare in depot for maintenance (M5) and ≥ 1 cold- reserve for I-category incidents. See RFC 0014 §4 for the formula.
2. Workforce sizing: the driverless financial model supplies group FTE; `lib/templates/workforce.toml` allocates those totals into city roles, skills and tasks. The city delivery generator writes the organisation and spreadsheet. Named duty/leave cover must still be checked against the rulebook's shift structure (§4.5).

Changing this prose does not change software parameters automatically. Update the controlled model/template inputs, regenerate and check fleet and workforce evidence when operating rules change.

10. Pitfalls and decisions

- One rulebook across deployments is a big bet. Legacy metros each have their own 200-page rulebook; consolidating to one is the simplicity gain. National NRSAs will initially resist. The counter: the process of NRSA approval works on a per-document basis, not per-system basis — a shorter document approves faster. The operator carries the regulatory overlay in a thin appendix.
- Six incident categories, not more. Operators unused to this regime sometimes want a I7 or I8 for niche scenarios. We absorb niche scenarios into the existing six rather than proliferate.
- Paper written orders are allowed in mode M2. This is not anti-digital — paper is explicitly the fallback when the digital system degrades. Written orders remove ambiguity about authority when the digital system can't issue it. Rare enough that the paperwork tax is tiny.
- No "caution mode" between M1 and M2. Legacy metros have a smorgasbord of half-ATP-half-manual modes; every one is a source of operator confusion. We commit to the hard boundary.
- ≤ 60 pages is a hard target. The rulebook is reviewed on every minor-version bump; if it exceeds 60 pages, something gets cut. Adding text is easier than subtracting.

11. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	`docs/operations/` scaffold with per-role stub files for dispatcher, station staff, maintenance, and control-centre roles, each with scope + cross-refs + proposed rule outline (done 2026-04-22)	v0
v2	Full unattended-operations rulebook drafted across all four role families. Every rule is one sentence + a `Why :` paragraph, cross-referenced to the relevant crate and safety-case solution. Practising-operator review is v2.1 per-deployment work.	v1
v3	NRSA submission pack for Samawah (rulebook + safety case cross-reference)	RFC 0003 §5, v2
v4	`osr-sim` scenario library covering every degraded-mode scenario for operator refresher training	v2

12. Relationship to existing work

- `lib/templates/workforce.toml` — shift-structure + headcount inputs. v3 of this RFC aligns the two documents.
- `lib/templates/service-hours.toml` — service hours drive the rosterable workforce; the rulebook sets the rostering envelope.
- `docs/safety-case/` — references specific rulebook rules as "solution" evidence for goals around fail- restrictive operation, evacuation readiness, and training competence. GSN goals for "operational fail-restrictive behaviour" land in v4.
- `crates/osr-occ` — the dispatcher's OCC UI embeds the dispatcher rulebook's decision-support prompts. Software change in v3.

13. Open questions

1. Bilingual rulebooks. Every target-region deployment operates in at least one local language plus English. Do we maintain both upstream? v2 commits on this: likely yes, with English as the canonical and translations as separate branches in `docs/operations/<lang>/`.
2. Printable-PDF vs living-document. Rail NRSAs typically want a static PDF they can archive. The project wants a live document. Compromise: every minor-version bump produces a sealed PDF, and the commit hash is the canonical reference.
3. Simulator-based competence testing. `osr-sim` can evaluate driver actions against MA envelope in software. Can NRSAs accept simulator certification in lieu of on-road supervised hours? Per-country, probably not at v1.

14. Done criteria

- [x] Table of contents per role (§4)
- [x] Degraded-mode operations (§5)
- [x] Weather rules (§6)
- [x] Incident categorisation (§7)
- [x] Training envelope (§8)
- [x] Auto-gen implications (§9)
- [x] Pitfalls (§10)
- [x] Rollout ordered (§11)
- [x] Relationship to existing work (§12)

The next session picks up at v1 — `docs/operations/` scaffold, short editorial PR that seeds the per-role files. Actual rule text lands in v2.

Reproducible city operating twins

Source: docs/operating/city-platform.md

Each city has a reproducible business operating baseline alongside its railway engineering baseline. ERPNext provides the native operating records. The OSR integration compiles planning inputs and returns permission-filtered execution snapshots; it never issues railway release authority.

Configuration layers

Layer	Location	Purpose
Shared configuration	generic.toml	Departments, warehouses, task categories, programme tasks and planning calendar defaults
City overrides	`cities/catalogue/<region>/<country>/<city>/operations/erpnext.toml`	Identity, local calendar, organisation, operating release and additional city tasks
Compiled package	`build/erpnext/cities/<slug>/`	Effective configuration, complete task plan, dependency graph and checksummed summary
ERP execution	Native Project, Task, Department, Warehouse, Cost Center	Actual work, assignments, dates, timesheets, costing and business document workflows
Twin feedback	Private `var/erpnext/operating-twins.json`	Observed task/category/asset status, recorded hours, task costs and project-linked business document counts

All 266 catalogue cities have specific override files. The compiler derives their actual task and asset scope from their own generated operations bundle; it does not reuse Samawah's tasks for another city. See the Samawah profile and Mosul profile for local timezones, illustrative six-day construction calendars and distinct city work packages. Other profiles inherit the shared planning assumptions until a city operator supplies overrides.

The catalogue readiness gate validates all 266 configurations against their own tracked asset, manifest and project-twin evidence. Supervision can compile directly from those reviewable assets for every city. Full ERP task plans additionally need the deterministic compressed operations payload: Samawah and Mosul keep it locally in Git, while the other 264 materialise it with `./osr city <slug>` before import. A manifest is not treated as a substitute for the task records it hashes.

Tables merge recursively. Lists explicitly replace the shared list. Unknown keys fail validation. `city_programme` adds city tasks to the shared programme; each task needs a unique ID. Removing a department requires updating every task category/programme reference that used it.

The generic company, calendar start date and timezone are unset. The shared Monday–Saturday calendar and the Iraq examples are planning assumptions, not claims about an operator's actual roster or approved public holidays. No employee, supplier, bank, payroll, tax or commissioned asset data is invented.

Repeatable deployment

```
# Creates missing profiles; existing overrides are preserved.
./osr erp city init-configs
./osr erp city validate
./osr readiness --check

# The target company must already exist in ERPNext.
./osr erp city prepare samawah --company 'OpenSourceRail Evaluation'
./osr erp import build/erpnext/cities/samawah/plan.json \
  --company 'OpenSourceRail Evaluation'

./osr erp city prepare mosul --company 'OpenSourceRail Evaluation'
./osr erp import build/erpnext/cities/mosul/plan.json \
  --company 'OpenSourceRail Evaluation'
./osr erp snapshot
```

Replace either slug with any catalogue city. If its generated operations bundle is missing, run `./osr city <slug>` first, or supply a generated bundle with `prepare --bundle <path>`. Generation and preparation do not contact ERP. Review `effective-config.json`, `summary.json` and `plan.json` before import.

To make company binding permanent, add `[organisation] company = "…"` to the city profile. CLI `--company`, `--start-date` and `--release` overrides are captured in the effective configuration and its checksum. No credentials belong in tracked city profiles; use separate private site configuration.

When updating the integration code on an existing site:

```
./osr erp backup
./osr erp build
./osr erp up
./osr erp bench migrate
```

Functions supplied by the city package

- Manufacturing delivery: native Tasks with OSR manufacturing identity, task types, baseline day offsets and native predecessor relationships.
- Maintenance planning: fleet work maps to Fleet Maintenance; other assets map to Infrastructure Maintenance. Source cadences, triggers and evidence remain attached. Operators turn approved schedules and actual commissioned assets into native Asset Maintenance/Repair workflows.
- Assurance coordination: native Tasks for QA actions retain the railway hold point, asset, evidence and release-authority references. Completing them in ERPNext does not close OSR NCRs or approve railway handback.
- Procurement packages: requirements are grouped by sourcing route, with their earliest order-by and latest required-by day. All original BOM and sourcing requirements remain in the source record. Use native Material Request/Quotation/Purchase Order forms after resolving real items, quantities, suppliers, currency and approvals. The procurement automation also converts a selected requirement into a native draft Material Request once the operator provides its Item, verified stock-unit quantity, date and city warehouse.
- Programme and people: tasks for governance, business masters, workforce planning, maintenance readiness, assurance responsibilities and recovery, plus the city's own additional tasks.
- Organisation: stable native departments, warehouse locations and a city cost centre, scoped by company and city. Task Types are shared. No existing roles or user permissions are expanded by the configuration process.
- Digital-twin feedback: native Project forms gain an OpenSourceRail Operating twin action. Workbench displays matching ERP execution in the Project Twin view, and the operating page compares city baselines and asset workload. Hours/costs come from native Task costing; these are not the full general ledger or the city's generated CAPEX estimate.

Project and Task forms keep their source evidence, checksums and configuration under an OpenSourceRail reference tab, alongside the native business tabs.

Relative working days map to dates only when `calendar.start_date` is set. Weekends and explicit holidays are skipped, including negative mobilisation offsets. A non-working anchor rolls forward to the next working day. The compiler rejects missing dependencies, cycles, duplicate IDs, invalid calendars, mixed-city records and checksum mismatches. Native ERP dependencies retain ERPNext's business semantics, including its treatment of cancelled tasks.

Baselines, upgrades and operator edits

The first operating release (`release = "1"`) upgrades the earlier Project/Task import in place when source identity and hashes match. It fills empty type, department and date fields, adds dependencies only when none exist, and preserves existing task IDs, status, progress, assignments, priorities and operator-entered planning fields. It also creates the new work categories. Subsequent identical imports skip configured tasks and reuse city masters.

A changed configuration or source package cannot silently replace an applied baseline. Increment the city's `release` and prepare again to create a separate reviewable project baseline. Existing records stay intact. There is no automatic transfer of actual costs or approval of a new release. Workbench's engineering view shows execution only when exactly one exported ERP project matches the city and engineering revision; multiple candidates require explicit review on the operating page. No release is silently selected as authoritative.

Imports use native document validation, one database transaction and a final project-total recalculation. A failure rolls back new masters/tasks and metadata. The importer preloads permitted task identities and uses ERPNext's native project bulk-update flag to avoid recalculating thousands of records repeatedly. Use the bench-backed CLI for large city packages; a synchronous HTTP import can exceed a deployment's request timeout. Native task-tree updates can hold database locks during large imports, so schedule them outside active task editing.

Refreshing the integrated twin

```
# One private, atomic export. Previous valid feedback survives export failures.
./osr erp snapshot

# Linux user timer: refresh every five minutes, using the local Docker/bench site.
./osr erp feedback start
./osr erp feedback status
./osr erp feedback stop
```

The timer and service are installed under `~/.config/systemd/user/` as `osr-erp-feedback`. The user session and container runtime must be running. The snapshot file is mode 0600 in a private directory; OSR's static server does not expose it. Only the loopback Workbench exposes the aggregate snapshot endpoint. Shared/network deployments need an authenticated integration service and their own permissions; this local bridge is not a shared-host auth scheme. The bundled single-site proxy validates browser WebSocket origins and routes ERP notification authentication through the internal Compose network. Review that proxy configuration when adding a public domain or TLS termination.

ERP callers can GET `/api/method/osr_erpnext.city_runtime.city_status?project=…`. The method checks Project access and uses native permission-filtered lists for Tasks and business documents. Unavailable document categories are marked unavailable, not reported as zero. No employee personal details are exported. The local bench export runs with the site's administrative visibility.

Snapshots show their observation time; the UI labels snapshots older than one hour. Refresh view reads the latest exported file; it does not claim to query ERP directly. ERP unavailability leaves the previous snapshot available with its original timestamp and does not affect railway control.

Verification

```
tools/automation/osr-python -m pytest -q \
  tools/automation/tests/test_erpnext_city.py \
  tools/automation/tests/test_erpnext_planning.py
./osr erp city validate
```

The tests cover configuration inheritance, every city's identity, working-day calendars, dependency rejection, deterministic packages, import fingerprints, legacy keys and company/release separation. Live validation should additionally exercise native imports, repeated imports, permission-filtered feedback and the Project/Workbench views against a private ERP site.

Upstream behaviour: ERPNext Tasks, Asset Maintenance.

Deployment Planning Reference

Source: docs/deployment-planning-reference.md

This method describes retained wheel/rail city and national calculations. Primary v0.5 city planning uses the maglev compiler and assumptions; old train counts, axle/track/civil quantities, costs and acceptance do not transfer. Existing geography and demand remain provisional inputs.

This page defines the common interpretation of every generated OpenSourceRail city README and national brief. Location folders contain only local inputs, calculated results, exceptions and evidence links. Shared assumptions belong here or in the linked machine-readable templates—not in hundreds of generated copies.

Generation And Review Boundary

The city catalogue is deterministic:

```
city catalogue + OSM/raster evidence + shared templates
  → routed network + stations + service + fleet
  → energy, civil, CAPEX and finance calculations
  → engineering and simulation evidence
  → concise city README and national aggregation
```

Generated READMEs are indexes and decision summaries. The authoritative local inputs and results remain `design.toml`, the expanded scenario TOML, and the JSON/GeoJSON evidence inside each city folder. Git records changes to inputs, generators and outputs together. Do not hand-edit a generated city or country README; change its source or a shared template and regenerate it.

Network And Station Planning

`osr_geo` rasterises source-locked OpenStreetMap roads, buildings, water, protected areas and demand anchors onto the planning grid. `osr-design` uses a demand-rewarded least-cost route, places stations against the demand surface, and classifies each segment as at-grade, elevated or bridge. Tunnels are not in the upstream catalogue.

The standard spacing policy is 1.6 km in central areas, 3 km in ordinary urban areas, and up to 7 km on low-demand outer approaches. Cross-line platforms inside the 600 m station-complex envelope are consolidated as one interchange. The retained `*.design-quality.yaml` coverage score is the fraction of high-demand cells within 20 routing cells of a track, irrespective of station access. It is not anchor-weighted resident coverage and must not be multiplied by the catalogue population. Each city's `engineering/access/README.md` instead separates direct transfers, reachability through intermediate lines, and native population counts within the union of station circles at 500/800/1,000/1,500/2,000 m. Population year, source-pixel resolution, bbox denominator and missing evidence remain explicit. Wider radii are access/feeder sensitivities; radial distance does not prove a walkable route across rivers or other barriers or create fare demand. No current population total is inferred by rebasing a 2020 raster.

Straight viaducts also require separate beam, support/foundation and vertical terrain checks. Every city retains a source-bound `engineering/clearance` screen; missing heights or mapping prevent an obstacle-clearance claim. See the viaduct obstacle-clearance policy.

Service, Fleet And Capacity

The default service is GoA 4 battery-electric operation with line-specific time windows and headways. Fleet sizing covers peak revenue service, planned maintenance spares and cold reserve. Off-peak surplus performs terminal or depot inspections without creating a second service-rotation fleet unless a local schedule proves one is required.

Healthy trainsets may stable overnight at protected powered stations near their first departure. Red defects, overdue heavy maintenance, isolation failures or security failures return a set to the main depot. Station stabling requires charging, CCTV, remote traction isolation, emergency access and an OCC-assigned track slot.

Capacity uses the selected trainset family's AW2 planning capacity, the actual generated timetable and a controlled practical load factor. The annual paid-trip range is a capacity-use scenario, not a ridership forecast. Replace it with a surveyed demand model before an investment decision.

Energy Method

Each city calculation combines:

- trainset energy per car-km and the generated timetable;
- climate and non-revenue movement factors;

- opportunity charging at powered stations;
- station/depot PV and stationary LFP storage;
- a dedicated solar plant or equivalent PPA asset for the remaining annual traction-energy requirement; and
- residual grid import after those assets.

The generator checks every line's powered-stop gap, traversal energy, delivered charge, protected onboard reserve and operating margin. These are feasibility screens; grid connection, protection, harmonics, thermal, fire and local solar yield studies remain deployment work.

Civil And Cost Method

Civil geometry is generated from the parametric source and exported to IFC4.3 for Bonsai coordination. Checked quantities feed the generated `civil-cost-model.toml`, which is then consumed by city CAPEX, finance, IFC properties and documentation. Current design targets are 2.584 M USD/route-km at-grade, 9.748 M USD/route-km elevated and 18 M USD/route-km for bridges. The retained 3/12/18 M benchmarks remain visible for audit.

These figures are unquoted planning targets. Land, utilities, tax and duty, site geotechnics, project risk, programme, independent checking and local supplier rates must be added or validated before procurement. The complete shared cost basis is in the generated cost model.

Capital And Finance Method

City CAPEX includes civil works, stations, depots, rolling stock, residual train control, charging interfaces, timetable-sized dedicated solar and EPC/project services. The national trainset factory is excluded from cities and counted once in each national brief, sized to the largest single-city fleet programme.

The catalogue's controlled core concepts adopt elevated land sections and straight radial tangents where analytical geometry permits. Water crossings remain bridges; retained unsuitable fragments keep their geometry review gates. Core boundaries are design-centroid planning screens requiring survey, property and obstacle confirmation. Immutable seeds preserve the original line inventory, including rings.

The current line-depot scope stores the full fleet, including spares, in one planning depot per line. Train length sets storage track quantities; measured maintenance workload remains a release requirement for workshop bays. Depot PV/storage reference equipment is included once in depot capital. Land, utility upgrades, installation and site acceptance remain open.

Station staffing uses two posts, two normal eight-hour shifts and additional cover for the actual service window. Paid hours are reduced for leave, training, sickness and handover. Country income proxies are retained estimates: basic wages start at 150% of that proxy, technical grades at 225% and management at 300%, plus 25% employer and 10% overtime allowances. Baghdad's dedicated revised scope retains its separate indexed Iraqi wage evidence.

City-order factory requirements use the actual train family and consist. Facility readiness is 18 months; qualification and serial manufacture follow it. Short civil programmes explicitly wait for fleet readiness. The national factory allowance is the larger of the module-based allowance and independent city-order physical envelopes, counted once. Concurrent national loading and sequencing require a separate accepted programme.

The procurement-origin model separates imported value—the minimum foreign currency or international-capital requirement—from locally supplied labour, materials, fabrication and services that can use domestic funding. Country finance parameters define construction period, debt rates, tenor and the local bond/equity split.

For Iraq, the primary proposal is the schedule-linked funding programme. It assigns proposed Chinese buyer credit only to assumed eligible imported component invoices, then splits the remainder among government appropriations, IQD bonds and IQD bank credit. The controlled assumptions remain uncommitted. Monthly native-currency debt schedules, interest on staged draws, fees, reserves, public support and downside cash gaps are explicit. Construction uses the actual resource-constrained procurement milestones under an assumed 260-working-day year; the generic five-year financing comparator does not define the Iraq construction schedule. Physical and operational release gates remain open even when all planning example artifacts are regenerated.

The foreign-turnkey comparison is an editable like-for-like sensitivity, not a vendor quotation. Both cases use the same network, fleet, service, energy scope and financing schedule. Replace the multiplier, external share and all funding terms with scope-normalised bids and signed lender terms before approval.

OPEX, revenue, NPV, IRR and DSCR outputs are planning screens. Capacity-use scenarios are not demand forecasts, and no grant, tax treatment, land-value capture or lender commitment exists unless explicit controlled evidence says otherwise.

QA, Maintenance And Assurance

Shared construction QA gates, manufacturing travellers and maintenance intervals are generated from the controlled templates. City-specific asset counts and schedules are materialised in the local operations bundle. The common system covers civil works, trackform, stations, rolling stock, passenger systems, energy, signalling/comms, depots and integrated trial running.

Generated evidence proves internal consistency and reproducibility—not railway certification. Survey, geotechnical release, structural design, fire/egress, product acceptance, calibrated operations modelling, independent safety assessment and authority approval remain deployment responsibilities.

Local Evidence Contract

Each city README links the current status of these local artifacts:

Artifact	Purpose
`design.toml`	Authoritative generated network, fleet and CAPEX design
`<slug>.toml`	Expanded simulator scenario
`<slug>.design-quality.yaml`	Coverage, routing and civil-quality gates
`engineering/finance/summary.json`	Reconciled CAPEX, OPEX and finance cases
`engineering/simulation/validation-summary.json`	Nominal and degraded simulation acceptance
`engineering/gis/summary.json`	GIS package provenance and layer counts
`engineering/sumo/summary.json`	Executed timetable screening
`engineering/energy/summary.json`	Grid, charging and solar screening
`operations/`	Asset, QA, manufacturing and maintenance evidence

The README generator verifies source hashes and passing status before publishing these evidence links. Repository health also regenerates city and country pages and rejects drift.

Regeneration

```
# One city
tools/automation/regenerate-city.sh samawah

# Complete catalogue, using current caches
tools/automation/regenerate-all.sh

# Documentation, links and generated drift
python3 tools/automation/check-readmes.py
python3 tools/automation/check-markdown-links.py
python3 tools/automation/repo-health.py --quiet
```

Changes to common assumptions should normally touch one template or this page, then regenerate affected outputs. Location folders should change only when their local input, calculated result, evidence or explicit exception changes.

Ballastless Slab Trackform Designs

Source: docs/civil/slab-trackforms.md

This note records the OSR planning-grade slab trackforms for urban at-grade and elevated sections. The parametric CAD source lives in `design/component-catalogue/src/osr_mech/civil/slab.py`.

The designs are reference envelopes for cost, clash, fabrication, and constructability review. Deployment partners still need local structural checks for concrete grade, reinforcement, geotechnical support, drainage, stray-current control, thermal movement, seismic load, flood/scour, and product acceptance of the direct-fixation fasteners.

At-Grade Urban Slab

Long open, machine-accessible corridors use a continuously slipformed slab by default. The table below defines the six-metre single-track ST6 product kept for constrained streets, utility crossings, short possessions, flood-prone and replaceable zones. Both methods retain the same controlled track support section and concrete quantity basis; a project method-zone schedule determines panel counts.

Item	Reference value
Trackform	OSR-ST6 single-track transportable panel; two rows form double track
Panel length	6.0 m
Overall panel width	2.9 m
Track-centre spacing	3.5 m
Base slab thickness	250 mm
Rail plinth width	380 mm
Rail plinth height above base slab	160 mm
Direct-fixation pitch	Maximum 650 mm; centred 600 mm on the 6 m reference module
Rail seats per 6 m panel	20
Nominal concrete per 6 m panel	5.08 m3
Planning bare panel mass	12.7 t

Use ST6 where the corridor is in a constrained street, utility or transition zone, short possession, flood-prone location, station interface, depot throat, or other place where ballast dust, tamping possessions, and loose aggregate are unwanted.

The slab uses two edge troughs for drainage, low-voltage/data routing, and maintainable cable access. The direct-fixation plinths are continuous so baseplates can recover small alignment tolerances without breaking out concrete.

Elevated Deck Slab

Item	Reference value
Trackform	Local direct-fixation plinths over thin non-structural alignment layer
Panel length	6.0 m
Overall alignment-layer width	2.7 m
Deck compatibility	Fits the 2.9 m OSR-Pi20/Pi25 flange
Alignment layer thickness	40 mm; may be omitted by project design
Rail plinth width	380 mm
Rail plinth height above base slab	160 mm
Direct-fixation pitch	Maximum 650 mm; centred 600 mm on the 6 m reference module
Rail seats per 6 m panel	20
Nominal concrete per 6 m panel	1.38 m3

Use one elevated trackform run per single-track Pi-beam. A double-track elevated section therefore uses two parallel Pi-beam/trackform assemblies.

The Pi-beam upper flange is the structural deck. The alignment layer provides local tolerance recovery only and may be omitted where surveyed casting plus adjustable baseplates can recover rail geometry. A full-width reinforced or floating slab is reserved for locations where a vibration/noise and structural study justifies its dead weight.

The cable/drainage trough is installed on the inner/non-egress side. The separate outer 1.0 m walkway/barrier cassette remains clear of service covers, cabinets, screen posts and other permanent obstructions.

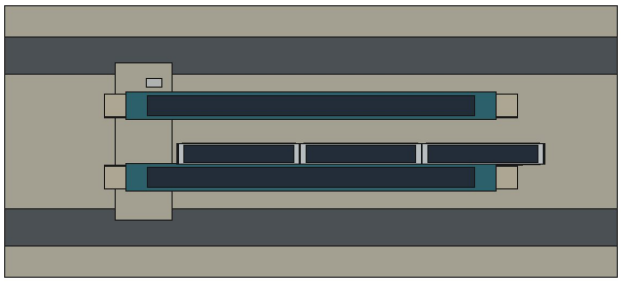
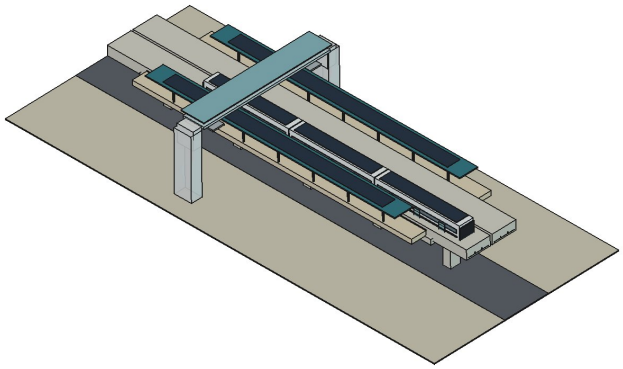
Iraqi manufacture and complete seat kits

The Baghdad detailed engineering plan adds the local slab production route, prototype inspection, lifting/bedding requirements and a complete fastener child schedule. The 6 m module has ten seats per rail centred at 300, 900, …, 5700 mm, avoiding the former pad projection beyond the panel ends. Supplier anchor edge distances still require release. Continuous-run quantities retain the existing 650 mm planning basis; precast method zones use their own panel/seat installation schedule.

Concrete volume and bare mass cover base/plinths only. The CAD service troughs are allocation envelopes; covers, drainage sections, steel, inserts, bedding and rigging need separate schedules. Make panels, cages, moulds, ducts and suitable metalwork in Iraq; source qualified fastener/anchor systems initially and qualify any localisation through system fatigue, stiffness, insulation and process tests. The catalogue does not issue a rebar schedule, mix grade, lifting approval or supplier drill pattern.

FreeCAD Scene Renders

The slab trackforms are shown below in station context with driverless rolling stock and the station access elements that affect civil cost.

At-grade ballastless station track	Elevated deck station track
	
At-grade ballastless station track with driverless train	Elevated deck station track with driverless train

The source scene document is `design/component-catalogue/models/cad/station-scenes.FCStd` and is regenerated with `design/component-catalogue/scripts/freecad_station_scenes.sh`.

Fabrication And Installation Rules

- Slipform long accessible runs over the released machine-controlled formation. Keep the 6 m ST6 module for constrained and replaceable zones; use closure pours only at turnouts, station ends, curves that require local survey correction, and transitions to bridge/special spans.
- Cast embedded inserts, earthing/bonding points, drainage scuppers, and lifting anchors in the precast yard.
- Require surveyed casting beds and a dimensional QA record for every batch.
- Use adjustable baseplates on all running rails.
- Keep prestressed elevated girders conservative; recycled aggregate and aggressive low-carbon mixes can be used first in non-prestressed slabs, piers, pile caps, and station modules.
- Treat direct-fixation pads, anchors, and baseplates as replaceable wear/interface parts, not as buried civil structure.

Cost Model Link

These slab designs feed the generated civil planning contract:

Civil class	Design-derived target	Retained benchmark
At-grade ballastless slab	2.584 M USD/route-km	3.0 M USD/route-km
Elevated guideway with local-plinth trackform	9.748 M USD/route-km	12.0 M USD/route-km
Bridge/water-crossing guideway	18.0 M USD/route-km	18.0 M USD/route-km

The active values come from `civil-cost-model.toml` and are planning targets, not tender or turnkey EPC estimates. Build the project estimate from `viaduct-quantity-cost-model.toml` plus site rates, utilities, access, risk, programme, independent checking, and contingency. Short elevated stubs, stations, soft ground, deep utilities, tight curves, or repeated

special spans carry project-specific premiums.

Civil works master plan

Source: docs/civil/civil-works-master-plan.md

Revision basis: 8 October 2026. Detailed construction planning and method-development package; project drawings, temporary works, permits, prices and site acceptance remain required before execution.

This plan connects viaducts, stations, ground-supported railway, special crossings, depots, energy sites, utilities, drainage, access and reinstatement. Use 25 m average span for strategic production and logistics sizing. Purchase and erect against the identified span schedule: shorter catalogue bays, station transitions and separately designed closures retain their own geometry. Do not replace that schedule with rounded kilometres multiplied by a standard beam count.

The generated Baghdad civil works package reconciles the current alignment, running-span register, station topology and depot scope. It provides line quantities, workfront and station registers, production/logistics sensitivities, package dependencies, inspection gates and source locks. Run the civil planning compiler after a controlled alignment or method revision; `--check` rejects stale outputs. Other cities can apply the methods after substituting their own identified assets and local evidence.

Package structure and responsibilities

Package	Accountable role	Controlled outputs and handover
CW-01 surveys, land and utilities	survey/consents lead	Survey control, corridor boundaries, utility trial-hole register, property/access permissions, verified site model
CW-02 enabling and temporary works	construction manager and temporary-works coordinator	Worksite/traffic layouts, haul roads, platforms, drainage, hoardings, design/check register, permits to load/use/dismantle
CW-03 foundations and ground treatment	geotechnical authority	Zone design, support schedule, tested method, actual installed lengths, verification and settlement records
CW-04 viaduct substructure	bridge construction lead	Columns, caps, abutments, bearing seats, dimensional/strength records and released support packets
CW-05 precast production and delivery	precast/logistics manager	Existing-facility audits, shop drawings, accepted serialized components, load/route plans and receiving records
CW-06 viaduct erection and completion	appointed lift planner and bridge lead	Configured plant, erection stages, temporary restraints, connections, waterproofing, barriers, drainage and structural handover
CW-07 station civil and access	station construction lead	Structure, platforms, shafts, stairs, concourse, roofs, street routes, equipment interfaces and public access acceptance
CW-08 at-grade track and approaches	track/earthworks lead	Released formation, treatment, retained approaches, slab/panel zones, crossings, track geometry and drainage
CW-09 special crossings and closures	special-structures authority	Site-specific bridge/closure design, access and erection study, water/road interfaces, independent check
CW-10 drainage and flood protection	drainage authority	Catchments, receiving outfalls, attenuation, cross drainage, scour/flood controls, clean/test records
CW-11 depot and stabling civil	depot design/construction lead	Surveyed yard, buildings, pits, pavements, tracks, charging foundations, workshop and fire/access interfaces
CW-12 energy and railway systems civil	systems interface manager	Ducts, foundations, equipment rooms, earthing interfaces, fire separation, pull/test and systems handover
CW-13 roads, public realm and reinstatement	urban works lead	Permanent traffic arrangements, accessible paths, road/utility restoration and local handback
CW-14 integrated acceptance	owner handover manager	Asset/as-built records, outstanding defect controls, independent acceptance and line-opening packet

Roles are proposed responsibilities, not appointments. The owner appoints designers, principal delivery leadership, independent checking and local approving bodies before corresponding releases. A contractor programme cannot grant its own independent acceptance.

Scope and 25 m average-span basis

The expanded Baghdad planning inventory has 54 lines and 863.9238 route-km: 273.8781 at grade, 545.9111 elevated and 44.1346 bridge/water-crossing. Running elevated intervals cover 407.4884 km; the remaining 138.4227 km of elevated station/transition scope requires station and interface design. These are distinct chainage scopes. Special bridge class is additional to unresolved short closures in running elevated intervals.

The identified running layout contains 15,888 catalogue bays (14,248 Pi25 and 1,640 Pi20), 31,776 catalogue single-track beams and 747 separately designed spans. Its 16,635 running spans average approximately 24.50 m. There are 17,782 proposed running support chainage records across 1,147 disconnected runs. The integrated line-based plan resolves closed-ring start/end aliases into physical support packets. Its candidate foundation count requires field design; support sharing, boundaries and specials prevent deriving foundations as bays plus one. The generated civil quantities and line

assembly summary control later revisions.

For a reference uninterrupted kilometre, 25 m means 40 double-track bays and 80 single-track full-span beams. Count 41 support positions only for that isolated kilometre; adjoining sections share boundaries. Four-span structural-continuity units and simple-span link slabs require different bearing counts and stage designs. Neither a 25 m average nor a reference-kilometre count establishes a foundation, pile length or released beam order.

The station scope includes 600 stations, 747 physical platforms and 1,303 boarding faces. There are 452 elevated and 148 at-grade station layouts, with 1,014 each of lifts, shafts, stairs and escalators. Layout counts do not size foundations, fire stairs, concourses or accessible street routes. Each station gets an individual work package and interface review.

Programme logic and working calendar

Use the programme and workface controls with viaduct, station, other civil works, logistics, and inspection/handover methods.

Start with two candidate erection fronts per line and an eighteen-machine study pool, subject to independently feasible released fronts. Expanded inventories queue later fronts behind the previous assignment of each physical machine, including transport/recommissioning gates. Foundation and substructure crews work ahead; precast stock is accepted before dispatch. Station structures and specials are parallel critical-path packages with explicit crossing/relocation interfaces. When radial fronts finish, evaluate additional ring fronts before moving machines to another city. Extra launchers help only when access, foundations, accepted stock, deliveries, crews and temporary works support them.

The resource envelope uses a proposed six-day erection week, an eight-hour shift and explicit holiday/maintenance/weather deductions. Production curing uses elapsed time, not an erection working-day counter. Illustrative rates of 0.5, 1.0 and 1.5 accepted bays per launcher-working-day test capacity; none is a measured production commitment. The minimum quantity/rate duration is a lower bound. Full dates require site-specific quantities, calendars, finite resources, release dates and the complete network of predecessors. Use the regenerated finite-pool programme for conditional running-bay comparisons; full line openings retain all station, special, depot, systems and acceptance gates.

Existing industry, procurement and commercial plan

Iraq has many existing precast facilities and production expertise. Begin with factory capability and spare-slot audits, adaptable beds/moulds, prestressing/handling equipment, concrete and reinforcement supply, and regional delivery routes. Use the potential vendor register for screening; a listing gives no compatible product, spare capacity or agreement. A dedicated new civil factory is an option only after comparing existing contracted capacity, upgrades, site and transport costs.

Issue configuration-specific packages for Pi20/Pi25, caps/columns, panels/walkways, bearings/expansion joints, lifting equipment, heavy haulage, station modules, drainage, track support and systems foundations. Separate design/IP support, tooling, qualification/first article, repeat supply, freight/permits, installation, commissioning, spares, maintenance and taxes. Require supplier drawings and declared complete mass, CG, lifting/support positions, release strengths, tolerances and traceability. Record rejected items, NCR rework and replacement capacity in delivery planning.

Cost, payment and opening interfaces

Build the estimate from named packages and measured quantities. Record foundations by actual type/count/length, columns by height, caps by selected type, beams by span ID, station structures by engineered take-off, and earthworks/ground treatment by surveyed volume/area. Include temporary works, access/traffic, factory tooling, spoil disposal, hauling/escorts, storage/handling, crane or launcher mobilisation, testing, utility diversions, special crossings, flood works, systems civil interfaces, overheads, escalation, risk and contingency.

The retained route-km rates and USD 9m launcher purchase allowance are comparator scopes; they are not a complete installed quotation. Credit embedded erection or plant allowances only after reconciling invoice scope. Milestone payments follow accepted releases/deliveries/installation and contractual certification, with retention and final handback separate. Unknown prices remain unknown; no sum of known rows becomes the full adopted CAPEX.

For each line, opening requires accepted running structures, stations, specials/transitions, track, energy, depot/stabling, fleet, testing and approvals. Construction handover includes safe access for following trades; opening additionally needs integrated railway and operating acceptance. The coupled programme study retains these gates and conditional finance months. No date, revenue or saving is adopted from a running-span finish alone.

Review and release plan

Review the first representative complete bay, station access/structure package, at-grade treatment/track zone, and depot/energy foundation package before repetition. Check design interfaces, sequencing, working platforms, ground/temporary loads, manufacturing and logistics capacity, tolerance stack-up, traffic/access, rescue arrangements and full installed costs. Feed measured first-cycle durations, rejected quantities and disruption back into the finite-resource programme.

The generated review register distinguishes repository quantity consistency from site approval. It currently flags unresolved survey/structural releases, no qualified supplier or delivery-route capacity, missing station/special take-offs, inconsistent illustrative buffer sizes, and incomplete depot/stabling provision. Close each through a dated configuration-bound record; do not turn an authoring review into independent construction acceptance.

Reference practice

International guidance supports method development; Iraqi authorities and the project design basis determine adoption. FHWA prefabricated bridge elements describes off-site fabrication, transport and site assembly. FHWA accelerated bridge construction treats planning, geotechnical and structural placement choices together. HSE temporary works guidance describes coordination, designs and checking. HSE lifting-operation planning supports competent planning and supervision. These references do not supply Baghdad geometry, capacities, acceptance values or permits.

Viaduct construction method

Source: docs/civil/viaduct-construction-method.md

Planning method for the civil master plan. Normal strategic span average is 25 m; use the identified Baghdad spans for orders and erection. Pi20, station approaches, closures, tight curves and special crossings stay individually resolved. Refer to the design basis, substructure kit, bearing schedule and transport envelope.

1. Release a complete workfront

The construction manager assembles a package for each continuous run: line/chainage, span and support IDs, survey datums, accepted design revision, land and access, trial holes and utility isolation/diversion, ground model, temporary works, traffic/pedestrian arrangements, environmental limits, approved lifting route and emergency access. Record station/junction/water/road boundaries and whether the launcher can pass, cross with a checked support arrangement, or dismantle/relocate. A visually continuous alignment is not proof of machine passage.

Stake support centres and offsets from redundant control; check against track alignment, utilities and adjoining structures. Establish settlement/vibration monitoring where excavations, piling or temporary loads affect property. Release piling/crane platforms with verified bearing, drainage, edges and underground-service exclusion. Install hoarding, vehicle/pedestrian segregation, lighting, welfare, spill response, washout and spoil management before production.

2. Select and construct foundations by ground zone

Select shallow foundations, bored shafts, driven bents, pile groups or ground treatment using the foundation system. The selection must consider actual axial/lateral actions, settlement, seismic/collision, groundwater, scour, nearby buildings and installation access. Record support-specific geometry and installed quantities. Do not assign one pile depth to the route or assume a generic foundation volume per kilometre.

Method	Proposed sequence	Release and retained records
Shallow footing/socket	Verify excavation/support and groundwater method; excavate to design level; inspect founding material; install blinding, reinforcement/inserts and formwork; cast, cure, survey and backfill in tested layers	Formation acceptance, level/bearing basis, reinforcement and concrete records, drainage/settlement protection
Bored shaft	Trial method and test foundation; prepare working platform; drill using design-selected casing/slurry; record strata and geometry; clean/inspect base; lower checked cage; place concrete by the released dry-hole or submerged method; inspect/test and trim for connection	Drilling and slurry records, cage position, concrete continuity/volume, integrity/load tests, accepted actual length
Driven bent/piles	Verify vibration/property constraints and platform; trial installation; lift and pitch with approved restraints; drive/splice to the accepted criteria; test and survey; trim and connect	Driving/hammer/penetration records, damage/splice checks, verification tests and final pile level/length
Pile group/cap	Install and verify group; excavate/shoring/dewater as designed; prepare pile heads; fit reinforcement/cap shell and column interface; cast/cure and restore drainage	Group effects and lateral design, pile/cap records, connection survey, excavation and backfill release

For submerged bored-shaft concreting, use the designer's tremie procedure, verified mix workability/retention and continuous placement records; no generic embedment, strength or acceptance number is issued here. FHWA's drilled-shaft manual is a method-development reference for drilling, concrete placement and inspection. Site trial, project specification and independent review determine the applicable procedure and test criteria.

Treat groundwater disposal and drilling spoil as planned waste streams. Maintain excavation support and dewatering until the designer permits removal; monitor drawdown and adjacent settlement. Failed integrity/strength/dimension results enter NCR control before a support is released.

3. Columns, caps, abutments and bearings

Compare in-situ columns with precast columns or shells using complete connections, temporary bracing, access and installed costs. Survey foundation/interface levels; prepare sockets or starter reinforcement; install the column/form with checked temporary restraint; inspect reinforcement, inserts, access and earthing; cast/grout and cure to the connection-specific strength. Maintain bracing until the designed permanent path is established and signed off.

For a precast or hollow-shell cap, verify complete mass, CG and lifting anchors against the configured crane/launcher and actual ground reactions. Survey seating and reinforcement/socket interfaces; erect, restrain, connect and complete infill/grouting under the approved stage design. Do not treat the 25–35 t catalogue cap target as a measured lifting weight. In-situ caps require independently checked formwork/falsework, pre-pour inspection and staged striking criteria.

Construct abutments, retained approaches and wing walls with released earth pressure, settlement, drainage and flood/scour design. Survey bearing seats and orientation. Install fixed/guided/free bearings, anchorages and grout to supplier/project

instructions, including temperature datum, permitted movements and replacement-jacking interfaces. A support-ready packet includes foundation verification, column/cap strength, geometry, bearing inspection, connection status and the actual next launcher/delivery reactions.

4. Produce and accept full-span beams

Use suitable existing Iraqi factories after auditing available production slots and the complete process. Freeze drawings, tendons/anchors, reinforcement, embeds, lifting/support locations and shop traveller against each beam/track/span ID. Survey mould and bed; inspect cages, ducts/inserts and calibrated stressing plant; cast the released mix with batch traceability and compaction method; retain curing temperature/maturity and test specimens.

Prestress transfer, demoulding, lifting, storage and installation are distinct strength gates. Release time comes from the qualified mix and test/maturity relationship, not elapsed hours alone. The existing 48-hour planning cycle stays a capacity assumption until trials validate it. A 24-hour target requires a new qualified cycle and equipment/thermal assessment. Sequence transfer and lifting to the supplier design; inspect cracks, camber, bearing/lifting zones and dimensions after transfer and handling.

The current Baghdad complete-member studies are approximately 69.68 t for Pi20 and 85.51 t for Pi25; with the illustrative 4 t gear, suspended studies are 73.68 and 89.51 t. These are unqualified budgeted masses from the connected beam register. Bare-section targets are a different scope. Require actual complete-member mass/CG and lifting design before selecting machinery; a nominal 100 t machine does not establish configured capacity.

Accepted members enter segregated serialized storage on designed supports with verified ground/stacking/restraint. Quarantine NCR items. Preserve strength, dimensions, lifting-anchor certificates, support positions and dispatch readiness in the traveller. Transport handling adds inspections at loading and receipt; acceptance at the factory does not clear transport damage.

5. Erect and commission the launcher

Deliver components against the logistics plan. Verify assembly space, ground bearing, traffic closure, lifting equipment and the launcher foundation/deck support arrangement. Assemble to supplier instructions; inspect structural/pin/bolted connections, electrical/hydraulic systems, brakes, limit devices and emergency arrangements. Complete required tests and documented commissioning at the actual configuration before use.

The stage design covers assembly, loaded/unloaded operation, launch/advance, turning or relocation, one-beam eccentricity, gust exposure, parked/storm state, delivery vehicle positions, breakdown and dismantling. Check every loaded pier, cap, bearing and previously erected span. Over-deck deliveries require construction load paths and protection against unauthorized additional plant. Ground-delivered erection requires a separately released access path and unloading position.

6. Repeat the bay cycle

1. Confirm daily/shift release: next span/support IDs, current beam traveller, work permit, weather/visibility within supplier limits, exclusion/traffic arrangements, healthy equipment, competent rested crew and communications. Recheck changes since the previous shift.
2. Verify seats, bearing orientation/levels, lifting gear and beam identity. Place transport in the permitted position; secure against movement. Review load, CG, lift path, rigging and contingency landing location with the team.
3. Attach only the released lifting points/gear. Apply the controlled trial-lift/check from the lift plan. Confirm balance, connections and clearances before travelling; stop for any unexpected movement or communication loss.
4. Place the first single-track beam on surveyed seats. Maintain the designed lateral/torsional restraints and required crane/launcher support. Do not release the load merely because it touches the bearings.
5. Place the second beam using the checked asymmetric sequence. Survey alignment, bearing contact and joint gaps; complete permanent/temporary restraints and necessary connection/grout strength gates before releasing the next stage.
6. Complete the bay inspection/accepted record. Advance only when the construction-load path is released, next supports are ready, delivery and next-machine positions are checked, and the worksite remains segregated.
7. Hand over a stable inspected bay to following trades; record outstanding restrictions, access/edge protection and permitted construction loads. Update stock, installed component IDs, cycle time, NCRs, weather and equipment hours.

Measure preparation, loading/arrival, each beam lift and securing, inspection, advance, handover and interruptions separately. Two lifts plus launcher advance are one double-track bay cycle. The plan does not prescribe wind limits, lifting capacity, rigging angles or concrete strengths: the approved equipment/stage design supplies them. Complex or tandem lifting needs

its own competent written plan; HSE lifting-operation guidance is a practice reference.

7. Connections and railway finish

Construct the selected continuity connection or link slab only to the released reinforcement, pour, cure, bearing and stage sequence. A link slab preserves independent girder-end support; it does not justify reducing bearing lines. Retain temporary restraints and movement controls until their removal is authorized. Inspect grout, concrete, joints, waterproofing continuity and drainage paths.

Install independent walkway/barrier cassettes, fall containment, cable/drainage troughs and earthing interfaces with their own handling and fixing designs. Keep the escape walkway clear. Survey track datum; prepare waterproofing/alignment interfaces; install plinths or the selected trackform, adjustable fasteners and rail; complete rail/structure interaction, movement and geometry checks. Install systems only under an agreed load/penetration/earthing interface; unreviewed drilling into prestressed members is prohibited by the proposed method.

8. Run boundaries, specials, recovery and demobilisation

Before each station, closure or disconnected run, decide whether to pass using a designed temporary support, bridge with an engineered launcher configuration, or dismantle and reassemble. Model closure permissions, access, transport, setup/commissioning time and resource reservations. A shorter span is not automatically launcher-compatible. Special spans require their own product and order; no Pi25 order is generated for a residual unsupported length.

Prepare recovery for hydraulic/electrical failure, suspended-load stall, damaged bearing, unseated beam, loss of support readiness and weather stop. Define a safe supported/parked condition, exclusion and communications, engineered rescue/repair, inspection and re-release. Stop deliveries when stock/access is saturated. No improvisational re-rigging, unreviewed props or moving a suspended load over the public is authorized by this plan.

After a front finishes, check the remaining critical path and candidate ring fronts. Inspect/dismantle, transport, verify new supports/access, recommission and reassess the crew before reassignment. Release a machine to another city only after the complete Baghdad programme and resource review. Close final bearing/structure surveys, as-built records, drainage and access, defects and monitoring before structural handover.

Station civil and access construction method

Source: docs/civil/station-construction-method.md

Apply with the civil master plan, other civil works, logistics and inspection plan. Each station has a generated individual package. Physical platforms, boarding faces and access assemblies are separate quantity bases. The current layouts identify equipment and interfaces; they do not release station structures, fire design or street entrances.

Station package and interface freeze

Freeze station ID/chainage, elevation and topology, physical platform lengths/widths, track spread/return, concourse and roof geometry, boarding faces, train/PSD clearance, shafts/stairs/escalators, accessible routes, utilities and drainage. Review actual boarding/alighting/transfer pulses, queues and equipment-outage capacity against the chosen layout. Include fire separation, evacuation and assisted rescue, cleaning/maintenance and emergency access. Confirm land rights for every entrance and pedestrian connection, rather than only the station centre point.

Coordinate structural spans/supports with the running viaduct and turnouts. Station widenings, lift shafts, concourse loads, roof/wind actions and launcher passage require a station-specific structural and temporary-stage design. Do not assume the running Pi25 deck or standard cap carries all station loads. Freeze load-transfer boundaries and dimensional datums before commissioning shop drawings.

The station lead prepares a delivery package with civil, architectural, MEP, fire/access, track, power/earthing, signalling/PSD and operations interface owners. Record sequencing constraints, tolerances and hold points. Select modular/precast platforms, stairs, shaft shells or steelwork only after installation access, lifting, connections, fire/durability and complete cost are reviewed.

Elevated station sequence

Stage	Construction method and coordination	Exit evidence
Enabling	Maintain roads, shop/pedestrian and emergency access; locate/divert utilities; construct isolated working platforms and approved crane/haul areas	Land/utility releases, traffic plan, ground/temporary-works checks
Foundations/substructure	Install station-specific foundations and verify; cast/erect columns, caps and shaft/concourse supports with temporary restraint	Actual foundation/test records, pre-pour inspections, strength and dimensional release
Primary structure	Erect beams/steelwork/platform support and concourse in checked stages; release bearings, connections and bracing	Stage-load checks, lifting records, welding/bolting/grout and survey acceptance
Platforms and edges	Install slabs/units, drainage, edge/PSD support and walkways; survey platform height/gap against track/train dynamic interface	Slab/connection records, tolerances and complete clear circulation envelope
Shafts and stairs	Build pits/shafts/landings and stair structures; fit temporary covers/barriers; keep access for equipment installation	Lift-maker interface, pit waterproofing, openings/loads, fire/access and fall-protection review
Roof and enclosure	Assemble checked roof/canopy/containment with restraints; complete flashings, drainage and weatherproofing	Structural/fixing and water tests, clear evacuation and maintenance routes
MEP and railway systems	Install ductwork, cables, earthing, plant bases, lighting, drainage, fire and passenger/PSD equipment against approved penetrations	Civil/system handover, isolation and inspection/test records
Access equipment	Supplier installs/commissions lifts and escalators after shaft/power/clearance acceptance; preserve stair access and protection during work	Certificates, functional/rescue/outage tests and independent access acceptance
Finishes and street routes	Install nonslip surfaces, wayfinding, lighting, barriers/crossings and accessible approaches; reinstate construction areas	Walk-through, drainage/lighting/finish checks and local street handback
Integrated handover	Test evacuation, assisted rescue, queues, PSD/train alignment, maintainability and outage operation with the final installed configuration	Accepted as-built station packet, resolved defects/restrictions and opening approval

Coordinate roof, platform-edge, shaft and track work so no crew works beneath another uncontrolled load or open edge. Temporary public routing cannot use an unfinished platform or uncommissioned lift. Define safe supplier installation paths for replacement motors/cabs/PSD modules; close permanent apertures only after large equipment has passed.

At-grade station sequence

Release land, crossings, utility isolation and traffic/pedestrian diversion. Construct drainage and reviewed earthworks/ground treatment; verify settlement and finished formation. Install services and foundations before slabs/platforms; provide protected draw pits and crossings. Cast or erect platforms and access structures with surveyed track/edge datum and bedding/support control.

Lay the selected slab/trackform and rail; complete platform faces, PSD support, canopy and systems. Maintain a continuous step-free street-to-platform path with drainage and crossings; isolate construction traffic. Complete lighting, signage, fencing, barriers and road/public-realm restoration. Test train/platform and access interfaces using the actual installed configuration and outage/rescue plan.

At-grade topology does not imply zero station civil cost. Soil treatment, retained levels, canopy/wind, service trenches, drainage/outfalls, safe street crossings and accessibility may govern the package. Elevated and ground-level interchange platforms need a reviewed transfer route and shared-resource capacity assessment.

Track spreading and running-viaduct interfaces

The generated running-span schedule excludes station/transition intervals. Resolve each excluded chainage with the station support plan, track-spread geometry, curves/cant, platform loads, movement joints and switch/closure interfaces. Confirm the station-end pier/cap boundary is counted once. Reconcile the approach lengths and support/beam take-off before ordering; do not fill the interval with rounded Pi25 bays.

Maintain a station-interface register covering datum, loads, tolerances, bearing/joint roles, waterproofing/drainage, egress, earthing/stray current, cable routes and erection access. The bridge and station authorities sign the structural/temporary-stage boundaries; track and systems leads sign their following-trade interfaces. Record whether the launcher crosses, parks short, or relocates around the station.

Logistics and workforce

Deliver bulky station structures and access equipment in installation order. Reserve off-road receiving space, checked crane/platform positions, shaft access and damage-free storage; keep lift/electrical components dry and secure to supplier requirements. Check dimensions and CG for stair/shaft/canopy modules independently of viaduct beam haulage. Plan returnable racks, packaging and rejection/replacement movements.

Reserve separate survey, concrete/precast, steelwork, roofing, access-equipment, MEP, track/PSD and independent inspection crews. Resource-share only when actual shift and task reservations allow it. Include work at height, rigging quantities, confined/pit access where applicable, lifting equipment authority, rescue competence, relief, heat management and handovers in the workload plan.

Hold points and handover

Stop before foundation production, primary lifting, temporary-support removal, covering reinforcement/ducts, waterproofing concealment, rail/PSD fixing and public access. Inspect platform equipment containment and continuous circulation paths after installation, including cabinets, covers and barriers. Test one-lift-out routing and throughput, assisted rescue, emergency power where selected, evacuation and smoke/fire assumptions; graph reachability alone is insufficient.

As-built deliverables include support/structure and service drawings, surveyed rail/platform geometry, installed component serials, material/connection/test records, operations/rescue and maintenance manuals, replacement access, defects, BCF/NCR closure and independently accepted station release. A civil handover for systems installation does not authorize passenger opening.

Other civil works construction methods

Source: docs/civil/other-civil-works-methods.md

These methods complete the civil master plan. Define project quantities and release criteria from survey, design and local approvals; route length is not an earthworks, pipe, pile or building take-off. Use the inspection and handover plan for concealed work and following-trade release.

Surveys, enabling, demolition and utilities

Establish verified horizontal/vertical control and a maintained survey network. Survey ground, existing roads/drains, water levels, structures, property, trees and utility routes; combine records with detection and trial holes. Resolve clashes in the common model and issue utility-owner approved protection/diversion/isolation plans. Locate working platforms, stockpiles and access outside protected services.

Before demolition or site clearance, establish ownership, hazardous-material/waste findings, structural sequence, adjacent support, service isolation, segregated removal route and disposal/recovery destinations. Maintain traffic, pedestrians and emergency access; record vibration, dust/noise and settlement where relevant. Protect retained assets and survey adjoining property before intrusive work. Restore survey control disturbed by construction.

Utility diversions precede conflicting excavation/piling. Construct replacement ducts/pipes and chambers, test, switch service under owner control, verify the new route, then remove/protect the obsolete asset. Record actual depths/offsets and access in as-built data before concealment. A drawing clash marked resolved is not field clearance.

Earthworks, formation and ground improvement

Strip and segregate reusable topsoil; test excavated material and decide reuse, treatment or disposal. Install temporary drainage and designed cut/excavation support before bulk works. Excavate/fill to surveyed profiles with inspected lifts, moisture/compaction control, test locations and proof/formation acceptance appropriate to the design. Track quantities using survey surfaces and truck/weighbridge records; account for swell, shrinkage and rejected material in haul planning.

Select rigid inclusions/load-transfer platforms, soil mixing, stone columns, stabilisation or lightweight fill by zone-specific settlement, drainage, cyclic loads, liquefaction and utilities/access constraints. Trial the method and verify equipment/material supply before production. Retain installation depths/geometry, mixing or compaction parameters, tests and settlement monitoring. The foundation/production system supplies selection interfaces, not site treatment depths.

Release formation only after drainage, bearing/settlement and survey acceptance. Protect it from delivery traffic and weather; repair and reverify damage before trackwork. Sequence treatment and trenches so later excavation does not remove or undermine accepted treatment.

Retaining structures, ramps and transitions

Review reinforced-soil walls, embankments, cast/precast retaining walls and abutments using land width, flood/scour, bearing/settlement, seismic and construction access. Install foundations and drainage, then facing/reinforcement and controlled fill in the designer's sequence. Keep heavy compaction/plant away from faces according to the approved method. Inspect hidden reinforcement, filters and outlets before each closure.

At grade-to-viaduct transitions, coordinate stiffness/settlement, rail geometry, movement, approach slabs, drainage and maintenance access. Survey as-built levels and monitor settlement through the selected stabilization/acceptance period. The 4.5 m retained-approach selection envelope is a catalogue planning rule, not permission to retain an unsuitable flood or soft-ground site.

At-grade and elevated track civil interfaces

For long accessible at-grade runs, compare the released continuous slipform method with constrained ST6 panel zones. Install tested formation, subbase and service/drainage crossings first. Survey line/level and reinforcement/inserts; cast and cure the selected slab or prepare bedding and erect panels on designed supports. Inspect bedding/grout, panel joints, anchors, drainage and stray-current interfaces. Use project-defined trial panels/slipform lengths before repetition.

Keep closure pours at station ends, turnouts, transitions and surveyed corrections separately designed. Record panel IDs and final coordinates. The slab-trackform catalogue has a 6 m single-track module; double track requires two rows. Panel mass is a bare study envelope; complete lifting/transport includes embeds, gear and configuration. Do not buy a rounded panel length that extends outside an identified method zone.

For elevated track, release deck geometry, movement, waterproofing and drainage before track supports. Fit local plinths/alignment layers and adjustable fasteners to the reviewed rail/structure design. Protect prestressing and

waterproofing during fixing. Rail welding/stressing, alignment/cant and fastening acceptance follow the railway specification; the civil plan supplies foundations and interfaces, not invented rail neutral temperatures or tolerances.

Drainage, water management and flood protection

Prepare catchments, storm basis, surveyed outfalls, receiving-system capacity, water-quality requirements and flood/scour scenarios. Design temporary drainage through all earthwork and foundation stages; avoid redirecting water into occupied streets or neighboring foundations. Sequence permanent culverts, cross-drains, attenuation and outfall protection ahead of dependent formation.

Construct foundations/bedding and pipes/channels, inspect joint levels, watertightness and access, then backfill in tested layers. Keep service separations and chamber access. Clean/CCTV or otherwise inspect the specified system; verify gravity falls and pumps/alarms/backups where required. Test before concealment and again before handover. Elevated deck drains, station roofs/shafts, abutment drainage and depot contaminated/washwater systems need separate catchments and controlled discharge.

Protect water crossings against construction runoff, debris, washout and accidental spills. Investigate erosion/scour, flood levels and safe construction windows. Temporary causeways/cofferdams, bank loading and navigational restrictions require hydrological/geotechnical design and permissions; no blanket dry-season assumption is adopted.

Special bridges, closure spans and road conflicts

The bridge/water civil class and unresolved running closures are separate inventories. Survey each conflict and compare rail bridge, road relocation/grade separation, longer-span steel/composite or segmental solutions, short verified products and alignment alternatives. Include road clearance, navigation/flood, utilities/property, foundations, track geometry, temporary works, traffic disruption and whole installed cost.

Issue a site-specific staged design and supplier package before buying special members. Define lifting/segmental or launching sequence, supports, stability, post-tensioning/joints where selected, closure geometry and rail/structure interaction. Coordinate permissions and shutdowns with adjacent running/station packages. A residual 23.3 m interval does not become a Pi25 catalogue beam, and a short segment does not automatically satisfy shipping width.

Depot, workshop and stabling civil

Freeze actual yard tracks, usable berth lengths, switches/leads, clearances, departure queues, charging positions and maintenance/recovery access. Reconcile declared storage against the depot scope. Baghdad's one declared heavy depot and 116 stalls do not establish physical overnight provision for all 772 trainsets; workshop bays and passenger platforms are not verified parking.

Sequence land/utility release, ground/drainage treatment, underground services, building/pit and equipment foundations, superstructure/enclosure, slab/pavements, yard track/turnouts, equipment installation, MEP/fire and commissioning. Inspect workshop pits for support, groundwater/waterproofing, ventilation/access/rescue and fall protection; coordinate jacking/bogie extraction and crane reactions with equipment suppliers. Do not load slabs or pits with plant before release.

Separate dirty washwater/oil drainage from stormwater; provide selected treatment and permitted discharge. Design stores, delivery/road turning, fire routes, staff access, waste, workshop extraction and equipment replacement. Fit charging/PV/battery bases, ducts and protected separation only after actual energy/fire/thermal requirements and usable layout are reviewed. Conduct yard-route and repeated-day commissioning with the final rail/charging graph.

Energy, equipment-room and systems civil

Survey every station/depot/wayside energy site. Freeze supplier foundation reactions, equipment dimensions, access/replacement envelopes, electrical isolation, fire separation, earthing and flood levels. Construct foundations, plinths, retaining and drainage; install ducts/chambers/cable bridges, sealed penetrations and equipment-room structure/enclosure. Inspect concealed earthing/ducts, reinforcement/inserts, water/weather protection and pull routes before backfill/closure.

Provide civil handover by equipment site: dry clean access, foundations and anchors surveyed, ducts cleaned/proved, load capacities and penetration rules, drainage, barriers, lighting and isolation boundaries. Grid energisation, BMS commissioning, signalling proving and fire acceptance are separate system gates. Protect battery equipment against construction dust, water, impact and unauthorized storage; supplier-specific fire/thermal data determines the permanent layout.

Roads, access, public realm and reinstatement

Create staged road/pedestrian plans for piling, deliveries and lifting, with emergency routes and local business/property access. Use booked off-peak windows only where approved; stop work if the approved segregation or access cannot be maintained. Plan diversions, crossing marshals and barrier/lighting inspections for every shift. Retain route access through future maintenance and recovery, not only construction.

After underground and structural completion, restore utility covers, pavements, drainage, kerbs, crossings, accessible paths, lighting, street furniture and landscape to local-owner standards. Survey and inspect reinstatement, remove temporary obstructions, repair monitored damage and close waste/land records. Hand back streets and property independently of railway opening; keep defect/maintenance responsibilities and warranties explicit.

Civil construction logistics plan

Source: docs/civil/construction-logistics-plan.md

The master plan uses 25 m average spans for logistics sizing and identified spans for orders. The generated logistics envelope distinguishes assumptions, required capacity, available evidence and adopted capacity. No surveyed delivery route or contracted supplier capacity is currently entered.

Factory and regional-yard network

Audit existing Iraqi precast facilities and production expertise before selecting expansion or a new factory. For each plant record bed usable length/width, mould positions, prestressing capacity, mix/curing control, cage production, handling/stacking capacity, available shifts, current commitments, inspection staff, power/water, raw-material suppliers, environmental permits and delivery access. Assess Pi20/Pi25, column/cap shells, walkways and ST6 separately; a suitable panel factory is not automatically a prestressed-beam supplier.

Select regional yards by permitted land, road/bridge access, geotechnical bearing/drainage, safe handling, flood risk, distance and delivery restrictions. Separate raw materials, cages/moulds, curing, accepted stock, quarantine, dispatch lanes and returnable gear. Layout access for the longest vehicle, fire/rescue and crane replacement. Design dunnage, restraint and storage stacking; the actual layout determines gross area and maximum stock. A count of beams cannot establish a yard area without supports, spacing and handling aisles.

Capacity equations and scenario boundaries

Requirement	Equation and qualification
Erection supply	`active launchers × accepted bays/launcher-working-day × 2 beams/bay`
Production positions	`ceil(accepted daily beam demand × mould cycle hours / production hours per day / first-pass yield)`; a position casts one complete beam per cycle
Weekly balance	`erection working days/week × daily demand`; compare with actual factory/haulage working days, cure time and outages
Fresh concrete	Beam-specific accepted count × released cast volume, adjusted for production rejects; foundation/station/other concrete is additional
Beam movements	One complete beam per proposed load; verify transporter configuration, axle/bridge capacity, load/support restraint and route permissions
Round-trip vehicle cycle	Loading + outward journey + unloading + return + delays/driver breaks; reserve loading/unloading equipment concurrently
Trailer requirement	`ceil(required trips/day / executable trips/vehicle/day)`; derive trips from the route's permitted window and complete cycle, then add independently justified resilience
Beam buffer	`2 × 10–15 released bays × active fronts`, limited by remaining continuous run, storage capacity and accepted handling/access
Ready support buffer	All unique supports for the next 10–15 bays, each released for the planned machine/beam load; endpoint sharing is explicit

The generated six-day scenarios use illustrative 48-hour production cycles, 24 production-calendar hours/day, 90% first-pass yield, a 12-hour haulage window and an eight-hour complete route cycle (including a declared allowance for return and delays). These are sizing assumptions, not supplier or route evidence. A 12-hour window is not permission for a driver to work without statutory rest. Verified driver/crew calendars further constrain executable trips.

Daily rates also need a compatible erection cycle. The programme controls deduct productivity, handover and maintenance from the proposed single shift; the existing eight-hour serial reference cycle does not support the higher one-shift targets. Resource rows show the cycle that would be required and retain achieved rate as unknown.

At eighteen active launchers and one accepted bay per launcher-working-day, the requirement is 36 accepted beams and 450 equivalent route-metres per erection day. Using the Pi25 complete-member study, beam cargo alone is about 3,078 t/day; trailer tare, rigging and all other materials are additional. Production at 90% yield needs 40 gross beams/day and 80 one-beam mould positions under the illustrative 48-hour/24-hour basis. These are capacity requirements to compare with available audited beds, not a proposal for eighty new moulds or a commitment to supply.

The 10–15-bay buffer requires 360–540 accepted beams across eighteen sufficiently long fronts. Existing conditional route examples use twelve beams per front, equivalent to only six double-track bays. The generated review flags that mismatch. Reconcile buffer policy, site/yard storage and front lengths before assigning a build rate; do not silently claim that six bays meets a ten-bay requirement.

Route survey and transport release

Create one route record per supplier–yard–front movement, with GIS path and direction, surveyed dimensions, gradients/crossfall, bends/swept path, weak bridge/culvert/pavement limits, overhead utilities, temporary widening, load permits, closures/escorts, delivery windows and alternates. Record the actual vehicle, axle spacing/load distribution, gross mass, loaded height/width/length, ground clearance, cargo CG, support/restraint design and braking/grade constraints. Gross tonnage alone cannot assess a bridge or pavement.

Load using inspected equipment and released beam/support points; verify member ID, mass, release strength, dimensions, restraints and travel documentation. Inspect after loading and at receipt. Use wheel/ground restraints and segregated receiving zones; place on designed storage supports or supply directly to the released erection position. Missing route, equipment or access evidence blocks dispatch even when a beam is accepted.

Release lift, transport and storage separately. The current Pi25 manufactured study is 85.51 t, versus 89.51 t suspended with illustrative gear. Transport gross mass also includes trailer/tractor and carried accessories. Neither the bare 75 t product target nor nominal equipment tonnage clears those loads. The handling envelope and supplier actual configuration govern release.

Booking, stock and material flow

Allocate each beam to span and track, factory slot, inspection gate, route, receiving yard/front, delivery appointment and installed record. Dispatch in erection sequence, including Pi20 and special interfaces; maintain location/condition and reservations in ERP. Reserve trailers, loading/unloading cranes, competent workers and route/closure slots before commitment. Keep rejected/awaiting-test stock separate from accepted available stock.

Coordinate concrete trucks, reinforcement/cage deliveries, spoil, station modules, lifts/escalators, rail/panels, fuel and systems equipment in the same access calendar. Check opposing flows and shared loading gates. Keep deliveries off pedestrian/emergency routes and avoid unplanned roadside queuing. Provide supplier/driver check-in, receiving inspection, safe standby, washout/spill and emergency procedures.

Record daily opening stock, accepted production, rejects/rework, loaded/dispatched, in transit, received, installed and closing stock by SKU/location. Balance mass and count; no item can be simultaneously available at the factory and workfront. Front saturation or a stopped crane/launcher pauses dispatch. Test loss of a supplier, road closure, trailer breakdown, rejected load, weather stop, buffer overflow and extended curing, with identified safe storage and controlled resequencing.

Material quantities, spoil and environment

Develop take-offs for concrete/rebar/prestress, foundations, station modules, track/panels/rails, bearings/joints, drainage/ducts, fill/ground treatment and road reinstatement. Convert them to delivery lots by approved vehicle/material constraints. The generated beam-only concrete quantity excludes other civil scope; do not present it as total project concrete or infer rebar from a generic kg/m³ study as a purchased schedule.

Plan permitted quarries/batching, water quality, standby concrete supply, curing/shade/temperature measures, pumping reach and rejected concrete disposal. Reuse suitable excavated material only after tests; measure outgoing spoil and transport conditions. Record disposal receipts, dust/noise/vibration controls, wheel cleaning, runoff/spill containment and packaging/returnable-rack recovery.

Launcher and station-module movements

Provide discrete transport/assembly/commissioning packages for each launcher transfer. Check dismantled component weights, unloading platforms, escorts/access, assembly crane and tested foundations/deck supports. Reserve transfer crew and equipment; assess additional ring fronts against remaining critical-path demand. The launcher is unavailable to both old and new fronts during transfer and recommissioning.

Station modules have their own load/route/installation packages and often share access with running beams. Reserve roof/platform/shaft lifts, maintain following-trade and emergency routes, and deliver access equipment before closing necessary apertures. Under-deck storage or commercial use requires fire, flooding, impact and inspection-access review; it is not automatically free logistics space.

Civil programme and workface controls

Source: docs/civil/civil-programme-and-controls.md

This is the scheduling and execution interface for the civil master plan. The generated package graph includes all fourteen civil packages per line and links the complete-line opening requirements for every current line. It supplies logic and quantities; dates stay unknown where surveyed scope, measured rates, contract calendars or release dates are missing.

Work-breakdown and asset records

Use programme city line/zone continuous run, station, special, depot or energy site work package approved Task serialized installed asset. Keep running span/support IDs, station/platform/access IDs and special/utility/drainage/yard references stable. Model station and transition scope explicitly rather than duplicating it in running bays. Bill quantities and costs by this hierarchy; link drawing/method/design revisions and suppliers.

Each Task carries location and IDs, method revision, design/temporary-works reference, predecessor release, measured quantity, required role quantities, reserved equipment/materials/crew, allowed shift/calendar, estimated/measured duration, hold points, completion evidence and independent acceptance requirements. Method authoring status cannot issue a construction permit. Use current native ERP eligibility and future reservations before assignment; revalidate at start and after a relevant change.

Programme stages

1. Mobilisation/design: appoint roles; land/survey/utility/ground investigations; approve worksite/temporary-works strategy and procurement scopes; qualify representative methods and trial products.
2. Enabling: create access, drainage and working platforms; divert/protect services; obtain road/water/land permissions and construction power/water.
3. Parallel production: qualify factories/first articles; construct verified foundations and substructure ahead; build station and special packages; book delivery and erection paths.
4. Running erection: draw from accepted stock and support buffers; complete two-beam bays with checked restraints and stage handover; track delivery/rejection and resource hours.
5. Interfaces and finishing: station approaches/special closures, continuity/joints, walkways/drainage/waterproofing, track support/rail, station enclosure/access, depot/energy civil and public realm.
6. Handover/testing: inspect complete assets, clear NCRs/restrictions, issue as-builts and systems access, complete integrated railway commissioning and independent/local approvals.

Foundations, columns/caps, beam supply and erection operate as linked crews and queues. Use a location/time chart to preserve the tested safe lead/lag and buffer, then a finite-resource CPM for stations, specials, utilities and handovers. Do not take the maximum of optimistic independent rates as the accepted programme.

Calendars and staffing

The proposed erection calendar is six days/week with an eight-hour shift before approved leave, holiday, heat/weather, maintenance and traffic-window deductions. No night work is credited without approval, supervision, lighting and relief. Factory curing and delivery operate on their own elapsed-time and permitted-shift calendars; a weekend erection stop can increase accepted stock and overflow a yard if production continues.

For the explicit single-shift capacity screen, 80% productive time less 0.5 h handover and 0.5 h maintenance leaves 5.4 productive hours/day. The existing connected reference has two 2 h placements, two 1 h securing operations and 2 h advance: an 8 h serial bay cycle, with two shifts in its original scenario. At one bay per day, a single-shift cycle would need to fit 5.4 h; at 1.5 bays, 3.6 h. Those targets therefore require measured method improvements or additional authorised shift hours. The generated review records this mismatch; it does not transfer the original two-shift rate into the one-shift calendar.

Calculate role-hours from actual task cycles and simultaneous fronts. Reserve qualified survey, foundation, pier/cap, launcher operator, two riggers where the method requires them, lift supervisor, independent inspection and handover resources. Add production/logistics, station trades and specialist installation crews by their own workloads. Protect supervisor/inspector independence, rest, breaks, leave, training, sickness and handover. A nominal role present does not satisfy required quantities or concurrency.

Recruit backwards from first-article/readiness dates through lead time, induction, training batches, supervised practice and practical equipment/task assessments. Supplier commissioning/technical support and equipment-authorised operators are separate dependencies. Training completion cannot substitute for work authorisation. The workload/recruitment model

leaves missing measurements and funded appointments open.

Running-front control and reassignment

Before the weekly plan, verify design/support/utility release horizons, accepted beam SKU stock, production slots, delivery appointments, equipment condition, crew competence and permissible work hours. Plan only tasks that can satisfy those constraints; show withheld packages and limiting resource. Review daily queue/buffer levels, cycle components, weather/traffic losses, NCRs and plant maintenance.

Treat each of the 548 disconnected Baghdad running intervals as a separate access/launcher-boundary review. Survey support spacing and actual relocation routes. Reuse a launcher sequentially only after dismantling or checked passage, transport, new-front readiness, reassembly/inspection and commissioning. Assess radial-to-ring redistribution with all resources and full-line milestones, then evaluate transfer to the next city. Record equipment exclusivity during every movement.

Cost and cash programme

Time-phase accepted quotation scopes against fabrication, factory acceptance, dispatch/arrival, installed milestones, tests and handover. Separate advance payments/security, retention, provisional items, recurring plant costs and final account. Finance ties to the same WBS and quantities; cash need may precede installation by months. Link delay/rejection/route changes to storage, demurrage, standby, interest and replacement, preserving unknown values rather than setting them to zero.

Use existing finance as a comparator until complete quantities/prices, scope credit and accepted dates are supplied. A higher launcher rate does not establish an installed-rate saving. Systems/fleet/testing/approval milestones remain separate from civil handover, and paid passenger revenue starts only through the accepted opening/service scenario.

Inspection, change and reporting

No next-stage release passes an open hold point or unresolved NCR affecting safety, load path or concealed work. Retain drawing/method/configuration hashes, observations, measured results, equipment/worker authority, reviewer identity/date and sign-off scope. Reinspect after transport damage, abnormal load, weather event, utility change or failed test. Quarantine changed/rejected items and reopen affected tasks/handovers.

Publish weekly planned/completed quantities by package, accepted stock, unique released supports, active fronts, installed span/beam IDs, station milestone state, labour/plant hours, rejects/rework, closures/disruption, forecast dependencies, costs/cash and decisions. Separate physical progress, documentary readiness, conditional forecast and accepted opening. Keep a reviewable change record showing input quantities, resource assumptions and conclusions that changed.

Complete opening

CW-14 civil handover requires CW-03/04/06/07/08/09/10/11/12/13 completion as applicable and controlled enabling/production records. Link it to running structures, stations, specials/transitions, track, energy and depot/stabling opening packages. Fleet, integrated testing and approvals are additional gates; the railway opens only after all are accepted for the same line/configuration. Missing package dates or costs keep complete opening date and cash unknown.

Civil inspection, testing and handover plan

Source: docs/civil/civil-inspection-and-handover.md

Apply to the master plan and method-specific asset register. The existing catalogue ITP retains reusable-product characteristics; this plan adds site/workface and full-line interfaces. Project designers define test methods, sampling, acceptance values and independent-review scope. Empty criteria or unsigned results cannot close a hold point.

H = stop until authorized release; W = notified witness opportunity under the agreed ITP; R = controlled record review. Each row must name responsible inspector, independent/owner role where required, current asset/design/method revision, actual test/result, acceptance basis, date and signed disposition. Proposed role titles are not approvals.

Stage	Type	Required evidence before following work
Survey/utility/land	H	Verified control, support/station/site coordinates, trial-hole and isolation records, rights and construction permissions
Ground and foundation method	H	Site ground/water model, released foundation/ground-treatment design, trial method and specified verification tests
Working platform/temporary works	H	Design/check, actual ground/setup inspection, limits and permit to use/load; subsequent change checks
Foundation installation	W/R	Strata/installation geometry, actual length/count, driving/drilling/treatment logs, materials and adverse events
Foundation acceptance	H	Required load/integrity/settlement results, dimensions and connection datum, resolved NCRs
Reinforcement/inserts before casting	H	Released drawing/bar schedule, cover/position, tendons/ducts/inserts, earthing and lifting interfaces
Concrete and grout	W/R	Mix/batch traceability, fresh properties, weather/temperature/curing, specimens/maturity, pour continuity and tests
Prestress transfer/demould/handling	H at each distinct release	Stress/calibration records, stage-specific strength, cracks/camber/dimensions, approved lifting/support design
Factory acceptance	H	Serialized complete member, materials/geometry, mass/CG and handling envelope, traveller and NCR closure
Transport dispatch	H	Accepted item, actual vehicle/axle/load restraint, surveyed route/permits, appointments and receiving readiness
Receiving	W/H for damage or discrepancy	Correct identity/condition, transport inspection, supported storage and traceability; quarantine discrepancies
Column/cap/bearing support ready	H	Foundation, connections/strength, survey, bearing type/orientation and next launcher/delivery-load release
Launcher assembly/commissioning	H	Supplier configuration, stage checks, assembly inspection/tests, maintained equipment and authorised crew
Each beam placement and bay release	H	Correct span/track/member, lift plan/load/gear, seats/contact/survey, temporary restraint and completed stage path
Continuity/joints and restraint removal	H	Selected structural scheme, reinforcement/grout/concrete /bolts, movement datum, strength and designer removal permission
Waterproofing/ducts/drainage concealment	H	Inspected continuity, protection, levels, joint and clean/pull/water tests, surveyed as-built routes
At-grade formation/track support	H	Treatment/compaction/settlement, drainage, line/level and bedding/slab/panel records
Station structure/access	H	Structure/fixing/shaft/roof records, platform/train/PSD interfaces, continuous accessible circulation, equipment commissioning
Special crossing	H	Site-specific independent design/stage acceptance, road/water permits, closure fit and final structural inspection
Depot/energy civil handover	H	Actual yard/equipment layout, foundations/pits/ducts/earthing/drainage, fire/replacement/maintenance access, tests and load restrictions
Reinstatement/public handback	H	Utility/road owner acceptance, accessible routes, drainage, waste/property records and defects
Structural/civil handover	H	As-builts, quantities, installed serials, complete inspections, monitoring, manuals, warranties and controlled remaining restrictions
Complete railway opening	H	Civil plus station/track/energy/depot/fleet/testing/approv al acceptance for the same installed configuration

Test planning and first repetition

Select representative foundation/ground zones and temporary-loading configurations, Pi20/Pi25/cap/column production and transport/lifts, complete first bay with connection/track/walkways, first station primary structure and access equipment, first at-grade slab/panel zone, special crossing and depot/pit/energy interfaces. Test both ordinary installation and credible degraded/recovery conditions. The representative unit does not qualify unlike ground zones, structural products or equipment configurations without a reviewed extension.

Record baseline calibration, test setup/instrumentation, operator/inspector competence, measured values, uncertainty, raw files, criteria and configuration. Review results before shortening cure cycles, increasing erection rates, reducing restraints/buffers or repeating a changed method. Failed results stay in the package and trigger investigation/rework/retest under a controlled NCR.

Following-trade and final handover

Use a signed boundary packet for each run/station/site: allowed loads/work/access, residual temporary works, edge protection, drainage/earthing, penetrations, installed component IDs, inspection state and outstanding restrictions. Systems contractors accept the civil interface before mounting, pulling cables or energising. No shared folder or completed-task flag substitutes for that packet.

At final handover, reconcile every purchased/installed item to the asset and drawing revision; retain survey/model/IFC references, supplier certificates, material/installation/test records and NCR/BCF closure. Transfer inspection/maintenance intervals, bearing/joint replacement and recovery access, drainage cleaning, warranties/spares and monitoring baselines. Record owner/operator/independent and local-authority acceptance separately.

Store native ERP references for Project, Task, Asset, construction release and competence/reservations, with access-controlled personal records. Public planning artifacts contain proposed roles and unaccepted evidence templates; they do not publish invented inspector identities or customer approvals.

OSR Rapid Viaduct Kit — Design Basis

Source: docs/civil/viaduct-design-basis.md

Status: controlled planning basis; not a construction design
Primary products: OSR-Pi25 and OSR-Pi20 prestressed decked single-track beams Special products: OSR-U25-SPECIAL acoustic/project-specific trough, OSR-US segmental U/box, and OSR-SP special crossing

Catalogue boundary

Two identical 2.9 m-wide decked pi-beams share a single-column pier at the same 3.5 m tangent track centres used at grade. Each beam has two stems beneath the rail lines and a full-depth upper flange forming the structural track deck. Separate 1 m outer walkway/barrier cassettes give a controlled 8.5–9.0 m twin-track envelope. Pi20 and Pi25 come from one long-line mould with movable stop ends. A 30 m beam is not in the catalogue.

Product	Controlled use
`OSR-Pi25`	Tangent and broad-curve normal span; target bare mass 65–75 t
`OSR-Pi20`	Closure bays, junctions and utility avoidance; target bare mass 50–60 t
`OSR-U25-SPECIAL`	Project-specific acoustic/containment trough after transport, lift and structural release
`OSR-US`	2.5–3.0 m match-cast segments where access or curvature prevents full-span erection
`OSR-SP`	Separately engineered I-girder, steel-composite, or segmental crossing over 30 m; turnouts and exceptional crossings

The OSR-US review model places each 2.5–3.0 m chord on a surveyed circular arc, with rotated match-cast joint planes and chord-following tendon corridors; it is no longer represented by a straight full-span extrusion with markers.

The default Python geometry is `osr_mech.civil.decked_pi`; legacy troughs stay in `osr_mech.civil.ugirder` as special products. Flange/stem proportions, diaphragms, anchorages, tendons, lifting zones, reinforcement, fatigue, derailment, vibration and rail–structure interaction remain supplier design and independent-check releases.

Actions and design situations

Use the deployment country's adopted standards and combinations. The design must include the complete deployment train: the LM3 reference has 12 axles; Baghdad's six-car baseline has 24 axles. Its supplier axle positions and loaded axle distribution must be frozen before moving-load analysis. A 16 t infrastructure axle allowance does not replace the full consist loading pattern. Also include dynamic amplification, braking, traction, nosing, centrifugal force, wind on empty and loaded trains, derailment/containment, collision, maintenance and rescue vehicles, temperature, creep, shrinkage, fatigue, seismic/foundation interaction, and every lifting, transport, temporary-bracing, and erection stage. The machine-readable seed is `viaduct-load-model.toml`.

Decked pi sections require three-dimensional analysis for torsion, transverse distribution, asymmetric loading, local flange/stem actions, end-zone bursting, camber, creep, shrinkage, fatigue and staged prestress. Final reinforcement, tendons, anchorages, bearings, foundations, rail/bridge interaction, and temporary works require independent checking by licensed bridge and geotechnical engineers.

Geometry and constructability gates

- A straight 25 m full-span unit is preferred at radii of about 300 m or more.
- Below about 300 m, use Pi20, OSR-US, a shorter verified product, an I-girder deck, or realign. A 90 m system curve is exceptional elevated geometry.
- Four normal spans may form a semi-continuous unit. Four bearings at internal piers require a checked structural continuity connection and its staged load path. A link slab joining simply supported decks retains eight bearings at internal piers; eliminating a deck joint does not eliminate bearing lines. Both configurations have four bearings at end supports. Price both using the bearing schedule, including connection, CWR, temperature, braking, seismic and foundation effects.
- Parapet/containment height is measured above the finished escape walkway, not above the structural soffit or pre-track floor.
- The trackform is local direct-fixation plinths over the waterproofed structural floor. A full structural topping is a project-specific acoustic or vibration option, not the default.
- The shared hollow/precast-shell cap is 6.5–7.5 m wide and targets 25–35 t.

- Foundation selection uses `foundation-catalog.toml`. Deep-element geometry and cost require actual site length; no six-metre placeholder is permitted.
- Foundations and accepted beam stock are held 10–15 bays ahead of erection.

Automated catalogue checks live in `osr_mech.civil.viaduct`. Passing them is necessary but not sufficient for design release.

Minimum release evidence

The release package must close survey and utilities, ground model, design basis and combinations, global and local analysis, prestress/reinforcement, fatigue and durability, bearings and movement, rail/bridge interaction, drainage and egress, transport/erection, first-article testing, independent check, and deployment-authority approvals.

Viaduct Bearing And Movement Schedule

Source: docs/civil/viaduct-bearing-and-movement-schedule.md

Compare two twin-track schemes over the same four-span movement unit. Bearings remain replaceable in both. The reduced arrangement is an unaccepted structural alternative, not a saving established by a link slab.

Support	Tracks	Bearing lines	Bearings
Abutment/end support	2	1	4
Internal pier, structural continuity connection	2	1	4
Internal pier, simple spans with link slab	2	2	8
Expansion-unit boundary	2	2	8

Internal bearings are in one longitudinal row; adjacent unit ends retain two independent rows at an expansion boundary. The supplier schedule must assign fixed, guided, and free directions for every span end and show design reactions, translations, rotations, uplift restraint if required, replacement loads, installation temperature, and identification marks. Permanent jacking shelves and a checked load path are mandatory.

The 25 m reference places a deck gap every four spans, nominally 100 m; the permitted four/five-span family stays within 80–125 m. A structural connection must establish the shared support load path, including construction stages, prestress, reinforcement and fatigue. A link slab may remove a deck joint while each girder retains its independent end bearings and permitted rotation. For the periodic 25 m reference kilometre these alternatives require 200 and 320 bearings respectively. A finite 40-span structural layout also has 200 bearings: 30 internal supports, nine boundaries and two end supports. It has 41 supports and nine expansion gaps; the periodic index has 40 supports and ten gaps. Do not mix periodic and finite corridor quantities. The project schedule must close thermal, creep, shrinkage, seismic, braking, bearing-friction, connection fatigue and differential-settlement actions; provide covers over walkways and cable routes; and complete continuous-welded-rail/bridge interaction. Rail expansion devices are installed only where that analysis requires them.

The FHWA UHPC connection guidance describes a link slab over simple-span girders with rotating/moving bearings. It is a highway precedent, not a released rail design. Connection cost, foundation forces, maintenance access and bearing replacement must accompany both bids before whole-life selection.

Viaduct Transport And Erection Envelope

Source: docs/civil/viaduct-transport-and-erection-envelope.md

The bare OSR-Pi25 envelope is 74,937.5 kg, leaving only 62.5 kg below the 75 t product target before net details. Pi20 is 59,950 kg. Complete-member and hook masses remain unresolved. Use the generated mass and equipment summary and its controlled budget before selecting transport or erection plant. The 3.0 m shipping-width and 75 t member targets require product compliance or an independently reviewed deviation bound to the actual mass budget.

Compare the Baghdad complete-package alternatives before selecting plant. Pi20's bare mass is also only 50 kg below its 60 t product target; smaller is not automatically compliant. Price the extra supports, foundations and complete-bay cycles alongside lighter handling. Overhead delivery over completed spans adds construction loads and launch reactions requiring their own deck, pier and temporary-works analysis.

A portal, launching frame or certified strand-jack arrangement is an erection option awaiting selection and qualification. Its required capacity comes from complete member, rigging and dynamic demand at the reviewed configuration; an approximate 80 t label cannot establish suitability. The number and weight of lifts come from the actual asset layout.

Before release, record the casting-yard route, swept path, bridge and culvert capacities, overhead/side clearances, pavement and temporary works, transporter axle configuration, escort/closure plan, launcher reactions, strand-jack synchronization, wind limits, lifting points, temporary bracing, bearing survey, contingency landing position and rescue/recovery method.

For isolated packages a mobile or crawler crane may be used only against its actual configured load chart. The release records load, rigging, lift radius, wind limit, outrigger/crawler reactions and verified ground-bearing pressure; nominal crane tonnage is not accepted as proof.

Where the route or worksite cannot accept Pi20/Pi25, use OSR-US match-cast 2.5–3.0 m segments with an overhead launcher only after verifying every transport constraint. The current OSR-US coordination section exceeds the primary 3 m shipping-width target: shorter segments do not resolve restricted width. A supplier redesign or reviewed oversize route is required. Its release adds segment geometry control, match-cast records, epoxy joints, shear keys, post-tensioning, grouting, launcher stages and specialist QA. The legacy OSR-U25-SPECIAL trough is retained only where project acoustic/containment needs justify its wider and heavier transport envelope.

Viaduct First-Article Test Plan

Source: docs/civil/viaduct-first-article-test-plan.md

The first production mould, first Pi20/Pi25 unit, first selected foundation in each geotechnical zone, first cap shell, first erected bay, and first completed track/egress bay are hold points. Production may not free-run until the responsible engineer and independent checker accept the applicable evidence.

Hold point	Minimum evidence
Foundation zone	Utilities/CPTu/boreholes, selected catalogue type, actual pile/shaft count and length, installation time and cost, concrete/rebar quantity, representative load test and as-built survey
Mould and casting bed	Survey, stop-end settings, tendon/rebar/inserts, release agent, concrete process, calibration
First girder	Materials, cube/cylinder strength, maturity, prestress force/elongation, transfer sequence, dimensions, camber, cover, NDT, drainage test
Trial lift and transport	Certified portal/launcher or actual crane load chart, rigging, radius, wind and ground-pressure release, load test where required, strain/deflection observations, transporter trial and route clearance
Cap/column connection	Reinforcement and connection record, geometry, bearing-seat survey, grout/concrete acceptance, jacking interface
First erected bay	Bearing identity/orientation, role-selected one/two-row survey, reactions/level, girder geometry, gaps, stability, diaphragm/end-zone inspection
First selected connection unit	Four-span staged analysis correlation; distinguish structural continuity/shared bearings from simple spans/link slab; cage and closure record, maturity/grout release, waterproofing, fatigue detail, CWR/braking/temperature/seismic/foundation-flexibility closure
Track and egress	Waterproofing, local plinth geometry, fastener proof/torque, drainage flow, 1.0 m clear walkway, 1.4 m containment datum, gap covers, earthing
Structural acceptance	Correlated analysis; static/dynamic proof test if required by the design basis; residual deflection/cracking criteria; independent-check closure

Nonconformities require disposition, repaired-item acceptance, root-cause review, and demonstrated recurrence controls before the production hold point is released.

Record the complete double-track bay cycle: accepted foundations and caps, casting/strength release, transport, both track beams or all segments, closures, survey and tolerance corrections, waterproofing, track/egress and acceptance. Include crew/equipment hours, curing delays, rejects/rework, stock buffers and traffic possessions. Beam lifts per day alone cannot validate the programme.

Viaduct building, support and terrain clearance

Source: docs/civil/viaduct-obstacle-clearance.md

Status: controlled planning screen; physical release blocked. Straight horizontal routes can overfly low buildings only with a verified vertical alignment and property/air rights. Elevation does not remove a tall obstacle or provide a place to build a pier.

`tools/automation/audit-viaduct-clearance.py --all --retain-inputs` retains source footprints near the actual full-resolution corridor and any available height tags and terrain. `--all --check` repeats the calculation offline and rejects altered sources or stale corridor evidence. Every city's `engineering/clearance/README.md` reports beam/building, support/foundation and terrain findings separately; the compressed register records actual chainages. Unknown heights, omitted footprints, invalid geometry, missing terrain, utilities and property approvals remain unresolved even if a reference screen finds no collision. These reports do not alter the accepted scenario or prices.

The reference swept envelope includes both tracks, deck/walkways and a lateral allowance. A building below a provisional soffit is a conditional overflight, not a released design. Each pier's entire foundation envelope is tested separately. A terrain crest is tested against the straight beam between its supports; provisional support elevations are checked for excessive railway grade. SRTM-derived tiles can contain surface/roof/canopy returns and must not be represented as a precise bare-earth survey.

The current screen uses these explicit planning assumptions:

Item	Assumption
Twin-track structure/walkway envelope	9.0 m
Extra lateral building allowance	2.0 m each side
Foundation reference radius / setback	3.0 m / 2.0 m
Rail height above support ground	12.0 m
Conservative rail-to-soffit depth allowance	1.6 m
Minimum provisional roof-to-soffit allowance	2.0 m
Pi25 / Pi20 reference span	25 m / 20 m
Reference maximum gradient	3.5%

The Pi20/Pi25 bare structural section is 1.155 m deep in `design/component-catalogue/src/osr_mech/civil/decked_pi.py`. The 1.6 m rail-to-soffit screening allowance leaves 0.445 m for the unverified track/rail stack. Special U/box/crossing products need their own supplier geometry; this Pi reference cannot establish their clearance.

These values require project approval and supplier/site design. They are not deployment-country statutory clearances. Building storeys times 3.5 m provide an explicitly uncertain estimate only; they do not replace a height survey.

To resolve a conflict, reroute or secure an approved property solution; raise the railway with grade-compliant approaches and revised structure costs; move support lines using verified Pi20/Pi25 spans; change independently checked foundation/cap arrangements; or release a project-specific crossing. The catalogue contains no ordinary 30 m full-span beam. Boundary and special span layouts remain unresolved until independently designed.

Release requires a surveyed building/roof/plant and ground model, longitudinal and cross-section profiles, utilities and protected zones, property/air rights, fire and rescue access, flood levels, vertical curves, railway/bridge movement, pile-group and cap layout, temporary works and each transport/erection stage. Any adopted geometry change requires regeneration of stations, quantities, costs, schedule, financial cash flows and native service evidence.

OSM height tags and Overture building/part height fields can supplement footprints, but neither source establishes mapping completeness. See the OSM height definition, Overture building documentation and Terrain Tiles source.

Civil allowances and routing scores

Source: docs/finance/civil-allowance-boundary.md

The civil estimate prices route length at controlled at-grade, elevated and bridge planning rates. A curvature deterrent is a route-search score, never a supplier price or a monetary multiplier. Local three-point radii use a 100 m route window on each side; elevated construction-method runs split at local product changes. The radius remains inferred raster geometry, rather than a surveyed design.

`refresh-local-civil-costs.py` calls the native classifier against the published full corridor geometry and preserves existing civil class boundaries, routes, station positions and fleets. It records an actual cost migration; it does not rewrite previous solver receipts. Changes to Rust or controlled designs require genuine new full-day and degraded planning runs.

The displayed capital is a base planning allowance, not a complete installed estimate. Tangent Pi25/Pi20 quantities can use their respective catalogue envelopes. OSR-US and exceptional curves require their own supplier geometry, connection, support, erection and foundation quantities. Missing rates or quantities remain unknown; they are never zero-cost scope or credited savings. Each special crossing is separately engineered. Actual foundations need surveyed ground zones and deep-element lengths. Transport and erection require complete mass budgets and configured equipment proposals.

Removing the old segment-wide search penalty corrects accounting. It establishes no saving against a buildable, priced alternative. Land, rights, utilities, special-structure increments, installed grid/charging upgrades, taxes/duties, escalation, risk and financing must close before the base allowance becomes a complete delivery budget. Maintenance charged on a smaller allowance is a conditional calculation, not evidence of reduced maintenance workload. Staffing, energy, train-km and replacement costs must retain their independent quantities.

Demand coverage, capacity and receipts are separate: population pixels describe residents near stations; network walksheds need verified entrances, accessible paths, crossings and vertical access; surveyed origins/destinations and jobs determine route choice and paid journeys. A transfer creates another boarding, not another paid journey under an integrated fare.

Baghdad cost and demand review — 5 October 2026

Source: docs/baghdad-cost-and-demand-review-2026-10-05.md

Controlled planning basis; construction prices, surveyed demand and committed financing remain unresolved.

The monetary model now excludes route-search curve penalties. Local curvature changes determine construction-product boundaries, while tangent length cannot spread a remote bend's penalty. This correction preserves Baghdad's controlled 479.0 km alignment, 186 station records, 772 six-car trainsets and nine full-fleet depots.

Same-alignment reference	Reviewed baseline	Corrected base case
Programme planning allowance, USD bn	15.233	8.312
Residual unfunded support, IQD tn	27.330	0.0326
Terminal debt, IQD tn	13.000	11.162
Company NPV before finance, USD bn	−10.591	−4.848
Permanent operating FTE	3,716	3,716

The lower allowance removes a pricing artifact. It is not a proven saving or a complete installed budget. Special structures, actual foundations, installed grid and charging upgrades, land, utilities and supplier qualification still require quantities and quotations. Maintenance is partly a capital-percentage proxy; its reduction does not demonstrate fewer physical maintenance tasks. Staffing, payroll and traction duty are retained. The source-bound reconciliation retains the original reviewed files for offline reproduction.

The reference allocates USD 1.983bn to modelled imported invoices. This includes origin shares inherited from benchmark allowances, rather than verified import quantities or supplier commitments. Government USD cash and Chinese USD credit each cover half, approximately USD 0.991bn. Other credit, bonds and cash are denominated in Iraqi dinars. Government's total capital contribution remains 25%. Imported invoices represent 23.85% of the corrected base allowance: the share rises when an inflated local-cost denominator is removed, even though modelled absolute imported exposure falls. These are uncommitted model allocations. Neither exchange-rate risk nor an outstanding IQD liability disappears through currency labelling.

Month 101, annualised from one nominal model month after the revenue ramp, gives USD 652.6m of fares and existing commercial receipts, USD 25.0m of additional assumed receipts, and USD 519.1m OPEX. The reference operating balance before debt service is USD 158.6m. This is not a calendar-year forecast or evidence of debt repayment: terminal debt remains, no debt-clearance date is recorded, and the financing case is not closed.

Integrated fares must charge a passenger journey once. The service-capacity reference assumes one boarding per paid journey and is a zero-transfer upper bound. Keeping trains, operating costs and non-fare receipts unchanged gives:

Mean boardings per paid journey	Month-101 annualised operating balance before debt, USD m
1.00 reference	158.6
1.25 sensitivity	34.2
1.50 sensitivity	−48.8
2.00 sensitivity	−152.5

These conversion factors are uncalibrated sensitivities. Population catchments do not establish fares or train loading. The demand handoff validates unique origin–destination cohorts, declared transfers and explicit tariffs, but adopts no surveyed forecast without evidence. Jobs, station entrances, pedestrian routes, route choice and peak section loads remain open. The retained road export is insufficient to claim verified entrance-based walking catchments.

The monthly financing ledger includes escalated fares and OPEX, commercial receipts, construction phasing, debt service and surplus sweeps. Mezzanine funding changes the liability profile; it does not establish sustainable repayment. Unknown construction increments must be priced before selecting a funded case.

Clearance results distinguish provisional support-to-support terrain gradients from a designed rail profile. DEM noise attribution, surveyed vertical alignment, building heights, foundation access and property rights remain unresolved. Event counts are not unique buildings, confirmed defects or procurement quantities. Depot launch and charging capacity remain separate acceptance gates. See the clearance screen and civil allowance boundary.

Across the catalogue, disconnected line networks receive candidate transfer links for survey, without invented walking paths or new accepted connections. Wau is filed under South Sudan; its retained `waw` slug is a compatibility alias. South Sudan's finance seed is explicitly generic and uncalibrated. Current city simulations and package checks bind their actual inputs and executable evidence; planning completeness does not release a physical design.

Repository Artifact Policy

Source: docs/repository-artifact-policy.md

Git contains the useful source and review artifacts needed to understand, inspect, build, test and change OpenSourceRail without specialist regeneration. That includes CAD, IFC, PDFs, images and animations when they help a GitHub user inspect or build the system. Reproducible scratch output still belongs in `build/`; CI enforces a straightforward 50 MiB limit per tracked file.

Tracked

- Rust, Python, shell, HTML, CAD-generator, and configuration source.
- RFCs, engineering requirements, safety evidence authored by people, and stable schemas.
- Small test fixtures with a named consumer.
- The checksum-pinned TLC assurance executable, retained as a named CI input because its upstream download URL is mutable. Its embedded licences and reviewed byte identity are preserved.
- The compact machine-readable city core for every entry in `lib/city-batches/world-sample.toml`: `design.toml`, simulator scenario, quality result, corridor GeoJSON, and station JSON. Routed designs are retained because regeneration can depend on changing external OSM and population data.
- Catalogue-wide ring/interchange and station-cluster validation reports. These reports are allowed to record inherited planning failures, but their source and validator hashes must remain current; reviewed references must pass both validators.
- Compact per-city soil/deployment and workforce/design review packages, including workforce CSVs and their input hashes. Large local operations bundles are bound through their tracked compressed-hash manifests; checking a compact report does not require downloading a complete operations bundle.
- One generated `NATIONAL-BRIEF.md` per country, aggregating city CAPEX, imported/external capital, local funding, and the single shared trainset factory.
- Two complete end-to-end acceptance references, Samawah (`light-metro-3car`) and Mosul (`metro-4car`), including map, README, alignment, simulation, energy, finance, QA, maintenance, manufacturing, acceptance evidence, and a fail-closed package manifest. Their simulation evidence includes nominal operation plus end-of-life battery, maximum planning-climate load, reduced charging-contact availability, all-site grid outage, and single-pad outage cases.
- One compact high-throughput architecture reference, Basra (`metro-6car`), adding a map and README without duplicating the complete acceptance bundles.
- Baghdad's complete planning proposal, supporting-data archive, source/output manifests and compact engineering/finance studies, including the manufactured viaduct comparison. Each tracked file remains below 50 MiB. The supporting archive preserves hash-bound ignored inputs for the clean-checkout bootstrap; it is not field acceptance or an independent source of revised assumptions. Current generated studies, explicitly superseded reviews and diagnostic sensitivities retain distinct labels; historical prices and returns do not describe a later scope.
- The Samawah OSR-ALN fixture used by converter tests.
- Compact FreeCAD and Blender review assemblies, including the LM3 selectable manufacturing-tool set; the named civil and LM3-manufacturing IFC4.3 review federations, native civil construction scene and complete sub-50-MiB review animations under `engineering/models/bim/reference/`; stable documentation images, and the root `OpenSourceRail-Book.pdf` reader edition. These are published review aids generated from authoritative sources, not parallel inputs.

Not tracked

- Rust/Python build products, logs, caches, temporary data, and screenshots that are not part of the stable documentation review set.
- High-volume generated city engineering and operations inputs except the named Mosul/Samawah acceptance payloads and Baghdad's hash-bound supporting archive. Baghdad's extracted ignored inputs remain local after bootstrap.
- Generated BOM exports, simulation traces, release-only brochures, unnamed or repeated videos, CAD/BIM job outputs, and other release binaries.
- Files that only preserve an obsolete schema, chemistry, voltage, or API name.

Output locations

Output	Location
Full city package	`cities/catalogue/`
Repeated simulation and engineering job output	`build/engineering/`
Public CAD review set	`design/component-catalogue/models/cad/`
Public BIM coordination review set	`engineering/models/bim/reference/`
BOM exports	`build/bom/`
Published reader book	`OpenSourceRail-Book.pdf`
Release-only media	`build/releases/` before tagged release storage
Local validation receipts and raw logs	`build/validation/` (ignored); reviewed public release evidence is packaged separately
Tagged deliverables	Release storage with version and checksum

Use `tools/automation/regenerate-city.sh <slug>` for a full city package in the canonical `cities/catalogue/` tree. Catalogue changes must include the corresponding current-generator scenario and pass the catalogue-completeness health check; generated engineering and operations directories remain tied to the city folder that produced them.

Rules

1. Change canonical source before regenerating output.
2. Do not hand-edit generated output.
3. Every retained fixture must name its generator and test consumer.
4. Local validation compares compact fixtures and regenerates larger evidence when required; GitHub Actions are disabled. See local processing.
5. Never commit credentials, caches, absolute local paths, or live operational databases.
6. A complete generated city package is not release-ready merely because its manifest passes: survey, calibrated demand, local ground/fire/utility data, supplier freeze, and independent safety/construction release remain external project gates.
7. Tracked reports link only to tracked review artifacts; local high-volume outputs are named as code, not presented as GitHub links.
8. The root reader PDF is generated, never hand-edited. Its Markdown and city model sources remain authoritative and are printed in the book.
9. Keep useful CAD/BIM/media artifacts in Git when each file is at most 50 MiB. `tools/automation/check-tracked-file-sizes.py` enforces the ceiling. A future history optimisation must preserve access and requires an explicit maintainer decision; it is not part of routine cleanup.

Documentation ownership

- The root `README.md` is the only human-facing navigation page and owns the common setup instructions and source-of-truth map.
- A discipline or folder README contains only material local to that subject; it links to common guidance instead of copying it.
- City and country pages contain only local values and evidence. Shared planning interpretation lives in `docs/deployment-planning-reference.md`.
- `docs/INDEX.md` is a generated inventory for search and CI, not a competing guide or technical authority.
- When a claim can be derived, generate it from the authoritative source and test for drift rather than maintaining a second prose value.

Baghdad delivery review — regenerated planning scope

Source: docs/baghdad-delivery-review-2026-10-04.md

Originally reviewed baseline: `135e249e20f0630f92f426764ea72d21e46c2a68`. The figures below are regenerated from current inputs, rather than attributed to that historical revision. This delivery-baseline accounting retains legacy depot and payroll allowances; the latest cost and demand review and expanded programme recalculation carry the current selected planning case. It is a targeted delivery review, not a line-by-line audit or investment/construction release.

The controlled city remains 14 lines, 513.3355 km, 159 stations and 809 six-car trains. The delivery-baseline city allowance is USD 7.626537bn, and one Baghdad factory including EPC brings that accounting basis to USD 7.951287bn. These amounts exclude the subsequent expanded-scope adjustments and are not the selected programme total. Its monthly capital table has 82 periods and conditional full-line openings are 40, 41, 43, 42, 44, 44, 50, 55, 59, 64, 69, 73, 78, 81 months; the facility's 18 months starts at proposed NTP. The older 347-month rollout and earlier funding totals are historical and superseded. Proposed government funding remains 25% of capital, imports 50% government USD / 50% Chinese USD loan, with other funds in IQD. No wider national revenue is added.

The delivery package reconciles baseline WBS, exclusions, depot quantities, six-car product, chronological energy, demand/fleet phasing and workload/competence. None of its unquoted alternatives is adopted into debt, fares or official capital. Depot layouts, land/utility/title, local pay/workload, OD, equipment quotations and six-car first-article release remain open. Named owners cannot be manufactured by a role generator.

The legacy depot allowance in this accounting is USD 8m. The expanded programme replaces it with nine quantity-based full-fleet depots; it must not be read as the current depot provision. Unresolved workshop sites, morning-direction checks, overlap and unpriced scope remain acceptance gates. This delivery study's storage/workshop alternatives are quantity-based, with explicit baseline-overlap and unpriced-cost fields. The legacy six-car allocation sums to USD 1.68m/train but does not imply qualified LM3 applicability or supplier quotations.

Hourly synthetic owned-solar reference purchases 942.7 GWh, costing USD 94.274m/year before separately identified wheeling/balancing/connection/O&M. These are sensitivity outputs from synthetic weather and aggregate pooling, not a Baghdad forecast. Unserved energy and per-site physical/contract constraints remain visible. Staffing uses explicit workload/shift cover rather than dividing a legacy labour allowance. The expanded programme's establishment and payroll supersede that allowance; pay, productive hours and roster authorisations remain unaccepted.

Fresh checkout: run `.venv/bin/python tools/automation/bootstrap_baghdad_tests.py`, then `.venv/bin/python -m pytest tools/automation/tests design/component-catalogue/tests -q`. The bootstrap validates the tracked proposal archive and selectively restores the exact ignored operations payload. It preserves differing workspace data and does not resynthesise designs. Install the existing test/render dependencies, including `mistune` and `reportlab`, before publication tests. Validate the delivery package and proposal with their generators' `--check` modes.

Six local ERP evidence drafts carry estimator work budgets, accountable functions, source revisions and independent exit evidence. No appointment, quote, lease, lender commitment, physical permission or railway authority is granted. Existing observation-only integration remains intact. The full Rust/browser/hardware suites are outside this targeted calculation change; verification results are reported with its commit.

Baghdad continuation: physical packages and revised financial conclusion

Source: docs/baghdad-continuation-review-2026-10-04.md

The delivery continuation completes the repository calculations and reviewable work packages following the first delivery reconciliation. Surveyed designs, quotations, actual demand, appointments, subscriptions and independent field acceptance remain open. The original 831-train, nine-line programme and factory opening dates are unchanged; new cases are explicit alternatives.

Financial finding

The full-fleet cost sensitivity replaces the USD 8m depot allowance with the USD 175.932m workload package and adds an explicit 7% incremental EPC allowance. Programme capital becomes USD 8.060bn, before still-unpriced scope. It replaces the original annual labour and solar allowances once, adds paid pre-opening mobilisation, and includes only replacement-reserve inflation shortfalls above contributions already in maintenance. Firm energy assumes physical upgrades whose capital is still unpriced.

Government covers 25%, USD 2.015bn equivalent. Its assumed imported-component downpayment is USD 913.123m; proposed Chinese credit is another USD 913.123m. Government's remaining cash, ordinary/green bonds, bank credit, supplemental credit, fares and commercial receipts are IQD. Total USD-denominated capital is USD 1.826bn, 22.7% of this scenario. At the historical model FX of IQD 1,300/USD this is about 89.9% below the requested USD 18bn all-foreign-currency comparator; distinct scope, price dates and unverified financing prohibit a like-for-like bid saving claim. Imported shares and Chinese eligibility require actual supplier/invoice evidence.

With existing variable pricing and annual 5% fare/OPEX/income growth, peak supplemental debt is IQD 11.360tn, and IQD 7.532tn remains outstanding at the 30-year full-network operating horizon. Before-finance company cash NPV is USD −4.692bn at the model's nominal discount rate. Surpluses repay USD 515.831m equivalent of eligible core principal early, after reserves/buffers and applicable call premiums, but do not eliminate the gap debt. Restricted maintenance cash cannot repay bonds or loans.

The monthly and six-month ledgers retain native currencies, bond-face/rounded-unit placement requirements, loan draws and principal/cash residuals. Conditional climate grants, rights proceeds, additional local income and cheap gap credit are uncommitted. A separate case fixes the original absolute government cash ceiling; another removes additional income and green pricing benefits. Opening-fleet and contracted-solar cases expose reduced receipts and transferred provider costs rather than treating them as free savings.

An initial IQD 1,750 base-fare sensitivity clears debt at month 378 after financial close, conditional on funding placement and the assumed demand elasticity. It still has negative company cash NPV, and 44 monthly trips consume 15.6% of the historical monthly income proxy before variable-price weighting. Higher-fare trials show repayment and affordability together; no tariff or commercial first-corridor selection is adopted. Actual OD, paid transfers, disposable-income distribution and concession funding remain required.

Prepared physical and operating packages

- Nine depot planning drawings label 533 storage positions, individual tracks/slots and 119 workload workshop bays. The yard-only slab production calculation meets cumulative installation deadlines using assumed mould cycles; its plant cost, curing qualification, reinforcement/foundations, turnouts, drainage, fire access and actual land remain open. Rectangular workshop packing is not an accepted accessible bay layout.
- Twenty-three six-car child-part/RFQ rows expand eight parent allocations within USD 1.68m/train. Unknown supplier quantities, prices, masses and drawings remain null. First article, LM3 applicability and qualification/design-training overlap remain unaccepted.
- Per-site chronological energy retains each installed storage/grid/charger limit. Reference imports are approximately 938.5GWh/year, with 6.6GWh unserved charging; firm energy services plus owned-solar maintenance cost USD 139.835m/year in the sensitivity. Synthetic allocation does not prove charger queues, charging events, outages or physical upgrade delivery.
- Task-based rolling maintenance and all 26 controlled intervals cover vehicles, stations, civil/track/structures, energy, wayside, tooling and finishes. Separate full/opening-fleet reserve ledgers show replacement dates, inflation top-ups and restricted cash without double-counting the replacement bill.
- Nine standalone opening-corridor cases compare operating break-even on a consistent basis; OD, actual catchments, shared dependencies and allocated debt remain unknown. Startup alternatives retain the frozen original dispatch points:

line 9 needs five extra revenue trains plus one spare, USD 10.08m vehicle reference before consequential costs, or an accepted smaller dispatch-point policy. Neither alternative closes the failed startup/access gate.

- Seven practical lesson cards and rest-limited anonymous pilot slots retain independent assessment and worker/task/location authorisation. No worker, assessor, permit or operational release is invented. Twelve scope-inclusion forms and supplier RFQs identify the remaining external inputs.

Native ERP and recovery evidence

The local ERP image was rebuilt with the workforce observer and bench-only administrative probes; the activated module hashes match repository files. Six existing DL90 Task descriptions were refreshed while preserving native Task identity, owner and operator-managed status. There are 34 Baghdad evidence Tasks in total; preparing or refreshing a Task accepts no evidence.

Five System Manager read probes each returned 54 permission-filtered rows from bounded native samples. The first took 0.455 seconds and subsequent samples approximately 0.008 seconds. This is a local read probe, not proof of complete-city throughput or the permissions of every production role. The workforce observer's permission and identity gates are separately tested.

Native Guest calls to the read probe, recovery fingerprint and workforce administration readiness were denied. These checks changed no worker, role, assignment or work-release record.

A native SQL backup was restored into a new MariaDB container with no network or published ports. Six data/schema/role/permission/File-inventory fingerprints matched; the temporary database was removed and the source database was never a restore destination. This verifies SQL recovery of the sampled inventories, not file decryption or production RTO/RPO. Private backups, import/deployment receipts, fingerprints and recovery receipts remain in ignored `var/erpnext/`; no native personal records or credentials enter the proposal archive.

Reproduction and acceptance

The checks below describe the preceding continuation. The CI and controls follow-up adds source-verified TLC, dependency, native-click and workforce fixes, earlier leadership/training cash and installed-throughput sensitivities.

Validation passed 1,747 Python tests and 12 subtests across city generation, components, engineering analysis, alignment/reference converters and automation. One Samawah planner test was skipped because its OSM cache is absent. Repository health, local Markdown links, source/output checks, proposal/archive checks and exact-input restoration in an empty temporary checkout passed. The full Rust, browser and target-hardware suites were not rerun for this continuation.

Run the exact-input bootstrap, then the delivery baseline/continuation generators and proposal builder in the documented dependency order. Both delivery generators and the publication support `--check`. The continuation binds source and output hashes; the proposal binds every packaged source and archive member. The complete proposal includes these packages and this review.

The six work packages remain field-open. Actual owners, surveyed parcels, supplier offers, OD/affordability evidence, Iraqi approvals, investor subscriptions and accepted engineering/operating evidence must replace the null fields. The delivered repository package is not a complete investment budget or railway release.

Baghdad review follow-up: controls, CI and deliverable-service sensitivity

Source: docs/baghdad-ci-controls-review-2026-10-04.md

> Historical revision review. Its figures describe the revision reviewed below. The reworked central alignment, current staffing/depot scope and current cashflows are in the Baghdad city summary and programme recalculation.

The attached review examined `aff4fbc9`; the subsequent continuation already integrated depot, energy, payroll and replacement-reserve cash. This follow-up strengthens that implementation and preserves the original financing case as a comparator. None of the engineering alternatives has field acceptance or committed finance.

Financing and service

The integrated monthly and six-month ledgers now include interim development leadership from NTP and separately purchased mentoring before opening. Existing contracted teaching/assessment is scheduled once. Added development/mentoring cash is IQD 2.582bn before inflation. The full-fleet sensitivity has USD 8.060bn capital, IQD 11.360tn peak gap debt, IQD 7.532tn terminal gap debt, and USD −4.692bn company cash NPV before finance. Approximately USD 515.831m equivalent of eligible core principal is prepaid after reserves, buffers and assumed call premiums; this does not clear gap debt.

Government remains 25% of scenario capital. Imports retain 50% government USD / 50% proposed Chinese USD credit. Remaining government cash, ordinary/green bonds, bank and gap credit, fares and commercial receipts remain IQD. There are no new commitments, subscriptions or guaranteed property proceeds. Existing variable fares and 5% annual fare/OPEX/income growth remain explicit assumptions. The IQD 1,750 sensitivity still clears debt in month 378, with negative company cash NPV and unverified demand/affordability.

The new installed-energy supply bound retains existing electrical capacity, removes purchases/connection charges for the unbuilt expansion, and reduces fares, station commercial receipts and additional commercial income. Synthetic installed charger/energy throughput is about 87.27% of allocated traction demand. Its terminal gap debt is approximately IQD 12.967tn, with IQD 2.787tn uncovered support. This is an energy-throughput upper bound; actual charging events, train circulation and a feasible timetable remain unaccepted. The firm-service case instead purchases shortage energy and requires unpriced physical upgrades. Neither may be described as a complete bankable budget or achieved service.

All integrated cases model zero dividends. Cash left after the debt/reserve/buffer waterfall stays in the company; audited after-tax distributable profit, covenant permission and shareholder approval remain unresolved. Earlier holding-company returns describe their original scope and must not be transferred to these higher-cost sensitivities.

Engineering and mobilisation

The 28 former pooled shortage hours remain available as a comparator. Site-hour diagnostics trace grid/charger limits, opening/closing stored energy and generation paths. On-site PV receives no remote wheeling loss or charge. The reference has 2,194 distinct hours with at least one bus shortfall; the larger site-hour count also includes charger overload. Those counts measure different scopes.

Retained workshop sites explicitly require line-local covered lifting, bogie exchange, wheel servicing and body overhaul. Eight zero-bay sites still need additional facilities and quotes; mobile tools do not replace covered bays, and a disconnected central workshop provides no train-transfer route. Road transport covers removed components only. The integrated case uses workload bays at all nine sites. The five missing morning-direction requirements remain an open gate: frozen-start alternatives require either an accepted smaller start-point policy or five revenue trains plus one spare and consequential engineering/funding.

The 450-train factory replay removes deferred orders and rebuilds factory resource queues, including the extra line-9 spare. Manufacturing selection, procurement cash, energy, operating payroll, maintenance and battery reserves follow the lower supply. Original expansion-capable factory/depot capital and civil opening dates remain funded; no earlier receipts or cheaper plant are invented. The case still has approximately IQD 6.525tn uncovered support, and deferred expansion remains unfunded.

Development and training schedules distinguish interim authorities, operating successors and signed handovers. Training leadership starts at month 18 and rigs must be ready by month 24, before the first operating cohorts join at month 34. Minimum purchased peak capacity is 19 teaching, 35 assessment and 90 mentoring posts, at different times. Operating recruits supply zero qualified training capacity. All 21 operating roles have lesson/coverage requirements; actual equipment, assessors, daily/rest rosters and named appointments remain absent. The anonymous OCC roster is only a pilot.

The editable metro-six-car price now feeds fleet procurement and the parent BOM allocation. Workforce conversion uses its configured FX reference; publication rejects inconsistent funding/workforce/corridor FX assumptions.

Workforce controls and CI corrections

The evaluator now checks task/location/asset/isolation scope, named authorised supervisors, qualified independent verifiers, correct calibrated tools and released part/revision/quantity. Duplicate stock records cannot count twice. Regression tests reject the review's combined wrong-line permit, missing supervisor and unqualified verifier example.

The native workforce preview validates a private attached snapshot. `snapshot_eligible` describes that snapshot; `eligible` and `live_start_check` remain false because current authoritative revocation is unresolved. Rechecking a timestamp detects expiry, not later revocation. Human task start still requires current authoritative evidence and work-release authority. No worker or assignment was created.

The six observed CI failures have corresponding changes:

- The Python test environment installs PyYAML. The hash-verified bootstrap restores declared energy/GIS/SUMO inputs and preserves differing workspace data.
- Portfolio browser expectations use the authoritative API and the UI's formatting rather than stale capital literals.
- Disabling unused postcard default features removes the heapless-CAS dependency path containing unmaintained `atomic-polyfill` and yanked `spin`; the wire fixture remains unchanged.
- TLC 1.8.0 is pinned to SHA256 `c2fe4e56e43bde19f213b4a7e441d037297fda733e503579623e859b79348239`, independently matched to upstream asset metadata and the published SHA1. A changed artifact must fail verification rather than run unverified.
- Workbench uses a viewport-sized flex layout instead of an oversized minimum frame. The native test dismisses an asynchronous ERP update notice through its visible close button when present. Proposal, review and verification retain real clicks without forced clicks.

5 October follow-up: upstream replaced the `v1.8.0/tla2tools.jar` asset with a different build, and CI correctly rejected its changed checksum. The retained TLC executable preserves the previously verified SHA256 and published SHA1 above. CI copies and verifies those exact bytes before executing the model; accepting a replacement requires a separate tool review and fresh assurance execution.

Local verification includes 124 focused tests, all 43 frontend acceptance tests, native disposition clicks with a deliberately obstructing Frappe notice, the source-bound TACS twin and bounded formal execution, and the Rust workspace/audit checks. The broad Python run passed 1,779 tests and twelve subtests, with one cache-dependent skip; its sole stale-offer failure was fixed and the four offer tests then passed. A further 124 publication/finance tests passed after regeneration. Source/output, archive, repository-health and link checks pass. Passing software checks provides no construction, railway-operation or investment release.

ERPNext now runs app 0.5.1 in the reproducible `osr14` image; deployed workforce/rules/probe hashes match repository files. Refreshing six packages updated 34 unique existing evidence Tasks, preserving native identity and operator-managed fields. A fresh SQL backup matched six inventory/schema/permission fingerprints in an isolated disposable database, which was removed. Private backups and receipts remain outside the publication.

For a source change, restore the exact ignored inputs first, then regenerate in this dependency order: Iraqi funding programme; delivery stress; viaduct rentals; qualification; financing redesign; equity; delivery baseline; delivery closure; reference offer; complete proposal; reader book. Source-bound checks reject stale intermediate packages. The proposal bootstrap also preserves differing local solver/GIS data rather than overwriting it.

Baghdad manufactured viaduct and cost review

Source: docs/baghdad-manufactured-viaduct-review-2026-10-04.md

The two supplied reviews cover a manufactured-viaduct proposal and the older `135e249e` cost baseline. The latter predates the integrated delivery and financing follow-up. Its USD 7.88bn is retained as a historical planning comparator; it is not a complete budget.

The new comparison package provides Pi20 and Pi25 with both structural-continuity and simple-span/link-slab schemes, plus an OSR-US constrained-access alternative. All use the same complete double-track comparison scope. The longest current Pi25 run is an investigation candidate; the separate 1 km study is not a surveyed alignment or selected project. No 30 m product or literature-based saving is introduced.

The civil load seed now distinguishes the 12-axle LM3 from Baghdad's six-car, 24-axle consist. Its 111 m length and tare/passenger masses reconcile with the controlled rolling-stock profile. Supplier axle positions and loaded axle distribution remain absent; repeating LM3 spacing would invent geometry. The 16 t infrastructure allowance must be applied with the complete loading pattern and adopted dynamic/combination rules.

The LM3 seed's passenger masses now use the same explicit 75 kg/person planning assumption: its previous AW2/AW3 masses silently used 60 kg. Reference axle geometry and both passenger-mass assumptions remain subject to deployment design acceptance; no physical load case is released by this correction.

Catalogue code now distinguishes a structural continuity connection from a link slab. A link slab retains eight bearings at internal twin-track piers; four shared bearings require the structural connection's staged load path. The finite support register includes both ends and reconciles with the shared catalogue function. Connections, CWR, thermal/braking/seismic/foundation forces, fatigue, replacement access and prices remain design/check releases. The existing lower civil rate is explicitly conditional on the unaccepted structural arrangement.

The comparison separates the standard allowance and routing penalty for every elevated segment, with an individual form for each exceptional alignment. Alternatives retain survey, wider curves, station movements, rights-of-way, segmental, utility/land, traffic and whole-life costs as unresolved fields. Removing a routing penalty is not an achieved saving. Existing capital and financing scenarios are preserved. A separate bearing-index cashflow sensitivity prices only existing standard Pi25 length, adds incremental EPC once and recalculates monthly/six-month funding, reserves and repayments. Government remains 25%; assumed imports retain 50% government USD / 50% Chinese USD credit, and other funding remains IQD. Invoice dates and origin shares are inherited assumptions; segmental/special designs receive no Pi25 bearing quantity or price credit. Connection and consequential costs remain unpriced.

Installed-price forms include quantities, invoice currency, price dates, origin, exclusions, supplier references and credit eligibility. Foundations need actual support ground zones and deep-element lengths. Unknown costs cannot produce a total. Ground cranes/portals and overhead launchers require configured proposals, complete member/rigging masses, route and support checks, temporary/deck loads and recovery plans. Pi25 and Pi20 have almost no bare-mass margin within their respective product targets.

The production model uses all eight fronts through survey/correction and final acceptance to calculate the slowest complete double-track bay rate. Missing cycles or resources keep that rate unknown. The existing casting cycle and 10–15-bay buffer remain planning assumptions pending measured trials. First article beam, pier/cap/connection, transport, erection and finished bay hold points provide the required measurement scope; no trial has been performed. UHPC connections and special composite crossings remain bounded alternatives, without carbon percentages being converted to cost savings.

The older review's depot/storage, hourly/site energy, six-car allocations, workforce, demand/fare sensitivities, rental alternatives and bootstrap work already have controlled outputs. Supplier-backed prices, site design, demand counts and financing commitments remain open. The artifact policy now names Baghdad's proposal/archive and distinguishes current, historical and diagnostic evidence. The front-page README includes nobody should be priced out of dignity.

Regenerate the comparison with `.venv/bin/python tools/automation/baghdad_viaduct_comparison.py`; use `--check` to verify source/output hashes. The complete proposal includes this study and its machine-readable inputs. Automated checks verify arithmetic and data boundaries; physical trials and licensed independent checks remain necessary.

Baghdad scope and industrial review — 4 October 2026

Source: docs/baghdad-scope-and-industrial-review-2026-10-04.md

> Historical review: numeric quantities and funding figures below describe the reviewed revision. Use the current city README, proposal and source-bound catalogue review for the regenerated planning basis.

> Historical revision review. Its figures describe the revision reviewed below. The reworked central alignment, current staffing/depot scope and current cashflows are in the Baghdad city summary and programme recalculation.

The programme recalculation is the latest Baghdad staffing, depot, procurement-origin and financing study. The complete proposal publishes it alongside its source files, twelve monthly cashflows, contract registers and six-month placement schedules. It supersedes the earlier staffing/depot assumptions for this study; the original catalogue design, generic city pages and financing cases remain separately identified comparators. No national borrowing or future factory sales are credited to Baghdad.

People and operating coverage

There are two concurrent station posts during two normal eight-hour shifts. That gives 728 daily shift assignments for 182 stations, rather than 728 employees covering every day. The retained 05:30–02:00 service requires an additional 4.5 hours of late cover. After weekly, leave, training, sickness and handover allowances, station cover requires 1,800 FTE within 3,762 permanent railway FTE. Role, line and recruitment-cohort registers reconcile payroll and pre-opening training. A shorter station opening day would require a separate service/revenue scenario.

The pay basis uses the employee median of IQD 614,000 in the 2021 Labour Force Survey, reported by the IMF. An assumed 5% annual index gives the editable 2026 proxy; the general grade is 1.5 times that proxy, and technical, supervisor, senior and director grades are 2.25, 3, 4 and 5 times it. Employer costs and overtime are explicit. This is not a measured current median, and the old household-income proxy is not substituted for employee wages or altered to improve fare affordability.

Construction, train manufacture, upstream component manufacture and permanent operations are separate establishments. The construction register is a configured work-package crew screen; it is not a complete contractor payroll or a schedule for the new depot/factory sites. Contract rates already contain labour. A separate labour-content stress tests an assumed 20% content with 50% wage uplift and incremental EPC once, without inventing a measured overlap credit. Factory payroll funds the full establishment through the actual scheduled order; productive labour in unit costs is deducted before adding capacity payroll.

Depots and manufacture

Each of the nine lines receives one depot sized for its full six-car fleet, including spares and cold reserve. Storage totals 831 slots at 121 m per train position, using the actual 111 m train. Workshop bays follow annual bay-hour demand independently of storage tracks. The item register prices storage/workshop/access tracks, turnouts, drainage/access, workshop shells/equipment/services, lighting/fire services, offices/stores, wash plants, wheel lathes and isolation/rescue. Existing depot PV/BESS is counted once. Land, utilities, actual ground conditions, connected access alignments and installed charging/grid upgrades remain open; depot launch/throat conflicts still require a full-service replay.

The component study follows the actual six-car BOM quantities: 9,972 bogies and motor/inverter sets each, 4,986 battery packs, 19,944 door cassettes and 29,916 window cassettes. Process cells match the train factory's required rate and include Chinese process machinery, Iraqi facilities/materials, labour, support, maintenance, qualification and residual imported inputs. Bogie fabrication, motor assembly, battery-pack assembly and door manufacture show positive whole-order margins under the unquoted inputs. Window manufacture has a negative Baghdad-only margin; future national orders are not assumed to rescue it. Imported cells/BMS and other critical parts remain. Battery pack assembly does not imply local cell manufacture.

Serial qualification within the 18-month facility target remains an assumption. A six-month supplier-delay case shifts rolling-stock invoices, line openings and operating receipts, prices idle payroll and preserves the full operating horizon. A 25% raw-input price stress tests the same selected facilities. License, yield/rework, tooling overlap and actual vendor quotations require evidence before investment.

Alignment and finance

The study permits a 40% elevated ceiling and screens 29.148 km of candidate additional viaduct around exceptional segments, reaching 20.32% elevated. Intervals include approach allowances, merge overlap and exclude existing elevated/bridge lengths. Elevation alone removes no horizontal bend. Wider-radius geometry, station moves, land, gradients, egress, geotechnical and utility constraints need design work. The 0/25/50% routing-penalty-removal cases are counterfactuals; no penalty reduction is adopted as an achieved saving.

Government funds exactly 25% of each case's capital. Imported invoices are split 50% government USD cash and 50% proposed Chinese USD credit. The remaining government appropriation and all domestic bonds, bank/gap debt and mezzanine are IQD. Machinery-heavy months follow their invoice downpayments; the total government contribution, rather than each month's share, is constrained to 25%. Foreign procurement intensity, foreign debt and reporting equivalents are separate measures. The old 148 km / USD 18bn third-party example retains its distinct scope and assumed foreign-currency funding; it is not a matched tender comparison.

Mezzanine substitutes for 10% of domestic residual capital borrowing. The tested terms are project assumptions: 6% cash coupon, 4% PIK, 2% fee, 15-year balloon and senior DSCR 1.20 before junior cash payments. Deferred coupons and PIK increase debt until maturity; unpaid balloons become overdue balances, with simple-interest sensitivity and no automatic refinancing or conversion. Gap draws or unsourced support cannot fund junior distributions. Source/use, senior principal, junior principal/PIK and overdue-interest identities are checked independently. Semiannual schedules report flows separately from closing stocks and distinguish face-denomination rounding from actual cash raised.

The revised senior case still has unsourced support and unpaid debt at the horizon. Mezzanine reduces immediate funding pressure by deferring payments but increases terminal debt and leaves 85 junior draw vintages in default. It is not recommended as the solution to the operating deficit. Indexed fares, OPEX inflation, advertising/kiosks/rentals and inherited additional receipts are already included. Green terms, grants, rights income and 2% concessional gap credit remain uncommitted; the 8% gap-rate stress shows that financing dependence. There are no dividends. Verified capital, affordable revenue or accepted scope savings are still needed.

Reproduction and release limits

The city landing page is published from the current case rather than the earlier catalogue budget. The city regeneration, batch generation and controls-refresh commands use publish-city-summary.py. Its registry identifies the current study and seventeen earlier reference documents. Original finance, depot, staffing, national and portfolio figures remain traceable with an explicit original-scope notice; document receipt updates do not change financial values or release flags. A stale study stops publication. CI checks the committed page before regeneration, then verifies the publisher after national-brief regeneration. The reader book includes the current city summary and current study before the original funding appendices.

- Assumptions: baghdad-programme-recalculation.toml.
- Generate: `.venv/bin/python tools/automation/baghdad_programme_recalculation.py`; verify with `--check`.
- Cash engine: baghdad_recalculation_finance.py.
- Tests: test_baghdad_programme_recalculation.py.
- Earlier comparators: delivery continuation and original funding programme.

The ERP payload contains planning drafts for staffing, depot projects/assets and factory workstations. It creates no actual employees, appointments, financing commitments, operational transactions or permissions. Survey, quotations, demand/timetable acceptance, wage data, manufacturing qualification, complete installed budgets and executed funding terms remain release gates.